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BASICS of SQL

WE ARE HAVING TOTAL 5 DIFFERENT TYPES OF STATEMENTS IN SQL

- 1) DDL (DATA DEFINITION LANGUAGE) STATEMENTS
 - CREATE
 - ALTER
 - DROP
 - RENAME
 - TRUNCATE
- 2) DML (DATA MANIPULATION LANGUAGE) STATEMENTS
 - INSERT
 - UPDATE
 - DELETE
 - MERGE (INSERT/UPDATE/DELETE)
- 3) DCL (DATA CONTROL LANGUAGE) STATEMENTS
 - GRANT
 - REVOKE
- 4) TCL (TRANSACTION CONTROL LANGUAGE) STATEMENTS
 - COMMIT
 - ROLLBACK
 - SAVEPOINT
- 5) DRL (DATA RETRIEVAL LANGUAGE) STATEMENTS
 - SELECT

➤ **To connect to the SQL server** = SQL> connect system/sai369baba;

➤ **Clear the screen** = SQL> CL SCR

➤ **To see the Orders** = SQL> Describe orders;

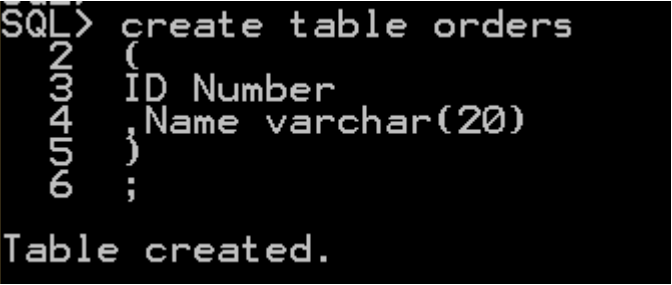
➤ **To create the table**

SQL> CREATE TABLE B26

```

2  (
3  ID NUMBER
4  ,NAME CHAR
5  );

```



```

SQL> create table orders
2  (
3  ID Number
4  ,Name varchar(20)
5  )
6  ;

Table created.

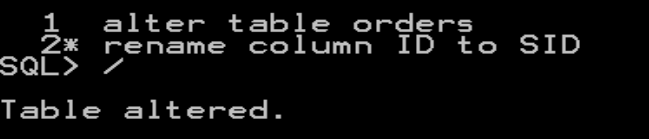
```

➤ **Add column in the table =**

SQL> alter table orders;

2 add gender varchar2(1);

➤ **To RENAME the Column Name .**



```

1  alter table orders
2* rename column ID to SID
SQL> /

Table altered.

```

➤ **To update the new row =**

insert into orders

values

(
8

, 'muthoot'

, 'chennai'

, 'm'

)

;

```
SQL> select *from orders;
```

ID	NAME	CITY	G
8	muthoot	chennai	M
6	pavani	blr	F
3	subha	hyd	f
4	rama	rajamundry	m
5	buss	hyd	

➤ **Modify - Change the Data type** of the column

```
SQL> alter table orders
2 modify name varchar2(20);
```

table altered.

```
SQL> desc orders
```

Name	Null?	Type
ID		NUMBER
NAME		VARCHAR2(20)
AGE		NUMBER(2)
FEES		NUMBER(6)
MAILID		VARCHAR2(20)

➤ **Remove column in the table =**

```
SQL> alter table orders
```

```
2 drop column gender;
```

➤ **to save the file of the commands =** commit;➤ **To delete the particular row =**

```
SQL> delete from orders
```

```
2 where id = 1;
```

➤ **To see the total table =** select *from orders ;➤ **To Update the row =**

Update orders

set city = 'rajamundry',

where ID =4 ;

➤ **To Delete the Table =** Drop table Orders;➤ **To update the specific Rows in the full column**

```
SQL> select *from Orders;
-----
ID  NAME          AGE  FEES
-----
1  Krishna      30   2000
2  pavani       28   5000
3  Rama         45   3000
4  subha        48   4000

SQL> update orders
2  set age=28,fees=5000
3  where Id=2;

1 row updated.

SQL> select*from orders;
-----
ID  NAME          AGE  FEES
-----
1  Krishna      30   2000
2  pavani       28   5000
3  Rama         45   3000
4  subha        48   4000
```

➤ Alias =AS

this is changing the header to the different name

```
NAME
-----
Krishna
pavani
Rama
subha

SQL> select name as sname from orders;

SNAME
-----
Krishna
pavani
Rama
subha
```

➤ Multiplication in the SQL

```
SQL> select *from orders;
-----
ID  NAME          AGE  FEES
-----
1  Krishna      30   2000
2  pavani       28   5000
3  Rama         45   3000
4  subha        48   4000
```

```
SQL> select fees, fees*12,fees+1000
2  from orders;

-----
FEES  FEES*12  FEES+1000
-----
2000  24000    3000
5000  60000    6000
3000  36000    4000
4000  48000    5000
```

Creating the sequences

```
SQL> create sequence s1
2 start with 1
3 increment by 1
4 minvalue 1
5 maxvalue 30;

Sequence created.
```

Using of the sequence commands

```
SQL> insert into orders
2 values
3 (
4 s1.nextval, '&name'
5 )
6 ;
```

➤ WHAT IS DUAL ?

DUAL IS A ORACLE PREDIFINED TABLE

IT CONTAINS ONE ROW AND ONE COLUMN ALWAYS

ROW VALUE WILL BE "X" AND COLUMN NAME WILL BE DUMMY

OWNER OF DUAL TABLE IS SYS USER

WE CAN PERFORM ONLY SELECT ON DUAL TABLE OTHER THAN THAT WE CAN NOT PERFORM ANY OTHER OPERATION (INSERT..UPDATE..DELETE..ETC IS NOT ALLOWED)

WHY WE NEED A DUAL ?

WHNEVER WE WANT TO SELECT SOMETHING WHICH IS NOT DEPEND ON ANY ONE OF BUSINESS TABLE THEN WE WILL GO FOR DUAL TABLE

```
SQL> select 'today date is'
2 ,sysdate as c1
3 from dual
4 ;

'TODAYDATEIS' C1
-----
today date is 12-MAY-18
```

```
1 SELECT 'TODAY DATE IS '
2 ||SYSDATE AS C1
3* FROM DUAL
SQL> /
```

C1

TODAY DATE IS 12-MAY-18

```
SQL> select 'todays date is' ||sysdate as c1
2 from dual;

C1
-----
todays date is12-MAY-18
```

Example

```
SQL> select 'Hi '
2      || 'your id is '
3      || id
4      || 'and your name is '
5      || name as wishes
6      from orders;

WISHES
-----
Hi your id is 1and your name is krishna
Hi your id is 2and your name is pavani
Hi your id is 3and your name is rama
Hi your id is 4and your name is subhardra
Hi your id is 5and your name is bhavani
```

```
1  UPDATE ORDERS
2  SET DOJ = '&DOJ'
3* WHERE ID = &ID
SQL> /
Enter value for doj: 11-NOV-18
old   2: SET DOJ = '&DOJ'
new   2: SET DOJ = '11-NOV-18'
Enter value for id: 2
old   3: WHERE ID = &ID
new   3: WHERE ID = 2
```

NULL Propagation Table**4-May-2018- ORACLE Basics class**

ORACLE IS A DATA BASE WHICH IS USED TO STORE THE DATA

ORACLE 2 TYPES OF ROLES

ADMINISTRATOR -- INSGTALLING ORACLE SOFTWARE,UPGRADING ORACLE SOFTWARE,DB BACKUP,RECOVERY DB..ETC

DEVELOPER-- WHO IS WRITTING SOME SQL AND PLSQL SCRIPTS TO FULL FILL THE CUSTOMER REQUIREMENT

(SQL + PLSQL +UNIX+SHELL SCRIPTING)

IF YOU WANT TO STORE THE DATA WE ARE HAVING LANGUAGE CALLED SQL (STRUCTURED QUERY LANGUAGE)

WE ARE HAVING TOTAL 5 DIFFERENCT TYPES OF STATEMENTS IN SQL

1) DDL(DATA DEFINITION LANGUAGE) STATEMENTS

CREATE

ALTER

DROP

RENAME

- TRUNCATE
- 2) DML (DATA MANIPULATION LANGUAGE) STATEMENTS
 - INSERT
 - UPDATE
 - DELETE
 - MERGE (INSERT/UPDATE/DELETE)
- 3) DCL (DATA CONTROL LANGUAGE) STATEMENTS
 - GRANT
 - REVOKE
- 4) TCL (TRANSACTION CONTROL LANGUAGE) STATEMENTS
 - COMMIT
 - ROLLBACK
 - SAVEPOINT
- 5) DRL (DATA RETRIEVAL LANGUAGE) STATEMENTS
 - SELECT

IN ANY DATABASE DATA WILL BE STORED IN TABLE FORMAT
TABLES CONSISTS OF ROWS AND COLUMNS

IF YOU WANT TO DEFINE STRUCTURE OF THE TABLE (I.E WHAT COLUMNS AND DATA TYPE IT SHOULD CONTAIN) WE HAVE TO USE CREATE COMMAND.

ALL DDL AND DCL COMMANDS CHANGES ARE AUTO SAVED (AUTO COMMIT)
ALL DML COMMANDS ARE NOT AUTO SAVED (NOT AUTO COMMIT)

Questions :

DIFF BETWEEN ALTER AND UPDATE
DIFF BETWEEN DELETE AND TRUNCATE
DIFF BETWEEN DROP AND TRUNCATE

5-May-2018 - Create table Notes

RULES TO CHOOSE THE TABLENAME :

- 1) IT SHOULD CONTAIN MAXIMUM 30 CHARACTERS
- 2) IT CAN BE A-Z a-z 0-9 _ ^ \$ # ..etc
- 3) it should start always with alphaebets
 - Student_1 -- right
 - 1student -- wrong
 - _student -- wrong --error
- 4) IT IS NOT CASE SENSITIVE
 - STUDENT =student -- both are same
- 5) in one user 2 objects names (table names) should not be same
- 6) IT SHOULD NOT CONTAIN SPACES
 - STUDENT INFORMATION -- NOT ALLOWED
 - STUDENT_INFORMATION -- ALLOWED
- 7) DONT USE ORACLE RESEREWED WORDS.

CREATE -- NOTALLOWED
INSERT -- ALLOWED

- 8) 3 4 6 7 BY DEFAULT NOT ALLOWED
BUT IF YOU KEEP WITH IN " " IT IS ALLOWED

```

1STUDENT --NOT ALLOWED
"1STUDENT"-- ALLOWED
Student-- you can refer however we want STUDENT ,STudent,STudenet
..same
"Student"-- then you should refere alwas "Student"

STUDENT INFORMATION--NOT ALLOWED
"STUDENT INFORMATION"-- ALLOWED

create -- not allowed
"CREATE"--allowed

```

but 8th point as per realtime coding standars not recomented

9) your table name should be as short as possible.

ALL ABOVE RULES ARE COMMON TO COLUMN NAMES ALSO
BUT HERE IN ONE TABLE WE CAN NOT HAVE 2 COLUMN NAMES SAME IRRESEPTIVE OF
THE DATATYPE.

EACH COLUMN SHOULD BE SEPARATED BY ,

PLSQL_STUDENT_INFORMATION

CREATE TABLE PLSQL_STD_INFO

```

(
  ID      NUMBER
,NAME     VARCHAR2(30)
,MOB      NUMBER(10)
,FEES     NUMBER(7,2)
,GEN      CHAR(1)
,DOJ      DATE
)

```

2 TO 4 YEARS--

FAKE-

DATA TYPES

NUMBER

IT IS USED TO STORE ONLY NUMBER(0-9)

IT IS HAVING WIDTH PROPERTY(HOW MANY DIGITS CAN BE STORED)

NUMBER(10)-- MAXIMUM 10 DIGITS WE CAN STORE

BUT IT IS OPTINAL

NUMBER-- BY DEFAULT WE CAN STORE UPTO 38 DIGITS

NUMBER(P1,P2)

IN TOTAL P1 HOW MANY ARE FOR DECIMALS(P2)

NUMBER(7,2)

IN TOTAL 7 DIGITS 2 FOR DECIMALS

99999.99--ACCEPTED

9999999.99--BIT AKKIED

CHAR

IT IS USED TO STORE ALPHA NUMERICS AND SPECIAL CHARACTERS

A-Z,0-9-_^&*..ETC

WIDTH PROPERTY IS MANDATORY
MAXIMUM WE CAN STORE 2000 CHARACTERS/BYTES
IT IS A FIXED LENGTH MEMORY ALLOCATION

VARCHAR2

IT IS USED TO STORE ALPHA NUMERICS AND SPECIAL CHARACTERS
A-Z,0-9-_^&*..ETC
WIDTH PROPERTY IS MANDATORY
MAXIMUM WE CAN STORE 4000 CHARACTERS/BYTES
IT IS A VARIABLE LENGTH MEMORY ALLOCATION

DIFF BETWEEN CHAR AND VARCHAR2 ?

LONG

IT IS NOT HAVING ANY WIDTH PROPERTY
IT IS USED TO STORE ALPHA NUMERICS AND SPECIAL CHARACTERS
A-Z,0-9-_^&*..ETC
MAXIMUM 8000 CHARACTERS/BYTES WE CAN STORE

CLOB

CHARACTER LARGE OBJECT
IT IS NOT HAVING ANY WIDTH PROPERTY
IT IS USED TO STORE ALPHA NUMERICS AND SPECIAL CHARACTERS
A-Z,0-9-_^&*..ETC
HERE WE CAN STORE -23GB TO +23 GB (THIS IS THE MAXIMUM WE CAN STORE)

BLOB

BINARY LARGE OBJECT
IT IS NOT HAVING ANY WIDTH PROPERTY
IT IS USED TO STORE YOUR IMAGES..PDF FILES..ECT

DATE

IT IS NOT HAVING ANY WIDTH PROPERTY
DEFAULT DATE FORMAT WHICH WILL ACCEPT IS DD-MON-YY
WHENEVER WE WANT TO STORE
DATES (DOJ, DOB, DOP, PA+GI, P_E_d...ECT, ORDER_DATE, P_A_D, P_D_D, T_A_T)

PSI***

CREATE TABLE

SERVICE BASED

CUSTOMER (CLIENT)

BUSINESS ANALYST-- CUSTOMER (CLIENT) --

REQUIREMENT DOCUMENT--

DATA ARCHITECT -- (TECHNICAL DESIGN DOCUMENT)

TABLES -- DATA TYPES

DEVELOPER (PLSQL DEVELOPER)

WE WILL WRITE A CODE TO POPULATE DATA INTO THOSE TABLES

CODE REVIEW--TEAM LEAD (TEAM COLLEAGUES)

UNIT TESTING (UNIT TESTING DOCUMENT)

TECHNICAL DESIGN DOCUMENT _UPDATE

WE WILL BE SENDING DETAILS TO QA TEAM (TESTING TEAM) -- VERSION CONTROL TOOL
(VSS, SVN...ETC)

QA WILL TEST YOUR CODE

IF IT IS WORKING AS PER REQUIREMENT THEN THEY WILL GIVE SIGN OFF. (TEST
RESULTS)

IT IS NOT WORKING FINE THEN THEY WILL RAISE BUG (CASE..)

YOU HAVE TO FIX THAT BUT AGAIN ASK QA TO TEST

TO TYPE/EXECUTE YOUR SQL/PLSQL COMMAND WE HAVE 2 TYPES OF EDITORS

1) COMMAND PROMPT (CLASS ROOM)

2) GUI TOOL (WE WILL BE USING IN REAL TIME)

GRAPHICAL USER INTERFACE

TOAD

SQL DEVELOPER TOOL

PLSQL DEVELOPER TOOL

ONCE ORACLE SOFTWARE INSTALLED

START -- ALLPOROMGRAMS-- ORACLE SOFTWARE -- RUN SQL COMMAND LINE--RIGHT
CLICK ON RUN SQL COMMANDLINE
CHOOSE RUN AS ADMINISTRATOR

YES/NO--YES
IT WILL OPOEN COMMAND PROMPT

CONNECT SYSTEM/ARJUN
TO SEE WHICH USER YOU CONNECTED
SHOW USER

SPOOL

CREATING A FILE IN SPECIFIC DIRECTORY/FOLDER
WHATEVER WE TYPE ON COMMAND PROMPT ALL COMMANDS/STATEMETNS WILL BE SAVED IN
THAT FILE

SPOOL FOLDERPATH\FILENAME.DOC

SPOOL FILE IS A NOT A DATABSSE
IT IS A ENGLHST NOTES

HOW TO PASTE IN COMMAND PORMPT

CTRL + V
RIGHT CLICK
ALT + SPACEBAR + E+ P

CL SCR -- TO CLEAR THE SCREEN

DESC TABLENAME-- TO SEE THE TABLE STRUCTURE
DESCRIBE TABLENAME-- TO SEE THE TABLE STRUCTURE

EDIT

IN COMMAND PROMPT ALWAYS IN BUFFER JUST PREVIOUS EXECUTED COMMAND WILL BE THERE

IF YOU WANT TO SEE WHAT IS THERE IN BUFFER YOU HAVE TO SAY EDIT AND ENTER IT WILL OPEN ONE NOTEPAD(BUFFER) AND YOU CAN SEE PREVIOUS SQL COMMAND

CHANGE WHAT EVEREVER YOU WANT AND SAVE AND CLOSE
NOW YOU CAN SEE CHANGED QUERY IN COMMAND PROMPT

NOW IF YOU WANT TO RUN WHAT IS THERE IN BUFFER(SAVED QUERY)
R-- IT WILL WRITE AND EXECUTE
/-- IT WILL JUST EXECUTES

5-May-2018 -create table Exa

```
SQL> CREATE TABLE B26_AKS_STD
  2  (
  3
SQL> EDIT
Wrote file afiedt.buf
```

```
  1  CREATE TABLE B26_AKS_STD
  2  (
  3      ID          NUMBER
  4      ,NAME       VARCHAR2(50)
  5      ,MOB        NUMBER(10)
  6      ,FEES       NUMBER(7,2)
  7      ,DOJ        DATE
  8      ,GEN        CHAR(1)
  9* )
SQL> CREATE TABLE B26_AKS_STD
  2  (
  3      ID          NUMBER
  4      ,NAME       VARCHAR2(50)
  5      ,MOB        NUMBER(10)
  6      ,FEES       NUMBER(7,2)
  7      ,DOJ        DATE
  8      ,GEN        CHAR(1)
  9  );
```

Table created.

```
SQL> DESCRIBE B26_AKS_STD
```

Name	Null?	Type
ID		NUMBER
NAME		VARCHAR2(50)
MOB		NUMBER(10)
FEES		NUMBER(7,2)
DOJ		DATE
GEN		CHAR(1)

```
SQL> NOT NULL(CONSTRAINTS ) 2 TO 2.30
```

SP2-0734: unknown command beginning "NOT NULL(C..." - rest of line ignored.

```
SQL> CL SCR
```

```
SQL> CREATE TABLE B26_AKS_STD_1
  2  (
  3      ID NUMBER
  4      ,NAME VARCHAR2(20)
```

```

5 ,MOB NUMBER(10)
6 ,GEN CHAR(1)
7 ,DOJ DATE
8 PAN VARCHAR2(10)
9 );
PAN VARCHAR2(10)
*
```

ERROR at line 8:
ORA-00907: missing right parenthesis

```

SQL> CREATE TABLE B26_AKS_STD_1
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(20)
5 ,MOB NUMBER(10)
6 ,GEN CHAR(1)
7 ,DOJ DATE
8 ,PAN VARCHAR2(10)
9 );
,MOB NUMBER(10)
*
```

ERROR at line 5:
ORA-00907: missing right parenthesis

```

SQL> EDIT
Wrote file afiedt.buf
```

```

1 CREATE TABLE B26_AKS_STD_1
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(20)
5 ,MOB NUMBER(10)
6 ,GEN CHAR(1)
7 ,DOJ DATE
8 ,PAN VARCHAR2(10)
9* )
SQL> /
```

Table created.

```

SQL> CL SCR
```

```

SQL> /
CREATE TABLE B26_AKS_STD_1
*
```

ERROR at line 1:
ORA-00955: name is already used by an existing object

```

SQL> R
1 CREATE TABLE B26_AKS_STD_1
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(20)
5 ,MOB NUMBER(10)
6 ,GEN CHAR(1)
7 ,DOJ DATE
8 ,PAN VARCHAR2(10)
9* )
```

```

CREATE TABLE B26_AKS_STD_1
*
```

ERROR at line 1:

ORA-00955: name is already used by an existing object

SQL> CL SCR

SQL> R

```
1 CREATE TABLE B26_AKS_STD_1
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(20)
5 ,MOB NUMBER(10)
6 ,GEN CHAR(1)
7 ,DOJ DATE
8 ,PAN VARCHAR2(10)
9* )
```

```
CREATE TABLE B26_AKS_STD_1
*
```

ERROR at line 1:

ORA-00955: name is already used by an existing object

SQL> EDIT

Wrote file afiedt.buf

```
1 CREATE TABLE B26_AKS_STD_2
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(20)
5 ,MOB NUMBER(10)
6 ,GEN CHAR(1)
7 ,DOJ DATE
8 ,PAN VARCHAR2(10)
9* )
```

SQL> CL SCR

SQL> R

```
1 CREATE TABLE B26_AKS_STD_2
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(20)
5 ,MOB NUMBER(10)
6 ,GEN CHAR(1)
7 ,DOJ DATE
8 ,PAN VARCHAR2(10)
9* )
```

Table created.

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```
1 CREATE TABLE B26_AKS_STD_3
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(20)
5 ,MOB NUMBER(10)
6 ,GEN CHAR(1)
7 ,DOJ DATE
8 ,PAN VARCHAR2(10)
9* )
```

SQL> CL SCR

```
SQL> /
```

```
Table created.
```

```
SQL> R
```

```
1 CREATE TABLE B26_AKS_STD_3
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(20)
5 ,MOB NUMBER(10)
6 ,GEN CHAR(1)
7 ,DOJ DATE
8 ,PAN VARCHAR2(10)
9* )
```

```
CREATE TABLE B26_AKS_STD_3
*
```

```
ERROR at line 1:
```

```
ORA-00955: name is already used by an existing object
```

```
SQL> CL SCR
```

```
SQL> R
```

```
1 CREATE TABLE B26_AKS_STD_3
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(20)
5 ,MOB NUMBER(10)
6 ,GEN CHAR(1)
7 ,DOJ DATE
8 ,PAN VARCHAR2(10)
9* )
```

```
CREATE TABLE B26_AKS_STD_3
*
```

```
ERROR at line 1:
```

```
ORA-00955: name is already used by an existing object
```

```
SQL> CL SCR
```

```
SQL> /
```

```
CREATE TABLE B26_AKS_STD_3
*
```

```
ERROR at line 1:
```

```
ORA-00955: name is already used by an existing object
```

```
SQL> CL SCR
```

```
SQL> CREATE TABLE B26_AKS_STD_B26_AKS_STD_B26_AKS_STD_B26_AKS_STD
```

```
2 (
3 ID NUMBER
4 ,NAME VARAR2(10)
5 ,MOB NUMER(10)
6 ,DOJ DTE
7 ,FEES NUMBER(7,2)
8 );
```

```
CREATE TABLE B26_AKS_STD_B26_AKS_STD_B26_AKS_STD_B26_AKS_STD
*
```

```
ERROR at line 1:
```

```
ORA-00972: identifier is too long
```

```
SQL> EDIT
```

Wrote file afiedt.buf

```
1 CREATE TABLE B26_AKS_STD_B26_AKS_STD_B26_AKS_STD
2 (
3 ID NUMBER
4 ,NAME VARAR2(10)
5 ,MOB NUMER(10)
6 ,DOJ DTE
7 ,FEES NUMBER(7,2)
8* )
SQL> R
1 CREATE TABLE B26_AKS_STD_B26_AKS_STD_B26_AKS_STD
2 (
3 ID NUMBER
4 ,NAME VARAR2(10)
5 ,MOB NUMER(10)
6 ,DOJ DTE
7 ,FEES NUMBER(7,2)
8* )
CREATE TABLE B26_AKS_STD_B26_AKS_STD_B26_AKS_STD
*
ERROR at line 1:
ORA-00972: identifier is too long
```

SQL> EDIT

Wrote file afiedt.buf

```
1 CREATE TABLE B26_AKS_STD_B26_AKS_STD_B26
2 (
3 ID NUMBER
4 ,NAME VARAR2(10)
5 ,MOB NUMER(10)
6 ,DOJ DTE
7 ,FEES NUMBER(7,2)
8* )
SQL> /
,NAME VARAR2(10)
*
ERROR at line 4:
ORA-00907: missing right parenthesis
```

SQL> EDIT

Wrote file afiedt.buf

```
1 CREATE TABLE B26_AKS_STD_B26_AKS_STD_B26
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(10)
5 ,MOB NUMER(10)
6 ,DOJ DTE
7 ,FEES NUMBER(7,2)
8* )
SQL> /
,MOB NUMER(10)
*
ERROR at line 5:
ORA-00907: missing right parenthesis
```

SQL> EDIT

Wrote file afiedt.buf


```

1 CREATE TABLE B26_AKS_STD_B26_AKS_STD_B26
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(10)
5 ,MOB NUMBER(10)
6 ,DOJ DTE
7 ,FEES NUMBER(7,2)
8* )

```

SQL> /

```

,DOJ DTE
*
```

ERROR at line 6:

ORA-00902: invalid datatype

SQL> ED

Wrote file afiedt.buf

```

1 CREATE TABLE B26_AKS_STD_B26_AKS_STD_B26
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(10)
5 ,MOB NUMBER(10)
6 ,DOJ DATE
7 ,FEES NUMBER(7,2)
8* )

```

SQL> /

Table created.

SQL> R

```

1 CREATE TABLE B26_AKS_STD_B26_AKS_STD_B26
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(10)
5 ,MOB NUMBER(10)
6 ,DOJ DATE
7 ,FEES NUMBER(7,2)
8* )

```

```

CREATE TABLE B26_AKS_STD_B26_AKS_STD_B26
*
```

ERROR at line 1:

ORA-00955: name is already used by an existing object

SQL> CL SCR

SQL> USER_TABLES OR DBA_TABLES OR ALL_TABLES

SP2-0734: unknown command beginning "USER_TABLE..." - rest of line ignored.

SQL>

SQL>

```

SQL> SELECT TABLE_NAME
2 FROM USER_TABLES
3 WHERE TABLE_NAME LIKE '%B26%';

```

TABLE_NAME

```

B26_AKS_STD
B26_AKS_STD_1
B26_AKS_STD_2
B26_AKS_STD_3
B26_AKS_STD_B26_AKS_STD_B26

```

SQL> EDIT

Wrote file afiedt.buf

```
1  SELECT TABLE_NAME
2  FROM USER_TABLES
3* WHERE TABLE_NAME LIKE '%B26%'
SQL>
SQL> CL SCR
```

```
SQL> CREATE TABLE 4_B26_AKS_STD
2  (
3  ID NUMBER
4  ,NAME VARCHAR2(20)
5  );
CREATE TABLE 4_B26_AKS_STD
*
ERROR at line 1:
ORA-00903: invalid table name
```

SQL> ED
Wrote file afiedt.buf

```
1  CREATE TABLE _B26_AKS_STD
2  (
3  ID NUMBER
4  ,NAME VARCHAR2(20)
5* )
SQL> /
CREATE TABLE _B26_AKS_STD
*
ERROR at line 1:
ORA-00911: invalid character
```

SQL> ED
Wrote file afiedt.buf

```
1  CREATE TABLE B26_AKS_STD_4
2  (
3  ID NUMBER
4  ,NAME VARCHAR2(20)
5* )
SQL> /
```

Table created.

SQL> ED
Wrote file afiedt.buf

```
1  CREATE TABLE B26 AKS STD
2  (
3  ID NUMBER
4  ,NAME VARCHAR2(20)
5* )
SQL> /
CREATE TABLE B26 AKS STD
*
ERROR at line 1:
ORA-00922: missing or invalid option
```

SQL> ED
Wrote file afiedt.buf

```

1 CREATE TABLE "B26 AKS STD"
2 (
3 ID NUMBER
4 ,NAME VARCHAR2(20)
5* )
SQL> /

```

Table created.

```

SQL> DESCRIBE B26 AKS STD
Usage: DESCRIBE [schema.]object[@db_link]

```

```
SQL>
```

```
SQL>
```

```
SQL> DESCRIBE "B26 AKS STD"
```

Name	Null?	Type
-----	-----	-----
ID		NUMBER
NAME		VARCHAR2(20)

```
SQL> SELECT ENAME
```

```

2 ,SAL
3 ,COMM
4 ,SAL + COMM AS TOTINCOME
5 FROM EMP;

```

ENAME	SAL	COMM	TOTINCOME
-----	-----	-----	-----
KING	5000		
BLAKE	2850		
CLARK	2450		
JONES	2975		
SCOTT	3000		
FORD	3000		
SMITH	800		
ALLEN	1600	300	1900
WARD	1250	500	1750
MARTIN	1250	1400	2650
TURNER	1500	0	1500
ADAMS	1100		
JAMES	950		
MILLER	1300		
PUNAM	4200		
NULL	2200		

16 rows selected.

```
SQL> EDIT
```

Wrote file afiedt.buf

```

1 SELECT ENAME
2 ,SAL
3 ,NVL (COMM
4 ,10
5 )
6 ,SAL + COMM AS TOTINCOME
7* FROM EMP
8 /

```

ENAME	SAL	NVL (COMM,10)	TOTINCOME
-----	-----	-----	-----
KING	5000	10	
BLAKE	2850	10	

CLARK	2450	10	
JONES	2975	10	
SCOTT	3000	10	
FORD	3000	10	
SMITH	800	10	
ALLEN	1600	300	1900
WARD	1250	500	1750
MARTIN	1250	1400	2650
TURNER	1500	0	1500
ADAMS	1100	10	
JAMES	950	10	
MILLER	1300	10	
PUNAM	4200	10	
NULL	2200	10	

16 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,NVL (COMM
5      ,10
6      )
7  ,SAL + COMM AS TOTINCOME
8* FROM EMP
SQL> /

```

ENAME	SAL	COMM	NVL (COMM,10)	TOTINCOME
KING	5000		10	
BLAKE	2850		10	
CLARK	2450		10	
JONES	2975		10	
SCOTT	3000		10	
FORD	3000		10	
SMITH	800		10	
ALLEN	1600	300	300	1900
WARD	1250	500	500	1750
MARTIN	1250	1400	1400	2650
TURNER	1500	0	0	1500
ADAMS	1100		10	
JAMES	950		10	
MILLER	1300		10	
PUNAM	4200		10	
NULL	2200		10	

16 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,NVL (COMM
5      ,10
6      )
7  ,SAL + COMM AS TOTINCOME
8* FROM EMP

```

ENAME	SAL	COMM	NVL (COMM,10)	TOTINCOME
-------	-----	------	---------------	-----------

```

-----
KING          5000          10
BLAKE         2850          10
CLARK         2450          10
JONES         2975          10
SCOTT         3000          10
FORD          3000          10
SMITH         800          10
ALLEN         1600          300          300          1900
WARD          1250          500          500          1750
MARTIN        1250         1400         1400          2650
TURNER        1500           0           0          1500
ADAMS         1100          10
JAMES         950          10
MILLER        1300          10
PUNAM         4200          10
NULL          2200          10

```

16 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

 1  SELECT ENAME
 2    ,SAL
 3    ,COMM
 4    ,SAL + COMM TOTINCOME
 5    ,NVL (COMM
 6          ,10
 7        )
 8    ,SAL + NVL( COMM
 9          ,10
10        ) AS TOTINCOME1
11* FROM EMP
SQL> /

```

ENAME	SAL	COMM	TOTINCOME	NVL (COMM,10)	TOTINCOME1
KING	5000			10	5010
BLAKE	2850			10	2860
CLARK	2450			10	2460
JONES	2975			10	2985
SCOTT	3000			10	3010
FORD	3000			10	3010
SMITH	800			10	810
ALLEN	1600	300	1900	300	1900
WARD	1250	500	1750	500	1750
MARTIN	1250	1400	2650	1400	2650
TURNER	1500	0	1500	0	1500
ADAMS	1100			10	1110
JAMES	950			10	960
MILLER	1300			10	1310
PUNAM	4200			10	4210
NULL	2200			10	2210

16 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

 1  SELECT ENAME
 2    ,SAL
 3    ,COMM
 4    ,SAL + NVL( COMM

```

```

5          ,10
6          ) AS TOTINCOME
7* FROM EMP
SQL> /

```

ENAME	SAL	COMM	TOTINCOME
KING	5000		5010
BLAKE	2850		2860
CLARK	2450		2460
JONES	2975		2985
SCOTT	3000		3010
FORD	3000		3010
SMITH	800		810
ALLEN	1600	300	1900
WARD	1250	500	1750
MARTIN	1250	1400	2650
TURNER	1500	0	1500
ADAMS	1100		1110
JAMES	950		960
MILLER	1300		1310
PUNAM	4200		4210
NULL	2200		2210

16 rows selected.

Where ,And,Between

INSTR

```

INSTR(
    P1-- COLUMN/LITERAL
    ,P2--SEARCHED STRING
    ,P3-- FROM WHICH POSITION WE WANT TO SEARCH
    ,P4-- FROM P3 HOW MANY TIMES WE WANT TO SEARCH
)

```

OUTPUT: IT WILLGIVE YOU POSITION OF THE SEARCHED STRING

```

INSTR('KALPANA'
    , 'A'
    , 1
    , 1
)

```

- 1) FROM P3 IF P4 TIMES P2 VALUE NOT IDENTIED THEN OUTPUT WILL BE "0"
- 2) ONCE SEARCH IS SUCCESSFUL ALWAYS IT WILL GIVE YOU THAT POSITION FROM STARTING/BEGINING OF THE STRING
(NOTE : IRRESPECTIVE FROM WHERE YOU ARE SEARCHING/FROM WHICH SIDE YOU ARE SEARCHING)
- 3) IF P3 POSITION NOT ABLE TO IDENTIFY THEN OUTPUT WILL BE "0"
- 4) P3 AND P4 ARE OPTIONAL
IF YOU ARE NOT PROVIDED BY DEFAULT IT WILL CONSIDERES AS "1"

- 5) IF P3 VALUE IS "0" THEN OUTPUT WILL BE "0"
- 6) IF P3 VALUE IS -VE IT WILL FIX POSITION FROM RIGHT SIDE
ONCE POSITION IS FIXED WHAT YOU WANT TO SEARCH FOR THAT
IT WILL SEARCH FROM THAT POSITION TO LEFT SIDE

ONCE SEARCH IS SUCCCESSFULL THAT POSITION WILL GIVE YOU
FROM LEFT SIDE STARTING/BEGINING OF THE STRING
- 7) IF P3 OR P4 VALUE IS NULL OUTPUT WILL BE NULL
- 8) IF P4<=0 THEN IT WILL GIVE YOU ERROR

ARGMENT OUT OF RANGE ERROR
- 9) IF P2 VALUE IS A STRING(MORE THAN ONE CHARACTER)
IT WILL SEARCH AS STRING
ONCE SEARCH IS SUCCESFULL IT WILL GIVE YOU POSITION OF THE
FIRST CHARACTER OF A SEARCHED STRING

6-May-2018-Diff bet Char and Varchar2

DB INSTALATION---
HOW TO CONNECT TO THE DATABASE
TOOLS USED TO CONNECT TO THE DATABASE
HOW TO SEE WHICH USER YOU CONNECTED --- SHOW USER
HOW TO CREATE A TABLE
RULES TO CREATE A TABLE
HOW TO SEE TABLE STU-- DESCRIBE
WHAT IS SPOOL
WHAT IS EDIT -- IS USED FOR
R
/

TABLE CREATION

SYNTAX:

```
CREATE TABLE <TABLENAME>
(
COLUMNNAME1 DATATYPE
,COLUMNNAME2 DATATYPE
```

```

.
.
.
.
COLUMNMAE1000 DATATYPE
);

```

RULES TO CHOOSE/SPECIFY TABLENAME---

SAME RULES ARE APPLICABLE TO COLUMN NAMES ALSO .. IN ONE TABLE 2 COLUMN NAMES SHOULD NOT BE SAME

DATATYPES

IT WILL TELL WHAT KIND OF DATA IS ALLOWED FOR THAT COLUMN

NUMBER

IT ACCEPTS ONLY 0-9

WIDTH PROPERTY IS OPTIONAL FOR NUMBER DATA TYPE

NUMBER(10)-- IT ACCEPTS ONLY MAXIMUM 10 DIGIT NUMBERS

WITH IN PARENTHESIS WHATEVER WE ARE KEEPING THAT IS CALLED WIDTH PROPERTY

IF YOU JUST SAY NUMBER WITH OUT () THEN BY DEFAULT IT ACCEPTS MAXIMUM 38 DIGIT NUMBER

NUMBER --- MAXIMUM 32 DIGIT LENGTH NUMBER IT ACCEPTS

NUMBER(5)-- IT ACCEPTS MAXIMUM 5 DIGIT LENGTH NUMBER

NUMBER(7,2)-- IT ACCEPTS 2 DECIMALS

AND 5 WHOLE NUMBERS TOTAL WILL BE 7 DIGIT NUMBER----

NUMBER(10,4)--

SAL-NUMBER(

ID -- NUMBER

FEES -- NUMBER

TRANSACTION AMOUNT -- NUMBER

AGE--NUMBER

CHAR

IT ACCEPTS 0-9 ALPHABETS AND SPECIAL CHARACTERS

FOR CHAR WIDTH PROPERTY IS OPTIONAL

WITH OUT WIDTH PROPERTY IF YOU SPECIFY CHAR BY DEFAULT IT ACCEPTS ONLY 1 CHARACTER

MAXIMUM 2000 CHARACTERS IT ACCEPTS

** IT IS A FIXED LENGTH MEMORY ALLOCATION

VARCHAR2

IT ACCEPTS 0-9 ALPHABETS AND SPECIAL CHARACTERS

WIDTH PROPERTY IS MANDATORY

MAXIMUM IT ACCEPTS 4000 CHARACTERS

** IT IS A VARIABLE LENGTH MEMORY ALLOCATION

NAME -- CHAR(10)--- SURESH

NAME -- VARCHAR2(10)--SURESH

DIFF CHAR AND VARCHAR2

LONG

FOR LONG DATA TYPE WIDTH PROPERTY IS NOT THERE

IT ACCEPTS NUMBERS, ALPHABETS AND SPECIAL CHARACTERS

MAXIMUM WE CAN STORE 8000 CHARACTERS/BYTES

CLOB

CHARACTER LARGE OBJECT

FOR CLOB DATA TYPE WIDTH PROPERTY IS NOT THERE

IT ACCEPTS NUMBERS, ALPHABETS AND SPECIAL CHARACTERS

WE CAN STORE 46 GB DATA (-23 GB TO +23 GB)

BLOB

BINARY LARGE OBJECT

FOR BLOB DATA TYPE WIDTH PROPERTY IS NOT THERE

IT IS USED TO STORE YOUR IMAGES .JPEG .PDF

DATE

FOR DATE DATA TYPE WIDTH PROPERTY IS NOT THERE

BY DEFAULT DATE FORMAT WHICH WILL ACCEPT AND WHICH WILL GIVE YOU IS

DD-MON-YY FORMAT (I.E 06-MAY-18) DD-MM-YY(06-05-18)

15 HOURS

2 CONSTRAINTS--

6 HOURS (JOINS SUB SET OPERATOR)

8 HOURS (SEQUENCES VIEWS MVIEWS PIVOT AND UNPIVOT REGEXP INDEXES
PARTITIONS... OLAP ETC)

SQL

US
UK
CANADA (ADAR LINKING)

HIPPA

HOSPITAL

CREATE TABLE PATIENTS

(
PID
PNAME
PCITINUMBER
PMOB
PAGE
PGEN
PEDUT

)

CREATE TABLE ROOMS

(
RNO NUMBER
,ROOM_TYPE VARCHAR2(10)
)

ROOM_CHARGES

(
RNO NMBER
AMT NUMBER
)

DOCTORS_LIST

```
(
DNO
DNAME
DSPEICALA
DEXP_STARTED
)
DOCTORS_FEES
MEDICINET
TREAT_MENT_DETAILS
(
PID
DNO
RNO
MEDIICNE_NO
MEIDICNE_DATE
.
.
.
ETC
);
```

```
.
.
.
ETC
```

INDEXES---(30 HOURS)-- PERFORMAN TUNNING (DATAA ARCHICTE DBA)--
DEVELOPER

```
AWR
AS,M
SQL TRACE..
```

```
ETC
PROFILING
```

DBA

INSTALATION UPGRARE MIGRATIONN

RECOVERY AND BACKUP

PERFORMANCE UTNING TIWTH (DEVELOPERS) ----

DATA BASE SECURITY

.

.

.

10 --

HOW TO STORE DATA IN TABLE

BY USING INSERT COMMAND WE CAN DO

CHANGE REQUESTST (SCR)

INVESTIGATION

--

TEST

TESING SING OFF

PRODUCTION

6-May-2018-Diff bet Char and Varchar2(Exam)

SQL>

SQL> CL SCR

SQL> CONNECT SYSTEM/akshara

ERROR:

ORA-01017: invalid username/password; logon denied

SQL> connect system/AKSHARA

Connected.

SQL>

SQL>

SQL> SHOW USER

USER is "SYSTEM"

SQL> CL SCR

SQL> DESCRIBE B26_AKS_STD

Name	Null?	Type
-----	-----	-----
ID		NUMBER
NAME		VARCHAR2 (50)
MOB		NUMBER (10)
FEES		NUMBER (7,2)
DOJ		DATE
GEN		CHAR (1)

SQL> DESCRIBE B26_AKS_STD_1

Name	Null?	Type
-----	-----	-----
ID		NUMBER
NAME		VARCHAR2 (20)
MOB		NUMBER (10)
GEN		CHAR (1)
DOJ		DATE
PAN		VARCHAR2 (10)

SQL> DESCRIBE B26_AKS_STD_2

Name	Null?	Type
-----	-----	-----
ID		NUMBER
NAME		VARCHAR2 (20)
MOB		NUMBER (10)
GEN		CHAR (1)
DOJ		DATE
PAN		VARCHAR2 (10)

SQL> DESCRIBE B26_AKS_STD_3

Name	Null?	Type
-----	-----	-----
ID		NUMBER
NAME		VARCHAR2 (20)
MOB		NUMBER (10)
GEN		CHAR (1)
DOJ		DATE
PAN		VARCHAR2 (10)

SQL> DESCRIBE B26_AKS_STD_10

ERROR:

ORA-04043: object B26_AKS_STD_10 does not exist

```
SQL> CL SCR
```

```
SQL> CREATE TABLE B26_CHAR_EX_1
2  (
3  ID NUMBER
4  ,NAME CHAR
5  );
```

Table created.

```
SQL> DESCRIBE B26_CHAR_EX_1
```

Name	Null?	Type
-----	-----	-----
ID		NUMBER
NAME		CHAR(1)

```
SQL> EDIT
```

Wrote file afiedt.buf

```
1 CREATE TABLE B26_CHAR_EX_2
2  (
3  ID NUMBER
4  ,NAME CHAR(2001)
5* )
```

```
SQL> R
```

```
1 CREATE TABLE B26_CHAR_EX_2
2  (
3  ID NUMBER
4  ,NAME CHAR(2001)
5* )
```

```
,NAME CHAR(2001)
*
```

ERROR at line 4:

ORA-00910: specified length too long for its datatype

```
SQL> EDIT
```

Wrote file afiedt.buf

```
1 CREATE TABLE B26_CHAR_EX_2
2  (
3  ID NUMBER(40)
4  ,NAME CHAR(2000)
5* )
```

```
SQL> R
```

```
1 CREATE TABLE B26_CHAR_EX_2
2  (
3  ID NUMBER(40)
4  ,NAME CHAR(2000)
5* )
```

```
ID NUMBER(40)
*
```

ERROR at line 3:

ORA-01727: numeric precision specifier is out of range (1 to 38)

```
SQL> EDIT
```

Wrote file afiedt.buf

```
1 CREATE TABLE B26_CHAR_EX_2
2  (
3  ID NUMBER
4  ,NAME CHAR
```

```

5* )
SQL> /

```

Table created.

```
SQL> CL SCR
```

```

SQL> CREATE TABLE B26_CHAR_VARCHAR2_EX
2  (
3  FNAME  CHAR(10)
4  ,LNAME VARCHAR2(10)
5  );

```

Table created.

```

SQL> INSERT INTO B26_CHAR_VARCHAR2_EX
2  VALUES
3  (
4  'SURESH'
5  , 'KUMAR'
6  );

```

1 row created.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  INSERT INTO B26_CHAR_VARCHAR2_EX
2  VALUES
3  (
4  'ARJUN'
5  , 'V'
6* )
SQL> /

```

1 row created.

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B26_CHAR_VARCHAR2_EX;
```

FNAME	LNAME
SURESH	KUMAR
ARJUN	V

```

SQL> SELECT FNAME
2  ,LENGTH(FNMAE)
3  ,LNAME
4  ,LENGTH(LNAME)
5  FROM B26_CHAR_VARCHAR2_EX;
,LENGTH(FNMAE)
      *

```

```

ERROR at line 2:
ORA-00904: "FNMAE": invalid identifier

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT FNAME
2  ,LENGTH(FNAME)
3  ,LNAME
4  ,LENGTH(LNAME)

```

```
5* FROM B26_CHAR_VARCHAR2_EX
SQL> /
```

FNAME	LENGTH(FNAME)	LNAME	LENGTH(LNAME)
SURESH	10	KUMAR	5
ARJUN	10	V	1

```
SQL> DESC B26_CHAR_VARCHAR2_EX
```

Name	Null?	Type
FNAME		CHAR(10)
LNAME		VARCHAR2(10)

```
SQL> CL SCR
```

```
SQL> CL SCR
```

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> DESC B26_CHAR_VARCHAR2_EX
```

Name	Null?	Type
FNAME		CHAR(10)
LNAME		VARCHAR2(10)

```
SQL> INSERT INTO B26_CHAR_VARCHAR2_EX
2 VALUES
3 (
4 '123ABC*'
5 , '456DSFAS_'
6 );
```

1 row created.

```
SQL> CL SCR
```

```
SQL> DESC B26_AKS_STD
```

Name	Null?	Type
ID		NUMBER
NAME		VARCHAR2(50)
MOB		NUMBER(10)
FEES		NUMBER(7,2)
DOJ		DATE
GEN		CHAR(1)

```
SQL> CL SCR
```

```
SQL> CREATE TABBLE B26_AKSHARA_STUDENT
```

```
2 (
3 SNO NUMBER
4 ,NAME VARCHAR2(20)
5 ,MOB NUMBER(10)
6 ,FEES NUMBER(8,2)
7 ,DOJ DATE
8 ,GEN CHAR(1)
9 )
10
SQL> EDIT
```


Wrote file afiedt.buf

```

1  CREATE TABBLE B26_AKSHARA_STUDENT
2  (
3  SNO    NUMBER
4  ,NAME  VARCHAR2(20)
5  ,MOB   NUMBER(10)
6  ,FEES  NUMBER(8,2)
7  ,DOJ   DATE
8  ,GEN   CHAR(1)
9* )
10 /
CREATE TABBLE B26_AKSHARA_STUDENT
      *
ERROR at line 1:
ORA-00901: invalid CREATE command

```

SQL> EDIT

Wrote file afiedt.buf

```

1  CREATE TABLE B26_AKSHARA_STUDENT
2  (
3  SNO    NUMBER
4  ,NAME  VARCHAR2(20)
5  ,MOB   NUMBER(10)
6  ,FEES  NUMBER(8,2)
7  ,DOJ   DATE
8  ,GEN   CHAR(1)
9* )
SQL> R
1  CREATE TABLE B26_AKSHARA_STUDENT
2  (
3  SNO    NUMBER
4  ,NAME  VARCHAR2(20)
5  ,MOB   NUMBER(10)
6  ,FEES  NUMBER(8,2)
7  ,DOJ   DATE
8  ,GEN   CHAR(1)
9* )

```

Table created.

SQL> R

```

1  CREATE TABLE B26_AKSHARA_STUDENT
2  (
3  SNO    NUMBER
4  ,NAME  VARCHAR2(20)
5  ,MOB   NUMBER(10)
6  ,FEES  NUMBER(8,2)
7  ,DOJ   DATE
8  ,GEN   CHAR(1)
9* )
CREATE TABLE B26_AKSHARA_STUDENT
      *

```

ERROR at line 1:

ORA-00955: name is already used by an existing object

SQL> CL SCR

SQL> /

```

CREATE TABLE B26_AKSHARA_STUDENT
      *

```

ERROR at line 1:
ORA-00955: name is already used by an existing object

```
SQL> R
1 CREATE TABLE B26_AKSHARA_STUDENT
2 (
3 SNO NUMBER
4 ,NAME VARCHAR2(20)
5 ,MOB NUMBER(10)
6 ,FEES NUMBER(8,2)
7 ,DOJ DATE
8 ,GEN CHAR(1)
9* )
```

```
CREATE TABLE B26_AKSHARA_STUDENT
*
```

ERROR at line 1:
ORA-00955: name is already used by an existing object

```
SQL> CL SCR
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```
1 CREATE TABLE B26_AKSHARA_STUDENT_1
2 (
3 SNO NUMBER
4 ,NAME VARCHAR2(20)
5 ,MOB NUMBER(10)
6 ,FEES NUMBER(8,2)
7 ,DOJ DATE(10)
8 ,GEN CHAR(1)
9* )
```

```
SQL> /
,DOJ DATE(10)
*
```

ERROR at line 7:
ORA-00907: missing right parenthesis

```
SQL> EDIT
Wrote file afiedt.buf
```

```
1 CREATE TABLE B26_AKSHARA_STUDENT_1
2 (
3 SNO NUMBER
4 ,NAME VARCHAR2(4020)
5 ,MOB NUMBER(10)
6 ,FEES NUMBER(8,2)
7 ,DOJ DATE
8 ,GEN CHAR(1)
9* )
```

```
SQL> /
,NAME VARCHAR2(4020)
*
```

ERROR at line 4:
ORA-00910: specified length too long for its datatype

```
SQL> ED
Wrote file afiedt.buf
```

```
1 CREATE TABLE B26_AKSHARA_STUDENT_1
```

```

2  (
3  SNO    NUMBER
4  ,NAME  VARCHAR2(20)
5  ,MOB    NUMBER(10)
6  ,FEES   NUMBER(8,2)
7  ,DOJ    DATE
8  ,GEN    CHAR(1)
9* )

```

SQL>

SQL>

SQL> CL SCR

SQL> R

```

1  CREATE TABLE B26_AKSHARA_STUDENT_1
2  (
3  SNO    NUMBER
4  ,NAME  VARCHAR2(20)
5  ,MOB    NUMBER(10)
6  ,FEES   NUMBER(8,2)
7  ,DOJ    DATE
8  ,GEN    CHAR(1)
9* )

```

Table created.

SQL> CL SCR

SQL> DESCRIBE B26_AKSHARA_STUDENT

Name	Null?	Type
SNO		NUMBER
NAME		VARCHAR2(20)
MOB		NUMBER(10)
FEES		NUMBER(8,2)
DOJ		DATE
GEN		CHAR(1)

SQL> INSERT INTO B26_AKSHARA_STUDENT

```

2  VALUES
3  (
4  101
5  ,ABHISHEK
6  ,12345
7  ,5000
8  '03-MAY-18'
9  , 'M'
10 ) ;
'03-MAY-18'
*
```

ERROR at line 8:

ORA-00917: missing comma

SQL> EDIT

Wrote file afiedt.buf

```

1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  101
5  ,ABHISHEK
6  ,12345
7  ,5000
8  , '03-MAY-18'

```

```
9  , 'M'
10* )
SQL> R
1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  101
5  , ABHISHEK
6  , 12345
7  , 5000
8  , '03-MAY-18'
9  , 'M'
10* )
, ABHISHEK
*
ERROR at line 5:
ORA-00984: column not allowed here
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  101
5  , 'ABHISHEK'
6  , 12345
7  , 5000
8  , 'MAY-18-03'
9  , 'M'
10* )
SQL> /
, 'MAY-18-03'
*
```

```
ERROR at line 8:
ORA-01858: a non-numeric character was found where a numeric was expected
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  101
5  , 'ABHISHEK'
6  , 12345
7  , 5000
8  , '03-MAY-18'
9  , 'M'
10* )
SQL> /
```

1 row created.

```
SQL> EDIT
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  102
5  , "SURESH"
```

```
6 ,12346
7 ,6000
8 , SYSDATE - 3
9 , 'M'
10* )
SQL> /
,"SURESH"
*
ERROR at line 5:
ORA-00984: column not allowed here
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 INSERT INTO B26_AKSHARA_STUDENT
2 VALUES
3 (
4 102
5 , 'SURESH KUMAR SINGH SURES'
6 ,123461234789
7 ,6000
8 , SYSDATE - 3
9 , 'B'
10* )
SQL> /
,'SURESH KUMAR SINGH SURES'
*
ERROR at line 5:
ORA-12899: value too large for column "SYSTEM"."B26_AKSHARA_STUDENT"."NAME"
(actual: 24, maximum: 20)
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```
1 INSERT INTO B26_AKSHARA_STUDENT
2 VALUES
3 (
4 102
5 , 'SURESH KUMAR SINGH SR'
6 ,123461234789
7 ,6000
8 , SYSDATE - 3
9 , 'B'
10* )
SQL> /
,'SURESH KUMAR SINGH SR'
*
ERROR at line 5:
ORA-12899: value too large for column "SYSTEM"."B26_AKSHARA_STUDENT"."NAME"
(actual: 21, maximum: 20)
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```
1 INSERT INTO B26_AKSHARA_STUDENT
2 VALUES
3 (
4 102
5 , 'SURESH KUMAR SINGH S'
6 ,123461234789
7 ,6000
```

```
      8 , SYSDATE - 3
      9 , 'B'
    10* )
SQL> /
,123461234789
*
```

ERROR at line 6:

ORA-01438: value larger than specified precision allowed for this column

SQL> EDIT

Wrote file afiedt.buf

```
1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  102
5  , 'SURESH KUMAR SINGH S'
6  , 1234612347
7  , 6000
8  , SYSDATE - 3
9  , 'B'
10* )
SQL> /
```

1 row created.

SQL> EDIT

Wrote file afiedt.buf

```
1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  103
5  , 'KRISHNA'
6  , 6500
7  , SYSDATE
8  , 'M'
9* )
SQL> /
INSERT INTO B26_AKSHARA_STUDENT
*
```

ERROR at line 1:

ORA-00947: not enough values

SQL> EDIT

Wrote file afiedt.buf

```
1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  103
5  , 'KRISHNA'
6  , 98765
7  , 6500
8  , SYSDATE
9  , 'M'
10 , 'CRICKET'
11* )
SQL> /
INSERT INTO B26_AKSHARA_STUDENT
*
```

ERROR at line 1:

ORA-00913: too many values

SQL> EDIT

Wrote file afiedt.buf

```
1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  103
5  , 'KRISHNA'
6  , 6500
7  , SYSDATE
8  , 'M'
9* )
```

SQL> /

```
INSERT INTO B26_AKSHARA_STUDENT
      *
```

ERROR at line 1:

ORA-00947: not enough values

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  103
5  , 'KRISHNA'
6  , NULL
7  , ''
8  , SYSDATE
9  , 'M'
10* )
```

SQL> /

1 row created.

SQL> EDIT

Wrote file afiedt.buf

```
1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  104
5  ,                                'HIMANSHU'
6  , NULL
7  , NULL
8  , SYSDATE
9  , 'M'
10* )
```

SQL> /

```
,                                'HIMANSHU'
      *
```

ERROR at line 5:

ORA-12899: value too large for column "SYSTEM"."B26_AKSHARA_STUDENT"."NAME"
(actual: 22, maximum: 20)

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_AKSHARA_STUDENT
```

```

2  VALUES
3  (
4  104
5  ,
6  ,NULL
7  ,NULL
8  ,SYSDATE
9  , 'M'
10* )
SQL> /

```

1 row created.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  'M'
5  ,4000
6  ,105
7  ,SYSDATE - 10
8  , 'HIMANSHU'
9  ,98765
10* )
SQL>
SQL>
SQL> CL SCR

```

```
SQL> DESC B26_AKSHARA_STUDENT
```

Name	Null?	Type
SNO		NUMBER
NAME		VARCHAR2(20)
MOB		NUMBER(10)
FEES		NUMBER(8,2)
DOJ		DATE
GEN		CHAR(1)

```

SQL> EDIT
Wrote file afiedt.buf

```

```

1  INSERT INTO B26_AKSHARA_STUDENT
2  VALUES
3  (
4  'M'
5  ,4000
6  ,105
7  ,SYSDATE - 10
8  , 'HIMANSHU'
9  ,98765
10* )
SQL> /
,SYSDATE - 10
      *

```

ERROR at line 7:
ORA-00932: inconsistent datatypes: expected NUMBER got DATE

```

SQL> EDIT
Wrote file afiedt.buf

```

```

1  INSERT INTO B26_AKSHARA_STUDENT

```



```
2  (  
3  GEN  
4  ,FEES  
5  ,SNO  
6  ,DOJ  
7  ,NAME  
8  ,MOB  
9  )  
10 VALUES  
11 (  
12 'M'  
13 ,4000  
14 ,105  
15 ,SYSDATE - 10  
16 , 'HIMANSHU'  
17 ,98765  
18* )  
SQL> /
```

1 row created.

SQL> CL SCR

SQL> EDIT
Wrote file afiedt.buf

```
1  INSERT INTO B26_AKSHARA_STUDENT  
2  (  
3  GEN  
4  ,SNO  
5  ,DOJ  
6  ,NAME  
7  ,MOB  
8  )  
9  VALUES  
10 (  
11 'F'  
12 ,106  
13 ,SYSDATE - 10  
14 , 'PUNAM'  
15 ,98766  
16* )  
SQL> /
```

1 row created.

SQL> edit
Wrote file afiedt.buf

```
1  INSERT INTO B26_AKSHARA_STUDENT  
2  (  
3  GEN  
4  ,SNO  
5  ,DOJ  
6  ,NAME  
7  ,MOB  
8  )  
9  VALUES  
10 (  
11 'F'  
12 ,107  
13 ,SYSDATE - 10  
14 , 'Teja'  
15 ,NULL
```

```
16* )
SQL> /
```

1 row created.

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B26_AKSHARA_STUDENT;
```

SNO	NAME	MOB	FEEES	DOJ	G
101	ABHISHEK	12345	5000	03-MAY-18	M
102	SURESH KUMAR SINGH S	1234612347	6000	03-MAY-18	B
103	KRISHNA			06-MAY-18	M
104	ARJUN			06-MAY-18	M
105	HIMANSHU	98765	4000	26-APR-18	M
106	PUNAM	98766		26-APR-18	F
107	Teja			26-APR-18	F

7 rows selected.

```
SQL> INSERT INTO B26_AKSHARA_STUDENT
```

```
2 (
3 MOB
4 ,FEEES
5 ,ID
6
```

```
SQL>
```

```
SQL>
```

```
SQL> ED
```

Wrote file afiedt.buf

```
1 INSERT INTO B26_AKSHARA_STUDENT
2 (
3 MOB
4 ,FEEES
5 ,SNO
6 ,SNAME
7 )
8 VALUES
9 (
10 234324
11 ,4500
12 ,107
13 , 'SAI'
14 ,SYSDATE - 4
15* )
```

```
SQL> /
```

```
INSERT INTO B26_AKSHARA_STUDENT
*
```

ERROR at line 1:

ORA-00913: too many values

```
SQL> ED
```

Wrote file afiedt.buf

```
1 INSERT INTO B26_AKSHARA_STUDENT
2 (
3 MOB
4 ,FEEES
5 ,SNO
6 ,SNAME
7 )
```

```

      8  VALUES
      9  (
     10  234324
     11  ,4500
     12  ,107
     13* )
SQL> /
VALUES
*
```

ERROR at line 8:
ORA-00947: not enough values

SQL> CL SCR

SQL> DESC B26_AKSHARA_STUDENT

Name	Null?	Type
SNO		NUMBER
NAME		VARCHAR2(20)
MOB		NUMBER(10)
FEES		NUMBER(8,2)
DOJ		DATE
GEN		CHAR(1)

```

SQL> INSERT INTO B26_AKSHARA_STUDENT
      2  VALUES
      3
```

SQL>

```

SQL> INSERT INTO B26_AKSHARA_STUDENT
      2  (
      3  SNO
      4  ,SNAME
      5  )
      6  VALUES
      7  (
      8  108
      9  , 'AKSHARA'
     10  );
,SNAME
*
```

ERROR at line 4:
ORA-00904: "SNAME": invalid identifier

SQL> EDIT

Wrote file afiedt.buf

```

      1  INSERT INTO B26_AKSHARA_STUDENT
      2  (
      3  SNO
      4  ,NAME
      5  )
      6  VALUES
      7  (
      8  108
      9  , 'AKSHARA'
     10* )
SQL> /
```

1 row created.

SQL> SELECT * FROM B26_AKSHARA_STUDENT;

SNO	NAME	MOB	FEEES	DOJ	G
101	ABHISHEK	12345	5000	03-MAY-18	M
102	SURESH KUMAR SINGH S	1234612347	6000	03-MAY-18	B
103	KRISHNA			06-MAY-18	M
104	ARJUN			06-MAY-18	M
105	HIMANSHU	98765	4000	26-APR-18	M
106	PUNAM	98766		26-APR-18	F
107	Teja			26-APR-18	F
108	AKSHARA				

8 rows selected.

SQL> CL SCR

SQL> R

1* SELECT * FROM B26_AKSHARA_STUDENT

SNO	NAME	MOB	FEEES	DOJ	G
101	ABHISHEK	12345	5000	03-MAY-18	M
102	SURESH KUMAR SINGH S	1234612347	6000	03-MAY-18	B
103	KRISHNA			06-MAY-18	M
104	ARJUN			06-MAY-18	M
105	HIMANSHU	98765	4000	26-APR-18	M
106	PUNAM	98766		26-APR-18	F
107	Teja			26-APR-18	F
108	AKSHARA				

8 rows selected.

SQL> COMMIT;

Commit complete.

SQL> L SCR

SP2-0224: invalid starting line number

SQL> CL SCR

SQL> CREATE TABLE B26_NULL_SPACE_EXS

2

SQL> CREATE TABLE B26_NULL_SPACE_EX

2 (

3 FNAME VARCHAR2(10)

4 ,LNAME VARCHAR2(10)

5);

Table created.

SQL> INSERT INTO B26_NULL_SPACE_EX

2 VALUES

3 (

4 NULL

5 , ' ' ,

6);

1 row created.

SQL> SELECT * FROM B26_NULL_SPACE_EX;

FNAME	LNAME
-----	-----

```
SQL> INSERT INTO B26_NULL_SPACE_EX
  2  VALUES
  3  (
  4  'ARJUN'
  5  , 'REDDY'
  6  );
```

1 row created.

```
SQL> SELECT * FROM B26_NULL_SPACE_EX;
```

FNAME	LNAME
ARJUN	REDDY

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT FNAME
  2  , LENGTH(FNAME)
  3  , LNAME
  4  , LENGTH(LNAME)
  5* FROM B26_NULL_SPACE_EX
SQL> /
```

FNAME	LENGTH(FNAME)	LNAME	LENGTH(LNAME)
ARJUN	5	REDDY	5

```
SQL> CL SCR
```

```
SQL> R
  1  SELECT FNAME
  2  , LENGTH(FNAME)
  3  , LNAME
  4  , LENGTH(LNAME)
  5* FROM B26_NULL_SPACE_EX
```

FNAME	LENGTH(FNAME)	LNAME	LENGTH(LNAME)
ARJUN	5	REDDY	5

```
SQL> CL SCR
```

```
SQL> INSERT INTO B26_AKSHARA_STUDENT
  2  (
  3  SNON
  4
```

```
SQL>
```

```
SQL> INSERT INTO B26_AKSHARA_STUDENT
  2  (
  3  SNON
  4
```

```
SQL>
```

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  INSERT INTO B26_AKSHARA_STUDENT
  2  (
  3  SNO
  4  , NAME
```

```
5  ,MOB
6  )
7  VALUES
8  (
9  'NULL'
10 , 'NULL'
11 , 'NULL'
12* );
SQL> /
);
*
ERROR at line 12:
ORA-00911: invalid character
```

```
SQL> ED
Wrote file afiedt.buf

1  INSERT INTO B26_AKSHARA_STUDENT
2  (
3  SNO
4  ,NAME
5  ,MOB
6  )
7  VALUES
8  (
9  'NULL'
10 , 'NULL'
11 , 'NULL'
12* )
SQL> /
'NULL'
*
ERROR at line 9:
ORA-01722: invalid number
```

```
SQL> ED
Wrote file afiedt.buf

1  INSERT INTO B26_AKSHARA_STUDENT
2  (
3  SNO
4  ,NAME
5  ,MOB
6  )
7  VALUES
8  (
9  NULL
10 , 'NULL'
11 , 'NULL'
12* )
SQL> /
, 'NULL'
*
ERROR at line 11:
ORA-01722: invalid number
```

```
SQL> ED
Wrote file afiedt.buf

1  INSERT INTO B26_AKSHARA_STUDENT
2  (
```

```

3 SNO
4 ,NAME
5 ,MOB
6 )
7 VALUES
8 (
9 NULL
10 , 'NULL'
11 ,NULL
12* )
SQL> /

```

1 row created.

```
SQL> SELECT * FROM B26_AKSHARA_STUDENT;
```

SNO	NAME	MOB	FEEES	DOJ	G
101	ABHISHEK	12345	5000	03-MAY-18	M
102	SURESH KUMAR SINGH S	1234612347	6000	03-MAY-18	B
103	KRISHNA			06-MAY-18	M
104	ARJUN			06-MAY-18	M
105	HIMANSHU	98765	4000	26-APR-18	M
106	PUNAM	98766		26-APR-18	F
107	Teja			26-APR-18	F
108	AKSHARA				
	NULL				

9 rows selected.

```
SQL> SPOOL OFF
```

6-May-2018-Diff bet Null and Implicit Null(Exam)

```

HOW TO INSTALL
HOW TO CONNECT
CONNECT UAERNAME/PASSOWRD
HOW TO RESET THE PASSWORD
WHAT IS SPOOL
EDIT
R
/
SPOOL
CREATE
DATA TYPES
DESC TABLENAME

```

```
CREATE
```

```
-----
```

```
INSERT
```

```
-----
```

INSERT IS A DML COMMAND
IT IS USED TO ADD ROWS TO THE TABLES

INSERT SYNTAX 1:

IF YOU ARE PROVIDING ALL COLUMNS VALUES IN SAME ORDER HOW THE TABLE IS CREATED

```
INSERT INTO <TABLENAME>
VALUES
(
COL1VALUE
,COL2VALUE
,COL3VALUE
.
.
.
.
, LASTCOLUMNVALUE
);
```

- 1) EACH COLUMN VALUE SHOULD BE SEPARATED BY ,
- 2) EXCEPT FOR NUMBER DATA TYPE FOR OTHER DATA TYPES COLUMNS DATA SHOULD BE PROVIDED WITH IN '' (SINGLE QUOTES)
, '
- 3) DD-MON-YY IS DEFAULT DATE FORMAT IN ORACLE

4) FOR ANY COLUMN IF YOU DONT HAVE A DATA THEN WE CALL IT AS ABSENCE OF VALUE
THAT YOU CAN TELL BY USING NULL

NULL IS NOTHIGN BUT ABSENCE OF VALUE
NULL WE CAN GIVE IN 2 WAYS
NULL
''--- NO SPACES INBETWEEN

NULL--MOSTLY I WILL BE USING
-- 2 TO 3 HOURS

CONSTRAINTS--2 HOURS

CONVERSION FUNCTIONS
TO_CHAR
TO_NUMBER
TO_DATE--2 HOURS

N/A-NOT AVAILABLE

SYSDATE IS A FUNCTION IT WILL GIVE YOU CURRENT SYSTEM DATE

INSERT SYNTAX 2:

IF YOU ARE PROVIDING ALL COLUMNS VALUE IN YOUR OWN ORDER (NOT FOLLOWING TABLE CREATED ORDER)

```
INSERT INTO <TABLENAME>
(
COL6
, COL3
, COL1
, COL4
, COL5
, COL2
)
VALUES
(
COL6VALUE
, COL3VALUE
, COL1VALUE
, COL4VALUE
, COL5VALUE
, COL12VALUE
);
```

NULL WE CAN INSERT IN 2 WAYS

- 1) EXPLICIT
 NULL
 ,,
- 2) IMPLICIT

WHILE INSERTING IF YOU SPECIFY DATA FOR ONLY SOME COLUMNS
AND IF THAT COLUMNS LIST IT IS THERE IN INSERT STATEMENT THEN
FOR REMAINING COLUMNS
 ORACLE AUTOMATICALLY KEEP NULL VALUE.

DIFF BETWEEN NULL AND SPACE ?

NULL IS NOHTING BUT YOU DONT KNOW WHAT IS THERE
SPACE IS ONE CHARACTER YOU CAN SAY ONE..OR SOME SPACES ARE THERE C

SESSION AND TRANSACTION

EACH TIME WHEN YOU CONNECT TO THE DATABASE IT WILL OPEN ONE NEW
SESSION
EACH SESSION WILL HAVE ONE UNIQUE IDENTIFIER CALLED SID

WHATEVER CHANGES WE ARE DOING IN ONE SESSION WE WILL NOT BE ABLE TO
SEE FROM
OTHER SESSION UNTIL IT IS SAVED(COMMIT)

IF YOU DO SOME CHANGES IN ONE SESSION AND BY MISTAKENLY IF YOU
CLOSE THAT SESSION EDITOR
THOSE CHANGES WILL BE ROLLBACK AUTOMATICALLY*****

IF YOU ARE PROVIDING DATA FOR ONLY SOME COLUMNS ONLY THAT COLUMNS
YOU CAN KEEP
IN INSERT STATEMENT

```
INSERT INTO TABLENAEM(COL10,COL1,COL12)
VALUES
(
COL10VALUE,COL1VALEU,COL12VALUE
);
```

THEN BY DEFAULTL ORACLE WILL KEEP NULL VALUES FOR OTHER COLUMNS
WHICH ARE THERE IN TABLE

24-MAY-2018 Substr & INSTR

```
SUBSTR
(
P1-- COLUMN / LITERAL
,P2 -- FROM WHICH POSITION/FROM WHERE WE WANT IN P1
,P3 -- FROM P2 HOW MANY POSITIONS/CHARACTERS WE WANT
```

)

OUTPUT : IT WILL GIVE YOU PART OF GIVEIN STRING(P1)

```
SUBSTR(
    'KRISHNA'
    ,1
    ,3
) AS C1
```

- 1) IN P1 FROM P2 POSITION IF P3 NUMBER OF CHARACTERS ARE NOT FOUND THEN IT WILL GIVE YOU HOW MANY ARE AVAILABLE.
- 2) P3 PARAMETER IS OPTIONAL
IF P3 NOT PROVIDED FROM P2 TILL END IT WILL GIVE YOU
- 3) IF P2 POSITION NOT ABLE TO IDENTIFY THEN IT WILL GIVE YOU NULL
- 4) P1 AND P2 PARAMETERS ARE MANDATORY
- 5) IF P2 PARAMETER VALUE IS "0" IS NOTHING BUT 1
- 6) IF P2 VALUE IS -VE THEN
IT WILL FIX POSITION FROM RIGHT SIDE
ONCE POSITIONS IS FIXED HOW MANY(P3) CHARACTERS YOU WANT IT WILL
GIVE YOU FROM THAT POSITIONS TO RIGHT SIDE
- 7) IF P2 OR P3 PARAMETER VALUE IS NULL THEN OUTPUT WILL BE NULL
- 8) IF P3 <= 0 THEN OUTPUT WILL BE NULL

LENGTH

```
LENGTH(
    P1-- COLUMN/CHARACTERS
)
```

-- TOTAL NUMBER OF POSITIONS
--INCLUDING SPACES

IF P1 VALUE IS NULL THEN OUTPUT WILL BE NULL

INSTR

```
INSTR(  
    P1-- COLUMN/LITERAL  
    ,P2--SEARCHED STRING  
    ,P3-- FROM WHICH POSITION WE WANT TO SEARCH  
    ,P4-- FROM P3 HOW MANY TIMES WE WANT TO SEARCH  
)
```

OUTPUT: IT WILLGIVE YOU POSITION OF THE SEARCHED STRING

```
INSTR('KALPANA'  
    , 'A'  
    , 1  
    , 1  
)
```

- 1) FROM P3 IF P4 TIMES P2 VALUE NOT IDENTIED THEN OUTPUT WILL BE "0"
- 2) ONCE SEARCH IS SUCCESSFUL ALWAYS IT WILL GIVE YOU THAT POSITION FROM STARTING/BEGINING OF THE STRING
(NOTE : IRRESPECTIVE FROM WHERE YOU ARE SEARCHING/FROM WHICH SIDE YOU ARE SEARCHING
)
- 3) IF P3 POSITION NOT ABLE TO IDENTIFY THEN OUTPUT WILL BE "0"
- 4) P3 AND P4 ARE OPTIONAL
IF YOU ARE NOT PROVIDED BY DEFAULT IT WILL CONSIDERES AS "1"
- 5) IF P3 VALUE IS "0" THEN OUTPUT WILL BE "0"
- 6) IF P3 VALUE IS -VE IT WILL FIX POSITION FROM RIGHT SIDE
ONCE POSITION IS FIXED WHAT YOU WANT TO SEARCH FOR THAT
IT WILL SEARCH FROM THAT POSITION TO LEFT SIDE

ONCE SEARCH IS SUCCCESSFULL THAT POSITION WILL GIVE YOU
FROM LEFT SIDE STARTING/BEGINING OF THE STRING
- 7) IF P3 OR P4 VALUE IS NULL OUTPUT WILL BE NULL
- 8) IF P4<=0 THEN IT WILL GIVE YOU ERROR

ARGMENT OUT OF RANGE ERROR

- 9) IF P2 VALUE IS A STRING (MORE THAN ONE CHARACTER)
 IT WILL SEARCH AS STRING
 ONCE SEARCH IS SUCCESSFUL IT WILL GIVE YOU POSITION OF THE
 FIRST CHARACTER OF A SEARCHED STRING

UPPER LOWER INITCAP SUBSTR INSTR LENGTH

- 1) WHATEVER STRING YOU GIVE FROM THAT STRING FIRST 2 CHARACTERS
 SHOULD BE SHOWN IN LOWER CASE
 LAST 2 CHARACTERS SHOULD BE SHOWN IN initcap CASE

hiHu

ENAME	ENAME1

KING	kiNg
BLAKE	blKe
CLARK	clRk

- 2) first half in ename should be in lower case and second half
 should
 be in initcap case

EMP

ENAME	ENAME1

KING	kiNg
BLAKE	blake SUBSTR(ENAME,1,LENGTH(ENAME)/2)
	,SUBSTR(ENAME,LENGTH(ENAME)/2+1
CLARK	clark
MARTIN	marTin

ROUND

TRUNC

SUBSTR || UPPER LOWER INITCAP LENGTH (LOGICAL THINKING)

- 3) SYSDATE -- DD-MON-YY -- 20-MAY-18

OUTPUT : YY-20-MAY

OUTPUT : YY-MON-DD

DD-MON-YY

YY

DD-MON

4)

NAME

 ABHI SHEK KUMAR -- between 1st space and 2nd space what is there
 that middlename
 SURESH KUMAR NAIDU-- between 1st space and 2nd space what is there
 that middlename
 A B C-- between 1st space and 2nd space what is there that
 middlename
 ARJUN 12 34-- between 1st space and 2nd space what is there that
 middlename

?

how to get 1st sspace position ? INSTR

how to get 2nd space position ? INSTR

how to get inbetween 1st and snd ?

OUTPUT:

FNAME	MNAME	LNAME
ABHI	SHEK	KUMAR
SURESH	KUMAR	NAIDU
A	B	C
ARJUN	12	34

MAILD

 ABHISHEK@CAPGEMINI.COM
 SURESH@CTS.COM
 ARJUN@GMAIL.COM
 KRISHNA@GMAIL.CO.IN

NAME	PROVIDER	DOMAIN
ABHISHEK	CAPGEMINI	COM
SURESH	CTS	COM

ARJUN
KRISHNA

GMAIL
XYZ

COM
CO.IN

```
6) emp
   ename
-----
   KING
   ARJUN
   BLAKE
   HIMANSHU
   TEJASREE
   SURESH
```

FIRST 2 CHARACTERS IN LOWERCASE AND LAST 2 CHARACTERS IN INITCACP
CASE
AND INBETWEEN CHARACTERS SHOULD BE SHOWN AS IT IS

```
kiNg
arJUN
hiMANSHu
```

16 DIGIT
FIRST 4 ASIT
LAST 4 AS IT IS
INBETWEEN *

SUBSTR INSTR LENGTH(LGOICAL THINKING)

FUNCTIONS CAN BE NESTED

```
-----
T1
--
TIME
-
```

15:45:30

15 HOURS 45 MINUTS 30 SECNONDS

```
SUBSTR(TIME
      ,1
      ,INSTR(TIME,':',1,1) -1
      ) || ' HOURS ' AS HOURS
```

REPLACE

```
(
  P1-- COLUMN/LITERAL
,P2 -- SEARCHED STRING
,P3 -- REPLACED STRING
)
```

- 1) P2 STRING WILL BE REPLACED WITH P3 STRING IN P1 IF IT IS FOUND
- 2) P3 IS OPTIONAL IF P3 NOT PROVIDED P2 VALUE WILL BE REMOED FROM P1 IF IT IS FOUND
- 3) IT WILL DO STRING BY STRING
- 4) P1 AND P2 PARAMETERS ARE MANDATORY
- 5) IF P2 VALUE NOT FOUND IN P1 THEN IT WILL GIVE YOU WHATEVER IT IS THERE IN P1 WITH OOUT ANY CHANGES

NOTE:

IF YOU WANT HIGHLIGHT SOMETHING IN COMMAND PROMPT
ALT + SPACEBAR+EDIT+MARK

THEN TRY TO HIGHLIGH WHICH YOU WANT
IF YOU WANT TO KEEP WHICH IS HIGHLITED PRES ENTER

TRANSLATE

```
(
  P1-- COLUMN/LITERAL
,P2-- SEARCHED CHARACTER SET
,P3-- REPLACED CHARACTER SET
)
```

- 1) IT WILL DO CHARACTER BY CHARACTER .

2) IF LENGTH(P2) > LENGTH(P3) THEN THE EXTRA CHARACTERS WHAT IS THERE IN P2

WILL BE REMOVED FROM P1 IF IT IS FOUND

3) IF LENGTH(P3) > LENGTH(P2) NO ISSUES YOU ARE KEEPING UNNECESSARY EXTRAC CHARACTERS IN P3

4) ALL 3 PARAMETERS ARE MANDATORY

DIFF REPLACE AND TRANSLATE ?

```
(
  P1-- COLUMN/LITERAL
,P2-- HOW MANY NUMBER OF CHARACTERS WE ARE EXPECTING IN P1
,P3 -- IF LENGTH(P1 ) < P2 THEN WHAT CHARACTERS WE WANT TO KEEP
LEFT SIDE OF P1 TO MAKE
      LENGTH(P1) = P2
)
```

1) P3 IS OPTIONAL IF P3 NOT PROVIDED THEN IT WILL ADD SPACES

2) IF P2 < LENGTH(P1) THEN IT WILL GIVE YOU FIRST P2 CHARACTERS FROM P1

```
LPAD(
  'SURESH'
,10
,'* '
)
```

****SURESH

RPAD

```
(
  P1-- COLUMN/LITERAL
,P2-- HOW MANY NUMBER OF CHARACTERS WE ARE EXPECTING IN P1
```

```
,P3 -- IF LENGTH(P1 ) < P2 THEN WHAT CHARACTERS WE WANT TO KEEP  
RIGHT SIDE OF P1 TO MAKE  
        LENGTH(P1) = P2  
)
```

- 1) P3 IS OPTIONAL IF P3 NOT PROVIDED THEN IT WILL ADD SPACES
- 2) IF P2 < LENGTH(P1) THEN IT WILL GIVE YOU FIRST P2 CHARACTERS FROM P1

LTRIM
RTRIM
TRIM
NVL
NVL2
NULLIF
COALESCE
CONCAT
CHR
GREATEST
LEAST
DECODE
CASE

24-MAY- 2018 Substr & INSTR Spool

```
SQL> connect system/AKSHARA  
Connected.
```

```
SQL> SELECT SYSDATE  
        2  FROM DUAL;
```

```
SYSDATE  
-----  
24-MAY-18
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
        1  SELECT SYSDATE AS C1  
        2* FROM DUAL  
SQL> /
```

```
C1  
-----  
24-MAY-18
```

```
SQL> CL SCR
```

```
SQL> R  
        1  SELECT SYSDATE AS C1  
        2* FROM DUAL
```

```
C1
-----
24-MAY-18
```

```
SQL> ED
Wrote file afiedt.buf
```

```
 1  SELECT SYSDATE AS C1
 2  ,SUBSTR(SYSDATE
 3  , -2
 4  ) AS YY
 5* FROM DUAL
SQL> /
```

```
C1      YY
-----
24-MAY-18 18
```

```
SQL> ED
Wrote file afiedt.buf
```

```
 1  SELECT SYSDATE AS C1
 2  ,SUBSTR(SYSDATE
 3  , -2
 4  ) AS YY
 5  ,SUBSTR(SYSDATE
 6  , 1
 7  , 2
 8  ) AS DD
 9* FROM DUAL
SQL> /
```

```
C1      YY      DD
-----
24-MAY-18 18      24
```

```
SQL> ED
Wrote file afiedt.buf
```

```
 1  SELECT SYSDATE AS C1
 2  ,SUBSTR(SYSDATE
 3  , -2
 4  ) AS YY
 5  ,SUBSTR(SYSDATE
 6  , 1
 7  , 2
 8  ) AS DD
 9  ,SUBSTR(SYSDATE
10  , 4
11  , 3
12  ) AS MON
13* FROM DUAL
SQL> /
```

```
C1      YY      DD      MON
-----
24-MAY-18 18      24      MAY
```

```
SQL> CL SCR
```

```
SQL> R
 1  SELECT SYSDATE AS C1
 2  ,SUBSTR(SYSDATE
 3  , -2
```

```

4          ) AS YY
5  ,SUBSTR(SYSDATE
6          ,1
7          ,2
8          ) AS DD
9  ,SUBSTR(SYSDATE
10         ,4
11         ,3
12         ) AS MON
13* FROM DUAL

```

C1	YY	DD	MON
24-MAY-18	18	24	MAY

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS C1
2  ,SUBSTR(SYSDATE
3          , -2
4          )
5  || '-' || SUBSTR(SYSDATE
6          ,1
7          ,2
8          )
9  || '-' || SUBSTR(SYSDATE
10         ,4
11         ,3
12         ) AS REQUIRED_DATEFORMAT
13* FROM DUAL

```

SQL>

SQL> /

C1	REQUIRED_DATEFORMAT
24-MAY-18	18-24-MAY

SQL> CL SCR

SQL> R

```

1  SELECT SYSDATE AS C1
2  ,SUBSTR(SYSDATE
3          , -2
4          )
5  || '-' || SUBSTR(SYSDATE
6          ,1
7          ,2
8          )
9  || '-' || SUBSTR(SYSDATE
10         ,4
11         ,3
12         ) AS REQUIRED_DATEFORMAT
13* FROM DUAL

```

C1	REQUIRED_DATEFORMAT
24-MAY-18	18-24-MAY

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS C1
2  ,SUBSTR(SYSDATE

```

```

3          ,-2
4          )
5  ||'-'||SUBSTR(SYSDATE
6          ,1
7          ,6
8          )
9  AS REQUIRED_DATEFORMAT
10* FROM DUAL
SQL> /

```

```

C1          REQUIRED_DATEFORMAT
-----
24-MAY-18 18-24-MAY

```

SQL> CL SCR

```

SQL> R
1  SELECT SYSDATE AS C1
2  ,SUBSTR(SYSDATE
3          ,-2
4          )
5  ||'-'||SUBSTR(SYSDATE
6          ,1
7          ,6
8          )
9  AS REQUIRED_DATEFORMAT
10* FROM DUAL

```

```

C1          REQUIRED_DATEFORMAT
-----
24-MAY-18 18-24-MAY

```

SQL> ED
Wrote file afiedt.buf

```

1  SELECT SYSDATE AS C1
2  ,SUBSTR(SYSDATE
3          ,-2
4          )
5  ||'-'||SUBSTR(SYSDATE
6          ,1
7          ,6
8          )
9  AS REQUIRED_DATEFORMAT
10* FROM DUAL
SQL> /

```

```

C1          REQUIRED_DATEFORMAT
-----
24-MAY-18 18-24-MAY

```

SQL> CL SCR

```

SQL> R
1  SELECT SYSDATE AS C1
2  ,SUBSTR(SYSDATE
3          ,-2
4          )
5  ||'-'||SUBSTR(SYSDATE
6          ,1
7          ,6
8          )
9  AS REQUIRED_DATEFORMAT
10* FROM DUAL

```

```

C1          REQUIRED_DATEFORMAT
-----
24-MAY-18  18-24-MAY

```

```
SQL> CL SCR
```

```
SQL> SELECT SYSDATE
       2 FROM DUAL;
```

```

SYSDATE
-----
24-MAY-18

```

```
SQL> ED
Wrote file afiedt.buf
```

```

   1 SELECT SYSDATE C1
   2* FROM DUAL
SQL> /

```

```

C1
-----
24-MAY-18

```

```
SQL> ED
Wrote file afiedt.buf
```

```

   1 SELECT INSTR( SYSDATE
   2                , '-'
   3                , -1
   4                , 1
   5                )
   6* FROM DUAL
SQL> /

```

```

INSTR(SYSDATE, '-', -1, 1)
-----
                          7

```

```
SQL> ED
Wrote file afiedt.buf
```

```

   1 SELECT SYSDATE C1
   2 ,SUBSTR( SYSDATE
   3          ,INSTR( SYSDATE
   4                  , '-'
   5                  , -1
   6                  , 1
   7                  ) + 1
   8          ) AS YY
   9* FROM DUAL
SQL> /

```

```

C1
-----
YY
-----
24-MAY-18
18

```

```
SQL> COLUMN CL SCR
SP2-0158: unknown COLUMN option "SCR"
```

```
SQL> CL SCR
```

```
SQL> COLUMN YY FORMAT A10
```

```
SQL> R
```

```

1  SELECT SYSDATE C1
2    ,SUBSTR( SYSDATE
3            ,INSTR( SYSDATE
4                  , '-'
5                  ,-1
6                  ,1
7                  ) + 1
8            ) AS YY
9* FROM DUAL
```

```

C1          YY
-----
24-MAY-18 18
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE C1
2    ,SUBSTR( SYSDATE
3            ,INSTR( SYSDATE
4                  , '-'
5                  ,-1
6                  ,1
7                  ) + 1
8            ) AS YY
9    ,SUBSTR(SYSATE
10           ,1
11           ,INSTR(SYSDATE
12                 , '-'
13                 ,1
14                 ,2
15                 ) -1
16           ) AS MM
17* FROM DUAL
```

```
SQL> /
```

```

,SUBSTR(SYSATE
      *
```

```
ERROR at line 9:
```

```
ORA-00904: "SYSATE": invalid identifier
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE C1
2    ,SUBSTR( SYSDATE
3            ,INSTR( SYSDATE
4                  , '-'
5                  ,-1
6                  ,1
7                  ) + 1
8            ) AS YY
9    ,SUBSTR(SYSDATE
10           ,1
11           ,INSTR(SYSDATE
12                 , '-'
13                 ,1
14                 ,2
15                 ) -1
16           ) AS MM
```

```

17* FROM DUAL
SQL> /

```

```

C1          YY
-----
MM
-----

```

```

24-MAY-18 18
24-MAY

```

```

SQL> COLUMN CL SCR
SP2-0158: unknown COLUMN option "SCR"
SQL> CL SCR

```

```

SQL> COLUMN MM FORMAT A10
SQL> R

```

```

1  SELECT SYSDATE C1
2    ,SUBSTR( SYSDATE
3            ,INSTR( SYSDATE
4                    , '-'
5                    ,-1
6                    ,1
7                    ) + 1
8            ) AS YY
9    ,SUBSTR(SYSDATE
10           ,1
11           ,INSTR(SYSDATE
12                   , '-'
13                   ,1
14                   ,2
15                   ) -1
16           ) AS MM
17* FROM DUAL

```

```

C1          YY          MM
-----
24-MAY-18 18          24-MAY

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT SYSDATE C1
2    ,SUBSTR( SYSDATE
3            ,INSTR( SYSDATE
4                    , '-'
5                    ,-1
6                    ,1
7                    ) + 1
8            )
9    || '-' || SUBSTR(SYSDATE
10           ,1
11           ,INSTR(SYSDATE
12                   , '-'
13                   ,1
14                   ,2
15                   ) -1
16           ) AS MM
17* FROM DUAL
SQL> /

```

```

C1          MM
-----
24-MAY-18 18-24-MAY

```


Wrote file afiedt.buf

C1	MM
-----	-----
24-MAY-18	18-24-MAY

```
SQL> R
 1  SELECT SYSDATE C1
 2    ,SUBSTR( SYSDATE
 3              ,INSTR( SYSDATE
 4                        , '-'
 5                        ,-1
 6                        ,1
 7                        ) + 1
 8              )
 9    || '-' || SUBSTR(SYSDATE
10                      ,1
11                      ,INSTR(SYSDATE
12                              , '-'
13                              ,1
14                              ,2
15                              ) -1
16                      ) AS MM
17* FROM DUAL
```

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3           ,INSTR( SYSDATE
4                   , '-'
5                   ,-1
6                   ,1
7                   ) + 1
8           )
9  ,

```

```

10* FROM DUAL
SQL> /
FROM DUAL
*
ERROR at line 10:
ORA-00936: missing expression

```

```
SQL> CL SCR
```

```

SQL> R
1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3  ,INSTR( SYSDATE
4  , '-'
5  ,-1
6  ,1
7  ) + 1
8  )
9  ,
10* FROM DUAL
FROM DUAL
*
ERROR at line 10:
ORA-00936: missing expression

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3  ,INSTR( SYSDATE
4  , '-'
5  ,-1
6  ,1
7  ) + 1
8  )
9* FROM DUAL
10 /

```

```

C1
-----
SUBSTR(SYSDATE,INSTR(SYSDATE,'-',-1,1)+1)
-----
24-MAY-18
18

```

```
SQL> CL SCR
```

```

SQL> R
1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3  ,INSTR( SYSDATE
4  , '-'
5  ,-1
6  ,1
7  ) + 1
8  )
9* FROM DUAL

```

```

C1
-----

```

```
SUBSTR(SYSDATE, INSTR(SYSDATE, '-', -1, 1) + 1)
```

```
-----
24-MAY-18
18
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4      , '-'
5      ,-1
6      ,1
7      ) + 1
8      ) AS YY
9* FROM DUAL
SQL> /
```

```

C1      YY
-----
24-MAY-18 18
```

```
SQL> CL SCR
```

```
SQL> R
```

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4      , '-'
5      ,-1
6      ,1
7      ) + 1
8      ) AS YY
9* FROM DUAL
```

```

C1      YY
-----
24-MAY-18 18
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4      , '-'
5      ,-1
6      ,1
7      ) + 1
8      ) AS YY
9  , INSTR(SYSDATE, '-', 1, 1)
10 , INSTR(SYSDATE, '-', 1, 2)
11* FROM DUAL
12 /
```

```

C1      YY      INSTR(SYSDATE, '-', 1, 1)  INSTR(SYSDATE, '-', 1, 2)
-----
24-MAY-18 18                                3                        7
```

```
SQL> CL SCR
```

```
SQL> R
```

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4          , '-'
5          ,-1
6          ,1
7      ) + 1
8      ) AS YY
9  , INSTR(SYSDATE, '-',1,1)
10 , INSTR(SYSDATE, '-',1,2)
11* FROM DUAL

```

C1	YY	INSTR(SYSDATE, '-',1,1)	INSTR(SYSDATE, '-',1,2)
24-MAY-18	18	3	7

SQL> D

SP2-0042: unknown command "D" - rest of line ignored.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4          , '-'
5          ,-1
6          ,1
7      ) + 1
8      ) AS YY
9  ,SUBSTR(SYSDATE
10     ,INSTR(SYSDATE, '-',1,1) + 1
11     ,INSTR(SYSDATE, '-',1,2) - INSTR(SYSDATE, '-',1,1) -1
12     ) AS MM
13* FROM DUAL
SQL> /

```

C1	YY	MM
24-MAY-18	18	MAY

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4          , '-'
5          ,-1
6          ,1
7      ) + 1
8      ) AS YY
9  ,SUBSTR(SYSDATE
10     ,INSTR(SYSDATE, '-',1,1) + 1
11     ,INSTR(SYSDATE, '-',1,2) - INSTR(SYSDATE, '-',1,1) -1
12     ) AS MM
13 ,SUBSTR(SYSDATE
14     ,1
15     ,INSTR(SYSDATE, '-',1,1) -1
16     ) AS DD
17* FROM DUAL
SQL> /

```

C1	YY	MM
----	----	----

DD

 24-MAY-18 18 MAY
 24

SQL> COLUMN DD FORMAT A109

SQL> cl scr

SQL> r

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4      , '-'
5      ,-1
6      ,1
7      ) + 1
8      ) AS YY
9  ,SUBSTR(SYSDATE
10     ,INSTR(SYSDATE, '-',1,1) + 1
11     ,INSTR(SYSDATE, '-',1,2) - INSTR(SYSDATE, '-',1,1) -1
12     ) AS MM
13 ,SUBSTR(SYSDATE
14     ,1
15     ,INSTR(SYSDATE, '-',1,1) -1
16     ) AS DD
17* FROM DUAL
```

C1	YY	MM

DD

 24-MAY-18 18 MAY
 24

SQL> column dd format a10

SQL> r

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4      , '-'
5      ,-1
6      ,1
7      ) + 1
8      ) AS YY
9  ,SUBSTR(SYSDATE
10     ,INSTR(SYSDATE, '-',1,1) + 1
11     ,INSTR(SYSDATE, '-',1,2) - INSTR(SYSDATE, '-',1,1) -1
12     ) AS MM
13 ,SUBSTR(SYSDATE
14     ,1
15     ,INSTR(SYSDATE, '-',1,1) -1
16     ) AS DD
17* FROM DUAL
```

C1	YY	MM	DD

24-MAY-18 18	MAY	24
--------------	-----	----

SQL> cl scr

SQL> r

```

1  SELECT SYSDATE C1
```

```

2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4          , '-'
5          ,-1
6          ,1
7          ) + 1
8      ) AS YY
9  ,SUBSTR(SYSDATE
10     ,INSTR(SYSDATE, '-',1,1)  + 1
11     ,INSTR(SYSDATE, '-',1,2)  - INSTR(SYSDATE, '-',1,1)  -1
12     ) AS MM
13  ,SUBSTR(SYSDATE
14     ,1
15     ,INSTR(SYSDATE, '-',1,1)  -1
16     ) AS DD
17* FROM DUAL

```

C1	YY	MM	DD
24-MAY-18	18	MAY	24

SQL> ed

Wrote file afiedt.buf

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4          , '-'
5          ,-1
6          ,1
7          ) + 1
8      ) AS YY
9  ,SUBSTR(SYSDATE
10     ,INSTR(SYSDATE, '-',1,1)  +1
11     ,instr(sysdate, '-',1,2)  - instr(sysdate, '-',1,1)  -1
12     ) AS MM
13  ,SUBSTR(SYSDATE
14     ,1
15     ,INSTR(SYSDATE, '-',1,1)  -1
16     ) AS DD
17  FROM DUAL
18  /
19* dd-mon-yy
20  /
/
*

```

ERROR at line 18:

ORA-00933: SQL command not properly ended

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4          , '-'
5          ,-1
6          ,1
7          ) + 1
8      ) AS YY
9  ,SUBSTR(SYSDATE
10     ,INSTR(SYSDATE, '-',1,1)  +1
11     ,instr(sysdate, '-',1,2)  - instr(sysdate, '-',1,1)  -1

```

```

12          ) AS MM
13 ,SUBSTR(SYSDATE
14          ,1
15          ,INSTR(SYSDATE,'-',1,1)  -1
16          ) AS DD
17* FROM DUAL
SQL> /
          ,instr(sysdate,'-',1,2)  - instr(sysdate,'-',1,1) -1
          *

```

ERROR at line 11:
ORA-01722: invalid number

SQL> CL SCR

SQL> E
SP2-0042: unknown command "E" - rest of line ignored.
SQL> ED
Wrote file afiedt.buf

```

 1  SELECT SYSDATE C1
 2  ,SUBSTR( SYSDATE
 3          ,INSTR( SYSDATE
 4                  , '-'
 5                  ,-1
 6                  ,1
 7                  ) + 1
 8          ) AS YY
 9  ,SUBSTR(SYSDATE
10          ,INSTR(SYSDATE,'-',1,1) +1
11          ,instr(sysdate,'-',1,2)  - instr(sysdate,'-',1,1) -1
12          ) AS MM
13  ,SUBSTR(SYSDATE
14          ,1
15          ,INSTR(SYSDATE,'-',1,1)  -1
16          ) AS DD
17* FROM DUAL
SQL>
SQL> /
          ,instr(sysdate,'-',1,2)  - instr(sysdate,'-',1,1) -1
          *

```

ERROR at line 11:
ORA-01722: invalid number

SQL> ED
Wrote file afiedt.buf

```

 1  SELECT SYSDATE C1
 2  ,SUBSTR( SYSDATE
 3          ,INSTR( SYSDATE
 4                  , '-'
 5                  ,-1
 6                  ,1
 7                  ) + 1
 8          ) AS YY
 9  ,SUBSTR(SYSDATE
10          ,INSTR(SYSDATE,'-',1,1) +1
11          ,instr(sysdate,'-',1,2)  - instr(sysdate,'-',1,1) -1
12          ) AS MM
13  ,SUBSTR(SYSDATE
14          ,1
15          ,INSTR(SYSDATE,'-',1,1)  -1
16          ) AS DD

```

```
17* FROM DUAL
SQL> /
```

C1	YY	MM	DD
24-MAY-18	18	MAY	24

```
SQL> CL S CR
SP2-0158: unknown CLEAR option "S"
SP2-0158: unknown CLEAR option "CR"
```

```
SQL> R
 1 SELECT SYSDATE C1
 2 ,SUBSTR( SYSDATE
 3         ,INSTR( SYSDATE
 4                 , '-'
 5                 , -1
 6                 , 1
 7                 ) + 1
 8         ) AS YY
 9 ,SUBSTR(SYSDATE
10         ,INSTR(SYSDATE, '-', 1, 1) + 1
11         ,instr(sysdate, '-', 1, 2) - instr(sysdate, '-', 1, 1) - 1
12         ) AS MM
13 ,SUBSTR(SYSDATE
14         , 1
15         ,INSTR(SYSDATE, '-', 1, 1) - 1
16         ) AS DD
17* FROM DUAL
```

C1	YY	MM	DD
24-MAY-18	18	MAY	24

```
SQL> CL SCR
```

```
SQL> R
 1 SELECT SYSDATE C1
 2 ,SUBSTR( SYSDATE
 3         ,INSTR( SYSDATE
 4                 , '-'
 5                 , -1
 6                 , 1
 7                 ) + 1
 8         ) AS YY
 9 ,SUBSTR(SYSDATE
10         ,INSTR(SYSDATE, '-', 1, 1) + 1
11         ,instr(sysdate, '-', 1, 2) - instr(sysdate, '-', 1, 1) - 1
12         ) AS MM
13 ,SUBSTR(SYSDATE
14         , 1
15         ,INSTR(SYSDATE, '-', 1, 1) - 1
16         ) AS DD
17* FROM DUAL
```

C1	YY	MM	DD
24-MAY-18	18	MAY	24

```
SQL> ED
Wrote file afiedt.buf
```

```
 1 SELECT SYSDATE C1
 2 ,SUBSTR( SYSDATE
 3         ,INSTR( SYSDATE
```



```

4          , '-'
5          , -1
6          , 1
7        ) + 1
8      )
9    || '-' || SUBSTR(SYSDATE
10      , INSTR(SYSDATE, '-', 1, 1) + 1
11      , INSTR(sysdate, '-', 1, 2) - INSTR(sysdate, '-', 1, 1) - 1
12      ) AS MM
13    || '-' || SUBSTR(SYSDATE
14      , 1
15      , INSTR(SYSDATE, '-', 1, 1) - 1
16      ) AS DD
17* FROM DUAL
SQL> /
|| '-' || SUBSTR(SYSDATE
*
ERROR at line 13:
ORA-00923: FROM keyword not found where expected

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT SYSDATE C1
2    , SUBSTR( SYSDATE
3      , INSTR( SYSDATE
4        , '-'
5        , -1
6        , 1
7      ) + 1
8    )
9    || '-' || SUBSTR(SYSDATE
10      , INSTR(SYSDATE, '-', 1, 1) + 1
11      , INSTR(sysdate, '-', 1, 2) - INSTR(sysdate, '-', 1, 1) - 1
12      )
13    || '-' || SUBSTR(SYSDATE
14      , 1
15      , INSTR(SYSDATE, '-', 1, 1) - 1
16      ) AS REQ_DDATE_FORMAT
17* FROM DUAL
SQL> /

```

```
C1
```

```
-----
REQ_DDATE_FORMAT
```

```
-----
24-MAY-18
```

```
18-MAY-24
```

```
SQL> CL SCR
```

```

SQL> R
1  SELECT SYSDATE C1
2    , SUBSTR( SYSDATE
3      , INSTR( SYSDATE
4        , '-'
5        , -1
6        , 1
7      ) + 1
8    )
9    || '-' || SUBSTR(SYSDATE
10      , INSTR(SYSDATE, '-', 1, 1) + 1

```

```

11          ,instr(sysdate,'-',1,2) - instr(sysdate,'-',1,1) -1
12      )
13  ||'-'||SUBSTR(SYSDATE
14      ,1
15      ,INSTR(SYSDATE,'-',1,1) -1
16      ) AS REQ_DDATE_FORMAT
17* FROM DUAL

```

C1

REQ_DDATE_FORMAT

24-MAY-18
18-MAY-24

SQL> COLUMN REQ_DDATE_FORMAT FORMAT A15

SQL> R

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4          , '-'
5          ,-1
6          ,1
7      ) + 1
8      )
9  ||'-'||SUBSTR(SYSDATE
10      ,INSTR(SYSDATE,'-',1,1) +1
11      ,instr(sysdate,'-',1,2) - instr(sysdate,'-',1,1) -1
12      )
13  ||'-'||SUBSTR(SYSDATE
14      ,1
15      ,INSTR(SYSDATE,'-',1,1) -1
16      ) AS REQ_DDATE_FORMAT
17* FROM DUAL

```

C1 REQ_DDATE_FORMA

24-MAY-18 18-MAY-24

SQL> CL SCR

SQL> R

```

1  SELECT SYSDATE C1
2  ,SUBSTR( SYSDATE
3      ,INSTR( SYSDATE
4          , '-'
5          ,-1
6          ,1
7      ) + 1
8      )
9  ||'-'||SUBSTR(SYSDATE
10      ,INSTR(SYSDATE,'-',1,1) +1
11      ,instr(sysdate,'-',1,2) - instr(sysdate,'-',1,1) -1
12      )
13  ||'-'||SUBSTR(SYSDATE
14      ,1
15      ,INSTR(SYSDATE,'-',1,1) -1
16      ) AS REQ_DDATE_FORMAT
17* FROM DUAL

```

C1 REQ_DDATE_FORMA

24-MAY-18 18-MAY-24

```
SQL> CL SCR
```

```
SQL> SELECT SYSDATE AS C1
       2 FROM DUAL;
```

```
C1
-----
24-MAY-18
```

```
SQL> ED
Wrote file afiedt.buf
```

```
       1 SELECT SYSDATE AS C1
       2 ,TO_CHAR(SYSDATE
       3           , 'YY-MON-DD'
       4           ) AS C2
       5* FROM DUAL
SQL> /
```

```
C1           C2
-----
24-MAY-18 18-MAY-24
```

```
SQL> ED
Wrote file afiedt.buf
```

```
       1 SELECT SYSDATE AS C1
       2 ,TO_CHAR(SYSDATE
       3           , 'YY'
       4           ) AS C2
       5* FROM DUAL
SQL> /
```

```
C1           C2
-----
24-MAY-18 18
```

```
SQL> CL SCR
```

```
SQL> CL SCR
```

```
SQL> SELECT SYSDATE
       2
SQL>
SQL> SELECT SYSDATE AS C1
       2 FROM DUAL;
```

```
C1
-----
24-MAY-18
```

```
SQL> ED
Wrote file afiedt.buf
```

```
       1 SELECT TO_CHAR( SYSDATE
       2                   , 'HH24:MI:SS'
       3                   ) AS C1
       4* FROM DUAL
SQL> /
```

```
C1
-----
08:38:09
```

```
SQL> R
 1  SELECT TO_CHAR( SYSDATE
 2                , 'HH24:MI:SS'
 3                ) AS C1
4*  FROM  DUAL
```

```
C1
-----
08:38:12
```

```
SQL> R
 1  SELECT TO_CHAR( SYSDATE
 2                , 'HH24:MI:SS'
 3                ) AS C1
4*  FROM  DUAL
```

```
C1
-----
08:38:14
```

```
SQL> R
 1  SELECT TO_CHAR( SYSDATE
 2                , 'HH24:MI:SS'
 3                ) AS C1
4*  FROM  DUAL
```

```
C1
-----
08:38:16
```

```
SQL> ED
Wrote file afiedt.buf
```

```
 1  SELECT TO_CHAR( SYSDATE
 2                , 'HH24'
 3                ) AS C1
4*  FROM  DUAL
SQL> /
```

```
C1
--
08
```

```
SQL> ED
Wrote file afiedt.buf
```

```
 1  SELECT TO_CHAR( SYSDATE
 2                , 'HH24'
 3                ) AS C1
 4  ,TO_CHAR( SYSDATE
 5                , 'MI'
 6                ) AS C1
7*  FROM  DUAL
SQL> /
```

```
C1 C1
-- --
08 38
```

```
SQL> ED
Wrote file afiedt.buf
```

```
 1  SELECT TO_CHAR( SYSDATE
```

```

2          , 'HH24'
3          ) AS C1
4 ,TO_CHAR( SYSDATE
5          , 'MI'
6          ) AS C2
7 ,TO_CHAR( SYSDATE
8          , 'SS'
9          ) AS C3
10* FROM DUAL
SQL> /

```

```

C1 C2 C3
-- -- --
08 38 51

```

```
SQL> CL SCR
```

```

SQL> R
1 SELECT TO_CHAR( SYSDATE
2          , 'HH24'
3          ) AS C1
4 ,TO_CHAR( SYSDATE
5          , 'MI'
6          ) AS C2
7 ,TO_CHAR( SYSDATE
8          , 'SS'
9          ) AS C3
10* FROM DUAL

```

```

C1 C2 C3
-- -- --
08 38 52

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT TO_CHAR( SYSDATE
2          , 'HH24'
3          ) || ' HOURS ' AS C1
4 ,TO_CHAR( SYSDATE
5          , 'MI'
6          ) || ' MINUTES ' AS C2
7 ,TO_CHAR( SYSDATE
8          , 'SS'
9          ) || ' SECONDS ' AS C3
10* FROM DUAL
SQL> /

```

```

C1          C2          C3
-----
08 HOURS   39 MINUTES  15 SECONDS

```

```
SQL> CL SCR
```

```

SQL> R
1 SELECT TO_CHAR( SYSDATE
2          , 'HH24'
3          ) || ' HOURS ' AS C1
4 ,TO_CHAR( SYSDATE
5          , 'MI'
6          ) || ' MINUTES ' AS C2
7 ,TO_CHAR( SYSDATE
8          , 'SS'
9          ) || ' SECONDS ' AS C3

```

```
10* FROM DUAL
```

```

C1          C2          C3
-----
08 HOURS   39 MINUTES 16 SECONDS

```

```
SQL> R
```

```

1  SELECT TO_CHAR( SYSDATE
2                , 'HH24'
3                ) || ' HOURS ' AS C1
4  ,TO_CHAR( SYSDATE
5                , 'MI'
6                ) || ' MINUTES' AS C2
7  ,TO_CHAR( SYSDATE
8                , 'SS'
9                ) || ' SECONDS ' AS C3
10* FROM DUAL

```

```

C1          C2          C3
-----
08 HOURS   39 MINUTES 19 SECONDS

```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT TO_CHAR( SYSDATE
2                , 'HH24'
3                ) || ' HOURS '
4  ,TO_CHAR( SYSDATE
5                , 'MI'
6                ) || ' MINUTES'
7  ,TO_CHAR( SYSDATE
8                , 'SS'
9                ) || ' SECONDS ' AS HOURS_MIN_SEC
10* FROM DUAL
SQL> /

```

```

TO_CHAR(S TO_CHAR(SY HOURS_MIN_S
-----
08 HOURS   39 MINUTES 32 SECONDS

```

```
SQL> CL SCR
```

```
SQL> R
```

```

1  SELECT TO_CHAR( SYSDATE
2                , 'HH24'
3                ) || ' HOURS '
4  ,TO_CHAR( SYSDATE
5                , 'MI'
6                ) || ' MINUTES'
7  ,TO_CHAR( SYSDATE
8                , 'SS'
9                ) || ' SECONDS ' AS HOURS_MIN_SEC
10* FROM DUAL

```

```

TO_CHAR(S TO_CHAR(SY HOURS_MIN_S
-----
08 HOURS   39 MINUTES 33 SECONDS

```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT TO_CHAR( SYSDATE
2                , 'HH24'

```

```

3          ) || ' HOURS '
4  ||TO_CHAR( SYSDATE
5          , 'MI'
6          ) || ' MINUTES'
7  ||TO_CHAR( SYSDATE
8          , 'SS'
9          ) || ' SECONDS '  AS HOURS_MIN_SEC
10* FROM  DUAL
SQL> /

```

```

HOURS_MIN_SEC
-----
08 HOURS 39 MINUTES41 SECONDS

```

```
SQL> CL SCR
```

```

SQL> R
1  SELECT TO_CHAR( SYSDATE
2          , 'HH24'
3          ) || ' HOURS '
4  ||TO_CHAR( SYSDATE
5          , 'MI'
6          ) || ' MINUTES'
7  ||TO_CHAR( SYSDATE
8          , 'SS'
9          ) || ' SECONDS '  AS HOURS_MIN_SEC
10* FROM  DUAL

```

```

HOURS_MIN_SEC
-----
08 HOURS 39 MINUTES42 SECONDS

```

```

SQL> R
1  SELECT TO_CHAR( SYSDATE
2          , 'HH24'
3          ) || ' HOURS '
4  ||TO_CHAR( SYSDATE
5          , 'MI'
6          ) || ' MINUTES'
7  ||TO_CHAR( SYSDATE
8          , 'SS'
9          ) || ' SECONDS '  AS HOURS_MIN_SEC
10* FROM  DUAL

```

```

HOURS_MIN_SEC
-----
08 HOURS 39 MINUTES44 SECONDS

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT TO_CHAR( SYSDATE
2          , 'HH24'
3          ) || ' HOURS '
4  ||TO_CHAR( SYSDATE
5          , 'MI'
6          ) || ' MINUTES '
7  ||TO_CHAR( SYSDATE
8          , 'SS'
9          ) || ' SECONDS '  AS HOURS_MIN_SEC
10* FROM  DUAL
SQL> /

```

```

HOURS_MIN_SEC

```

```
-----
08 HOURS 39 MINUTES 50 SECONDS
```

```
SQL> R
1  SELECT TO_CHAR( SYSDATE
2              , 'HH24'
3              ) || ' HOURS '
4  || TO_CHAR( SYSDATE
5              , 'MI'
6              ) || ' MINUTES '
7  || TO_CHAR( SYSDATE
8              , 'SS'
9              ) || ' SECONDS '  AS HOURS_MIN_SEC
10* FROM DUAL
```

```
HOURS_MIN_SEC
-----
08 HOURS 39 MINUTES 51 SECONDS
```

```
SQL> R
1  SELECT TO_CHAR( SYSDATE
2              , 'HH24'
3              ) || ' HOURS '
4  || TO_CHAR( SYSDATE
5              , 'MI'
6              ) || ' MINUTES '
7  || TO_CHAR( SYSDATE
8              , 'SS'
9              ) || ' SECONDS '  AS HOURS_MIN_SEC
10* FROM DUAL
```

```
HOURS_MIN_SEC
-----
08 HOURS 39 MINUTES 51 SECONDS
```

```
SQL> R
1  SELECT TO_CHAR( SYSDATE
2              , 'HH24'
3              ) || ' HOURS '
4  || TO_CHAR( SYSDATE
5              , 'MI'
6              ) || ' MINUTES '
7  || TO_CHAR( SYSDATE
8              , 'SS'
9              ) || ' SECONDS '  AS HOURS_MIN_SEC
10* FROM DUAL
```

```
HOURS_MIN_SEC
-----
08 HOURS 39 MINUTES 54 SECONDS
```

```
SQL> CL SCR
```

```
SQL> SELECT REPLACE('SURESH IS A GOOD BOY'
2              , 'GOOD'
3              , 'BAD'
4              ) AS C1
5  FROM DUAL;
```

```
C1
-----
SURESH IS A BAD BOY
```

```
SQL> EDIT
```


Wrote file afiedt.buf

```

1  SELECT REPLACE('SURESH IS A GOOD BOY'
2                      , 'GOOD'
3                      , 'VERY VERY BAD'
4                      ) AS C1
5* FROM DUAL
SQL> /

```

C1

SURESH IS A VERY VERY BAD BOY

SQL> ED

Wrote file afiedt.buf

```

1  SELECT REPLACE('SURESH IS A GOOD BOY'
2                      , 'GOOD'
3                      , 'VERY VERY GOOD'
4                      ) AS C1
5* FROM DUAL
SQL> /

```

C1

SURESH IS A VERY VERY GOOD BOY

SQL> ED

Wrote file afiedt.buf

```

1  SELECT REPLACE('SURESH IS A GOOD BOY'
2                      , 'GOOD'
3                      ) AS C1
4* FROM DUAL
SQL> /

```

C1

SURESH IS A BOY

SQL> CL SCR

SQL> R

```

1  SELECT REPLACE('SURESH IS A GOOD BOY'
2                      , 'GOOD'
3                      ) AS C1
4* FROM DUAL

```

C1

SURESH IS A BOY

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT REPLACE('SURESH IS A GOOD BOY'
2                      ) AS C1
3* FROM DUAL

```

SQL> /

```

SELECT REPLACE('SURESH IS A GOOD BOY'
*

```

ERROR at line 1:

ORA-00938: not enough arguments for function

```
SQL> ED
Wrote file afiedt.buf

  1  SELECT REPLACE('SURESH IS A GOOD BOY'
  2                      , 'GIRL'
  3                      , 'BEAUTIGUL GIRL'
  4                      ) AS C1
  5* FROM DUAL
SQL> /
```

```
C1
-----
SURESH IS A GOOD BOY
```

```
SQL> CL SCR
```

```
SQL> R
  1  SELECT REPLACE('SURESH IS A GOOD BOY'
  2                      , 'GIRL'
  3                      , 'BEAUTIGUL GIRL'
  4                      ) AS C1
  5* FROM DUAL
```

```
C1
-----
SURESH IS A GOOD BOY
```

```
SQL> CL SCR
```

```
SQL> SET PAGESIZE 200
SQL> SET LINESIZE 200
SQL> R
  1  SELECT REPLACE('SURESH IS A GOOD BOY'
  2                      , 'GIRL'
  3                      , 'BEAUTIGUL GIRL'
  4                      ) AS C1
  5* FROM DUAL
```

```
C1
-----
SURESH IS A GOOD BOY
```

```
SQL> CL SCR
```

```
SQL> R
  1  SELECT REPLACE('SURESH IS A GOOD BOY'
  2                      , 'GIRL'
  3                      , 'BEAUTIGUL GIRL'
  4                      ) AS C1
  5* FROM DUAL
```

```
C1
-----
SURESH IS A GOOD BOY
```

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
  2      ,SAL
  3      ,JOB
  4  FROM EMP;
```

```
ENAME
```

```
SAL  JOB
```

```

-----
KING                5000  PRESIDENT
BLAKE               2850  MANAGER
CLARK               2450  MANAGER
JONES               2975  MANAGER
SCOTT               3000  ANALYST
FORD                3000  ANALYST
SMITH                800   CLERK
ALLEN               1600  SALESMAN
WARD                1250  SALESMAN
MARTIN              1250  SALESMAN
TURNER              1500  SALESMAN
ADAMS               1100  CLERK
JAMES                950   CLERK
MILLER              1300  CLERK
SHAIK               2000

```

15 rows selected.

SQL> COLUMN ENAME FORMAT A10

SQL> R

```

 1  SELECT ENAME
 2  ,SAL
 3  ,JOB
 4* FROM EMP

```

```

ENAME                SAL  JOB
-----
KING                5000  PRESIDENT
BLAKE               2850  MANAGER
CLARK               2450  MANAGER
JONES               2975  MANAGER
SCOTT               3000  ANALYST
FORD                3000  ANALYST
SMITH                800   CLERK
ALLEN               1600  SALESMAN
WARD                1250  SALESMAN
MARTIN              1250  SALESMAN
TURNER              1500  SALESMAN
ADAMS               1100  CLERK
JAMES                950   CLERK
MILLER              1300  CLERK
SHAIK               2000

```

15 rows selected.

SQL> CL SCR

SQL> R

```

 1  SELECT ENAME
 2  ,SAL
 3  ,JOB
 4* FROM EMP

```

```

ENAME                SAL  JOB
-----
KING                5000  PRESIDENT
BLAKE               2850  MANAGER
CLARK               2450  MANAGER
JONES               2975  MANAGER
SCOTT               3000  ANALYST
FORD                3000  ANALYST
SMITH                800   CLERK
ALLEN               1600  SALESMAN

```

```

WARD          1250 SALESMAN
MARTIN        1250 SALESMAN
TURNER        1500 SALESMAN
ADAMS         1100 CLERK
JAMES         950  CLERK
MILLER        1300 CLERK
SHAIK         2000

```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  ,REPLACE(JOB
5           , 'MAN'
6           , 'DAM'
7           ) AS JOB1
8* FROM EMP
SQL> /

```

ENAME	SAL	JOB	JOB1
KING	5000	PRESIDENT	PRESIDENT
BLAKE	2850	MANAGER	DAMAGER
CLARK	2450	MANAGER	DAMAGER
JONES	2975	MANAGER	DAMAGER
SCOTT	3000	ANALYST	ANALYST
FORD	3000	ANALYST	ANALYST
SMITH	800	CLERK	CLERK
ALLEN	1600	SALESMAN	SALESDAM
WARD	1250	SALESMAN	SALESDAM
MARTIN	1250	SALESMAN	SALESDAM
TURNER	1500	SALESMAN	SALESDAM
ADAMS	1100	CLERK	CLERK
JAMES	950	CLERK	CLERK
MILLER	1300	CLERK	CLERK
SHAIK	2000		

15 rows selected.

SQL> CL SCR

```

SQL> SELECT REPLACE('SURESH IS A GOOD BOY'
2           , 'BOY'
3           , '123'
4           ) AS C1
5           ,TRANSLATE('SURESH IS A GOOD BOY'
6           , 'BOY'
7           , '123'
8           ) AS C2
9 FROM DUAL;

```

C1	C2
SURESH IS A GOOD 123	SURESH IS A G22D 123

SQL> ED

Wrote file afiedt.buf

```

1  SELECT REPLACE('SURESH IS A GOOD BOY'
2           , 'BOYS'

```

```

3          , '1234'
4          ) AS C1
5      , TRANSLATE('SURESH IS A GOOD BOY'
6          , 'BOYS'
7          , '1234'
8          ) AS C2
9* FROM DUAL
SQL> /

```

```

C1          C2
-----
SURESH IS A GOOD BOY 4URE4H I4 A G22D 123

```

```

SQL> R
1  SELECT REPLACE('SURESH IS A GOOD BOY'
2          , 'BOYS'
3          , '1234'
4          ) AS C1
5      , TRANSLATE('SURESH IS A GOOD BOY'
6          , 'BOYS'
7          , '1234'
8          ) AS C2
9* FROM DUAL

```

```

C1          C2
-----
SURESH IS A GOOD BOY 4URE4H I4 A G22D 123

```

```
SQL> CL SCR
```

```

SQL> R
1  SELECT REPLACE('SURESH IS A GOOD BOY'
2          , 'BOYS'
3          , '1234'
4          ) AS C1
5      , TRANSLATE('SURESH IS A GOOD BOY'
6          , 'BOYS'
7          , '1234'
8          ) AS C2
9* FROM DUAL

```

```

C1          C2
-----
SURESH IS A GOOD BOY 4URE4H I4 A G22D 123

```

```
SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT TRANSLATE('SURESH IS A GOOD BOY'
2          , 'BOYS'
3          , '1234'
4          ) AS C2
5* FROM DUAL
SQL> /

```

```

C2
-----
4URE4H I4 A G22D 123

```

```
SQL> CL SCR
```

```

SQL> R
1  SELECT TRANSLATE('SURESH IS A GOOD BOY'
2          , 'BOYS'

```

```
3          , '1234'
4          ) AS C2
5* FROM DUAL
```

```
C2
-----
4URE4H I4 A G22D 123
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT TRANSLATE('SURESH IS A GOOD BOY'
2                    , 'BOYS'
3                    , '123'
4                    ) AS C2
5* FROM DUAL
SQL> /
```

```
C2
-----
UREH I A G22D 123
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT TRANSLATE('SURESH IS A GOOD BOY'
2                    , 'BOYS'
3                    , '12345678'
4                    ) AS C2
5* FROM DUAL
SQL> /
```

```
C2
-----
4URE4H I4 A G22D 123
```

```
SQL> CL SCR
```

```
SQL> R
1  SELECT TRANSLATE('SURESH IS A GOOD BOY'
2                    , 'BOYS'
3                    , '12345678'
4                    ) AS C2
5* FROM DUAL
```

```
C2
-----
4URE4H I4 A G22D 123
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT TRANSLATE('SURESH IS A GOOD BOY'
2                    , 'BOYS'
3                    ) AS C2
4* FROM DUAL
SQL> /
SELECT TRANSLATE('SURESH IS A GOOD BOY'
                *
```

```
ERROR at line 1:
ORA-00909: invalid number of arguments
```

```
SQL> CL SCR
```

```
SQL> CL SCR
```

```
SQL> SELECT LPAD(
2          'SURESH'
3          ,10
4          , '*'
5          ) AS C1
6  FROM MDUAL;
FROM MDUAL
*
```

ERROR at line 6:
ORA-00942: table or view does not exist

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT LPAD(
2          'SURESH'
3          ,10
4          , '*'
5          ) AS C1
6* FROM DUAL
SQL> /
```

```
C1
-----
****SURESH
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT LPAD(
2          'SURESH'
3          ,20
4          , '*'
5          ) AS C1
6* FROM DUAL
SQL> /
```

```
C1
-----
*****SURESH
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT LPAD(
2          'SURESH'
3          ,10
4          ) AS C1
5* FROM DUAL
SQL> /
```

```
C1
-----
SURESH
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT LPAD(
2          'KSURESH'
```

```

3          ,10
4          ,'^*'
5        ) AS C1
6* FROM DUAL
7 /

```

```

C1
-----
*^*KSURESH

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT LPAD(
2      'SURESH'
3      ,3
4      , '*'
5  ) AS C1
6* FROM DUAL
SQL> /

```

```

C1
---
SUR

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT RPAD(
2      'SURESH'
3      ,10
4      , '*'
5  ) AS C1
6* FROM DUAL
SQL> /

```

```

C1
-----
SURESH****

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT RPAD(
2      'SURESH'
3      ,20
4      , '*'
5  ) AS C1
6* FROM DUAL
SQL> /

```

```

C1
-----
SURESH*****

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT RPAD(
2      'KSURESH'
3      ,10
4      , '^*'
5  ) AS C1
6* FROM DUAL

```



```
SQL> /
```

```
C1
-----
KSURESH*^*
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT RPAD(
2      'KSURESH'
3      ,10
4      ) AS C1
5*  FROM DUAL
6  /
```

```
C1
-----
KSURESH
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT RPAD(
2      'SURESH'
3      ,3
4      , '*'
5      ) AS C1
6*  FROM DUAL
SQL> /
```

```
C1
---
SUR
```

```
SQL> CL SCR
```

```
SQL> R
1  SELECT RPAD(
2      'SURESH'
3      ,3
4      , '*'
5      ) AS C1
6*  FROM DUAL
```

```
C1
---
SUR
```

25-MAY-2018 LTRIM,RTIM, COALESCE

```
-----
UPPER
LOWER
INITCAP
SUBSTR
INSTR
LPAD
RPAD
REPLACE
TRANSLATE
```

```
-----  
GREATEST  
(  
P1  
,P2  
,P3  
,P4  
.  
.  
.  
PN  
)  
IT WILL GIVE YOU GREATEST VALUE WHAT IS THERE IN A ROW
```

```
GREATEST(10,20,30,40,50,60,90,15,115,25)
```

```
IF 999 > 15000 THEN  
    15000 IS YOUR PROCE
```

```
ELSE  
    999 IS HYOU RPROCESSIN GES
```

```
END IF;
```

```
999 1% WHICHE IS EVER HIGHER
```

```
GREATE3ST(999,15000)  -- 15000
```

```
LEAST(10,20,30,40,50,60,90,15,115,25)
```

```
T1  
-----  
C1    C2    C3  
10    25    40  
100   5     125  
120   135   110  
90    76    55
```

```
GREATEST(C1,C2,C3)  -- 40
```

```
MAX(40  
125  
135  
90  
)
```

```
135
```

1) IN T1 TABLE WHAT IS THE HIGHEST VALUE ?-- GREATEST + MAX

2) IN C1,C2,C3 COLUMNS WHAT IS THE HIGHEST VALUE-- MAX

3) IN EACH ROW IN C1,C2,C3 COLUMNS WHICH COLUMN IS HAVINNG HIGHEST DATA?-- GREATEST

```
LTRIM
(
  P1-- COLUMN /LITERAL
,P2-- TRIMMED CHARACTER SET(it can be combination of alphabets,numbers,special
characters
)
```

```
LTRIM('xyzXYZzyx'
      , 'y'
      )
```

- 1) it will take one by one character from p1 and it will check if that character is available in p2
- 2) if it is available it will remove that character from p1 and take next character and it will repeat step1
- 3) if it is not available it will stop removing from that character whatever is pending/remaining in p1 that will give you
as output.

NOTE:

IF ALL CHARACTERS ARE TRIMMED(REMOVED) THEN OUTPUT WILL BE NULL
P2 PARAMETER IS OPTIONAL IF P2 VALUE NOT PROVIDED BY DEFAULT IT WILL REMOVE ALL LEFT SIDE SPACES WHICH ARE THERE IN P1
(I.E IT WILL CONSIDER P2 VALUE AS SPACE(' '))

RTRIM

NOTE :RTRIM ALSO SAME AS LTRIM BUT IT WILL TRIM FROM RIGHT SIDE

TRIM

IF YOU WANT TO TRIM BOTH SIDES THEN WE WILL GO FOR TRIM

SYNTAX:

```
TRIM ( trimCHARACTER FROM COLUMN/LITERALS)
```

by using trim we can trim both sides but only one character we can trim
(i.e trimmed character should be only one character)

ORA-30001: trim set should have only one character

NVL

```
(
P1-- COLUMN/LITERAL
,P2-- COLUMN/LITERAL
)
```

IF P1 VALUE IS NULL THEN IT WILL GIVE YOU P2 VALUE
IF P1 VALUE IS NOT NULL THEN IT WILL GIVE YOU P1 ONLY

P1 AND P2 DATA TYPE SHOULD BE SAME
P1 AND P2 BOTH PARAMETERS ARE MANDATORY

NVL2

(
 P1-- COLUMN/LITERAL
 ,P2-- COLUMN/LITERAL
 ,P3-- COLUMN/LITERAL
)

IF P1 VALUE IS NULL THEN IT WILL GIVE YOU P3 VALUE
IF P1 VALUE IS NOT NULL THEN IT WILL GIVE YOU P2 VALUE
P1 ,P2,P3 DATA TYPE CAN BE ANY DATA TYPE

NOTE :

WHENEVER YOU GET A REQUIREMENT OF IF P1 VALUE IS NULL DO SOME THING ELSE DO SOME
THIHGG IN THAT SCENARIO WE CAN
USE NVL2

DIFF BETWEEN NVL AND NVL2 ?

NULLIF

NULLIF
(
 P1-- COLUMN/LITERAL
 ,P2--COLUMN/LITERAL
)

IF P1 = P2 THEN OUTPUT WILL BE NULL
IF P1 <> P2 THEN OUTPUT WILL BE P1 VALUE
P1 AND P2 DATA TYPE SHOULD BE SAME

COALESCE

COALESCE

(
 P1
 ,P2
 ,P3
 .
 .
 PN
)

IT WILL GIVE YOU FIRST NOT NULL PARAMETER VALUE.

IF P1 IS A NUMBER DATA TYPE ALL OTHER PARAMETERS VALUES SHOULD BE NUMBER
(I.E ALL PARAMETERS DATA TYPE SHOULD BE SAME)

FIRST NAME MIDDLENAME LAST NAME

'MR ' || COALESCE(FIRSTNAME,MNAME,LANME)

ARJUN_FUC

```
CONCAT
(
P1
,P2
)
```

IT WILL COMBINE P1 AND P2 VALUE

CHR

EMPLOYEE_INFORMATION

```
-----
King SALARY IS 5000 AND WORKING AS PRESIDENT
Blake SALARY IS 2850 AND WORKING AS MANAGER
```

25-MAY-2018 - LTRIM,RTIM, COALESCE Examples

```
1  SELECT GREATEST(10,20,30,40,50,60,90,15,115,25) AS C1
2*  FROM DUAL
```

```
      C1
-----
      115
```

```
1  SELECT GREATEST(10,20,30,40,50,60,90,15,115,25,160) AS C1
2*  FROM DUAL
SQL> /
```

```
      C1
-----
      160
```

```
SQL> COLUMN ENAME FORMAT A10
SQL> CL SCR
```

```
SQL> SELECT ENAME
2      ,SAL
3      ,COMM
4  FROM EMP;
```

no rows selected

```
SQL> @C:\PROJECT\PROJECT\Personal\Inst\EMP_DATA.SQL
```

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
2      ,SAL
3      ,COMM
4  FROM EMP;
```

ENAME	SAL	COMM
KING	5000	
BLAKE	2850	
CLARK	2450	
JONES	2975	
SCOTT	3000	
FORD	3000	
SMITH	800	
ALLEN	1600	300
WARD	1250	500
MARTIN	1250	1400
TURNER	1500	0
ADAMS	1100	
JAMES	950	
MILLER	1300	

14 rows selected.

SQL> CL SCR

SQL> SELECT GREATEST(100,200) AS C1
2 FROM DUAL;

C1
200

SQL> ED
Wrote file afiedt.buf

1 SELECT GREATEST('SURESH','KRISHNA') AS C1
2* FROM DUAL
SQL> /

C1
SURESH

SQL> ED
Wrote file afiedt.buf

1 SELECT GREATEST('SURESH','KRISHNA','Z') AS C1
2* FROM DUAL
SQL> /

C
Z

SQL> CL SCR

SQL> ED
Wrote file afiedt.buf

1 SELECT GREATEST(SYSDATE , SYSDATE +10, SYSDATE - 10) AS C1
2* FROM DUAL
SQL> /

C1
04-JUN-18

```
SQL> ED
Wrote file afiedt.buf

  1  SELECT LEAST(SYSDATE , SYSDATE +10, SYSDATE - 10 ) AS C1
  2* FROM DUAL
SQL> /
```

```
C1
-----
15-MAY-18
```

```
SQL> ED
Wrote file afiedt.buf

  1  SELECT LEAST ('SURESH','KRISHNA','ARJUN') AS C1
  2* FROM DUAL
SQL> /
```

```
C1
-----
ARJUN
```

```
SQL> ED
Wrote file afiedt.buf

  1  SELECT LEAST (10,20,5,125,145,110,67) AS C1
  2* FROM DUAL
SQL> /
```

```
          C1
-----
          5
```

```
SQL> CL SCR
```

```
SQL> R
  1  SELECT LEAST (10,20,5,125,145,110,67) AS C1
  2* FROM DUAL
```

```
          C1
-----
          5
```

```
SQL> CL SCR
```

```
SQL> select LTRIM(/
  2          , 'y'
  3          ) as c1
  4 from dual;
```

```
C1
-----
xyzXYZzyx
```

```
SQL> edit
Wrote file afiedt.buf
```

```
  1  select LTRIM('xyzXYZzyx'
  2                , 'x'
  3                ) as c1
  4* from dual
  5  /
```

```
C1
```

```
-----  
yzXYZzyx
```

```
SQL> edit  
Wrote file afiedt.buf
```

```
 1  select LTRIM('xyzXYZzyx'  
 2                      , 'yx'  
 3                      ) as c1  
 4* from dual  
SQL> /
```

```
C1  
-----  
zXYZzyx
```

```
SQL> ed  
Wrote file afiedt.buf
```

```
 1  select LTRIM('xyzXYZzyx'  
 2                      , '*yx'  
 3                      ) as c1  
 4* from dual  
SQL> /
```

```
C1  
-----  
zXYZzyx
```

```
SQL> ed  
Wrote file afiedt.buf
```

```
 1  select LTRIM('xyzXYZzyx'  
 2                      , 'z*yx'  
 3                      ) as c1  
 4* from dual  
SQL> /
```

```
C1  
-----  
XYZzyx
```

```
SQL> ed  
Wrote file afiedt.buf
```

```
 1  select LTRIM('xyzXYZzyx'  
 2                      , 'z*yxZ'  
 3                      ) as c1  
 4* from dual  
SQL> /
```

```
C1  
-----  
XYZzyx
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
 1  select LTRIM('xyzXYZzyx'  
 2                      , 'Yz*yxZ'  
 3                      ) as c1  
 4* from dual  
SQL> /
```



```
C1
-----
XYZzyx
```

```
SQL> cl scr
```

```
SQL> r
 1 select LTRIM('xyzXYZzyx'
 2             , 'Yz*yxZ'
 3             ) as c1
 4* from dual
```

```
C1
-----
XYZzyx
```

```
SQL> ed
Wrote file afiedt.buf
```

```
 1 select LTRIM('xyzXYZzyx'
 2             , 'Yz*yxZX'
 3             ) as c1
 4* from dual
SQL> /
```

```
C
-
```

```
SQL> COLUMN C1 FORMAT A10
SQL> CLS CR
SP2-0042: unknown command "CLS CR" - rest of line ignored.
SQL> CL SCR
```

```
SQL> R
 1 select LTRIM('xyzXYZzyx'
 2             , 'Yz*yxZX'
 3             ) as c1
 4* from dual
```

```
C1
-----
```

```
SQL> ED
Wrote file afiedt.buf
```

```
 1 select LTRIM('9xyzXYZzyx'
 2             , 'Yz*yxZX'
 3             ) as c1
 4* from dual
SQL> /
```

```
C1
-----
9xyzXYZzyx
```

```
SQL> cl scr
```

```
SQL> R
 1 select LTRIM('9xyzXYZzyx'
 2             , 'Yz*yxZX'
 3             ) as c1
```

```
4* from dual

C1
-----
9xyzXYZzyx

SQL> EDIT
Wrote file afiedt.buf

 1 select LTRIM('      xyzXYZzyx'
 2                , 'Yz*yxZX'
 3                ) as c1
 4* from dual
SQL> /
```

```
C1
-----
      xyzX
YZzyx
```

```
SQL> COLUMN C1 FORMAT A20
SQL> R
 1 select LTRIM('      xyzXYZzyx'
 2                , 'Yz*yxZX'
 3                ) as c1
 4* from dual
```

```
C1
-----
      xyzXYZzyx
```

```
SQL> CL SCR
```

```
SQL> R
 1 select LTRIM('      xyzXYZzyx'
 2                , 'Yz*yxZX'
 3                ) as c1
 4* from dual
```

```
C1
-----
      xyzXYZzyx
```

```
SQL> ED
Wrote file afiedt.buf
```

```
 1 select LTRIM('      xyzXYZzyx'
 2                ) as c1
 3* from dual
SQL> /
```

```
C1
-----
xyzXYZzyx
```

```
SQL> ED
Wrote file afiedt.buf
```

```
 1 select LTRIM('      xyzXYZzyx'
 2                , ' '
 3                ) as c1
 4* from dual
SQL> /
```

```
C1
-----
xyzXYZzyx
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  select LTRIM('      xyzXYZzyx'
2              ) as c1
3* from dual
4  /
```

```
C1
-----
xyzXYZzyx
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  select LTRIM('7      xyzXYZzyx'
2              , ' '
3              ) as c1
4* from dual
SQL> /
```

```
C1
-----
7      xyzXYZzyx
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  select LTRIM('      xyzXYZzyx'
2              , 'X'
3              ) as c1
4* from dual
SQL> /
```

```
C1
-----
      xyzXYZzyx
```

```
SQL> CL  SCR
```

```
SQL> R
1  select LTRIM('      xyzXYZzyx'
2              , 'X'
3              ) as c1
4* from dual
```

```
C1
-----
      xyzXYZzyx
```

```
SQL> CL SCR
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```
1  select RTRIM('xyzXYZzyx'
2              , 'x'
3              ) as c1
4* from dual
```

```
SQL> /
```

```
C1
-----
xyzXYZzy
```

```
SQL> ed
Wrote file afiedt.buf
```

```
1  select RTRIM('xyzXYZzyx'
2              , 'xX'
3              ) as c1
4*  from dual
SQL> /
```

```
C1
-----
xyzXYZzy
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  select RTRIM('xyzXYZzyx'
2              , 'YxX'
3              ) as c1
4*  from dual
SQL> /
```

```
C1
-----
xyzXYZzy
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  select RTRIM('xyzXYZzyx'
2              , 'YxXy'
3              ) as c1
4*  from dual
SQL> /
```

```
C1
-----
xyzXYZz
```

```
SQL> ed
Wrote file afiedt.buf
```

```
1  select RTRIM('xyzXYZzyx'
2              , 'zYxXy'
3              ) as c1
4*  from dual
SQL> /
```

```
C1
-----
xyzXYZ
```

```
SQL> ed
Wrote file afiedt.buf
```

```
1  select RTRIM('xyzXYZzyx'
2              , 'zYxXyZ'
3              ) as c1
```

```
4* from dual
SQL> /
```

```
C1
-----
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  select RTRIM('xyzXYZzyx'
2              , 'zYxXyZARJUN'
3              ) as c1
4* from dual
SQL> /
```

```
C1
-----
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  select RTRIM('xyzXYZzyx'
2              , 'zYxXyZARJUN45347053475323SDFDSFASFSDSASFSDAFADSF'
3              ) as c1
4* from dual
SQL> /
```

```
C1
-----
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  select RTRIM('xyzXYZzyx'
2              , 'zYxXyZARJUN45347053475323SDFDSFASFSDSASFSDAFADSF'
3              ) as c1
4* from dual
SQL>
SQL> /
```

```
C1
-----
```

```
SQL> CL SCR
```

```
SQL> R
1  select RTRIM('xyzXYZzyx'
2              , 'zYxXyZARJUN45347053475323SDFDSFASFSDSASFSDAFADSF'
3              ) as c1
4* from dual
```

```
C1
-----
```

```
SQL> CL SCR
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```

1  select  TRIM('x'
2          FROM
3          'xyzXYZzyx'
4          ) as c1
5* from dual
SQL> /

```

```

C1
-----
yzXYZzy

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  select  TRIM('x'
2          FROM
3          'xxxxxxxxxxxxxxxxxxxxxxxxyzXYZzyxxxxxxxxxxxxxxxxxxxxxxxxxxxxx'
4          ) as c1
5* from dual
SQL> '/
SP2-0042: unknown command "'/'" - rest of line ignored.
SQL> /

```

```

C1
-----
yzXYZzy

```

```

SQL> ed
Wrote file afiedt.buf

```

```

1  select  TRIM('x'
2          FROM
3          '7xxxxxxxxxxxxxxxxxxxxxxxxyzXYZzyxxxxxxxxxxxxxxxxxxxxxxxxxxxxx'
4          ) as c1
5* from dual
SQL> /

```

```

C1
-----
7xxxxxxxxxxxxxxxxxxxxxxx
xyzXYZzy

```

```

SQL> column c1 format a30
SQL> r

```

```

1  select  TRIM('x'
2          FROM
3          '7xxxxxxxxxxxxxxxxxxxxxxxxyzXYZzyxxxxxxxxxxxxxxxxxxxxxxxxxxxxx'
4          ) as c1
5* from dual

```

```

C1
-----
7xxxxxxxxxxxxxxxxxxxxxxxxyzXYZzy

```

```

SQL> ed
Wrote file afiedt.buf

```

```

1  select  TRIM('x'
2          FROM
3          '7xxxxxxxxxxxxxxxxxxxxxxxxyzXYZzy'
4          ) as c1
5* from dual
SQL> /

```

Wrote file afiedt.buf

SQL> /

ERROR at line 1:

ORA-30001: trim set should have only one character

Wrote file afiedt.buf

[illegible]

C1

xxxxxxxxxxzXyZzxxxxxxxx

Wrote file afiedt.buf

[illegible]

C1

zXYZz

```
SQL> CL SCR
```

```

1  select  TRIM ('y'
2          FROM
3          TRIM ('x'
4          FROM
5
6  'xxxxxxxxxxxxxxxxxxxxxyyyyyyyzXYZzyyyyyyyxxxxxxxxxxxxxxxxxxxxxx'
7  )

```

```

7          ) as c1
8* from dual

```

```

C1
-----
zXYZzz

```

```
SQL> CL SCR
```

```

SQL> SELECT NVL(NULL
2          ,10
3          ) AS C1
4  FROM DUAL;

```

```

          C1
-----
#####

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT NVL(NULL
2          ,10
3          ) AS C2
4* FROM DUAL
SQL> /

```

```

          C2
-----
10

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT NVL(5
2          ,10
3          ) AS C2
4* FROM DUAL
SQL> /

```

```

          C2
-----
5

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT NVL(NULL
2          ,NULL
3          ) AS C2
4* FROM DUAL
SQL> /

```

```

C
-

```

```
SQL> CL SCR
```

```

SQL> SELECT ENAME
2  ,SAL
3  FROM EMP;

```

```

ENAME          SAL

```



```

-----
KING          5000
BLAKE         2850
CLARK         2450
JONES         2975
SCOTT         3000
FORD          3000
SMITH         800
ALLEN         1600
WARD          1250
MARTIN        1250
TURNER        1500
ADAMS         1100
JAMES         950
MILLER        1300

```

14 rows selected.

SQL> COLUMN SAL FORMAT A10

SQL> R

```

1  SELECT ENAME
2  ,SAL
3* FROM EMP

```

```

ENAME          SAL
-----
KING          #####
BLAKE         #####
CLARK         #####
JONES         #####
SCOTT         #####
FORD          #####
SMITH         #####
ALLEN         #####
WARD          #####
MARTIN        #####
TURNER        #####
ADAMS         #####
JAMES         #####
MILLER        #####

```

14 rows selected.

SQL> COLUMN SAL FORMAT 99999

SQL> R

```

1  SELECT ENAME
2  ,SAL
3* FROM EMP

```

```

ENAME          SAL
-----
KING          5000
BLAKE         2850
CLARK         2450
JONES         2975
SCOTT         3000
FORD          3000
SMITH         800
ALLEN         1600
WARD          1250
MARTIN        1250
TURNER        1500
ADAMS         1100
JAMES         950

```

MILLER 1300

14 rows selected.

SQL> CL SCR

SQL> R

```
1  SELECT ENAME
2  ,SAL
3* FROM EMP
```

ENAME	SAL
KING	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	3000
FORD	3000
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

SQL> CL SCR

```
SQL> SELECT NVL2(NULL
2      ,10
3      ,20
4      ) AS C2
5 FROM DUAL;
```

C2
20

SQL> ED

Wrote file afiedt.buf

```
1  SELECT NVL2(5
2      ,10
3      ,20
4      ) AS C2
5* FROM DUAL
SQL> /
```

C2
10

SQL> CL SCR

```
SQL> SELECT ENAME
2  ,SAL
3  ,COMM
4  FROM EMP;
```

ENAME	SAL	COMM
-------	-----	------

```

-----
KING          5000
BLAKE         2850
CLARK         2450
JONES         2975
SCOTT         3000
FORD          3000
SMITH         800
ALLEN         1600          300
WARD          1250          500
MARTIN        1250        1400
TURNER        1500          0
ADAMS         1100
JAMES         950
MILLER        1300

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2    ,SAL
 3    ,COMM
 4    , NVL2 ( COMM
 5              ,1000
 6              ,2000
 7            )
 8*  FROM EMP
SQL> /

```

ENAME	SAL	COMM	NVL2 (COMM,1000,2000)
KING	5000		2000
BLAKE	2850		2000
CLARK	2450		2000
JONES	2975		2000
SCOTT	3000		2000
FORD	3000		2000
SMITH	800		2000
ALLEN	1600	300	1000
WARD	1250	500	1000
MARTIN	1250	1400	1000
TURNER	1500	0	1000
ADAMS	1100		2000
JAMES	950		2000
MILLER	1300		2000

14 rows selected.

SQL> CL SCR

SQL> R

```

 1  SELECT ENAME
 2    ,SAL
 3    ,COMM
 4    , NVL2 ( COMM
 5              ,1000
 6              ,2000
 7            )
 8*  FROM EMP

```

ENAME	SAL	COMM	NVL2 (COMM,1000,2000)
-------	-----	------	-----------------------

KING	5000		2000
BLAKE	2850		2000
CLARK	2450		2000
JONES	2975		2000
SCOTT	3000		2000
FORD	3000		2000
SMITH	800		2000
ALLEN	1600	300	1000
WARD	1250	500	1000
MARTIN	1250	1400	1000
TURNER	1500	0	1000
ADAMS	1100		2000
JAMES	950		2000
MILLER	1300		2000

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  , NVL2( COMM
5           ,1000
6           ,2000
7         )
8  SAL + NVL2( COMM
9           ,1000
10          ,2000
11         ) AS NSAL
12* FROM EMP

```

SQL> /

```

SAL + NVL2( COMM
*

```

ERROR at line 8:

ORA-00923: FROM keyword not found where expected

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  , NVL2( COMM
5           ,1000
6           ,2000
7         )
8  SAL + NVL2( COMM
9           ,1000
10          ,2000
11         ) AS NSAL
12* FROM EMP

```

```

SAL + NVL2( COMM
*

```

ERROR at line 8:

ORA-00923: FROM keyword not found where expected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME

```

```

2  ,SAL
3  ,COMM
4  , NVL2( COMM
5              ,1000
6              ,2000
7              )
8  ,SAL + NVL2( COMM
9              ,1000
10             ,2000
11             ) AS NSAL
12* FROM EMP
SQL> /

```

ENAME	SAL	COMM	NVL2 (COMM,1000,2000)	NSAL
KING	5000		2000	7000
BLAKE	2850		2000	4850
CLARK	2450		2000	4450
JONES	2975		2000	4975
SCOTT	3000		2000	5000
FORD	3000		2000	5000
SMITH	800		2000	2800
ALLEN	1600	300	1000	2600
WARD	1250	500	1000	2250
MARTIN	1250	1400	1000	2250
TURNER	1500	0	1000	2500
ADAMS	1100		2000	3100
JAMES	950		2000	2950
MILLER	1300		2000	3300

14 rows selected.

```

SQL> CL L SCR
SP2-0158: unknown CLEAR option "L"

```

```

SQL> R
1  SELECT ENAME
2  ,SAL
3  ,COMM
4  , NVL2( COMM
5              ,1000
6              ,2000
7              )
8  ,SAL + NVL2( COMM
9              ,1000
10             ,2000
11             ) AS NSAL
12* FROM EMP

```

ENAME	SAL	COMM	NVL2 (COMM,1000,2000)	NSAL
KING	5000		2000	7000
BLAKE	2850		2000	4850
CLARK	2450		2000	4450
JONES	2975		2000	4975
SCOTT	3000		2000	5000
FORD	3000		2000	5000
SMITH	800		2000	2800
ALLEN	1600	300	1000	2600
WARD	1250	500	1000	2250
MARTIN	1250	1400	1000	2250
TURNER	1500	0	1000	2500
ADAMS	1100		2000	3100
JAMES	950		2000	2950

MILLER	1300	2000	3300
--------	------	------	------

```
14 rows selected.
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  , NVL2( COMM
5              , 'COMMISSION IS THERE'
6              , 'NO COMMISSION'
7          )
8* FROM EMP
9  /

```

ENAME	SAL	COMM	NVL2 (COMM, 'COMMISSION')
KING	5000		NO COMMISSION
BLAKE	2850		NO COMMISSION
CLARK	2450		NO COMMISSION
JONES	2975		NO COMMISSION
SCOTT	3000		NO COMMISSION
FORD	3000		NO COMMISSION
SMITH	800		NO COMMISSION
ALLEN	1600	300	COMMISSION IS THERE
WARD	1250	500	COMMISSION IS THERE
MARTIN	1250	1400	COMMISSION IS THERE
TURNER	1500	0	COMMISSION IS THERE
ADAMS	1100		NO COMMISSION
JAMES	950		NO COMMISSION
MILLER	1300		NO COMMISSION

```
14 rows selected.
```

```
SQL> CL SCR
```

```
SQL> R
1  SELECT ENAME
2  ,SAL
3  ,COMM
4  , NVL2( COMM
5          , 'COMMISSION IS THERE'
6          , 'NO COMMISSION'
7          )
8* FROM EMP
```

ENAME	SAL	COMM	NVL2 (COMM, 'COMMISSION')
KING	5000		NO COMMISSION
BLAKE	2850		NO COMMISSION
CLARK	2450		NO COMMISSION
JONES	2975		NO COMMISSION
SCOTT	3000		NO COMMISSION
FORD	3000		NO COMMISSION
SMITH	800		NO COMMISSION
ALLEN	1600	300	COMMISSION IS THERE
WARD	1250	500	COMMISSION IS THERE
MARTIN	1250	1400	COMMISSION IS THERE
TURNER	1500	0	COMMISSION IS THERE
ADAMS	1100		NO COMMISSION
JAMES	950		NO COMMISSION
MILLER	1300		NO COMMISSION

14 rows selected.

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CL SCR

```
SQL> SELECT NULLIF(10,20) AS C1
       2 FROM DUAL;
```

```
          C1
-----
#####
```

SQL> ED

Wrote file afiedt.buf

```
       1 SELECT NULLIF(10,20) AS C2
       2* FROM DUAL
SQL> /
```

```
          C2
-----
          10
```

SQL> ED

Wrote file afiedt.buf

```
       1 SELECT NULLIF(10,10) AS C2
       2* FROM DUAL
SQL> /
```

```
          C2
-----
```

SQL> ED

Wrote file afiedt.buf

```
       1 SELECT NULLIF(10,'ARJUN') AS C2
       2* FROM DUAL
SQL> /
```

```
SELECT NULLIF(10,'ARJUN') AS C2
                        *
```

ERROR at line 1:

ORA-00932: inconsistent datatypes: expected NUMBER got CHAR

SQL> CL S CR

SP2-0158: unknown CLEAR option "S"

SP2-0158: unknown CLEAR option "CR"

SQL> CL SCR

```
SQL> SELECT COALESCE(10,20,30,40,50,60,80,100,5,110) AS C2
       2 FROM DUAL;
```

```
          C2
-----
          10
```

SQL> ED

Wrote file afiedt.buf

```
       1 SELECT COALESCE(10
```

```

2          ,200
3          ,30
4          ,40
5          ,50
6          ,60
7          ,80
8          ,100
9          ,5
10         ,110
11        ) AS C2
12* FROM DUAL
SQL> /

```

```

          C2
-----
          10

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COALESCE(NULL
2          ,200
3          ,30
4          ,40
5          ,50
6          ,60
7          ,80
8          ,100
9          ,5
10         ,110
11        ) AS C2
12* FROM DUAL
SQL> /

```

```

          C2
-----
          200

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COALESCE(NULL
2          ,NULL
3          ,30
4          ,40
5          ,50
6          ,60
7          ,80
8          ,100
9          ,5
10         ,110
11        ) AS C2
12* FROM DUAL
SQL> /

```

```

          C2
-----
          30

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COALESCE(NULL
2          ,NULL

```



```

3          ,NULL
4          ,40
5          ,50
6          ,60
7          ,80
8          ,100
9          ,5
10         ,110
11       ) AS C2
12* FROM DUAL
SQL> /

```

```

          C2
-----
          40

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COALESCE(NULL
2          ,NULL
3          ,NULL
4          ,NULL
5          ,50
6          ,60
7          ,80
8          ,100
9          ,5
10         ,110
11       ) AS C2
12* FROM DUAL
SQL> /

```

```

          C2
-----
          50

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COALESCE(NULL
2          ,NULL
3          ,NULL
4          ,NULL
5          ,NULL
6          ,60
7          ,80
8          ,100
9          ,5
10         ,110
11       ) AS C2
12* FROM DUAL
SQL> /

```

```

          C2
-----
          60

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COALESCE(NULL
2          ,NULL
3          ,NULL

```

```

4          ,NULL
5          ,NULL
6          ,60
7          ,80
8          ,NULL
9          ,5
10         ,110
11        ) AS C2
12* FROM DUAL
SQL> /

```

```

          C2
-----
          60

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COALESCE(NULL
2          ,NULL
3          ,NULL
4          ,NULL
5          ,NULL
6          ,60
7          ,80
8          ,NULL
9          ,5
10         ,NULL
11        ) AS C2
12* FROM DUAL
SQL> /

```

```

          C2
-----
          60

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COALESCE(NULL
2          ,NULL
3          ,NULL
4          ,NULL
5          ,NULL
6          ,60
7          ,80
8          ,NULL
9          ,NULL
10         ,NULL
11        ) AS C2
12* FROM DUAL
SQL> /

```

```

          C2
-----
          60

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COALESCE(NULL
2          ,NULL
3          ,NULL
4          ,NULL

```

```

5          ,NULL
6          ,60
7          ,NULL
8          ,NULL
9          ,NULL
10         ,NULL
11       ) AS C2
12* FROM DUAL
SQL> /

```

```

          C2
-----
          60

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COALESCE(NULL
2          ,NULL
3          ,NULL
4          ,NULL
5          ,NULL
6          ,NULL
7          ,NULL
8          ,NULL
9          ,NULL
10         ,NULL
11       ) AS C2
12* FROM DUAL
SQL> /

```

```

C
-

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COALESCE(10
2          , 'ARJUN'
3          ,SYSDATE
4          ,NULL
5          ,NULL
6          ,NULL
7          ,NULL
8          ,NULL
9          ,NULL
10         ,NULL
11       ) AS C2
12* FROM DUAL
SQL> /

```

```

          , 'ARJUN'
          *

```

```

ERROR at line 2:
ORA-00932: inconsistent datatypes: expected NUMBER got CHAR

```

```

SQL> CL SCR

```

```

SQL> SELECT ENAEM
2
SQL>
SQL> SELECT ENAME
2 ,SAL

```

```

3  ,JOB
4  FROM EMP;

```

ENAME	SAL	JOB
KING	5000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	3000	ANALYST
FORD	3000	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

SQL> CL SCR

```

SQL> R
1  SELECT ENAME
2  ,SAL
3  ,JOB
4* FROM EMP

```

ENAME	SAL	JOB
KING	5000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	3000	ANALYST
FORD	3000	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  , ' SALARY IS '
3  ,SAL
4  , ' AND WORKING AS '
5  ,JOB
6* FROM EMP
SQL> /

```

ENAME	'SALARYIS'	SAL	'ANDWORKINGAS'	JOB
KING	SALARY IS	5000	AND WORKING AS	PRESIDENT
BLAKE	SALARY IS	2850	AND WORKING AS	MANAGER

```

CLARK      SALARY IS      2450 AND WORKING AS  MANAGER
JONES      SALARY IS      2975 AND WORKING AS  MANAGER
SCOTT      SALARY IS      3000 AND WORKING AS  ANALYST
FORD       SALARY IS      3000 AND WORKING AS  ANALYST
SMITH      SALARY IS        800 AND WORKING AS  CLERK
ALLEN      SALARY IS      1600 AND WORKING AS  SALESMAN
WARD       SALARY IS      1250 AND WORKING AS  SALESMAN
MARTIN     SALARY IS      1250 AND WORKING AS  SALESMAN
TURNER     SALARY IS      1500 AND WORKING AS  SALESMAN
ADAMS      SALARY IS      1100 AND WORKING AS  CLERK
JAMES      SALARY IS        950 AND WORKING AS  CLERK
MILLER     SALARY IS      1300 AND WORKING AS  CLERK

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2  ||' SALARY IS '
 3  ||SAL
 4  ||' AND WORKING AS '
 5  ||JOB AS EMPLOYEE_INFORMATION
 6* FROM EMP
SQL> /

```

EMPLOYEE_INFORMATION

```

-----
KING SALARY IS 5000 AND WORKING AS PRESIDENT
BLAKE SALARY IS 2850 AND WORKING AS MANAGER
CLARK SALARY IS 2450 AND WORKING AS MANAGER
JONES SALARY IS 2975 AND WORKING AS MANAGER
SCOTT SALARY IS 3000 AND WORKING AS ANALYST
FORD SALARY IS 3000 AND WORKING AS ANALYST
SMITH SALARY IS 800 AND WORKING AS CLERK
ALLEN SALARY IS 1600 AND WORKING AS SALESMAN
WARD SALARY IS 1250 AND WORKING AS SALESMAN
MARTIN SALARY IS 1250 AND WORKING AS SALESMAN
TURNER SALARY IS 1500 AND WORKING AS SALESMAN
ADAMS SALARY IS 1100 AND WORKING AS CLERK
JAMES SALARY IS 950 AND WORKING AS CLERK
MILLER SALARY IS 1300 AND WORKING AS CLERK

```

14 rows selected.

SQL> CL SCR

SQL> R

```

 1  SELECT ENAME
 2  ||' SALARY IS '
 3  ||SAL
 4  ||' AND WORKING AS '
 5  ||JOB AS EMPLOYEE_INFORMATION
 6* FROM EMP

```

EMPLOYEE_INFORMATION

```

-----
KING SALARY IS 5000 AND WORKING AS PRESIDENT
BLAKE SALARY IS 2850 AND WORKING AS MANAGER
CLARK SALARY IS 2450 AND WORKING AS MANAGER
JONES SALARY IS 2975 AND WORKING AS MANAGER
SCOTT SALARY IS 3000 AND WORKING AS ANALYST

```

```
FORD SALARY IS 3000 AND WORKING AS ANALYST
SMITH SALARY IS 800 AND WORKING AS CLERK
ALLEN SALARY IS 1600 AND WORKING AS SALESMAN
WARD SALARY IS 1250 AND WORKING AS SALESMAN
MARTIN SALARY IS 1250 AND WORKING AS SALESMAN
TURNER SALARY IS 1500 AND WORKING AS SALESMAN
ADAMS SALARY IS 1100 AND WORKING AS CLERK
JAMES SALARY IS 950 AND WORKING AS CLERK
MILLER SALARY IS 1300 AND WORKING AS CLERK
```

14 rows selected.

SQL> CL SCR

```
SQL> SELECT 'HI TODAY DATE IS '||SYSDATE AS C1
2 FROM DUAL;
```

C1

```
-----
HI TODAY DATE IS 25-MAY-18
```

SQL> ED

Wrote file afiedt.buf

```
1 SELECT CONCAT( 'HI TODAY DATE IS '
2                ,SYSDATE
3                ) AS C1
4* FROM DUAL
SQL> /
```

C1

```
-----
HI TODAY DATE IS 25-MAY-18
```

SQL> ED

Wrote file afiedt.buf

```
1 SELECT ENAME
2        ,SAL
3        ,JOB
4* FROM EMP
SQL> /
```

ENAME	SAL	JOB
KING	5000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	3000	ANALYST
FORD	3000	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

```
SQL> R
  1  SELECT ENAME
  2  ,SAL
  3  ,JOB
  4* FROM EMP
```

ENAME	SAL	JOB
KING	5000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	3000	ANALYST
FORD	3000	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

```
SQL> CL S CR
SP2-0158: unknown CLEAR option "S"
SP2-0158: unknown CLEAR option "CR"
SQL> R
```

```
  1  SELECT ENAME
  2  ,SAL
  3  ,JOB
  4* FROM EMP
```

ENAME	SAL	JOB
KING	5000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	3000	ANALYST
FORD	3000	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

```
SQL> CL SCR
```

```
SQL> R
  1  SELECT ENAME
  2  ,SAL
  3  ,JOB
  4* FROM EMP
```

ENAME	SAL	JOB
KING	5000	PRESIDENT

```

BLAKE          2850  MANAGER
CLARK          2450  MANAGER
JONES          2975  MANAGER
SCOTT          3000  ANALYST
FORD           3000  ANALYST
SMITH          800   CLERK
ALLEN          1600  SALESMAN
WARD           1250  SALESMAN
MARTIN         1250  SALESMAN
TURNER         1500  SALESMAN
ADAMS          1100  CLERK
JAMES          950   CLERK
MILLER         1300  CLERK

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT CONCAT( ENAME
 2                  , ' SALARY IS '
 3                  )
 4* FROM EMP
SQL> /

```

CONCAT(ENAME,'SALARYIS')

```

-----
KING SALARY IS
BLAKE SALARY IS
CLARK SALARY IS
JONES SALARY IS
SCOTT SALARY IS
FORD SALARY IS
SMITH SALARY IS
ALLEN SALARY IS
WARD SALARY IS
MARTIN SALARY IS
TURNER SALARY IS
ADAMS SALARY IS
JAMES SALARY IS
MILLER SALARY IS

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT CONCAT( ENAME
 2                  , ' SALARY IS '
 3                  , SAL
 4                  )
 5* FROM EMP
SQL> /

```

```

SELECT CONCAT( ENAME
               *

```

ERROR at line 1:

ORA-00909: invalid number of arguments

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT CONCAT( ENAME
 2                  , ' SALARY IS '

```



```
3          )
4* FROM EMP
SQL> /
```

```
CONCAT(ENAME, 'SALARYIS')
```

```
-----
KING SALARY IS
BLAKE SALARY IS
CLARK SALARY IS
JONES SALARY IS
SCOTT SALARY IS
FORD SALARY IS
SMITH SALARY IS
ALLEN SALARY IS
WARD SALARY IS
MARTIN SALARY IS
TURNER SALARY IS
ADAMS SALARY IS
JAMES SALARY IS
MILLER SALARY IS
```

14 rows selected.

```
SQL> CL SCR
```

```
SQL> R
1  SELECT CONCAT( ENAME
2                  , ' SALARY IS '
3                  )
4* FROM EMP
```

```
CONCAT(ENAME, 'SALARYIS')
```

```
-----
KING SALARY IS
BLAKE SALARY IS
CLARK SALARY IS
JONES SALARY IS
SCOTT SALARY IS
FORD SALARY IS
SMITH SALARY IS
ALLEN SALARY IS
WARD SALARY IS
MARTIN SALARY IS
TURNER SALARY IS
ADAMS SALARY IS
JAMES SALARY IS
MILLER SALARY IS
```

14 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT CONCAT(
2              CONCAT( ENAME
3                      , ' SALARY IS '
4                      )
5              , SAL
6              )
7* FROM EMP
SQL> /
```

```
CONCAT(CONCAT(ENAME, 'SALARYIS'), SAL)
```

```

KING SALARY IS 5000
BLAKE SALARY IS 2850
CLARK SALARY IS 2450
JONES SALARY IS 2975
SCOTT SALARY IS 3000
FORD SALARY IS 3000
SMITH SALARY IS 800
ALLEN SALARY IS 1600
WARD SALARY IS 1250
MARTIN SALARY IS 1250
TURNER SALARY IS 1500
ADAMS SALARY IS 1100
JAMES SALARY IS 950
MILLER SALARY IS 1300

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CONCAT(
2      CONCAT(
3          CONCAT( ENAME
4              , ' SALARY IS '
5          )
6      , SAL
7  )
8  , ' AND WORKING AS '
9  )
10* FROM EMP
SQL> /

```

```

CONCAT(CONCAT(CONCAT(ENAME, 'SALARYIS'), SAL), 'ANDWORKINGAS')
-----

```

```

KING SALARY IS 5000 AND WORKING AS
BLAKE SALARY IS 2850 AND WORKING AS
CLARK SALARY IS 2450 AND WORKING AS
JONES SALARY IS 2975 AND WORKING AS
SCOTT SALARY IS 3000 AND WORKING AS
FORD SALARY IS 3000 AND WORKING AS
SMITH SALARY IS 800 AND WORKING AS
ALLEN SALARY IS 1600 AND WORKING AS
WARD SALARY IS 1250 AND WORKING AS
MARTIN SALARY IS 1250 AND WORKING AS
TURNER SALARY IS 1500 AND WORKING AS
ADAMS SALARY IS 1100 AND WORKING AS
JAMES SALARY IS 950 AND WORKING AS
MILLER SALARY IS 1300 AND WORKING AS

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CONCAT(
2      CONCAT(
3          CONCAT(
4              CONCAT( ENAME
5                  , ' SALARY IS '
6              )
7          , SAL
8      )
9      , ' AND WORKING AS '

```

```

10          )
11          ,JOB
12          ) AS EMPLOYEE_INFORMATION
13* FROM EMP
SQL> /

```

EMPLOYEE_INFORMATION

```

KING SALARY IS 5000 AND WORKING AS PRESIDENT
BLAKE SALARY IS 2850 AND WORKING AS MANAGER
CLARK SALARY IS 2450 AND WORKING AS MANAGER
JONES SALARY IS 2975 AND WORKING AS MANAGER
SCOTT SALARY IS 3000 AND WORKING AS ANALYST
FORD SALARY IS 3000 AND WORKING AS ANALYST
SMITH SALARY IS 800 AND WORKING AS CLERK
ALLEN SALARY IS 1600 AND WORKING AS SALESMAN
WARD SALARY IS 1250 AND WORKING AS SALESMAN
MARTIN SALARY IS 1250 AND WORKING AS SALESMAN
TURNER SALARY IS 1500 AND WORKING AS SALESMAN
ADAMS SALARY IS 1100 AND WORKING AS CLERK
JAMES SALARY IS 950 AND WORKING AS CLERK
MILLER SALARY IS 1300 AND WORKING AS CLERK

```

14 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT CONCAT(
2      CONCAT(
3          CONCAT(
4              CONCAT( ENAME
5                  , ' SALARY IS '
6              )
7          , SAL
8      )
9      , ' AND WORKING AS '
10 )
11 ,JOB
12 ) AS EMPLOYEE_INFORMATION
13* FROM EMP

```

EMPLOYEE_INFORMATION

```

KING SALARY IS 5000 AND WORKING AS PRESIDENT
BLAKE SALARY IS 2850 AND WORKING AS MANAGER
CLARK SALARY IS 2450 AND WORKING AS MANAGER
JONES SALARY IS 2975 AND WORKING AS MANAGER
SCOTT SALARY IS 3000 AND WORKING AS ANALYST
FORD SALARY IS 3000 AND WORKING AS ANALYST
SMITH SALARY IS 800 AND WORKING AS CLERK
ALLEN SALARY IS 1600 AND WORKING AS SALESMAN
WARD SALARY IS 1250 AND WORKING AS SALESMAN
MARTIN SALARY IS 1250 AND WORKING AS SALESMAN
TURNER SALARY IS 1500 AND WORKING AS SALESMAN
ADAMS SALARY IS 1100 AND WORKING AS CLERK
JAMES SALARY IS 950 AND WORKING AS CLERK
MILLER SALARY IS 1300 AND WORKING AS CLERK

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CONCAT(
2      CONCAT(
3          CONCAT(
4              CONCAT( initcap( ENAME)
5                  , ' SALARY IS '
6              )
7          , SAL
8      )
9      , ' AND WORKING AS '
10 )
11 , lower(JOB)
12 ) AS EMPLOYEE_INFORMATION
13* FROM EMP
SQL> /

```

EMPLOYEE_INFORMATION

```

-----
King SALARY IS 5000 AND WORKING AS president
Blake SALARY IS 2850 AND WORKING AS manager
Clark SALARY IS 2450 AND WORKING AS manager
Jones SALARY IS 2975 AND WORKING AS manager
Scott SALARY IS 3000 AND WORKING AS analyst
Ford SALARY IS 3000 AND WORKING AS analyst
Smith SALARY IS 800 AND WORKING AS clerk
Allen SALARY IS 1600 AND WORKING AS salesman
Ward SALARY IS 1250 AND WORKING AS salesman
Martin SALARY IS 1250 AND WORKING AS salesman
Turner SALARY IS 1500 AND WORKING AS salesman
Adams SALARY IS 1100 AND WORKING AS clerk
James SALARY IS 950 AND WORKING AS clerk
Miller SALARY IS 1300 AND WORKING AS clerk

```

14 rows selected.

SQL> cl scr

SQL> SELECT INITCAP(enaem)

```

2  |
3

```

SQL>

SQL> ed

Wrote file afiedt.buf

```

1  SELECT INITCAP(ename)
2  || ' salary is '
3  || SAL
4  || ' AND WORKING AS '
5  || LOWER(JOB) AS E_I
6* FROM EMP/

```

SQL> /

FROM EMP/

*

ERROR at line 6:

ORA-00933: SQL command not properly ended

SQL> ED

Wrote file afiedt.buf

```

1  SELECT INITCAP(ename)
2  || ' salary is '

```

```

3  ||SAL
4  ||' AND  WORKING AS '
5  ||LOWER(JOB) AS E_I
6* FROM EMP
SQL> /

```

```

E_I
-----
-----
King salary is 5000 AND  WORKING AS president
Blake salary is 2850 AND  WORKING AS manager
Clark salary is 2450 AND  WORKING AS manager
Jones salary is 2975 AND  WORKING AS manager
Scott salary is 3000 AND  WORKING AS analyst
Ford salary is 3000 AND  WORKING AS analyst
Smith salary is 800 AND  WORKING AS clerk
Allen salary is 1600 AND  WORKING AS salesman
Ward salary is 1250 AND  WORKING AS salesman
Martin salary is 1250 AND  WORKING AS salesman
Turner salary is 1500 AND  WORKING AS salesman
Adams salary is 1100 AND  WORKING AS clerk
James salary is 950 AND  WORKING AS clerk
Miller salary is 1300 AND  WORKING AS clerk

```

14 rows selected.

```
SQL> CL SCR
```

```

SQL> R
1  SELECT INITCAP(ename)
2  ||' salary is '
3  ||SAL
4  ||' AND  WORKING AS '
5  ||LOWER(JOB) AS E_I
6* FROM EMP

```

```

E_I
-----
-----
King salary is 5000 AND  WORKING AS president
Blake salary is 2850 AND  WORKING AS manager
Clark salary is 2450 AND  WORKING AS manager
Jones salary is 2975 AND  WORKING AS manager
Scott salary is 3000 AND  WORKING AS analyst
Ford salary is 3000 AND  WORKING AS analyst
Smith salary is 800 AND  WORKING AS clerk
Allen salary is 1600 AND  WORKING AS salesman
Ward salary is 1250 AND  WORKING AS salesman
Martin salary is 1250 AND  WORKING AS salesman
Turner salary is 1500 AND  WORKING AS salesman
Adams salary is 1100 AND  WORKING AS clerk
James salary is 950 AND  WORKING AS clerk
Miller salary is 1300 AND  WORKING AS clerk

```

14 rows selected.

28-MAY-2018 -DECODE,CASE,CHR,ASCII

DIFF DECODE AND CASE ?

DECODE

```
(  
P1  
,P2  
,P3  
,P4  
.  
.  
PN  
)
```

DECODE IS SINGLE ROW FUNCTION
WE CAN NOT USED DIRECTLY IN PLSQL
WE CAN CHECK ONLY ONE CONDITON THAT CONDITION SHOULD BE BASED ON EQUALITY
DECODE WE CAN USE WHERE AND ORDER BY

- 1) IT ACCEPTS N PARAMETERS
- 2) DECODE WILL WORK ON EQUALITY

TO UNDERSTAND HOW DECODE IS WORKINNG?

1) AFTER 1ST PARAMETER KEEP ALL REMAINING PARAMETERS AS A PAIR IN SEPARATE LINES
IF YOU HAVE ANY PARAMETER WHICH DOESNOT HAVE A PAIR THAT WILL CONSIDER AS ELSE
VALUE

2) IT WILL TAKE FIRST PARAMETER VALUE AND IT WILL CHECK WITH EACH PAIR 1ST
PARAMETER VALUE
FROMM STARTING.IF IT IS MATCHING FROM THAT PAIR IT WILL GIVE YOU 2ND
PARAMENTER VALUE

IF IT IS NOT MATCHINNG IT WILL CONTINUES WITH 2ND PAIR.

IF IT IS MATCHING IT WILL STOP AND GIVES YOU THE 2ND PARAMENTER VALUE

IF IT IS NOT MATCHING ANY ONE OF THE PAID THEN OUTPUT WILL BE
NULL OR IT WILL GIVE YOU ELSE PARAMETER VALUE.

(IF ELSE PARAMETER VALUE IS THERE IT WILL GIVE YOU THAT ELSE OUTPUT WILL BE
NULL)

NOTE :

ALL FIRST PARAMENTERS IN PAIR DATA TYPE SHOULD BE SAME AS 1ST PARAMETER.

ALL 2ND PARAMERTS IN A PAIR DATA TYPE SHOULD BE SAME.

DECODE WE CAN USE IN ORDER BY CLAUSE

```
FOR DEPTNO 10 EMPLOYEES 1000  
FOR DEPTNO 20 EMPLOYEES 2000  
FOR DEPTNO 30 EMPLOYEES 3000
```

DECODE AND CASE

CASE

CASE IS EXPRESSIONN
WE CAN USE DIRECTLY IN PLSQL

WE CAN CHECK N NUMBER OF CONDITIONS BASED ON OUR REQUIREMENT
CASE WE CAN USE WHERE AND ORDER BY

SYNTAX:

```
CASE
    WHEN CONDITITONS THEN WHATYOUWANTTODO
    WHEN CONDITITONS THEN WHATYOUWANTTODO
    WHEN CONDITITONS THEN WHATYOUWANTTODO
    WHEN CONDITITONS THEN WHATYOUWANTTODO
    WHEN CONDITITONS THEN WHATYOUWANTTODO
    WHEN CONDITITONS THEN WHATYOUWANTTODO
ELSE    WHEN CONDITITONS THEN WHATYOUWANTTODO
END
```

```
FOR DEPTNO 10  MANAGER EMPLOYEES 1000
FOR DEPTNO 10  NOT MANAGER EMPLOYEES 1500
FOR DEPTNO 20 MANAGER EMPLOYEES 2000
FOR DEPTNO 20 NOT MANAGER EMPLOYEES 2500
FOR DEPTNO 30 MANAGE3R EMPLOYEES 3000
FOR DEPTNO 30 NOT MANAGE3R EMPLOYEES 3500
```

```
CHAR-- DATA TYPE
CHR-- FUNCTION
(
P1-- COLUMN/LITERAL
)
IT ACCEPTS NUMBER AS INPUT PARAMENTER
FOR THAT NUMBER EQUIVALNET CHARACTER IT WILL GIVE YOU AS OUTPUT.
```

ASCII NUMBER TO CHARACTER

```
ASCII
----
CHARACTER TO ASCII NUMBER
```

```
QUESTIONS
-----
NUMBER FUNCTIONS
-----
```

```
EMPLOYEE_INFORMATION
-----
```

```
KING SALARY IS 5000
AND WORKING AS PRESIDNET
```

```
BLAKE SALARY IS 2850
AND WORKING AS MANAGER
```

```
CLARK SALRY IS 2450
AND WORKING AS MANAGER
```

SMTP SERVER

UTL_MAIL
.ADDRESS
.ADD
.

HI TEAM||CHR(10)||CHR(10)||TODAY LOADS ARE
STARTED.||CHR(10)||REGARDS||CHR(10)SUPPORT TEAM.

Q1) WRITE A SELECT STATEMENT TO GET IN EACH NAME HOW MANY A'S ARE THERE

ENAME	NUMBERA

KING	0
BLAKE	1
CLARK	1
JONES	
SCOTT	
FORD	
SMITH	
ALLEN	
WARD	
MARTIN	
TURNER	
ADAMS	
JAMES	
MILLER	
KRISHNA	1

2)

ENAME	OUTPUT
-----	-----
KING	*****KING*****
BLAKE	*****BLAKE*****
CLARK	
JONES	
SCOTT	
FORD	
SMITH	
ALLEN	
WARD	
MARTIN	
TURNER	
ADAMS	
JAMES	
MILLER	
KRISHNA	***KRISHNA***

3)

ENAME	ENAME1
KING	K**G
BLAKE	B***E
CLARK	C***E
JONES	
SCOTT	
FORD	
SMITH	
ALLEN	
WARD	
MARTIN	
TURNER	
ADAMS	
JAMES	
MILLER	
KRISHNA	K*****A

4)

ENAME	ENAME1
KING	****
BLAKE	*****
CLARK	*****
JONES	*****
SCOTT	
FORD	
SMITH	
ALLEN	
WARD	
MARTIN	
TURNER	
ADAMS	
JAMES	
MILLER	
KRISHNA	*****

28-MAY-2018 -DECODE,CASE,CHR,ASCII,CASE Spool File

```
SQL> SELECT CHR(65)
      2  FROM DUAL;
FROM DUAL
      *
```

```
      1  SELECT CHR(65) AS C1
      2* FROM DUAL
SQL> /
```

```
C
-
A
```

```
SQL> COLUMN C1 FORMAT A10
SQL> CL SCR
```

```
SQL> R
      1  SELECT CHR(65) AS C1
      2* FROM DUAL
```

```
C1
-----
```

A

SQL> ED

Wrote file afiedt.buf

```
1  SELECT CHR(66) AS C1
2* FROM DUAL
```

SQL> /

C1

B

SQL> ED

Wrote file afiedt.buf

```
1  SELECT CHR(67) AS C1
2* FROM DUAL
```

SQL> /

C1

C

SQL> CL SCR

SQL> ED

Wrote file afiedt.buf

```
1  SELECT CHR(1) AS C1
2* FROM DUAL
```

SQL> /

C1

SQL> ED

Wrote file afiedt.buf

```
1  SELECT CHR(2) AS C1
2* FROM DUAL
```

SQL> /

C1

SQL> ED

Wrote file afiedt.buf

```
1  SELECT CHR(5) AS C1
2* FROM DUAL
```

SQL> /

C1

SQL> SET PAGESIZE 200

SQL> SET LINESIZE 200

SQL> SELECT ENAME

```
2  ,SAL
```

```

3  ,JOB
4  FROM EMP;

```

ENAME	SAL	JOB
KING	5000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	3000	ANALYST
FORD	3000	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK
KRISHNA	2000	

15 rows selected.

```
SQL> COLUMN ENAME FORMAT A10
```

```
SQL> CL SCR
```

```
SQL> R
```

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4* FROM EMP

```

ENAME	SAL	JOB
KING	5000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	3000	ANALYST
FORD	3000	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK
KRISHNA	2000	

15 rows selected.

```
SQL> ED
```

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ||' SALARY IS '
3  ||SAL
4  ||' AND WORKING AS '
5  ||JOB AS EMPLOYEE_INFORMATION
6* FROM EMP

```

```
SQL> /
```

EMPLOYEE_INFORMATION

```

-----
KING SALARY IS 5000 AND WORKING AS PRESIDENT
BLAKE SALARY IS 2850 AND WORKING AS MANAGER
CLARK SALARY IS 2450 AND WORKING AS MANAGER
JONES SALARY IS 2975 AND WORKING AS MANAGER
SCOTT SALARY IS 3000 AND WORKING AS ANALYST
FORD SALARY IS 3000 AND WORKING AS ANALYST
SMITH SALARY IS 800 AND WORKING AS CLERK
ALLEN SALARY IS 1600 AND WORKING AS SALESMAN
WARD SALARY IS 1250 AND WORKING AS SALESMAN
MARTIN SALARY IS 1250 AND WORKING AS SALESMAN
TURNER SALARY IS 1500 AND WORKING AS SALESMAN
ADAMS SALARY IS 1100 AND WORKING AS CLERK
JAMES SALARY IS 950 AND WORKING AS CLERK
MILLER SALARY IS 1300 AND WORKING AS CLERK
KRISHNA SALARY IS 2000 AND WORKING AS

```

15 rows selected.

```

1  SELECT ENAME
2  || ' SALARY IS '
3  ||SAL
4  , ' AND WORKING AS '
5  ||JOB AS EMPLOYEE_INFORMATION
6* FROM EMP
SQL> /

```

```

ENAME||'SALARYIS' ||SAL
EMPLOYEE_INFORMATION

```

```

-----
KING SALARY IS 5000
AND WORKING AS PRESIDENT
BLAKE SALARY IS 2850
AND WORKING AS MANAGER
CLARK SALARY IS 2450
AND WORKING AS MANAGER
JONES SALARY IS 2975
AND WORKING AS MANAGER
SCOTT SALARY IS 3000
AND WORKING AS ANALYST
FORD SALARY IS 3000
AND WORKING AS ANALYST
SMITH SALARY IS 800
AND WORKING AS CLERK
ALLEN SALARY IS 1600
AND WORKING AS SALESMAN
WARD SALARY IS 1250
AND WORKING AS SALESMAN
MARTIN SALARY IS 1250
AND WORKING AS SALESMAN
TURNER SALARY IS 1500
AND WORKING AS SALESMAN
ADAMS SALARY IS 1100
AND WORKING AS CLERK
JAMES SALARY IS 950
AND WORKING AS CLERK
MILLER SALARY IS 1300
AND WORKING AS CLERK
KRISHNA SALARY IS 2000
AND WORKING AS

```

15 rows selected.

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT ENAME||' SALARY IS '||SAL
2  ||' AND WORKING AS '||JOB AS EMPLOYEE_INFORMATION
3* FROM EMP
SQL> /

```

EMPLOYEE_INFORMATION

```

-----
KING SALARY IS 5000 AND WORKING AS PRESIDENT
BLAKE SALARY IS 2850 AND WORKING AS MANAGER
CLARK SALARY IS 2450 AND WORKING AS MANAGER
JONES SALARY IS 2975 AND WORKING AS MANAGER
SCOTT SALARY IS 3000 AND WORKING AS ANALYST
FORD SALARY IS 3000 AND WORKING AS ANALYST
SMITH SALARY IS 800 AND WORKING AS CLERK
ALLEN SALARY IS 1600 AND WORKING AS SALESMAN
WARD SALARY IS 1250 AND WORKING AS SALESMAN
MARTIN SALARY IS 1250 AND WORKING AS SALESMAN
TURNER SALARY IS 1500 AND WORKING AS SALESMAN
ADAMS SALARY IS 1100 AND WORKING AS CLERK
JAMES SALARY IS 950 AND WORKING AS CLERK
MILLER SALARY IS 1300 AND WORKING AS CLERK
KRISHNA SALARY IS 2000 AND WORKING AS

```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME||' SALARY IS '||SAL
2  ||CHR(10)||' AND WORKING AS '||JOB AS EMPLOYEE_INFORMATION
3* FROM EMP
SQL> /

```

EMPLOYEE_INFORMATION

```

-----
KING SALARY IS 5000
AND WORKING AS PRESIDENT

BLAKE SALARY IS 2850
AND WORKING AS MANAGER

CLARK SALARY IS 2450
AND WORKING AS MANAGER

JONES SALARY IS 2975
AND WORKING AS MANAGER

SCOTT SALARY IS 3000
AND WORKING AS ANALYST

FORD SALARY IS 3000
AND WORKING AS ANALYST

SMITH SALARY IS 800

```

```
AND WORKING AS CLERK

ALLEN SALARY IS 1600
AND WORKING AS SALESMAN

WARD SALARY IS 1250
AND WORKING AS SALESMAN

MARTIN SALARY IS 1250
AND WORKING AS SALESMAN

TURNER SALARY IS 1500
AND WORKING AS SALESMAN

ADAMS SALARY IS 1100
AND WORKING AS CLERK

JAMES SALARY IS 950
AND WORKING AS CLERK

MILLER SALARY IS 1300
AND WORKING AS CLERK

KRISHNA SALARY IS 2000
AND WORKING AS
```

```
15 rows selected.
```

```
SQL> CL SCR
```

```
SQL> SELECT CHR(65) AS C1
       2 FROM DUAL;
```

```
C1
-----
A
```

```
SQL> SELECT ASCII('A') AS C1
       2 FROM DUAL;
```

```
          C1
-----
#####
```

```
1 SELECT ASCII('A') AS C2
2* FROM DUAL
SQL> /
```

```
          C2
-----
        65
```

```
SQL> SELECT DECODE(10,20,30,40,50,60,10,80,70,10,95,115,120) AS C1
       2 FROM DUAL;
```

```
          C1
-----
#####          ed
```

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT DECODE(10,20,30,40,50,60,10,80,70,10,95,115,120) AS C2
2*  FROM DUAL
SQL> /

```

```

          C2
-----
          95

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT DECODE(10
2              ,20,30
3              ,40,50
4              ,60,10
5              ,80,70
6              ,10,95
7              ,115,120
8              ) AS C2
9*  FROM DUAL
SQL> /

```

```

          C2
-----
          95

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT DECODE(10
2              ,20,30
3              ,40,50
4              ,60,10
5              ,80,70
6              ,11,95
7              ,115,120
8              ) AS C2
9*  FROM DUAL
SQL> /

```

```

          C2
-----

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT DECODE(10
2              ,20,30
3              ,40,50
4              ,60,10
5              ,80,70
6              ,11,95
7              ,115,120
8              ,125
9              ) AS C2
10* FROM DUAL
SQL> '/

```

SP2-0042: unknown command "'/" - rest of line ignored.

```

SQL> /

          C2
-----

```

125

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DECODE(10
2                ,20,30
3                ,40,50
4                ,60,10
5                ,80,70
6                ,11,95
7                ,115,120
8                ,125
9                ) AS C2
10* FROM DUAL
SQL> R
1  SELECT DECODE(10
2                ,20,30
3                ,40,50
4                ,60,10
5                ,80,70
6                ,11,95
7                ,115,120
8                ,125
9                ) AS C2
10* FROM DUAL

```

C2

```

-----
125

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DECODE(10,20,30,40,50,60,10,80,70,11,95,115,120,125) AS C2
2* FROM DUAL
SQL> /

```

C2

```

-----
125

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DECODE(10,(20,30),(40,50),60,10,80,70,11,95,115,120,125) AS C2
2* FROM DUAL
SQL> /
SELECT DECODE(10,(20,30),(40,50),60,10,80,70,11,95,115,120,125) AS C2
*
```

ERROR at line 1:

ORA-00907: missing right parenthesis

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DECODE(10,20,30,40,50,60,10,80,70,11,95,115,120,125) AS C2
2* FROM DUAL
SQL> /

```

C2

```

-----
125

```


SQL> ED

Wrote file afiedt.buf

```

1  SELECT DECODE(10
2                ,20,30
3                ,40,50
4                ,60,10
5                ,80,70
6                ,11,95
7                ,115,120
8                ,125
9                ) AS C2
10* FROM DUAL
11 /

```

```

          C2
-----
          125

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DECODE(10
2                ,20,30
3                ,40,50
4                ,10,10
5                ,80,70
6                ,11,95
7                ,115,120
8                ,125
9                ) AS C2
10* FROM DUAL
SQL> /

```

```

          C2
-----
          10

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DECODE(10
2                ,20,30
3                ,40,50
4                ,10,'MATCHING'
5                ,80,70
6                ,11,95
7                ,115,120
8                ,125
9                ) AS C2
10* FROM DUAL
SQL> /

```

```

          ,10,'MATCHING'
          *

```

ERROR at line 4:

ORA-01722: invalid number

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DECODE(10
2                ,20,'NM'

```

```

3          ,40,'NM'
4          ,10,'MATCHING'
5          ) AS C2
6* FROM DUAL
SQL> /

```

```

C2
-----
MATCHING

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT DECODE(10
2                ,20,'NM'
3                ,40,'NM'
4                ,10,'MATCHING'
5                ) AS C2
6* FROM DUAL
SQL> C
SP2-0025: Invalid change string.
SQL> /

```

```

C2
-----
MATCHING

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT DECODE(10
2                ,20,30
3                ,40,50
4                ,10,'MATCHING'
5                ,80,70
6                ,11,95
7                ,115,120
8                ,125
9                ) AS C2
10* FROM DUAL
SQL> /
          ,10,'MATCHING'
          *

```

```

ERROR at line 4:
ORA-01722: invalid number

```

```

SQL> CL SCR

```

```

SQL> R
1  SELECT DECODE(10
2                ,20,30
3                ,40,50
4                ,10,'MATCHING'
5                ,80,70
6                ,11,95
7                ,115,120
8                ,125
9                ) AS C2
10* FROM DUAL
          ,10,'MATCHING'
          *

```

```

ERROR at line 4:
ORA-01722: invalid number

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DECODE(10
2                ,20,30
3                ,40,50
4                ,10,15
5                ,80,70
6                ,11,95
7                ,115,120
8                ,125
9                ) AS C2
10* FROM DUAL
SQL> /

```

```

          C2
-----
          15

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DECODE(10
2                ,20,30
3                ,40,50
4                ,10,15
5                ,80,70
6                ,10,95
7                ,115,120
8                ,125
9                ) AS C2
10* FROM DUAL
SQL> //

```

```

          C2
-----
          15

```

SQL> CL SCCR

SP2-0158: unknown CLEAR option "SCCR"

SQL> R

```

1  SELECT DECODE(10
2                ,20,30
3                ,40,50
4                ,10,15
5                ,80,70
6                ,10,95
7                ,115,120
8                ,125
9                ) AS C2
10* FROM DUAL

```

```

          C2
-----
          15

```

SQL> CL SCR

SQL> R

```

1  SELECT DECODE(10
2                ,20,30
3                ,40,50

```

```

4          ,10,15
5          ,80,70
6          ,10,95
7          ,115,120
8          ,125
9          ) AS C2
10* FROM DUAL

```

```

C2
-----

```

```

15

```

```
SQL> CL SCR
```

```

SQL> COLUMN ENAME FORMAT A1
SQL> COLUMN ENAME FORMAT A10
SQL> CL SCR

```

```

SQL> SELECT ENAME
2      ,SAL
3      ,DEPTNO
4  FROM EMP;

```

ENAME	SAL	DEPTNO
-----	-----	-----
KING	5000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	3000	20
FORD	3000	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10
KRISHNA	2000	

```
15 rows selected.
```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2      ,SAL
3      ,DEPTNO
4      ,SAL + 1000 AS NSAL
5* FROM EMP
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
-----	-----	-----	-----
KING	5000	10	6000
BLAKE	2850	30	3850
CLARK	2450	10	3450
JONES	2975	20	3975
SCOTT	3000	20	4000
FORD	3000	20	4000
SMITH	800	20	1800
ALLEN	1600	30	2600
WARD	1250	30	2250

MARTIN	1250	30	2250
TURNER	1500	30	2500
ADAMS	1100	20	2100
JAMES	950	30	1950
MILLER	1300	10	2300
KRISHNA	2000		3000

15 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL + 1000 AS NSAL
5* FROM EMP

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
BLAKE	2850	30	3850
CLARK	2450	10	3450
JONES	2975	20	3975
SCOTT	3000	20	4000
FORD	3000	20	4000
SMITH	800	20	1800
ALLEN	1600	30	2600
WARD	1250	30	2250
MARTIN	1250	30	2250
TURNER	1500	30	2500
ADAMS	1100	20	2100
JAMES	950	30	1950
MILLER	1300	10	2300
KRISHNA	2000		3000

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL + 1000 AS NSAL
5  FROM EMP
6* WHERE DEPTNO =10
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
CLARK	2450	10	3450
MILLER	1300	10	2300

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL + 1000 AS NSAL
5  FROM EMP
6* WHERE DEPTNO =10

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
CLARK	2450	10	3450
MILLER	1300	10	2300

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL + 2000 AS NSAL
5  FROM EMP
6* WHERE DEPTNO =20
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
JONES	2975	20	4975
SCOTT	3000	20	5000
FORD	3000	20	5000
SMITH	800	20	2800
ADAMS	1100	20	3100

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL + 3000 AS NSAL
5  FROM EMP
6* WHERE DEPTNO =30
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
BLAKE	2850	30	5850
ALLEN	1600	30	4600
WARD	1250	30	4250
MARTIN	1250	30	4250
TURNER	1500	30	4500
JAMES	950	30	3950

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL + 1000 AS NSAL
5  FROM EMP
6  WHERE DEPTNO =10
7  UNION ALL
8  SELECT ENAME
9  ,SAL
10 ,DEPTNO
11 ,SAL + 2000 AS NSAL
12 FROM EMP
13 WHERE DEPTNO =10

```

```

14 UNION ALL
15 SELECT ENAME
16 ,SAL
17 ,DEPTNO
18 ,SAL + 3000 AS NSAL
19 FROM EMP
20* WHERE DEPTNO =30
21 /

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
CLARK	2450	10	3450
MILLER	1300	10	2300
KING	5000	10	7000
CLARK	2450	10	4450
MILLER	1300	10	3300
BLAKE	2850	30	5850
ALLEN	1600	30	4600
WARD	1250	30	4250
MARTIN	1250	30	4250
TURNER	1500	30	4500
JAMES	950	30	3950

12 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1 SELECT ENAME
2 ,SAL
3 ,DEPTNO
4 ,SAL + 1000 AS NSAL
5 FROM EMP
6 WHERE DEPTNO =10
7 UNION ALL
8 SELECT ENAME
9 ,SAL
10 ,DEPTNO
11 ,SAL + 2000 AS NSAL
12 FROM EMP
13 WHERE DEPTNO =20
14 UNION ALL
15 SELECT ENAME
16 ,SAL
17 ,DEPTNO
18 ,SAL + 3000 AS NSAL
19 FROM EMP
20* WHERE DEPTNO =30
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
CLARK	2450	10	3450
MILLER	1300	10	2300
JONES	2975	20	4975
SCOTT	3000	20	5000
FORD	3000	20	5000
SMITH	800	20	2800
ADAMS	1100	20	3100
BLAKE	2850	30	5850
ALLEN	1600	30	4600
WARD	1250	30	4250

MARTIN	1250	30	4250
TURNER	1500	30	4500
JAMES	950	30	3950

14 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL + 1000 AS NSAL
5  FROM EMP
6  WHERE DEPTNO =10
7  UNION ALL
8  SELECT ENAME
9  ,SAL
10 ,DEPTNO
11 ,SAL + 2000 AS NSAL
12 FROM EMP
13 WHERE DEPTNO =20
14 UNION ALL
15 SELECT ENAME
16 ,SAL
17 ,DEPTNO
18 ,SAL + 3000 AS NSAL
19 FROM EMP
20* WHERE DEPTNO =30

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
CLARK	2450	10	3450
MILLER	1300	10	2300
JONES	2975	20	4975
SCOTT	3000	20	5000
FORD	3000	20	5000
SMITH	800	20	2800
ADAMS	1100	20	3100
BLAKE	2850	30	5850
ALLEN	1600	30	4600
WARD	1250	30	4250
MARTIN	1250	30	4250
TURNER	1500	30	4500
JAMES	950	30	3950

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL + 1000 AS NSAL
5* FROM EMP
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
BLAKE	2850	30	3850
CLARK	2450	10	3450

JONES	2975	20	3975
SCOTT	3000	20	4000
FORD	3000	20	4000
SMITH	800	20	1800
ALLEN	1600	30	2600
WARD	1250	30	2250
MARTIN	1250	30	2250
TURNER	1500	30	2500
ADAMS	1100	20	2100
JAMES	950	30	1950
MILLER	1300	10	2300
KRISHNA	2000		3000

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL +DECODE(DEPTNO
5      ,10,1000
6      ,20,2000
7      ,30,3000
8      ,500
9      )AS NSAL
10* FROM EMP
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
BLAKE	2850	30	5850
CLARK	2450	10	3450
JONES	2975	20	4975
SCOTT	3000	20	5000
FORD	3000	20	5000
SMITH	800	20	2800
ALLEN	1600	30	4600
WARD	1250	30	4250
MARTIN	1250	30	4250
TURNER	1500	30	4500
ADAMS	1100	20	3100
JAMES	950	30	3950
MILLER	1300	10	2300
KRISHNA	2000		2500

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL +DECODE(DEPTNO
5      ,10,1000
6      ,20,2000
7      ,30,3000
8      ,10,10000
9      ,500
10 )AS NSAL
11* FROM EMP

```

SQL> /

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
BLAKE	2850	30	5850
CLARK	2450	10	3450
JONES	2975	20	4975
SCOTT	3000	20	5000
FORD	3000	20	5000
SMITH	800	20	2800
ALLEN	1600	30	4600
WARD	1250	30	4250
MARTIN	1250	30	4250
TURNER	1500	30	4500
ADAMS	1100	20	3100
JAMES	950	30	3950
MILLER	1300	10	2300
KRISHNA	2000		2500

15 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL +DECODE(DEPTNO
5      ,10,1000
6      ,20,2000
7      ,30,3000
8      ,10,10000
9      ,500
10 )AS NSAL
11* FROM EMP

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
BLAKE	2850	30	5850
CLARK	2450	10	3450
JONES	2975	20	4975
SCOTT	3000	20	5000
FORD	3000	20	5000
SMITH	800	20	2800
ALLEN	1600	30	4600
WARD	1250	30	4250
MARTIN	1250	30	4250
TURNER	1500	30	4500
ADAMS	1100	20	3100
JAMES	950	30	3950
MILLER	1300	10	2300
KRISHNA	2000		2500

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL +DECODE(DEPTNO

```

```

5          ,10,1000
6          ,20,2000
7          ,30,3000
8          ,10,10000
9          ,500
10         )AS NSAL
11 FROM EMP
12 ORDER BY DECODE(DEPTNO
13          ,10,1000
14          ,20,2000
15          ,30,3000
16          ,10,10000
17          ,500
18*        )
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
KRISHNA	2000		2500
MILLER	1300	10	2300
CLARK	2450	10	3450
KING	5000	10	6000
SCOTT	3000	20	5000
FORD	3000	20	5000
SMITH	800	20	2800
JONES	2975	20	4975
ADAMS	1100	20	3100
JAMES	950	30	3950
TURNER	1500	30	4500
MARTIN	1250	30	4250
WARD	1250	30	4250
ALLEN	1600	30	4600
BLAKE	2850	30	5850

15 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL +DECODE(DEPTNO
5          ,10,1000
6          ,20,2000
7          ,30,3000
8          ,10,10000
9          ,500
10         )AS NSAL
11 FROM EMP
12 ORDER BY SAL, DECODE(DEPTNO
13          ,10,1000
14          ,20,2000
15          ,30,3000
16          ,10,10000
17          ,500
18*        )
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
SMITH	800	20	2800
JAMES	950	30	3950
ADAMS	1100	20	3100

MARTIN	1250	30	4250
WARD	1250	30	4250
MILLER	1300	10	2300
TURNER	1500	30	4500
ALLEN	1600	30	4600
KRISHNA	2000		2500
CLARK	2450	10	3450
BLAKE	2850	30	5850
JONES	2975	20	4975
FORD	3000	20	5000
SCOTT	3000	20	5000
KING	5000	10	6000

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL +DECODE(DEPTNO
5              ,10,1000
6              ,20,2000
7              ,30,3000
8              ,10,10000
9              ,500
10             )AS NSAL
11 FROM EMP
12 ORDER BY SAL
13 , DECODE(DEPTNO
14         ,10,1000
15         ,20,2000
16         ,30,3000
17         ,10,10000
18         ,500
19*      )
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
SMITH	800	20	2800
JAMES	950	30	3950
ADAMS	1100	20	3100
MARTIN	1250	30	4250
WARD	1250	30	4250
MILLER	1300	10	2300
TURNER	1500	30	4500
ALLEN	1600	30	4600
KRISHNA	2000		2500
CLARK	2450	10	3450
BLAKE	2850	30	5850
JONES	2975	20	4975
FORD	3000	20	5000
SCOTT	3000	20	5000
KING	5000	10	6000

15 rows selected.

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL

```

```

3  ,DEPTNO
4  ,SAL +DECODE(10
5      ,10,1000
6      ,20,2000
7      ,30,3000
8      ,10,10000
9      ,500
10 )AS NSAL
11* FROM EMP
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
BLAKE	2850	30	3850
CLARK	2450	10	3450
JONES	2975	20	3975
SCOTT	3000	20	4000
FORD	3000	20	4000
SMITH	800	20	1800
ALLEN	1600	30	2600
WARD	1250	30	2250
MARTIN	1250	30	2250
TURNER	1500	30	2500
ADAMS	1100	20	2100
JAMES	950	30	1950
MILLER	1300	10	2300
KRISHNA	2000		3000

15 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  DECODE(10
2      ,10,1000
3      ,20,2000
4      ,30,3000
5      ,10,10000
6      ,500
7  )AS NSAL
8* FROM EMP
SQL> /

```

```

DECODE(10
*

```

```

ERROR at line 1:
ORA-00900: invalid SQL statement

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT DECODE(10
2      ,10,1000
3      ,20,2000
4      ,30,3000
5      ,10,10000
6      ,500
7  )AS NSAL
8* FROM EMP
SQL> /

```

```

      NSAL
-----

```

```
1000
1000
1000
1000
1000
1000
1000
1000
1000
1000
1000
1000
1000
1000
1000
1000
```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT DECODE(DEPTNO
2              ,10,1000
3              ,20,2000
4              ,30,3000
5              ,10,10000
6              ,500
7              )AS NSAL
8* FROM EMP
SQL> /
```

```
      NSAL
-----
1000
3000
1000
2000
2000
2000
2000
3000
3000
3000
3000
3000
3000
2000
3000
1000
500
```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2      ,SAL
3      ,DEPTNO
4      ,SAL + DECODE(DEPTNO
5                  ,10,1000
6                  ,20,2000
7                  ,30,3000
8                  ,10,10000
9                  ,500
10             )AS NSAL
```

```

11* FROM EMP
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
BLAKE	2850	30	5850
CLARK	2450	10	3450
JONES	2975	20	4975
SCOTT	3000	20	5000
FORD	3000	20	5000
SMITH	800	20	2800
ALLEN	1600	30	4600
WARD	1250	30	4250
MARTIN	1250	30	4250
TURNER	1500	30	4500
ADAMS	1100	20	3100
JAMES	950	30	3950
MILLER	1300	10	2300
KRISHNA	2000		2500

15 rows selected.

```
SQL> CL SCR
```

```
SQL> R
```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL + DECODE(DEPTNO
5      ,10,1000
6      ,20,2000
7      ,30,3000
8      ,10,10000
9      ,500
10 )AS NSAL
11* FROM EMP

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
BLAKE	2850	30	5850
CLARK	2450	10	3450
JONES	2975	20	4975
SCOTT	3000	20	5000
FORD	3000	20	5000
SMITH	800	20	2800
ALLEN	1600	30	4600
WARD	1250	30	4250
MARTIN	1250	30	4250
TURNER	1500	30	4500
ADAMS	1100	20	3100
JAMES	950	30	3950
MILLER	1300	10	2300
KRISHNA	2000		2500

15 rows selected.

```
SQL> EDIT
```

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO

```

```

4  ,SAL +CASE
5      WHEN DEPTNO = 10 THEN 1000
6      WHEN DEPTNO = 20 THEN 2000
7      WHEN DEPTNO = 30 TEHN 3000
8      WHEN DEPTNO = 10 THEN 10000
9      ELSE 500
10     END AS NSAL
11* FROM EMP
SQL> /

```

```

      WHEN DEPTNO = 30 TEHN 3000
      *

```

ERROR at line 7:
ORA-00905: missing keyword

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL +CASE
5      WHEN DEPTNO = 10 THEN 1000
6      WHEN DEPTNO = 20 THEN 2000
7      WHEN DEPTNO = 30 THEN 3000
8      WHEN DEPTNO = 10 THEN 10000
9      ELSE 500
10     END AS NSAL
11* FROM EMP
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
KING	5000	10	6000
BLAKE	2850	30	5850
CLARK	2450	10	3450
JONES	2975	20	4975
SCOTT	3000	20	5000
FORD	3000	20	5000
SMITH	800	20	2800
ALLEN	1600	30	4600
WARD	1250	30	4250
MARTIN	1250	30	4250
TURNER	1500	30	4500
ADAMS	1100	20	3100
JAMES	950	30	3950
MILLER	1300	10	2300
KRISHNA	2000		2500

15 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL +CASE
5      WHEN DEPTNO = 10 THEN 1000
6      WHEN DEPTNO = 20 THEN 2000
7      WHEN DEPTNO = 30 THEN 3000
8      WHEN DEPTNO = 10 THEN 10000
9      ELSE 500
10     END AS NSAL

```



```

11 FROM EMP
12 ORDER BY
13 CASE
14         WHEN DEPTNO = 10 THEN 1000
15         WHEN DEPTNO = 20 THEN 2000
16         WHEN DEPTNO = 30 THEN 3000
17         WHEN DEPTNO = 10 THEN 10000
18     ELSE 500
19*     END
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
KRISHNA	2000		2500
MILLER	1300	10	2300
CLARK	2450	10	3450
KING	5000	10	6000
SCOTT	3000	20	5000
FORD	3000	20	5000
SMITH	800	20	2800
JONES	2975	20	4975
ADAMS	1100	20	3100
JAMES	950	30	3950
TURNER	1500	30	4500
MARTIN	1250	30	4250
WARD	1250	30	4250
ALLEN	1600	30	4600
BLAKE	2850	30	5850

15 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,SAL +CASE
5      WHEN DEPTNO = 10 THEN 1000
6      WHEN DEPTNO = 20 THEN 2000
7      WHEN DEPTNO = 30 THEN 3000
8      WHEN DEPTNO = 10 THEN 10000
9      ELSE 500
10     END AS NSAL
11 FROM EMP
12 ORDER BY
13 DEPTNO
14 ,CASE
15     WHEN DEPTNO = 10 THEN 1000
16     WHEN DEPTNO = 20 THEN 2000
17     WHEN DEPTNO = 30 THEN 3000
18     WHEN DEPTNO = 10 THEN 10000
19     ELSE 500
20*    END
SQL> /

```

ENAME	SAL	DEPTNO	NSAL
CLARK	2450	10	3450
MILLER	1300	10	2300
KING	5000	10	6000
SCOTT	3000	20	5000
ADAMS	1100	20	3100
JONES	2975	20	4975

FORD	3000	20	5000
SMITH	800	20	2800
JAMES	950	30	3950
TURNER	1500	30	4500
MARTIN	1250	30	4250
WARD	1250	30	4250
ALLEN	1600	30	4600
BLAKE	2850	30	5850
KRISHNA	2000		2500

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,JOB
5* FROM EMP
SQL> /

```

ENAME	SAL	DEPTNO	JOB
KING	5000	10	PRESIDENT
BLAKE	2850	30	MANAGER
CLARK	2450	10	MANAGER
JONES	2975	20	MANAGER
SCOTT	3000	20	ANALYST
FORD	3000	20	ANALYST
SMITH	800	20	CLERK
ALLEN	1600	30	SALESMAN
WARD	1250	30	SALESMAN
MARTIN	1250	30	SALESMAN
TURNER	1500	30	SALESMAN
ADAMS	1100	20	CLERK
JAMES	950	30	CLERK
MILLER	1300	10	CLERK
KRISHNA	2000		

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,JOB
5  ,SAL +
6      CASE
7          WHEN DEPTNO = 10 AND JOB = 'MANAGER' THEN 1000
8          WHEN DEPTNO = 10 AND JOB <> 'MANAGER' THEN 1500
9          WHEN DEPTNO = 20 AND JOB = 'MANAGER' THEN 2000
10         WHEN DEPTNO = 20 AND JOB <> 'MANAGER' THEN 2500
11         WHEN DEPTNO = 30 AND JOB = 'MANAGER' THEN 3000
12         WHEN DEPTNO = 30 AND JOB <> 'MANAGER' THEN 3500
13     ELSE
14         0
15     END AS NSAL
16* FROM EMP
SQL> /

```

ENAME	SAL	DEPTNO	JOB	NSAL
-------	-----	--------	-----	------

```

-----
KING          5000          10 PRESIDENT          6500
BLAKE         2850          30 MANAGER           5850
CLARK         2450          10 MANAGER           3450
JONES         2975          20 MANAGER           4975
SCOTT         3000          20 ANALYST           5500
FORD          3000          20 ANALYST           5500
SMITH          800          20 CLERK             3300
ALLEN         1600          30 SALESMAN          5100
WARD          1250          30 SALESMAN          4750
MARTIN        1250          30 SALESMAN          4750
TURNER        1500          30 SALESMAN          5000
ADAMS         1100          20 CLERK             3600
JAMES          950          30 CLERK             4450
MILLER        1300          10 CLERK             2800
KRISHNA       2000

```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,JOB
5  ,SAL  +
6      CASE
7      WHEN DEPTNO = 10 AND JOB = 'MANAGER' THEN 1000
8      WHEN DEPTNO = 10 AND JOB <> 'MANAGER' THEN 1500
9      WHEN DEPTNO = 20 AND JOB = 'MANAGER' THEN 2000
10     WHEN DEPTNO = 20 AND JOB <> 'MANAGER' THEN 2500
11     WHEN DEPTNO = 30 AND JOB = 'MANAGER' THEN 3000
12     WHEN DEPTNO = 30 AND JOB <> 'MANAGER' THEN 3500
13     ELSE
14         0
15     END AS NSAL
16 FROM EMP
17* ORDER BY DEPTNO
SQL> /
/

```

```

-----
ENAME          SAL          DEPTNO  JOB          NSAL
-----
CLARK          2450          10  MANAGER          3450
MILLER         1300          10  CLERK            2800
KING           5000          10  PRESIDENT          6500
SCOTT          3000          20  ANALYST           5500
ADAMS          1100          20  CLERK             3600
JONES          2975          20  MANAGER           4975
FORD           3000          20  ANALYST           5500
SMITH           800          20  CLERK             3300
JAMES           950          30  CLERK             4450
TURNER         1500          30  SALESMAN          5000
MARTIN         1250          30  SALESMAN          4750
WARD           1250          30  SALESMAN          4750
ALLEN          1600          30  SALESMAN          5100
BLAKE          2850          30  MANAGER           5850
KRISHNA        2000

```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,JOB
5  ,SAL +
6      CASE
7          WHEN DEPTNO = 10 AND JOB ='MANAGER' AND LENGTH(ENAME) > 4 AND
ENAME LIKE '%A%' THEN 1000
8          WHEN DEPTNO = 10 AND JOB <> 'MANAGER' THEN 1500
9          WHEN DEPTNO = 20 AND JOB ='MANAGER' THEN 2000
10         WHEN DEPTNO = 20 AND JOB <>'MANAGER' THEN 2500
11         WHEN DEPTNO = 30 AND JOB ='MANAGER' THEN 3000
12         WHEN DEPTNO = 30 AND JOB <> 'MANAGER' THEN 3500
13     ELSE
14         0
15     END AS NSAL
16 FROM EMP
17* ORDER BY DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	JOB	NSAL
CLARK	2450	10	MANAGER	3450
MILLER	1300	10	CLERK	2800
KING	5000	10	PRESIDENT	6500
SCOTT	3000	20	ANALYST	5500
ADAMS	1100	20	CLERK	3600
JONES	2975	20	MANAGER	4975
FORD	3000	20	ANALYST	5500
SMITH	800	20	CLERK	3300
JAMES	950	30	CLERK	4450
TURNER	1500	30	SALESMAN	5000
MARTIN	1250	30	SALESMAN	4750
WARD	1250	30	SALESMAN	4750
ALLEN	1600	30	SALESMAN	5100
BLAKE	2850	30	MANAGER	5850
KRISHNA	2000			2000

15 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,JOB
5  ,SAL +
6      CASE
7          WHEN DEPTNO = 10 AND JOB ='MANAGER' AND LENGTH(ENAME) > 4 AND
ENAME LIKE '%S%' THEN 1000
8          WHEN DEPTNO = 10 AND JOB <> 'MANAGER' THEN 1500
9          WHEN DEPTNO = 20 AND JOB ='MANAGER' THEN 2000
10         WHEN DEPTNO = 20 AND JOB <>'MANAGER' THEN 2500
11         WHEN DEPTNO = 30 AND JOB ='MANAGER' THEN 3000
12         WHEN DEPTNO = 30 AND JOB <> 'MANAGER' THEN 3500
13     ELSE
14         0
15     END AS NSAL
16 FROM EMP
17* ORDER BY DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	JOB	NSAL
-------	-----	--------	-----	------

```

-----
CLARK          2450          10  MANAGER          2450
MILLER         1300          10  CLERK           2800
KING           5000          10  PRESIDENT        6500
SCOTT          3000          20  ANALYST          5500
ADAMS          1100          20  CLERK           3600
JONES          2975          20  MANAGER          4975
FORD           3000          20  ANALYST          5500
SMITH           800          20  CLERK           3300
JAMES           950          30  CLERK           4450
TURNER         1500          30  SALESMAN         5000
MARTIN         1250          30  SALESMAN         4750
WARD           1250          30  SALESMAN         4750
ALLEN           1600          30  SALESMAN         5100
BLAKE          2850          30  MANAGER          5850
KRISHNA        2000

```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2  ,SAL
 3  ,DEPTNO
 4  ,JOB
 5  ,SAL  +
 6          CASE
 7          WHEN DEPTNO = 10 AND JOB = 'MANAGER' AND LENGTH(ENAME) > 4 AND
ENAME LIKE '%S%' THEN 1000
 8          WHEN DEPTNO = 10 AND JOB <> 'MANAGER' THEN 1500
 9          WHEN DEPTNO = 20 AND JOB = 'MANAGER' THEN 2000
10          WHEN DEPTNO = 20 AND JOB <> 'MANAGER' THEN 2500
11          WHEN DEPTNO = 30 AND JOB = 'MANAGER' THEN 3000
12          WHEN DEPTNO = 30 AND JOB <> 'MANAGER' THEN 3500
13          ELSE
14              10
15          END AS NSAL
16  FROM EMP
17* ORDER BY DEPTNO
SQL> /

```

```

ENAME          SAL          DEPTNO  JOB          NSAL
-----
CLARK          2450          10  MANAGER          2460
MILLER         1300          10  CLERK           2800
KING           5000          10  PRESIDENT        6500
SCOTT          3000          20  ANALYST          5500
ADAMS          1100          20  CLERK           3600
JONES          2975          20  MANAGER          4975
FORD           3000          20  ANALYST          5500
SMITH           800          20  CLERK           3300
JAMES           950          30  CLERK           4450
TURNER         1500          30  SALESMAN         5000
MARTIN         1250          30  SALESMAN         4750
WARD           1250          30  SALESMAN         4750
ALLEN           1600          30  SALESMAN         5100
BLAKE          2850          30  MANAGER          5850
KRISHNA        2000

```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,JOB
5  ,SAL +
6      CASE
7          WHEN DEPTNO = 10 AND JOB ='MANAGER' AND LENGTH(ENAME) > 4 AND
ENAME LIKE '%S%' THEN 1000
8          WHEN DEPTNO = 10 AND JOB <> 'MANAGER' THEN 1500
9          WHEN DEPTNO = 20 AND JOB ='MANAGER' THEN 2000
10         WHEN DEPTNO = 20 AND JOB <>'MANAGER' THEN 2500
11         WHEN DEPTNO = 30 AND JOB ='MANAGER' THEN 3000
12         WHEN DEPTNO = 30 AND JOB <> 'MANAGER' THEN 3500
13     END AS NSAL
14 FROM EMP
15* ORDER BY DEPTNO
16 /

```

ENAME	SAL	DEPTNO	JOB	NSAL
CLARK	2450	10	MANAGER	
MILLER	1300	10	CLERK	2800
KING	5000	10	PRESIDENT	6500
SCOTT	3000	20	ANALYST	5500
ADAMS	1100	20	CLERK	3600
JONES	2975	20	MANAGER	4975
FORD	3000	20	ANALYST	5500
SMITH	800	20	CLERK	3300
JAMES	950	30	CLERK	4450
TURNER	1500	30	SALESMAN	5000
MARTIN	1250	30	SALESMAN	4750
WARD	1250	30	SALESMAN	4750
ALLEN	1600	30	SALESMAN	5100
BLAKE	2850	30	MANAGER	5850
KRISHNA	2000			

15 rows selected.

SQL> ED
Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  ,JOB
5  ,SAL +
6      CASE
7          WHEN DEPTNO = 10 AND JOB ='MANAGER' AND LENGTH(ENAME) > 4 AND
ENAME LIKE '%S%' THEN 1000
8          WHEN DEPTNO = 10 AND JOB <> 'MANAGER' THEN 1500
9          WHEN DEPTNO = 20 AND JOB ='MANAGER' THEN 2000
10         WHEN DEPTNO = 20 AND JOB <>'MANAGER' THEN 2500
11         WHEN DEPTNO = 30 AND JOB ='MANAGER' THEN 3000
12         WHEN DEPTNO = 30 AND JOB <> 'MANAGER' THEN 3500
13     ELSE
14         0
15     END AS NSAL
16 FROM EMP
17* ORDER BY DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	JOB	NSAL
-------	-----	--------	-----	------

CLARK	2450	10	MANAGER	2450
MILLER	1300	10	CLERK	2800
KING	5000	10	PRESIDENT	6500
SCOTT	3000	20	ANALYST	5500
ADAMS	1100	20	CLERK	3600
JONES	2975	20	MANAGER	4975
FORD	3000	20	ANALYST	5500
SMITH	800	20	CLERK	3300
JAMES	950	30	CLERK	4450
TURNER	1500	30	SALESMAN	5000
MARTIN	1250	30	SALESMAN	4750
WARD	1250	30	SALESMAN	4750
ALLEN	1600	30	SALESMAN	5100
BLAKE	2850	30	MANAGER	5850
KRISHNA	2000			2000

15 rows selected.

SQL> CL SCR

```
SQL> SELECT ENAME
  2  ,SAL
  3  ,DEPTNO
  4  FROM EMP
  5  WHERE CASE
  6      WHEN DEPTNO = 10 THEN 1000
  7      ELSE 500
  8      END + SAL > 3000;
```

ENAME	SAL	DEPTNO
KING	5000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	3000	20
FORD	3000	20

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT ENAME
  2  ,SAL
  3  ,DEPTNO
  4  FROM EMP
  5  WHERE CASE
  6      WHEN DEPTNO = 10 THEN 1000
  7      ELSE 500
  8*      END + SAL > 3000
```

SQL>

SQL> /

ENAME	SAL	DEPTNO
KING	5000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	3000	20
FORD	3000	20

6 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE CASE
6      WHEN DEPTNO = 10 THEN 1000
7      ELSE 500
8      END + SAL > 3000
9* AND JOB = 'MANAGER'
10 /
```

ENAME	SAL	DEPTNO
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE DECODE(DEPTNO
6      ,10,3000
7      ,500
8*      ) + 1000 > 3000
SQL> /
```

ENAME	SAL	DEPTNO
KING	5000	10
CLARK	2450	10
MILLER	1300	10

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
2  FROM EMP;
```

```

ENAME
-----
KING
BLAKE
CLARK
JONES
SCOTT
FORD
SMITH
ALLEN
WARD
MARTIN
TURNER
ADAMS
JAMES
MILLER
KRISHNA
```

```
15 rows selected.
```



```
SQL> SELECT SAL
      2 FROM EMP;
```

```
      SAL
-----
      5000
      2850
      2450
      2975
      3000
      3000
       800
      1600
      1250
      1250
      1500
      1100
       950
      1300
      2000
```

15 rows selected.

```
SQL>
```

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
      2 ,SAL
      3
```

```
SQL> SELECT ENAME
      2 ,SAL
      3 FROM EMP;
```

ENAME	SAL
-----	-----
KING	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	3000
FORD	3000
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300
KRISHNA	2000

15 rows selected.

31-May-2018 - CHARACTER FUNCTIONS

CHARACTER FUNCTIONS

U L I C S I N R E T R L P R P L T R T T G L N N2 N I F C O A L E S E C H R C O N C A T
 D E C O D E C A S E . . . E T C

NUMBER FUNCTIONS

LENGTH

```
(
P1
)
```

ROUND

```
(
  P1-- NUMBER COLUMN/NUMBER LITERAL
, P2-- NUMBER COLUMN/NUMBER LITERAL
)
```

P2 IS OPTIONALL IF P2 NOT PROVIDED THEN IT WILL ROUND IT TO WHOLE NUMBER

IF DECIMAL VALUE IS ≥ 5 THEN IT WILL GIVE YOU NEXT NUMBER(VALUE)
 IF DECIMAL VALUE IS ≤ 4 THEN IT WILL GIVE YOU SAME NUMBER(VALUE)

IF P2 VALUE PROVIDED THAT MEANS IN P1 HOW MANY DECIAML POSITIONS YOU ARE EXPECTING

NOW LET SAY WE HAVE 5 DECIMAL POSITIONS BUT YOU WANT ONLY 2 DECIMAL POSITIONS

NOW ROUND WILL CHECK 3 DECIMAL VALUE IS IT ≥ 5 IF YES THEN IT WILL ADD ONE NUMBER 2 DECIMAL POSITION VALUE

IF NOT IT WILL GIVE YOU 2ND DECIMAL VLAUE AS IT IS.

ROUND(15.4)

.

TRUNC

```
(
  P1-- NUMBER COLUMN/NUMBER LITERAL
, P2-- NUMBER COLUMN/NUMBER LITERAL
)
```

IF P2 VALUE NOT PROVIDED

IRRESPECTIVE OF WHETHER DECIMAL VALUE IS ≥ 5 OR ≤ 4 IT WILL GIVE YOU ALWAYS SAME WHOLE NUMBER

IF YOU WANT DECIMAL PLACES THEN WE HAVE TO SPECIFY P2 VALUE

ENAME	SALARY	PERDAYSALARY
-------	--------	--------------

ROUND
TRUNC

95.67

95
96

TRUNC
RPOU

CTS

4 LACKS

5000	167	20	=167*20=3340
	166	20	-- 20=3320 = 20

4LACKS*20

CEIL
(
P1
)

CEIL WILL GIVE YOU ALWAYS NEAREST HIGHEST WHOLE NUMBER.
IT WILL NOT GIVE YOU ANY DECIAL VALUES IN FINAL RESULTS.

15.10	-- CEIL(16)
'	ROUND(15)
	TRUNC(15)

15.60	-- CEIL(16)
	ROUND(16)
	TRUNC(15)

FLOOR

```
(  
    P1  
)
```

IT WILL GIVE YOU ALWAYS NEAREST LEAST WHOLE NUMBER
IT WILL NOT GIVE YOU ANY DECIMAL PLACES

MOD

```
(  
    P1  
,P2  
)
```

IT WILL GIVE YOU REMINDER VALUE.

```
2)15(7  
    14  
    --  
    1 -- REMINDER
```

IF P1 VALUE < P2 VALUE THEN P1 VALUE ONLY IT WILL GIVE AS REMINDER
VALUE.

SIGN

```
(  
    P1  
)
```

IF P1 VALUE IS -VE THEN IT WILL GIVE YOU -1
IF P1 VALUE IS +VE THEN IT WILL GIVE YOU 1
IF P1 VALUE IS 0 THEN IT WILL GIVE YOU 0
IF P1 VALUE IS NULL THEN IT WILL GIVE YOU NULL

10000	8000	LOSS
10000	12000	PROFIT

```
CASE WHEN SIGN(PC-SC) = -1 THEN 'LOSS'
WHEN SIGN(PC-SC) = 1 THEN 'PROFIT'
END LOSS_PROF
```

```
ABS
(
P1
)
```

IT WILL NOT SHOW YOU ANY SIGN(-VE OR +VE) ALWAYS IT WILL GIVE YOU AS +VE VALUE.

```
LOSS    3000    1000 -2000
```

```
LOSS -2000
```

```
LOSS 2000
```

```
SQRT
(
P1
,P2
)
```

```
-----
|4
```

```
2
```

```
POWER
(
P1
,P2
)
```

IT WILL GIVE YOU P1 POWER OF P2 VALUE

```
POWER(2,5)
```

IT WILL GIVE YOU 2 POWER 5 VALUE.

```
VSIZE
(
P1
)
```

IT WILL GIVE YOU HOW MANY BYTES OF MEMORY IT IS GOING TO USE/ HOW MANY BYTES THAT GIVE STRING REQUIRED/CONTAINS

DIFF BETWEEN CHAR AND BYTE ?

LENGTH (

ARJUN --5

VSIZE ARJUN
5

2+1+1+1+4-- 9
LENGTH --5

ROUND
TRUNC
CEIL
FLOOR
SIGN
ABS
MOD
LENGTH
SQRT
POWER
VSIZE

DATE FUNCTIONS

ADD_MONTHS
MONTHS_BETWEEN
LAST_DAY
NEXT_DAY
ROUND
TRUNC

DATE + NUMBER
DATE - NUMBER
DATE * NUMBER
DATE / NUMBER

DATE - DATE
DATE + DATE
DATE * DATE
DATE / DATE

HOW TO ADD HOURS
 HOW TO ADD SECONDS
 HOW TO ADD MINUTES

EXTRACT

CONVERSTION FUNCTIONS

31-May-2018 - CHARACTER FUNCTIONS-Example

```
SQL> SELECT LENGTH('SDAFFDSAKJLFDSFJDSA;FDSAFDSFDS')
      2 FROM DUAL;
```

```
LENGTH('SDAFFDSAKJLFDSFJDSA;FDSAFDSFDS')
-----
                                         30
```

```
SQL> ED
Wrote file afiedt.buf
```

```
      1 SELECT LENGTH('SDAFFDSAKJLFDSFJDSA; FDSAFDSFDS')
      2* FROM DUAL
SQL> /
```

```
LENGTH('SDAFFDSAKJLFDSFJDSA;FDSAFDSFDS')
-----
                                         33
```

```
SQL> CL SCR
```

```
SQL> R
      1 SELECT LENGTH('SDAFFDSAKJLFDSFJDSA; FDSAFDSFDS')
      2* FROM DUAL
```

```
LENGTH('SDAFFDSAKJLFDSFJDSA;FDSAFDSFDS')
-----
                                         33
```

```
SQL> SELECT ENAME
      2 ,SAL
      3 FROM EMP;
```

ENAME	SAL
KING	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	3000
FORD	3000
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  FROM EMP
4* WHERE LENGTH(ENAME) = 4
SQL> /
```

ENAME	SAL
KING	5000
FORD	3000
WARD	1250

SQL> CL SCR

SQL> R

```
1  SELECT ENAME
2  ,SAL
3  FROM EMP
4* WHERE LENGTH(ENAME) = 4
```

ENAME	SAL
KING	5000
FORD	3000
WARD	1250

```
SQL> SELECT ROUND(15.4) AS C1
2  FROM DUAL;
```

C1
15

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ROUND(15.5) AS C1
2* FROM DUAL
SQL> /
```

C1
16

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ROUND(15.5) AS C1
2* FROM DUAL
SQL> /
```

C1
16

SQL> ED

Wrote file afiedt.buf


```
1  SELECT ROUND(15.5) AS C1
2*  FROM DUAL
SQL> /
```

```
      C1
-----
      16
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT ROUND(15.5678) AS C1
2*  FROM DUAL
SQL> /
```

```
      C1
-----
      16
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT ROUND(15.5678
2      ,2
3      ) AS C1
4*  FROM DUAL
SQL> /
```

```
      C1
-----
     15.57
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT ROUND(15.5678
2      ,2
3      ) AS C1
4*  FROM DUAL
SQL>
SQL> /
```

```
      C1
-----
     15.57
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT ROUND(15.5649
2      ,2
3      ) AS C1
4*  FROM DUAL
SQL> /
```

```
      C1
-----
     15.56
```

```
SQL> R
1  SELECT ROUND(15.5649
2      ,2
3      ) AS C1
4*  FROM DUAL
```

```
          C1
-----
        15.56
```

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT ROUND(15.5699
  2                      ,2
  3                      ) AS C1
  4* FROM DUAL
SQL> /
```

```
          C1
-----
        15.57
```

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT ROUND(15.9999
  2                      ,2
  3                      ) AS C1
  4* FROM DUAL
SQL> /
```

```
          C1
-----
         16
```

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT ROUND(15.9999
  2                      ,2
  3                      ) AS C1
  4* FROM DUAL
  5  /
```

```
          C1
-----
         16
```

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT ROUND(15.123456789
  2                      ,4
  3                      ) AS C1
  4* FROM DUAL
SQL> /
```

```
          C1
-----
        15.1235
```

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT ROUND(15.123456789
  2                      ,4
  3                      ) AS C1
  4* FROM DUAL
```

SQL>

SQL> /

```
          C1
-----
    15.1235
```

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT ROUND(15.123456789
  2                      ) AS C1
  3* FROM DUAL
SQL>
SQL> /
```

```
          C1
-----
          15
```

SQL> CL SCR

SQL> R

```
  1  SELECT ROUND(15.123456789
  2                      ) AS C1
  3* FROM DUAL
```

```
          C1
-----
          15
```

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT ROUND(15.5678
  2                      ,1.5
  3                      ) AS C1
  4* FROM DUAL
SQL> /
```

```
          C1
-----
        15.6
```

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT ROUND(15.5678
  2                      ,1.9
  3                      ) AS C1
  4* FROM DUAL
SQL> /
```

```
          C1
-----
        15.6
```

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT ROUND(15.5678
  2                      ,1.92
  3                      ) AS C1
  4* FROM DUAL
```

```
SQL> /
```

```
          C1
-----
        15.6
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
 1  SELECT ROUND(15.5678
 2                      ,ROUND(1.92)
 3                      ) AS C1
 4* FROM DUAL
SQL> /
```

```
          C1
-----
        15.57
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
 1  SELECT ROUND(15.5678
 2                      ,1.999999999999999
 3                      ) AS C1
 4* FROM DUAL
SQL> /
```

```
          C1
-----
        15.6
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
 1  SELECT ROUND(15.5678
 2                      ,1.999999999999999
 3                      ) AS C1
 4* FROM DUAL
SQL>
SQL> /
```

```
          C1
-----
        15.6
```

```
SQL> CL SCR
```

```
SQL> SELECT ROUND(15.99999) AS C1
 2          ,TRUNC(15.99999) AS C2
 3 FROM DUAL;
```

```
          C1          C2
-----
        16          15
```

```
SQL> EDIT
```

```
Wrote file afiedt.buf
```

```
 1  SELECT ROUND(15.99999) AS C1
 2          ,TRUNC(15.99999) AS C2
 3* FROM DUAL
SQL>
SQL> /
```

C1	C2
-----	-----
16	15

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ROUND(15.56789) AS C1
2      ,TRUNC(15.56789) AS C2
3*  FROM DUAL
SQL> /
```

C1	C2
-----	-----
16	15

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ROUND(15.56789,2) AS C1
2      ,TRUNC(15.56789,2) AS C2
3*  FROM DUAL
SQL> /
```

C1	C2
-----	-----
15.57	15.56

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ROUND(15.56999,2) AS C1
2      ,TRUNC(15.56999,2) AS C2
3*  FROM DUAL
SQL> /
```

C1	C2
-----	-----
15.57	15.56

SQL> CL SCR

SQL> R

```
1  SELECT ROUND(15.56999,2) AS C1
2      ,TRUNC(15.56999,2) AS C2
3*  FROM DUAL
```

C1	C2
-----	-----
15.57	15.56

SQL> CL SCR

SQL> SELECT ENAME

```
2  ,SAL
3  FROM EMP;
```

ENAME	SAL
-----	-----
KING	5000
BLAKE	2850
CLARK	2450
JONES	2975

SCOTT	3000
FORD	3000
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,SAL / 30 AS PERDAYSALARY
4* FROM EMP
SQL> /
```

ENAME	SAL	PERDAYSALARY
KING	5000	166.666667
BLAKE	2850	95
CLARK	2450	81.6666667
JONES	2975	99.1666667
SCOTT	3000	100
FORD	3000	100
SMITH	800	26.6666667
ALLEN	1600	53.3333333
WARD	1250	41.6666667
MARTIN	1250	41.6666667
TURNER	1500	50
ADAMS	1100	36.6666667
JAMES	950	31.6666667
MILLER	1300	43.3333333

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,SAL / 30 AS PERDAYSALARY
4* FROM EMP
SQL>
SQL> /
```

ENAME	SAL	PERDAYSALARY
KING	5000	166.666667
BLAKE	2850	95
CLARK	2450	81.6666667
JONES	2975	99.1666667
SCOTT	3000	100
FORD	3000	100
SMITH	800	26.6666667
ALLEN	1600	53.3333333
WARD	1250	41.6666667
MARTIN	1250	41.6666667
TURNER	1500	50

```

ADAMS          1100    36.6666667
JAMES          950     31.6666667
MILLER         1300    43.3333333

```

14 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,SAL / 30 AS PERDAYSALARY
4* FROM EMP

```

ENAME	SAL	PERDAYSALARY
KING	5000	166.6666667
BLAKE	2850	95
CLARK	2450	81.6666667
JONES	2975	99.1666667
SCOTT	3000	100
FORD	3000	100
SMITH	800	26.6666667
ALLEN	1600	53.3333333
WARD	1250	41.6666667
MARTIN	1250	41.6666667
TURNER	1500	50
ADAMS	1100	36.6666667
JAMES	950	31.6666667
MILLER	1300	43.3333333

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,ROUND( SAL / 30 ) AS RPERDAYSALARY
4* FROM EMP
SQL> /

```

ENAME	SAL	RPERDAYSALARY
KING	5000	167
BLAKE	2850	95
CLARK	2450	82
JONES	2975	99
SCOTT	3000	100
FORD	3000	100
SMITH	800	27
ALLEN	1600	53
WARD	1250	42
MARTIN	1250	42
TURNER	1500	50
ADAMS	1100	37
JAMES	950	32
MILLER	1300	43

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,ROUND( SAL / 30 )  AS RPERDAYSALARY
4  ,TRUNC( SAL / 30 )  AS TPERDAYSALARY
5* FROM EMP
SQL> /

```

ENAME	SAL	RPERDAYSALARY	TPERDAYSALARY
KING	5000	167	166
BLAKE	2850	95	95
CLARK	2450	82	81
JONES	2975	99	99
SCOTT	3000	100	100
FORD	3000	100	100
SMITH	800	27	26
ALLEN	1600	53	53
WARD	1250	42	41
MARTIN	1250	42	41
TURNER	1500	50	50
ADAMS	1100	37	36
JAMES	950	32	31
MILLER	1300	43	43

14 rows selected.

```
SQL> CL SCR
```

```

SQL> R
1  SELECT ENAME
2  ,SAL
3  ,ROUND( SAL / 30 )  AS RPERDAYSALARY
4  ,TRUNC( SAL / 30 )  AS TPERDAYSALARY
5* FROM EMP

```

ENAME	SAL	RPERDAYSALARY	TPERDAYSALARY
KING	5000	167	166
BLAKE	2850	95	95
CLARK	2450	82	81
JONES	2975	99	99
SCOTT	3000	100	100
FORD	3000	100	100
SMITH	800	27	26
ALLEN	1600	53	53
WARD	1250	42	41
MARTIN	1250	42	41
TURNER	1500	50	50
ADAMS	1100	37	36
JAMES	950	32	31
MILLER	1300	43	43

14 rows selected.

```
SQL> CLS CR
```

SP2-0042: unknown command "CLS CR" - rest of line ignored.

```
SQL> CL SCR
```

```

SQL> SELECT ROUND(15.0000000000000001) AS C1
2    FROM DUAL;

```

C1

15


```

3* FROM DUAL
SQL> /

```

C1	C2
15	15

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT TRUNC(15.9999999999999999999999999999,2) AS C1
2      ,FLOOR(15.9999999999999999999999999999,2) AS C2
3* FROM DUAL
SQL> /
      ,FLOOR(15.9999999999999999999999999999,2) AS C2
*
```

```

ERROR at line 2:
ORA-00909: invalid number of arguments

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT TRUNC(15.9999999999999999999999999999,2) AS C1
2      ,FLOOR(15.9999999999999999999999999999) AS C2
3* FROM DUAL
SQL> /

```

C1	C2
15.99	15

```

SQL> CL SCR

```

```

SQL> CL SCR

```

```

SQL> SELECT MOD(15,2) AS C1
2 FROM DUAL;

```

C1
1

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT MOD(15,3) AS C1
2* FROM DUAL
SQL> /

```

C1
0

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT MOD(15,4) AS C1
2* FROM DUAL
SQL> /

```

C1
3

```
SQL> CL SCR
```

```
SQL> R
```

```
1  SELECT MOD(15,4) AS C1
2*  FROM DUAL
```

```
      C1
-----
      3
```

```
SQL> EDIT
```

```
Wrote file afiedt.buf
```

```
1  SELECT MOD(4,15) AS C1
2*  FROM DUAL
```

```
SQL> /
```

```
      C1
-----
      4
```

```
SQL> R
```

```
1  SELECT MOD(4,15) AS C1
2*  FROM DUAL
```

```
      C1
-----
      4
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT MOD(5,15) AS C1
2*  FROM DUAL
```

```
SQL> /
```

```
      C1
-----
      5
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT MOD(14,15) AS C1
2*  FROM DUAL
```

```
SQL> /
```

```
      C1
-----
     14
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT MOD(15,15) AS C1
2*  FROM DUAL
```

```
SQL> /
```

```
      C1
-----
      0
```

```
SQL> CL SCR
```

```
SQL> SELECT ENAEM
      2  ,SA
      3
SQL> CL SCR
```

```
SQL> SELECT ENAME
      2  ,SAL
      3  ,HIREDATE
      4  FROM EMP;
```

ENAME	SAL	HIREDATE
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

14 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
      1  SELECT ENAME
      2  ,SAL
      3  ,HIREDATE
      4  ,MOD(SAL , 2)
      5* FROM EMP
SQL> /
```

ENAME	SAL	HIREDATE	MOD(SAL,2)
KING	5000	17-NOV-81	0
BLAKE	2850	01-MAY-81	0
CLARK	2450	09-JUN-81	0
JONES	2975	02-APR-81	1
SCOTT	3000	19-APR-87	0
FORD	3000	03-DEC-81	0
SMITH	800	17-DEC-80	0
ALLEN	1600	20-FEB-81	0
WARD	1250	22-FEB-81	0
MARTIN	1250	28-SEP-81	0
TURNER	1500	08-SEP-81	0
ADAMS	1100	23-MAY-87	0
JAMES	950	03-DEC-81	0
MILLER	1300	23-JAN-82	0

14 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
      1  SELECT ENAME
      2  ,SAL
```

```

3  ,HIREDATE
4  ,MOD(SAL , 2)
5  FROM EMP
6* WHERE MOD(SAL , 2) = 0
SQL> /

```

ENAME	SAL	HIREDATE	MOD(SAL,2)
KING	5000	17-NOV-81	0
BLAKE	2850	01-MAY-81	0
CLARK	2450	09-JUN-81	0
SCOTT	3000	19-APR-87	0
FORD	3000	03-DEC-81	0
SMITH	800	17-DEC-80	0
ALLEN	1600	20-FEB-81	0
WARD	1250	22-FEB-81	0
MARTIN	1250	28-SEP-81	0
TURNER	1500	08-SEP-81	0
ADAMS	1100	23-MAY-87	0
JAMES	950	03-DEC-81	0
MILLER	1300	23-JAN-82	0

13 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,MOD(SAL , 2)
5  FROM EMP
6* WHERE MOD(SAL , 2) = 1
SQL> /

```

ENAME	SAL	HIREDATE	MOD(SAL,2)
JONES	2975	02-APR-81	1

```
SQL> CL SCR
```

```

SQL> R
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,MOD(SAL , 2)
5  FROM EMP
6* WHERE MOD(SAL , 2) = 1

```

ENAME	SAL	HIREDATE	MOD(SAL,2)
JONES	2975	02-APR-81	1

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4* FROM EMP
5  /

```

ENAME	SAL	HIREDATE
-------	-----	----------

KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

14 rows selected.

SQL> CL SCR

SQL> R

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4* FROM EMP
```

ENAME	SAL	HIREDATE
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4* FROM EMP
5  /
```

ENAME	SAL	HIREDATE
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81

```

MARTIN          1250 28-SEP-81
TURNER          1500 08-SEP-81
ADAMS           1100 23-MAY-87
JAMES           950  03-DEC-81
MILLER          1300 23-JAN-82

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2    ,SAL
 3    ,HIREDATE
 4    ,SUBSTR(HIREDATE,1,2)
 5* FROM EMP
SQL> /

```

ENAME	SAL	HIREDATE	SUBSTR(H
-----	-----	-----	-----
KING	5000	17-NOV-81	17
BLAKE	2850	01-MAY-81	01
CLARK	2450	09-JUN-81	09
JONES	2975	02-APR-81	02
SCOTT	3000	19-APR-87	19
FORD	3000	03-DEC-81	03
SMITH	800	17-DEC-80	17
ALLEN	1600	20-FEB-81	20
WARD	1250	22-FEB-81	22
MARTIN	1250	28-SEP-81	28
TURNER	1500	08-SEP-81	08
ADAMS	1100	23-MAY-87	23
JAMES	950	03-DEC-81	03
MILLER	1300	23-JAN-82	23

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2    ,SAL
 3    ,HIREDATE
 4    ,MOD ( SUBSTR(HIREDATE,1,2)
 5          ,2
 6          )
 7* FROM EMP
SQL> /

```

ENAME	SAL	HIREDATE	MOD (SUBSTR(HIREDATE,1,2),2)
-----	-----	-----	-----
KING	5000	17-NOV-81	1
BLAKE	2850	01-MAY-81	1
CLARK	2450	09-JUN-81	1
JONES	2975	02-APR-81	0
SCOTT	3000	19-APR-87	1
FORD	3000	03-DEC-81	1
SMITH	800	17-DEC-80	1
ALLEN	1600	20-FEB-81	0
WARD	1250	22-FEB-81	0
MARTIN	1250	28-SEP-81	0
TURNER	1500	08-SEP-81	0
ADAMS	1100	23-MAY-87	1
JAMES	950	03-DEC-81	1

MILLER 1300 23-JAN-82 1

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,MOD ( SUBSTR(HIREDATE,1,2)
5      ,2
6      )
7  FROM EMP
8  WHERE MOD ( SUBSTR(HIREDATE,1,2)
9      ,2
10*      ) = 0

```

SQL> /

ENAME	SAL	HIREDATE	MOD(SUBSTR(HIREDATE,1,2),2)
JONES	2975	02-APR-81	0
ALLEN	1600	20-FEB-81	0
WARD	1250	22-FEB-81	0
MARTIN	1250	28-SEP-81	0
TURNER	1500	08-SEP-81	0

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,MOD ( SUBSTR(HIREDATE,1,2)
5      ,2
6      )
7  FROM EMP
8  WHERE MOD ( SUBSTR(HIREDATE,1,2)
9      ,2
10*      ) = 1

```

SQL> /

ENAME	SAL	HIREDATE	MOD(SUBSTR(HIREDATE,1,2),2)
KING	5000	17-NOV-81	1
BLAKE	2850	01-MAY-81	1
CLARK	2450	09-JUN-81	1
SCOTT	3000	19-APR-87	1
FORD	3000	03-DEC-81	1
SMITH	800	17-DEC-80	1
ADAMS	1100	23-MAY-87	1
JAMES	950	03-DEC-81	1
MILLER	1300	23-JAN-82	1

9 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4* FROM EMP

```

SQL> /

ENAME	SAL	HIREDATE
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

14 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4* FROM EMP

```

ENAME	SAL	HIREDATE
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5  WHERE MOD( SUBSTR(SAL ,1,2)
6            ,2
7            ) =0
8  AND MOD(SUBSTR(SAL,-2)
9          ,2
10*        ) =1
11  /

```

no rows selected

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5  WHERE MOD( SUBSTR(SAL ,1,2)
6             ,2
7             ) =1
8  AND MOD(SUBSTR(SAL,-2)
9           ,2
10*        ) =0
```

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5  WHERE MOD( SUBSTR(SAL ,1,2)
6             ,2
7             ) =1
8  AND MOD(SUBSTR(SAL,-2)
9           ,2
10*        ) =0
```

```
SQL> /
```

ENAME	SAL	HIREDATE
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

```
SQL> /
```

ENAME	SAL	HIREDATE
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5  WHERE MOD( SUBSTR(SAL ,1,2)
6             ,2
7             ) =1
8  AND MOD(SUBSTR(SAL,-2)
9           ,2
10*        ) =0
11* AND LENGTH(SAL ) =4
```

```
SQL> /
```

ENAME	SAL	HIREDATE
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87

MILLER 1300 23-JAN-82

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5  WHERE MOD( SUBSTR(SAL ,1,2)
6             ,2
7             ) =1
8  AND MOD(SUBSTR(SAL,-2)
9           ,2
10          ) =0
11* AND LENGTH(SAL ) =4

```

SQL>

SQL> /

ENAME	SAL	HIREDATE
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
MILLER	1300	23-JAN-82

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5  WHERE MOD( SUBSTR(SAL ,1,2)
6             ,2
7             ) =1
8  AND MOD(SUBSTR(SAL,-2)
9           ,2
10          ) =0
11* AND LENGTH(SAL ) =4

```

ENAME	SAL	HIREDATE
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
MILLER	1300	23-JAN-82

SQL> CL SCR

SQL> SELLECT ENAME

SP2-0734: unknown command beginning "SELLECT EN..." - rest of line ignored.

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CL SCR

SQL> SELECT ENAME

```

2  ,SAL
3  FROM EMP;

```

ENAME	SAL
KING	5000
BLAKE	2850
CLARK	2450
JONES	2975

SCOTT	3000
FORD	3000
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE MOD( LENGTH(ENAME)
5             ,2
6*          ) =0
SQL> /

```

ENAME	SAL
KING	5000
FORD	3000
WARD	1250
MARTIN	1250
TURNER	1500
MILLER	1300

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE MOD( LENGTH(ENAME)
5             ,2
6*          ) =1
SQL> /

```

ENAME	SAL
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	3000
SMITH	800
ALLEN	1600
ADAMS	1100
JAMES	950

8 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT (15,4) AS C1
2* FROM DUAL

```

```
3 /
SELECT (15,4) AS C1
      *
ERROR at line 1:
ORA-00907: missing right parenthesis
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT MOD(15,4) AS C1
2* FROM DUAL
SQL> /
```

```
      C1
-----
      3
```

```
SQL> CL SCR
```

```
SQL> R
1 SELECT MOD(15,4) AS C1
2* FROM DUAL
```

```
      C1
-----
      3
```

```
SQL> SELECT SIGN( 100 - 200) AS C1
2 FROM DUAL;
```

```
      C1
-----
     -1
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT SIGN( 300 - 200) AS C1
2* FROM DUAL
SQL> /
```

```
      C1
-----
      1
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT SIGN( 300 - 300) AS C1
2* FROM DUAL
SQL> /
```

```
      C1
-----
      0
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT SIGN( 300 - NULL) AS C1
2* FROM DUAL
SQL> /
```

C1

SQL> ED

Wrote file afiedt.buf

```
1  SELECT SIGN(-3242423432432432432) AS C1
2* FROM DUAL
SQL> /
```

C1

-1

SQL> ED

Wrote file afiedt.buf

```
1  SELECT SIGN(3242423432432432432) AS C1
2* FROM DUAL
SQL> /
```

C1

1

SQL> ED

Wrote file afiedt.buf

```
1  SELECT SIGN(0) AS C1
2* FROM DUAL
SQL> /
```

C1

0

SQL> ED

Wrote file afiedt.buf

```
1  SELECT SIGN(NULL) AS C1
2* FROM DUAL
SQL> /
```

C1

SQL> ED

Wrote file afiedt.buf

```
1  SELECT SIGN(-100) AS C1
2* FROM DUAL
SQL> /
```

C1

-1

SQL> ED

Wrote file afiedt.buf

```
1  SELECT SIGN(100 - 200) AS C1
2* FROM DUAL
```

SQL> /

```

          C1
-----
        -1

```

SQL> CL SCR

```

SQL> SELECT ENAME
       2  ,SAL
       3  ,COMM
       4  FROM EMP;

```

ENAME	SAL	COMM
KING	5000	
BLAKE	2850	
CLARK	2450	
JONES	2975	
SCOTT	3000	
FORD	3000	
SMITH	800	
ALLEN	1600	300
WARD	1250	500
MARTIN	1250	1400
TURNER	1500	0
ADAMS	1100	
JAMES	950	
MILLER	1300	

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

   1  SELECT ENAME
   2  ,SAL
   3  ,COMM
   4  FROM EMP
   5* WHERE COMM > SAL
SQL> /

```

ENAME	SAL	COMM
MARTIN	1250	1400

```

SQL> ED
Wrote file afiedt.buf

```

```

   1  SELECT ENAME
   2  ,SAL
   3  ,COMM
   4  FROM EMP
   5* WHERE SAL < COMM
SQL> /

```

ENAME	SAL	COMM
MARTIN	1250	1400

```

SQL> ED
Wrote file afiedt.buf

```

```

   1  SELECT ENAME

```

```

2  ,SAL
3  ,COMM
4  FROM EMP
5* WHERE SIGN(SAL - COMM ) =-1
SQL> /

```

ENAME	SAL	COMM
MARTIN	1250	1400

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  FROM EMP
5* WHERE SIGN(SAL - COMM ) =-1
6  /

```

ENAME	SAL	COMM
MARTIN	1250	1400

```
SQL>
```

```
SQL>
```

```
SQL> 10000      8000      LOSS
```

```
SQL> 10000      12000     PROFIT
```

```
SQL>
```

```
SQL>
```

```
SQL> CASE WHEN SIGN(PC-SC) = -1 THEN 'LOSS'
```

```
SP2-0734: unknown command beginning "CASE WHEN ..." - rest of line ignored.
```

```
SQL> WHEN SIGN(PC-SC) = 1 THEN 'PROFIT'
```

```
SP2-0734: unknown command beginning "WHEN SIGN..." - rest of line ignored.
```

```
SQL> END LOSS_PROF
```

```
SP2-0734: unknown command beginning "END LOSS_P..." - rest of line ignored.
```

```
SQL>
```

```
SQL>
```

```
SQL> CL SCR
```

```
SQL> SELECT 100 -300 AS C1
```

```
2 FROM DUAL;
```

C1
-200

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT ABS ( 100 -300 ) AS C1
```

```
2* FROM DUAL
```

```
SQL> /
```

C1
200

```
SQL> CL SCR
```

```
SQL> R
```

```
1  SELECT ABS ( 100 -300 ) AS C1
```

```
2* FROM DUAL
```



```
          C1
-----
          200
```

```
SQL> SELECT SQRT(4) AS C1
       2 FROM DUAL;
```

```
          C1
-----
          2
```

```
SQL> ED
Wrote file afiedt.buf
```

```
       1 SELECT SQRT(9) AS C1
       2* FROM DUAL
SQL> /
```

```
          C1
-----
          3
```

```
SQL> ED
Wrote file afiedt.buf
```

```
       1 SELECT SQRT(16) AS C1
       2* FROM DUAL
SQL> /
```

```
          C1
-----
          4
```

```
SQL> ED
Wrote file afiedt.buf
```

```
       1 SELECT SQRT(18) AS C1
       2* FROM DUAL
SQL> /
```

```
          C1
-----
4.24264069
```

```
SQL> ED
Wrote file afiedt.buf
```

```
       1 SELECT ROUND ( SQRT(18) ) AS C1
       2* FROM DUAL
SQL> /
```

```
          C1
-----
          4
```

```
SQL> ED
Wrote file afiedt.buf
```

```
       1 SELECT ROUND ( SQRT(18) ) AS C1
       2* FROM DUAL
SQL>
SQL> /
```

```
          C1
```

```
-----  
4
```

```
SQL> CL SCR
```

```
SQL> SELECT POWER(2,10) AS C1  
2 FROM DUAL;
```

```
      C1  
-----  
    1024
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
1 SELECT POWER(2,100) AS C1  
2* FROM DUAL
```

```
SQL> /
```

```
      C1  
-----  
1.2677E+30
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
1 SELECT POWER(2,20) AS C1  
2* FROM DUAL
```

```
SQL> /
```

```
      C1  
-----  
    1048576
```

```
SQL> CL SCR
```

```
SQL> SELECT LENGTH('ARJUN') AS C1  
2 FROM DUAL;
```

```
      C1  
-----  
5
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
1 SELECT LENGTH('ARJUN') AS C1  
2      ,VSIZE('ARJUN') AS C2  
3* FROM DUAL
```

```
SQL> /
```

```
      C1      C2  
-----  
5          5
```

```
SQL> R
```

```
1 SELECT LENGTH('ARJUN') AS C1  
2      ,VSIZE('ARJUN') AS C2  
3* FROM DUAL
```

```
      C1      C2  
-----  
5          5
```

```
SQL> R
1  SELECT LENGTH('ARJUN') AS C1
2      ,VSIZE('ARJUN') AS C2
3* FROM DUAL
```

```

      C1      C2
-----
      5      5
```

```
SQL> CL SCR
```

```
SQL> R
1  SELECT LENGTH('ARJUN') AS C1
2      ,VSIZE('ARJUN') AS C2
3* FROM DUAL
```

```

      C1      C2
-----
      5      5
```

```
SQL> CL SCR
```

```
SQL> SELECT SYSDATE AS C1
2 FROM DUAL;
```

```

C1
-----
31-MAY-18
```

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE + 40 AS C1
2* FROM DUAL
SQL> /
```

```

C1
-----
10-JUL-18
```

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE-20 AS C1
2* FROM DUAL
SQL> /
```

```

C1
-----
11-MAY-18
```

1-Jun-2018 Date Functions

CHARACTER FUNCTIONS

NUMBER FUNCTIONS

DATE FUNCTIONS

ADD_MONTHS

```
(
P1-- DATE COLUMN/DATE LITERAL
,P2 -- HOW MANY MONTHS WE WANT TO ADD/SUBTRACT TO/FROM P1
```

)

IF P2 VALUE IS +VE IT WILL ADD MONTHS TO P1
IF P2 VALUE IS -VE IT WILL SUBSTRACT MONTHS FROM P1

FLIPKART

6 MONTHS EMI

ADD_MONTHS (STARTIMGEMIDATE
 , 6
)

04-JUN-18
04-JUL-18
04-AUG-18
04-SEP-18
04-OCT-18
04-NOV-18

LEVEL

TODAY

ADD_MONTHS (DOJ
 , 60
) AS GRATUTI_ELI_DATE

TODAY LOAN PERSONAL 12/24/36/48/60

ADD_MONTHS (LOANFIRSTEMIDATE
 , 60
)

MONTHS_BETWEEN
(
 P1-- DATE COLUMN /DATE LITERAL
 , P2 DATE COLUMN /DATE LITERAL
)

IT WILL GIVE YOU BETWEEN P1 AND P2 DATES HOW MANY MONTHS ARE THERE

```
ENAME          SAL JOB          HIREDATE          EXP_INMONTHS
EXP_INMONTHS    EXP_INYEARS    EXP_INMONTHS    EXP_INYEARS
```

```
LAST_DAY
(
P1-- DATE COLUMN/DATE LITERAL
)
```

IT WILL GIVE YOU LAST DAY OF THE GIVEN YEAR AND MONTH

LETs SAY

```
LAST_DAY('05-FEB-1985')
```

OUTPUT : IN 1985 FEB MONTH WHAT IS THE LAST DATE

```
ENAME          SAL HIREDATE    JOIN_MONTH_SALARY
TOTALNUMBEROFWORKINGDAY * PERDAYSALRY
-----
```

1) WRITE A SELECT STATEMENT TO KNOW HOW MANY DAYS ARE THERE IN EACH WEEKDAY IN CURRENT MONTH ?

01-JUN-18

IN 2018 JUN MONTH

MON	TUE	WED	THU	FRI	SAT
SUN					
5	4	4	5	5	5
4					

01-JUN-18 IS A FRIDAY

02-JUN-18 IS A SATUREDAY

.
.
 .
.
 .

30-JUN-18 IS A WEDNESDAY

```
LAST_DAY
TO_CHAR
COUNT
CASE
LEVEL
```

```
NEXT_DAY
(
  P1-- DATE COLUMN/DATE LITERAL
,P2 -- WEEKDAY NAME  SUN MON TUE WED THU FRI SAT
)
```

IT WILL GIVE YOU AFTER P1 DATE COMING P2 DAY DATE

```
NEXT_DAY(SYSDATE
          , 'THU'
        )
```

```
USER_SCHEDULER_JOBS
-----
```

```
1) CRONTAB/CRONLIST
* * * * *
```

```
WHICH DAY
WHICH HOUR
WHHICMONT
```

```
2)
SCHEDULER -- DBMS_SCHEDULER
           DBMS_JOBS
-----
```

```
NEXT RUN "NEXT_DAY(SYSDATE
              , 'WED'
            )
```

```
3) AUTO SYS /APPWORX
```

```
DEV    UAT    PRODCUTIONS
```

```
VERSION CONTROL TOOL
DBA
TESTING
BUG/HOW TO FIEX
TOOLS
```

```
ROUND
(
  P1 -- DATE COLUMN/DATE LITERAL
,P2  -- DAY/MONTH/YEAR
)
```

```
TRUNC
(
```

```
P1 -- DATE COLUMN/DATE LITERAL
,P2 -- DAY/MONTH/YEAR
)
```

IN BOTH ROUND AND TRUNC P2 IS OPTIONAL

IF P2 VALUE NOT PROVIDED

TILL AM ROUND WILL GIVE YOU SAME DAY DATE WITH 00 HOURS :00
MINTUES : 00 SECONDS
FROM PM ROUND WILL GIVE YOU NEXT DAY DATE WITH 00 HOURS :00
MINTUES : 00 SECONDS

BUT TRUNC ALWAYS WILL GIVE YOU SAME DAY DATE WITH 00 HOURS :00
MINTUES : 00 SECONDS

IF P2 VALUE IS DAY

TILL WEDNESDAY AM ROUND WILL GIVE YOU PREVIOUS SUNDAY DATE
FROM WEDNESDAY PM ROUND WILL GIVE YOU COMING SUNDAY DATE

BUT TRUNC ALWAYS WILL GIVE YOU PREVIOUS SUNDAY DATE

IF P2 VALUE IS MONTH

TILL 15 DAYS ROUND WILL GIVE YOU SAME MONTH STARTING DATE
FROM 16TH DAY ROUND WILL GIVE YOU NEXT MONTH STARTING DATE

BUT TRUNC ALWAYS WILL GIVE YOU SAME MONTH STARTING DATE

IF P2 VALUE IS YEAR

TILL 6 MONTHS ROUND WILL GIVE YOU SAME YEAR STARTING DATE
FROM 7TH MONTH ROUND WILL GIVE YOU NEXT YEAR STARTING DATE

BUT TRUNC ALWAYS WILL GIVE YOU SAME YEAR STARTING DATE

EXTRACT

1) WRITE A SELECT STATEMENT TO GET HOW MANY DAYS ARE REMAINING IN
CURRENT MONTH ?

1) WRITE A SELECT STATEMENT TO GET HOW MANY DAYS ARE REMAINING IN CURRENT YEAR ?

EXAMPLE :

```
01-JUN-2018  -- 29 DAYS MONTH
-- 180 DAYS ARE REMING IN A YEAR
```

```
01-JAN -2018 -- 30 DAYS
365 DAYS
```

```
SELECT SYSDATE AS C1
      ,TO_CHAR(SYSDATE,'DD-MON-YYYY HH24:MI:SS') AS C11
      ,ROUND(SYSDATE) AS RC1
      ,TO_CHAR(ROUND(SYSDATE),'DD-MON-YYYY HH24:MI:SS') AS RC11
      ,TRUNC(SYSDATE) AS TC1
      ,TO_CHAR(TRUNC(SYSDATE),'DD-MON-YYYY HH24:MI:SS') AS TC11
FROM DUAL
/
```

1) HOW TO GET TODAY'S DATE ?
SYSDATE (DD-MON-YY)

```
DATE + NUMBER -- FUTURE DATE
DATE - NUMBER  -- PAST DATE
DATE * NUMBER  -- ERROR
DATE / NUMBER  -- ERROR
```

IF ANY ONE OPERANAD VALUE IS DATE AND ANOTHER OPERAND VALUE IS NUMBER THEN
WE CAN DO ONLY ADDITION OR SUBSTRACTION ON DATES OTHER THAT WE CAN NOT DO

```
DATE - DATE  -- NUMBER OF DAYS BETWEEN THE DATES
DATE + DATE  -- NOT ALLOWED
DATE * DATE  -- NOT ALLOWED
DATE / DATE  -- NOT ALLOWED
```

```
DATE + NUMBER
DATE - NUMBER
DATE - DATE
```

2) HOW MANY DAYS OLD IS YOUR BABY ?

3) WRITE A SELECT STATEMENT TO GET HOW MANY DAYS OLD YOU ARE ?

```
SYSDATE - DOJ
SYSDATE - DOB
```

HOW TO ADD HOURS TO A DATE
MINUTE

1-Jun-2018 Date Functions (Exa)

```
SQL> SELECT SYSDATE AS C1
       2 FROM DUAL;
```

```
C1
-----
01-JUN-18
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT SYSDATE AS C1
2       ,SYSDATE + 45 AS C2
3* FROM DUAL
SQL> /
```

```
C1          C2
-----
01-JUN-18 16-JUL-18
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT SYSDATE AS C1
2       ,SYSDATE - 45 AS C2
3* FROM DUAL
SQL> /
```

```
C1          C2
-----
01-JUN-18 17-APR-18
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT SYSDATE AS C1
2       ,SYSDATE * 2 AS C2
3* FROM DUAL
SQL> /
```

```
       ,SYSDATE * 2 AS C2
       *
```

```
ERROR at line 2:
ORA-00932: inconsistent datatypes: expected NUMBER got DATE
```

```
SQL> ED
```

Wrote file afiedt.buf

```
1  SELECT SYSDATE AS C1
2      ,SYSDATE / 2 AS C2
3* FROM DUAL
SQL> /
      ,SYSDATE / 2 AS C2
      *
```

ERROR at line 2:

ORA-00932: inconsistent datatypes: expected NUMBER got DATE

SQL> ED

Wrote file afiedt.buf

```
1  SELECT SYSDATE - (SYSDATE - 20) AS C1
2* FROM DUAL
SQL> /

      C1
-----
      20
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT SYSDATE + (SYSDATE - 20) AS C1
2* FROM DUAL
SQL> /
SELECT SYSDATE + (SYSDATE - 20) AS C1
      *
```

ERROR at line 1:
ORA-00975: date + date not allowed

SQL> ED

Wrote file afiedt.buf

```
1  SELECT SYSDATE + (SYSDATE + 1) AS C1
2* FROM DUAL
SQL> /
SELECT SYSDATE + (SYSDATE + 1) AS C1
      *
```

ERROR at line 1:
ORA-00975: date + date not allowed

SQL> ED

Wrote file afiedt.buf

```
1  SELECT SYSDATE * (SYSDATE + 1) AS C1
2* FROM DUAL
SQL> /
SELECT SYSDATE * (SYSDATE + 1) AS C1
      *
```

ERROR at line 1:
ORA-00932: inconsistent datatypes: expected NUMBER got DATE

SQL> ED

Wrote file afiedt.buf

```
1  SELECT SYSDATE / (SYSDATE + 1) AS C1
2* FROM DUAL
```

```
SQL> /
SELECT  SYSDATE / (SYSDATE + 1) AS C1
      *
ERROR at line 1:
ORA-00932: inconsistent datatypes: expected NUMBER got DATE
```

```
SQL> CL SCR
```

```
SQL> SELECT SYSDATE
```

```
2
```

```
SQL>
```

```
SQL> SELECT SYSDATE AS C1
```

```
2      ,SYSDATE + 5/24 AS C2
```

```
3 FROM DUAL;
```

```
C1      C2
-----
01-JUN-18 01-JUN-18
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT SYSDATE AS C1
```

```
2      ,TO_CHAR(SYSDATE,'DD-MON-YYYY HH24') AS C2
```

```
3      ,SYSDATE + 5/24 AS C3
```

```
4      ,TO_CHAR(SYSDATE + 5/24 , 'DD-MON-YYYY HH24') AS C4
```

```
5* FROM DUAL
```

```
SQL> /
```

```
ERROR:
```

```
ORA-01756: quoted string not properly terminated
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT SYSDATE AS C1
```

```
2      ,TO_CHAR(SYSDATE,'DD-MON-YYYY HH24') AS C2
```

```
3      ,SYSDATE + 5/24 AS C3
```

```
4      ,TO_CHAR(SYSDATE + 5/24 , 'DD-MON-YYYY HH24') AS C4
```

```
5* FROM DUAL
```

```
SQL> /
```

```
C1      C2      C3      C4
-----
01-JUN-18 01-JUN-2018 07      01-JUN-18 01-JUN-2018 12
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT SYSDATE AS C1
```

```
2      ,TO_CHAR(SYSDATE,'DD-MON-YYYY HH24') AS C2
```

```
3      ,SYSDATE + 23/24 AS C3
```

```
4      ,TO_CHAR(SYSDATE + 23/24 , 'DD-MON-YYYY HH24') AS C4
```

```
5* FROM DUAL
```

```
SQL> /
```

```
C1      C2      C3      C4
-----
01-JUN-18 01-JUN-2018 07      02-JUN-18 02-JUN-2018 06
```

```
SQL> CL SCR
```

```
SQL> R
```

```

1  SELECT SYSDATE  AS C1
2      ,TO_CHAR(SYSDATE,'DD-MON-YYYY HH24') AS C2
3      ,SYSDATE + 23/24 AS C3
4      ,TO_CHAR(SYSDATE + 23/24 ,'DD-MON-YYYY HH24') AS C4
5* FROM DUAL

```

C1	C2	C3	C4
01-JUN-18	01-JUN-2018 07	02-JUN-18	02-JUN-2018 06

```

SQL> SELECT SYSDATE AS C1
2      ,ADD_MONTHS(SYSDATE
3      ,2
4      ) AS C1
5  FROM DUAL;

```

C1	C1
01-JUN-18	01-AUG-18

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT SYSDATE AS C1
2      ,ADD_MONTHS(SYSDATE
3      ,-2
4      ) AS C1
5* FROM DUAL
SQL> /

```

C1	C1
01-JUN-18	01-APR-18

```

SQL> CL SCR

```

```

SQL> EDIT
Wrote file afiedt.buf

```

```

1  SELECT SYSDATE AS C1
2      ,ADD_MONTHS(SYSDATE
3      ,-2
4      ) AS C2
5      ,SYSDATE - 60 AS C3
6* FROM DUAL
SQL> /

```

C1	C2	C3
01-JUN-18	01-APR-18	02-APR-18

```

SQL> SELECT MONTHS_BETWEEN( SYDATE
2      , SYSDATE - 60
3      ) AS C1
4  FROM DUAL;
SELECT MONTHS_BETWEEN( SYDATE
*

```

```

ERROR at line 1:
ORA-00904: "SYDATE": invalid identifier

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT MONTHS_BETWEEN( SYSDATE
2                          , SYSDATE - 60
3                          ) AS C1
4* FROM DUAL
SQL> /

```

```

          C1
-----
1.96774194

```

```
SQL> CL SCR
```

```
SQL> COLUMN ENAME FORMAT A10
```

```
SQL>
```

```
SQL> SET LINESIZE 200
```

```
SQL> SET PAGESIZE 200
```

```
SQL> CLS CR
```

```
SP2-0042: unknown command "CLS CR" - rest of line ignored.
```

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
```

```
2  ,SAL
```

```
3  ,JOB
```

```
4  ,HIREDATE
```

```
5  FROM EMP;
```

ENAME	SAL	JOB	HIREDATE
KING	5000	PRESIDENT	17-NOV-81
BLAKE	2850	MANAGER	01-MAY-81
CLARK	2450	MANAGER	09-JUN-81
JONES	2975	MANAGER	02-APR-81
SCOTT	3000	ANALYST	19-APR-87
FORD	3000	ANALYST	03-DEC-81
SMITH	800	CLERK	17-DEC-80
ALLEN	1600	SALESMAN	20-FEB-81
WARD	1250	SALESMAN	22-FEB-81
MARTIN	1250	SALESMAN	28-SEP-81
TURNER	1500	SALESMAN	08-SEP-81
ADAMS	1100	CLERK	23-MAY-87
JAMES	950	CLERK	03-DEC-81
MILLER	1300	CLERK	23-JAN-82

```
14 rows selected.
```

```
SQL> EEWD
```

```
SP2-0042: unknown command "EEWD" - rest of line ignored.
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  ,HIREDATE
5  ,SYSDATE - HIREDATE AS EXP_INDDAYS
6* FROM EMP
SQL> /

```

ENAME	SAL	JOB	HIREDATE	EXP_INDDAYS
KING	5000	PRESIDENT	17-NOV-81	13345.3206
BLAKE	2850	MANAGER	01-MAY-81	13545.3206
CLARK	2450	MANAGER	09-JUN-81	13506.3206
JONES	2975	MANAGER	02-APR-81	13574.3206

```

SCOTT          3000 ANALYST   19-APR-87 11366.3206
FORD           3000 ANALYST   03-DEC-81 13329.3206
SMITH          800  CLERK     17-DEC-80 13680.3206
ALLEN          1600 SALESMAN  20-FEB-81 13615.3206
WARD           1250 SALESMAN  22-FEB-81 13613.3206
MARTIN         1250 SALESMAN  28-SEP-81 13395.3206
TURNER         1500 SALESMAN  08-SEP-81 13415.3206
ADAMS          1100 CLERK     23-MAY-87 11332.3206
JAMES          950  CLERK     03-DEC-81 13329.3206
MILLER         1300 CLERK     23-JAN-82 13278.3206

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  ,HIREDATE
5  ,ROUND (SYSDATE - HIREDATE) AS EXP_IN DAYS
6* FROM EMP
SQL> /

```

ENAME	SAL	JOB	HIREDATE	EXP_IN DAYS
KING	5000	PRESIDENT	17-NOV-81	13345
BLAKE	2850	MANAGER	01-MAY-81	13545
CLARK	2450	MANAGER	09-JUN-81	13506
JONES	2975	MANAGER	02-APR-81	13574
SCOTT	3000	ANALYST	19-APR-87	11366
FORD	3000	ANALYST	03-DEC-81	13329
SMITH	800	CLERK	17-DEC-80	13680
ALLEN	1600	SALESMAN	20-FEB-81	13615
WARD	1250	SALESMAN	22-FEB-81	13613
MARTIN	1250	SALESMAN	28-SEP-81	13395
TURNER	1500	SALESMAN	08-SEP-81	13415
ADAMS	1100	CLERK	23-MAY-87	11332
JAMES	950	CLERK	03-DEC-81	13329
MILLER	1300	CLERK	23-JAN-82	13278

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  ,HIREDATE
5  ,ROUND (SYSDATE - HIREDATE) AS EXP_IN DAYS
6  ,MONTHS_BETWEEN (SYSDATE
7  ,HIREDATE
8  ) AS EXP_IN MONTHS
9* FROM EMP
SQL> /

```

ENAME	SAL	JOB	HIREDATE	EXP_IN DAYS	EXP_IN MONTHS
KING	5000	PRESIDENT	17-NOV-81	13345	438.494236
BLAKE	2850	MANAGER	01-MAY-81	13545	445
CLARK	2450	MANAGER	09-JUN-81	13506	443.752301
JONES	2975	MANAGER	02-APR-81	13574	445.978107
SCOTT	3000	ANALYST	19-APR-87	11366	373.42972

FORD	3000	ANALYST	03-DEC-81	13329	437.945849
SMITH	800	CLERK	17-DEC-80	13680	449.494236
ALLEN	1600	SALESMAN	20-FEB-81	13615	447.397462
WARD	1250	SALESMAN	22-FEB-81	13613	447.332946
MARTIN	1250	SALESMAN	28-SEP-81	13395	440.139397
TURNER	1500	SALESMAN	08-SEP-81	13415	440.784559
ADAMS	1100	CLERK	23-MAY-87	11332	372.300688
JAMES	950	CLERK	03-DEC-81	13329	437.945849
MILLER	1300	CLERK	23-JAN-82	13278	436.300688

14 rows selected.

SQL> EDED

SP2-0042: unknown command "EDED" - rest of line ignored.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  ,HIREDATE
5  ,ROUND (SYSDATE - HIREDATE) AS EXP_IN DAYS
6  ,ROUND( MONTHS_BETWEEN(SYSDATE
7      ,HIREDATE
8      )
9      ,2
10     ) AS EXP_INMONTHS
11* FROM EMP
SQL> /

```

ENAME	SAL	JOB	HIREDATE	EXP_IN DAYS	EXP_INMONTHS
KING	5000	PRESIDENT	17-NOV-81	13345	438.49
BLAKE	2850	MANAGER	01-MAY-81	13545	445
CLARK	2450	MANAGER	09-JUN-81	13506	443.75
JONES	2975	MANAGER	02-APR-81	13574	445.98
SCOTT	3000	ANALYST	19-APR-87	11366	373.43
FORD	3000	ANALYST	03-DEC-81	13329	437.95
SMITH	800	CLERK	17-DEC-80	13680	449.49
ALLEN	1600	SALESMAN	20-FEB-81	13615	447.4
WARD	1250	SALESMAN	22-FEB-81	13613	447.33
MARTIN	1250	SALESMAN	28-SEP-81	13395	440.14
TURNER	1500	SALESMAN	08-SEP-81	13415	440.78
ADAMS	1100	CLERK	23-MAY-87	11332	372.3
JAMES	950	CLERK	03-DEC-81	13329	437.95
MILLER	1300	CLERK	23-JAN-82	13278	436.3

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  ,HIREDATE
5  ,ROUND (SYSDATE - HIREDATE) AS EXP_IN DAYS
6  ,ROUND( MONTHS_BETWEEN(SYSDATE
7      ,HIREDATE
8      )
9      ,2
10     ) AS EXP_INMONTHS
11 ,MONTHS_BETWEEN(SYSDATE,HIREDATE)/12 AS EXP_INYEARS
12* FROM EMP

```

SQL> /

ENAME	SAL	JOB	HIREDATE	EXP_IN DAYS	EXP_IN MONTHS	EXP_IN YEARS
KING	5000	PRESIDENT	17-NOV-81	13345	438.49	36.5411885
BLAKE	2850	MANAGER	01-MAY-81	13545	445	37.0833333
CLARK	2450	MANAGER	09-JUN-81	13506	443.75	36.9793606
JONES	2975	MANAGER	02-APR-81	13574	445.98	37.1648444
SCOTT	3000	ANALYST	19-APR-87	11366	373.43	31.1191455
FORD	3000	ANALYST	03-DEC-81	13329	437.95	36.4954896
SMITH	800	CLERK	17-DEC-80	13680	449.49	37.4578552
ALLEN	1600	SALESMAN	20-FEB-81	13615	447.4	37.283124
WARD	1250	SALESMAN	22-FEB-81	13613	447.33	37.2777477
MARTIN	1250	SALESMAN	28-SEP-81	13395	440.14	36.6782853
TURNER	1500	SALESMAN	08-SEP-81	13415	440.78	36.7320487
ADAMS	1100	CLERK	23-MAY-87	11332	372.3	31.0250595
JAMES	950	CLERK	03-DEC-81	13329	437.95	36.4954896
MILLER	1300	CLERK	23-JAN-82	13278	436.3	36.3583928

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  ,HIREDATE
5  ,ROUND (SYSDATE - HIREDATE) AS EXP_IN DAYS
6  ,ROUND( MONTHS_BETWEEN(SYSDATE
7  ,HIREDATE
8  )
9  ,2
10 ) AS EXP_IN MONTHS
11 ,ROUND(MONTHS_BETWEEN(SYSDATE,HIREDATE)/12) AS EXP_IN YEARS
12* FROM EMP
SQL> /

```

ENAME	SAL	JOB	HIREDATE	EXP_IN DAYS	EXP_IN MONTHS	EXP_IN YEARS
KING	5000	PRESIDENT	17-NOV-81	13345	438.49	37
BLAKE	2850	MANAGER	01-MAY-81	13545	445	37
CLARK	2450	MANAGER	09-JUN-81	13506	443.75	37
JONES	2975	MANAGER	02-APR-81	13574	445.98	37
SCOTT	3000	ANALYST	19-APR-87	11366	373.43	31
FORD	3000	ANALYST	03-DEC-81	13329	437.95	36
SMITH	800	CLERK	17-DEC-80	13680	449.49	37
ALLEN	1600	SALESMAN	20-FEB-81	13615	447.4	37
WARD	1250	SALESMAN	22-FEB-81	13613	447.33	37
MARTIN	1250	SALESMAN	28-SEP-81	13395	440.14	37
TURNER	1500	SALESMAN	08-SEP-81	13415	440.78	37
ADAMS	1100	CLERK	23-MAY-87	11332	372.3	31
JAMES	950	CLERK	03-DEC-81	13329	437.95	36
MILLER	1300	CLERK	23-JAN-82	13278	436.3	36

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  ,HIREDATE

```



```

5  ,ROUND (SYSDATE - HIREDATE) AS EXP_IN DAYS
6  ,ROUND( MONTHS_BETWEEN(SYSDATE
7      ,HIREDATE
8      )
9      ,2
10     ) AS EXP_INMONTHS
11  ,ROUND(MONTHS_BETWEEN(SYSDATE,HIREDATE)/12) AS EXP_INYEARS
12  FROM EMP
13* WHERE ROUND(MONTHS_BETWEEN(SYSDATE,HIREDATE) > 440
SQL> /
WHERE ROUND(MONTHS_BETWEEN(SYSDATE,HIREDATE) > 440
          *
ERROR at line 13:
ORA-00907: missing right parenthesis

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  ,HIREDATE
5  ,ROUND (SYSDATE - HIREDATE) AS EXP_IN DAYS
6  ,ROUND( MONTHS_BETWEEN(SYSDATE
7      ,HIREDATE
8      )
9      ,2
10     ) AS EXP_INMONTHS
11  ,ROUND(MONTHS_BETWEEN(SYSDATE,HIREDATE)/12) AS EXP_INYEARS
12  FROM EMP
13* WHERE ROUND(MONTHS_BETWEEN(SYSDATE,HIREDATE) )> 440
SQL> /

```

ENAME	SAL	JOB	HIREDATE	EXP_IN DAYS	EXP_INMONTHS	EXP_INYEARS
BLAKE	2850	MANAGER	01-MAY-81	13545	445	37
CLARK	2450	MANAGER	09-JUN-81	13506	443.75	37
JONES	2975	MANAGER	02-APR-81	13574	445.98	37
SMITH	800	CLERK	17-DEC-80	13680	449.49	37
ALLEN	1600	SALESMAN	20-FEB-81	13615	447.4	37
WARD	1250	SALESMAN	22-FEB-81	13613	447.33	37
TURNER	1500	SALESMAN	08-SEP-81	13415	440.78	37

7 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  ,HIREDATE
5  ,ROUND (SYSDATE - HIREDATE) AS EXP_IN DAYS
6  ,ROUND( MONTHS_BETWEEN(SYSDATE
7      ,HIREDATE
8      )
9      ,2
10     ) AS EXP_INMONTHS
11  ,ROUND(MONTHS_BETWEEN(SYSDATE,HIREDATE)/12) AS EXP_INYEARS
12  FROM EMP
13  WHERE ROUND(MONTHS_BETWEEN(SYSDATE,HIREDATE) )> 440
14* ORDER BY SAL ASC
SQL> /

```

ENAME	SAL	JOB	HIREDATE	EXP_IN DAYS	EXP_IN MONTHS	EXP_IN YEARS
SMITH	800	CLERK	17-DEC-80	13680	449.49	37
WARD	1250	SALESMAN	22-FEB-81	13613	447.33	37
TURNER	1500	SALESMAN	08-SEP-81	13415	440.78	37
ALLEN	1600	SALESMAN	20-FEB-81	13615	447.4	37
CLARK	2450	MANAGER	09-JUN-81	13506	443.75	37
BLAKE	2850	MANAGER	01-MAY-81	13545	445	37
JONES	2975	MANAGER	02-APR-81	13574	445.98	37

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  ,HIREDATE
5  ,ROUND (SYSDATE - HIREDATE) AS EXP_IN DAYS
6  ,ROUND( MONTHS_BETWEEN(SYSDATE
7  ,HIREDATE
8  )
9  ,2
10 ) AS EXP_IN MONTHS
11 ,ROUND(MONTHS_BETWEEN(SYSDATE,HIREDATE)/12) AS EXP_IN YEARS
12 FROM EMP
13 WHERE ROUND(MONTHS_BETWEEN(SYSDATE,HIREDATE) )> 440
14* ORDER BY SAL ASC
SQL> /

```

ENAME	SAL	JOB	HIREDATE	EXP_IN DAYS	EXP_IN MONTHS	EXP_IN YEARS
SMITH	800	CLERK	17-DEC-80	13680	449.49	37
WARD	1250	SALESMAN	22-FEB-81	13613	447.33	37
TURNER	1500	SALESMAN	08-SEP-81	13415	440.78	37
ALLEN	1600	SALESMAN	20-FEB-81	13615	447.4	37
CLARK	2450	MANAGER	09-JUN-81	13506	443.75	37
BLAKE	2850	MANAGER	01-MAY-81	13545	445	37
JONES	2975	MANAGER	02-APR-81	13574	445.98	37

7 rows selected.

SQL> CL SCR

```

SQL> SELECT SYSDATE AS C1
2  ,LAST_DAY(SYSDATE) AS C1
3  FROM DUAL;

```

C1	C1
01-JUN-18	30-JUN-18

SQL> CL SCR

```

SQL> SELECT ENAME
2  ,SAL
3  FROM EMP;

```

ENAME	SAL
KING	5000
BLAKE	2850

CLARK	2450
JONES	2975
SCOTT	3000
FORD	3000
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

```
SQL> SELECT ENAME
      2  ,SAL
      3  ,HIREDATE
      4  FROM EMP;
```

ENAME	SAL	HIREDATE
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

14 rows selected.

```
SQL> INSERT INTO EMP (EMPNO, ENAME, SAL, HIREDATE)
      2  VALUES
      3  (
      4  101, 'KRISHNA', 2500, '17-MAY-2018');
```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CLS CR
SP2-0042: unknown command "CLS CR" - rest of line ignored.
SQL> CL SCR
```

```
SQL> SELECT ENAME
      2  ,SAL
      3  ,HIREDATE
      4  FROM EMP;
```

ENAME	SAL	HIREDATE
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81

```

CLARK          2450 09-JUN-81
JONES          2975 02-APR-81
SCOTT          3000 19-APR-87
FORD           3000 03-DEC-81
SMITH          800  17-DEC-80
ALLEN          1600 20-FEB-81
WARD           1250 22-FEB-81
MARTIN         1250 28-SEP-81
TURNER         1500 08-SEP-81
ADAMS          1100 23-MAY-87
JAMES          950  03-DEC-81
MILLER         1300 23-JAN-82

```

15 rows selected.

SQL> CL SCR

```

SQL> R
  1  SELECT ENAME
  2    ,SAL
  3    ,HIREDATE
  4*  FROM EMP

```

ENAME	SAL	HIREDATE
-----	-----	-----
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

15 rows selected.

SQL> CL SCR

```

SQL> SELECT ENAEM
  2
SQL>
SQL> CL SCR

```

```

SQL> SELECT ENAME
  2    ,SAL
  3    ,HIREDATE
  4  FROM EMP;

```

ENAME	SAL	HIREDATE
-----	-----	-----
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81

```
SMITH          800 17-DEC-80
ALLEN          1600 20-FEB-81
WARD           1250 22-FEB-81
MARTIN         1250 28-SEP-81
TURNER         1500 08-SEP-81
ADAMS          1100 23-MAY-87
JAMES          950 03-DEC-81
MILLER         1300 23-JAN-82
```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,LAST_DAY(HIREDATE)
5* FROM EMP
SQL> /
```

ENAME	SAL	HIREDATE	LAST_DAY (
KRISHNA	2500	17-MAY-18	31-MAY-18
KING	5000	17-NOV-81	30-NOV-81
BLAKE	2850	01-MAY-81	31-MAY-81
CLARK	2450	09-JUN-81	30-JUN-81
JONES	2975	02-APR-81	30-APR-81
SCOTT	3000	19-APR-87	30-APR-87
FORD	3000	03-DEC-81	31-DEC-81
SMITH	800	17-DEC-80	31-DEC-80
ALLEN	1600	20-FEB-81	28-FEB-81
WARD	1250	22-FEB-81	28-FEB-81
MARTIN	1250	28-SEP-81	30-SEP-81
TURNER	1500	08-SEP-81	30-SEP-81
ADAMS	1100	23-MAY-87	31-MAY-87
JAMES	950	03-DEC-81	31-DEC-81
MILLER	1300	23-JAN-82	31-JAN-82

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,LAST_DAY(HIREDATE)
5  ,LAST_DAY(HIREDATE) - HIREDATE AS NUMOFWORKEDGDAYS
6* FROM EMP
SQL> /
```

ENAME	SAL	HIREDATE	LAST_DAY (	NUMOFWORKEDGDAYS
KRISHNA	2500	17-MAY-18	31-MAY-18	14
KING	5000	17-NOV-81	30-NOV-81	13
BLAKE	2850	01-MAY-81	31-MAY-81	30
CLARK	2450	09-JUN-81	30-JUN-81	21
JONES	2975	02-APR-81	30-APR-81	28
SCOTT	3000	19-APR-87	30-APR-87	11
FORD	3000	03-DEC-81	31-DEC-81	28
SMITH	800	17-DEC-80	31-DEC-80	14
ALLEN	1600	20-FEB-81	28-FEB-81	8
WARD	1250	22-FEB-81	28-FEB-81	6

MARTIN	1250	28-SEP-81	30-SEP-81	2
TURNER	1500	08-SEP-81	30-SEP-81	22
ADAMS	1100	23-MAY-87	31-MAY-87	8
JAMES	950	03-DEC-81	31-DEC-81	28
MILLER	1300	23-JAN-82	31-JAN-82	8

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,LAST_DAY(HIREDATE)
5  ,LAST_DAY(HIREDATE) - HIREDATE AS NUMOFWORKEDGDAYS
6  ,SAL / SUBSTR( LAST_DAY(HIREDATE)
7              ,1
8              ,2
9              ) AS PERDAYSAL
10* FROM EMP
SQL> /

```

ENAME	SAL	HIREDATE	LAST_DAY(	NUMOFWORKEDGDAYS	PERDAYSAL
KRISHNA	2500	17-MAY-18	31-MAY-18	14	80.6451613
KING	5000	17-NOV-81	30-NOV-81	13	166.666667
BLAKE	2850	01-MAY-81	31-MAY-81	30	91.9354839
CLARK	2450	09-JUN-81	30-JUN-81	21	81.6666667
JONES	2975	02-APR-81	30-APR-81	28	99.1666667
SCOTT	3000	19-APR-87	30-APR-87	11	100
FORD	3000	03-DEC-81	31-DEC-81	28	96.7741935
SMITH	800	17-DEC-80	31-DEC-80	14	25.8064516
ALLEN	1600	20-FEB-81	28-FEB-81	8	57.1428571
WARD	1250	22-FEB-81	28-FEB-81	6	44.6428571
MARTIN	1250	28-SEP-81	30-SEP-81	2	41.6666667
TURNER	1500	08-SEP-81	30-SEP-81	22	50
ADAMS	1100	23-MAY-87	31-MAY-87	8	35.483871
JAMES	950	03-DEC-81	31-DEC-81	28	30.6451613
MILLER	1300	23-JAN-82	31-JAN-82	8	41.9354839

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,LAST_DAY(HIREDATE)
5  ,LAST_DAY(HIREDATE) - HIREDATE AS NUMOFWORKEDGDAYS
6  ,SUBSTR( LAST_DAY(HIREDATE)
7              ,1
8              ,2
9              ) AS TOTDAYSINTTJOINMONT
10 ,SAL / SUBSTR( LAST_DAY(HIREDATE)
11              ,1
12              ,2
13              ) AS PERDAYSAL
14* FROM EMP
SQL> /

```

ENAME	SAL	HIREDATE	LAST_DAY(	NUMOFWORKEDGDAYS	TOTDAYSI	PERDAYSAL
-------	-----	----------	-----------	------------------	----------	-----------

KRISHNA	2500	17-MAY-18	31-MAY-18	14	31	80.6451613
KING	5000	17-NOV-81	30-NOV-81	13	30	166.666667
BLAKE	2850	01-MAY-81	31-MAY-81	30	31	91.9354839
CLARK	2450	09-JUN-81	30-JUN-81	21	30	81.6666667
JONES	2975	02-APR-81	30-APR-81	28	30	99.1666667
SCOTT	3000	19-APR-87	30-APR-87	11	30	100
FORD	3000	03-DEC-81	31-DEC-81	28	31	96.7741935
SMITH	800	17-DEC-80	31-DEC-80	14	31	25.8064516
ALLEN	1600	20-FEB-81	28-FEB-81	8	28	57.1428571
WARD	1250	22-FEB-81	28-FEB-81	6	28	44.6428571
MARTIN	1250	28-SEP-81	30-SEP-81	2	30	41.6666667
TURNER	1500	08-SEP-81	30-SEP-81	22	30	50
ADAMS	1100	23-MAY-87	31-MAY-87	8	31	35.483871
JAMES	950	03-DEC-81	31-DEC-81	28	31	30.6451613
MILLER	1300	23-JAN-82	31-JAN-82	8	31	41.9354839

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,LAST_DAY(HIREDATE)
5  ,LAST_DAY(HIREDATE) - HIREDATE AS NUMOFWORKEDGDAYS
6  ,SUBSTR( LAST_DAY(HIREDATE)
7  ,1
8  ,2
9  ) AS TOTDAYSINTTJOINMONT
10 ,ROUND(SAL / SUBSTR( LAST_DAY(HIREDATE)
11 ,1
12 ,2
13 )
14 ) AS PERDAYSAL
15* FROM EMP
SQL> /

```

ENAME	SAL	HIREDATE	LAST_DAY(	NUMOFWORKEDGDAYS	TOTDAYSI	PERDAYSAL
KRISHNA	2500	17-MAY-18	31-MAY-18	14	31	81
KING	5000	17-NOV-81	30-NOV-81	13	30	167
BLAKE	2850	01-MAY-81	31-MAY-81	30	31	92
CLARK	2450	09-JUN-81	30-JUN-81	21	30	82
JONES	2975	02-APR-81	30-APR-81	28	30	99
SCOTT	3000	19-APR-87	30-APR-87	11	30	100
FORD	3000	03-DEC-81	31-DEC-81	28	31	97
SMITH	800	17-DEC-80	31-DEC-80	14	31	26
ALLEN	1600	20-FEB-81	28-FEB-81	8	28	57
WARD	1250	22-FEB-81	28-FEB-81	6	28	45
MARTIN	1250	28-SEP-81	30-SEP-81	2	30	42
TURNER	1500	08-SEP-81	30-SEP-81	22	30	50
ADAMS	1100	23-MAY-87	31-MAY-87	8	31	35
JAMES	950	03-DEC-81	31-DEC-81	28	31	31
MILLER	1300	23-JAN-82	31-JAN-82	8	31	42

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE

```

```

4  ,LAST_DAY(HIREDATE)
5  ,LAST_DAY(HIREDATE) - HIREDATE AS NUMOFWORKEDGDAYS
6  ,SUBSTR( LAST_DAY(HIREDATE)
7      ,1
8      ,2
9      ) AS TOTDAYSINTTJOINMONT
10 ,ROUND(SAL / SUBSTR( LAST_DAY(HIREDATE)
11      ,1
12      ,2
13      )
14      ) AS PERDAYSAL
15 ,PERDAYSAL * NUMOFWORKEDGDAYS AS JOINED_MONTHS_SALARY
16* FROM EMP
17 /
,PERDAYSAL * NUMOFWORKEDGDAYS AS JOINED_MONTHS_SALARY
*
```

ERROR at line 15:

ORA-00904: "NUMOFWORKEDGDAYS": invalid identifier

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,LAST_DAY(HIREDATE)
5  ,LAST_DAY(HIREDATE) - HIREDATE AS NUMOFWORKEDGDAYS
6  ,SUBSTR( LAST_DAY(HIREDATE)
7      ,1
8      ,2
9      ) AS TOTDAYSINTTJOINMONT
10 ,ROUND(SAL / SUBSTR( LAST_DAY(HIREDATE)
11      ,1
12      ,2
13      )
14      ) AS PERDAYSAL
15 ,ROUND(SAL / SUBSTR( LAST_DAY(HIREDATE)
16      ,1
17      ,2
18      )
19      ) * (LAST_DAY(HIREDATE) - HIREDATE) AS JOINED_MONTHS_SALARY
20* FROM EMP
SQL> /
```

ENAME	SAL	HIREDATE	LAST_DAY(	NUMOFWORKEDGDAYS	TOTDAYSI	PERDAYSAL
JOINED_MONTHS_SALARY						

KRISHNA	2500	17-MAY-18	31-MAY-18	14	31	81
1134						
KING	5000	17-NOV-81	30-NOV-81	13	30	167
2171						
BLAKE	2850	01-MAY-81	31-MAY-81	30	31	92
2760						
CLARK	2450	09-JUN-81	30-JUN-81	21	30	82
1722						
JONES	2975	02-APR-81	30-APR-81	28	30	99
2772						
SCOTT	3000	19-APR-87	30-APR-87	11	30	100
1100						
FORD	3000	03-DEC-81	31-DEC-81	28	31	97
2716						

SMITH 364	800	17-DEC-80	31-DEC-80	14	31	26
ALLEN 456	1600	20-FEB-81	28-FEB-81	8	28	57
WARD 270	1250	22-FEB-81	28-FEB-81	6	28	45
MARTIN 84	1250	28-SEP-81	30-SEP-81	2	30	42
TURNER 1100	1500	08-SEP-81	30-SEP-81	22	30	50
ADAMS 280	1100	23-MAY-87	31-MAY-87	8	31	35
JAMES 868	950	03-DEC-81	31-DEC-81	28	31	31
MILLER 336	1300	23-JAN-82	31-JAN-82	8	31	42

15 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,LAST_DAY(HIREDATE)
5  ,LAST_DAY(HIREDATE) - HIREDATE AS NUMOFWORKEDGDAYS
6  ,SUBSTR( LAST_DAY(HIREDATE)
7  ,1
8  ,2
9  ) AS TOTDAYSINTTJOINMONT
10 ,ROUND(SAL / SUBSTR( LAST_DAY(HIREDATE)
11 ,1
12 ,2
13 )
14 ) AS PERDAYSAL
15 ,ROUND(SAL / SUBSTR( LAST_DAY(HIREDATE)
16 ,1
17 ,2
18 )
19 ) * (LAST_DAY(HIREDATE) - HIREDATE) AS JOINED_MONTHS_SALARY
20* FROM EMP

```

ENAME JOINED_MONTHS_SALARY	SAL	HIREDATE	LAST_DAY(	NUMOFWORKEDGDAYS	TOTDAYSI	PERDAYSAL
-----	-----	-----	-----	-----	-----	-----
-----	-----	-----	-----	-----	-----	-----
KRISHNA 1134	2500	17-MAY-18	31-MAY-18	14	31	81
KING 2171	5000	17-NOV-81	30-NOV-81	13	30	167
BLAKE 2760	2850	01-MAY-81	31-MAY-81	30	31	92
CLARK 1722	2450	09-JUN-81	30-JUN-81	21	30	82
JONES 2772	2975	02-APR-81	30-APR-81	28	30	99
SCOTT 1100	3000	19-APR-87	30-APR-87	11	30	100
FORD 2716	3000	03-DEC-81	31-DEC-81	28	31	97
SMITH 364	800	17-DEC-80	31-DEC-80	14	31	26

ALLEN 456	1600 20-FEB-81 28-FEB-81	8 28	57
WARD 270	1250 22-FEB-81 28-FEB-81	6 28	45
MARTIN 84	1250 28-SEP-81 30-SEP-81	2 30	42
TURNER 1100	1500 08-SEP-81 30-SEP-81	22 30	50
ADAMS 280	1100 23-MAY-87 31-MAY-87	8 31	35
JAMES 868	950 03-DEC-81 31-DEC-81	28 31	31
MILLER 336	1300 23-JAN-82 31-JAN-82	8 31	42

15 rows selected.

SQL> CL SCR

```
SQL> SELECT NEXT_DAY(SYSDATE
2      , 'THU'
3      )
4      AS C1
5      FROM DUAL;
```

```
C1
-----
07-JUN-18
```

SQL> ED
Wrote file afiedt.buf

```
1  SELECT SYSDATE AS C1
2  ,NEXT_DAY(SYSDATE
3      , 'THU'
4      )
5      AS C2
6* FROM DUAL
SQL> /
```

```
C1      C2
-----
01-JUN-18 07-JUN-18
```

SQL> ED
Wrote file afiedt.buf

```
1  SELECT SYSDATE AS C1
2  ,NEXT_DAY(SYSDATE
3      , 'SAT'
4      )
5      AS C2
6* FROM DUAL
SQL> /
```

```
C1      C2
-----
01-JUN-18 02-JUN-18
```

SQL> ED
Wrote file afiedt.buf

```
1  SELECT SYSDATE AS C1
2  ,NEXT_DAY(SYSDATE
```

```

3          , 'FRI'
4        )
5        AS C2
6* FROM DUAL
SQL> /

```

```

C1          C2
-----
01-JUN-18  08-JUN-18

```

```
SQL> CL SCR
```

```

SQL> R
1  SELECT SYSDATE AS C1
2    ,NEXT_DAY(SYSDATE
3              , 'FRI'
4            )
5        AS C2
6* FROM DUAL

```

```

C1          C2
-----
01-JUN-18  08-JUN-18

```

```

SQL> SELECT SYSDATE AS C1
2        ,ROUND(SYSDATE) AS RC1
3        ,TRUNC(SYSDATE) AS TC1
4  FROM DUAL;

```

```

C1          RC1          TC1
-----
01-JUN-18  01-JUN-18  01-JUN-18

```

```

SQL> R
1  SELECT SYSDATE AS C1
2        ,ROUND(SYSDATE) AS RC1
3        ,TRUNC(SYSDATE) AS TC1
4* FROM DUAL

```

```

C1          RC1          TC1
-----
01-JUN-18  02-JUN-18  01-JUN-18

```

```
SQL> CL SCR
```

Once the time is changed to PM Manually , the Round will reflect the next day

```

SQL> R
1  SELECT SYSDATE AS C1
2        ,ROUND(SYSDATE) AS RC1
3        ,TRUNC(SYSDATE) AS TC1
4* FROM DUAL

```

```

C1          RC1          TC1
-----
01-JUN-18  02-JUN-18  01-JUN-18

```

```
SQL> EDIT
```

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS C1
2        ,ROUND(SYSDATE) AS RC1
3        ,TO_CHAR(ROUND(SYSDATE), 'DD-MON-YYYY HH24:MI:SS') AS RC11
4        ,TRUNC(SYSDATE) AS TC1
5        ,TO_CHAR(TRUNC(SYSDATE), 'DD-MON-YYYY HH24:MI:SS') AS TC11

```

```

6* FROM DUAL
SQL> /

```

C1	RC1	RC11	TC1	TC11
01-JUN-18	02-JUN-18	02-JUN-2018	00:00:00	01-JUN-18 01-JUN-2018 00:00:00

```

SQL> D
SP2-0042: unknown command "D" - rest of line ignored.
SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT SYSDATE AS C1
2      ,TO_CHAR(SYSDATE,'DD-MON-YYYY HH24:MI:SS') AS C11
3      ,ROUND(SYSDATE) AS RC1
4      ,TO_CHAR(ROUND(SYSDATE),'DD-MON-YYYY HH24:MI:SS') AS RC11
5      ,TRUNC(SYSDATE) AS TC1
6      ,TO_CHAR(TRUNC(SYSDATE),'DD-MON-YYYY HH24:MI:SS') AS TC11
7* FROM DUAL
SQL> /

```

C1	C11	RC1	RC11	TC1	TC11
01-JUN-18	01-JUN-2018 15:15:17	02-JUN-18	02-JUN-2018 00:00:00	01-JUN-18	01-JUN-2018 00:00:00

```

SQL> R
1 SELECT SYSDATE AS C1
2      ,TO_CHAR(SYSDATE,'DD-MON-YYYY HH24:MI:SS') AS C11
3      ,ROUND(SYSDATE) AS RC1
4      ,TO_CHAR(ROUND(SYSDATE),'DD-MON-YYYY HH24:MI:SS') AS RC11
5      ,TRUNC(SYSDATE) AS TC1
6      ,TO_CHAR(TRUNC(SYSDATE),'DD-MON-YYYY HH24:MI:SS') AS TC11
7* FROM DUAL

```

C1	C11	RC1	RC11	TC1	TC11
01-JUN-18	01-JUN-2018 15:15:26	02-JUN-18	02-JUN-2018 00:00:00	01-JUN-18	01-JUN-2018 00:00:00

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1
4* FROM DUAL
SQL> /

```

C1	RC1	TC1
01-JUN-18	03-JUN-18	27-MAY-18

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1

```

```

4* FROM DUAL
SQL> RF/
SP2-0042: unknown command "RF/" - rest of line ignored.
SQL> /

```

C1	RC1	TC1
30-MAY-18	03-JUN-18	27-MAY-18

```

SQL> R
1  SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1
4* FROM DUAL

```

C1	RC1	TC1
30-MAY-18	03-JUN-18	27-MAY-18

```

SQL> R
1  SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1
4* FROM DUAL

```

C1	RC1	TC1
30-MAY-18	03-JUN-18	27-MAY-18

```

SQL> R
1  SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1
4* FROM DUAL

```

C1	RC1	TC1
30-MAY-18	03-JUN-18	27-MAY-18

```

SQL> R
1  SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1
4* FROM DUAL

```

C1	RC1	TC1
30-MAY-18	27-MAY-18	27-MAY-18

```

SQL> R
1  SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1
4* FROM DUAL

```

C1	RC1	TC1
30-MAY-18	27-MAY-18	27-MAY-18

```

SQL> R
1  SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1
4* FROM DUAL

```

C1	RC1	TC1
-----	-----	-----
30-MAY-18	27-MAY-18	27-MAY-18

```
SQL> R
  1  SELECT SYSDATE AS C1
  2          ,ROUND(SYSDATE,'DAY') AS RC1
  3          ,TRUNC(SYSDATE,'DAY') AS TC1
  4* FROM DUAL
```

C1	RC1	TC1
-----	-----	-----
30-MAY-18	27-MAY-18	27-MAY-18

```
SQL> R
  1  SELECT SYSDATE AS C1
  2          ,ROUND(SYSDATE,'DAY') AS RC1
  3          ,TRUNC(SYSDATE,'DAY') AS TC1
  4* FROM DUAL
```

C1	RC1	TC1
-----	-----	-----
30-MAY-18	27-MAY-18	27-MAY-18

```
SQL> R
  1  SELECT SYSDATE AS C1
  2          ,ROUND(SYSDATE,'DAY') AS RC1
  3          ,TRUNC(SYSDATE,'DAY') AS TC1
  4* FROM DUAL
```

C1	RC1	TC1
-----	-----	-----
30-MAY-18	27-MAY-18	27-MAY-18

```
SQL> R
  1  SELECT SYSDATE AS C1
  2          ,ROUND(SYSDATE,'DAY') AS RC1
  3          ,TRUNC(SYSDATE,'DAY') AS TC1
  4* FROM DUAL
```

C1	RC1	TC1
-----	-----	-----
30-MAY-18	27-MAY-18	27-MAY-18

```
SQL> R
  1  SELECT SYSDATE AS C1
  2          ,ROUND(SYSDATE,'DAY') AS RC1
  3          ,TRUNC(SYSDATE,'DAY') AS TC1
  4* FROM DUAL
```

C1	RC1	TC1
-----	-----	-----
30-MAY-18	27-MAY-18	27-MAY-18

```
SQL> R
  1  SELECT SYSDATE AS C1
  2          ,ROUND(SYSDATE,'DAY') AS RC1
  3          ,TRUNC(SYSDATE,'DAY') AS TC1
  4* FROM DUAL
```

C1	RC1	TC1
-----	-----	-----
30-MAY-18	27-MAY-18	27-MAY-18

```
SQL> R
1  SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1
4*  FROM DUAL
```

C1	RC1	TC1
30-MAY-18	27-MAY-18	27-MAY-18

```
SQL> R
1  SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1
4*  FROM DUAL
```

C1	RC1	TC1
30-MAY-18	27-MAY-18	27-MAY-18

```
SQL> R
1  SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1
4*  FROM DUAL
```

C1	RC1	TC1
30-MAY-18	27-MAY-18	27-MAY-18

```
SQL> R
1  SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1
4*  FROM DUAL
```

C1	RC1	TC1
30-MAY-18	27-MAY-18	27-MAY-18

```
SQL> R
1  SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'DAY') AS RC1
3      ,TRUNC(SYSDATE,'DAY') AS TC1
4*  FROM DUAL
```

C1	RC1	TC1
30-MAY-18	03-JUN-18	27-MAY-18

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS C1
2      ,ROUND(SYSDATE,'MONTH') AS RC1
3      ,TRUNC(SYSDATE,'MONTH') AS TC1
4*  FROM DUAL
SQL> /
```

C1	RC1	TC1
30-MAY-18	01-JUN-18	01-MAY-18

```
SQL> R
  1  SELECT SYSDATE AS C1
  2      ,ROUND(SYSDATE,'MONTH') AS RC1
  3      ,TRUNC(SYSDATE,'MONTH') AS TC1
  4* FROM DUAL
```

C1	RC1	TC1
14-MAY-18	01-MAY-18	01-MAY-18

```
SQL> R
  1  SELECT SYSDATE AS C1
  2      ,ROUND(SYSDATE,'MONTH') AS RC1
  3      ,TRUNC(SYSDATE,'MONTH') AS TC1
  4* FROM DUAL
```

C1	RC1	TC1
15-MAY-18	01-MAY-18	01-MAY-18

```
SQL> R
  1  SELECT SYSDATE AS C1
  2      ,ROUND(SYSDATE,'MONTH') AS RC1
  3      ,TRUNC(SYSDATE,'MONTH') AS TC1
  4* FROM DUAL
```

C1	RC1	TC1
16-MAY-18	01-JUN-18	01-MAY-18

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT SYSDATE AS C1
  2      ,ROUND(SYSDATE,'YEAR') AS RC1
  3      ,TRUNC(SYSDATE,'YEAR') AS TC1
  4* FROM DUAL
SQL> /
```

C1	RC1	TC1
16-MAY-18	01-JAN-18	01-JAN-18

```
SQL> R
  1  SELECT SYSDATE AS C1
  2      ,ROUND(SYSDATE,'YEAR') AS RC1
  3      ,TRUNC(SYSDATE,'YEAR') AS TC1
  4* FROM DUAL
```

C1	RC1	TC1
16-AUG-18	01-JAN-19	01-JAN-18

```
SQL> CL SCR
```

```
SQL> SPOOL OFF
```

3-jun-2018-Conversion functions

CHARACTER FUNCTIONS
NUMBER FUNCTIONS

DATE FUNCTIONS

CONVERSION FUNCTIONS

TO_CHAR

TO_NUMBER

TO_DATE

FROM DB IF YOU WANT YOUR OWN FORMAT THEN YOU WILL GO FOR TO_CHAR

WHILE INSERTING/UPDATING IF YOU DONT WANT TO PROVIDED DEFAULT DATE
FORMAT WHAT DB ACCEPTS

THEN WE HAVE TO GO FOR TO_DATE

WHILE INSERTING/UPDATING IF YOU DONT WANT TO PROVIDE DEFAULT NUMBER
FORMAT WHAT DB ACCEPTS

THEN WE HAVE TO GO FOR TO_NUMBER.

TO_CHAR

(
 P1-- COLUMN /LITERAL
,P2 -- FORMAT SPECIFIERS
,P3 -- NLS FORMAT CHANGES
)

1) P2 AND P3 ARE OPTIONAL

DATE TO CHARACTER FORMAT SPECIFIERS

D -- IN A WEEK WHICH DAY WE ARE IN
 SUN 1 MON 2 .. SAT 7

DD -- IN A MONTH WHICH DAY WE ARE IN

DDD --- IN A YEAR WHICH DAY WE AR EIN

DAY -- COMPLETE WEEKDAY NAME

DY -- ABBRIVATED WEEKDAY NAME

MONTH -- COMPLETE MONTH NAME

MON -- ABBRIVATED MONTH NAME

MM -- NUMERIC MONTH INDICATOR

HH24 -- RAILWAY TIME FORMAT 0-23

HH12 -- 12 HOURS TIME FORMAT

MI -- MINUTES

SS -- SECONDS

SSSSS -- SECONDS FROM PAST MIDNIGHT

AM OR A.M. -- IF TIME IS IS AM IT WILL GIVE YOU AM ELSE IT WILL
GIVE YOU PM

PM OR P.M. -- IF TIME IS IS AM IT WILL GIVE YOU AM ELSE IT WILL GIVE YOU PM

CC -- IT WILL GIVE YOU CENTURY INFORMATION

SCC -- IT WILL GIVE YOU CENTURY INFORMATION ONLY BUT IF THE DATE IS BELONGS TO B.D THEN IT WILL ADD - BEFORE CENTURY

AD OR A.D. -- IF THE DATE IS AFTER CHRIST THEN IT WILL GIVE YOU AC ELSE IT WILL GIVE YOU B.C

BC OR B.C. -- IF THE DATE IS AFTER CHRIST THEN IT WILL GIVE YOU AC ELSE IT WILL GIVE YOU B.

YYYY-- FOUR DIGIT YEAR

YYY -- LAST 3 DIGIT YEAR

YY --- LAST 2 DIGIT YEAR

Y -- LAST 1 DIGIT YEAR

YEAR -- IT WILL SPELL THE YEAR INFORMATION

SP -- BEFORE SP WHATEVER WE KEEP IT WILL SPELL

TH -- STANDARD TH AT END OF THE NUMBER BASED ON HOW WE PRONOUNCED

FM --- FILL MODE

IT IS USED TO REMOVE EXTRA SPACES WHICH CAME BECAUSE OF TO_CHAR

FM WILL BE USED ONLY WITH TO_CHAR

RRRR -- LAST FOUR DIGIT YEAR

RR --- LAST 2 DIGIT YEAR

-- YY AND RR --- (10 MINS)

RR AND YY IT WILL WORK AS IT IS SAME IF WE ARE USING IN SELECT STATEMENT

BUT IF YOU ARE USING IN INSERT/UPDATE STATEMENT RR BEHAVES DIFFERENTLY

(CENTURY INFORMATION WILL BE CHANGED IF YOU USE RR)

Q --- QUARTER INFORMATION

J -- JULIAN NUMBER INDICATOR

AS PER ORACLE STARTING DATE IS 01 JAN -4712 BC
2458272 DAY
2446985

DATE

3-jun-2018-Conversion functions (Example)

```
SQL> SELECT SYSDATE AS C1
       2 FROM DUAL;
```

```
C1
-----
02-JUN-18
```

```
ed
  1 SELECT SYSDATE AS C1
  2           ,TO_CHAR(SYSDATE
  3                     , 'DAY'
  4                     )
  5* FROM DUAL
SQL> /
```

```
C1           TO_CHAR(SYSDATE, 'DAY')
-----
02-JUN-18 SATURDAY
```

```
SQL> ED
Wrote file afiedt.buf
```

```
  1 SELECT SYSDATE AS C1
  2           ,TO_CHAR(SYSDATE
  3                     , 'DAY'
  4                     ) AS C1
  5* FROM DUAL
SQL> /
```

```
C1           C1
-----
02-JUN-18 SATURDAY
```

```
SQL> CL SCR
```

```
SQL> R
  1 SELECT SYSDATE AS C1
  2           ,TO_CHAR(SYSDATE
  3                     , 'DAY'
  4                     ) AS C1
  5* FROM DUAL
```

```
C1           C1
-----
02-JUN-18 SATURDAY
```

```
SQL> ED
Wrote file afiedt.buf
```

```
  1 SELECT SYSDATE AS C1
  2           ,TO_CHAR(SYSDATE
  3                     , 'DY'
  4                     ) AS C1
  5* FROM DUAL
SQL> /
```

```
C1           C1
-----
02-JUN-18 SAT
```

```
SQL> ED
Wrote file afiedt.buf
```

```
  1 SELECT SYSDATE AS C1
```

```

2          ,TO_CHAR(SYSDATE
3              , 'MONTH'
4              ) AS C1
5* FROM DUAL
SQL> /

```

```

C1          C1
-----
02-JUN-18  JUNE

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT SYSDATE AS C1
2          ,TO_CHAR(SYSDATE
3              , 'YYYY'
4              ) AS C1
5* FROM DUAL
SQL> /

```

```

C1          C1
-----
02-JUN-18  2018

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT SYSDATE AS C1
2          ,TO_CHAR(SYSDATE
3              , 'Q'
4              ) AS C1
5* FROM DUAL
SQL> /

```

```

C1          C
-----
02-JUN-18  2

```

```

SQL> COLUMN C1 FORMAT A10
SQL> R
1  SELECT SYSDATE AS C1
2          ,TO_CHAR(SYSDATE
3              , 'Q'
4              ) AS C1
5* FROM DUAL

```

```

C1          C1
-----
02-JUN-18  2

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT SYSDATE AS C1
2          ,TO_CHAR(SYSDATE
3              , 'D'
4              ) AS C1
5* FROM DUAL
SQL> /

```

```

C1          C1
-----
02-JUN-18  7

```

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE AS C1
2      ,TO_CHAR(SYSDATE
3              , 'DAY D'
4              ) AS C1
5* FROM DUAL
SQL> /
```

C1	C1
02-JUN-18	SATURDAY
	7

```
SQL> COLUMN C1 FORMAT A30
```

```
SQL> R
1  SELECT SYSDATE AS C1
2      ,TO_CHAR(SYSDATE
3              , 'DAY D'
4              ) AS C1
5* FROM DUAL
```

C1	C1
02-JUN-18	SATURDAY 7

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE AS C1
2      ,TO_CHAR(SYSDATE
3              , 'DAY DD'
4              ) AS C1
5* FROM DUAL
SQL> /
```

C1	C1
02-JUN-18	SATURDAY 02

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE AS C1
2      ,TO_CHAR(SYSDATE
3              , 'DAY DDD'
4              ) AS C1
5* FROM DUAL
SQL> /
```

C1	C1
02-JUN-18	SATURDAY 153

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE AS C1
2      ,TO_CHAR(SYSDATE
3              , 'DAY DDD'
4              ) AS C1
5* FROM DUAL
```

```
SQL> CL SCR
```

```
SQL> SELECT SYSDATE AS C2
       2 FROM DUAL;
```

```
C2
-----
02-JUN-18
```

```
SQL> ED
Wrote file afiedt.buf
```

```
       1 SELECT SYSDATE AS SD
       2           ,TO_CHAR(SYSDATE
       3                       , 'YYYY-MM-DD HH24 MI SS DAY'
       4                       ) AS C1
       5* FROM DUAL
SQL> /
```

```
SD           C1
-----
02-JUN-18 2018-06-02 08 20 20 SATURDAY
```

```
SQL> R
       1 SELECT SYSDATE AS SD
       2           ,TO_CHAR(SYSDATE
       3                       , 'YYYY-MM-DD HH24 MI SS DAY'
       4                       ) AS C1
       5* FROM DUAL
```

```
SD           C1
-----
02-JUN-18 2018-06-02 08 20 35 SATURDAY
```

```
SQL> CL SCR
```

```
SQL> R
       1 SELECT SYSDATE AS SD
       2           ,TO_CHAR(SYSDATE
       3                       , 'YYYY-MM-DD HH24 MI SS DAY'
       4                       ) AS C1
       5* FROM DUAL
```

```
SD           C1
-----
02-JUN-18 2018-06-02 08 20 48 SATURDAY
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```
       1 SELECT SYSDATE AS SD
       2           ,TO_CHAR(SYSDATE
       3                       , 'YYYY'
       4                       ) AS C1
       5* FROM DUAL
SQL> /
```

```
SD           C1
-----
02-JUN-18 2018
```

```
SQL> SELECT ENAME
       2 ,SAL
```

```

3  ,HIREDATE
4  FROM EMP;

```

ENAME	SAL	HIREDATE
-----	-----	-----
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

15 rows selected.

SQL> COLUMN ENAME FORMAT A10

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4* FROM EMP

```

ENAME	SAL	HIREDATE
-----	-----	-----
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

15 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4* FROM EMP

```

ENAME	SAL	HIREDATE
-----	-----	-----
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81

BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

15 rows selected.

```

-----
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE HIREDATE LIKE '%DEC%'
SQL> /

```

ENAME	SAL	HIREDATE
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
JAMES	950	03-DEC-81

SQL> ED
Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE SUBSTR(HIREDATE,4,3) ='DEC'
SQL> /

```

ENAME	SAL	HIREDATE
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
JAMES	950	03-DEC-81

SQL> ED
Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MM') =12
SQL> /

```

ENAME	SAL	HIREDATE
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
JAMES	950	03-DEC-81

SQL> ED
Wrote file afiedt.buf


```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MON') ='DEC'
SQL> /

```

ENAME	SAL	HIREDATE
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
JAMES	950	03-DEC-81

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MONTH') ='DECEMBER'
SQL> /

```

no rows selected

```
SQL> CL SCR
```

```

SQL> R
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MONTH') ='DECEMBER'

```

no rows selected

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MON') ='DEC'
SQL> /

```

ENAME	SAL	HIREDATE
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
JAMES	950	03-DEC-81

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MM') =12
SQL> /

```

ENAME	SAL	HIREDATE
-------	-----	----------

```
FORD          3000 03-DEC-81
SMITH         800  17-DEC-80
JAMES         950  03-DEC-81
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MONTH') ='DECEMBER'
SQL> /
```

```
no rows selected
```

```
-----
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE 'DECEMBER' = 'DECEMBER'
SQL> /
```

ENAME	SAL	HIREDATE
-----	-----	-----
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

```
15 rows selected.
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE 'DECEMBER' = 'DECEMBER'
SQL> /
```

ENAME	SAL	HIREDATE
-----	-----	-----
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81

```

WARD          1250 22-FEB-81
MARTIN        1250 28-SEP-81
TURNER        1500 08-SEP-81
ADAMS         1100 23-MAY-87
JAMES         950  03-DEC-81
MILLER        1300 23-JAN-82

```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE 'DECEMBER' = 'DECEMBER'
SQL> /

```

ENAME	SAL	HIREDATE
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE 'APPLE'='APPLE'
SQL> /

```

ENAME	SAL	HIREDATE
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87

```
JAMES          950 03-DEC-81
MILLER         1300 23-JAN-82
```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE 'APPLE' ='APPLE'
SQL> /
```

ENAME	SAL	HIREDATE
-----	-----	-----
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'DAY') ='DECEMBER'
SQL> /
```

no rows selected

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'DAY') ='DECEMBER '
SQL> /
```

no rows selected

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
```

```

3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MONTH') ='DECEMBER '
SQL> /

```

ENAME	SAL	HIREDATE
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
JAMES	950	03-DEC-81

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MONTH') ='DECEMBER'
SQL> /

```

no rows selected

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4* FROM EMP
5  /

```

ENAME	SAL	HIREDATE
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

15 rows selected.

```

SQL> CL SCR

```

```

SQL> R
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4* FROM EMP

```

ENAME	SAL	HIREDATE
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81

```
SCOTT          3000 19-APR-87
FORD           3000 03-DEC-81
SMITH          800  17-DEC-80
ALLEN          1600 20-FEB-81
WARD           1250 22-FEB-81
MARTIN         1250 28-SEP-81
TURNER         1500 08-SEP-81
ADAMS          1100 23-MAY-87
JAMES          950  03-DEC-81
MILLER         1300 23-JAN-82
```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MON') ='JUN'
SQL> /
```

```
ENAME          SAL HIREDATE
-----
CLARK          2450 09-JUN-81
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MONTH') ='JUNE'
SQL> /
```

no rows selected

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MONTH') ='JUNE'
6  /
```

```
ENAME          SAL HIREDATE
-----
CLARK          2450 09-JUN-81
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE RTRIM ( TO_CHAR(HIREDATE,'MONTH') ) ='JUNE'
SQL> /
```

```
ENAME          SAL HIREDATE
```

```
-----
CLARK                2450 09-JUN-81
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE RTRIM ( TO_CHAR(HIREDATE,'MONTH') ) ='DECEMBER'
SQL> /
```

```

ENAME                SAL HIREDATE
-----
FORD                  3000 03-DEC-81
SMITH                 800  17-DEC-80
JAMES                 950  03-DEC-81
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE RTRIM ( TO_CHAR(HIREDATE,'MONTH') ) ='MARCH'
SQL> /
```

```
no rows selected
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE ( TO_CHAR(HIREDATE,'MONTH') ) ='MARCH      '
SQL> /
```

```
no rows selected
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MONTH') ='MARCH      '
SQL> /
```

```
no rows selected
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MON') ='MARCH'
SQL> /
```

no rows selected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE   TO_CHAR(HIREDATE,'MON')   ='MAR'
SQL> /

```

no rows selected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4* FROM EMP
5  /

```

ENAME	SAL	HIREDATE
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE   TO_CHAR(HIREDATE,'MON')   ='MAY'
SQL> /

```

ENAME	SAL	HIREDATE
KRISHNA	2500	17-MAY-18
BLAKE	2850	01-MAY-81
ADAMS	1100	23-MAY-87

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME

```



```

2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'MONTH')  ='MAY'
SQL> /

```

no rows selected

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  FROM EMP
5* WHERE TO_CHAR(HIREDATE,'FMMONTH')  ='MAY'
SQL> /

```

ENAME	SAL	HIREDATE
KRISHNA	2500	17-MAY-18
BLAKE	2850	01-MAY-81
ADAMS	1100	23-MAY-87

```
SQL> CL SCR
```

```

SQL> SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'DD-MON-YYYY DAY-MONTH-Q') AS C2
3  FROM DUAL;

```

SD	C2
02-JUN-18	02-JUN-2018 SATURDAY -JUNE -2

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT SYSDATE AS SD
2  ,RTRIM ( TO_CHAR(SYSDATE,'DD-MON-YYYY DAY-MONTH-Q') ) AS C2
3* FROM DUAL
SQL> /

```

SD	C2
02-JUN-18	02-JUN-2018 SATURDAY -JUNE -2

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT SYSDATE AS SD
2  , TO_CHAR(SYSDATE,'FMDD-MON-YYYY DAY-MONTH-Q') AS C2
3* FROM DUAL
SQL> /

```

SD	C2
02-JUN-18	2-JUN-2018 SATURDAY-JUNE-2

```
SQL> CL SCR
```

```
SQL> R
```

```

1  SELECT SYSDATE AS SD
2  , TO_CHAR(SYSDATE, 'FMDD-MON-YYYY DAY-MONTH-Q') AS C2
3* FROM DUAL

```

```
SD          C2
```

```
-----
02-JUN-18 2-JUN-2018 SATURDAY-JUNE-2
```

```
SQL> CL SCR
```

```
SQL> CL SCR
```

```
SQL> SELECT SYSDATE AS SD
      2 ,TO_CHAR(SYSDATE, 'DDD DD D DAY DY MONTH MON MM HH24 MI SS YYYY YYY YY Y Q')
AS C2
      3 FROM DUAL;
,TO_CHAR(SYSDATE, 'DDD DD D DAY DY MONTH MON MM HH24 MI SS YYYY YYY YY Y Q') AS C2
      *
```

```
ERROR at line 2:
```

```
ORA-01801: date format is too long for internal buffer
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE, 'DDD DD D DAY DY MON MM HH24 MI SS YYYY YYY YY Y Q') AS C2
3* FROM DUAL
SQL> /

```

```
SD          C2
```

```
-----
02-JUN-18 153 02 7 SATURDAY SAT JUN 06 08 36 02 2018 018 18 8 2
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE, 'DDD DD D DAY DY MON MM HH24 MI SS YYYY YYY YY Y Q')
3  ||TO_CHAR(SYSDATE, 'MONTH') AS C2
4* FROM DUAL
SQL> /

```

```
SD          C2
```

```
-----
02-JUN-18 153 02 7 SATURDAY SAT JUN 06 08 36 24 2018 018 18 8 2JUNE
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE, 'DDD DD D DAY DY MON MM HH24 MI SS YYYY YYY YY Y Q') AS C2
3* FROM DUAL
SQL> /

```

```
SD          C2
```

```
-----
02-JUN-18 153 02 7 SATURDAY SAT JUN 06 08 36 33 2018 018 18 8 2
```

```
SQL> CL SCR
```

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE, 'DDD DD D DAY DY MON MM HH24 MI SS YYYY YY Y Q') AS C2
  3* FROM DUAL
```

```
SD          C2
```

```
-----
02-JUN-18 153 02 7 SATURDAY  SAT JUN 06 08 37 09 2018 018 18 8 2
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE, 'DDD DD D DAY DY MON MM HH24 MI SS YYYY YY Y Q') AS C2
  3* FROM DUAL
```

```
SQL>
```

```
SQL>
```

```
SQL>
```

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
```

```
  2  ,SAL
  3  ,HIREDATE
  4  FROM EMP;
```

ENAME	SAL	HIREDATE
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

```
15 rows selected.
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
  1  SELECT ENAME
  2  ,SAL
  3  ,HIREDATE
  4  ,SUBSTR(HIREDATE,-1)
  5* FROM EMP
```

```
SQL> /
```

ENAME	SAL	HIREDATE	SUBS
KRISHNA	2500	17-MAY-18	8
KING	5000	17-NOV-81	1
BLAKE	2850	01-MAY-81	1
CLARK	2450	09-JUN-81	1
JONES	2975	02-APR-81	1

```

SCOTT          3000 19-APR-87 7
FORD           3000 03-DEC-81 1
SMITH          800  17-DEC-80 0
ALLEN          1600 20-FEB-81 1
WARD           1250 22-FEB-81 1
MARTIN         1250 28-SEP-81 1
TURNER         1500 08-SEP-81 1
ADAMS          1100 23-MAY-87 7
JAMES          950  03-DEC-81 1
MILLER         1300 23-JAN-82 2

```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,SUBSTR(HIREDATE,-1)
5  FROM EMP
6* WHERE MOD(SUBSTR(HIREDATE,-1),2) =0
SQL> /

```

ENAME	SAL	HIREDATE	SUBS
KRISHNA	2500	17-MAY-18	8
SMITH	800	17-DEC-80	0
MILLER	1300	23-JAN-82	2

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,SUBSTR(HIREDATE,-1)
5  FROM EMP
6* WHERE MOD ( TO_CHAR(HIREDATE,'Y') ,2)=0
SQL> /

```

ENAME	SAL	HIREDATE	SUBS
KRISHNA	2500	17-MAY-18	8
SMITH	800	17-DEC-80	0
MILLER	1300	23-JAN-82	2

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4* FROM EMP
5  /

```

ENAME	SAL	HIREDATE
KRISHNA	2500	17-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87

FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

15 rows selected.

SQL> CL SCR

```
SQL> SELECT SYSDATE AS SD
       2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
       3  FROM DUAL;
```

SD	C1
-----	-----
02-JUN-18	31195

```
SQL> R
     1  SELECT SYSDATE AS SD
     2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
     3* FROM DUAL
```

SD	C1
-----	-----
02-JUN-18	31197

```
SQL> R
     1  SELECT SYSDATE AS SD
     2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
     3* FROM DUAL
```

SD	C1
-----	-----
02-JUN-18	31199

```
SQL> R
     1  SELECT SYSDATE AS SD
     2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
     3* FROM DUAL
```

SD	C1
-----	-----
02-JUN-18	86359

SQL> CL SCR

```
SQL> R
     1  SELECT SYSDATE AS SD
     2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
     3* FROM DUAL
```

SD	C1
-----	-----
02-JUN-18	86361

```
SQL> R
     1  SELECT SYSDATE AS SD
     2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
     3* FROM DUAL
```

```
SD          C1
-----
02-JUN-18  86362
```

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

```
SD          C1
-----
02-JUN-18  86362
```

```
SQL> CL SCR
```

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

```
SD          C1
-----
02-JUN-18  86364
```

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

```
SD          C1
-----
02-JUN-18  86388
```

```
SQL> CLS CR
SP2-0042: unknown command "CLS CR" - rest of line ignored.
SQL> CLS CR
SP2-0042: unknown command "CLS CR" - rest of line ignored.
SQL> CL SCR
```

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

```
SD          C1
-----
02-JUN-18  86394
```

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

```
SD          C1
-----
02-JUN-18  86395
```

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

```
SD          C1
```

02-JUN-18 86396

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

SD C1

02-JUN-18 86397

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

SD C1

02-JUN-18 86398

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

SD C1

02-JUN-18 86399

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

SD C1

03-JUN-18 00000

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

SD C1

03-JUN-18 00009

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

SD C1

03-JUN-18 00009

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSSS') AS C1
  3* FROM DUAL
```

SD C1

03-JUN-18 00010

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SSSS') AS C1
  3* FROM DUAL
SQL> /
```

```
SD          C1
-----
03-JUN-18 3535
```

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SS SS') AS C1
  3* FROM DUAL
SQL> /
```

```
SD          C1
-----
03-JUN-18 47 47
```

SQL> R

```
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SS SS') AS C1
  3* FROM DUAL
```

```
SD          C1
-----
03-JUN-18 48 48
```

SQL> R

```
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SS SS') AS C1
  3* FROM DUAL
```

```
SD          C1
-----
03-JUN-18 48 48
```

SQL> R

```
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'SS SS') AS C1
  3* FROM DUAL
```

```
SD          C1
-----
03-JUN-18 49 49
```

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'DDDDDD') AS C1
  3* FROM DUAL
SQL> /
```

```
SD          C1
-----
03-JUN-18 154154
```



```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'DDDDDD') AS C1
  3* FROM DUAL
```

```
SD          C1
-----
03-JUN-18 154154
```

```
SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'DDDDDD') AS C1
  3* FROM DUAL
```

```
SD          C1
-----
03-JUN-18 154154
```

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'DDD DD D') AS C1
  3* FROM DUAL
SQL> /
```

```
SD          C1
-----
03-JUN-18 154 03 1
```

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'DDD') AS C1
  3* FROM DUAL
SQL> /
```

```
SD          C1
-----
03-JUN-18 154
```

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'DDD')||TO_CHAR(SYSDATE,'DD') AS C1
  3* FROM DUAL
SQL> /
```

```
SD          C1
-----
03-JUN-18 15403
```

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'DDD')||TO_CHAR(SYSDATE,'DD')||TO_CHAR(SYSDATE,'D') AS C1
  3* FROM DUAL
SQL> /
```

```
SD          C1
```

```
-----
03-JUN-18 154031

SQL> CL SCR

SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'DDD')||TO_CHAR(SYSDATE,'DD')||TO_CHAR(SYSDATE,'D') AS C1
  3* FROM DUAL

SD          C1
-----
03-JUN-18 154031

SQL> ED
Wrote file afiedt.buf

  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'DDDDDD') AS C2
  3* FROM DUAL
SQL> /

SD          C2
-----
03-JUN-18 154154

SQL> ED
Wrote file afiedt.buf

  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'DDD')||TO_CHAR(SYSDATE,'DD')||TO_CHAR(SYSDATE,'D') AS C1 AS
C2
  3* FROM DUAL
SQL> /
,TO_CHAR(SYSDATE,'DDD')||TO_CHAR(SYSDATE,'DD')||TO_CHAR(SYSDATE,'D') AS C1 AS C2
                                                                *

ERROR at line 2:
ORA-00923: FROM keyword not found where expected

SQL> ED
Wrote file afiedt.buf

  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'DDD')||TO_CHAR(SYSDATE,'DD')||TO_CHAR(SYSDATE,'D') AS C1
  3* FROM DUAL
SQL> /

SD          C1
-----
03-JUN-18 154031

SQL> CL SCR

SQL> R
  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'DDD')||TO_CHAR(SYSDATE,'DD')||TO_CHAR(SYSDATE,'D') AS C1
  3* FROM DUAL

SD          C1
-----
03-JUN-18 154031

SQL> ED
```

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'MONTH-DAY-Q') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
03-JUN-18 JUNE      -SUNDAY      -2

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'MONTH-DAY-QFM') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
03-JUN-18 JUNE      -SUNDAY      -2

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'MONTH-DAYFM-Q') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
03-JUN-18 JUNE      -SUNDAY      -2

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'MONTHFM-DAY-Q') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
03-JUN-18 JUNE      -SUNDAY-2

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'MONTH-FMDAY-Q') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
03-JUN-18 JUNE      -SUNDAY-2

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'FMMONTH-FMDAY-Q') AS C1

```

```
3* FROM DUAL
SQL> /
```

```
SD          C1
-----
03-JUN-18  JUNE-SUNDAY    -2
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'FMMONTH-DAY-Q') AS C1
3* FROM DUAL
SQL> /
```

```
SD          C1
-----
03-JUN-18  JUNE-SUNDAY-2
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'FMMONTH-FMDAY-Q') AS C1
3* FROM DUAL
SQL> /
```

```
SD          C1
-----
03-JUN-18  JUNE-SUNDAY    -2
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'FMMONTH-DAY-Q') AS C1
3* FROM DUAL
SQL> /
```

```
SD          C1
-----
03-JUN-18  JUNE-SUNDAY-2
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'FMMONTH-DAY-Q') AS C1
3* FROM DUAL
SQL> /
```

```
SD          C1
-----
03-JUN-18  JUNE-SUNDAY-2
```

```
SQL> CL SCR
```

```
SQL> R
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'FMMONTH-DAY-Q') AS C1
3* FROM DUAL
```

```
SD          C1
-----
```

02-JUN-18 JUNE-SATURDAY-2

```
SQL> SELECT SYSDATE AS SD
      2 ,TO_CHAR(SYSDATE,'PM') AS C1
      3 FROM DUAL;
```

```
SD          C1
-----
```

02-JUN-18 AM

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT SYSDATE AS SD
2* FROM DUAL
3 /
```

```
SD
-----
```

02-JUN-18

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT SYSDATE AS SD
2 ,TO_CHAR(SYSDATE,'DD-MON-YY:PM') AS C1
3* FROM DUAL
SQL> /
```

```
SD          C1
-----
```

02-JUN-18 02-JUN-18:AM

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT SYSDATE AS SD
2 ,TO_CHAR(SYSDATE,'DD-MON-YY:P.M.') AS C1
3* FROM DUAL
SQL> /
```

```
SD          C1
-----
```

02-JUN-18 02-JUN-18:A.M.

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT SYSDATE AS SD
2 ,TO_CHAR(SYSDATE,'DD-MON-YY:P.M') AS C1
3* FROM DUAL
```

```
SQL> /
,TO_CHAR(SYSDATE,'DD-MON-YY:P.M') AS C1
      *
```

```
ERROR at line 2:
ORA-01821: date format not recognized
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT SYSDATE AS SD
2 ,TO_CHAR(SYSDATE,'DD-MON-YY:AM') AS C1
3* FROM DUAL
```

```
SQL> /
```

```
SD          C1
-----
02-JUN-18  02-JUN-18:AM
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'DD-MON-YY:PM') AS C1
3* FROM DUAL
```

```
SQL> /
```

```
SD          C1
-----
02-JUN-18  02-JUN-18:AM
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'DD-MON-YY') AS C1
3* FROM DUAL
```

```
SQL> /
```

```
SD          C1
-----
02-JUN-18  02-JUN-18
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'DD-MON-YY AM') AS C1
3* FROM DUAL
```

```
SQL> /
```

```
SD          C1
-----
02-JUN-18  02-JUN-18 AM
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'DD-MON-YY CC') AS C1
3* FROM DUAL
```

```
SQL> /
```

```
SD          C1
-----
02-JUN-18  02-JUN-18 21
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'CC') AS C1
3* FROM DUAL
```

```
SQL> /
```

```
SD          C1
-----
```

02-JUN-18 21

SQL> ED

Wrote file afiedt.buf

```
 1  SELECT SYSDATE AS SD
 2  ,TO_CHAR(SYSDATE,'SCC') AS C1
 3* FROM DUAL
SQL> /
```

```
SD          C1
-----
```

02-JUN-18 21

SQL> ED

Wrote file afiedt.buf

```
 1  SELECT SYSDATE AS SD
 2  ,TO_CHAR(SYSDATE,'AC') AS C1
 3* FROM DUAL
SQL> /
```

```
,TO_CHAR(SYSDATE,'AC') AS C1
                        *
```

ERROR at line 2:

ORA-01821: date format not recognized

SQL> ED

Wrote file afiedt.buf

```
 1  SELECT SYSDATE AS SD
 2  ,TO_CHAR(SYSDATE,'AD') AS C1
 3* FROM DUAL
SQL> /
```

```
SD          C1
-----
```

02-JUN-18 AD

SQL> ED

Wrote file afiedt.buf

```
 1  SELECT SYSDATE AS SD
 2  ,TO_CHAR(SYSDATE,'BC') AS C1
 3* FROM DUAL
SQL> /
```

```
SD          C1
-----
```

02-JUN-18 AD

SQL> ED

Wrote file afiedt.buf

```
 1  SELECT SYSDATE AS SD
 2  ,TO_CHAR(SYSDATE,'AD') AS C1
 3* FROM DUAL
SQL> /
```

```
SD          C1
-----
```

02-JUN-18 AD

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'BC') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
02-JUN-18 AD

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'B.C.') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
02-JUN-18 A.D.

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'YEAR') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
02-JUN-18 TWENTY EIGHTEEN

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'YYYYSP') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
02-JUN-18 TWO THOUSAND EIGHTEEN

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'QSP') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
02-JUN-18 TWO

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'DDD') AS C1

```



```
3* FROM DUAL
SQL> /
```

```
SD          C1
-----
02-JUN-18 153
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'DDDSP') AS C1
3* FROM DUAL
SQL> /
```

```
SD          C1
-----
02-JUN-18 ONE HUNDRED FIFTY-THREE
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'DDD DDDSP') AS C1
3* FROM DUAL
SQL> /
```

```
SD          C1
-----
02-JUN-18 153 ONE HUNDRED FIFTY-THREE
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE,'D') AS C1
3* FROM DUAL
SQL> /
```

```
SD          C1
-----
02-JUN-18 7
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE -1 AS SD
2  ,TO_CHAR(SYSDATE -1 , 'DD') AS C1
3* FROM DUAL
SQL> /
```

```
SD          C1
-----
01-JUN-18 01
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE AS SD
2  ,TO_CHAR(SYSDATE , 'DD') AS C1
3* FROM DUAL
SQL> /
```

```
SD          C1
```

```
-----  
02-JUN-18 02
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE  AS SD  
2  ,TO_CHAR(SYSDATE , 'DDTH') AS C1  
3* FROM DUAL  
SQL> /
```

```
SD          C1  
-----  
02-JUN-18 02ND
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE -1  AS SD  
2  ,TO_CHAR(SYSDATE -1 , 'DDTH') AS C1  
3* FROM DUAL  
SQL> /
```

```
SD          C1  
-----  
01-JUN-18 01ST
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE +1  AS SD  
2  ,TO_CHAR(SYSDATE + 1 , 'DDTH') AS C1  
3* FROM DUAL  
SQL> /
```

```
SD          C1  
-----  
03-JUN-18 03RD
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE +2  AS SD  
2  ,TO_CHAR(SYSDATE + 2 , 'DDTH') AS C1  
3* FROM DUAL  
SQL> /
```

```
SD          C1  
-----  
04-JUN-18 04TH
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE +2  AS SD  
2  ,TO_CHAR(SYSDATE + 2 , 'DDSP') AS C1  
3* FROM DUAL  
SQL> /
```

```
SD          C1  
-----  
04-JUN-18 FOUR
```

```
SQL> ED
```

Wrote file afiedt.buf

```

1  SELECT SYSDATE +2  AS SD
2  ,TO_CHAR(SYSDATE + 2  ,'DDSPTH') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
04-JUN-18  FOURTH

```

SQL> ED
Wrote file afiedt.buf

```

1  SELECT SYSDATE  AS SD
2  ,TO_CHAR(SYSDATE  ,'DDSPTH') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
02-JUN-18  SECOND

```

SQL> ED
Wrote file afiedt.buf

```

1  SELECT SYSDATE  AS SD
2  ,TO_CHAR(SYSDATE  ,'DDSP') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
02-JUN-18  TWO

```

SQL> ED
Wrote file afiedt.buf

```

1  SELECT SYSDATE  AS SD
2  ,TO_CHAR(SYSDATE  ,'DDSPTH') AS C1
3* FROM DUAL
SQL> /

```

```

SD          C1
-----
02-JUN-18  SECOND

```

SQL> CL SCR

```

SQL> R
1  SELECT SYSDATE  AS SD
2  ,TO_CHAR(SYSDATE  ,'DDSPTH') AS C1
3* FROM DUAL

```

```

SD          C1
-----
02-JUN-18  SECOND

```

SQL> ED
Wrote file afiedt.buf

```

1  SELECT SYSDATE  AS SD
2  ,TO_CHAR(SYSDATE  ,'RRRR') AS C1
3* FROM DUAL

```

```
SQL> /
```

```
SD          C1
```

```
-----  
02-JUN-18 2018
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE    AS SD  
2  ,TO_CHAR(SYSDATE  ,'RR') AS C1  
3* FROM DUAL
```

```
SQL> /
```

```
SD          C1
```

```
-----  
02-JUN-18 18
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE    AS SD  
2  ,TO_CHAR(SYSDATE  ,'R') AS C1  
3* FROM DUAL
```

```
SQL> /
```

```
,TO_CHAR(SYSDATE  ,'R') AS C1  
      *
```

```
ERROR at line 2:
```

```
ORA-01821: date format not recognized
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT SYSDATE    AS SD  
2  ,TO_CHAR(SYSDATE  ,'RRR') AS C1  
3* FROM DUAL
```

```
SQL> /
```

```
,TO_CHAR(SYSDATE  ,'RRR') AS C1  
      *
```

```
ERROR at line 2:
```

```
ORA-01821: date format not recognized
```

```
SQL> CL SCR
```

```
SQL> SELECT TO_DATE(1,'J') AS C1  
2  FROM DUAL;
```

```
C1
```

```
-----  
01-JAN-12
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT TO_CHAR(TO_DATE(1,'J'),'DD-MON-YYYY') AS C1  
2* FROM DUAL
```

```
SQL> /
```

```
C1
```

```
-----  
01-JAN-4712
```

```
SQL> ED
Wrote file afiedt.buf

  1  SELECT TO_CHAR(TO_DATE(1,'J'),'DD-MON-SYYYY') AS C1
  2* FROM DUAL
SQL> /

C1
-----
01-JAN--4712

SQL> ED
Wrote file afiedt.buf

  1  SELECT TO_CHAR(TO_DATE(1,'J'),'DD-MON-SYYYY AD') AS C1
  2* FROM DUAL
SQL> /

C1
-----
01-JAN--4712 BC

SQL> ED
Wrote file afiedt.buf

  1  SELECT SYSDATE AS SD
  2  ,TO_CHAR(SYSDATE,'J') AS C1
  3* FROM DUAL
SQL> /

SD          C1
-----
02-JUN-18  2458272

SQL> CL SCR

SQL> SELECT TO_CHAR( TO_DATE('08-JUL-1987'),'J') AS C1
  2  FROM DUAL;

C1
-----
2446985

SQL> SELECT 2458272 - 2446985 AS C1
  2  FROM DUAL;

          C1
-----
#####

SQL> ED
Wrote file afiedt.buf

  1  SELECT 2458272 - 2446985 AS C2
  2* FROM DUAL
SQL> /

          C2
-----
11287

SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT (2458272 - 2446985)/365 AS C2
2* FROM DUAL
SQL> /

```

```

      C2
-----
30.9232877

```

```
SQL> CL SCR
```

```

SQL> SELECT ENAME
2      ,SAL
3      ,HIREDATE
4      ,TO_CHAR(HIREDATE,'YYYY-MON') AS C1
5      ,TO_CHAR(HIREDATE,'DDD DD D') AS C2
6      ,TO_CHAR(HIREDATE,'Q HH24:MI:SS') AS C3
7  FROM EMP;

```

ENAME	SAL	HIREDATE	C1	C2	C3
KRISHNA	2500	17-MAY-18	2018-MAY	137 17 5 2	00:00:00
KING	5000	17-NOV-81	1981-NOV	321 17 3 4	00:00:00
BLAKE	2850	01-MAY-81	1981-MAY	121 01 6 2	00:00:00
CLARK	2450	09-JUN-81	1981-JUN	160 09 3 2	00:00:00
JONES	2975	02-APR-81	1981-APR	092 02 5 2	00:00:00
SCOTT	3000	19-APR-87	1987-APR	109 19 1 2	00:00:00
FORD	3000	03-DEC-81	1981-DEC	337 03 5 4	00:00:00
SMITH	800	17-DEC-80	1980-DEC	352 17 4 4	00:00:00
ALLEN	1600	20-FEB-81	1981-FEB	051 20 6 1	00:00:00
WARD	1250	22-FEB-81	1981-FEB	053 22 1 1	00:00:00
MARTIN	1250	28-SEP-81	1981-SEP	271 28 2 3	00:00:00
TURNER	1500	08-SEP-81	1981-SEP	251 08 3 3	00:00:00
ADAMS	1100	23-MAY-87	1987-MAY	143 23 7 2	00:00:00
JAMES	950	03-DEC-81	1981-DEC	337 03 5 4	00:00:00
MILLER	1300	23-JAN-82	1982-JAN	023 23 7 1	00:00:00

```
15 rows selected.
```

```
SQL> CL SCR
```

```

SQL> R
1  SELECT ENAME
2      ,SAL
3      ,HIREDATE
4      ,TO_CHAR(HIREDATE,'YYYY-MON') AS C1
5      ,TO_CHAR(HIREDATE,'DDD DD D') AS C2
6      ,TO_CHAR(HIREDATE,'Q HH24:MI:SS') AS C3
7* FROM EMP

```

ENAME	SAL	HIREDATE	C1	C2	C3
KRISHNA	2500	17-MAY-18	2018-MAY	137 17 5 2	00:00:00
KING	5000	17-NOV-81	1981-NOV	321 17 3 4	00:00:00
BLAKE	2850	01-MAY-81	1981-MAY	121 01 6 2	00:00:00
CLARK	2450	09-JUN-81	1981-JUN	160 09 3 2	00:00:00
JONES	2975	02-APR-81	1981-APR	092 02 5 2	00:00:00
SCOTT	3000	19-APR-87	1987-APR	109 19 1 2	00:00:00
FORD	3000	03-DEC-81	1981-DEC	337 03 5 4	00:00:00
SMITH	800	17-DEC-80	1980-DEC	352 17 4 4	00:00:00
ALLEN	1600	20-FEB-81	1981-FEB	051 20 6 1	00:00:00
WARD	1250	22-FEB-81	1981-FEB	053 22 1 1	00:00:00
MARTIN	1250	28-SEP-81	1981-SEP	271 28 2 3	00:00:00
TURNER	1500	08-SEP-81	1981-SEP	251 08 3 3	00:00:00
ADAMS	1100	23-MAY-87	1987-MAY	143 23 7 2	00:00:00

```
JAMES          950 03-DEC-81 1981-DEC          337 03 5 4 00:00:00
MILLER         1300 23-JAN-82 1982-JAN          023 23 7 1 00:00:00
```

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,TO_CHAR(HIREDATE,'YYYY-MON') AS C1
5  ,TO_CHAR(HIREDATE,'DDD DD D') AS C2
6  ,TO_CHAR(HIREDATE,'Q HH24:MI:SS') AS C3
7* FROM EMP
```

SQL>

SQL>

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CL SCR

SQL> R

```
1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,TO_CHAR(HIREDATE,'YYYY-MON') AS C1
5  ,TO_CHAR(HIREDATE,'DDD DD D') AS C2
6  ,TO_CHAR(HIREDATE,'Q HH24:MI:SS') AS C3
7* FROM EMP
```

ENAME	SAL	HIREDATE	C1	C2	C3
KRISHNA	2500	17-MAY-18	2018-MAY	137 17 5 2	00:00:00
KING	5000	17-NOV-81	1981-NOV	321 17 3 4	00:00:00
BLAKE	2850	01-MAY-81	1981-MAY	121 01 6 2	00:00:00
CLARK	2450	09-JUN-81	1981-JUN	160 09 3 2	00:00:00
JONES	2975	02-APR-81	1981-APR	092 02 5 2	00:00:00
SCOTT	3000	19-APR-87	1987-APR	109 19 1 2	00:00:00
FORD	3000	03-DEC-81	1981-DEC	337 03 5 4	00:00:00
SMITH	800	17-DEC-80	1980-DEC	352 17 4 4	00:00:00
ALLEN	1600	20-FEB-81	1981-FEB	051 20 6 1	00:00:00
WARD	1250	22-FEB-81	1981-FEB	053 22 1 1	00:00:00
MARTIN	1250	28-SEP-81	1981-SEP	271 28 2 3	00:00:00
TURNER	1500	08-SEP-81	1981-SEP	251 08 3 3	00:00:00
ADAMS	1100	23-MAY-87	1987-MAY	143 23 7 2	00:00:00
JAMES	950	03-DEC-81	1981-DEC	337 03 5 4	00:00:00
MILLER	1300	23-JAN-82	1982-JAN	023 23 7 1	00:00:00

15 rows selected.

SQL> INSERT INTO EMP (EMPNO, ENAME, SAL, HIREDATE)

```
2  VALUES
3  (
4  101, 'ARJUN'
5  ,2000
6  , '02-JUN-18'
7  );
```

1 row created.

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO EMP (EMPNO, ENAME, SAL, HIREDATE)
```

```

2  VALUES
3  (
4  102,'SURESH'
5  ,2100
6  ,SYSDATE
7* )
SQL> /

```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> EDIT
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,TO_CHAR(HIREDATE,'YYYY-MON') AS C1
5  ,TO_CHAR(HIREDATE,'DDD DD D') AS C2
6  ,TO_CHAR(HIREDATE,'Q HH24:MI:SS') AS C3
7* FROM EMP
SQL> /

```

ENAME	SAL	HIREDATE	C1	C2	C3
SURESH	2100	02-JUN-18	2018-JUN	153 02 7 2	09:04:07
KRISHNA	2500	17-MAY-18	2018-MAY	137 17 5 2	00:00:00
ARJUN	2000	02-JUN-18	2018-JUN	153 02 7 2	00:00:00
KING	5000	17-NOV-81	1981-NOV	321 17 3 4	00:00:00
BLAKE	2850	01-MAY-81	1981-MAY	121 01 6 2	00:00:00
CLARK	2450	09-JUN-81	1981-JUN	160 09 3 2	00:00:00
JONES	2975	02-APR-81	1981-APR	092 02 5 2	00:00:00
SCOTT	3000	19-APR-87	1987-APR	109 19 1 2	00:00:00
FORD	3000	03-DEC-81	1981-DEC	337 03 5 4	00:00:00
SMITH	800	17-DEC-80	1980-DEC	352 17 4 4	00:00:00
ALLEN	1600	20-FEB-81	1981-FEB	051 20 6 1	00:00:00
WARD	1250	22-FEB-81	1981-FEB	053 22 1 1	00:00:00
MARTIN	1250	28-SEP-81	1981-SEP	271 28 2 3	00:00:00
TURNER	1500	08-SEP-81	1981-SEP	251 08 3 3	00:00:00
ADAMS	1100	23-MAY-87	1987-MAY	143 23 7 2	00:00:00
JAMES	950	03-DEC-81	1981-DEC	337 03 5 4	00:00:00
MILLER	1300	23-JAN-82	1982-JAN	023 23 7 1	00:00:00

17 rows selected.

```
SQL> CL SCR
```

```

SQL> INSERT INTO EMP
2  (
3  EMPNO
4  ,ENAME
5  ,SAL
6  ,HIREDATE
7  )
8  VALUES
9  (
10 105
11 , 'HIMANSHU'
12 ,2300
13 ,06012018105541
14 );

```



```
,06012018105541
```

```
*
```

```
ERROR at line 13:
```

```
ORA-00932: inconsistent datatypes: expected DATE got NUMBER
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  INSERT INTO EMP
2  (
3  EMPNO
4  ,ENAME
5  ,SAL
6  ,HIREDATE
7  )
8  VALUES
9  (
10 105
11 , 'HIMANSHU'
12 ,2300
13 ,TO_DATE('06012018105541'
14           ,MMDDYYYYHH24MISS'
15           )
16* )
SQL> /
```

```
ERROR:
```

```
ORA-01756: quoted string not properly terminated
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  INSERT INTO EMP
2  (
3  EMPNO
4  ,ENAME
5  ,SAL
6  ,HIREDATE
7  )
8  VALUES
9  (
10 105
11 , 'HIMANSHU'
12 ,2300
13 ,TO_DATE('06012018105541'
14           , 'MMDDYYYYHH24MISS'
15           )
16* )
SQL> /
```

```
1 row created.
```

```
SQL> SELECT ENAME
```

```
2 ,SAL
3 ,HIREDATE
4 FROM EMP;
```

ENAME	SAL	HIREDATE
SURESH	2100	02-JUN-18
HIMANSHU	2300	01-JUN-18
KRISHNA	2500	17-MAY-18
ARJUN	2000	02-JUN-18

```

KING          5000 17-NOV-81
BLAKE         2850 01-MAY-81
CLARK         2450 09-JUN-81
JONES         2975 02-APR-81
SCOTT         3000 19-APR-87
FORD          3000 03-DEC-81
SMITH         800  17-DEC-80
ALLEN         1600 20-FEB-81
WARD          1250 22-FEB-81
MARTIN        1250 28-SEP-81
TURNER        1500 08-SEP-81
ADAMS         1100 23-MAY-87
JAMES         950  03-DEC-81
MILLER        1300 23-JAN-82

```

18 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2  ,SAL
 3  ,TO_CHAR(HIREDATE,'DD-MON-YYYY HH24 MI SS ') AS C1
 4* FROM EMP
SQL> /

```

ENAME	SAL	C1
SURESH	2100	02-JUN-2018 09 04 07
HIMANSHU	2300	01-JUN-2018 10 55 41
KRISHNA	2500	17-MAY-2018 00 00 00
ARJUN	2000	02-JUN-2018 00 00 00
KING	5000	17-NOV-1981 00 00 00
BLAKE	2850	01-MAY-1981 00 00 00
CLARK	2450	09-JUN-1981 00 00 00
JONES	2975	02-APR-1981 00 00 00
SCOTT	3000	19-APR-1987 00 00 00
FORD	3000	03-DEC-1981 00 00 00
SMITH	800	17-DEC-1980 00 00 00
ALLEN	1600	20-FEB-1981 00 00 00
WARD	1250	22-FEB-1981 00 00 00
MARTIN	1250	28-SEP-1981 00 00 00
TURNER	1500	08-SEP-1981 00 00 00
ADAMS	1100	23-MAY-1987 00 00 00
JAMES	950	03-DEC-1981 00 00 00
MILLER	1300	23-JAN-1982 00 00 00

18 rows selected.

SQL> CL SCR

```

SQL> SELECT TO_CHAR( TO_DATE(19900602,'YYYYMMDD'),'J') AS C1
 2  FROM DUAL;

```

```

C1
-----
2448045

```

```

SQL> SELECT ENAME
 2  ,SAL
 3  ,HIREDATE
 4  ,TO_CHAR(HIREDATE,'J') AS C1
 5  FROM DUAL;
,TO_CHAR(HIREDATE,'J') AS C1

```

*

ERROR at line 4:
ORA-00904: "HIREDATE": invalid identifier

SQL> ED
Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,HIREDATE
4  ,TO_CHAR(HIREDATE,'J') AS C1
5* FROM EMP
SQL> /

```

ENAME	SAL	HIREDATE	C1
SURESH	2100	02-JUN-18	2458272
HIMANSHU	2300	01-JUN-18	2458271
KRISHNA	2500	17-MAY-18	2458256
ARJUN	2000	02-JUN-18	2458272
KING	5000	17-NOV-81	2444926
BLAKE	2850	01-MAY-81	2444726
CLARK	2450	09-JUN-81	2444765
JONES	2975	02-APR-81	2444697
SCOTT	3000	19-APR-87	2446905
FORD	3000	03-DEC-81	2444942
SMITH	800	17-DEC-80	2444591
ALLEN	1600	20-FEB-81	2444656
WARD	1250	22-FEB-81	2444658
MARTIN	1250	28-SEP-81	2444876
TURNER	1500	08-SEP-81	2444856
ADAMS	1100	23-MAY-87	2446939
JAMES	950	03-DEC-81	2444942
MILLER	1300	23-JAN-82	2444993

18 rows selected.

03-jun-2018

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '9,999.99'
5  ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00

SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

18 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4           , '9,999.99'
5           ) AS SAL1
6* FROM EMP
SQL>
SQL> INSERT INTO EMP(EMPNO,ENAME,SAL)
2  VALUES
3  (
4  110,'SAI',25540);

```

1 row created.

SQL> COMMIT;

Commit complete.

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4           , '9,999.99'
5           ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
-----	-----	-----
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	#####
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '99,999.99'
5  ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	25,540.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

SQL> DESCRIBE EMP

Name

Null? Type

```

-----
-----
-----
EMPNO
NUMBER(4)
ENAME
VARCHAR2(30)
JOB
VARCHAR2(9)
MGR
NUMBER(4)
HIREDATE
DATE
SAL
NUMBER(8,2)
COMM
NUMBER(7,2)
DEPTNO
NUMBER(2)

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4      ,',99,99,999,99,999.99'
5      ) AS SAL1
6* FROM EMP
SQL> /
      ,',99,99,999,99,999.99'
      *
ERROR at line 4:
ORA-01481: invalid number format model

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4      ,',99,99,999,99,999.99'
5      ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	25,540.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4      ,',9,999.99'
5      ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00

ARJUN	2000	2,000.00
SAI	25540	#####
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '99,999.99'
5  ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	25,540.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '99,99,999.99'
5  ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	25,540.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

SQL> DESC EMP

Name	Null?	Type
EMPNO		NUMBER(4)
ENAME		VARCHAR2(30)
JOB		VARCHAR2(9)
MGR		NUMBER(4)
HIREDATE		DATE
SAL		NUMBER(8,2)
COMM		NUMBER(7,2)
DEPTNO		NUMBER(2)

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '99,99,999.99'
5  ) AS SAL1
6* FROM EMP

```

SQL>

SQL> /

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00

KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	25,540.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4           , '9,99,999.99'
5           ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	25,540.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4           , '99,999.99'
5           ) AS SAL1
6* FROM EMP

```

SQL> /

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	25,540.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '9,999.99'
5  ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	#####
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

SQL> DESC EMP

Name	Type
Null?	

```

-----
-----
EMPNO
NUMBER(4)
ENAME
VARCHAR2(30)
JOB
VARCHAR2(9)
MGR
NUMBER(4)
HIREDATE
DATE
SAL
NUMBER(8,2)
COMM
NUMBER(7,2)
DEPTNO
NUMBER(2)

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '99999,999.99'
5  ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	25,540.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '99,999.99'
5  ) AS SAL1
6* FROM EMP

```

SQL> /

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	25,540.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '9,9,9,9,9.99'
5  ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,1,0,0.00
HIMANSHU	2300	2,3,0,0.00
KRISHNA	2500	2,5,0,0.00
ARJUN	2000	2,0,0,0.00
SAI	25540	2,5,5,4,0.00
KING	5000	5,0,0,0.00
BLAKE	2850	2,8,5,0.00
CLARK	2450	2,4,5,0.00
JONES	2975	2,9,7,5.00
SCOTT	3000	3,0,0,0.00
FORD	3000	3,0,0,0.00
SMITH	800	8,0,0,0.00
ALLEN	1600	1,6,0,0.00
WARD	1250	1,2,5,0.00
MARTIN	1250	1,2,5,0.00
TURNER	1500	1,5,0,0.00
ADAMS	1100	1,1,0,0.00
JAMES	950	9,5,0,0.00
MILLER	1300	1,3,0,0.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4      , '9,9,9,9,9,9.9.9'
5      ) AS SAL1
6* FROM EMP
SQL> /

```

```

      , '9,9,9,9,9,9.9.9'
      *

```

ERROR at line 4:
ORA-01481: invalid number format model

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4      , '99,999.9.9'
5      ) AS SAL1
6* FROM EMP
SQL> /

```

```

      , '99,999.9.9'
      *

```

ERROR at line 4:
ORA-01481: invalid number format model

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4      , '99,999.99'
5      ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	25,540.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4           , '99G999D99'
5           ) AS SAL1
6* FROM EMP
SQL> /
```

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	25,540.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4           , '9G9G9G9G9D99'
5           ) AS SAL1
6* FROM EMP
SQL> /
```

ENAME	SAL	SAL1
SURESH	2100	2,1,0,0.00
HIMANSHU	2300	2,3,0,0.00
KRISHNA	2500	2,5,0,0.00
ARJUN	2000	2,0,0,0.00
SAI	25540	2,5,5,4,0.00
KING	5000	5,0,0,0.00
BLAKE	2850	2,8,5,0.00
CLARK	2450	2,4,5,0.00
JONES	2975	2,9,7,5.00
SCOTT	3000	3,0,0,0.00
FORD	3000	3,0,0,0.00
SMITH	800	8,0,0,0.00
ALLEN	1600	1,6,0,0.00
WARD	1250	1,2,5,0.00
MARTIN	1250	1,2,5,0.00
TURNER	1500	1,5,0,0.00

```

ADAMS          1100      1,1,0,0.00
JAMES          950       9,5,0.00
MILLER         1300      1,3,0,0.00

```

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '9G9G9G9G9D9D9'
5  ) AS SAL1
6* FROM EMP
SQL> /

```

```

      , '9G9G9G9G9D9D9'
      *

```

ERROR at line 4:

ORA-01481: invalid number format model

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '9G9G9G9G9D99'
5  ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,1,0,0.00
HIMANSHU	2300	2,3,0,0.00
KRISHNA	2500	2,5,0,0.00
ARJUN	2000	2,0,0,0.00
SAI	25540	2,5,5,4,0.00
KING	5000	5,0,0,0.00
BLAKE	2850	2,8,5,0.00
CLARK	2450	2,4,5,0.00
JONES	2975	2,9,7,5.00
SCOTT	3000	3,0,0,0.00
FORD	3000	3,0,0,0.00
SMITH	800	8,0,0,0.00
ALLEN	1600	1,6,0,0.00
WARD	1250	1,2,5,0.00
MARTIN	1250	1,2,5,0.00
TURNER	1500	1,5,0,0.00
ADAMS	1100	1,1,0,0.00
JAMES	950	9,5,0.00
MILLER	1300	1,3,0,0.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '99G999.99'

```

```

5          ) AS SAL1
6* FROM EMP
SQL> /

```

```

      , '99G999.99'
      *

```

ERROR at line 4:

ORA-01481: invalid number format model

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '00G000.00'
5  ) AS SAL1
6* FROM EMP
SQL> /

```

```

      , '00G000.00'
      *

```

ERROR at line 4:

ORA-01481: invalid number format model

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,TO_CHAR(SAL
4  , '00G000D00'
5  ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	02,100.00
HIMANSHU	2300	02,300.00
KRISHNA	2500	02,500.00
ARJUN	2000	02,000.00
SAI	25540	25,540.00
KING	5000	05,000.00
BLAKE	2850	02,850.00
CLARK	2450	02,450.00
JONES	2975	02,975.00
SCOTT	3000	03,000.00
FORD	3000	03,000.00
SMITH	800	00,800.00
ALLEN	1600	01,600.00
WARD	1250	01,250.00
MARTIN	1250	01,250.00
TURNER	1500	01,500.00
ADAMS	1100	01,100.00
JAMES	950	00,950.00
MILLER	1300	01,300.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME

```



```

2  ,SAL
3  ,TO_CHAR(SAL
4      ,'99G999D99'
5      ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	2,100.00
HIMANSHU	2300	2,300.00
KRISHNA	2500	2,500.00
ARJUN	2000	2,000.00
SAI	25540	25,540.00
KING	5000	5,000.00
BLAKE	2850	2,850.00
CLARK	2450	2,450.00
JONES	2975	2,975.00
SCOTT	3000	3,000.00
FORD	3000	3,000.00
SMITH	800	800.00
ALLEN	1600	1,600.00
WARD	1250	1,250.00
MARTIN	1250	1,250.00
TURNER	1500	1,500.00
ADAMS	1100	1,100.00
JAMES	950	950.00
MILLER	1300	1,300.00

19 rows selected.

```
SQL> CL SCR
```

```

SQL> SELECT ENAME
2  ,SAL
3  ,COMM
4  FROM EMP;

```

ENAME	SAL	COMM
SURESH	2100	
HIMANSHU	2300	
KRISHNA	2500	
ARJUN	2000	
SAI	25540	
KING	5000	
BLAKE	2850	
CLARK	2450	
JONES	2975	
SCOTT	3000	
FORD	3000	
SMITH	800	
ALLEN	1600	300
WARD	1250	500
MARTIN	1250	1400
TURNER	1500	0
ADAMS	1100	
JAMES	950	
MILLER	1300	

19 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5* FROM EMP
SQL> /

```

ENAME	SAL	COMM	DIFF
SURESH	2100		
HIMANSHU	2300		
KRISHNA	2500		
ARJUN	2000		
SAI	25540		
KING	5000		
BLAKE	2850		
CLARK	2450		
JONES	2975		
SCOTT	3000		
FORD	3000		
SMITH	800		
ALLEN	1600	300	1300
WARD	1250	500	750
MARTIN	1250	1400	-150
TURNER	1500	0	1500
ADAMS	1100		
JAMES	950		
MILLER	1300		

19 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL - COMM
6            , 'S999999'
7            )
8* FROM EMP
SQL> /

```

ENAME	SAL	COMM	DIFF	TO_CHAR
SURESH	2100			
HIMANSHU	2300			
KRISHNA	2500			
ARJUN	2000			
SAI	25540			
KING	5000			
BLAKE	2850			
CLARK	2450			
JONES	2975			
SCOTT	3000			
FORD	3000			
SMITH	800			
ALLEN	1600	300	1300	+1300
WARD	1250	500	750	+750
MARTIN	1250	1400	-150	-150
TURNER	1500	0	1500	+1500
ADAMS	1100			

```
JAMES          950
MILLER         1300
```

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  , CASE
6      WHEN SIGN(SAL - COMM) = 1 THEN '+' || (SAL - COMM)
7      WHEN SIGN(SAL - COMM) = -1 THEN '-' || (SAL - COMM)
8  END SA1
9* FROM EMP
10 /
```

ENAME	SAL	COMM	DIFF	SA1
-----	-----	-----	-----	-----
--				
SURESH	2100			
HIMANSHU	2300			
KRISHNA	2500			
ARJUN	2000			
SAI	25540			
KING	5000			
BLAKE	2850			
CLARK	2450			
JONES	2975			
SCOTT	3000			
FORD	3000			
SMITH	800			
ALLEN	1600	300	1300	+1300
WARD	1250	500	750	+750
MARTIN	1250	1400	-150	--150
TURNER	1500	0	1500	+1500
ADAMS	1100			
JAMES	950			
MILLER	1300			

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  , CASE
6      WHEN SIGN(SAL - COMM) = 1 THEN '+' || (SAL - COMM)
7      ELSE (SAL - COMM)
8  END SA1
9* FROM EMP
SQL> /
      ELSE (SAL - COMM)
      *
```

ERROR at line 7:

ORA-00932: inconsistent data types: expected CHAR got NUMBER

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  , CASE
6      WHEN SIGN(SAL - COMM) = 1 THEN '+' || (SAL - COMM)
7      ELSE SAL - COMM
8  END SA1
9* FROM EMP
SQL> /
      ELSE SAL - COMM
          *
```

ERROR at line 7:

ORA-00932: inconsistent datatypes: expected CHAR got NUMBER

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  , CASE
6      WHEN SIGN(SAL - COMM) = 1 THEN '+' || (SAL - COMM)
7      ELSE TO_CHAR((SAL - COMM))
8  END SA1
9* FROM EMP
SQL> /
```

ENAME	SAL	COMM	DIFF	SA1
-----	-----	-----	-----	-----
--				
SURESH	2100			
HIMANSHU	2300			
KRISHNA	2500			
ARJUN	2000			
SAI	25540			
KING	5000			
BLAKE	2850			
CLARK	2450			
JONES	2975			
SCOTT	3000			
FORD	3000			
SMITH	800			
ALLEN	1600	300	1300	+1300
WARD	1250	500	750	+750
MARTIN	1250	1400	-150	-150
TURNER	1500	0	1500	+1500
ADAMS	1100			
JAMES	950			
MILLER	1300			

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
```

```

5  , CASE
6      WHEN SIGN(SAL - COMM) = 1 THEN '+' || (SAL - COMM)
7      ELSE TO_CHAR((SAL - COMM))
8      END SA1
9* FROM EMP
SQL> /

```

ENAME	SAL	COMM	DIFF	SA1
SURESH	2100			
HIMANSHU	2300			
KRISHNA	2500			
ARJUN	2000			
SAI	25540			
KING	5000			
BLAKE	2850			
CLARK	2450			
JONES	2975			
SCOTT	3000			
FORD	3000			
SMITH	800			
ALLEN	1600	300	1300	+1300
WARD	1250	500	750	+750
MARTIN	1250	1400	-150	-150
TURNER	1500	0	1500	+1500
ADAMS	1100			
JAMES	950			
MILLER	1300			

19 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'S99999') AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	COMM	DIFF	SAL1
SURESH	2100			
HIMANSHU	2300			
KRISHNA	2500			
ARJUN	2000			
SAI	25540			
KING	5000			
BLAKE	2850			
CLARK	2450			
JONES	2975			
SCOTT	3000			
FORD	3000			
SMITH	800			
ALLEN	1600	300	1300	+1300
WARD	1250	500	750	+750
MARTIN	1250	1400	-150	-150
TURNER	1500	0	1500	+1500
ADAMS	1100			
JAMES	950			
MILLER	1300			

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'S99999') AS SAL1
6* FROM EMP

```

SQL>

SQL> /

ENAME	SAL	COMM	DIFF	SAL1
SURESH	2100			
HIMANSHU	2300			
KRISHNA	2500			
ARJUN	2000			
SAI	25540			
KING	5000			
BLAKE	2850			
CLARK	2450			
JONES	2975			
SCOTT	3000			
FORD	3000			
SMITH	800			
ALLEN	1600	300	1300	+1300
WARD	1250	500	750	+750
MARTIN	1250	1400	-150	-150
TURNER	1500	0	1500	+1500
ADAMS	1100			
JAMES	950			
MILLER	1300			

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'99999S') AS SAL1
6* FROM EMP

```

SQL> /

ENAME	SAL	COMM	DIFF	SAL1
SURESH	2100			
HIMANSHU	2300			
KRISHNA	2500			
ARJUN	2000			
SAI	25540			
KING	5000			
BLAKE	2850			
CLARK	2450			
JONES	2975			
SCOTT	3000			
FORD	3000			
SMITH	800			

ALLEN	1600	300	1300	1300+
WARD	1250	500	750	750+
MARTIN	1250	1400	-150	150-
TURNER	1500	0	1500	1500+
ADAMS	1100			
JAMES	950			
MILLER	1300			

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'999S99') AS SAL1
6* FROM EMP

```

SQL> /

ERROR:

ORA-01481: invalid number format model

no rows selected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'S99999S') AS SAL1
6* FROM EMP

```

SQL> /

ERROR:

ORA-01481: invalid number format model

no rows selected

SQL> CL SCR

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'S99999') AS SAL1
6* FROM EMP

```

SQL> /

ENAME	SAL	COMM	DIFF	SAL1
SURESH	2100			
HIMANSHU	2300			
KRISHNA	2500			
ARJUN	2000			
SAI	25540			

KING	5000			
BLAKE	2850			
CLARK	2450			
JONES	2975			
SCOTT	3000			
FORD	3000			
SMITH	800			
ALLEN	1600	300	1300	+1300
WARD	1250	500	750	+750
MARTIN	1250	1400	-150	-150
TURNER	1500	0	1500	+1500
ADAMS	1100			
JAMES	950			
MILLER	1300			

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'99999') AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	COMM	DIFF	SAL1
SURESH	2100			
HIMANSHU	2300			
KRISHNA	2500			
ARJUN	2000			
SAI	25540			
KING	5000			
BLAKE	2850			
CLARK	2450			
JONES	2975			
SCOTT	3000			
FORD	3000			
SMITH	800			
ALLEN	1600	300	1300	1300
WARD	1250	500	750	750
MARTIN	1250	1400	-150	-150
TURNER	1500	0	1500	1500
ADAMS	1100			
JAMES	950			
MILLER	1300			

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'99999MI') AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	COMM	DIFF	SAL1
-------	-----	------	------	------


```

-----
SURESH          2100
HIMANSHU        2300
KRISHNA         2500
ARJUN           2000
SAI             25540
KING            5000
BLAKE           2850
CLARK           2450
JONES           2975
SCOTT           3000
FORD            3000
SMITH           800
ALLEN           1600          300          1300  1300
WARD            1250          500          750   750
MARTIN           1250        1400         -150  150-
TURNER           1500           0         1500  1500
ADAMS            1100
JAMES            950
MILLER           1300

```

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'MI99999') AS SAL1
6* FROM EMP

```

SQL> /

ERROR:

ORA-01481: invalid number format model

no rows selected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'99999') AS SAL1
6* FROM EMP

```

SQL> /

```

ENAME          SAL          COMM          DIFF SAL1
-----
SURESH          2100
HIMANSHU        2300
KRISHNA         2500
ARJUN           2000
SAI             25540
KING            5000
BLAKE           2850
CLARK           2450
JONES           2975
SCOTT           3000
FORD            3000

```

SMITH	800			
ALLEN	1600	300	1300	1300
WARD	1250	500	750	750
MARTIN	1250	1400	-150	-150
TURNER	1500	0	1500	1500
ADAMS	1100			
JAMES	950			
MILLER	1300			

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,ABS (TO_CHAR(SAL-COMM,'99999')) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	COMM	DIFF	SAL1
SURESH	2100			
HIMANSHU	2300			
KRISHNA	2500			
ARJUN	2000			
SAI	25540			
KING	5000			
BLAKE	2850			
CLARK	2450			
JONES	2975			
SCOTT	3000			
FORD	3000			
SMITH	800			
ALLEN	1600	300	1300	1300
WARD	1250	500	750	750
MARTIN	1250	1400	-150	150
TURNER	1500	0	1500	1500
ADAMS	1100			
JAMES	950			
MILLER	1300			

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'99999') AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL	COMM	DIFF	SAL1
SURESH	2100			
HIMANSHU	2300			
KRISHNA	2500			
ARJUN	2000			
SAI	25540			

KING	5000			
BLAKE	2850			
CLARK	2450			
JONES	2975			
SCOTT	3000			
FORD	3000			
SMITH	800			
ALLEN	1600	300	1300	1300
WARD	1250	500	750	750
MARTIN	1250	1400	-150	-150
TURNER	1500	0	1500	1500
ADAMS	1100			
JAMES	950			
MILLER	1300			

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'PR99999') AS SAL1
6* FROM EMP

```

SQL> /

ERROR:

ORA-01481: invalid number format model

no rows selected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,SAL - COMM AS DIFF
5  ,TO_CHAR(SAL-COMM,'99999PR') AS SAL1
6* FROM EMP

```

SQL> /

ENAME	SAL	COMM	DIFF	SAL1
SURESH	2100			
HIMANSHU	2300			
KRISHNA	2500			
ARJUN	2000			
SAI	25540			
KING	5000			
BLAKE	2850			
CLARK	2450			
JONES	2975			
SCOTT	3000			
FORD	3000			
SMITH	800			
ALLEN	1600	300	1300	1300
WARD	1250	500	750	750
MARTIN	1250	1400	-150	<150>
TURNER	1500	0	1500	1500
ADAMS	1100			

```
JAMES          950
MILLER         1300
```

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,TO_CHAR(SAL,'9999999') AS SAL1
5* FROM EMP
SQL> /
```

ENAME	SAL	COMM	SAL1
SURESH	2100		2100
HIMANSHU	2300		2300
KRISHNA	2500		2500
ARJUN	2000		2000
SAI	25540		25540
KING	5000		5000
BLAKE	2850		2850
CLARK	2450		2450
JONES	2975		2975
SCOTT	3000		3000
FORD	3000		3000
SMITH	800		800
ALLEN	1600	300	1600
WARD	1250	500	1250
MARTIN	1250	1400	1250
TURNER	1500	0	1500
ADAMS	1100		1100
JAMES	950		950
MILLER	1300		1300

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,TO_CHAR(SAL,'99,999.99') AS SAL1
5* FROM EMP
SQL> /
```

ENAME	SAL	COMM	SAL1
SURESH	2100		2,100.00
HIMANSHU	2300		2,300.00
KRISHNA	2500		2,500.00
ARJUN	2000		2,000.00
SAI	25540		25,540.00
KING	5000		5,000.00
BLAKE	2850		2,850.00
CLARK	2450		2,450.00
JONES	2975		2,975.00
SCOTT	3000		3,000.00
FORD	3000		3,000.00
SMITH	800		800.00
ALLEN	1600	300	1,600.00

WARD	1250	500	1,250.00
MARTIN	1250	1400	1,250.00
TURNER	1500	0	1,500.00
ADAMS	1100		1,100.00
JAMES	950		950.00
MILLER	1300		1,300.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,TO_CHAR(SAL,'99,999.99L') AS SAL1
5* FROM EMP
SQL> /

```

ENAME	SAL	COMM	SAL1
SURESH	2100		2,100.00\$
HIMANSHU	2300		2,300.00\$
KRISHNA	2500		2,500.00\$
ARJUN	2000		2,000.00\$
SAI	25540		25,540.00\$
KING	5000		5,000.00\$
BLAKE	2850		2,850.00\$
CLARK	2450		2,450.00\$
JONES	2975		2,975.00\$
SCOTT	3000		3,000.00\$
FORD	3000		3,000.00\$
SMITH	800		800.00\$
ALLEN	1600	300	1,600.00\$
WARD	1250	500	1,250.00\$
MARTIN	1250	1400	1,250.00\$
TURNER	1500	0	1,500.00\$
ADAMS	1100		1,100.00\$
JAMES	950		950.00\$
MILLER	1300		1,300.00\$

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,TO_CHAR(SAL,'L99,999.99') AS SAL1
5* FROM EMP
SQL> /

```

ENAME	SAL	COMM	SAL1
SURESH	2100		\$2,100.00
HIMANSHU	2300		\$2,300.00
KRISHNA	2500		\$2,500.00
ARJUN	2000		\$2,000.00
SAI	25540		\$25,540.00
KING	5000		\$5,000.00
BLAKE	2850		\$2,850.00
CLARK	2450		\$2,450.00
JONES	2975		\$2,975.00

SCOTT	3000		\$3,000.00
FORD	3000		\$3,000.00
SMITH	800		\$800.00
ALLEN	1600	300	\$1,600.00
WARD	1250	500	\$1,250.00
MARTIN	1250	1400	\$1,250.00
TURNER	1500	0	\$1,500.00
ADAMS	1100		\$1,100.00
JAMES	950		\$950.00
MILLER	1300		\$1,300.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,TO_CHAR(SAL, 'L99,999.99', 'NLS_CURRENCY=RS') AS SAL1
5* FROM EMP
SQL> /

```

ENAME	SAL	COMM	SAL1
SURESH	2100		RS2,100.00
HIMANSHU	2300		RS2,300.00
KRISHNA	2500		RS2,500.00
ARJUN	2000		RS2,000.00
SAI	25540		RS25,540.00
KING	5000		RS5,000.00
BLAKE	2850		RS2,850.00
CLARK	2450		RS2,450.00
JONES	2975		RS2,975.00
SCOTT	3000		RS3,000.00
FORD	3000		RS3,000.00
SMITH	800		RS800.00
ALLEN	1600	300	RS1,600.00
WARD	1250	500	RS1,250.00
MARTIN	1250	1400	RS1,250.00
TURNER	1500	0	RS1,500.00
ADAMS	1100		RS1,100.00
JAMES	950		RS950.00
MILLER	1300		RS1,300.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,TO_CHAR(SAL, '99,999.99L', 'NLS_CURRENCY=RS') AS SAL1
5* FROM EMP
SQL> /

```

ENAME	SAL	COMM	SAL1
SURESH	2100		2,100.00RS
HIMANSHU	2300		2,300.00RS
KRISHNA	2500		2,500.00RS
ARJUN	2000		2,000.00RS
SAI	25540		25,540.00RS

KING	5000		5,000.00RS
BLAKE	2850		2,850.00RS
CLARK	2450		2,450.00RS
JONES	2975		2,975.00RS
SCOTT	3000		3,000.00RS
FORD	3000		3,000.00RS
SMITH	800		800.00RS
ALLEN	1600	300	1,600.00RS
WARD	1250	500	1,250.00RS
MARTIN	1250	1400	1,250.00RS
TURNER	1500	0	1,500.00RS
ADAMS	1100		1,100.00RS
JAMES	950		950.00RS
MILLER	1300		1,300.00RS

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,TO_CHAR(SAL, '99,999.99L', 'NLS_CURRENCY=      RS') AS SAL1
5* FROM EMP
SQL> /

```

ENAME	SAL	COMM	SAL1
SURESH	2100		2,100.00RS
HIMANSHU	2300		2,300.00RS
KRISHNA	2500		2,500.00RS
ARJUN	2000		2,000.00RS
SAI	25540		25,540.00RS
KING	5000		5,000.00RS
BLAKE	2850		2,850.00RS
CLARK	2450		2,450.00RS
JONES	2975		2,975.00RS
SCOTT	3000		3,000.00RS
FORD	3000		3,000.00RS
SMITH	800		800.00RS
ALLEN	1600	300	1,600.00RS
WARD	1250	500	1,250.00RS
MARTIN	1250	1400	1,250.00RS
TURNER	1500	0	1,500.00RS
ADAMS	1100		1,100.00RS
JAMES	950		950.00RS
MILLER	1300		1,300.00RS

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,TO_CHAR(SAL, '99,999.99') AS SAL1
5* FROM EMP
SQL> /

```

ENAME	SAL	COMM	SAL1
SURESH	2100		2,100.00

HIMANSHU	2300		2,300.00
KRISHNA	2500		2,500.00
ARJUN	2000		2,000.00
SAI	25540		25,540.00
KING	5000		5,000.00
BLAKE	2850		2,850.00
CLARK	2450		2,450.00
JONES	2975		2,975.00
SCOTT	3000		3,000.00
FORD	3000		3,000.00
SMITH	800		800.00
ALLEN	1600	300	1,600.00
WARD	1250	500	1,250.00
MARTIN	1250	1400	1,250.00
TURNER	1500	0	1,500.00
ADAMS	1100		1,100.00
JAMES	950		950.00
MILLER	1300		1,300.00

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,TO_CHAR(SAL,'99,999.99')||' RS' AS SAL1
5* FROM EMP
SQL> /

```

ENAME	SAL	COMM	SAL1
SURESH	2100		2,100.00 RS
HIMANSHU	2300		2,300.00 RS
KRISHNA	2500		2,500.00 RS
ARJUN	2000		2,000.00 RS
SAI	25540		25,540.00 RS
KING	5000		5,000.00 RS
BLAKE	2850		2,850.00 RS
CLARK	2450		2,450.00 RS
JONES	2975		2,975.00 RS
SCOTT	3000		3,000.00 RS
FORD	3000		3,000.00 RS
SMITH	800		800.00 RS
ALLEN	1600	300	1,600.00 RS
WARD	1250	500	1,250.00 RS
MARTIN	1250	1400	1,250.00 RS
TURNER	1500	0	1,500.00 RS
ADAMS	1100		1,100.00 RS
JAMES	950		950.00 RS
MILLER	1300		1,300.00 RS

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COMM
4  ,TO_CHAR(SAL,'99,999.99L','NLS_CURRENCY=      RS') AS SAL1
5* FROM EMP
SQL> /

```


ENAME	SAL	COMM	SAL1
SURESH	2100		2,100.00RS
HIMANSHU	2300		2,300.00RS
KRISHNA	2500		2,500.00RS
ARJUN	2000		2,000.00RS
SAI	25540		25,540.00RS
KING	5000		5,000.00RS
BLAKE	2850		2,850.00RS
CLARK	2450		2,450.00RS
JONES	2975		2,975.00RS
SCOTT	3000		3,000.00RS
FORD	3000		3,000.00RS
SMITH	800		800.00RS
ALLEN	1600	300	1,600.00RS
WARD	1250	500	1,250.00RS
MARTIN	1250	1400	1,250.00RS
TURNER	1500	0	1,500.00RS
ADAMS	1100		1,100.00RS
JAMES	950		950.00RS
MILLER	1300		1,300.00RS

19 rows selected.

SQL> cl scr

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT TO_CHAR(5,'RM') AS C1
2* FROM DUAL
SQL> /

```

```

C1
-----
          V

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT TO_CHAR(5,'rm') AS C1
2* FROM DUAL
SQL> /

```

```

C1
-----
          v

```

SQL> cl scr

SQL> DESC EMP

SP2-0042: unknown command "DESCS EMP" - rest of line ignored.

SQL>

SQL> DESC EMP

Name

Null? Type

```

-----
-----
-----
EMPNO
NUMBER(4)
ENAME
VARCHAR2(30)

```

```
JOB
VARCHAR2(9)
MGR
NUMBER(4)
HIREDATE
DATE
SAL
NUMBER(8,2)
COMM
NUMBER(7,2)
DEPTNO
NUMBER(2)
```

```
SQL> INSERT INTO EMP (EMPNO, ENAME, SAL, HIREDATE)
  2  VALUES
  3  (
  4  111, 'SUBHASH'
  5
```

```
SQL>
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
  1  INSERT INTO EMP (EMPNO
  2                      , ENAME
  3                      , SAL
  4                      , HIREDATE
  5                      )
  6  VALUES
  7  (
  8  111
  9  , 'SUBHASH'
 10  , 22,590.00RS
 11  , 05302018
 12* )
SQL> /
,22,590.00RS
      *
```

```
ERROR at line 10:
ORA-00917: missing comma
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
  1  INSERT INTO EMP (EMPNO
  2                      , ENAME
  3                      , SAL
  4                      , HIREDATE
  5                      )
  6  VALUES
  7  (
  8  111
  9  , 'SUBHASH'
 10  , 22,590.00
 11  , 05302018
 12* )
SQL> /
INSERT INTO EMP (EMPNO
      *
```

```
ERROR at line 1:
ORA-00913: too many values
```

```
SQL> ED
```

Wrote file afiedt.buf

```

1  INSERT INTO EMP(EMPNO
2      ,ENAME
3      ,SAL
4      ,HIREDATE
5      )
6  VALUES
7  (
8  111
9  , 'SUBHASH'
10 ,22
11 ,590.00
12 ,05302018
13* )

```

SQL> /

```

INSERT INTO EMP(EMPNO
                *

```

ERROR at line 1:

ORA-00913: too many values

SQL> ED

Wrote file afiedt.buf

```

1  INSERT INTO EMP(EMPNO
2      ,ENAME
3      ,SAL
4      ,HIREDATE
5      )
6  VALUES
7  (
8  111
9  , 'SUBHASH'
10 , '22,590.00'
11 ,05302018
12* )

```

SQL> /

```

,05302018
*

```

ERROR at line 11:

ORA-00932: inconsistent datatypes: expected DATE got NUMBER

SQL> ED

Wrote file afiedt.buf

```

1  INSERT INTO EMP(EMPNO
2      ,ENAME
3      ,SAL
4      ,HIREDATE
5      )
6  VALUES
7  (
8  111
9  , 'SUBHASH'
10 , '22,590.00'
11 ,TO_DATE('05302018','MMDDYYYY')
12* )

```

SQL> /

```

, '22,590.00'
*

```

ERROR at line 10:

ORA-01722: invalid number

SQL> ED

Wrote file afiedt.buf

```

1  INSERT INTO EMP(EMPNO
2      ,ENAME
3      ,SAL
4      ,HIREDATE
5      )
6  VALUES
7  (
8  111
9  , 'SUBHASH'
10 ,TO_NUMBER('22,590.00','99G999D99')
11 ,TO_DATE('05302018','MMDDYYYY')
12* )
SQL> /

```

1 row created.

```

SQL> SELECT ENAME,SAL,HIREDATE
2  FROM EMP;

```

ENAME	SAL	HIREDATE
SURESH	2100	02-JUN-18
HIMANSHU	2300	01-JUN-18
KRISHNA	2500	17-MAY-18
ARJUN	2000	02-JUN-18
SAI	25540	
SUBHASH	22590	30-MAY-18
KING	5000	17-NOV-81
BLAKE	2850	01-MAY-81
CLARK	2450	09-JUN-81
JONES	2975	02-APR-81
SCOTT	3000	19-APR-87
FORD	3000	03-DEC-81
SMITH	800	17-DEC-80
ALLEN	1600	20-FEB-81
WARD	1250	22-FEB-81
MARTIN	1250	28-SEP-81
TURNER	1500	08-SEP-81
ADAMS	1100	23-MAY-87
JAMES	950	03-DEC-81
MILLER	1300	23-JAN-82

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,TO_CHAR(SAL,'99,999.99L') SAL
3      ,TO_CHAR(HIREDATE,'DD-MON-YYYY DAY Q MONTH') AS HIREDATEC
4* FROM EMP
SQL> /

```

ENAME	SAL	HIREDATEC
SURESH	2,100.00\$	02-JUN-2018 SATURDAY 2 JUNE
HIMANSHU	2,300.00\$	01-JUN-2018 FRIDAY 2 JUNE
KRISHNA	2,500.00\$	17-MAY-2018 THURSDAY 2 MAY

ARJUN	2,000.00\$	02-JUN-2018	SATURDAY	2	JUNE
SAI	25,540.00\$				
SUBHASH	22,590.00\$	30-MAY-2018	WEDNESDAY	2	MAY
KING	5,000.00\$	17-NOV-1981	TUESDAY	4	NOVEMBER
BLAKE	2,850.00\$	01-MAY-1981	FRIDAY	2	MAY
CLARK	2,450.00\$	09-JUN-1981	TUESDAY	2	JUNE
JONES	2,975.00\$	02-APR-1981	THURSDAY	2	APRIL
SCOTT	3,000.00\$	19-APR-1987	SUNDAY	2	APRIL
FORD	3,000.00\$	03-DEC-1981	THURSDAY	4	DECEMBER
SMITH	800.00\$	17-DEC-1980	WEDNESDAY	4	DECEMBER
ALLEN	1,600.00\$	20-FEB-1981	FRIDAY	1	FEBRUARY
WARD	1,250.00\$	22-FEB-1981	SUNDAY	1	FEBRUARY
MARTIN	1,250.00\$	28-SEP-1981	MONDAY	3	SEPTEMBER
TURNER	1,500.00\$	08-SEP-1981	TUESDAY	3	SEPTEMBER
ADAMS	1,100.00\$	23-MAY-1987	SATURDAY	2	MAY
JAMES	950.00\$	03-DEC-1981	THURSDAY	4	DECEMBER
MILLER	1,300.00\$	23-JAN-1982	SATURDAY	1	JANUARY

20 rows selected.

SQL> CL SCR

```
SQL> SELECT LAST_DAY(TO_CHAR('1987FEB','YYYYMON'))
      2 FROM DUAL;
```

```
SELECT LAST_DAY(TO_CHAR('1987FEB','YYYYMON'))
      *
```

ERROR at line 1:

ORA-01722: invalid number

SQL> ED

Wrote file afiedt.buf

```
      1 SELECT LAST_DAY('02-FEB-1987') AS C1
      2* FROM DUAL
SQL> /
```

```
C1
-----
28-FEB-87
```

SQL> ED

Wrote file afiedt.buf

```
      1 SELECT LAST_DAY('02-FEB-1987') - '02-FEB-1987' AS C1
      2* FROM DUAL
SQL> /
SELECT LAST_DAY('02-FEB-1987') - '02-FEB-1987' AS C1
      *
```

ERROR at line 1:

ORA-01722: invalid number

SQL> ED

Wrote file afiedt.buf

```
      1 SELECT LAST_DAY('02-FEB-1987') - '02-FEB-87' AS C1
      2* FROM DUAL
SQL> /
SELECT LAST_DAY('02-FEB-1987') - '02-FEB-87' AS C1
      *
```

ERROR at line 1:

ORA-01722: invalid number

```
SQL> EDED
SP2-0042: unknown command "EDED" - rest of line ignored.
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT LAST_DAY('02-FEB-1987') - ('02-FEB-87') AS C1
2* FROM DUAL
```

```
SQL> /
```

```
SELECT LAST_DAY('02-FEB-1987') - ('02-FEB-87') AS C1
*
```

```
ERROR at line 1:
ORA-01722: invalid number
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT LAST_DAY('02-FEB-1987') - TO_DATE('02-FEB-87') AS C1
2* FROM DUAL
```

```
SQL> /
```

```
          C1
-----
          26
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT LAST_DAY('02-FEB-1987') - TO_DATE('02-FEB-87','DD-MON-YY') AS C1
2* FROM DUAL
```

```
SQL> /
```

```
          C1
-----
        -36499
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT LAST_DAY('02-FEB-1987') - TO_DATE('02-FEB-87','DD-MON-YYYY') AS C1
2* FROM DUAL
```

```
SQL> /
```

```
          C1
-----
        693988
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT LAST_DAY('02-FEB-1987') - TO_DATE('02-FEB-1987','DD-MON-YYYY') AS C1
2* FROM DUAL
```

```
SQL> /
```

```
          C1
-----
          26
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT LAST_DAY('02-FEB-1987') - TO_DATE('02-FEB-1987','DD-MON-YY') AS C1
2* FROM DUAL
```

```
SQL> /
```

```
      C1
-----
      26
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
      1  SELECT  LAST_DAY('02-FEB-1987') - TO_DATE('02-FEB-87','DD-MON-YY') AS C1
      2* FROM DUAL
SQL> /
```

```
      C1
-----
     -36499
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
      1  SELECT  LAST_DAY('02-FEB-1987') - TO_CHAR(TO_DATE('02-FEB-87','DD-MON-
YY'),'YYYY') AS C1
      2* FROM DUAL
SQL> /
```

```
      C1
-----
    12-JUN-81
```

```
SQL> E3D
```

```
SP2-0042: unknown command "E3D" - rest of line ignored.
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
      1  SELECT  TO_CHAR(TO_DATE('02-FEB-87','DD-MON-YY'),'YYYY') AS C1
      2* FROM DUAL
SQL> /
```

```
      C1
-----
     2087
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
      1  SELECT  '02-FEB-1987' + 7 AS C1
      2* FROM DUAL
SQL> /
```

```
SELECT  '02-FEB-1987' + 7 AS C1
      *
```

```
ERROR at line 1:
```

```
ORA-01722: invalid number
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
      1  SELECT  TO_DATE('02-FEB-1987') + 7 AS C1
      2* FROM DUAL
SQL> /
```

```
      C1
-----
    09-FEB-87
```

```
SQL> CL SCR
```

```
SQL> CL SCR
```

```
SQL> SELECT TO_DATE('JUL1987','MONYYYY') AS C1
2 FROM DUAL;
```

```
C1
-----
01-JUL-87
```

```
SQL> CL SCR
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```
1 SELECT TO_DATE('987','YYYY') AS C1
2* FROM DUAL
SQL> /
```

```
C1
-----
01-JUN-87
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT TO_DATE('1987','YYYY') AS C1
2* FROM DUAL
SQL> /
```

```
C1
-----
01-JUN-87
```

```
SQL> R
1 SELECT TO_DATE('1987','YYYY') AS C1
2* FROM DUAL
```

```
C1
-----
01-MAY-87
```

```
SQL> CL SCR
```

```
SQL> SELECT TO_DATE('DEC2018','MONYYYY') AS C1
2 FROM DUAL;
```

```
C1
-----
01-DEC-18
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT LAST_DAY( TO_DATE('DEC2018','MONYYYY')
2 ) AS C1
3* FROM DUAL
SQL> /
```

```
C1
-----
31-DEC-18
```



```
SQL> ED
Wrote file afiedt.buf

  1  SELECT LAST_DAY( TO_DATE('DEC2018','MONYYYY')
  2                      ) - SYSDATE AS C1
  3* FROM DUAL
SQL> /
```

```
          C1
-----
241.621285
```

```
SQL> R
  1  SELECT LAST_DAY( TO_DATE('DEC2018','MONYYYY')
  2                      ) - SYSDATE AS C1
  3* FROM DUAL
```

```
          C1
-----
-123.37888
```

```
SQL> ED
Wrote file afiedt.buf

  1  SELECT LAST_DAY( TO_DATE('DEC2018','MONYYYY')
  2                      ) - SYSDATE AS C1
  3* FROM DUAL
SQL>
SQL> /
```

```
          C1
-----
-123.37917
```

```
SQL> ED
Wrote file afiedt.buf

  1  SELECT LAST_DAY( TO_DATE('DEC2019','MONYYYY')
  2                      ) - SYSDATE AS C1
  3* FROM DUAL
SQL> /
```

```
          C1
-----
241.620764
```

```
SQL> ED
Wrote file afiedt.buf

  1  SELECT SYSDATE
  2* FROM DUAL
SQL> /
```

```
SYSDATE
-----
03-MAY-19
```

```
SQL> ED
Wrote file afiedt.buf

  1  SELECT TO_CHAR(SYSDATE,'YYYY')
  2* FROM DUAL
SQL> /
```

TO_C

2019

SQL> ED

Wrote file afiedt.buf

```
1  SELECT  'DEC' || TO_CHAR(SYSDATE, 'YYYY')
2* FROM DUAL
```

SQL> /

'DEC' ||

DEC2019

SQL> ED

Wrote file afiedt.buf

```
1  SELECT  TO_DATE('DEC' || TO_CHAR(SYSDATE, 'YYYY')
2          , 'MONYYYY'
3          )
```

```
4* FROM DUAL
```

SQL> /

TO_DATE('

01-DEC-19

SQL> ED

Wrote file afiedt.buf

```
1  SELECT  LAST_DAY(TO_DATE('DEC' || TO_CHAR(SYSDATE, 'YYYY')
2          , 'MONYYYY'
3          )
```

```
4          )
```

```
5* FROM DUAL
```

SQL> /

LAST_DAY(

31-DEC-19

SQL> ED

Wrote file afiedt.buf

```
1  SELECT  LAST_DAY(TO_DATE('DEC' || TO_CHAR(SYSDATE, 'YYYY')
2          , 'MONYYYY'
3          )
```

```
4          ) - SYSDATE
```

```
5* FROM DUAL
```

SQL> /

LAST_DAY(TO_DATE('DEC' || TO_CHAR(SYSDATE, 'YYYY'), 'MONYYYY')) - SYSDATE

241.619618

SQL> R

```
1  SELECT  LAST_DAY(TO_DATE('DEC' || TO_CHAR(SYSDATE, 'YYYY')
2          , 'MONYYYY'
3          )
```

```
4          ) - SYSDATE
```

```
5* FROM DUAL
```

```
LAST_DAY(TO_DATE('DEC' || TO_CHAR(SYSDATE, 'YYYY'), 'MONYYYY')) - SYSDATE
```

```
-----  
241.619514
```

```
SQL> CL SCR
```

```
SQL> SELECT SYSDATE  
2 FROM DUAL;
```

```
SYSDATE  
-----  
03-JUN-18
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
1 SELECT TRUNC( SYSDATE  
2             , 'YEAR'  
3             )  
4* FROM DUAL  
SQL> /
```

```
TRUNC(SYS  
-----  
01-JAN-18
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
1 SELECT ADD_MONTHS(  
2             TRUNC( SYSDATE  
3             , 'YEAR'  
4             )  
5             , 12  
6             )  
7* FROM DUAL  
SQL> /
```

```
ADD_MONTH  
-----  
01-JAN-19
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
1 SELECT ADD_MONTHS(  
2             TRUNC( SYSDATE  
3             , 'YEAR'  
4             )  
5             , 12  
6             ) - 1  
7* FROM DUAL  
SQL> /
```

```
ADD_MONTH  
-----  
31-DEC-18
```

```
SQL> ED  
Wrote file afiedt.buf
```

```
1 SELECT ( ADD_MONTHS(  
2             TRUNC( SYSDATE  
3             , 'YEAR'
```

```
SQL> CL SCR
```

```
SQL> R
1  SELECT  ( ADD_MONTHS (
2              TRUNC ( SYSDATE
3                      , 'YEAR'
4                  )
5              , 12
6          ) - 1
7      ) - SYSDATE
8* FROM DUAL

(ADD_MONTHS (TRUNC (SYSDATE, 'YEAR') , 12) -1) -SYSDATE
-----
88.6168519
```

```
SQL> R
1  SELECT  (  ADD_MONTHS (
2              TRUNC ( SYSDATE
3                      , 'YEAR'
4                  )
5              , 12
6          ) - 1
7      ) - SYSDATE
8* FROM DUAL

(ADD_MONTHS (TRUNC (SYSDATE, 'YEAR'), 12) - 1) - SYSDATE
-----
88.6167593
```

```
SQL> R
1  SELECT  (  ADD_MONTHS (
2              TRUNC ( SYSDATE
3                      , 'YEAR'
4                      )
5              , 12
6              ) - 1
7          ) - SYSDATE
8* FROM DUAL
```

```
(ADD_MONTHS (TRUNC (SYSDATE, 'YEAR'), 12) - 1) - SYSDATE
-----
149.61647
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ( ADD_MONTHS (
2              TRUNC ( SYSDATE
3                  , 'YEAR'
4              )
5              , 12
6          ) - 1
7      ) - SYSDATE
8* FROM DUAL
```

SQL>

SQL> /

```
(ADD_MONTHS (TRUNC (SYSDATE, 'YEAR'), 12) - 1) - SYSDATE
-----
149.616123
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT TO_CHAR (
2              ( ADD_MONTHS (
3                  TRUNC ( SYSDATE
4                      , 'YEAR'
5                  )
6                  , 12
7              )
8              - 1
9              )
10         , 'DDD'
11         ) - TO_CHAR (SYSDATE, 'DDD')
12* FROM DUAL
SQL> /
```

```
TO_CHAR ( (ADD_MONTHS (TRUNC (SYSDATE, 'YEAR'), 12) - 1) , 'DDD') - TO_CHAR (SYSDATE, 'DDD')
-----
150
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT TO_CHAR (
2              ( ADD_MONTHS (
3                  TRUNC ( SYSDATE
4                      , 'YEAR'
5                  )
6                  , 12
7              )
8              - 1
9              )
10         , 'DDD'
11         ) - TO_CHAR (SYSDATE, 'DDD')
12* FROM DUAL
```

SQL>

SQL> /

```
TO_CHAR ( (ADD_MONTHS (TRUNC (SYSDATE, 'YEAR'), 12) - 1) , 'DDD') - TO_CHAR (SYSDATE, 'DDD')
-----
```

```
SQL> CL SCR
```

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
       2 FROM EMP;
```

```
ENAME
-----
```

```
SURESH
HIMANSHU
KRISHNA
ARJUN
SAI
SUBHASH
KING
BLAKE
CLARK
JONES
SCOTT
FORD
SMITH
ALLEN
WARD
MARTIN
TURNER
ADAMS
JAMES
MILLER
```

```
20 rows selected.
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT ENAME
2           ,SUBSTR(ENAME,1,1) AS C1
3           ,SUBSTR(ENAME,-1,1) AS C1
4* FROM EMP
SQL> /
```

ENAME	C1	C1
-----	----	----
SURESH	S	H
HIMANSHU	H	U
KRISHNA	K	A
ARJUN	A	N
SAI	S	I
SUBHASH	S	H
KING	K	G
BLAKE	B	E
CLARK	C	K
JONES	J	S
SCOTT	S	T
FORD	F	D
SMITH	S	H
ALLEN	A	N
WARD	W	D
MARTIN	M	N
TURNER	T	R
ADAMS	A	S
JAMES	J	S

MILLER M R

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT LPAD('*'
2             ,4
3             , '*'
4             ) AS C1
5* FROM DUAL
SQL> /
```

```
C1
----
****
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT LPAD('*'
2             ,LENGTH('SURESH') - 2
3             , '*'
4             ) AS C1
5* FROM DUAL
SQL> /
```

```
C1
----
****
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT LPAD('*'
2             ,LENGTH('HIMANSHU') - 2
3             , '*'
4             ) AS C1
5* FROM DUAL
SQL> /
```

```
C1
-----
*****
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT LPAD('*'
2             ,LENGTH('AB') - 2
3             , '*'
4             ) AS C1
5* FROM DUAL
SQL> /
```

```
C
-
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT LPAD('*'
```

```

2          ,LENGTH('ACB') - 2
3          , '*'
4        ) AS C1
5* FROM DUAL
SQL> /

```

```

C
-
*

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2          ,SUBSTR(ENAME,1,1) AS C1
3          ,LPAD(' ',
4              ,LENGTH(ENAME) - 2
5              , '*'
6          )
7          ,SUBSTR(ENAME,-1,1) AS C2
8* FROM EMP
SQL> /

```

```

ENAME          C1
-----
LPAD(' ', LENGTH(ENAME)-2, '*')
-----
C2
----
SURESH          S
*****
H

HIMANSHU        H
*****
U

KRISHNA         K
*****
A

ARJUN           A
***
N

SAI             S
*
I

SUBHASH         S
*****
H

KING            K
**
G

BLAKE           B
***
E

CLARK           C

```

K

JONES J

S

SCOTT S

T

FORD F

**

D

SMITH S

H

ALLEN A

N

WARD W

**

D

MARTIN M

N

TURNER T

R

ADAMS A

S

JAMES J

S

MILLER M

R

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2      ,SUBSTR(ENAME,1,1) AS C1
3      ,LPAD(' ',
4          ,LENGTH(ENAME) - 2
5          , ' ')
6          ) C3
7      ,SUBSTR(ENAME,-1,1) AS C2
8* FROM EMP
SQL> /
```

ENAME	C1
-------	----

C3

C2

SURESH S

H

HIMANSHU H

U

KRISHNA K

A

ARJUN A

N

SAI S
*

I

SUBHASH S

H

KING K
**

G

BLAKE B

E

CLARK C

K

JONES J

S

SCOTT S

T

FORD F
**

D

SMITH S

H

ALLEN A

N

WARD W

**

D

MARTIN M

N

TURNER T

R

ADAMS A

S

JAMES J

S

MILLER M

R

20 rows selected.

SQL> COLUMN C3 FORMAT A10

SQL> R

```

1  SELECT ENAME
2      ,SUBSTR(ENAME,1,1) AS C1
3      ,LPAD(' '
4          ,LENGTH(ENAME) - 2
5          , ' '
6          ) C3
7      ,SUBSTR(ENAME,-1,1) AS C2
8* FROM EMP

```

ENAME	C1	C3	C2
SURESH	S	****	H
HIMANSHU	H	*****	U
KRISHNA	K	*****	A
ARJUN	A	***	N
SAI	S	*	I
SUBHASH	S	*****	H
KING	K	**	G
BLAKE	B	***	E
CLARK	C	***	K
JONES	J	***	S
SCOTT	S	***	T
FORD	F	**	D
SMITH	S	***	H
ALLEN	A	***	N
WARD	W	**	D
MARTIN	M	****	N
TURNER	T	****	R
ADAMS	A	***	S
JAMES	J	***	S
MILLER	M	****	R

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2      , SUBSTR(ENAME,1,1)
3      || LPAD(' ',
4      , LENGTH(ENAME) - 2
5      , ' ')
6      )
7      || SUBSTR(ENAME,-1,1) AS ENAME1
8* FROM EMP
SQL> /
```

ENAME

ENAME1

SURESH

S****H

HIMANSHU

H*****U

KRISHNA

K*****A

ARJUN

A***N

SAI

S*I

SUBHASH

S*****H

KING

K**G

BLAKE

B***E

CLARK

C***K

JONES

J***S

SCOTT

S***T

FORD

F**D

SMITH

S***H

ALLEN

A***N

WARD

W**D

MARTIN

M*****N

TURNER
T****R

ADAMS
A***S

JAMES
J***S

MILLER
M****R

20 rows selected.

```
SQL> COLUMN ENAME1 FORMAT A1090
SQL> COLUMN ENAME1 FORMAT A10
SQL> CL SCR
```

```
SQL> R
 1  SELECT ENAME
 2          , SUBSTR(ENAME,1,1)
 3          || LPAD('*',
 4              , LENGTH(ENAME) - 2
 5              , '*'
 6              )
 7          || SUBSTR(ENAME,-1,1) AS ENAME1
 8* FROM EMP
```

ENAME	ENAME1
SURESH	S****H
HIMANSHU	H*****U
KRISHNA	K*****A
ARJUN	A***N
SAI	S*I
SUBHASH	S*****H
KING	K**G
BLAKE	B***E
CLARK	C***K
JONES	J***S
SCOTT	S***T
FORD	F**D
SMITH	S***H
ALLEN	A***N
WARD	W**D
MARTIN	M****N
TURNER	T****R
ADAMS	A***S
JAMES	J***S
MILLER	M****R

20 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
 1  SELECT ENAME
 2          , SUBSTR(ENAME,1,1)
 3          || RPAD('*',
 4              , LENGTH(ENAME) - 2
 5              , '*'
 6              )
```

```

      7      ||SUBSTR(ENAME,-1,1) AS ENAME1
      8* FROM EMP
SQL> /

```

ENAME	ENAME1
SURESH	S****H
HIMANSHU	H*****U
KRISHNA	K*****A
ARJUN	A***N
SAI	S*I
SUBHASH	S*****H
KING	K**G
BLAKE	B***E
CLARK	C***K
JONES	J***S
SCOTT	S***T
FORD	F**D
SMITH	S***H
ALLEN	A***N
WARD	W**D
MARTIN	M*****N
TURNER	T****R
ADAMS	A***S
JAMES	J***S
MILLER	M****R

20 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

      1  SELECT SUBSTR(ENAME,1,1)||'***'||SUBSTR(ENAME-1) AS ENAME1
      2* FROM EMP
SQL> /
SELECT SUBSTR(ENAME,1,1)||'***'||SUBSTR(ENAME-1) AS ENAME1
                                     *

```

ERROR at line 1:
ORA-00938: not enough arguments for function

```

SQL> ED
Wrote file afiedt.buf

```

```

      1  SELECT SUBSTR(ENAME,1,1)||'***'||SUBSTR(ENAME,-1) AS ENAME1
      2* FROM EMP
SQL> /

```

ENAME1
S**H
H**U
K**A
A**N
S**I
S**H
K**G
B**E
C**K
J**S
S**T
F**D
S**H
A**N

```

W**D
M**N
T**R
A**S
J**S
M**R

```

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  select ename
2      , replace(ename
3                , SUBSTR(ENAME,2,3)
4                , '**'
5                )
6* from emp
7  /

```

ENAME	REPLACE(ENAME, SUBSTR(ENAME,2,3), '**')
SURESH	S**SH
HIMANSHU	H**NSHU
KRISHNA	K**HNA
ARJUN	A**N
SAI	S**
SUBHASH	S**ASH
KING	K**
BLAKE	B**E
CLARK	C**K
JONES	J**S
SCOTT	S**T
FORD	F**
SMITH	S**H
ALLEN	A**N
WARD	W**
MARTIN	M**IN
TURNER	T**ER
ADAMS	A**S
JAMES	J**S
MILLER	M**ER

SURESH	S**SH
HIMANSHU	H**NSHU
KRISHNA	K**HNA
ARJUN	A**N
SAI	S**
SUBHASH	S**ASH
KING	K**
BLAKE	B**E
CLARK	C**K
JONES	J**S
SCOTT	S**T
FORD	F**
SMITH	S**H
ALLEN	A**N
WARD	W**
MARTIN	M**IN
TURNER	T**ER
ADAMS	A**S
JAMES	J**S
MILLER	M**ER

20 rows selected.

SQL> CL SCR

```

SQL> SELECT ENAME
2  FROM EMP;

```

```

ENAME
-----
SURESH
HIMANSHU
KRISHNA
ARJUN
SAI
SUBHASH
KING
BLAKE
CLARK
JONES
SCOTT
FORD

```

SMITH
 ALLEN
 WARD
 MARTIN
 TURNER
 ADAMS
 JAMES
 MILLER

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,LENGTH(ENAME) AS C1
3* FROM EMP
SQL> /

```

ENAME	C1
SURESH	6
HIMANSHU	8
KRISHNA	7
ARJUN	5
SAI	3
SUBHASH	7
KING	4
BLAKE	5
CLARK	5
JONES	5
SCOTT	5
FORD	4
SMITH	5
ALLEN	5
WARD	4
MARTIN	6
TURNER	6
ADAMS	5
JAMES	5
MILLER	6

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,LENGTH(ENAME) AS C1
3  ,REPLACE(ENAME
4  , 'A'
5  ) AS C2
6* FROM EMP
SQL> /

```

ENAME	C1	C2
SURESH	6	SURESH
HIMANSHU	8	HIMNSHU
KRISHNA	7	KRISHN
ARJUN	5	RJUN
SAI	3	SI
SUBHASH	7	SUBHSH
KING	4	KING

BLAKE	5	BLKE
CLARK	5	CLRK
JONES	5	JONES
SCOTT	5	SCOTT
FORD	4	FORD
SMITH	5	SMITH
ALLEN	5	LLEN
WARD	4	WRD
MARTIN	6	MRTIN
TURNER	6	TURNER
ADAMS	5	DMS
JAMES	5	JMES
MILLER	6	MILLER

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,LENGTH(ENAME) AS C1
3  ,LENGTH(REPLACE(ENAME
4  , 'A'
5  )) AS C2
6* FROM EMP
SQL> /

```

ENAME	C1	C2
SURESH	6	6
HIMANSHU	8	7
KRISHNA	7	6
ARJUN	5	4
SAI	3	2
SUBHASH	7	6
KING	4	4
BLAKE	5	4
CLARK	5	4
JONES	5	5
SCOTT	5	5
FORD	4	4
SMITH	5	5
ALLEN	5	4
WARD	4	3
MARTIN	6	5
TURNER	6	6
ADAMS	5	3
JAMES	5	4
MILLER	6	6

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,LENGTH(ENAME)
3  -LENGTH(REPLACE(ENAME
4  , 'A'
5  )) AS C2
6* FROM EMP
SQL> /

```

ENAME	C2
-------	----

```

-----
SURESH                0
HIMANSHU              1
KRISHNA               1
ARJUN                 1
SAI                   1
SUBHASH               1
KING                  0
BLAKE                 1
CLARK                 1
JONES                 0
SCOTT                 0
FORD                  0
SMITH                 0
ALLEN                 1
WARD                  1
MARTIN                1
TURNER                0
ADAMS                 2
JAMES                 1
MILLER                0

```

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,LENGTH(ENAME)
3  -LENGTH(REPLACE(ENAME
4  , 'A'
5  )) AS C2
6  FROM EMP
7  WHERE LENGTH(ENAME)
8  -LENGTH(REPLACE(ENAME
9  , 'A'
10* )) = 2
SQL> /

```

```

ENAME                C2
-----
ADAMS                 2

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,LENGTH(ENAME)
3  -LENGTH(REPLACE(ENAME
4  , 'A'
5  )) AS C2
6  FROM EMP
7  WHERE LENGTH(ENAME)
8  -LENGTH(REPLACE(ENAME
9  , 'A'
10* )) = 0
SQL> /

```

```

ENAME                C2
-----
SURESH               0
KING                 0
JONES                0
SCOTT                0

```

```

FORD                0
SMITH               0
TURNER             0
MILLER             0

```

8 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2    ,LENGTH(ENAME)
 3  -LENGTH(REPLACE(ENAME
 4                , 'A'
 5                )) AS C2
 6  FROM EMP
 7  WHERE LENGTH(ENAME)
 8  -LENGTH(REPLACE(ENAME
 9                , 'A'
10*                )) >1
SQL> /

```

ENAME	C2
ADAMS	2

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2* FROM EMP
SQL> /

```

ENAME
SURESH
HIMANSHU
KRISHNA
ARJUN
SAI
SUBHASH
KING
BLAKE
CLARK
JONES
SCOTT
FORD
SMITH
ALLEN
WARD
MARTIN
TURNER
ADAMS
JAMES
MILLER

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1* select ename, rpad(lpad(ename,10,'*'),15,'*') as output from emp;
SQL> /
select ename, rpad(lpad(ename,10,'*'),15,'*') as output from emp;

```

*

ERROR at line 1:
ORA-00911: invalid character

SQL> ED
Wrote file afiedt.buf

```
1* select ename, rpad(lpad(ename,10,''),15,'') as output from emp
SQL> /
```

ENAME	OUTPUT
SURESH	*****SURESH*****
HIMANSHU	**HIMANSHU*****
KRISHNA	***KRISHNA*****
ARJUN	*****ARJUN*****
SAI	*****SAI*****
SUBHASH	***SUBHASH*****
KING	*****KING*****
BLAKE	*****BLAKE*****
CLARK	*****CLARK*****
JONES	*****JONES*****
SCOTT	*****SCOTT*****
FORD	*****FORD*****
SMITH	*****SMITH*****
ALLEN	*****ALLEN*****
WARD	*****WARD*****
MARTIN	***MARTIN*****
TURNER	*****TURNER*****
ADAMS	*****ADAMS*****
JAMES	*****JAMES*****
MILLER	*****MILLER*****

20 rows selected.

SQL> ED
Wrote file afiedt.buf

```
1* select ename, rpad(lpad(ename,10,''),10,'') as output from emp
SQL> /
```

ENAME	OUTPUT
SURESH	*****SURESH
HIMANSHU	**HIMANSHU
KRISHNA	***KRISHNA
ARJUN	*****ARJUN
SAI	*****SAI
SUBHASH	***SUBHASH
KING	*****KING
BLAKE	*****BLAKE
CLARK	*****CLARK
JONES	*****JONES
SCOTT	*****SCOTT
FORD	*****FORD
SMITH	*****SMITH
ALLEN	*****ALLEN
WARD	*****WARD
MARTIN	***MARTIN
TURNER	*****TURNER
ADAMS	*****ADAMS
JAMES	*****JAMES
MILLER	*****MILLER

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1* select ename, rpad(lpad(ename,10,'*'),15,'*') as output from emp
SQL> /
```

ENAME	OUTPUT
SURESH	*****SURESH*****
HIMANSHU	**HIMANSHU*****
KRISHNA	***KRISHNA*****
ARJUN	*****ARJUN*****
SAI	*****SAI*****
SUBHASH	***SUBHASH*****
KING	*****KING*****
BLAKE	*****BLAKE*****
CLARK	*****CLARK*****
JONES	*****JONES*****
SCOTT	*****SCOTT*****
FORD	*****FORD*****
SMITH	*****SMITH*****
ALLEN	*****ALLEN*****
WARD	*****WARD*****
MARTIN	***MARTIN*****
TURNER	*****TURNER*****
ADAMS	*****ADAMS*****
JAMES	*****JAMES*****
MILLER	***MILLER*****

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2      ,LPAD(ENAME
3          , '*'
4          ,10
5          )
6* FROM EMP
SQL> /
```

```
, '*'
*
```

ERROR at line 3:
ORA-01722: invalid number

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2      ,LPAD(ENAME
3          ,10
4          , '*'
5          )
6* FROM EMP
SQL> /
```

ENAME	LPAD(ENAME,10,'*')
SURESH	*****SURESH

```

HIMANSHU  **HIMANSHU
KRISHNA   ***KRISHNA
ARJUN     *****ARJUN
SAI       *****SAI
SUBHASH   ***SUBHASH
KING      *****KING
BLAKE     *****BLAKE
CLARK     *****CLARK
JONES     *****JONES
SCOTT     *****SCOTT
FORD      *****FORD
SMITH     *****SMITH
ALLEN     *****ALLEN
WARD      *****WARD
MARTIN    ****MARTIN
TURNER    ****TURNER
ADAMS     *****ADAMS
JAMES     *****JAMES
MILLER    ****MILLER

```

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2      ,LPAD (ENAME
 3          ,10
 4          , '*'
 5          )
 6      ,RPAD (ENAME
 7          ,10
 8          , '*'
 9          )
10* FROM EMP
SQL> /

```

ENAME	LPAD (ENAME,10, '*')	RPAD (ENAME,10, '*')
SURESH	****SURESH	SURESH****
HIMANSHU	**HIMANSHU	HIMANSHU**
KRISHNA	***KRISHNA	KRISHNA***
ARJUN	*****ARJUN	ARJUN*****
SAI	*****SAI	SAI*****
SUBHASH	***SUBHASH	SUBHASH***
KING	*****KING	KING*****
BLAKE	*****BLAKE	BLAKE*****
CLARK	*****CLARK	CLARK*****
JONES	*****JONES	JONES*****
SCOTT	*****SCOTT	SCOTT*****
FORD	*****FORD	FORD*****
SMITH	*****SMITH	SMITH*****
ALLEN	*****ALLEN	ALLEN*****
WARD	*****WARD	WARD*****
MARTIN	****MARTIN	MARTIN****
TURNER	****TURNER	TURNER****
ADAMS	*****ADAMS	ADAMS*****
JAMES	*****JAMES	JAMES*****
MILLER	****MILLER	MILLER****

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,LPAD(ENAME
3          ,10
4          , '*'
5          )
6      ||RPAD(ENAME
7          ,10
8          , '*'
9          )
10* FROM EMP
SQL> /

```

ENAME	LPAD(ENAME,10,'*') RPAD(ENAME,10,'*')

SURESH	*****SURESHSURESH*****
HIMANSHU	**HIMANSHUHIMANSHU**
KRISHNA	***KRISHNAKRISHNA***
ARJUN	*****ARJUNARJUN*****
SAI	*****SAISAI*****
SUBHASH	***SUBHASHSUBHASH***
KING	*****KINGKING*****
BLAKE	*****BLAKEBLAKE*****
CLARK	*****CLARKCLARK*****
JONES	*****JONESJONES*****
SCOTT	*****SCOTTSCOTT*****
FORD	*****FORDFORD*****
SMITH	*****SMITHSMITH*****
ALLEN	*****ALLENALLEN*****
WARD	*****WARDWARD*****
MARTIN	*****MARTINMARTIN*****
TURNER	*****TURNERTURNER*****
ADAMS	*****ADAMSADAMS*****
JAMES	*****JAMESJAMES*****
MILLER	*****MILLERMILLER*****

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,LPAD(ENAME
3          ,10
4          , '*'
5          )
6      ,RPAD(ENAME
7          ,10
8          , '*'
9          )
10* FROM EMP
SQL> /

```

ENAME	LPAD(ENAME,10,'*')	RPAD(ENAME,10,'*')

SURESH	*****SURESH	SURESH*****
HIMANSHU	**HIMANSHU	HIMANSHU**
KRISHNA	***KRISHNA	KRISHNA***
ARJUN	*****ARJUN	ARJUN*****
SAI	*****SAI	SAI*****
SUBHASH	***SUBHASH	SUBHASH***

KING	*****KING	KING*****
BLAKE	*****BLAKE	BLAKE*****
CLARK	*****CLARK	CLARK*****
JONES	*****JONES	JONES*****
SCOTT	*****SCOTT	SCOTT*****
FORD	*****FORD	FORD*****
SMITH	*****SMITH	SMITH*****
ALLEN	*****ALLEN	ALLEN*****
WARD	*****WARD	WARD*****
MARTIN	*****MARTIN	MARTIN*****
TURNER	*****TURNER	TURNER*****
ADAMS	*****ADAMS	ADAMS*****
JAMES	*****JAMES	JAMES*****
MILLER	*****MILLER	MILLER*****

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,LPAD(ENAME
3          ,10
4          , '*'
5          )
6      ,REPLACE( RPAD(ENAME
7                  ,10
8                  , '*'
9                  )
10             ,ENAME
11             )
12* FROM EMP
SQL> /

```

```

ENAME      LPAD(ENAME,10,'*')
REPLACE(RPAD(ENAME,10,'*'),ENAME)
-----

```

SURESH	*****SURESH	*****
HIMANSHU	**HIMANSHU	**
KRISHNA	***KRISHNA	***
ARJUN	*****ARJUN	*****
SAI	*****SAI	*****
SUBHASH	***SUBHASH	***
KING	*****KING	*****
BLAKE	*****BLAKE	*****
CLARK	*****CLARK	*****
JONES	*****JONES	*****
SCOTT	*****SCOTT	*****
FORD	*****FORD	*****
SMITH	*****SMITH	*****
ALLEN	*****ALLEN	*****
WARD	*****WARD	*****
MARTIN	*****MARTIN	*****
TURNER	*****TURNER	*****
ADAMS	*****ADAMS	*****
JAMES	*****JAMES	*****
MILLER	*****MILLER	*****

20 rows selected.

SQL> ED

Wrote file afiedt.buf


```

1  SELECT ENAME
2      ,LPAD(ENAME
3          ,10
4          , '*'
5          )
6      ||REPLACE( RPAD(ENAME
7                  ,10
8                  , '*'
9                  )
10             ,ENAME
11             )
12* FROM EMP
SQL> /

```

ENAME	LPAD(ENAME,10,'*') REPLACE(RPAD(ENAME,10,'*'), ENAME)
-------	--

SURESH	*****SURESH*****
HIMANSHU	**HIMANSHU**
KRISHNA	***KRISHNA***
ARJUN	*****ARJUN*****
SAI	*****SAI*****
SUBHASH	***SUBHASH***
KING	*****KING*****
BLAKE	*****BLAKE*****
CLARK	*****CLARK*****
JONES	*****JONES*****
SCOTT	*****SCOTT*****
FORD	*****FORD*****
SMITH	*****SMITH*****
ALLEN	*****ALLEN*****
WARD	*****WARD*****
MARTIN	*****MARTIN*****
TURNER	*****TURNER*****
ADAMS	*****ADAMS*****
JAMES	*****JAMES*****
MILLER	*****MILLER*****

20 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2      ,LPAD(ENAME
3          ,10
4          , '*'
5          )
6      ||SUBSTR(RPAD(ENAME
7                  ,10
8                  , '*'
9                  )
10             ,LENGTH(ENAME) +1
11             )
12* FROM EMP
SQL> /

```

ENAME	LPAD(ENAME,10,'*') SUBSTR(RPAD(ENAME,10,'*'), LENGTH(ENAME)+1)
-------	---

SURESH	*****SURESH*****
HIMANSHU	**HIMANSHU**
KRISHNA	***KRISHNA***
ARJUN	*****ARJUN*****

```

SAI          *****SAI*****
SUBHASH      ***SUBHASH***
KING         *****KING*****
BLAKE        *****BLAKE*****
CLARK        *****CLARK*****
JONES        *****JONES*****
SCOTT        *****SCOTT*****
FORD         *****FORD*****
SMITH        *****SMITH*****
ALLEN        *****ALLEN*****
WARD         *****WARD*****
MARTIN       *****MARTIN*****
TURNER       *****TURNER*****
ADAMS        *****ADAMS*****
JAMES        *****JAMES*****
MILLER       *****MILLER*****

```

20 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2      ,LPAD (ENAME
3          ,10
4          , '*'
5          )
6      || SUBSTR (RPAD (ENAME
7          ,10
8          , '*'
9          )
10         , LENGTH (ENAME) +1
11        )
12* FROM EMP

```

ENAME	LPAD (ENAME,10,'*') SUBSTR (RPAD (ENAME,10,'*') , LENGTH (ENAME) +1)
-------	---

SURESH	*****SURESH*****
HIMANSHU	**HIMANSHU**
KRISHNA	***KRISHNA***
ARJUN	*****ARJUN*****
SAI	*****SAI*****
SUBHASH	***SUBHASH***
KING	*****KING*****
BLAKE	*****BLAKE*****
CLARK	*****CLARK*****
JONES	*****JONES*****
SCOTT	*****SCOTT*****
FORD	*****FORD*****
SMITH	*****SMITH*****
ALLEN	*****ALLEN*****
WARD	*****WARD*****
MARTIN	*****MARTIN*****
TURNER	*****TURNER*****
ADAMS	*****ADAMS*****
JAMES	*****JAMES*****
MILLER	*****MILLER*****

20 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2      ,LPAD (ENAME
3          ,10
4          , '*'
5          )
6      || SUBSTR (RPAD (ENAME
7          ,10
8          , '*'
9          )
10         , LENGTH (ENAME) +1
11     )
12* FROM EMP

```

```

ENAME      LPAD (ENAME,10,'*') || SUBSTR (RPAD (ENAME,10,'*') , LENGTH (ENAME) +1)
-----

```

```

-----
SURESH      *****SURESH*****
HIMANSHU    **HIMANSHU**
KRISHNA     ***KRISHNA***
ARJUN       *****ARJUN*****
SAI         *****SAI*****
SUBHASH     ***SUBHASH***
KING        *****KING*****
BLAKE       *****BLAKE*****
CLARK       *****CLARK*****
JONES       *****JONES*****
SCOTT       *****SCOTT*****
FORD        *****FORD*****
SMITH       *****SMITH*****
ALLEN       *****ALLEN*****
WARD        *****WARD*****
MARTIN      *****MARTIN*****
TURNER      *****TURNER*****
ADAMS       *****ADAMS*****
JAMES       *****JAMES*****
MILLER      *****MILLER*****

```

20 rows selected.

SQL> CL SCR

```

SQL> SELECT ENAME
2      ,SAL
3  FROM EMP;

```

```

ENAME      SAL
-----
SURESH      2100
HIMANSHU    2300
KRISHNA     2500
ARJUN       2000
SAI         25540
SUBHASH     22590
KING        5000
BLAKE       2850
CLARK       2450
JONES       2975
SCOTT       3000
FORD        3000
SMITH       800
ALLEN       1600
WARD        1250
MARTIN      1250
TURNER      1500

```

```

ADAMS          1100
JAMES          950
MILLER         1300

```

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2  ,SAL
 3  ,LPAD('*',
 4      ,LEGNTH(SAL)
 5      , '*'
 6      )
 7* FROM EMP

```

SQL> /

```

      ,LEGNTH(SAL)
      *

```

ERROR at line 4:

ORA-00904: "LEGNTH": invalid identifier

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2  ,SAL
 3  ,LPAD('*',
 4      ,LENGTH(SAL)
 5      , '*'
 6      )
 7* FROM EMP

```

SQL> /

```

ENAME          SAL
-----
LPAD('*',LENGTH(SAL), '*')
-----
-----

```

```

SURESH          2100
****

```

```

HIMANSHU        2300
****

```

```

KRISHNA         2500
****

```

```

ARJUN           2000
****

```

```

SAI             25540
*****

```

```

SUBHASH         22590
*****

```

```

KING            5000
****

```

```

BLAKE           2850
****

```

CLARK *****	2450
JONES *****	2975
SCOTT *****	3000
FORD *****	3000
SMITH ***	800
ALLEN *****	1600
WARD *****	1250
MARTIN *****	1250
TURNER *****	1500
ADAMS *****	1100
JAMES ***	950
MILLER *****	1300

20 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,LPAD('*',
4      ,LENGTH(SAL)
5      ,'*)
6      ) SAL
7* FROM EMP
SQL> /
```

ENAME	SAL
-------	-----

SAL

SURESH *****	2100
-----------------	------

HIMANSHU *****	2300
-------------------	------

KRISHNA *****	2500
ARJUN *****	2000
SAI *****	25540
SUBHASH *****	22590
KING *****	5000
BLAKE *****	2850
CLARK *****	2450
JONES *****	2975
SCOTT *****	3000
FORD *****	3000
SMITH ***	800
ALLEN *****	1600
WARD *****	1250
MARTIN *****	1250
TURNER *****	1500
ADAMS *****	1100
JAMES ***	950
MILLER *****	1300

20 rows selected.

```
SQL> COLUMN SAL1 FORMAT 10
SP2-0246: Illegal FORMAT string "10"
SQL> R
 1  SELECT ENAME
 2    ,SAL
 3    ,LPAD(' ',
 4          ,LENGTH(SAL))
```

```
5      , '*'
6      ) SAL
7* FROM EMP
```

ENAME	SAL
-------	-----

SAL

SURESH	2100

HIMANSHU	2300

KRISHNA	2500

ARJUN	2000

SAI	25540

SUBHASH	22590

KING	5000

BLAKE	2850

CLARK	2450

JONES	2975

SCOTT	3000

FORD	3000

SMITH	800

ALLEN	1600

WARD	1250

MARTIN	1250

TURNER	1500

ADAMS	1100

JAMES 950

MILLER 1300

20 rows selected.

SQL> COLUMN SAL1 FORMAT A10

SQL> R

```
1  SELECT ENAME
2    ,SAL
3    ,LPAD(' ',
4          ,LENGTH(SAL)
5          , ' '*
6          ) SAL
7*  FROM EMP
```

ENAME	SAL
-------	-----

-----	-----
SAL	

SURESH	2100

HIMANSHU	2300

KRISHNA	2500

ARJUN	2000

SAI	25540

SUBHASH	22590

KING	5000

BLAKE	2850

CLARK	2450

JONES	2975

SCOTT	3000

FORD	3000

SMITH	800

```

ALLEN                1600
*****

WARD                 1250
*****

MARTIN               1250
*****

TURNER               1500
*****

ADAMS                1100
*****

JAMES                950
***

MILLER               1300
*****

```

20 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

 1  SELECT ENAME
 2    ,SAL
 3    ,LPAD('*',
 4          ,LENGTH(SAL)
 5          , '*')
 6          ) SAL1
 7* FROM EMP
SQL> /

```

ENAME	SAL	SAL1
SURESH	2100	*****
HIMANSHU	2300	*****
KRISHNA	2500	*****
ARJUN	2000	*****
SAI	25540	*****
SUBHASH	22590	*****
KING	5000	*****
BLAKE	2850	*****
CLARK	2450	*****
JONES	2975	*****
SCOTT	3000	*****
FORD	3000	*****
SMITH	800	***
ALLEN	1600	*****
WARD	1250	*****
MARTIN	1250	*****
TURNER	1500	*****
ADAMS	1100	*****
JAMES	950	***
MILLER	1300	*****

20 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,LPAD(' '
3      ,LENGTH(SAL)
4      , ' '
5      ) SAL1
6* FROM EMP
7  /

```

ENAME	SAL1
-----	-----
SURESH	****
HIMANSHU	****
KRISHNA	****
ARJUN	****
SAI	*****
SUBHASH	*****
KING	****
BLAKE	****
CLARK	****
JONES	****
SCOTT	****
FORD	****
SMITH	***
ALLEN	****
WARD	****
MARTIN	****
TURNER	****
ADAMS	****
JAMES	***
MILLER	****

20 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,TRANSLATE(SAL
3      , '0123456789'
4      , '*****')
5      ) AS SAL1
6* FROM EMP
SQL> /

```

ENAME	SAL1
-----	-----
SURESH	****
HIMANSHU	****
KRISHNA	****
ARJUN	****
SAI	*****
SUBHASH	*****
KING	****
BLAKE	****
CLARK	****
JONES	****
SCOTT	****
FORD	****
SMITH	***
ALLEN	****
WARD	****
MARTIN	****
TURNER	****
ADAMS	****

```
JAMES      ***
MILLER     ****
```

20 rows selected.

```
SQL> CL SCRCCL SC
SP2-0158: unknown CLEAR option "SCRCCL"
SP2-0158: unknown CLEAR option "SC"
SQL> R
  1  SELECT ENAME
  2    ,TRANSLATE(SAL
  3                , '0123456789'
  4                , '*****')
  5                ) AS SAL1
  6* FROM EMP
```

ENAME	SAL1
SURESH	****
HIMANSHU	****
KRISHNA	****
ARJUN	****
SAI	*****
SUBHASH	*****
KING	****
BLAKE	****
CLARK	****
JONES	****
SCOTT	****
FORD	****
SMITH	***
ALLEN	****
WARD	****
MARTIN	****
TURNER	****
ADAMS	****
JAMES	***
MILLER	****

20 rows selected.

```
SQL> R
  1  SELECT ENAME
  2    ,TRANSLATE(SAL
  3                , '0123456789'
  4                , '*****')
  5                ) AS SAL1
  6* FROM EMP
```

ENAME	SAL1
SURESH	****
HIMANSHU	****
KRISHNA	****
ARJUN	****
SAI	*****
SUBHASH	*****
KING	****
BLAKE	****
CLARK	****
JONES	****
SCOTT	****
FORD	****
SMITH	***

```
ALLEN      ****
WARD       ****
MARTIN     ****
TURNER     ****
ADAMS      ****
JAMES      ***
MILLER     ****
```

20 rows selected.

SQL> CL SCR

SQL> SPOOL OFF

4-Jun-2018 (GROUP FUNCTIONS/AGGR FUNCTIONS/ MULTI ROW FUNCTIONS)

COUNT

```
(
DISTINCT P1 -- COLUMN/LITERAL/*
)
```

COUNT(LITERAL) OR COUNT(*) OUTPUT WILL GIVE YOU SAME RESULTS BUT HOW IT EXECUTES INTERNALLY WILL DIFFER

COUNT(COLUMN) -- IT MAY OR MAY NOT GIVE SAME RESULTES

I.E IF COLUMN CONTAINS NULL VAUES IT WILL NOT CONSIER THOESE NULL VALUES.

WHENEVER WE ARE DOING COUNT(LITERAL) IF YOU ARE USING DISTINCT KEY WORD WE SHOULD KNOW HOW ITS EXECUTE

COUNT(LITERAL) AND COUNT(DISTINCT LITERAL)

DIFF BETWEEN COUNT(*) ,COUNT(LITERAL)

TOTCNT	TOTCNT1	TOTCNT2
14	14	4

MIN

```
(
DISTINCT P1-- COLUMN/LITERAL
)
```

MAX

```
(  
DISTINCT P1--COLUMN/LITERAL  
)  
SUM  
(  
DISTINCT P1-- COLUMN/LITERAL  
)  
AVG  
(  
DISTINCT P1-- COLUMN/LITERAL  
)
```

```
AVG(SAL) = SUM(SAL) / COUNT(SAL)  
AVG(DISTINCT SAL) = SUM(DISTINCT SAL) / COUNT(DISTINCT SAL)
```

WITH MIN AND MAX WHETHER WE ARE USING DISTINCT ON COLUMN OR NOT
OUTPUT WILL BE THE SAME

group by
having

RULE 1 :

AGGR FUNCTIONS RESULTS ON WHAT BASIS YOU WANT SPLIT THOSE COLUMNS
SHOULD BE THERE IN GROUP BY CLAUSE
TO UNDERSTAND RESULT PROPERLY SAME COLUMNS WE HAVE TO KEEP IN SELECT
STATEMENT ALSO--
IT IS NOT MANDATORY BUT IF YOU ARE NOT KEEPING YOU WILL NOT
UNDERSTAND THE FINAL RESULTS**

RULE 2:

WHENEVER GROUP FUNCTIONS ARE THERE IN SELECT STATEMENT IF ANY
OTHER COLUMNS/SINGLE ROW FUNCTIONS ARE
THERE IN SAME SELECT STATEMENT THOSE COLUMNS/SINGLE ROW FUNCTIONS
SHOULD BE THERE IN GROUP BY CLAUSE
ELSE IT WILL GIVE YOU AN ERROR

DIFF BETWEEN WHERE AND HAVING

WHERE CLAUSE IS USED TO FILTER NORMAL ROWS
IT WILL EXECUTE BEFORE GROUP BY CLAUSE
IF YOU WANT TO HAVE WHERE CLAUSE GROUP BY CLAUSE IS NOT MANDATORY

HAVING CALUSE IS USED TO FILTER ON GROUP FUNCTIONS/AGGR FUNCTIONS RESULTS

IT WILL EXECUTE AFTER GROUP BY CLAUSE

WITH OUT GROUP BY CLAUSE HAVING CLAUSE WILL NOT COME.

TOTCNT

14 EMPLOYEES ARE THERE

TOTCNT	TOTSAL	LOSAL	HISAL
AVGSAL			
14	40000	800	5000
2320			

ADAR

ADARNUM

GEN

AGE

MOB

DOB

STATE

DISTICT

QUAL

```
SELECT COUNT(1)--120 CR
FROM ADAR;
```

```
SELECT COUNT(1)--2 CR
FROM ADAR
WHERE STATE ='KARNATAKA'
```

```
SELECT COUNT(1)--2 CR
FROM ADAR
WHERE STATE ='KARNATAKA'
AND GEN ='F'
```

```
SELECT COUNT(1)--2 CR
FROM ADAR
WHERE STATE ='KARNATAKA'
AND GEN ='F'
AND QUAL IS NOT NULL;
```

```
SELECT COUNT(1)--120 CR
FROM ADAR;
```

KARNAKTAKA	MALE	1.5CR
KARNAKTA	FEMALAE	.5CR
ANDAHRA	MALE	2CR
ANDHARA	FEMAL	1CR

```
SELECT STATE
      ,GEN
      ,COUNT(1)
FROM ADAR
GROUP BY
STATE
      ,GEN
```

KARNAKTAKA	MALE	QUAL 1.5CR
KARNAKTA	FEMALAE	.5CR
ANDAHRA	MALE	2CR
ANDHARA	FEMAL	1CR

```
SELECT STATE
      , GEN
      , QUAL
      , COUNT (1)
```

```
FROM ADAR
GROUP BY
STATE
      , GEN
, QUAL
```

-- STATE GEN HOW MANY PEOPLE ARE NOT EDUCATED ?

```
SELECT STATE
      , GEN
      , QUAL
      , COUNT (1)
```

```
FROM ADAR
WHERE QUAL IS NULL-- 50CR
GROUP BY
STATE
      , GEN
, QUAL
```

KARNATAKA	2CR
ANSHARA	3 CR
TELANAG	4CR
TAMAIL	5CR
MAHAR	6CR

```
GROUP BY
-----
```

WHENVER WE WANT AGGR RESULTS FOR EACH GROUP SEPARATELY THEN WE HAVE
TO GO FOR GROUP BY CLAUSE
IT WILL COME AFTER WHERE CLAUSE

```
SELECT COUNT (1)
FROM ADAR
GROUP BY STATE;
```


2CR
3CR
5CR
7CR

```
SELECT STATE,COUNT(1)
FROM ADAR
GROUP BY STATE;
```

KARANTATAK 2CR
ANDARANRA 3CR
TELANGALAN5CR
TAMILNADU 7CR

4-Jun-2018 (GROUP FUNCTIONS/AGGR FUNCTIONS/ MULTI ROW FUNCTIONS)-Exam

```
SQL> DESCRIBE EMP
```

Name	Null?	Type
EMPNO		NUMBER (4)
ENAME		VARCHAR2 (30)
JOB		VARCHAR2 (9)
MGR		NUMBER (4)
HIREDATE		DATE
SAL		NUMBER (8,2)
COMM		NUMBER (7,2)
DEPTNO		NUMBER (2)

```
SQL> SELECT * FROM EMP;
```

EMPNO	ENAME	JOB	MGR	HIREDATE
102	SURESH			02-JUN-18
2100				
105	HIMANSHU			01-JUN-18
2300				
101	KRISHNA			17-MAY-18
2500				

EMPNO	ENAME	JOB	MGR	HIREDATE
101	ARJUN			02-JUN-18
2000				
110	SAI			
25540				

111 SUBHASH
22590

30-MAY-18

EMPNO	ENAME		JOB	MGR	HIREDATE
SAL	COMM	DEPTNO			
7839	KING	10	PRESIDENT		17-NOV-81
5000					
7698	BLAKE	30	MANAGER	7839	01-MAY-81
2850					
7782	CLARK	10	MANAGER	7839	09-JUN-81
2450					
EMPNO	ENAME		JOB	MGR	HIREDATE
SAL	COMM	DEPTNO			
7566	JONES	20	MANAGER	7839	02-APR-81
2975					
7788	SCOTT	20	ANALYST	7566	19-APR-87
3000					
7902	FORD	20	ANALYST	7566	03-DEC-81
3000					
EMPNO	ENAME		JOB	MGR	HIREDATE
SAL	COMM	DEPTNO			
7369	SMITH	20	CLERK	7902	17-DEC-80
800					
7499	ALLEN	30	SALESMAN	7698	20-FEB-81
1600	300				
7521	WARD	30	SALESMAN	7698	22-FEB-81
1250	500				
EMPNO	ENAME		JOB	MGR	HIREDATE
SAL	COMM	DEPTNO			
7654	MARTIN	30	SALESMAN	7698	28-SEP-81
1250	1400				
7844	TURNER	30	SALESMAN	7698	08-SEP-81
1500	0				
7876	ADAMS	20	CLERK	7788	23-MAY-87
1100					
EMPNO	ENAME		JOB	MGR	HIREDATE
SAL	COMM	DEPTNO			

7900 JAMES 950	30	CLERK	7698 03-DEC-81
7934 MILLER 1300	10	CLERK	7782 23-JAN-82

20 rows selected.

```
SQL> SET PAGESIZE 200
SQL> SET LINESIZE 200
SQL> R
1* SELECT * FROM EMP
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL
COMM	DEPTNO				
102	SURESH			02-JUN-18	2100
105	HIMANSHU			01-JUN-18	2300
101	KRISHNA			17-MAY-18	2500
101	ARJUN			02-JUN-18	2000
110	SAI				25540
111	SUBHASH			30-MAY-18	22590
7839	KING	PRESIDENT		17-NOV-81	5000
7698	BLAKE	MANAGER	7839	01-MAY-81	2850
7782	CLARK	MANAGER	7839	09-JUN-81	2450
7566	JONES	MANAGER	7839	02-APR-81	2975
7788	SCOTT	ANALYST	7566	19-APR-87	3000
7902	FORD	ANALYST	7566	03-DEC-81	3000
7369	SMITH	CLERK	7902	17-DEC-80	800
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600
7521	WARD	SALESMAN	7698	22-FEB-81	1250
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250
7844	TURNER	SALESMAN	7698	08-SEP-81	1500
7876	ADAMS	CLERK	7788	23-MAY-87	1100
7900	JAMES	CLERK	7698	03-DEC-81	950
7934	MILLER	CLERK	7782	23-JAN-82	1300

20 rows selected.

```
SQL> CL SCR
```

```
SQL> R
1* SELECT * FROM EMP
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL
COMM	DEPTNO				

	102	SURESH		02-JUN-18	2100
	105	HIMANSHU		01-JUN-18	2300
	101	KRISHNA		17-MAY-18	2500
	101	ARJUN		02-JUN-18	2000
	110	SAI			25540
	111	SUBHASH		30-MAY-18	22590
	7839	KING	PRESIDENT	17-NOV-81	5000
10	7698	BLAKE	MANAGER	7839 01-MAY-81	2850
30	7782	CLARK	MANAGER	7839 09-JUN-81	2450
10	7566	JONES	MANAGER	7839 02-APR-81	2975
20	7788	SCOTT	ANALYST	7566 19-APR-87	3000
20	7902	FORD	ANALYST	7566 03-DEC-81	3000
20	7369	SMITH	CLERK	7902 17-DEC-80	800
20	7499	ALLEN	SALESMAN	7698 20-FEB-81	1600
300	30				
500	7521	WARD	SALESMAN	7698 22-FEB-81	1250
500	30				
1400	7654	MARTIN	SALESMAN	7698 28-SEP-81	1250
1400	30				
0	7844	TURNER	SALESMAN	7698 08-SEP-81	1500
0	30				
20	7876	ADAMS	CLERK	7788 23-MAY-87	1100
20	7900	JAMES	CLERK	7698 03-DEC-81	950
30	7934	MILLER	CLERK	7782 23-JAN-82	1300
10					

20 rows selected.

SQL> DELETE FROM EMP WHERE JOB IS NULL;

6 rows deleted.

SQL> COMMIT;

Commit complete.

SQL> CL SCR

SQL> SELECT * FROM EMP;

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL
COMM	DEPTNO				
	7839	KING	PRESIDENT	17-NOV-81	5000
10	7698	BLAKE	MANAGER	7839 01-MAY-81	2850
30	7782	CLARK	MANAGER	7839 09-JUN-81	2450
10	7566	JONES	MANAGER	7839 02-APR-81	2975
20	7788	SCOTT	ANALYST	7566 19-APR-87	3000
20					

20	7902 FORD	ANALYST	7566 03-DEC-81	3000
20	7369 SMITH	CLERK	7902 17-DEC-80	800
300	7499 ALLEN 30	SALESMAN	7698 20-FEB-81	1600
500	7521 WARD 30	SALESMAN	7698 22-FEB-81	1250
1400	7654 MARTIN 30	SALESMAN	7698 28-SEP-81	1250
0	7844 TURNER 30	SALESMAN	7698 08-SEP-81	1500
20	7876 ADAMS	CLERK	7788 23-MAY-87	1100
30	7900 JAMES	CLERK	7698 03-DEC-81	950
10	7934 MILLER	CLERK	7782 23-JAN-82	1300

14 rows selected.

SQL> CL SCR

SQL> SELECT *
2 FROM EMP;

COMM	EMPNO	ENAME	DEPTNO	JOB	MGR	HIREDATE	SAL
-----	-----	-----	-----	-----	-----	-----	-----
10	7839	KING		PRESIDENT		17-NOV-81	5000
30	7698	BLAKE		MANAGER	7839	01-MAY-81	2850
10	7782	CLARK		MANAGER	7839	09-JUN-81	2450
20	7566	JONES		MANAGER	7839	02-APR-81	2975
20	7788	SCOTT		ANALYST	7566	19-APR-87	3000
20	7902	FORD		ANALYST	7566	03-DEC-81	3000
20	7369	SMITH		CLERK	7902	17-DEC-80	800
300	7499	ALLEN 30		SALESMAN	7698	20-FEB-81	1600
500	7521	WARD 30		SALESMAN	7698	22-FEB-81	1250
1400	7654	MARTIN 30		SALESMAN	7698	28-SEP-81	1250
0	7844	TURNER 30		SALESMAN	7698	08-SEP-81	1500
20	7876	ADAMS		CLERK	7788	23-MAY-87	1100
30	7900	JAMES		CLERK	7698	03-DEC-81	950
10	7934	MILLER		CLERK	7782	23-JAN-82	1300

14 rows selected.

SQL> CL SCR

SQL> SELECT COUNT(1)

```
2 FROM EMP;
```

```
COUNT(1)
-----
14
```

```
SQL> CL SCR
```

```
SQL> SELECT COUNT(*)
2 FROM EMP;
```

```
COUNT(*)
-----
14
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT COUNT(*) AS TOTCNT
2* FROM EMP
SQL> /
```

```
TOTCNT
-----
14
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT * AS TOTCNT
2* FROM EMP
SQL> /
SELECT * AS TOTCNT
*
```

```
ERROR at line 1:
ORA-00923: FROM keyword not found where expected
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT *
2* FROM EMP
SQL> /
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL
COMM	DEPTNO				
	7839 KING	PRESIDENT		17-NOV-81	5000
10	7698 BLAKE	MANAGER	7839	01-MAY-81	2850
30	7782 CLARK	MANAGER	7839	09-JUN-81	2450
10	7566 JONES	MANAGER	7839	02-APR-81	2975
20	7788 SCOTT	ANALYST	7566	19-APR-87	3000
20	7902 FORD	ANALYST	7566	03-DEC-81	3000
20	7369 SMITH	CLERK	7902	17-DEC-80	800

300	7499 ALLEN 30	SALESMAN	7698 20-FEB-81	1600
500	7521 WARD 30	SALESMAN	7698 22-FEB-81	1250
1400	7654 MARTIN 30	SALESMAN	7698 28-SEP-81	1250
0	7844 TURNER 30	SALESMAN	7698 08-SEP-81	1500
20	7876 ADAMS	CLERK	7788 23-MAY-87	1100
30	7900 JAMES	CLERK	7698 03-DEC-81	950
10	7934 MILLER	CLERK	7782 23-JAN-82	1300

14 rows selected.

SQL> ED
Wrote file afiedt.buf

```

1  SELECT COUNT(* )
2*  FROM EMP
SQL> /

```

```

COUNT(*)
-----
14

```

SQL> ED
Wrote file afiedt.buf

```

1  SELECT 1
2*  FROM EMP
SQL> /

```

```

1
-----
1
1
1
1
1
1
1
1
1
1
1
1
1
1
1
1

```

14 rows selected.

SQL> ED
Wrote file afiedt.buf

```

1  SELECT COUNT(1)
2*  FROM EMP
SQL> /

```

```

COUNT(1)
-----
14

```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT COUNT(1),COUNT(*)
2*  FROM EMP
SQL> /
```

```
      COUNT(1)      COUNT(*)
-----
          14          14
```

```
SQL> CL SCR
```

```
SQL> SELECT COUNT(*)
2  FROM EMP;
```

```
      COUNT(*)
-----
          14
```

```
SQL> SELECT *
2  FROM EMP;
```

COMM	EMPNO	ENAME	DEPTNO	JOB	MGR	HIREDATE	SAL
	7839	KING		PRESIDENT		17-NOV-81	5000
10	7698	BLAKE		MANAGER	7839	01-MAY-81	2850
30	7782	CLARK		MANAGER	7839	09-JUN-81	2450
10	7566	JONES		MANAGER	7839	02-APR-81	2975
20	7788	SCOTT		ANALYST	7566	19-APR-87	3000
20	7902	FORD		ANALYST	7566	03-DEC-81	3000
20	7369	SMITH		CLERK	7902	17-DEC-80	800
300	7499	ALLEN	30	SALESMAN	7698	20-FEB-81	1600
500	7521	WARD	30	SALESMAN	7698	22-FEB-81	1250
1400	7654	MARTIN	30	SALESMAN	7698	28-SEP-81	1250
0	7844	TURNER	30	SALESMAN	7698	08-SEP-81	1500
20	7876	ADAMS		CLERK	7788	23-MAY-87	1100
30	7900	JAMES		CLERK	7698	03-DEC-81	950
10	7934	MILLER		CLERK	7782	23-JAN-82	1300

```
14 rows selected.
```

```
SQL> CL SCR
```

```
SQL> SELECT *
2  FROM EMP;
```


COMM	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL
	DEPTNO					
10	7839	KING	PRESIDENT		17-NOV-81	5000
30	7698	BLAKE	MANAGER	7839	01-MAY-81	2850
10	7782	CLARK	MANAGER	7839	09-JUN-81	2450
20	7566	JONES	MANAGER	7839	02-APR-81	2975
20	7788	SCOTT	ANALYST	7566	19-APR-87	3000
20	7902	FORD	ANALYST	7566	03-DEC-81	3000
20	7369	SMITH	CLERK	7902	17-DEC-80	800
300	7499	ALLEN	SALESMAN	7698	20-FEB-81	1600
500	7521	WARD	SALESMAN	7698	22-FEB-81	1250
1400	7654	MARTIN	SALESMAN	7698	28-SEP-81	1250
0	7844	TURNER	SALESMAN	7698	08-SEP-81	1500
20	7876	ADAMS	CLERK	7788	23-MAY-87	1100
30	7900	JAMES	CLERK	7698	03-DEC-81	950
10	7934	MILLER	CLERK	7782	23-JAN-82	1300

14 rows selected.

```
SQL> SELECT COUNT(*)
      2 ,COUNT(1)
      3 FROM EMP;
```

COUNT(*)	COUNT(1)
14	14

```
SQL> CL SCR
```

```
SQL> DESC B26_AKS_STD
```

Name

Null? Type

```
-----
ID
NUMBER
NAME
VARCHAR2(50)
MOB
NUMBER(10)
FEES
NUMBER(7,2)
DOJ
DATE
GEN
CHAR(1)
```

```
SQL> SELECT COUNT(1)
      2 ,COUNT(*)
      3 FROM B26_AKS_STD;
```

```
      COUNT(1)      COUNT(*)
-----
0
```

```
SQL> ED
Wrote file afiedt.buf
```

```
      1 SELECT COUNT(1)
      2 ,COUNT(*)
      3* FROM AKS_B26_STD
```

```
SQL> /
```

```
FROM AKS_B26_STD
      *
```

```
ERROR at line 3:
ORA-00942: table or view does not exist
```

```
SQL> ED
Wrote file afiedt.buf
```

```
      1 SELECT COUNT(1)
      2 ,COUNT(*)
      3* FROM B26_AKSHARA_STD
```

```
SQL> /
```

```
FROM B26_AKSHARA_STD
      *
```

```
ERROR at line 3:
ORA-00942: table or view does not exist
```

```
SQL> CL SCR
```

```
SQL> SELECT TABLE_NAME
      2 FROM USER_TABLES
      3 WHERE TABLE_NAME LIKE '%B26%';
```

```
TABLE_NAME
-----
B26_AKS_STD
B26_AKS_STD_1
B26_CHAR_EX_1
B26_CHAR_EX_2
B26_CHAR_VARCHAR2_EX
B26_AKSHARA_STUDENT
B26_AKSHARA_STUDENT_1
B26_NULL_SPACE_EX
B26_T1
B26_T2
B26_AKS_STD_7
B26_T3
B26_T4
B26_T5
B26_T7
B26_78
B26_AKS_STD_2
B26_AKS_STD_3
B26_AKS_STD_B26_AKS_STD_B26
B26_AKS_STD_4
B26_AKS_STD
```

21 rows selected.

```
SQL> SELECT COUNT(1)
      2 FROM B26_AKSHARA_STUDENT;
```

```
      COUNT(1)
-----
          9
```

SQL> CL SCR

```
SQL> R
      1 SELECT COUNT(1)
      2* FROM B26_AKSHARA_STUDENT
```

```
      COUNT(1)
-----
          9
```

```
SQL> SELECT *
      2 FROM B26_AKSHARA_STUDENT;
```

SNO	NAME	MOB	FEEES	DOJ	G
101	ABHISHEK	12345	5000	03-MAY-18	M
102	SURESH KUMAR SINGH S	1234612347	6000	03-MAY-18	B
103	KRISHNA			06-MAY-18	M
104	ARJUN			06-MAY-18	M
105	HIMANSHU	98765	4000	26-APR-18	M
106	PUNAM	98766		26-APR-18	F
107	Teja			26-APR-18	F
108	AKSHARA				
	NULL				

9 rows selected.

SQL> CL SCR

```
SQL> R
      1 SELECT *
      2* FROM B26_AKSHARA_STUDENT
```

SNO	NAME	MOB	FEEES	DOJ	G
101	ABHISHEK	12345	5000	03-MAY-18	M
102	SURESH KUMAR SINGH S	1234612347	6000	03-MAY-18	B
103	KRISHNA			06-MAY-18	M
104	ARJUN			06-MAY-18	M
105	HIMANSHU	98765	4000	26-APR-18	M
106	PUNAM	98766		26-APR-18	F
107	Teja			26-APR-18	F
108	AKSHARA				
	NULL				

9 rows selected.

```
SQL> SELECT COUNT(*) AS TOTCNT
      2 FROM B26_AKSHARA_STUDENT;
```

```
      TOTCNT
-----
          9
```

SQL> CL SCR

```
SQL> R
1  SELECT COUNT(*) AS TOTCNT
2*  FROM B26_AKSHARA_STUDENT
```

```
TOTCNT
-----
          9
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT *
2*  FROM B26_AKSHARA_STUDENT
SQL> /
```

SNO	NAME	MOB	FEEES	DOJ	G
101	ABHISHEK	12345	5000	03-MAY-18	M
102	SURESH KUMAR SINGH S	1234612347	6000	03-MAY-18	B
103	KRISHNA			06-MAY-18	M
104	ARJUN			06-MAY-18	M
105	HIMANSHU	98765	4000	26-APR-18	M
106	PUNAM	98766		26-APR-18	F
107	Teja			26-APR-18	F
108	AKSHARA				
	NULL				

9 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT 100
2*  FROM B26_AKSHARA_STUDENT
SQL> /
```

```
100
-----
100
100
100
100
100
100
100
100
100
100
100
```

9 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT COUNT(100)
2*  FROM B26_AKSHARA_STUDENT
SQL> /
```

```
COUNT(100)
-----
          9
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT COUNT(1)
2* FROM B26_AKSHARA_STUDENT
SQL> /
```

```
COUNT(1)
-----
          9
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT 1
2* FROM B26_AKSHARA_STUDENT
SQL> /
```

```
          1
-----
          1
          1
          1
          1
          1
          1
          1
          1
          1
          1
          1
```

9 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT COUNT(1)
2* FROM B26_AKSHARA_STUDENT
SQL> /
```

```
COUNT(1)
-----
          9
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT 0
2* FROM B26_AKSHARA_STUDENT
SQL> /
```

```
          0
-----
          0
          0
          0
          0
          0
          0
          0
          0
          0
          0
```

9 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT COUNT(0)
2* FROM B26_AKSHARA_STUDENT
SQL> /
```

```
COUNT(0)
-----
          9
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT 'SURESH'
2* FROM B26_AKSHARA_STUDENT
SQL> /
```

```
'SURES
-----
SURESH
SURESH
SURESH
SURESH
SURESH
SURESH
SURESH
SURESH
SURESH
SURESH
```

9 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT COUNT('SURESH')
2* FROM B26_AKSHARA_STUDENT
SQL> /
```

```
COUNT('SURESH')
-----
          9
```

```
SQL> CL SCR
```

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
2  ,SAL
3  ,COMM
4  FROM EMP;
```

ENAME	SAL	COMM
-----	-----	-----
KING	5000	
BLAKE	2850	
CLARK	2450	
JONES	2975	
SCOTT	3000	
FORD	3000	
SMITH	800	
ALLEN	1600	300
WARD	1250	500
MARTIN	1250	1400
TURNER	1500	0
ADAMS	1100	

```
JAMES          950
MILLER         1300
```

14 rows selected.

```
SQL> SELECT COUNT(ENAME) AS TOTCNT
      2 ,COUNT(SAL) AS TOTCNT1
      3 ,COUNT(COMM) AS TOTCNT2
      4 FROM EMP;
```

TOTCNT	TOTCNT1	TOTCNT2
14	14	4

SQL>

SQL> CL SCR

SQL> R

```
1 SELECT COUNT(ENAME) AS TOTCNT
2 ,COUNT(SAL) AS TOTCNT1
3 ,COUNT(COMM) AS TOTCNT2
4* FROM EMP
```

TOTCNT	TOTCNT1	TOTCNT2
14	14	4

SQL> SELECT ENAME

```
2 ,SAL
3 ,COMM
4 FROM EMP;
```

ENAME	SAL	COMM
KING	5000	
BLAKE	2850	
CLARK	2450	
JONES	2975	
SCOTT	3000	
FORD	3000	
SMITH	800	
ALLEN	1600	300
WARD	1250	500
MARTIN	1250	1400
TURNER	1500	0
ADAMS	1100	
JAMES	950	
MILLER	1300	

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1 SELECT ENAME
2 ,SAL
3 ,COMM
4 FROM EMP
5* WHERE COMM IS NULL
```

SQL> /

ENAME	SAL	COMM
KING	5000	

BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	3000
FORD	3000
SMITH	800
ADAMS	1100
JAMES	950
MILLER	1300

10 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT COUNT(1)
2  FROM EMP
3* WHERE COMM IS NULL
SQL> /
```

```
      COUNT(1)
-----
          10
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT 1
2  FROM EMP
3* WHERE COMM IS NULL
SQL> /
```

```
      1
-----
      1
      1
      1
      1
      1
      1
      1
      1
      1
      1
      1
      1
```

10 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT COUNT(1)
2  FROM EMP
3* WHERE COMM IS NULL
SQL> /
```

```
      COUNT(1)
-----
          10
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT COUNT(1)
2* FROM EMP
```


3 /

COUNT(1)

```
-----
      14
```

SQL> ED

Wrote file afiedt.buf

1 SELECT COUNT(1) , COUNT(COMM)

2* FROM EMP

SQL> /

COUNT(1) COUNT(COMM)

```
-----
      14          4
```

SQL> ED

Wrote file afiedt.buf

1 SELECT COUNT(1) - COUNT(COMM) AS TOT_EMPCNT_WHONOTHAVINGCOMM

2* FROM EMP

SQL> /

TOT_EMPCNT_WHONOTHAVINGCOMM

```
-----
                        10
```

SQL> ED

Wrote file afiedt.buf

1 SELECT COUNT(*)

2* FROM EMP

SQL> /

COUNT(*)

```
-----
      14
```

SQL> ED

Wrote file afiedt.buf

1 SELECT COUNT(COMM)

2* FROM EMP

SQL> /

COUNT(COMM)

```
-----
      4
```

SQL> CL SCR

SQL> SELECT ENAME, SAL

2 FROM EMP;

ENAME	SAL
KING	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	3000
FORD	3000
SMITH	800

ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

```
SQL> SELECT DISTINCT SAL
      2 FROM EMP;
```

SAL
5000
2450
1300
2850
1250
2975
1100
3000
800
1600
1500
950

12 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
      1 SELECT COUNT( DISTINCT SAL)
      2* FROM EMP
SQL> /
```

COUNT(DISTINCTSAL)
12

```
SQL> CL SCR
```

```
SQL> SELECT COUNT(SAL)
      2 ,COUNT(DISTINCT SAL)
      3 FROM EMPED
      4
```

```
SQL>
```

```
SQL> .
```

```
SQL>
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
      1 SELECT COUNT(SAL) AS C1
      2 ,COUNT(DISTINCT SAL) AS C2
      3* FROM EMP
SQL> /
```

C1	C2
14	12

```
SQL> CL SCR
```

```
SQL> R
  1  SELECT COUNT(SAL) AS C1
  2  ,COUNT(DISTINCT SAL) AS C2
  3* FROM EMP
```

C1	C2
14	12

```
SQL> CL SCR
```

```
SQL> SELECT 1
  2  FROM EMP;
```

1
1
1
1
1
1
1
1
1
1
1
1
1
1
1

14 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT 1
  2* FROM EMP
SQL> /
```

1
1
1
1
1
1
1
1
1
1
1
1
1
1
1

14 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT DISTINCT 1
  2* FROM EMP
```

SQL> /

1
1

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COUNT(DISTINCT 1)
2* FROM EMP
SQL> /
```

COUNT(DISTINCT1)
1

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COUNT(DISTINCT 1)
2  ,COUNT(1)
3* FROM EMP
SQL> /
```

COUNT(DISTINCT1)	COUNT(1)
1	14

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COUNT(DISTINCT 1)
2  ,COUNT(1)
3* FROM EMP
```

SQL> CL SC

SP2-0158: unknown CLEAR option "SC"

SQL>

SQL>

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL>

SQL> CL SCR

```

SQL> SELECT COUNT('SURESH') AS C1
2  ,COUNT(DISTINCT 'SURESH') AS C2
3  FROM EMP;
```

C1	C2
14	1

SQL>

SQL>

SQL> CL SCR

SQL> SELECT ENAME

```

2  ,SAL
3  ,COMM
4  FROM EMP;
```

ENAME	SAL	COMM
KING	5000	

BLAKE	2850	
CLARK	2450	
JONES	2975	
SCOTT	3000	
FORD	3000	
SMITH	800	
ALLEN	1600	300
WARD	1250	500
MARTIN	1250	1400
TURNER	1500	0
ADAMS	1100	
JAMES	950	
MILLER	1300	

14 rows selected.

```
SQL> SELECT COUNT(ENAME) AS C1
      2 ,COUNT(DISTINCT ENAME) AS C21
      3 ,COUNT(SAL) AS C3
      4 ,COUNT(DISTINNCT SAL) AS C4
      5 ,COUNT(COMM) AS C4
      6 ,COUNT(DISTINCT COMM) AS C6
      7
```

SQL>

SQL> ED

Wrote file afiedt.buf

```
      1 SELECT COUNT(ENAME) AS C1
      2 ,COUNT(DISTINCT ENAME) AS C2
      3 ,COUNT(SAL) AS C3
      4 ,COUNT(DISTINCT SAL) AS C4
      5 ,COUNT(COMM) AS C5
      6 ,COUNT(DISTINCT COMM) AS C6
      7* FROM EMP
SQL> /
```

C1	C2	C3	C4	C5	C6
14	14	14	12	4	4

SQL> CL SCR

```
SQL> SELLECT COUNT(1) AS TOTCNT
SP2-0734: unknown command beginning "SELLECT CO..." - rest of line ignored.
```

SQL>

SQL>

```
SQL> SELECT COUNT(1) AS TOTCNT
      2 FROM EMP;
```

TOTCNT
14

SQL> ED

Wrote file afiedt.buf

```
      1 SELECT COUNT(1)
      2      || ' EMPLOYEES ARE THERE' AS TOTCNT
      3* FROM EMP
SQL> /
```

TOTCNT

14 EMPLOYEES ARE THERE

SQL> CL SCR

SQL> R

```
1  SELECT COUNT(1)
2      || ' EMPLOYEES ARE THERE' AS TOTCNT
3*  FROM EMP
```

TOTCNT

14 EMPLOYEES ARE THERE

```
SQL> SELECT COUNT(1) AS TOTCNT
2      ,SUM(SAL) AS TOTSAL
3      ,MIN(SAL) AS LOSAL
4      ,MAX(SAL) AS HISAL
5      ,AVG(SAL) AS AVG_SAL
6
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT COUNT(1) AS TOTCNT
2      ,SUM(SAL) AS TOTSAL
3      ,MIN(SAL) AS LOSAL
4      ,MAX(SAL) AS HISAL
5      ,AVG(SAL) AS AVG_SAL
6*  FROM EMP
SQL> /
```

TOTCNT	TOTSAL	LOSAL	HISAL	AVG_SAL
14	29025	800	5000	2073.21429

SQL> ED

Wrote file afiedt.buf

```
1  SELECT COUNT(1) AS TOTCNT
2      ,SUM(SAL) AS TOTSAL
3      ,MIN(SAL) AS LOSAL
4      ,MAX(SAL) AS HISAL
5      ,TRUNC( AVG(SAL) ) AS AVG_SAL
6*  FROM EMP
SQL> /
```

TOTCNT	TOTSAL	LOSAL	HISAL	AVG_SAL
14	29025	800	5000	2073

SQL> CL SCR

SQL> R

```
1  SELECT COUNT(1) AS TOTCNT
2      ,SUM(SAL) AS TOTSAL
3      ,MIN(SAL) AS LOSAL
4      ,MAX(SAL) AS HISAL
5      ,TRUNC( AVG(SAL) ) AS AVG_SAL
6*  FROM EMP
```

TOTCNT	TOTSAL	LOSAL	HISAL	AVG_SAL
14	29025	800	5000	2073

SQL> CL SCR

```
SQL> SELECT COUNT(1) AS TOTCNT
2          ,SUM(1) AS
3
```

```
SQL> .
```

```
SQL>
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT COUNT(1) AS C1
2          ,SUM(1) AS C2
3* FROM EMP
SQL> /
```

C1	C2
14	14

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT 1 AS C1
2          ,1 AS C2
3* FROM EMP
SQL> /
```

C1	C2
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1

```
14 rows selected.
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT COUNT(1) AS C1
2          ,1 AS C2
3* FROM EMP
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT COUNT(1) AS C1
2          ,SUM(1) AS C2
3* FROM EMP
SQL> /
```

C1	C2
14	14

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT COUNT(100) AS C1
2      ,SUM(100) AS C2
3* FROM EMP
4  /

```

C1	C2
14	1400

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COUNT(100) AS C1
2      ,SUM(100) AS C2
3* FROM EMP

```

```

SQL>
SQL> /

```

C1	C2
14	1400

```

SQL> CL SCR

```

```

SQL> R
1  SELECT COUNT(100) AS C1
2      ,SUM(100) AS C2
3* FROM EMP

```

C1	C2
14	1400

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COUNT(DISTINCT 100) AS C1
2      ,SUM(DISTINCT 100) AS C2
3* FROM EMP
SQL> /

```

C1	C2
1	100

```

SQL> CL SCR

```

```

SQL> SELECT ENAME
2      ,SAL
3

```

```

SQL>
SQL> SELECT ENAME
2      ,SAL
3  FROM EMP;

```

ENAME	SAL
KING	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	3000
FORD	3000

SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

```
SQL> SELECT SUM(SAL) AS TOTSAL
  2      ,SUM(DISTINCT SAL) AS TOTSAL1
  3 FROM EMP;
```

TOTSAL	TOTSAL1
29025	24775

```
SQL> CL SCR
```

```
SQL> SELECT ENAME,SAL
  2 FROM EMP;
```

ENAME	SAL
KING	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	3000
FORD	3000
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

```
SQL> SELECT MIN(SAL)
  2      ,MIN(DISTINCT SAL)
  3 FROM EMP;
```

MIN(SAL)	MIN(DISTINCTSAL)
800	800

```
SQL> ED
Wrote file afiedt.buf
```

```
  1 SELECT MIN(SAL) AS C1
  2      ,MIN(DISTINCT SAL) AS C2
  3* FROM EMP
SQL> /
```

C1	C2
800	800

```
SQL> ED
```

Wrote file afiedt.buf

```

1  SELECT MAX(SAL)  AS C1
2  ,MAX(DISTINCT SAL) AS C2
3* FROM EMP
SQL> /

```

C1	C2
5000	5000

SQL> CL SCR

```

SQL> SELECT MIN(100) AS C1
2  ,MAX(100) AS C2
3  FROM EMP;

```

C1	C2
100	100

SQL> CL SCR

```

SQL> SELECT AVG(SAL) AS C1
2  ,SUM(SAL)/AVG(SAL) AS C2
3  ,AVG(DISTINCT SAL) AS C3
4  ,SUM(DISTINCT SAL)/COUNT(DISTINCT SAL) AS C4
5

```

SQL>

SQL> ED

Wrote file afiedt.buf

```

1  SELECT AVG(SAL) AS C1
2  ,SUM(SAL)/COUNT(SAL) AS C2
3  ,AVG(DISTINCT SAL) AS C3
4  ,SUM(DISTINCT SAL)/COUNT(DISTINCT SAL) AS C4
5* FROM EMP
6  /

```

C1	C2	C3	C4
2073.21429	2073.21429	2064.58333	2064.58333

SQL> ED

Wrote file afiedt.buf

```

1  SELECT AVG(DISTINCT SAL) AS C1
2  ,SUM(DISTINCT SAL) AS C2
3  ,COUNT(DISTINCT SAL) AS C3
4* FROM EMP
SQL> /

```

C1	C2	C3
2064.58333	24775	12

SQL> CL SCR

```

SQL> SELECT AVG(1) AS C1
2  FROM EMP;

```

C1
1

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT AVG(100) AS C1
  2*   FROM EMP
SQL> /
```

```
      C1
-----
      100
```

```
SQL> CL SCR
```

```
SQL> SELECT COUNT(1)
  2  FROM EMP;
```

```
      COUNT(1)
-----
          14
```

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT COUNT(1)
  2  FROM EMP
  3* GROUP BY DEPTNO
SQL> /
```

```
      COUNT(1)
-----
          6
          5
          3
```

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT DEPTNO,COUNT(1)
  2  FROM EMP
  3* GROUP BY DEPTNO
SQL> /
```

```
      DEPTNO      COUNT(1)
-----
          30          6
          20          5
          10          3
```

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT DEPTNO
  2      ,COUNT(1) AS TOTCNT
  3  FROM EMP
  4* GROUP BY DEPTNO
SQL> /
```

```
      DEPTNO      TOTCNT
-----
          30          6
          20          5
          10          3
```

SQL> CL SCR

SQL> R

```
1  SELECT DEPTNO
2      ,COUNT(1) AS TOTCNT
3  FROM EMP
4*  GROUP BY DEPTNO
```

DEPTNO	TOTCNT
30	6
20	5
10	3

SQL> ED

Wrote file afiedt.buf

```
1  SELECT COUNT(1) AS TOTCNT
2  FROM EMP
3* GROUP BY DEPTNO
```

SQL> /

TOTCNT
6
5
3

SQL> CL SCR

SQL>

SQL> R

```
1  SELECT COUNT(1) AS TOTCNT
2  FROM EMP
3* GROUP BY DEPTNO
```

TOTCNT
6
5
3

SQL> ED

Wrote file afiedt.buf

```
1  SELECT DEPTNO
2      ,COUNT(1) AS TOTCNT
3  FROM EMP
4*  GROUP BY DEPTNO
```

SQL> /

DEPTNO	TOTCNT
30	6
20	5
10	3

SQL> ED

Wrote file afiedt.buf

```
1  SELECT 10 AS DEPTNO
2      ,COUNT(1) AS TOTCNT
3  FROM EMP
4*  WHERE DEPTNO = 10
```

SQL> /

DEPTNO	TOTCNT
-----	-----
10	3

SQL> CL SCR

SQL> R

```
1  SELECT 10 AS DEPTNO
2      ,COUNT(1) AS TOTCNT
3  FROM EMP
4* WHERE DEPTNO = 10
```

DEPTNO	TOTCNT
-----	-----
10	3

SQL> ED

Wrote file afiedt.buf

```
1  SELECT COUNT(1) AS TOTCNT
2  FROM EMP
3* WHERE DEPTNO IN( 10,20,30)
```

SQL> /

TOTCNT

14

SQL> ED

Wrote file afiedt.buf

```
1  SELECT DEPTNO
2      ,COUNT(1) AS TOTCNT
3  FROM EMP
4* GROUP BY DEPTNO
```

SQL> /

DEPTNO	TOTCNT
-----	-----
30	6
20	5
10	3

SQL> ED

Wrote file afiedt.buf

```
1  SELECT 10 AS DEPTNO
2      ,COUNT(1) AS TOTCNT
3  FROM EMP
4* WHERE DEPTNO = 10
```

SQL> /

DEPTNO	TOTCNT
-----	-----
10	3

SQL> ED

Wrote file afiedt.buf

```
1  SELECT 20 AS DEPTNO
2      ,COUNT(1) AS TOTCNT
3  FROM EMP
```

```

4* WHERE DEPTNO = 20
SQL> /

```

DEPTNO	TOTCNT
20	5

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT 30 AS DEPTNO
2      ,COUNT(1) AS TOTCNT
3  FROM EMP
4* WHERE DEPTNO = 30
SQL> /

```

DEPTNO	TOTCNT
30	6

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT 10 AS DEPTNO
2      ,COUNT(1) AS TOTCNT
3  FROM EMP
4  WHERE DEPTNO = 10
5  UNION ALL
6  SELECT 20 AS DEPTNO
7      ,COUNT(1) AS TOTCNT
8  FROM EMP
9  WHERE DEPTNO = 20
10 UNION ALL
11 SELECT 30 AS DEPTNO
12      ,COUNT(1) AS TOTCNT
13 FROM EMP
14* WHERE DEPTNO = 30
15 /

```

DEPTNO	TOTCNT
10	3
20	5
30	6

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT 10 AS DEPTNO
2      ,COUNT(1) AS TOTCNT
3  FROM EMP
4  WHERE DEPTNO = 10
5  UNION ALL
6  SELECT 20 AS DEPTNO
7      ,COUNT(1) AS TOTCNT
8  FROM EMP
9  WHERE DEPTNO = 20
10 UNION ALL
11 SELECT 30 AS DEPTNO
12      ,COUNT(1) AS TOTCNT
13 FROM EMP
14 WHERE DEPTNO = 30
15 UNION ALL
16 SELECT 40 AS DEPTNO

```

```

17      ,COUNT(1) AS TOTCNT
18 FROM EMP
19 WHERE DEPTNO = 40
20 UNION ALL
21 SELECT 50 AS DEPTNO
22      ,COUNT(1) AS TOTCNT
23 FROM EMP
24* WHERE DEPTNO = 50
SQL> /

```

DEPTNO	TOTCNT
10	3
20	5
30	6
40	0
50	0

```
SQL> CL SCR
```

```

SQL> SELECT DEPTNO
2      ,COUNT(1)
3 FROM EMP
4 GROUP BY DEPTNO;

```

DEPTNO	COUNT(1)
30	6
20	5
10	3

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT DEPTNO
2      ,COUNT(1)
3 FROM EMP
4 WHERE COUNT(1) > 3
5* GROUP BY DEPTNO
6 /
WHERE COUNT(1) > 3
*
ERROR at line 4:
ORA-00934: group function is not allowed here

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT DEPTNO
2      ,COUNT(1)
3 FROM EMP
4 GROUP BY DEPTNO
5* HAVING COUNT(1) > 3
SQL> /

```

DEPTNO	COUNT(1)
30	6
20	5

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT DEPTNO
2  ,COUNT(1)
3  FROM EMP
4  GROUP BY DEPTNO
5  HAVING COUNT(1) > 3
6* ORDER BY COUNT(1) ASC
SQL> /

```

DEPTNO	COUNT(1)
20	5
30	6

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT DEPTNO
2  ,COUNT(1)
3  FROM EMP
4  GROUP BY DEPTNO
5  HAVING COUNT(1) > 3
6* ORDER BY COUNT(1) ASC
7  /

```

DEPTNO	COUNT(1)
20	5
30	6

```
SQL> CL SCR
```

```

SQL> SELECT *
2  FROM EMP;

```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL
7839	KING	PRESIDENT		17-NOV-81	5000
7698	BLAKE	MANAGER	7839	01-MAY-81	2850
7782	CLARK	MANAGER	7839	09-JUN-81	2450
7566	JONES	MANAGER	7839	02-APR-81	2975
7788	SCOTT	ANALYST	7566	19-APR-87	3000
7902	FORD	ANALYST	7566	03-DEC-81	3000
7369	SMITH	CLERK	7902	17-DEC-80	800
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600
7521	WARD	SALESMAN	7698	22-FEB-81	1250
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250
7844	TURNER	SALESMAN	7698	08-SEP-81	1500
7876	ADAMS	CLERK	7788	23-MAY-87	1100
7900	JAMES	CLERK	7698	03-DEC-81	950

7934 MILLER CLERK 7782 23-JAN-82 1300

10

14 rows selected.

```
SQL> SELECT JOB
      2      ,COUNT(1)
      3      FROM EMP;
SELECT JOB
      *
```

ERROR at line 1:
ORA-00937: not a single-group group function

```
SQL> ED
Wrote file afiedt.buf
```

```
      1  SELECT JOB
      2      ,COUNT(1)
      3*   FROM EMP
```

```
SQL> /
SELECT JOB
      *
```

ERROR at line 1:
ORA-00937: not a single-group group function

```
SQL> ED
Wrote file afiedt.buf
```

```
      1  SELECT JOB
      2      ,COUNT(1)
      3      FROM EMP
      4*  GROUP BY JOB
```

```
SQL> /
```

JOB	COUNT(1)
CLERK	4
SALESMAN	4
PRESIDENT	1
MANAGER	3
ANALYST	2

```
SQL> ED
Wrote file afiedt.buf
```

```
      1  SELECT COUNT(1)
      2  FROM EMP
      3*  GROUP BY JOB
```

```
SQL> /
```

COUNT(1)
4
4
1
3
2

```
SQL> ED
Wrote file afiedt.buf
```

```
      1  SELECT JOB, COUNT(1)
```

```

2 FROM EMP
3* GROUP BY JOB
SQL> /

```

JOB	COUNT (1)
CLERK	4
SALESMAN	4
PRESIDENT	1
MANAGER	3
ANALYST	2

```
SQL> SPOOL OFF
```

8-jun-2018 Constraints Notes

NOT NULL

- IT WILL NOT ACCEPT NULL VALUE
- IT WILL ACCEPT DUPLICATE VALUES
- IN ONE TABLE WE CAN HAVE N NUMBER OF NOT NULL CONSTRAINTS

UNIQUE

- IT WILL ACCEPT NULL VALUES
- IT WILL NOT ACCEPT DUPLICATE VALUES
- IN ONE TABLE WE CAN HAVE N NUMBER OF UNIQUE
- WE CAN INSERT N NUMBER OF NULL VALUES INTO A UNIQUE CONSTRAINT COLUMN
- BECAUSE NULL = NULL IS A FALSE

PRIMARY KEY(NOT NULL+UNIQUE)

- IT WILL NOT ACCEPT NULL VALUES
- IT WILL NOT ACCEPT DUPLICATE VALUES
- IN ONE TABLE WE CAN HAVE ONLY ONE PRIMARY KEY

CHECK

- CHECKING THE DATA IF IT IS VALID THEN IT WILL ALLOW
- WE CAN HAVE N NUMBER OF CHECK CONSTRAINTS

DEFAULT

- BY DEFAULT WHAT VALUE WE WANT TO KEEP IT WHEN WE ARE NOT PROVIDING
- ANY VALUE TO COLUMN
- WE CAN HAVE N NUMBER OF DEFAULT CONSTRAINTS

unique and primary key

WHEN WE CRATE A TABLE IF WE ARE NOT GIVINIG CONSTRAINT NAME THEN

ORACLE WILL GIVE CONSTRAINT NAME THAT WILL START WITH SYS_C(SOME AUTO GENERATED NUMBER)

IT WILL BE DIFFICULT TO IDENTIFY ON WHICH COLUMN THIS CONSTRAINT IS

VIOLATED

SO DON'T DEPEND ON ORACLE GIVEN NAMES GIVE ALWAYS YOUR OWN NAME

FOR CONSTRAINTS.

HOW TO GIVE THE CONSTRAINTNAME :

CONSTRAINTNAME:

CONSTRAINT CONSTRAINTNAME CONSTRAINTTYPE

CONSTRAINT NAME CAN BE COMBNICATION OF ALL BELOW(REALT TIME STANDARD)

ABBREVATEDTABLENAME

ABBREVATEDCOLUMNNAME

ABBREVATEDCONSTRAINTTYPE

EXAMPLE :MOB NUMBER(10) CONSTRAINT B26STD2_MOB_CHK CHECK (LENGTH(MOB) =10)

Z

****FOR DEFAULT CONSTRAINTS WE CAN NOT GIVE ANY CONSTRAINT NAME**

FOR NOT NULL CONSTRAINTS IT WILL GIVE YOU CLEAR ERROR MESSAGE..

ERROR at line 5:

ORA-01400: cannot insert NULL into ("SYSTEM"."B22_CON3"."NAME")

ERROR at line 1:

ORA-02290: check constraint (SYSTEM.SYS_C007382) violated

SEQUENCES-- 40 MINS

CURRVAL

NEXT

WHATEVER WE CREATE IN THE DATABASE THAT INFORMATION ORACLE WILL BE STORING INTO SOME TABLES
THOSE TABLES WE CALL ORACLE METADATA TABLES.

USER_TABLES OR DBA_TABLES OR ALL_TABLES

USER_TAB_COLUMNS OR DBA_TAB_COLUMNS OR ALL_TAB_COLUMNS

USER_CONSTRAINTS

USER_CONS_COLUMNS

USER_SEQUENCES

USER_VIEWS

USER_SOURCE

..ETC

PRIMARY KEY

ON ONE COLUMN HOW MANY CONSTRAINTS WE CAN SPECIFY ?

WE CAN HAVE N NUMBER OF CONSGTRAINTS

How many ways we can specify constraints

COLUMN LEVEL

after each column immediately if you specify constraint that we call it as column level constraint specification

TABLEL LEVEL

after all columns if you specify constraints that we call it as table level constraint specification

NOTE :

COMPOSITE CONSTRAINTS WE SHOULD SPECIFY ONLY AT TABLE LEVEL

NOT NULL AND DEFAULT CONSTRAINTS WE CAN'T SPECIFY AT TABLE LEVEL

COMPOSITE CONSTRAINT

MORE THAN ONE COLUMN TO GETHERE (32 COLUMNS)

IN ONE USER 2 CONSTRAINTS NAMES SHOULD NOT BE SAME

TUESDAY

FOREIGN KEY

WED

THU

FRI SUBQ +SUBQ

8-jun-2018 Constraints Examples

```
SQL> SHOW USER
USER is "SYSTEM"
SQL>
SQL> CL SCR
```

```
SQL> CREATE TABLE B26_STD1
 2  (
 3  ID NUMBER
 4  ,NAME VARCHAR2(20)
 5  ,MOB NUMBER(10)
 6  ,FEES NUMBER(5)
 7  ,GEN CHAR(1)
 8  ,DOJ DATE
 9  );
```

Table created.

```
SQL> INSERT INTO B26_STD1
 2  VALUES
 3  (
 4  1
 5  , 'SURESH'
 6  ,12345
 7  ,99000
 8  , 'L'
 9  , '10-MAR-1989'
10  );
```

1 row created.

```
SQL> EDIT
Wrote file afiedt.buf
```

```
 1  INSERT INTO B26_STD1
 2  VALUES
 3  (
 4  1
 5  , 'KRISHNA'
 6  ,12346
 7  ,98000
 8  , 'S'
 9  , '11-MAR-1989'
10* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
 1  INSERT INTO B26_STD1
 2  VALUES
 3  (
 4  3
 5  ,NULL
 6  ,12347
 7  ,95000
 8  , 'S'
 9  , '11-MAR-1989'
10* )
SQL> /
```

1 row created.

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD1
2  VALUES
3  (
4  4
5  , 'AKBAR'
6  , 12348
7  , 93000
8  , 'H'
9* )
```

SQL> /

```
INSERT INTO B26_STD1
          *
```

ERROR at line 1:

ORA-00947: not enough values

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD1
2  (
3  ID
4  , NAME
5  , MOB
6  , FEES
7  , GEN
8  )
9  VALUES
10 (
11 4
12 , 'AKBAR'
13 , 12348
14 , 93000
15 , 'H'
16* )
```

SQL> /

1 row created.

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD1
2  (
3  ID
4  , NAME
5  , MOB
6  , FEES
7  , GEN
8  )
9  VALUES
10 (
11 4
12 , 'ARJUN'
13 , 12349
14 , 91000
15 , 'I'
16* )
```

```
SQL>
SQL> /
```

1 row created.

```
SQL> ed
Wrote file afiedt.buf
```

```

 1  INSERT INTO B26_STD1
 2  (
 3  ID
 4  ,NAME
 5  ,MOB
 6  ,FEES
 7  ,GEN
 8  )
 9  VALUES
10  (
11  null
12  , 'ABHISHEK'
13  ,12340
14  ,92000
15  , 'Y'
16* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```

 1  INSERT INTO B26_STD1
 2  (
 3  ID
 4  ,NAME
 5  ,MOB
 6  ,FEES
 7  ,GEN
 8  )
 9  VALUES
10  (
11  4
12  , 'ABHISHEK'
13  ,12340
14  ,92000
15  , 'Y'
16* )
SQL> /
```

1 row created.

```
SQL> CL SCR
```

```
SQL> COMMIT;
```

Commit complete.

```
SQL> DESC B26_STD1
```

Name	Null?	Type
ID		NUMBER
NAME		VARCHAR2(20)
MOB		NUMBER(10)
FEES		NUMBER(5)

GEN	CHAR(1)
DOJ	DATE

SQL> CL SCR

SQL> CREATE TABLE B26_STD2

```

2  (
3  ID NUMBER
4  ,NAME VARCHAR2(20)
5  ,MOB NUMBER(10)
6  ,FEES NUMBER(5)
7  ,GEN CHAR(1)
8  ,DOJ DATE
9  )
10 .

```

SQL> ED

Wrote file afiedt.buf

```

1 CREATE TABLE B26_STD2
2 (
3 ID NUMBER UNIQUE
4 ,NAME VARCHAR2(20) NOT NULL
5 ,MOB NUMBER(10) CHECK ( LENGTH(MOB) = 10)
6 ,FEES NUMBER(5) CHECK (FEES <=30000)
7 ,GEN CHAR(1) CHECK ( GEN = 'M' OR GEN ='F')
8 ,DOJ DATE DEFAULT SYSDATE
9* )

```

SQL> /

Table created.

SQL> DESCRIBE B26_STD1

Name	Null?	Type
ID		NUMBER
NAME		VARCHAR2(20)
MOB		NUMBER(10)
FEES		NUMBER(5)
GEN		CHAR(1)
DOJ		DATE

SQL> DESCRIBE B26_STD2

Name	Null?	Type
ID		NUMBER
NAME	NOT NULL	VARCHAR2(20)
MOB		NUMBER(10)
FEES		NUMBER(5)
GEN		CHAR(1)
DOJ		DATE

SQL> CL SCR

SQL> DESCRIBE B26_STD2

Name	Null?	Type
ID		NUMBER
NAME	NOT NULL	VARCHAR2(20)
MOB		NUMBER(10)
FEES		NUMBER(5)
GEN		CHAR(1)
DOJ		DATE

SQL> ED

Wrote file afiedt.buf

```

1 CREATE TABLE B26_STD2
2 (
3 ID NUMBER UNIQUE
4 ,NAME VARCHAR2(20) NOT NULL
5 ,MOB NUMBER(10) CHECK ( LENGTH(MOB) = 10)
6 ,FEES NUMBER(5) CHECK (FEES <=30000)
7 ,GEN CHAR(1) CHECK ( GEN = 'M' OR GEN = 'F')
8 ,DOJ DATE DEFAULT SYSDATE
9* )
SQL> CL S CR
SP2-0158: unknown CLEAR option "S"
SP2-0158: unknown CLEAR option "CR"
SQL> CL SCR

```

```

SQL> DESC B26_STD2
Name Null? Type
-----
ID NUMBER
NAME NOT NULL VARCHAR2(20)
MOB NUMBER(10)
FEES NUMBER(5)
GEN CHAR(1)
DOJ DATE

```

```
SQL> SELECT * FROM B26_STD2;
```

no rows selected

```

SQL> INSERT INTO B26_STD2
2 VALUES
3 (
4 1
5 , 'SURESH'
6 ,12345
7 ,99000
8 , 'L'
9 , '19-MAR-1989'
10 );
INSERT INTO B26_STD2
*
```

```

ERROR at line 1:
ORA-02290: check constraint (SYSTEM.SYS_C007109) violated

```

```
SQL> ED
Wrote file afiedt.buf
```

```

1 INSERT INTO B26_STD2
2 VALUES
3 (
4 1
5 , 'SURESH'
6 ,1234512346
7 ,99000
8 , 'L'
9 , '19-MAR-1989'
10* )
SQL> /
INSERT INTO B26_STD2
*
ERROR at line 1:
ORA-02290: check constraint (SYSTEM.SYS_C007109) violated

```

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD2
2  VALUES
3  (
4  1
5  , 'SURESH'
6  , 1234512346
7  , 9000
8  , 'L'
9  , '19-MAR-1989'
10* )
```

SQL> /

INSERT INTO B26_STD2

*

ERROR at line 1:

ORA-02290: check constraint (SYSTEM.SYS_C007109) violated

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD2
2  VALUES
3  (
4  1
5  , 'SURESH'
6  , 12345123
7  , 99000
8  , 'M'
9  , '19-MAR-1989'
10* )
```

SQL> /

INSERT INTO B26_STD2

*

ERROR at line 1:

ORA-02290: check constraint (SYSTEM.SYS_C007108) violated

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD2
2  VALUES
3  (
4  1
5  , 'SURESH'
6  , 12345123
7  , 9000
8  , 'M'
9  , '19-MAR-1989'
10* )
```

SQL> /

INSERT INTO B26_STD2

*

ERROR at line 1:

ORA-02290: check constraint (SYSTEM.SYS_C007107) violated

SQL> ED

Wrote file afiedt.buf

```

1  INSERT INTO B26_STD2
2  VALUES
3  (
4  1
5  , 'SURESH'
6  , 1234512345
7  , 9000
8  , 'M'
9  , '19-MAR-1989'
10* )
SQL> /

```

1 row created.

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  INSERT INTO B26_STD2
2  VALUES
3  (
4  2
5  , NULL
6  , 1234512346
7  , 9500
8  , 'M'
9  , '10-MAR-1989'
10* )
SQL> /

```

```

, NULL
*

```

ERROR at line 5:

ORA-01400: cannot insert NULL into ("SYSTEM"."B26_STD2"."NAME")

SQL> DESC B26_STD2

Name	Null?	Type
ID		NUMBER
NAME	NOT NULL	VARCHAR2(20)
MOB		NUMBER(10)
FEES		NUMBER(5)
GEN		CHAR(1)
DOJ		DATE

SQL> ED

Wrote file afiedt.buf

```

1  INSERT INTO B26_STD2
2  VALUES
3  (
4  2
5  , NULL
6  , 123451
7  , 95000
8  , 'L'
9  , '10-MAR-1989'
10* )
SQL> /

```

```

, NULL
*

```

ERROR at line 5:

ORA-01400: cannot insert NULL into ("SYSTEM"."B26_STD2"."NAME")

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD2
2  VALUES
3  (
4  2
5  , 'ABC'
6  , 123451
7  , 95000
8  , 'L'
9  , '10-MAR-1989'
10* )
```

SQL> /

INSERT INTO B26_STD2

*

ERROR at line 1:

ORA-02290: check constraint (SYSTEM.SYS_C007109) violated

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD2
2  VALUES
3  (
4  2
5  , 'ABC'
6  , 123451
7  , 95000
8  , 'M'
9  , '10-MAR-1989'
10* )
```

SQL> /

INSERT INTO B26_STD2

*

ERROR at line 1:

ORA-02290: check constraint (SYSTEM.SYS_C007108) violated

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD2
2  VALUES
3  (
4  2
5  , 'ABC'
6  , 123451
7  , 9500
8  , 'M'
9  , '10-MAR-1989'
10* )
```

SQL> /

INSERT INTO B26_STD2

*

ERROR at line 1:

ORA-02290: check constraint (SYSTEM.SYS_C007107) violated

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD2
2  VALUES
3  (
4  2
5  , 'ABC'
6  , 1234511232
7  , 9500
8  , 'M'
9  , '10-MAR-1989'
10* )
SQL> /
```

1 row created.

SQL> CL SCR

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD2
2  VALUES
3  (
4  1
5  , 'KRISHNA'
6  , 1234511233
7  , 9600
8  , 'M'
9  , '11-MAR-1989'
10* )
SQL> /
INSERT INTO B26_STD2
*
```

ERROR at line 1:

ORA-00001: unique constraint (SYSTEM.SYS_C007110) violated

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD2
2  VALUES
3  (
4  NULL
5  , 'KRISHNA'
6  , 1234511233
7  , 9600
8  , 'M'
9  , '11-MAR-1989'
10* )
SQL> /
```

1 row created.

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_STD2
2  VALUES
3  (
4  NULL
5  , 'TEJASREE'
6  , 1234511234
```

```
7 ,9600
8 , 'M'
9 , '11-MAR-1989'
10* )
SQL> /
```

1 row created.

```
SQL> R
1 INSERT INTO B26_STD2
2 VALUES
3 (
4 NULL
5 , 'TEJASREE'
6 ,1234511234
7 ,9600
8 , 'M'
9 , '11-MAR-1989'
10* )
```

1 row created.

```
SQL> R
1 INSERT INTO B26_STD2
2 VALUES
3 (
4 NULL
5 , 'TEJASREE'
6 ,1234511234
7 ,9600
8 , 'M'
9 , '11-MAR-1989'
10* )
```

1 row created.

```
SQL> R
1 INSERT INTO B26_STD2
2 VALUES
3 (
4 NULL
5 , 'TEJASREE'
6 ,1234511234
7 ,9600
8 , 'M'
9 , '11-MAR-1989'
10* )
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1 INSERT INTO B26_STD2
2 VALUES
3 (
4 NULL
5 , 'TEJASREE'
6 ,1234511234
7 ,9600
8 , 'M'
9 , '11-MAR-1989'
10* )
SQL>
```

```
SQL>
SQL>
SQL> SELECT * FROM B26_STD2;
```

ID	NAME	MOB	FEES	G	DOJ
1	SURESH	1234512345	9000	M	19-MAR-89
2	ABC	1234511232	9500	M	10-MAR-89
	KRISHNA	1234511233	9600	M	11-MAR-89
	TEJASREE	1234511234	9600	M	11-MAR-89
	TEJASREE	1234511234	9600	M	11-MAR-89
	TEJASREE	1234511234	9600	M	11-MAR-89
	TEJASREE	1234511234	9600	M	11-MAR-89

7 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_STD2
2  VALUES
3  (
4  4
5  , 'ABHISHEK'
6  , 1234511235
7  , 9700
8  , 'M'
9* )
```

```
SQL> /
INSERT INTO B26_STD2
      *
ERROR at line 1:
ORA-00947: not enough values
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_STD2
2  (
3  ID
4  , NAME
5  , MOB
6  , FEES
7  , GEN
8  )
9  VALUES
10 (
11 4
12 , 'ABHISHEK'
13 , 1234511235
14 , 9700
15 , 'M'
16* )
SQL> /
```

1 row created.

```
SQL> SELECT * FROM B26_STD2;
```

ID	NAME	MOB	FEES	G	DOJ
1	SURESH	1234512345	9000	M	19-MAR-89
2	ABC	1234511232	9500	M	10-MAR-89

KRISHNA	1234511233	9600 M 11-MAR-89
TEJASREE	1234511234	9600 M 11-MAR-89
TEJASREE	1234511234	9600 M 11-MAR-89
TEJASREE	1234511234	9600 M 11-MAR-89
TEJASREE	1234511234	9600 M 11-MAR-89
4 ABHISHEK	1234511235	9700 M 08-JUN-18

8 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1 CREATE TABLE B26_STD3
2 (
3 ID NUMBER CONSTRAINT B26STD3_ID_PK PRIMARY KEY
4 ,NAME VARCHAR2(20) CONSTRAINT B26STD3_NAME_NN NOT NULL
5 ,MOB NUMBER(10) CONSTRAINT B26STD3_MOB_CHK CHECK ( LENGTH(MOB) = 10)
6 CONSTRAINT B26STD3_MOB_UQ UNIQUE
7 CONSTRAINT B26STD3_MOB_NN NOT NULL
8 ,FEES NUMBER(5) CONSTRAINT B26STD3_FEES_CHK CHECK (FEES <=30000)
9 ,GEN CHAR(1) CONSTRAINT B26STD3_GEN_CHK CHECK ( GEN = 'M' OR GEN ='F')
10 ,DOJ DATE CONSTRAINT B26STD3_DOJ_DF DEFAULT SYSDATE
11* )
12 /
,DOJ DATE CONSTRAINT B26STD3_DOJ_DF DEFAULT SYSDATE
*
```

ERROR at line 10:

ORA-02253: constraint specification not allowed here

SQL> ED

Wrote file afiedt.buf

```

1 CREATE TABLE B26_STD3
2 (
3 ID NUMBER CONSTRAINT B26STD3_ID_PK PRIMARY KEY
4 ,NAME VARCHAR2(20) CONSTRAINT B26STD3_NAME_NN NOT NULL
5 ,MOB NUMBER(10) CONSTRAINT B26STD3_MOB_CHK CHECK ( LENGTH(MOB) = 10)
6 CONSTRAINT B26STD3_MOB_UQ UNIQUE
7 CONSTRAINT B26STD3_MOB_NN NOT NULL
8 ,FEES NUMBER(5) CONSTRAINT B26STD3_FEES_CHK CHECK (FEES <=30000)
9 ,GEN CHAR(1) CONSTRAINT B26STD3_GEN_CHK CHECK ( GEN = 'M' OR GEN ='F')
10 ,DOJ DATE DEFAULT SYSDATE
11* )
SQL> /
```

Table created.

SQL> ED

Wrote file afiedt.buf

```

1 CREATE TABLE B26_STD3
2 (
3 ID NUMBER CONSTRAINT B26STD3_ID_PK PRIMARY KEY
4 ,NAME VARCHAR2(20) CONSTRAINT B26STD3_NAME_NN NOT NULL
5 ,MOB NUMBER(10) CONSTRAINT B26STD3_MOB_CHK CHECK ( LENGTH(MOB) = 10)
6 CONSTRAINT B26STD3_MOB_UQ UNIQUE
7 CONSTRAINT B26STD3_MOB_NN NOT NULL
8 ,FEES NUMBER(5) CONSTRAINT B26STD3_FEES_CHK CHECK (FEES <=30000)
9 ,GEN CHAR(1) CONSTRAINT B26STD3_GEN_CHK CHECK ( GEN = 'M' OR GEN ='F')
10 ,DOJ DATE DEFAULT SYSDATE
11* )
SQL> CL SCR
```

```

SQL> R
1  CREATE TABLE B26_STD3
2  (
3  ID NUMBER CONSTRAINT B26STD3_ID_PK PRIMARY KEY
4  ,NAME VARCHAR2(20) CONSTRAINT B26STD3_NAME_NN NOT NULL
5  ,MOB NUMBER(10) CONSTRAINT B26STD3_MOB_CHK CHECK ( LENGTH(MOB) = 10)
6  CONSTRAINT B26STD3_MOB_UQ UNIQUE
7  CONSTRAINT B26STD3_MOB_NN NOT NULL
8  ,FEES NUMBER(5) CONSTRAINT B26STD3_FEES_CHK CHECK (FEES <=30000)
9  ,GEN CHAR(1) CONSTRAINT B26STD3_GEN_CHK CHECK ( GEN = 'M' OR GEN ='F')
10 ,DOJ DATE DEFAULT SYSDATE
11* )
CREATE TABLE B26_STD3
*
ERROR at line 1:
ORA-00955: name is already used by an existing object

```

```

SQL> DESC B26_STD3
Name                                         Null?      Type
-----
ID                                           NOT NULL   NUMBER
NAME                                         NOT NULL   VARCHAR2(20)
MOB                                           NOT NULL   NUMBER(10)
FEES                                           NUMBER(5)
GEN                                           CHAR(1)
DOJ                                           DATE

```

```

SQL> INSERT INTO B26_STD3
2  VALUES
3  (
4  1
5  , 'SURESH'
6  ,12345
7  ,99000
8  , 'L'
9  , '19-MAR-1989'
10 );
INSERT INTO B26_STD3
*
ERROR at line 1:
ORA-02290: check constraint (SYSTEM.B26STD3_GEN_CHK) violated

```

```

SQL> EDIT
Wrote file afiedt.buf

```

```

1  INSERT INTO B26_STD3
2  VALUES
3  (
4  1
5  , 'SURESH'
6  ,12345
7  ,99000
8  , 'M'
9  , '19-MAR-1989'
10* )
SQL> /
INSERT INTO B26_STD3
*
ERROR at line 1:
ORA-02290: check constraint (SYSTEM.B26STD3_FEES_CHK) violated

```

```
SQL> ED
Wrote file afiedt.buf

  1  INSERT INTO B26_STD3
  2  VALUES
  3  (
  4  1
  5  , 'SURESH'
  6  , 12345
  7  , 9000
  8  , 'M'
  9  , '19-MAR-1989'
10* )
SQL> /
INSERT INTO B26_STD3
*
ERROR at line 1:
ORA-02290: check constraint (SYSTEM.B26STD3_MOB_CHK) violated
```

```
SQL> ED
Wrote file afiedt.buf

  1  INSERT INTO B26_STD3
  2  VALUES
  3  (
  4  1
  5  , 'SURESH'
  6  , 1234512345
  7  , 9000
  8  , 'M'
  9  , '19-MAR-1989'
10* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf

  1  INSERT INTO B26_STD3
  2  VALUES
  3  (
  4  1
  5  , 'KRISHNA'
  6  , 1234512345
  7  , 9000
  8  , 'M'
  9  , '20-MAR-1989'
10* )
SQL> /
INSERT INTO B26_STD3
*
ERROR at line 1:
ORA-00001: unique constraint (SYSTEM.B26STD3_ID_PK) violated
```

```
SQL> ED
Wrote file afiedt.buf

  1  INSERT INTO B26_STD3
  2  VALUES
  3  (
  4  2
```

```
5  , 'KRISHNA'
6  , 1234512345
7  , 9000
8  , 'M'
9  , '20-MAR-1989'
10* )
SQL> /
INSERT INTO B26_STD3
*
ERROR at line 1:
ORA-00001: unique constraint (SYSTEM.B26STD3_MOB_UQ) violated
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_STD3
2  VALUES
3  (
4  2
5  , 'KRISHNA'
6  , 1234512346
7  , 9000
8  , 'M'
9  , '20-MAR-1989'
10* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_STD3
2  VALUES
3  (
4  NULL
5  , 'TEAJSREE'
6  , NULL
7  , 9000
8  , 'F'
9  , '20-MAR-1989'
10* )
SQL> /
NULL
*
```

```
ERROR at line 4:
ORA-01400: cannot insert NULL into ("SYSTEM"."B26_STD3"."ID")
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_STD3
2  VALUES
3  (
4  3
5  , 'TEAJSREE'
6  , NULL
7  , 9000
8  , 'F'
9  , '20-MAR-1989'
10* )
SQL> /
```

```
,NULL
*
ERROR at line 6:
ORA-01400: cannot insert NULL into ("SYSTEM"."B26_STD3"."MOB")
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_STD3
2  VALUES
3  (
4  3
5  , 'TEAJSREE'
6  , 12345
7  , 9000
8  , 'F'
9  , '20-MAR-1989'
10* )
```

```
SQL> /
```

```
INSERT INTO B26_STD3
```

```
*
```

```
ERROR at line 1:
ORA-02290: check constraint (SYSTEM.B26STD3_MOB_CHK) violated
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_STD3
2  VALUES
3  (
4  3
5  , 'TEAJSREE'
6  , 1234512345
7  , 9000
8  , 'F'
9  , '20-MAR-1989'
10* )
```

```
SQL> /
```

```
INSERT INTO B26_STD3
```

```
*
```

```
ERROR at line 1:
ORA-00001: unique constraint (SYSTEM.B26STD3_MOB_UQ) violated
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_STD3
2  VALUES
3  (
4  3
5  , 'TEAJSREE'
6  , 1234512349
7  , 9000
8  , 'F'
9  , '20-MAR-1989'
10* )
```

```
SQL> /
```

```
1 row created.
```

```
SQL> CL SCR
```

```
SQL> SELECT *
      2 FROM B26_STD10;
FROM B26_STD10
      *
```

ERROR at line 2:
ORA-00942: table or view does not exist

```
SQL> SELECT *
      2 FROM B26_STD1;
```

ID	NAME	MOB	FEES	G	DOJ
1	SURESH	12345	99000	L	10-MAR-89
1	KRISHNA	12346	98000	S	11-MAR-89
3		12347	95000	S	11-MAR-89
4	AKBAR	12348	93000	H	
4	ARJUN	12349	91000	I	
	ABHISHEK	12340	92000	Y	
4	ABHISHEK	12340	92000	Y	

7 rows selected.

```
SQL> CL SCR
```

```
SQL> DESC B26_STD1
```

Name	Null?	Type
ID		NUMBER
NAME		VARCHAR2(20)
MOB		NUMBER(10)
FEES		NUMBER(5)
GEN		CHAR(1)
DOJ		DATE

```
SQL> DESC B26_STD2
```

Name	Null?	Type
ID		NUMBER
NAME	NOT NULL	VARCHAR2(20)
MOB		NUMBER(10)
FEES		NUMBER(5)
GEN		CHAR(1)
DOJ		DATE

```
SQL> INSERT INTO B26_STD2
```

```
      2 VALUES
      3 (
      4 10
      5 , 'SUBHASH'
      6 , 98765
      7 , 55000
      8 , 'H'
      9 , SYSDATE
     10 );
```

```
INSERT INTO B26_STD2
```

```
*
```

ERROR at line 1:
ORA-02290: check constraint (SYSTEM.SYS_C007109) violated

```
SQL> DESC USER_CONSTRAINTS
```

Name	Null?	Type
------	-------	------

```

-----
OWNER                                VARCHAR2(120)
CONSTRAINT_NAME                     NOT NULL VARCHAR2(30)
CONSTRAINT_TYPE                     VARCHAR2(1)
TABLE_NAME                           NOT NULL VARCHAR2(30)
SEARCH_CONDITION                     LONG
R_OWNER                             VARCHAR2(120)
R_CONSTRAINT_NAME                   VARCHAR2(30)
DELETE_RULE                          VARCHAR2(9)
STATUS                              VARCHAR2(8)
DEFERRABLE                          VARCHAR2(14)
DEFERRED                            VARCHAR2(9)
VALIDATED                           VARCHAR2(13)
GENERATED                           VARCHAR2(14)
BAD                                 VARCHAR2(3)
RELY                                VARCHAR2(4)
LAST_CHANGE                         DATE
INDEX_OWNER                         VARCHAR2(30)
INDEX_NAME                          VARCHAR2(30)
INVALID                             VARCHAR2(7)
VIEW_RELATED                        VARCHAR2(14)

SQL> SELECT CONSTRAINT_TYPE, TABLE_NAME, SEARCH_CONDITION
      2 FROM USER_CONSTRAINTS
      3 WHERE CONSTRAINT_NAME='SYS_C007109';

C TABLE_NAME
-----
SEARCH_CONDITION
-----

C B26_STD2
  GEN = 'M' OR GEN = 'F'

SQL> SET LINESIZE 200
SQL> R
      1 SELECT CONSTRAINT_TYPE, TABLE_NAME, SEARCH_CONDITION
      2 FROM USER_CONSTRAINTS
      3* WHERE CONSTRAINT_NAME='SYS_C007109'

C TABLE_NAME                      SEARCH_CONDITION
-----
-----

C B26_STD2                        GEN = 'M' OR GEN = 'F'

SQL> COLUMN CONSTRAINT_TYPE FORMAT A10
SQL> R
      1 SELECT CONSTRAINT_TYPE, TABLE_NAME, SEARCH_CONDITION
      2 FROM USER_CONSTRAINTS
      3* WHERE CONSTRAINT_NAME='SYS_C007109'

CONSTRAINT TABLE_NAME                      SEARCH_CONDITION
-----
-----

C          B26_STD2                        GEN = 'M' OR GEN = 'F'

SQL> COLUMN CONSTRAINT_TYPE FORMAT A15
SQL> R
      1 SELECT CONSTRAINT_TYPE, TABLE_NAME, SEARCH_CONDITION
      2 FROM USER_CONSTRAINTS
      3* WHERE CONSTRAINT_NAME='SYS_C007109'

CONSTRAINT_TYPE TABLE_NAME                      SEARCH_CONDITION

```

```

-----
C                B26_STD2                GEN = 'M' OR GEN = 'F'

```

```
SQL> DESC USER_CONS_COLUMNS
```

```
  Name
```

```
Null?      Type
```

```
-----
OWNER
```

```
NOT NULL VARCHAR2(30)
```

```
  CONSTRAINT_NAME
```

```
NOT NULL VARCHAR2(30)
```

```
  TABLE_NAME
```

```
NOT NULL VARCHAR2(30)
```

```
  COLUMN_NAME
```

```
VARCHAR2(4000)
```

```
  POSITION
```

```
NUMBER
```

```
SQL> SELECT TABLE_NAME,COLUMN_NAME,CONSTRAINT_NAME
```

```
2 FROM USER_CONS_COLUMNS
```

```
3 WHERE TABLE_NAME LIKE '%B26%';
```

```
TABLE_NAME
```

```
-----
COLUMN_NAME
```

```
-----
CONSTRAINT_NAME
```

```
-----
B26_STD2
```

```
ID
```

```
SYS_C007110
```

```
B26_STD2
```

```
NAME
```

```
SYS_C007106
```

```
TABLE_NAME
```

```
-----
COLUMN_NAME
```

```
-----
CONSTRAINT_NAME
```

```
-----
B26_STD2
```

```
MOB
```

```
SYS_C007107
```

```
B26_STD2
```

```
FEES
```

```
TABLE_NAME
```

```
-----
COLUMN_NAME
```

```
-----
CONSTRAINT_NAME
```

SYS_C007108

B26_STD2
GEN
SYS_C007109

B26_STD3

TABLE_NAME

COLUMN_NAME

CONSTRAINT_NAME

ID
B26STD3_ID_PK

B26_STD3
NAME
B26STD3_NAME_NN

TABLE_NAME

COLUMN_NAME

CONSTRAINT_NAME

B26_STD3
MOB
B26STD3_MOB_UQ

B26_STD3
MOB
B26STD3_MOB_CHK

TABLE_NAME

COLUMN_NAME

CONSTRAINT_NAME

B26_STD3
MOB
B26STD3_MOB_NN

B26_STD3
FEES

TABLE_NAME

COLUMN_NAME

CONSTRAINT_NAME

```
-----
B26STD3_FEES_CHK
```

```
B26_STD3
```

```
GEN
```

```
B26STD3_GEN_CHK
```

```
12 rows selected.
```

```
SQL> COLUMN COLUMN_NAME FORMAT A20
```

```
SQL> R
```

```
1 SELECT TABLE_NAME,COLUMN_NAME,CONSTRAINT_NAME
2 FROM USER_CONS_COLUMNS
3* WHERE TABLE_NAME LIKE '%B26%'
```

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME
B26_STD2	ID	SYS_C007110
B26_STD2	NAME	SYS_C007106
B26_STD2	MOB	SYS_C007107
B26_STD2	FEES	SYS_C007108
B26_STD2	GEN	SYS_C007109
B26_STD3	ID	B26STD3_ID_PK
B26_STD3	NAME	B26STD3_NAME_NN
B26_STD3	MOB	B26STD3_MOB_UQ
B26_STD3	MOB	B26STD3_MOB_CHK
B26_STD3	MOB	B26STD3_MOB_NN
B26_STD3	FEES	B26STD3_FEES_CHK

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME
B26_STD3	GEN	B26STD3_GEN_CHK

```
12 rows selected.
```

```
SQL> SET PAGESIZE 200
```

```
SQL> R
```

```
1 SELECT TABLE_NAME,COLUMN_NAME,CONSTRAINT_NAME
2 FROM USER_CONS_COLUMNS
3* WHERE TABLE_NAME LIKE '%B26%'
```

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME
B26_STD2	ID	SYS_C007110
B26_STD2	NAME	SYS_C007106
B26_STD2	MOB	SYS_C007107
B26_STD2	FEES	SYS_C007108
B26_STD2	GEN	SYS_C007109
B26_STD3	ID	B26STD3_ID_PK
B26_STD3	NAME	B26STD3_NAME_NN
B26_STD3	MOB	B26STD3_MOB_UQ
B26_STD3	MOB	B26STD3_MOB_CHK
B26_STD3	MOB	B26STD3_MOB_NN
B26_STD3	FEES	B26STD3_FEES_CHK
B26_STD3	GEN	B26STD3_GEN_CHK

```
12 rows selected.
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT TABLE_NAME,COLUMN_NAME,CONSTRAINT_NAME
2 FROM USER_CONS_COLUMNS
```

```

3* WHERE CONSTRAINT_NAME ='SYS_C007109'
SQL> /

```

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME
B26_STD2	GEN	SYS_C007109

```

SQL> EDIT
Wrote file afiedt.buf

```

To Findout All the constraints from all the table

```

1  SELECT TABLE_NAME,COLUMN_NAME,CONSTRAINT_NAME
2* FROM USER_CONS_COLUMNS
3  /

```

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME
LOGMNR_SESSION_EVOLVE\$	SESSION#	LOGMNR_SESSION_EVOLVE\$_PK
LOGMNR_SESSION_EVOLVE\$	DB_ID	LOGMNR_SESSION_EVOLVE\$_PK
LOGMNR_SESSION_EVOLVE\$	RESET_SCN	LOGMNR_SESSION_EVOLVE\$_PK
LOGMNR_SESSION_EVOLVE\$	RESET_TIMESTAMP	LOGMNR_SESSION_EVOLVE\$_PK
LOGMNR_UID\$	SESSION#	LOGMNR_UID\$_PK
LOGMNR_DBNAME_UID_MAP	GLOBAL_NAME	SYS_C001360
LOGMNR_DBNAME_UID_MAP	GLOBAL_NAME	LOGMNR_DBNAME_UID_MAP_PK
LOGMNR_LOG\$	SESSION#	LOGMNR_LOG\$_PK
LOGMNR_LOG\$	THREAD#	LOGMNR_LOG\$_PK
LOGMNR_LOG\$	SEQUENCE#	LOGMNR_LOG\$_PK
LOGMNR_LOG\$	FIRST_CHANGE#	LOGMNR_LOG\$_PK
LOGMNR_LOG\$	DB_ID	LOGMNR_LOG\$_PK
LOGMNR_LOG\$	RESETLOGS_CHANGE#	LOGMNR_LOG\$_PK
LOGMNR_LOG\$	RESET_TIMESTAMP	LOGMNR_LOG\$_PK
LOGMNR_PROCESSED_LOG\$	SESSION#	LOGMNR_PROCESSED_LOG\$_PK
LOGMNR_PROCESSED_LOG\$	THREAD#	LOGMNR_PROCESSED_LOG\$_PK
LOGMNR_SPILL\$	SESSION#	LOGMNR_SPILL\$_PK
LOGMNR_SPILL\$	XIDUSN	LOGMNR_SPILL\$_PK
LOGMNR_SPILL\$	XIDSLT	LOGMNR_SPILL\$_PK
LOGMNR_SPILL\$	XIDSQN	LOGMNR_SPILL\$_PK
LOGMNR_SPILL\$	CHUNK	LOGMNR_SPILL\$_PK
LOGMNR_SPILL\$	STARTIDX	LOGMNR_SPILL\$_PK
LOGMNR_SPILL\$	ENDIDX	LOGMNR_SPILL\$_PK
LOGMNR_SPILL\$	FLAG	LOGMNR_SPILL\$_PK
LOGMNR_SPILL\$	SEQUENCE#	LOGMNR_SPILL\$_PK
LOGMNR_AGE_SPILL\$	SESSION#	LOGMNR_AGE_SPILL\$_PK
LOGMNR_AGE_SPILL\$	XIDUSN	LOGMNR_AGE_SPILL\$_PK
LOGMNR_AGE_SPILL\$	XIDSLT	LOGMNR_AGE_SPILL\$_PK
LOGMNR_AGE_SPILL\$	XIDSQN	LOGMNR_AGE_SPILL\$_PK
LOGMNR_AGE_SPILL\$	CHUNK	LOGMNR_AGE_SPILL\$_PK

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME
B26_STD2	NAME	SYS_C007106
B26_STD2	MOB	SYS_C007107
B26_STD2	FEES	SYS_C007108
B26_STD2	GEN	SYS_C007109
B26_STD3	ID	B26STD3_ID_PK
B26_STD3	NAME	B26STD3_NAME_NN
B26_STD3	MOB	B26STD3_MOB_NN
B26_STD3	MOB	B26STD3_MOB_CHK
B26_STD3	MOB	B26STD3_MOB_UQ
B26_STD3	FEES	B26STD3_FEES_CHK
B26_STD3	GEN	B26STD3_GEN_CHK

765 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT TABLE_NAME,COLUMN_NAME,CONSTRAINT_NAME
 2  FROM USER_CONS_COLUMNS
 3* WHERE TABLE_NAME LIKE '%B26%'
SQL> /

```

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME
B26_STD2	ID	SYS_C007110
B26_STD2	NAME	SYS_C007106
B26_STD2	MOB	SYS_C007107
B26_STD2	FEES	SYS_C007108
B26_STD2	GEN	SYS_C007109
B26_STD3	ID	B26STD3_ID_PK
B26_STD3	NAME	B26STD3_NAME_NN
B26_STD3	MOB	B26STD3_MOB_UQ
B26_STD3	MOB	B26STD3_MOB_CHK
B26_STD3	MOB	B26STD3_MOB_NN
B26_STD3	FEES	B26STD3_FEES_CHK
B26_STD3	GEN	B26STD3_GEN_CHK

12 rows selected.

SQL> CL SCR

SQL> R

```

 1  SELECT TABLE_NAME,COLUMN_NAME,CONSTRAINT_NAME
 2  FROM USER_CONS_COLUMNS
 3* WHERE TABLE_NAME LIKE '%B26%'

```

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME
B26_STD2	ID	SYS_C007110
B26_STD2	NAME	SYS_C007106
B26_STD2	MOB	SYS_C007107
B26_STD2	FEES	SYS_C007108
B26_STD2	GEN	SYS_C007109
B26_STD3	ID	B26STD3_ID_PK
B26_STD3	NAME	B26STD3_NAME_NN
B26_STD3	MOB	B26STD3_MOB_UQ
B26_STD3	MOB	B26STD3_MOB_CHK
B26_STD3	MOB	B26STD3_MOB_NN
B26_STD3	FEES	B26STD3_FEES_CHK
B26_STD3	GEN	B26STD3_GEN_CHK

12 rows selected.

SQL> spool off

9-Jun-2018 -Foreign KEY Example2

SQL>

SQL>

SQL> CL SCR

```
SQL> ED
Wrote file afiedt.buf
```

```

1 CREATE TABLE B19_PRODUCT
2 (
3   PID          NUMBER CONSTRAINT B19PROD_PID_PK  PRIMARY KEY
4   ,PNAME       VARCHAR2(20) CONSTRAINT B19PROD_PNM_NN NOT NULL
5   ,PMFR        VARCHAR2(20)
6   ,PCOST       NUMBER(5)  CONSTRAINT B19PROD_PCost_NN NOT NULL
7* )
SQL> /
```

Table created.

```
SQL> ED
Wrote file afiedt.buf
```

```

1 CREATE TABLE B19_CUSTOMER
2 (
3   CID          NUMBER CONSTRAINT B19CUST_CID_PK  PRIMARY KEY
4   ,CNAME       VARCHAR2(20) CONSTRAINT B19CUST_CNAME_NN NOT NULL
5   ,CMOB        NUMBER(10) CONSTRAINT B19CUST_CMOB_NN NOT NULL
6               CONSTRAINT B19CUST_CMOB_CHK CHECK(LENGTH(MOB)=10)
7               CONSTRAINT B19_CUST_CMOB_UNQ UNIQUE
8* )
SQL> /

                CONSTRAINT B19_CUST_CMOB_UNQ UNIQUE
*
```

```
ERROR at line 7:
ORA-02438: Column check constraint cannot reference other columns
```

```
SQL> ED
Wrote file afiedt.buf
```

```

1 CREATE TABLE B19_CUSTOMER
2 (
3   CID          NUMBER CONSTRAINT B19CUST_CID_PK  PRIMARY KEY
4   ,CNAME       VARCHAR2(20) CONSTRAINT B19CUST_CNAME_NN NOT NULL
5   ,CMOB        NUMBER(10) CONSTRAINT B19CUST_CMOB_NN NOT NULL
6               CONSTRAINT B19CUST_CMOB_CHK CHECK(LENGTH(CMOB)=10)
7               CONSTRAINT B19_CUST_CMOB_UNQ UNIQUE
8* )
SQL> /
```

Table created.

```
SQL> ED
Wrote file afiedt.buf
```

```

1 CREATE TABLE B19_LOCATION
2 (
3   LOC_ID       NUMBER CONSTRAINT B19LOCATION_LID_PK  PRIMARY KEY
4   ,AREA        VARCHAR2(20)
5   ,POST_CODE   NUMBER(6)
6   ,CITY        VARCHAR2(20)
7   ,STATE       VARCHAR2(20)
8* )
SQL> /
```

Table created.

```
SQL> CL SCR
```

SQL> DESCRIBE B19_PRODUCT

Name	Null?	Type
PID	NOT NULL	NUMBER
PNAME	NOT NULL	VARCHAR2(20)
PMFR		VARCHAR2(20)
PCOST	NOT NULL	NUMBER(5)

SQL> DESCRIBE B19_CUSTOMER

Name	Null?	Type
CID	NOT NULL	NUMBER
CNAME	NOT NULL	VARCHAR2(20)
CMOB	NOT NULL	NUMBER(10)

SQL> DESCRIBE B19_LOCATION

Name	Null?	Type
LOC_ID	NOT NULL	NUMBER
AREA		VARCHAR2(20)
POST_CODE		NUMBER(6)
CITY		VARCHAR2(20)
STATE		VARCHAR2(20)

SQL> EDIT

Wrote file afiedt.buf

```

1 CREATE TABLE B19_ORDER_DETAILS
2 (
3   ORD_ID NUMBER
4   ,PID NUMBER CONSTRAINT ORDDET_PID_FK REFERENCES B19_PRODUCT(PID)
5   ,CID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_CUSTOMER(CID)
6   ,LOC_ID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_LOCATION(LOC_ID)
7   ,QTY NUMBER NOT NULL
8   ,ORD_DATE DATE DEFAULT SYSDATE
9* )

```

SQL> /

```

,LOC_ID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_LOCATION(LOC_ID)
*

```

ERROR at line 6:

ORA-02264: name already used by an existing constraint

SQL> CL SCR

SQL> R

```

1 CREATE TABLE B19_ORDER_DETAILS
2 (
3   ORD_ID NUMBER
4   ,PID NUMBER CONSTRAINT ORDDET_PID_FK REFERENCES B19_PRODUCT(PID)
5   ,CID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_CUSTOMER(CID)
6   ,LOC_ID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_LOCATION(LOC_ID)
7   ,QTY NUMBER NOT NULL
8   ,ORD_DATE DATE DEFAULT SYSDATE
9* )
,LOC_ID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_LOCATION(LOC_ID)
*

```

ERROR at line 6:

ORA-02264: name already used by an existing constraint

SQL> EED

SP2-0042: unknown command "EED" - rest of line ignored.

SQL> ED

Wrote file afiedt.buf

```

1 CREATE TABLE B19_ORDER_DETAILS
2 (
3 ORD_ID NUMBER
4 ,PID NUMBER CONSTRAINT ORDDET_PID_FK REFERENCES B19_PRODUCT(PID)
5 ,CID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_CUSTOMER(CID)
6 ,LOC_ID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_LOCATION(LOC_ID)
7 ,QTY NUMBER NOT NULL
8 ,ORD_DATE DATE DEFAULT SYSDATE
9* )
SQL>
SQL> /
,LOC_ID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_LOCATION(LOC_ID)
*
ERROR at line 6:
ORA-02264: name already used by an existing constraint

```

SQL> CL SCR

```

SQL> R
1 CREATE TABLE B19_ORDER_DETAILS
2 (
3 ORD_ID NUMBER
4 ,PID NUMBER CONSTRAINT ORDDET_PID_FK REFERENCES B19_PRODUCT(PID)
5 ,CID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_CUSTOMER(CID)
6 ,LOC_ID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_LOCATION(LOC_ID)
7 ,QTY NUMBER NOT NULL
8 ,ORD_DATE DATE DEFAULT SYSDATE
9* )
,LOC_ID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_LOCATION(LOC_ID)
*
ERROR at line 6:
ORA-02264: name already used by an existing constraint

```

SQL> ED

Wrote file afiedt.buf

```

1 CREATE TABLE B19_ORDER_DETAILS
2 (
3 ORD_ID NUMBER
4 ,PID NUMBER CONSTRAINT ORDDET_PID_FK REFERENCES B19_PRODUCT(PID)
5 ,CID NUMBER CONSTRAINT ORDDET_CID_FK REFERENCES B19_CUSTOMER(CID)
6 ,LOC_ID NUMBER CONSTRAINT ORDDET_LOCID_FK REFERENCES B19_LOCATION(LOC_ID)
7 ,QTY NUMBER NOT NULL
8 ,ORD_DATE DATE DEFAULT SYSDATE
9* )
SQL> /

```

Table created.

SQL> CL SCR

SQL> DESCRIBE B19_PRODUCT

Name	Null?	Type
PID	NOT NULL	NUMBER
PNAME	NOT NULL	VARCHAR2(20)
PMFR		VARCHAR2(20)
PCOST	NOT NULL	NUMBER(5)

SQL> DESCRIBE B19_CUSTOMER

Name	Null?	Type
CID	NOT NULL	NUMBER
CNAME	NOT NULL	VARCHAR2(20)
CMOB	NOT NULL	NUMBER(10)

```
SQL> DESCRIBE B19_LOCATION
```

Name	Null?	Type
LOC_ID	NOT NULL	NUMBER
AREA		VARCHAR2(20)
POST_CODE		NUMBER(6)
CITY		VARCHAR2(20)
STATE		VARCHAR2(20)

```
SQL> DESCRIBE B19_ORDER_DETAILS
```

Name	Null?	Type
ORD_ID		NUMBER
PID		NUMBER
CID		NUMBER
LOC_ID		NUMBER
QTY	NOT NULL	NUMBER
ORD_DATE		DATE

```
SQL> INSERT INTO B19_PRODUCUT
```

```
2 VALUES
3 (
4  &PID
5  , '&PNAME'
6  , '&PMFR'
7  , &PCOST
8 );
```

```
Enter value for pid: 1
```

```
old 4: &PID
```

```
new 4: 1
```

```
Enter value for pname: CURD
```

```
old 5: , '&PNAME'
```

```
new 5: , 'CURD'
```

```
Enter value for pmfr: NANDINI
```

```
old 6: , '&PMFR'
```

```
new 6: , 'NANDINI'
```

```
Enter value for pcost: 30
```

```
old 7: , &PCOST
```

```
new 7: , 30
```

```
INSERT INTO B19_PRODUCUT
```

```
*
```

```
ERROR at line 1:
```

```
ORA-00942: table or view does not exist
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 INSERT INTO B19_PRODUCT
2 VALUES
3 (
4  &PID
5  , '&PNAME'
6  , '&PMFR'
7  , &PCOST
8* )
```

```
SQL> /
```

```
Enter value for pid: 1
```



```
old    4: &PID
new    4: 1
Enter value for pname: CURD
old    5: , '&PNAME'
new    5: , 'CURD'
Enter value for pmfr: NANDINI
old    6: , '&PMFR'
new    6: , 'NANDINI'
Enter value for pcost: 30
old    7: , &PCOST
new    7: , 30
```

1 row created.

```
SQL> /
Enter value for pid: 2
old    4: &PID
new    4: 2
Enter value for pname: MILK
old    5: , '&PNAME'
new    5: , 'MILK'
Enter value for pmfr: NANDINI
old    6: , '&PMFR'
new    6: , 'NANDINI'
Enter value for pcost: 20
old    7: , &PCOST
new    7: , 20
```

1 row created.

```
SQL> /
Enter value for pid: 3
old    4: &PID
new    4: 3
Enter value for pname: ICE
old    5: , '&PNAME'
new    5: , 'ICE'
Enter value for pmfr: NESTLEY
old    6: , '&PMFR'
new    6: , 'NESTLEY'
Enter value for pcost: 40
old    7: , &PCOST
new    7: , 40
```

1 row created.

```
SQL> /
Enter value for pid: 4
old    4: &PID
new    4: 4
Enter value for pname: CHOCOLATES
old    5: , '&PNAME'
new    5: , 'CHOCOLATES'
Enter value for pmfr: NESTLEY
old    6: , '&PMFR'
new    6: , 'NESTLEY'
Enter value for pcost: 20
old    7: , &PCOST
new    7: , 20
```

1 row created.

```
SQL> /
Enter value for pid: 5
```

```

old 4: &PID
new 4: 5
Enter value for pname: BISCUITS
old 5: , '&PNAME'
new 5: , 'BISCUITS'
Enter value for pmfr: NESTLEY
old 6: , '&PMFR'
new 6: , 'NESTLEY'
Enter value for pcost: 10
old 7: , &PCOST
new 7: , 10

```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT *
      2 FROM B19_PRODUCT;
```

PID	PNAME	PMFR	PCOST
1	CURD	NANDINI	30
2	MILK	NANDINI	20
3	ICE	NESTLEY	40
4	CHOCOLATES	NESTLEY	20
5	BISCUITS	NESTLEY	10

```
SQL> DESC B19_CUSTOMER
```

Name	Null?	Type
CID	NOT NULL	NUMBER
CNAME	NOT NULL	VARCHAR2(20)
CMOB	NOT NULL	NUMBER(10)

```

SQL> INSERT INTO B19_CUSTOMER
      2 VALUES
      3 (
      4 &CID
      5 , '&CNAME'
      6 , &CMOB
      7 );

```

Enter value for cid: 1

```
old 4: &CID
```

```
new 4: 1
```

Enter value for cname: BALU

```
old 5: , '&CNAME'
```

```
new 5: , 'BALU'
```

Enter value for cmob: 12345

```
old 6: , &CMOB
```

```
new 6: , 12345
```

```
INSERT INTO B19_CUSTOMER
```

```
*
```

ERROR at line 1:

ORA-02290: check constraint (SYSTEM.B19CUST_CMOB_CHK) violated

```
SQL> /
```

Enter value for cid: 1

```
old 4: &CID
```

```
new 4: 1
```

```
Enter value for cname: BALUE
old 5: , '&CNAME'
new 5: , 'BALUE'
Enter value for cmob: 1234512345
old 6: , '&CMOB'
new 6: , '1234512345'
```

1 row created.

```
SQL> /
Enter value for cid: 2
old 4: &CID
new 4: 2
Enter value for cname: RAGHAVA
old 5: , '&CNAME'
new 5: , 'RAGHAVA'
Enter value for cmob: 1234512346
old 6: , '&CMOB'
new 6: , '1234512346'
```

1 row created.

```
SQL> /
Enter value for cid: 3
old 4: &CID
new 4: 3
Enter value for cname: BASWANTH
old 5: , '&CNAME'
new 5: , 'BASWANTH'
Enter value for cmob: 1234512347
old 6: , '&CMOB'
new 6: , '1234512347'
```

1 row created.

```
SQL> /
Enter value for cid: 4
old 4: &CID
new 4: 4
Enter value for cname: SURENDRA
old 5: , '&CNAME'
new 5: , 'SURENDRA'
Enter value for cmob: 1234512348
old 6: , '&CMOB'
new 6: , '1234512348'
```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT *
      2 FROM B19_CUSTOMER;
```

	CID	CNAME	CMOB
1		BALUE	1234512345
2		RAGHAVA	1234512346
3		BASWANTH	1234512347
4		SURENDRA	1234512348

```
SQL> DESC B19_LOCATION
```

Name	Null?	Type
-----	-----	-----
LOC_ID	NOT NULL	NUMBER
AREA		VARCHAR2(20)
POST_CODE		NUMBER(6)
CITY		VARCHAR2(20)
STATE		VARCHAR2(20)

```
SQL> INSERT INTO B19_LOCATION
```

```
2 VALUES
```

```
3 (
```

```
4 &LID
```

```
5 , '&ARE'
```

```
6 , &PC
```

```
7 , '&CITY'
```

```
8 , '&STA'
```

```
9 );
```

```
Enter value for lid: 1
```

```
old 4: &LID
```

```
new 4: 1
```

```
Enter value for are: HSR
```

```
old 5: , '&ARE'
```

```
new 5: , 'HSR'
```

```
Enter value for pc: 560034
```

```
old 6: , &PC
```

```
new 6: , 560034
```

```
Enter value for city: BANGALORE
```

```
old 7: , '&CITY'
```

```
new 7: , 'BANGALORE'
```

```
Enter value for sta: KARNATAKA
```

```
old 8: , '&STA'
```

```
new 8: , 'KARNATAKA'
```

```
1 row created.
```

```
SQL> /
```

```
Enter value for lid: 2
```

```
old 4: &LID
```

```
new 4: 2
```

```
Enter value for are: BTM
```

```
old 5: , '&ARE'
```

```
new 5: , 'BTM'
```

```
Enter value for pc: 560033
```

```
old 6: , &PC
```

```
new 6: , 560033
```

```
Enter value for city: BANGALORE
```

```
old 7: , '&CITY'
```

```
new 7: , 'BANGALORE'
```

```
Enter value for sta: KARNAKATAKA
```

```
old 8: , '&STA'
```

```
new 8: , 'KARNAKATAKA'
```

```
1 row created.
```

```
SQL> /
```

```
Enter value for lid: 3
```

```
old 4: &LID
```

```
new 4: 3
```

```
Enter value for are: AGARA
```

```
old 5: , '&ARE'
```

```
new 5: , 'AGARA'
```

```
Enter value for pc: 560031
```

```
old 6: , &PC
```

```

new      6: ,560031
Enter value for city: AGARA
old      7: , '&CITY'
new      7: , 'AGARA'
Enter value for sta: KARANATAKA
old      8: , '&STA'
new      8: , 'KARANATAKA'

```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT *
      2  FROM B19_LOCATION;
```

LOC_ID	AREA	POST_CODE	CITY	STATE
1	HSR	560034	BANGALORE	KARNATAKA
2	BTM	560033	BANGALORE	KARNAKTAKA
3	AGARA	560031	AGARA	KARANATAKA

```
SQL> SET LINESIZE 200
```

```
SQL> R
```

```

1  SELECT *
2* FROM B19_LOCATION

```

LOC_ID	AREA	POST_CODE	CITY	STATE
1	HSR	560034	BANGALORE	KARNATAKA
2	BTM	560033	BANGALORE	KARNAKTAKA
3	AGARA	560031	AGARA	KARANATAKA

```

SQL> UPDATE B19_LOCATION
      2  SET CITY = 'BANGALORE'
      3  WHERE LOC_ID=3;

```

1 row updated.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL S CR
```

SP2-0158: unknown CLEAR option "S"

SP2-0158: unknown CLEAR option "CR"

```
SQL> CL SCR
```

```

SQL> SELECT *
      2  FROM B19_PRODUCT;

```

PID PNAME

PMFR

PCOST

```

-----
      1 CURD                      NANDINI                      30
      2 MILK                      NANDINI                      20
      3 ICE                      NESTLEY                      40
      4 CHOCOLATES              NESTLEY                      20
      5 BISCUITS                NESTLEY                      10

```

```

SQL> SELECT *
      2 FROM B19_CUSTOMER;

```

```

      CID CNAME                      CMOB
-----
      1 BALUE                    1234512345
      2 RAGHAVA                  1234512346
      3 BASWANTH                 1234512347
      4 SURENDRA                 1234512348

```

```

SQL> SELECT *
      2 FROM B19_LOCATION;

```

```

      LOC_ID AREA                      POST_CODE CITY                      STATE
-----
      1 HSR                      560034 BANGALORE                      KARNATAKA
      2 BTM                      560033 BANGALORE                      KARNAKTAKA
      3 AGARA                    560031 BANGALORE                      KARANATAKA

```

```

SQL> DESC B19_ORDER_DETAILS

```

```

Name
Null?   Type
-----
ORD_ID
NUMBER
PID
NUMBER
CID
NUMBER
LOC_ID
NUMBER
QTY
NOT NULL NUMBER
ORD_DATE
DATE

```

```

SQL> EDIT
Wrote file afiedt.buf

```

```

      1 SELECT *
      2* FROM B19_LOCATION
SQL> CL SCR

```

```

SQL> DESC B19_ORDER_DETAILS

```

```

Name
Null?   Type
-----

```

```

ORD_ID
NUMBER
PID
NUMBER

```

```

CID
NUMBER
LOC_ID
NUMBER
QTY
NOT NULL NUMBER
ORD_DATE
DATE

```

```

SQL> SELECT *
      2 FROM B19_PRODUCT;

```

PID	PNAME	PMFR	PCOST
1	CURD	NANDINI	30
2	MILK	NANDINI	20
3	ICE	NESTLEY	40
4	CHOCOLATES	NESTLEY	20
5	BISCUITS	NESTLEY	10

```

SQL> SELECT *
      2 FROM B19_CUSTOMER;

```

CID	CNAME	CMOB
1	BALUE	1234512345
2	RAGHAVA	1234512346
3	BASWANATH	1234512347
4	SURENDRA	1234512348

```

SQL> SELECT *
      2 FROM B19_LOCATION;

```

LOC_ID	AREA	POST_CODE	CITY	STATE
1	HSR	560034	BANGALORE	KARNATAKA
2	BTM	560033	BANGALORE	KARNAKTAKA
3	AGARA	560031	BANGALORE	KARANATAKA

```

SQL> INSERT INTO B19_ORDER_DETAILS
      2 VALUES
      3 (
      4 101
      5 ,1
      6 ,1
      7 ,1
      8 ,3
      9 ,SYSDATE
     10 );

```

1 row created.

```

SQL> ED
Wrote file afiedt.buf

```

```

1 INSERT INTO B19_ORDER_DETAILS
2 VALUES
3 (
4 101
5 ,2
6 ,1
7 ,1
8 ,5

```

```
9 ,SYSDATE
10* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1 INSERT INTO B19_ORDER_DETAILS
2 VALUES
3 (
4 101
5 ,10
6 ,1
7 ,1
8 ,5
9 ,SYSDATE
10* )
SQL> /
```

```
INSERT INTO B19_ORDER_DETAILS
*
```

```
ERROR at line 1:
ORA-02291: integrity constraint (SYSTEM.ORDDET_PID_FK) violated - parent key not
found
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 INSERT INTO B19_ORDER_DETAILS
2 VALUES
3 (
4 101
5 ,NULL
6 ,1
7 ,1
8 ,5
9 ,SYSDATE
10* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1 INSERT INTO B19_ORDER_DETAILS
2 VALUES
3 (
4 102
5 ,2
6 ,2
7 ,2
8 ,3
9 ,SYSDATE
10* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```



```
1  INSERT INTO B19_ORDER_DETAILS
2  VALUES
3  (
4  102
5  ,3
6  ,2
7  ,2
8  ,10
9  ,SYSDATE
10* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO B19_ORDER_DETAILS
2  VALUES
3  (
4  102
5  ,4
6  ,2
7  ,2
8  ,5
9  ,SYSDATE
10* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO B19_ORDER_DETAILS
2  VALUES
3  (
4  102
5  ,2
6  ,2
7  ,5
8  ,10
9  ,SYSDATE
10* )
SQL> /
```

```
INSERT INTO B19_ORDER_DETAILS
*
```

ERROR at line 1:
ORA-02291: integrity constraint (SYSTEM.ORDDET_LOCID_FK) violated - parent key not found

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO B19_ORDER_DETAILS
2  VALUES
3  (
4  102
5  ,2
6  ,2
7  ,2
8  ,10
9  ,SYSDATE
```

```
10* )
SQL> /
```

1 row created.

```
SQL> CL SCR
```

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT *
      2  FROM B19_PRODUCT;
```

PID	PNAME	PMFR	PCOST
1	CURD	NANDINI	30
2	MILK	NANDINI	20
3	ICE	NESTLEY	40
4	CHOCOLATES	NESTLEY	20
5	BISCUITS	NESTLEY	10

```
SQL> SELECT *
      2  FROM B19_CUSTOMER;
```

CID	CNAME	CMOB
1	BALUE	1234512345
2	RAGHAVA	1234512346
3	BASWANTH	1234512347
4	SURENDRA	1234512348

```
SQL> SELECT *
      2  FROM B19_LOCATION;
```

LOC_ID	AREA	POST_CODE	CITY	STATE
1	HSR	560034	BANGALORE	KARNATAKA
2	BTM	560033	BANGALORE	KARNAKTAKA
3	AGARA	560031	BANGALORE	KARANATAKA

```
SQL> SELECT *
      2  FROM B19_ORDER_DETAILS;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101	2	1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102	2	2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102	2	2	2	10	30-AUG-17

7 rows selected.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> DELETE FROM B19_PRODUCT WHERE PID=2;
```

```
DELETE FROM B19_PRODUCT WHERE PID=2
```

```
*
```

```
ERROR at line 1:
```

```
ORA-02292: integrity constraint (SYSTEM.ORDDET_PID_FK) violated - child record found
```

```
SQL> DELETE FROM B19_PRODUCT WHERE PID=5;
```

```
1 row deleted.
```

```
SQL> DELETE FROM B19_PRODUCT WHERE PID=4;
```

```
DELETE FROM B19_PRODUCT WHERE PID=4
```

```
*
```

```
ERROR at line 1:
```

```
ORA-02292: integrity constraint (SYSTEM.ORDDET_PID_FK) violated - child record found
```

```
SQL> DELETE FROM B19_PRODUCT WHERE PID=3;
```

```
DELETE FROM B19_PRODUCT WHERE PID=3
```

```
*
```

```
ERROR at line 1:
```

```
ORA-02292: integrity constraint (SYSTEM.ORDDET_PID_FK) violated - child record found
```

```
SQL> DELETE FROM B19_PRODUCT WHERE PID=2;
```

```
DELETE FROM B19_PRODUCT WHERE PID=2
```

```
*
```

```
ERROR at line 1:
```

```
ORA-02292: integrity constraint (SYSTEM.ORDDET_PID_FK) violated - child record found
```

```
SQL> DELETE FROM B19_PRODUCT WHERE PID=1;
```

```
DELETE FROM B19_PRODUCT WHERE PID=1
```

```
*
```

```
ERROR at line 1:
```

```
ORA-02292: integrity constraint (SYSTEM.ORDDET_PID_FK) violated - child record found
```

```
SQL>
```

```
SQL> COMMIT;
```

```
Commit complete.
```

```
SQL> CL SCR
```

```
SQL> SELECT *
```

```
2 FROM B19_CUSTOMER;
```

CID	CNAME	CMOB
1	BALUE	1234512345
2	RAGHAVA	1234512346
3	BASWANTH	1234512347
4	SURENDRA	1234512348

```
SQL> SELECT *
```

```
2 FROM B19_ORDER_DETAILS;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
--------	-----	-----	--------	-----	----------

```

-----
      101          1          1          1          3 30-AUG-17
      101          2          1          1          5 30-AUG-17
      101          1          1          1          5 30-AUG-17
      102          2          2          2          3 30-AUG-17
      102          3          2          2         10 30-AUG-17
      102          4          2          2          5 30-AUG-17
      102          2          2          2         10 30-AUG-17

```

7 rows selected.

```
SQL> DELETE FROM B19_CUSTOMER WHERE CID=2;
```

```
DELETE FROM B19_CUSTOMER WHERE CID=2
```

```
*
```

```
ERROR at line 1:
```

```
ORA-02292: integrity constraint (SYSTEM.ORDDET_CID_FK) violated - child record
found
```

```
SQL> DELETE FROM B19_CUSTOMER WHERE CID=3;
```

1 row deleted.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> CREATE TABLE B19_ORDER_DETAILS_1
```

```

2  (
3  ORD_ID NUMBER
4  ,PID NUMBER CONSTRAINT ORDDDET_PID_FK1 REFERENCES B19_PRODUCT(PID) ON DELETE
CASCADE
5  ,CID NUMBER CONSTRAINT ORDDDET_CID_FK1 REFERENCES B19_CUSTOMER(CID) ON DELETE
CASCADE
6  ,LOC_ID NUMBER CONSTRAINT ORDDDET_CID_FK1 REFERENCES B19_LOCATION(LOC_ID) ON
DELETE CASCADE
7  ,QTY NUMBER NOT NULL
8  ,ORD_DATE DATE DEFAULT SYSDATE
9  );

```

```
,LOC_ID NUMBER CONSTRAINT ORDDDET_CID_FK1 REFERENCES B19_LOCATION(LOC_ID) ON DELETE
CASCADE
```

```
*
```

```
ERROR at line 6:
```

```
ORA-02264: name already used by an existing constraint
```

```
SQL> ED
```

Wrote file afiedt.buf

```

1  CREATE TABLE B19_ORDER_DETAILS_1
2  (
3  ORD_ID NUMBER
4  ,PID NUMBER CONSTRAINT ORDDDET_PID_FK1 REFERENCES B19_PRODUCT(PID) ON DELETE
CASCADE
5  ,CID NUMBER CONSTRAINT ORDDDET_CID_FK1 REFERENCES B19_CUSTOMER(CID) ON DELETE
CASCADE
6  ,LOC_ID NUMBER CONSTRAINT ORDDDET_LOCID_FK1 REFERENCES B19_LOCATION(LOC_ID) ON
DELETE CASCADE
7  ,QTY NUMBER NOT NULL
8  ,ORD_DATE DATE DEFAULT SYSDATE
9* )
SQL> /

```

Table created.

```
SQL>
SQL>
SQL> CREATE TABLE B19_ORDER_DETAILS_2
  2  (
  3  ORD_ID NUMBER
  4  ,PID NUMBER CONSTRAINT ORDDET_PID_FK2 REFERENCES B19_PRODUCT(PID) ON DELETE
SET NULL
  5  ,CID NUMBER CONSTRAINT ORDDET_CID_FK2 REFERENCES B19_CUSTOMER(CID) ON DELETE
SET NULL
  6  ,LOC_ID NUMBER CONSTRAINT ORDDET_LOCID_FK2 REFERENCES B19_LOCATION(LOC_ID) ON
DELETE SET NULL
  7  ,QTY NUMBER NOT NULL
  8  ,ORD_DATE DATE DEFAULT SYSDATE
  9  );
```

Table created.

```
SQL> CL SCR
```

```
SQL> SELECT *
  2  FROM B19_ORDER_DETAILS;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101	2	1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102	2	2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102	2	2	2	10	30-AUG-17

7 rows selected.

```
SQL> SELECT *
  2  FROM B19_ORDER_DETAILS_1;
```

no rows selected

```
SQL> INSERT INTO B19_ORDER_DETAILS_1
  2  SELECT *
  3  FROM B19_ORDER_DETAILS;
```

7 rows created.

```
SQL> EDIT
Wrote file afiedt.buf
```

```
  1  INSERT INTO B19_ORDER_DETAILS_2
  2  SELECT *
  3* FROM B19_ORDER_DETAILS
SQL> /
```

7 rows created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT *
      2 FROM B19_PRODUCT;
```

PID	PNAME	PMFR	PCOST
1	CURD	NANDINI	30
2	MILK	NANDINI	20
3	ICE	NESTLEY	40
4	CHOCOLATES	NESTLEY	20

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101	2	1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102	2	2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102	2	2	2	10	30-AUG-17

7 rows selected.

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS_1;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101	2	1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102	2	2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102	2	2	2	10	30-AUG-17

7 rows selected.

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS_12;
FROM B19_ORDER_DETAILS_12
      *
```

ERROR at line 2:
ORA-00942: table or view does not exist

```
SQL> SELECT *
      2 ..
      3
```

```
SQL>
```

```
SQL> CL SCR
```

```
SQL> SELECT *
      2 FROM B19_PRODUCT;
```

PID	PNAME	PMFR	PCOST
1	CURD	NANDINI	30
2	MILK	NANDINI	20
3	ICE	NESTLEY	40
4	CHOCOLATES	NESTLEY	20

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101	2	1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102	2	2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102	2	2	2	10	30-AUG-17

7 rows selected.

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS_1;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101	2	1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102	2	2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102	2	2	2	10	30-AUG-17

7 rows selected.

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS_2;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101	2	1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102	2	2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102	2	2	2	10	30-AUG-17

7 rows selected.

```
SQL> DELETE FROM B19_PRODUCT WHERE PID=2;
DELETE FROM B19_PRODUCT WHERE PID=2
*
```

ERROR at line 1:

ORA-02292: integrity constraint (SYSTEM.ORDDET_PID_FK) violated - child record found

```
SQL> CL SCR
```

```
SQL> SELECT *
      2 FROM B19_PRODUCT;
```

PID	PNAME	PMFR	PCOST
1	CURD	NANDINI	30
2	MILK	NANDINI	20
3	ICE	NESTLEY	40
4	CHOCOLATES	NESTLEY	20

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101	2	1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102	2	2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102	2	2	2	10	30-AUG-17

7 rows selected.

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS_1;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101	2	1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102	2	2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102	2	2	2	10	30-AUG-17

7 rows selected.

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS_2;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101	2	1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102	2	2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102	2	2	2	10	30-AUG-17

7 rows selected.

```
SQL> DELETE FROM B19_ORDER_DETAILS;
```

7 rows deleted.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT *
      2 FROM B19_PRODUCT;
```

PID	PNAME	PMFR	PCOST
1	CURD	NANDINI	30
2	MILK	NANDINI	20
3	ICE	NESTLEY	40

4 CHOCOLATES

NESTLEY

20

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS;
```

no rows selected

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS_1;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101	2	1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102	2	2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102	2	2	2	10	30-AUG-17

7 rows selected.

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS_2;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101	2	1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102	2	2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102	2	2	2	10	30-AUG-17

7 rows selected.

```
SQL> DELETE FROM B19_PRODUCT WHERE PID=2;
```

1 row deleted.

```
SQL> SELECT *
      2 FROM B19_PRODUCT;
```

PID	PNAME	PMFR	PCOST
1	CURD	NANDINI	30
3	ICE	NESTLEY	40
4	CHOCOLATES	NESTLEY	20

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS;
```

no rows selected

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS_1;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101		1	1	5	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17

```
SQL> SELECT *
      2 FROM B19_ORDER_DETAILS_2;
```

ORD_ID	PID	CID	LOC_ID	QTY	ORD_DATE
101	1	1	1	3	30-AUG-17
101		1	1	5	30-AUG-17
101		1	1	5	30-AUG-17
102		2	2	3	30-AUG-17
102	3	2	2	10	30-AUG-17
102	4	2	2	5	30-AUG-17
102		2	2	10	30-AUG-17

7 rows selected.

```
SQL> SPOOL OFF
```

9-Jun-2018 -Example1

```
SQL> CREATE TABLE B24_STUDENT
      2 (
      3 SID NUMBER
      4 ,SNAME VARCHAR2(20)
      5 ,SMOB NUMBER(10)
      6 ,SLOC VARCHAR2(10)
      7 ,SQUAL VARCHAR2(10)
      8 .
```

```
SQL>
```

```
SQL> ED
```

Wrote file afiedt.buf

```
      1 CREATE TABLE B24_STUDENT
      2 (
      3 SID NUMBER CONSTRAINT B24STD_SID_PK PRIMARY KEY
      4 ,SNAME VARCHAR2(20) CONSTRAINT B24STD_SNAME_NN NOT NULL
      5 ,SMOB NUMBER(10) CONSTRAINT B24STD_SMOB_CHK CHECK (LENGTH(MOB) = 10)
      6 ,SLOC VARCHAR2(10)
      7 ,SQUAL VARCHAR2(10)
      8* )
```

```
SQL> /
```

SP2-0640: Not connected

```
SQL> CONNECT SYSTEM/ARJUN
```

Connected.

```
SQL> CL SCR
```

```
SQL> R
```

```
      1 CREATE TABLE B24_STUDENT
      2 (
      3 SID NUMBER CONSTRAINT B24STD_SID_PK PRIMARY KEY
      4 ,SNAME VARCHAR2(20) CONSTRAINT B24STD_SNAME_NN NOT NULL
      5 ,SMOB NUMBER(10) CONSTRAINT B24STD_SMOB_CHK CHECK (LENGTH(MOB) = 10)
      6 ,SLOC VARCHAR2(10)
      7 ,SQUAL VARCHAR2(10)
      8* )
```

```
,SLOC VARCHAR2(10)
```

```
*
```

ERROR at line 6:

ORA-02438: Column check constraint cannot reference other columns

```
SQL> ED
```

Wrote file afiedt.buf

```

1 CREATE TABLE B24_STUDENT
2 (
3   SID NUMBER CONSTRAINT B24STD_SID_PK PRIMARY KEY
4   ,SNAME VARCHAR2(20) CONSTRAINT B24STD_SNAME_NN NOT NULL
5   ,SMOB NUMBER(10) CONSTRAINT B24STD_SMOB_CHK CHECK (LENGTH(SMOB) = 10)
6   ,SLOC VARCHAR2(10)
7   ,SQUAL VARCHAR2(10)
8* )
SQL> /

```

Table created.

SQL> CL SCR

```

SQL> DESC CONSTRAINT B24STD_SID_PK
Usage: DESCRIBE [schema.]object[@db_link]
SQL> DESC B24_STUDENT

```

Name	Null?	Type
SID	NOT NULL	NUMBER
SNAME	NOT NULL	VARCHAR2(20)
SMOB		NUMBER(10)
SLOC		VARCHAR2(10)
SQUAL		VARCHAR2(10)

SQL> CL SCR

```

SQL> DESC B24_STUDENT

```

Name	Null?	Type
SID	NOT NULL	NUMBER
SNAME	NOT NULL	VARCHAR2(20)
SMOB		NUMBER(10)
SLOC		VARCHAR2(10)
SQUAL		VARCHAR2(10)

```

SQL> CREATE TABLE B24_COURSE
2 (
3   CID NUMBER
4   ,CNAME VARCHAR2(20)
5   ,CDUATION NUMBER(3)
6   ,CFEES NUMBER(5)
7   ,CONSTRAINT B24CRS_CID_PK PRIMARY KEY(CID)
8 );

```

Table created.

SQL> CL SCR

```

SQL> DESC B24_COURSE

```

Name	Null?	Type
CID	NOT NULL	NUMBER
CNAME		VARCHAR2(20)
CDUATION		NUMBER(3)
CFEES		NUMBER(5)

```

SQL> DESC B24_STUDENT

```

Name	Null?	Type
SID	NOT NULL	NUMBER
SNAME	NOT NULL	VARCHAR2(20)

SMOB
SLOC
SQUAL

NUMBER(10)
VARCHAR2(10)
VARCHAR2(10)

SQL> SELECT * FROM B24_STUDENT;

no rows selected

SQL> SELECT * FROM B24_COURSE;

no rows selected

SQL> INSERT INTO B24_COURSE
2 VALUES
3 (
4 &CID
5 , '&CNAME'
6 , &CD'
7 , '&CFEES'
8);

Enter value for cid: 1

old 4: &CID

new 4: 1

Enter value for cname: PLSQL

old 5: , '&CNAME'

new 5: , 'PLSQL'

Enter value for cd: 60

old 6: , &CD'

new 6: , 60'

Enter value for cfees: 10000

old 7: , '&CFEES'

new 7: , '10000'

ERROR:

ORA-01756: quoted string not properly terminated

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B24_COURSE
2  VALUES
3  (
4    &CID
5    , '&CNAME'
6    , &CD
7    , '&CFEES'
8* )
```

SQL> /

Enter value for cid: 1

old 4: &CID

new 4: 1

Enter value for cname: PLSQL

old 5: , '&CNAME'

new 5: , 'PLSQL'

Enter value for cd: 60

old 6: , &CD

new 6: , 60

Enter value for cfees: 10000

old 7: , '&CFEES'

new 7: , '10000'

1 row created.

SQL> R

```
1  INSERT INTO B24_COURSE
2  VALUES
3  (
4    &CID
5    , '&CNAME'
6    , &CD
7    , '&CFEES'
8* )
```

Enter value for cid: 2

old 4: &CID

new 4: 2

Enter value for cname: EXCEL

old 5: , '&CNAME'

new 5: , 'EXCEL'

Enter value for cd: 15

old 6: , &CD

new 6: , 15

Enter value for cfees: 2000

old 7: , '&CFEES'

new 7: , '2000'

1 row created.

SQL> R

```
1  INSERT INTO B24_COURSE
2  VALUES
3  (
4    &CID
5    , '&CNAME'
6    , &CD
7    , '&CFEES'
8* )
```

Enter value for cid: 3

old 4: &CID

new 4: 3

Enter value for cname: MACROS

old 5: , '&CNAME'

new 5: , 'MACROS'

Enter value for cd: 30

old 6: , &CD

new 6: , 30

Enter value for cfees: 6000

old 7: , '&CFEES'

new 7: , '6000'

1 row created.

SQL> R

```
1  INSERT INTO B24_COURSE
2  VALUES
3  (
4    &CID
5    , '&CNAME'
6    , &CD
7    , '&CFEES'
8* )
```

Enter value for cid: 4

old 4: &CID

new 4: 4

Enter value for cname: TABLEAU

old 5: , '&CNAME'

new 5: , 'TABLEAU'

Enter value for cd: 60

old 6: , &CD

```

new    6: ,60
Enter value for cfees: 11000
old    7: , '&CFEES'
new    7: , '11000'

```

1 row created.

```

SQL> R
1  INSERT INTO B24_COURSE
2  VALUES
3  (
4    &CID
5    , '&CNAME'
6    , &CD
7    , '&CFEES'
8* )
Enter value for cid: 5
old    4:  &CID
new    4:  5
Enter value for cname: DATA SCIENCE
old    5: , '&CNAME'
new    5: , 'DATA SCIENCE'
Enter value for cd: 120
old    6: , &CD
new    6: , 120
Enter value for cfees: 35000
old    7: , '&CFEES'
new    7: , '35000'

```

1 row created.

```

SQL> R
1  INSERT INTO B24_COURSE
2  VALUES
3  (
4    &CID
5    , '&CNAME'
6    , &CD
7    , '&CFEES'
8* )
Enter value for cid: 6
old    4:  &CID
new    4:  6
Enter value for cname: JAVA
old    5: , '&CNAME'
new    5: , 'JAVA'
Enter value for cd: 45
old    6: , &CD
new    6: , 45
Enter value for cfees: 6000
old    7: , '&CFEES'
new    7: , '6000'

```

1 row created.

SQL> CL SCR

SQL> SELECT * FROM B24_COURSE;

	CID	CNAME	CDUATION	CFEES
1	PLSQL		60	10000
2	EXCEL		15	2000
3	MACROS		30	6000

4	TABLEAU	60	11000
5	DATA SCIENCE	120	35000
6	JAVA	45	6000

6 rows selected.

SQL> COMMIT;

Commit complete.

SQL> CL SCR

SQL> DESC B24_STUDENT

Name	Null?	Type
SID	NOT NULL	NUMBER
SNAME	NOT NULL	VARCHAR2(20)
SMOB		NUMBER(10)
SLOC		VARCHAR2(10)
SQUAL		VARCHAR2(10)

SQL> SELECT * FROM B24_STUDENT;

no rows selected

SQL> INSERT INTO B24_STUDENT

2 VALUES

3 (

4 &SID

5 , '&SNAME'

6 , &MO

7 , '&LC'

8 , '&QAL'

9);

Enter value for sid: 1

old 4: &SID

new 4: 1

Enter value for sname: AKBAR

old 5: , '&SNAME'

new 5: , 'AKBAR'

Enter value for mo: 9876598765

old 6: , &MO

new 6: , 9876598765

Enter value for lc: HAL

old 7: , '&LC'

new 7: , 'HAL'

Enter value for qal: MBA

old 8: , '&QAL'

new 8: , 'MBA'

1 row created.

SQL> R

1 INSERT INTO B24_STUDENT

2 VALUES

3 (

4 &SID

5 , '&SNAME'

6 , &MO

7 , '&LC'

8 , '&QAL'

9*)

Enter value for sid: 2

old 4: &SID

```

new      4: 2
Enter value for sname: NIVEDAN
old      5: , '&SNAME'
new      5: , 'NIVEDAN'
Enter value for mo: 9876598766
old      6: , &MO
new      6: , 9876598766
Enter value for lc: SANTHINAGAR
old      7: , '&LC'
new      7: , 'SANTHINAGAR'
Enter value for qal: BE
old      8: , '&QAL'
new      8: , 'BE'
, 'SANTHINAGAR'
*
ERROR at line 7:
ORA-12899: value too large for column "SYSTEM"."B24_STUDENT"."SLOC" (actual:
11, maximum: 10)

```

```

SQL> R
1  INSERT INTO B24_STUDENT
2  VALUES
3  (
4  &SID
5  , '&SNAME'
6  , &MO
7  , '&LC'
8  , '&QAL'
9* )
Enter value for sid: 2
old      4: &SID
new      4: 2
Enter value for sname: NIVEDAN
old      5: , '&SNAME'
new      5: , 'NIVEDAN'
Enter value for mo: 9876598766
old      6: , &MO
new      6: , 9876598766
Enter value for lc: SANTHINAGA
old      7: , '&LC'
new      7: , 'SANTHINAGA'
Enter value for qal: BE
old      8: , '&QAL'
new      8: , 'BE'

```

1 row created.

```

SQL> R
1  INSERT INTO B24_STUDENT
2  VALUES
3  (
4  &SID
5  , '&SNAME'
6  , &MO
7  , '&LC'
8  , '&QAL'
9* )
Enter value for sid: 3
old      4: &SID
new      4: 3
Enter value for sname: SURYA
old      5: , '&SNAME'
new      5: , 'SURYA'

```



```

Enter value for mo: 9876598768
old   6: ,&MO
new   6: ,9876598768
Enter value for lc: HSR
old   7: , '&LC'
new   7: , 'HSR'
Enter value for qal: BE
old   8: , '&QAL'
new   8: , 'BE'

```

1 row created.

```

SQL> R
1  INSERT INTO B24_STUDENT
2  VALUES
3  (
4  &SID
5  , '&SNAME'
6  , &MO
7  , '&LC'
8  , '&QAL'
9* )

```

```

Enter value for sid: 4
old   4: &SID
new   4: 4
Enter value for sname: PRAVEEN
old   5: , '&SNAME'
new   5: , 'PRAVEEN'
Enter value for mo: 1234512345
old   6: , &MO
new   6: , 1234512345
Enter value for lc: TINFACTORY
old   7: , '&LC'
new   7: , 'TINFACTORY'
Enter value for qal: BE
old   8: , '&QAL'
new   8: , 'BE'

```

1 row created.

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B24_STUDENT;
```

SID	SNAME	SMOB	SLOC	SQUAL
1	AKBAR	9876598765	HAL	MBA
2	NIVEDAN	9876598766	SANTHINAGA	BE
3	SURYA	9876598768	HSR	BE
4	PRAVEEN	1234512345	TINFACTORY	BE

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B24_STUDENT;
```

SID	SNAME	SMOB	SLOC	SQUAL
1	AKBAR	9876598765	HAL	MBA
2	NIVEDAN	9876598766	SANTHINAGA	BE
3	SURYA	9876598768	HSR	BE

4 PRAVEEN

1234512345 TINFACTORY BE

```
SQL> SELECT * FROM B24_COURSE;
```

	CID	CNAME	CDUATION	CFEES
1	PLSQL		60	10000
2	EXCEL		15	2000
3	MACROS		30	6000
4	TABLEAU		60	11000
5	DATA SCIENCE		120	35000
6	JAVA		45	6000

```
6 rows selected.
```

```
SQL> CREATE TABLE B24_STD_CRS_DET
```

```
2  (
3  SID    NUMBER CONSTRAINT B24STDCRSDET_SID_FK REFERENCES B24_STUDENT(SID)
4  ,CID    NUMBER CONSTRAINT B24STDCRSDET_CID_FK REFERENCES B24_COURSE(CID)
5  );
```

```
Table created.
```

```
SQL> INSERT INTO B24_STD_CRS_DET
```

```
2  VALUES
3  (
4  1
5  ,1
6  );
```

```
1 row created.
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  INSERT INTO B24_STD_CRS_DET
2  VALUES
3  (
4  2
5  ,1
6* )
SQL> /
```

```
1 row created.
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  INSERT INTO B24_STD_CRS_DET
2  VALUES
3  (
4  3
5  ,1
6* )
SQL> /
```

```
1 row created.
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  INSERT INTO B24_STD_CRS_DET
2  VALUES
3  (
```

```
4 4
5 ,1
6* )
SQL> /
```

1 row created.

SQL> ED

Wrote file afiedt.buf

```
1 INSERT INTO B24_STD_CRS_DET
2 VALUES
3 (
4 7
5 ,1
6* )
SQL> /
```

INSERT INTO B24_STD_CRS_DET

*

ERROR at line 1:

ORA-02291: integrity constraint (SYSTEM.B24STDCRSDET_SID_FK) violated - parent key not found

SQL> EDIT

Wrote file afiedt.buf

```
1 INSERT INTO B24_STD_CRS_DET
2 VALUES
3 (
4 1
5 ,2
6* )
SQL> /
```

1 row created.

SQL> ED

Wrote file afiedt.buf

```
1 INSERT INTO B24_STD_CRS_DET
2 VALUES
3 (
4 1
5 ,3
6* )
SQL> /
```

1 row created.

SQL> ED

Wrote file afiedt.buf

```
1 INSERT INTO B24_STD_CRS_DET
2 VALUES
3 (
4 1
5 ,10
6* )
SQL> /
```

INSERT INTO B24_STD_CRS_DET

*

ERROR at line 1:

ORA-02291: integrity constraint (SYSTEM.B24STDCRSDET_CID_FK) violated - parent

key not found

```
SQL> RFIY
SP2-0042: unknown command "RFIY" - rest of line ignored.
SQL> EDIT
Wrote file afiedt.buf
```

```
1  INSERT INTO B24_STD_CRS_DET
2  VALUES
3  (
4  1
5  ,4
6* )
SQL> /
```

1 row created.

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B24_STD_CRS_DET;
```

SID	CID
1	1
2	1
3	1
4	1
1	2
1	3
1	4

7 rows selected.

```
SQL> INSERT INTO B24_STD_CRS_DET
2  VALUES
3  (
4  4
5  ,4
6  );
```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B24_STD_CRS_dET;
```

SID	CID
1	1
2	1
3	1
4	1
1	2
1	3
1	4
4	4

8 rows selected.

```
SQL> SELECT * FROM B24_STUDENT;
```

SID	SNAME	SMOB	SLOC	SQUAL
1	AKBAR	9876598765	HAL	MBA
2	NIVEDAN	9876598766	SANTHINAGA	BE
3	SURYA	9876598768	HSR	BE
4	PRAVEEN	1234512345	TINFACTORY	BE

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B24_COURSE;
```

CID	CNAME	CDUATION	CFEES
1	PLSQL	60	10000
2	EXCEL	15	2000
3	MACROS	30	6000
4	TABLEAU	60	11000
5	DATA SCIENCE	120	35000
6	JAVA	45	6000

6 rows selected.

```
SQL> SELECT * FROM B24_STD_CRS_DET;
```

SID	CID
1	1
2	1
3	1
4	1
1	2
1	3
1	4
4	4

8 rows selected.

```
SQL> DELETE FROM B24_COURSE WHERE CID IN (5,6);
```

2 rows deleted.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B24_COURSE;
```

CID	CNAME	CDUATION	CFEES
1	PLSQL	60	10000
2	EXCEL	15	2000
3	MACROS	30	6000
4	TABLEAU	60	11000

```
SQL> SELECT * FROM B24_STD_CRS_DET;
```

SID	CID
1	1
2	1

3	1
4	1
1	2
1	3
1	4
4	4

8 rows selected.

```
SQL> DELETEE FROM B24_COURSE WHERE CID = 3;
SP2-0734: unknown command beginning "DELETEE FR..." - rest of line ignored.
SQL>
```

```
SQL> DELETE FROM B24_COURSE WHERE CID = 3;
DELETE FROM B24_COURSE WHERE CID = 3
*
```

```
ERROR at line 1:
ORA-02292: integrity constraint (SYSTEM.B24STDCRSDET_CID_FK) violated - child
record found
```

```
SQL>
SQL> CREATE TABLE B24_STD_CRS_DET_1
2      (
3      SID      NUMBER CONSTRAINT B24STDCRSDET_SID_FK REFERENCES B24_STUDENT(SID) ON
DELETE CASCADE
4      ,CID      NUMBER CONSTRAINT B24STDCRSDET_CID_FK REFERENCES B24_COURSE(CID) ON
DELETE CASCADE
5      );
      SID      NUMBER CONSTRAINT B24STDCRSDET_SID_FK REFERENCES B24_STUDENT(SID) ON
DELETE CASCADE
      *
```

```
ERROR at line 3:
ORA-02264: name already used by an existing constraint
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  CREATE TABLE B24_STD_CRS_DET_1
2  (
3  SID      NUMBER CONSTRAINT B24STDCRSDET1_SID_FK REFERENCES B24_STUDENT(SID)
ON DELETE CASCADE
4  ,CID      NUMBER CONSTRAINT B24STDCRSDET1_CID_FK REFERENCES B24_COURSE(CID) ON
DELETE CASCADE
5*      )
SQL> /
```

Table created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  CREATE TABLE B24_STD_CRS_DET_2
2  (
3  SID      NUMBER CONSTRAINT B24STDCRSDET2_SID_FK REFERENCES B24_STUDENT(SID)
ON DELETE SET NULL
4  ,CID      NUMBER CONSTRAINT B24STDCRSDET2_CID_FK REFERENCES B24_COURSE(CID) ON
DELETE SET NULL
5*      )
SQL> /
```

Table created.

```
SQL> CL SCR
```

```
SQL> DESC B24_STD_CRS_DET
```

Name	Null?	Type
SID		NUMBER
CID		NUMBER

```
SQL> DESC B24_STD_CRS_DET_1
```

Name	Null?	Type
SID		NUMBER
CID		NUMBER

```
SQL> DESC B24_STD_CRS_DET_2
```

Name	Null?	Type
SID		NUMBER
CID		NUMBER

```
SQL> SELECT * FROM B24_STD_CRS_DET;
```

SID	CID
1	1
2	1
3	1
4	1
1	2
1	3
1	4
4	4

8 rows selected.

```
SQL> ED
```

Wrote file afiedt.buf

```

1  INSERT INTO  B24_STD_CRS_DET_1
2  SELECT *
3* FROM B24_STD_CRS_DET
SQL> /
```

8 rows created.

```
SQL> ED
```

Wrote file afiedt.buf

```

1  INSERT INTO  B24_STD_CRS_DET_2
2  SELECT *
3* FROM B24_STD_CRS_DET
SQL> /
```

8 rows created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B24_COURSE;
```

CID	CNAME	CDUATION	CFEES
-----	-------	----------	-------

1	PLSQL	60	10000
2	EXCEL	15	2000
3	MACROS	30	6000
4	TABLEAU	60	11000

```
SQL> SELECT * FROM B24_STD_CRS_DET;
```

SID	CID
1	1
2	1
3	1
4	1
1	2
1	3
1	4
4	4

8 rows selected.

```
SQL> SELECT * FROM B24_STD_CRS_DET_1;
```

SID	CID
1	1
2	1
3	1
4	1
1	2
1	3
1	4
4	4

8 rows selected.

```
SQL> SELECT * FROM B24_STD_CRS_DET_2;
```

SID	CID
1	1
2	1
3	1
4	1
1	2
1	3
1	4
4	4

8 rows selected.

```
SQL> DELETE FROM B24_COURSE WHERE CID =3;
```

```
DELETE FROM B24_COURSE WHERE CID =3
```

```
*
```

```
ERROR at line 1:
```

```
ORA-02292: integrity constraint (SYSTEM.B24STDCRSDET_CID_FK) violated - child
record found
```

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B24_COURSE;
```

CID	CNAME	CDUATION	CFEES
-----	-------	----------	-------

1	PLSQL	60	10000
2	EXCEL	15	2000
3	MACROS	30	6000
4	TABLEAU	60	11000

```
SQL> SELECT * FROM B24_STD_CRS_DET;
```

SID	CID
1	1
2	1
3	1
4	1
1	2
1	3
1	4
4	4

8 rows selected.

```
SQL> SELECT * FROM B24_STD_CRS_DET_1;
```

SID	CID
1	1
2	1
3	1
4	1
1	2
1	3
1	4
4	4

8 rows selected.

```
SQL> SELECT * FROM B24_STD_CRS_DET_2;
```

SID	CID
1	1
2	1
3	1
4	1
1	2
1	3
1	4
4	4

8 rows selected.

```
SQL> DELETE FROM B24_STD_CRS_DET;
```

8 rows deleted.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B24_COURSE;
```

CID	CNAME	CDUATION	CFEES
-----	-------	----------	-------

1	PLSQL	60	10000
2	EXCEL	15	2000
3	MACROS	30	6000
4	TABLEAU	60	11000

```
SQL> SELECT * FROM B24_STD_CRS_DET;
```

no rows selected

```
SQL> SELECT * FROM B24_STD_CRS_DET_1;
```

SID	CID
1	1
2	1
3	1
4	1
1	2
1	3
1	4
4	4

8 rows selected.

```
SQL> SELECT * FROM B24_STD_CRS_DET_2;
```

SID	CID
1	1
2	1
3	1
4	1
1	2
1	3
1	4
4	4

8 rows selected.

```
SQL> DELETE FROM B24_COURSE WHERE CID =3;
```

1 row deleted.

```
SQL> SELECT * FROM B24_COURSE;
```

CID	CNAME	CDUATION	CFEES
1	PLSQL	60	10000
2	EXCEL	15	2000
4	TABLEAU	60	11000

```
SQL> SELECT * FROM B24_STD_CRS_DET;
```

no rows selected

```
SQL> SELECT * FROM B24_STD_CRS_DET_1;
```

SID	CID
1	1
2	1
3	1
4	1
1	2

1	4
4	4

7 rows selected.

SQL> SELECT * FROM B24_STD_CRS_DET_2;

SID	CID
1	1
2	1
3	1
4	1
1	2
1	2
1	4
4	4

8 rows selected.

SQL> cl scr

SQL> DESC USER_CONSTRAINTS

Name	Null?	Type
OWNER		VARCHAR2(120)
CONSTRAINT_NAME	NOT NULL	VARCHAR2(30)
CONSTRAINT_TYPE		VARCHAR2(1)
TABLE_NAME	NOT NULL	VARCHAR2(30)
SEARCH_CONDITION		LONG
R_OWNER		VARCHAR2(120)
R_CONSTRAINT_NAME		VARCHAR2(30)
DELETE_RULE		VARCHAR2(9)
STATUS		VARCHAR2(8)
DEFERRABLE		VARCHAR2(14)
DEFERRED		VARCHAR2(9)
VALIDATED		VARCHAR2(13)
GENERATED		VARCHAR2(14)
BAD		VARCHAR2(3)
RELY		VARCHAR2(4)
LAST_CHANGE		DATE
INDEX_OWNER		VARCHAR2(30)
INDEX_NAME		VARCHAR2(30)
INVALID		VARCHAR2(7)
VIEW_RELATED		VARCHAR2(14)

SQL> SELECT TABLE_NAME, CONSTRAINT_NAME, CONSTRAINT_TYPE, DELETE_RULE
2 FROM USER_CONSTRAINTS
3 WHERE TABLE_NAME LIKE 'B24_STD_CRS_DET%';

TABLE_NAME	CONSTRAINT_NAME	C DELETE_RULE
B24_STD_CRS_DET	B24STDCRSDET_SID_FK	R NO ACTION
B24_STD_CRS_DET	B24STDCRSDET_CID_FK	R NO ACTION
B24_STD_CRS_DET_1	B24STDCRSDET1_SID_FK	R CASCADE
B24_STD_CRS_DET_1	B24STDCRSDET1_CID_FK	R CASCADE
B24_STD_CRS_DET_2	B24STDCRSDET2_SID_FK	R SET NULL
B24_STD_CRS_DET_2	B24STDCRSDET2_CID_FK	R SET NULL

6 rows selected.

SQL> COLUMN DELETE_RULE FORMAT A20

SQL> CL SCR

```
SQL> R
  1  SELECT TABLE_NAME,CONSTRAINT_NAME,CONSTRAINT_TYPE,DELETE_RULE
  2  FROM USER_CONSTRAINTS
  3* WHERE TABLE_NAME LIKE 'B24_STD_CRS_DET%'
```

TABLE_NAME	CONSTRAINT_NAME	C	DELETE_RULE
B24_STD_CRS_DET	B24STDCRSDET_SID_FK	R	NO ACTION
B24_STD_CRS_DET	B24STDCRSDET_CID_FK	R	NO ACTION
B24_STD_CRS_DET_1	B24STDCRSDET1_SID_FK	R	CASCADE

TABLE_NAME	CONSTRAINT_NAME	C	DELETE_RULE
B24_STD_CRS_DET_1	B24STDCRSDET1_CID_FK	R	CASCADE
B24_STD_CRS_DET_2	B24STDCRSDET2_SID_FK	R	SET NULL
B24_STD_CRS_DET_2	B24STDCRSDET2_CID_FK	R	SET NULL

6 rows selected.

```
SQL> SET LINESIZE 200
SQL> R
  1  SELECT TABLE_NAME,CONSTRAINT_NAME,CONSTRAINT_TYPE,DELETE_RULE
  2  FROM USER_CONSTRAINTS
  3* WHERE TABLE_NAME LIKE 'B24_STD_CRS_DET%'
```

TABLE_NAME	CONSTRAINT_NAME	C	DELETE_RULE
B24_STD_CRS_DET	B24STDCRSDET_SID_FK	R	NO ACTION
B24_STD_CRS_DET	B24STDCRSDET_CID_FK	R	NO ACTION
B24_STD_CRS_DET_1	B24STDCRSDET1_SID_FK	R	CASCADE
B24_STD_CRS_DET_1	B24STDCRSDET1_CID_FK	R	CASCADE
B24_STD_CRS_DET_2	B24STDCRSDET2_SID_FK	R	SET NULL
B24_STD_CRS_DET_2	B24STDCRSDET2_CID_FK	R	SET NULL

6 rows selected.

```
SQL> CL SCR
```

```
SQL> R
  1  SELECT TABLE_NAME,CONSTRAINT_NAME,CONSTRAINT_TYPE,DELETE_RULE
  2  FROM USER_CONSTRAINTS
  3* WHERE TABLE_NAME LIKE 'B24_STD_CRS_DET%'
```

TABLE_NAME	CONSTRAINT_NAME	C	DELETE_RULE
B24_STD_CRS_DET	B24STDCRSDET_SID_FK	R	NO ACTION

B24_STD_CRS_DET	B24STDCRSDET_CID_FK	R NO ACTION
B24_STD_CRS_DET_1	B24STDCRSDET1_SID_FK	R CASCADE
B24_STD_CRS_DET_1	B24STDCRSDET1_CID_FK	R CASCADE
B24_STD_CRS_DET_2	B24STDCRSDET2_SID_FK	R SET NULL
B24_STD_CRS_DET_2	B24STDCRSDET2_CID_FK	R SET NULL

6 rows selected.

SQL> DESC USER_CONSTRAINTS

```

Name
Null?      Type
-----
OWNER
VARCHAR2(120)
CONSTRAINT_NAME
NOT NULL VARCHAR2(30)
CONSTRAINT_TYPE
VARCHAR2(1)
TABLE_NAME
NOT NULL VARCHAR2(30)
SEARCH_CONDITION
LONG
R_OWNER
VARCHAR2(120)
R_CONSTRAINT_NAME
VARCHAR2(30)
DELETE_RULE
VARCHAR2(9)
STATUS
VARCHAR2(8)
DEFERRABLE
VARCHAR2(14)
DEFERRED
VARCHAR2(9)
VALIDATED
VARCHAR2(13)
GENERATED
VARCHAR2(14)
BAD
VARCHAR2(3)
RELY
VARCHAR2(4)
LAST_CHANGE
DATE
INDEX_OWNER
VARCHAR2(30)
INDEX_NAME
VARCHAR2(30)
INVALID
VARCHAR2(7)
VIEW_RELATED
VARCHAR2(14)

```

SQL> CL SCR

SQL> DESC USER_

ERROR:

ORA-04043: object USER_ does not exist

SQL> DESC USER_CONS_COLUMNS


```

B24_STD_CRS_DET_1          B24STDCRSDET1_SID_FK
SID

B24_STD_CRS_DET_1          B24STDCRSDET1_CID_FK
CID

B24_STD_CRS_DET_2          B24STDCRSDET2_SID_FK
SID

```

```

TABLE_NAME                  CONSTRAINT_NAME
-----

```

```

COLUMN_NAME
-----

```

```

B24_STD_CRS_DET_2          B24STDCRSDET2_CID_FK
CID

```

10 rows selected.

```

SQL> SET PAGESIZE 200
SQL> COLUMN COLUMN_NAME FORMAT A20
SQL> CL SCR

```

```

SQL> R
 1  SELECT TABLE_NAME,CONSTRAINT_NAME,COLUMN_NAME
 2  FROM USER_CONS_COLUMNS
 3  WHERE TABLE_NAME IN (
 4
 5              'B24_STUDENT'
 6              , 'B24_COURSE'
 7              , 'B24_STD_CRS_DET'
 8              , 'B24_STD_CRS_DET_1'
 9              , 'B24_STD_CRS_DET_2'
10              )

```

TABLE_NAME	CONSTRAINT_NAME	COLUMN_NAME
B24_STUDENT	B24STD_SNAME_NN	SNAME
B24_STUDENT	B24STD_SMOB_CHK	SMOB
B24_STUDENT	B24STD_SID_PK	SID
B24_COURSE	B24CRS_CID_PK	CID
B24_STD_CRS_DET	B24STDCRSDET_SID_FK	SID
B24_STD_CRS_DET	B24STDCRSDET_CID_FK	CID
B24_STD_CRS_DET_1	B24STDCRSDET1_SID_FK	SID
B24_STD_CRS_DET_1	B24STDCRSDET1_CID_FK	CID
B24_STD_CRS_DET_2	B24STDCRSDET2_SID_FK	SID
B24_STD_CRS_DET_2	B24STDCRSDET2_CID_FK	CID

10 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

 1  SELECT TABLE_NAME,CONSTRAINT_NAME,COLUMN_NAME
 2  FROM USER_CONS_COLUMNS
 3  WHERE TABLE_NAME IN (
 4
 5              'B24_STUDENT'
 6              , 'B24_COURSE'
 7              , 'B24_STD_CRS_DET'
 8              , 'B24_STD_CRS_DET_1'
 9              , 'B24_STD_CRS_DET_2'
10              , 'T1'

```

```

10          , 'T2'
11          , 'T3'
12          , 'T4'
13          , 'T5'
14*        )
SQL> /

```

TABLE_NAME	CONSTRAINT_NAME	COLUMN_NAME
B24_STUDENT	B24STD_SNAME_NN	SNAME
B24_STUDENT	B24STD_SMOB_CHK	SMOB
B24_STUDENT	B24STD_SID_PK	SID
B24_COURSE	B24CRS_CID_PK	CID
B24_STD_CRS_DET	B24STDCRSDET_SID_FK	SID
B24_STD_CRS_DET	B24STDCRSDET_CID_FK	CID
B24_STD_CRS_DET_1	B24STDCRSDET1_SID_FK	SID
B24_STD_CRS_DET_1	B24STDCRSDET1_CID_FK	CID
B24_STD_CRS_DET_2	B24STDCRSDET2_SID_FK	SID
B24_STD_CRS_DET_2	B24STDCRSDET2_CID_FK	CID

10 rows selected.

SQL> CL SCR

SQL> SPOOL OFF

10-Jun-2018 Joins.1 Notes

JOINS

WHY ?

to get data from more than one table

we will go for joins

JOINS ARE USED TO GET DATA FROM 2 OR MORE TABLES

TYPES OF JOINS

1) CROSS JOIN/CORTESIAN JOIN

IF N TABLES ARE THERE IN FROM CLAUSE

N-1 JOIN CONDITIONS SHOULD BE THERE

ELSE

IT WILL GO TO CORTESIAN

EX : IF 4 TABLES 3 CONDISOINS.

3 TABLES 2 CONDITONS

10 TABLES ARE 9 CONDITIONS

NUMBEROF RECORDSIN ONE TABLE * NUMBER
OF REOCRDS IN ANOTHER TABLE

CORTESAIN RESULTS ARE NOT USEFUL
SOMETIMES IT IS VERY USEFUL

WHEN IT WILL BE USEFULL:

IF ONE OF THE TABLE IS HAVING ONELY ONE RECORD

RULE :

WHNEVER WE ARE DOING JOIN CONDITONS

FIRST WE HAVE TO SEE IN TRNASACTIONAL TABLE HOW MANY REOCRDS ARE THERE (1 LACK RECORDS ARE THERE)
TO FULL FILL YOUR REQUIREMENT THIS TRANSACTIONAL TABLE YOU ARE JOINING WITH ANOTHER TABLE

AFTER JOINING WITH ANOTHER TABLE THEN YOU SHOULD GET RECORDS $\leq$ (TOTAL RECORDS WHICH ARE IN TRANSACTIONAL TABLE)

EQUI JOIN

IF WE ARE ABLE TO MAKE A RELATIONSHIP BETWEEN 2 TABLES

BY USING = OPERATOR THEN WE CALL IT AS EQUI JOIN

MATCHING RECORDS BETWEEN 2 TABLES.

NON EQUI JOIN

IF WE ARE ABLE TO MAKE A RELATIONSHIP BETWEEN 2 TABLES

BY USING RANGE(BETWEEN AND) THEN WE CALL IT AS NON EQUI JOIN

LEFT OUTER JOIN

MATCHING RECORDS BETWEEN 2 TABLES PLUS NON MATCHING RECORDS FROM ANY ONE TABLE

(+) OUTER JOIN OPERATOR

IF (+) OUTER JOIN OPERATOR IS THERE RIGHT SIDE OF THE = OPERATOR

THEN WE CALL LEFT OUTER JOIN

I.E FROM LEFT SIDE OF THE = OPERATOR IT WILL GIVE YOU NON MATCHING RECORDS

IN ANSI JOIN IT WILL GIVE YOU NON

MATCHING RECORDS FROM TABLE WHICH IS THERE LEFT SIDE OF LEFT KEY WORD

RIGHT OUTER JOIN

MATCHING RECORDS BETWEEN 2 TABLES NON MATCHING RECORDS FROM ANY ONE TABLE

IF (+) OUTER JOIN OPERATOR IS THERE LEFT SIDE OF THE = OPEARTOR

THEN WE CALL RIGHT OUTER JOIN

I.E FROM RIGHT SIDE OF THE = OPERATOR IT WILL GIVE YOU NON MATCHING RECORDS

IN ANSI JOIN IT WILL GIVE YOU NON

MATCHING RECORDS FROM TABLE WHICH IS RIGHT SIDE OF RIGHT KEY WORD

FULL OUTER JOIN

MATCHING RECORDS BETWEEN 2 TABLES

NON MATCHING RECORDS FROM BOTH 2 TABLES

THIS JOIN IS NOT AVAILABLE IN ORACLE JOINS

IF YOU WANT TO DO BY USING ORACLE JOINS THEN WE HAVE TO GO FOR "SET OPERATORS"

IT IS AVAILABLE IN ANSI JOINS

--

(+) OUTER JOIN OPERATORS

SELF JOIN / SELF RELATION

ANSI JOINS

ORACLE JOINS

IF N TABLES ARE THERE IN FROM CLAUSE

N-1 JOIN CONDITIONS SHOULD BE THERE

EX : IF 4 TABLES 3 CONDISOINS.

ELSE

IT WILL GO TO CORTESIAN

NUMBEROF RECORDSIN ONE TABLE * NUMBER

OF REOCRDS IN ANOTHER TABLE

CORTESAIN RESULTS ARE NOT USEFUL

SOMETIMES IT IS VERY USEFUL

WHEN IT WILL BE USEFULL:

IF ONE OF THE TABLE IS HAVING ONLY ONE RECORD

10-Jun-2018 Joins.1 Examples

```
SQL> SELECT * FROM B26_PROD;
```

PID	PNAME	PMFR	PRICE
1	MILK	NANDINI	20
2	CURD	DODLA	30

```
SQL> SELECT * FROM B26_CUST;
```

CID	CNAME	CMOB	CADDR	CGEN
1	SURESH	12345	JAKKASANDRA	MALE
2	KRISHNA	12346	HSR	MALE
3	TEJASREE	12348	TEACHERSCOLONY	FEMALE

```
SQL> SELECT * FROM B26_ORD_CUST_DET;
```

```
SELECT * FROM B26_ORD_CUST_DET
      *
```

```
ERROR at line 1:
```

```
ORA-00942: table or view does not exist
```

```
SQL> SELECT * FROM B26_CUST_ORD_DET;
```

CID	PID	QTY	ORD_DATE
1	1	4	07-JUN-18
1	2	2	07-JUN-18
1		100	07-JUN-18
2	2	3	07-JUN-18
2	1	5	07-JUN-18
	1	20	07-JUN-18
2	1	10	08-JUN-18

```
7 rows selected.
```

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B26_PROD;
```

PID	PNAME	PMFR	PRICE
1	MILK	NANDINI	20
2	CURD	DODLA	30

```
SQL> SELECT * FROM B26_CUST;
```

CID	CNAME	CMOB	CADDR	CGEN
1	SURESH	12345	JAKKASANDRA	MALE
2	KRISHNA	12346	HSR	MALE
3	TEJASREE	12348	TEACHERSCOLONY	FEMALE

```
SQL> SELECT * FROM B26_CUST_ORD_DET;
```

CID	PID	QTY	ORD_DATE
1	1	4	07-JUN-18
1	2	2	07-JUN-18
1		100	07-JUN-18
2	2	3	07-JUN-18
2	1	5	07-JUN-18
	1	20	07-JUN-18
2	1	10	08-JUN-18

```
7 rows selected.
```

```
SQL> SELECT PNAME
```

```
2 ,PRICE
```

```
3 FROM B26_PROD;
```

PNAME	PRICE
MILK	20
CURD	30

```
SQL> SELECT CNAME
      2      ,PNAME
      3      ,PRICE
      4      ,QTY
      5      ,QTY * PRICE AS TOTALPRICE
      6 FROM B26_CUST
      7      ,B26_PROD
      8      ,B26_CUST_ORD_DET;
```

CNAME	PNAME	PRICE	QTY	TOTALPRICE
SURESH	MILK	20	4	80
SURESH	MILK	20	2	40
SURESH	MILK	20	100	2000
SURESH	MILK	20	3	60
SURESH	MILK	20	5	100
SURESH	MILK	20	20	400
SURESH	MILK	20	10	200
KRISHNA	MILK	20	4	80
KRISHNA	MILK	20	2	40
KRISHNA	MILK	20	100	2000
KRISHNA	MILK	20	3	60

CNAME	PNAME	PRICE	QTY	TOTALPRICE
KRISHNA	MILK	20	5	100
KRISHNA	MILK	20	20	400
KRISHNA	MILK	20	10	200
TEJASREE	MILK	20	4	80
TEJASREE	MILK	20	2	40
TEJASREE	MILK	20	100	2000
TEJASREE	MILK	20	3	60
TEJASREE	MILK	20	5	100
TEJASREE	MILK	20	20	400
TEJASREE	MILK	20	10	200
SURESH	CURD	30	4	120

CNAME	PNAME	PRICE	QTY	TOTALPRICE
SURESH	CURD	30	2	60
SURESH	CURD	30	100	3000
SURESH	CURD	30	3	90
SURESH	CURD	30	5	150
SURESH	CURD	30	20	600
SURESH	CURD	30	10	300
KRISHNA	CURD	30	4	120
KRISHNA	CURD	30	2	60
KRISHNA	CURD	30	100	3000
KRISHNA	CURD	30	3	90
KRISHNA	CURD	30	5	150

CNAME	PNAME	PRICE	QTY	TOTALPRICE
KRISHNA	CURD	30	20	600
KRISHNA	CURD	30	10	300
TEJASREE	CURD	30	4	120
TEJASREE	CURD	30	2	60
TEJASREE	CURD	30	100	3000

TEJASREE	CURD	30	3	90
TEJASREE	CURD	30	5	150
TEJASREE	CURD	30	20	600
TEJASREE	CURD	30	10	300

42 rows selected.

SQL> SET PAGESIZE 200

SQL> R

```

1  SELECT CNAME
2      ,PNAME
3      ,PRICE
4      ,QTY
5      ,QTY * PRICE AS TOTALPRICE
6  FROM B26_CUST
7      ,B26_PROD
8*   ,B26_CUST_ORD_DET

```

CNAME	PNAME	PRICE	QTY	TOTALPRICE
SURESH	MILK	20	4	80
SURESH	MILK	20	2	40
SURESH	MILK	20	100	2000
SURESH	MILK	20	3	60
SURESH	MILK	20	5	100
SURESH	MILK	20	20	400
SURESH	MILK	20	10	200
KRISHNA	MILK	20	4	80
KRISHNA	MILK	20	2	40
KRISHNA	MILK	20	100	2000
KRISHNA	MILK	20	3	60
KRISHNA	MILK	20	5	100
KRISHNA	MILK	20	20	400
KRISHNA	MILK	20	10	200
TEJASREE	MILK	20	4	80
TEJASREE	MILK	20	2	40
TEJASREE	MILK	20	100	2000
TEJASREE	MILK	20	3	60
TEJASREE	MILK	20	5	100
TEJASREE	MILK	20	20	400
TEJASREE	MILK	20	10	200
SURESH	CURD	30	4	120
SURESH	CURD	30	2	60
SURESH	CURD	30	100	3000
SURESH	CURD	30	3	90
SURESH	CURD	30	5	150
SURESH	CURD	30	20	600
SURESH	CURD	30	10	300
KRISHNA	CURD	30	4	120
KRISHNA	CURD	30	2	60
KRISHNA	CURD	30	100	3000
KRISHNA	CURD	30	3	90
KRISHNA	CURD	30	5	150
KRISHNA	CURD	30	20	600
KRISHNA	CURD	30	10	300
TEJASREE	CURD	30	4	120
TEJASREE	CURD	30	2	60
TEJASREE	CURD	30	100	3000
TEJASREE	CURD	30	3	90
TEJASREE	CURD	30	5	150
TEJASREE	CURD	30	20	600
TEJASREE	CURD	30	10	300

42 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT *
2  (
3  SELECT CNAME
4          ,PNAME
5          ,PRICE
6          ,QTY
7          ,QTY * PRICE AS TOTALPRICE
8  FROM B26_CUST
9          ,B26_PROD
10         ,B26_CUST_ORD_DET
11 )
12* WHERE CNAME ='SURESH'
SQL> /

```

```

(
*

```

ERROR at line 2:

ORA-00923: FROM keyword not found where expected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT *
2  FROM (
3  SELECT CNAME
4          ,PNAME
5          ,PRICE
6          ,QTY
7          ,QTY * PRICE AS TOTALPRICE
8  FROM B26_CUST
9          ,B26_PROD
10         ,B26_CUST_ORD_DET
11 )
12* WHERE CNAME ='SURESH'
SQL> /

```

CNAME	PNAME	PRICE	QTY	TOTALPRICE
SURESH	MILK	20	4	80
SURESH	MILK	20	2	40
SURESH	MILK	20	100	2000
SURESH	MILK	20	3	60
SURESH	MILK	20	5	100
SURESH	MILK	20	20	400
SURESH	MILK	20	10	200
SURESH	CURD	30	4	120
SURESH	CURD	30	2	60
SURESH	CURD	30	100	3000
SURESH	CURD	30	3	90
SURESH	CURD	30	5	150
SURESH	CURD	30	20	600
SURESH	CURD	30	10	300

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT *
2  FROM (

```



```

3  SELECT CNAME
4      ,PNAME
5      ,PRICE
6      ,QTY
7      ,QTY * PRICE AS TOTALPRICE
8  FROM B26_CUST
9      ,B26_PROD
10     ,B26_CUST_ORD_DET
11 )
12* WHERE CNAME ='KRISHNA'
SQL> /

```

CNAME	PNAME	PRICE	QTY	TOTALPRICE
KRISHNA	MILK	20	4	80
KRISHNA	MILK	20	2	40
KRISHNA	MILK	20	100	2000
KRISHNA	MILK	20	3	60
KRISHNA	MILK	20	5	100
KRISHNA	MILK	20	20	400
KRISHNA	MILK	20	10	200
KRISHNA	CURD	30	4	120
KRISHNA	CURD	30	2	60
KRISHNA	CURD	30	100	3000
KRISHNA	CURD	30	3	90
KRISHNA	CURD	30	5	150
KRISHNA	CURD	30	20	600
KRISHNA	CURD	30	10	300

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT *
2  FROM (
3  SELECT CNAME
4      ,PNAME
5      ,PRICE
6      ,QTY
7      ,QTY * PRICE AS TOTALPRICE
8  FROM B26_CUST
9      ,B26_PROD
10     ,B26_CUST_ORD_DET
11 )
12* WHERE CNAME ='TEJASREE'
SQL> /

```

CNAME	PNAME	PRICE	QTY	TOTALPRICE
TEJASREE	MILK	20	4	80
TEJASREE	MILK	20	2	40
TEJASREE	MILK	20	100	2000
TEJASREE	MILK	20	3	60
TEJASREE	MILK	20	5	100
TEJASREE	MILK	20	20	400
TEJASREE	MILK	20	10	200
TEJASREE	CURD	30	4	120
TEJASREE	CURD	30	2	60
TEJASREE	CURD	30	100	3000
TEJASREE	CURD	30	3	90
TEJASREE	CURD	30	5	150
TEJASREE	CURD	30	20	600
TEJASREE	CURD	30	10	300

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CNAME
2      ,PNAME
3      ,PRICE
4      ,QTY
5      ,QTY * PRICE AS TOTALPRICE
6  FROM B26_CUST
7      ,B26_PROD
8*   ,B26_CUST_ORD_DET
SQL> /

```

CNAME	PNAME	PRICE	QTY	TOTALPRICE
SURESH	MILK	20	4	80
SURESH	MILK	20	2	40
SURESH	MILK	20	100	2000
SURESH	MILK	20	3	60
SURESH	MILK	20	5	100
SURESH	MILK	20	20	400
SURESH	MILK	20	10	200
KRISHNA	MILK	20	4	80
KRISHNA	MILK	20	2	40
KRISHNA	MILK	20	100	2000
KRISHNA	MILK	20	3	60
KRISHNA	MILK	20	5	100
KRISHNA	MILK	20	20	400
KRISHNA	MILK	20	10	200
TEJASREE	MILK	20	4	80
TEJASREE	MILK	20	2	40
TEJASREE	MILK	20	100	2000
TEJASREE	MILK	20	3	60
TEJASREE	MILK	20	5	100
TEJASREE	MILK	20	20	400
TEJASREE	MILK	20	10	200
SURESH	CURD	30	4	120
SURESH	CURD	30	2	60
SURESH	CURD	30	100	3000
SURESH	CURD	30	3	90
SURESH	CURD	30	5	150
SURESH	CURD	30	20	600
SURESH	CURD	30	10	300
KRISHNA	CURD	30	4	120
KRISHNA	CURD	30	2	60
KRISHNA	CURD	30	100	3000
KRISHNA	CURD	30	3	90
KRISHNA	CURD	30	5	150
KRISHNA	CURD	30	20	600
KRISHNA	CURD	30	10	300
TEJASREE	CURD	30	4	120
TEJASREE	CURD	30	2	60
TEJASREE	CURD	30	100	3000
TEJASREE	CURD	30	3	90
TEJASREE	CURD	30	5	150
TEJASREE	CURD	30	20	600
TEJASREE	CURD	30	10	300

42 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CNAME
2      ,PNAME
3      ,PRICE
4      ,QTY
5      ,QTY * PRICE AS TOTALPRICE
6  FROM B26_CUST
7      ,B26_PROD
8      ,B26_CUST_ORD_DET
9  WHERE PID = PID
10*   AND CID = CID
SQL> /
      AND CID = CID
      *
```

ERROR at line 10:
ORA-00918: column ambiguously defined

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT CNAME
2      ,PNAME
3      ,PRICE
4      ,QTY
5      ,QTY * PRICE AS TOTALPRICE
6  FROM B26_CUST
7      ,B26_PROD
8      ,B26_CUST_ORD_DET
9  WHERE PID = PID
10*   AND B26_CUST_ORD_DET.CID = B26_CUST.CID
SQL> /
WHERE PID = PID
      *
```

ERROR at line 9:
ORA-00918: column ambiguously defined

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT CNAME
2      ,PNAME
3      ,PRICE
4      ,QTY
5      ,QTY * PRICE AS TOTALPRICE
6  FROM      B26_CUST_ORD_DET
7            ,B26_PROD
8            ,B26_CUST
9  WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
10*   AND B26_CUST_ORD_DET.CID = B26_CUST.CID
11  /
```

CNAME	PNAME	PRICE	QTY	TOTALPRICE
SURESH	MILK	20	4	80
SURESH	CURD	30	2	60
KRISHNA	CURD	30	3	90
KRISHNA	MILK	20	5	100
KRISHNA	MILK	20	10	200

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CNAME
2      ,PNAME
3      ,QTY
4      ,PRICE
5      ,QTY * PRICE AS TOTALPRICE
6  FROM      B26_CUST_ORD_DET
7      ,B26_PROD
8      ,B26_CUST
9  WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
10*   AND  B26_CUST_ORD_DET.CID = B26_CUST.CID
11  /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE
SURESH	MILK	4	20	80
SURESH	CURD	2	30	60
KRISHNA	CURD	3	30	90
KRISHNA	MILK	5	20	100
KRISHNA	MILK	10	20	200

```

SQL> EDIT
Wrote file afiedt.buf

```

```

1  SELECT CNAME
2      ,PNAME
3      ,QTY
4      ,PRICE
5      ,QTY * PRICE AS TOTALPRICE
6  FROM      B26_CUST_ORD_DET
7      ,B26_PROD
8      ,B26_CUST
9  WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
10*   AND  B26_CUST_ORD_DET.CID = B26_CUST.CID
SQL>
SQL>
SQL>
SQL> CL SCR

```

```

SQL> SELECT *
2  FROM B26_CUST_ORD_DET;

```

CID	PID	QTY	ORD_DATE
1	1	4	07-JUN-18
1	2	2	07-JUN-18
1		100	07-JUN-18
2	2	3	07-JUN-18
2	1	5	07-JUN-18
	1	20	07-JUN-18
2	1	10	08-JUN-18

7 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT CNAME
2      ,PNAME
3      ,QTY
4      ,PRICE
5      ,QTY * PRICE AS TOTALPRICE
6  FROM      B26_CUST_ORD_DET
7      ,B26_PROD

```

```

      8          ,B26_CUST
      9 WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID(F)
     10*   AND B26_CUST_ORD_DET.CID = B26_CUST.CID
SQL> /
WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID(F)
      *
```

ERROR at line 9:
ORA-00904: "B26_PROD"."PID": invalid identifier

```
SQL> ED
Wrote file afiedt.buf
```

```

      1 SELECT CNAME
      2          ,PNAME
      3          ,QTY
      4          ,PRICE
      5          ,QTY * PRICE AS TOTALPRICE
      6 FROM      B26_CUST_ORD_DET
      7          ,B26_PROD
      8          ,B26_CUST
      9 WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
     10*   AND B26_CUST_ORD_DET.CID = B26_CUST.CID
SQL> /
```

CNAME	PNAME	QTY	PRICE	TOTALPRICE
SURESH	MILK	4	20	80
SURESH	CURD	2	30	60
KRISHNA	CURD	3	30	90
KRISHNA	MILK	5	20	100
KRISHNA	MILK	10	20	200

```
SQL> EDIT
Wrote file afiedt.buf
```

```

      1 SELECT CNAME
      2          ,PNAME
      3          ,QTY
      4          ,PRICE
      5          ,QTY * PRICE AS TOTALPRICE
      6 FROM      B26_CUST_ORD_DET
      7          ,B26_PROD
      8          ,B26_CUST
      9 WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
     10   AND B26_CUST_ORD_DET.CID = B26_CUST.CID
     11*   AND B26_CUST_ORD_DET.ORDER_DATE = '07-JUN-18'
SQL> /
      AND B26_CUST_ORD_DET.ORDER_DATE = '07-JUN-18'
      *
```

ERROR at line 11:
ORA-00904: "B26_CUST_ORD_DET"."ORDER_DATE": invalid identifier

```
SQL> DESC B26_CUST_ORD_DET
```

Name	Null?	Type
CID		NUMBER
PID		NUMBER
QTY	NOT NULL	NUMBER
ORD_DATE		DATE

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT CNAME
2         ,PNAME
3         ,QTY
4         ,PRICE
5         ,QTY * PRICE AS TOTALPRICE
6  FROM    B26_CUST_ORD_DET
7         ,B26_PROD
8         ,B26_CUST
9  WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
10     AND B26_CUST_ORD_DET.CID = B26_CUST.CID
11*    AND B26_CUST_ORD_DET.ORD_DATE = '07-JUN-18'
SQL> /

```

no rows selected

```

SQL> EDIT
Wrote file afiedt.buf

```

```

1  SELECT CNAME
2         ,PNAME
3         ,QTY
4         ,PRICE
5         ,QTY * PRICE AS TOTALPRICE
6  FROM    B26_CUST_ORD_DET
7         ,B26_PROD
8         ,B26_CUST
9  WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
10     AND B26_CUST_ORD_DET.CID = B26_CUST.CID
11*    AND B26_CUST_ORD_DET.ORD_DATE = '07-JUN-18'
SQL>
SQL>
SQL> CL SCR

```

```

SQL> SELECT *
2  FROM B26_CUST_ORD_DET;

```

CID	PID	QTY	ORD_DATE
1	1	4	07-JUN-18
1	2	2	07-JUN-18
1		100	07-JUN-18
2	2	3	07-JUN-18
2	1	5	07-JUN-18
	1	20	07-JUN-18
2	1	10	08-JUN-18

7 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT CNAME
2         ,PNAME
3         ,QTY
4         ,PRICE
5         ,QTY * PRICE AS TOTALPRICE
6  FROM    B26_CUST_ORD_DET
7         ,B26_PROD
8         ,B26_CUST
9  WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
10     AND B26_CUST_ORD_DET.CID = B26_CUST.CID
11*    AND B26_CUST_ORD_DET.ORD_DATE = '07-JUN-18'
SQL> /

```

no rows selected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CNAME
2      ,PNAME
3      ,QTY
4      ,PRICE
5      ,QTY * PRICE AS TOTALPRICE
6      ,ORD_DATE
7  FROM      B26_CUST_ORD_DET
8            ,B26_PROD
9            ,B26_CUST
10 WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
11*    AND B26_CUST_ORD_DET.CID = B26_CUST.CID
12 /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE
ORD_DATE				
SURESH 07-JUN-18	MILK	4	20	80
SURESH 07-JUN-18	CURD	2	30	60
KRISHNA 07-JUN-18	CURD	3	30	90
KRISHNA 07-JUN-18	MILK	5	20	100
KRISHNA 08-JUN-18	MILK	10	20	200

SQL> SET LINESIZE 200

SQL> R

```

1  SELECT CNAME
2      ,PNAME
3      ,QTY
4      ,PRICE
5      ,QTY * PRICE AS TOTALPRICE
6      ,ORD_DATE
7  FROM      B26_CUST_ORD_DET
8            ,B26_PROD
9            ,B26_CUST
10 WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
11*    AND B26_CUST_ORD_DET.CID = B26_CUST.CID

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
SURESH 18	MILK	4	20	80	07-JUN-
SURESH 18	CURD	2	30	60	07-JUN-
KRISHNA 18	CURD	3	30	90	07-JUN-
KRISHNA 18	MILK	5	20	100	07-JUN-

```

KRISHNA          MILK          10          20          200 08-JUN-
18

```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

 1  SELECT CNAME
 2          ,PNAME
 3          ,QTY
 4          ,PRICE
 5          ,QTY * PRICE AS TOTALPRICE
 6          ,ORD_DATE
 7  FROM      B26_CUST_ORD_DET
 8            ,B26_PROD
 9            ,B26_CUST
10 WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
11      AND B26_CUST_ORD_DET.CID = B26_CUST.CID
12*      AND B26_CUST_ORD_DET.ORD_DATE ='07-JUN-18'
SQL> ./
SP2-0042: unknown command "./" - rest of line ignored.
SQL> /

```

```
no rows selected
```

```
SQL> CL SCR
```

```
SQL> EDIT
```

```
Wrote file afiedt.buf
```

```

 1  SELECT CNAME
 2          ,PNAME
 3          ,QTY
 4          ,PRICE
 5          ,QTY * PRICE AS TOTALPRICE
 6          ,ORD_DATE
 7  FROM      B26_CUST_ORD_DET
 8            ,B26_PROD
 9            ,B26_CUST
10 WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
11*      AND B26_CUST_ORD_DET.CID = B26_CUST.CID
12  /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
SURESH	MILK	4	20	80	07-JUN-
18					
SURESH	CURD	2	30	60	07-JUN-
18					
KRISHNA	CURD	3	30	90	07-JUN-
18					
KRISHNA	MILK	5	20	100	07-JUN-
18					
KRISHNA	MILK	10	20	200	08-JUN-
18					

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

 1  SELECT CNAME
 2          ,PNAME
 3          ,QTY
 4          ,PRICE
 5          ,QTY * PRICE AS TOTALPRICE

```



```

6          ,ORD_DATE
7 FROM      B26_CUST_ORD_DET
8          ,B26_PROD
9          ,B26_CUST
10 WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
11       AND B26_CUST_ORD_DET.CID = B26_CUST.CID
12*      AND B26_CUST_ORD_DET.ORD_dATE='07-JUN-18'
SQL> /

```

no rows selected

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

SP2-0223: No lines in SQL buffer.

SQL>

SQL> CL SCR

```

SQL> SELECT *
2 FROM B26_CUST_ORD_DET;

```

CID	PID	QTY	ORD_DATE
1	1	4	07-JUN-18
1	2	2	07-JUN-18
1		100	07-JUN-18
2	2	3	07-JUN-18
2	1	5	07-JUN-18
	1	20	07-JUN-18
2	1	10	08-JUN-18

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1 SELECT *
2 FROM B26_CUST_ORD_DET
3* WHERE ORD_dATE ='07-JUN-18'
SQL> /

```

no rows selected

SQL> ED

Wrote file afiedt.buf

```

1 SELECT *
2 FROM B26_CUST_ORD_DET
3* WHERE TO_CHAR(ORD_dATE,'DD-MON-YY') ='07-JUN-18'
SQL> /

```

CID	PID	QTY	ORD_DATE
1	1	4	07-JUN-18
1	2	2	07-JUN-18
1		100	07-JUN-18
2	2	3	07-JUN-18
2	1	5	07-JUN-18
	1	20	07-JUN-18

6 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT CID
2  ,PID
3  ,QTY
4  ,ORD_date
5  ,TO_CHAR(ORD_DATE,'DD-MON-YY HH24 MI SS') DD1
6  FROM B26_CUST_ORD_DET
7* WHERE ORD_date ='07-JUN-18'
SQL> /
```

no rows selected

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT CID
2  ,PID
3  ,QTY
4  ,ORD_date
5  ,TO_CHAR(ORD_DATE,'DD-MON-YY HH24 MI SS') DD1
6  FROM B26_CUST_ORD_DET
7* --WHERE ORD_date ='07-JUN-18'
SQL> /
```

CID	PID	QTY	ORD_DATE	DD1
1	1	4	07-JUN-18	07-JUN-18 08 43 27
1	2	2	07-JUN-18	07-JUN-18 08 43 49
1		100	07-JUN-18	07-JUN-18 08 44 58
2	2	3	07-JUN-18	07-JUN-18 08 45 13
2	1	5	07-JUN-18	07-JUN-18 08 45 18
	1	20	07-JUN-18	07-JUN-18 08 46 33
2	1	10	08-JUN-18	08-JUN-18 08 46 45

7 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT CID
2  ,PID
3  ,QTY
4  ,ORD_date
5  ,TO_CHAR(ORD_DATE,'DD-MON-YY HH24 MI SS') DD1
6  FROM B26_CUST_ORD_DET
7* WHERE ORD_date ='07-JUN-18'
SQL> /
```

no rows selected

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT CID
2  ,PID
3  ,QTY
4  ,ORD_date
5  ,TO_CHAR(ORD_DATE,'DD-MON-YY HH24 MI SS') DD1
6  FROM B26_CUST_ORD_DET
7* WHERE TO_CHAR(ORD_date,'DD-MON-YY') ='07-JUN-18'
SQL> /
```

CID	PID	QTY	ORD_DATE	DD1
1	1	4	07-JUN-18	07-JUN-18 08 43 27
1	2	2	07-JUN-18	07-JUN-18 08 43 49
1		100	07-JUN-18	07-JUN-18 08 44 58
2	2	3	07-JUN-18	07-JUN-18 08 45 13
2	1	5	07-JUN-18	07-JUN-18 08 45 18
	1	20	07-JUN-18	07-JUN-18 08 46 33

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CNAME
2      ,PNAME
3      ,QTY
4      ,PRICE
5      ,QTY * PRICE AS TOTALPRICE
6      ,ORD_DATE
7  FROM    B26_CUST_ORD_DET
8          ,B26_PROD
9          ,B26_CUST
10 WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
11      AND B26_CUST_ORD_DET.CID = B26_CUST.CID
12*      AND B26_CUST_ORD_DET.ORD_dATE='07-JUN-18'
SQL> /

```

no rows selected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CNAME
2      ,PNAME
3      ,QTY
4      ,PRICE
5      ,QTY * PRICE AS TOTALPRICE
6      ,ORD_DATE
7  FROM    B26_CUST_ORD_DET
8          ,B26_PROD
9          ,B26_CUST
10 WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
11      AND B26_CUST_ORD_DET.CID = B26_CUST.CID
12*      AND TO_CHAR(B26_CUST_ORD_DET.ORD_dATE,'DD-MON-YY')='07-JUN-18'
SQL> /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
SURESH	MILK	4	20	80	07-JUN-18
SURESH	CURD	2	30	60	07-JUN-18
KRISHNA	CURD	3	30	90	07-JUN-18
KRISHNA	MILK	5	20	100	07-JUN-18

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CNAME
2      ,PNAME

```

```

3      ,QTY
4      ,PRICE
5      ,QTY * PRICE AS TOTALPRICE
6      ,ORD_DATE
7 FROM      B26_CUST_ORD_DET
8           ,B26_PROD
9           ,B26_CUST
10 WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
11*    AND B26_CUST_ORD_DET.CID = B26_CUST.CID
12 /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
SURESH	MILK	4	20	80	07-JUN-
SURESH	CURD	2	30	60	07-JUN-
KRISHNA	CURD	3	30	90	07-JUN-
KRISHNA	MILK	5	20	100	07-JUN-
KRISHNA	MILK	10	20	200	08-JUN-

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT CNAME
2      ,PNAME
3      ,QTY
4      ,PRICE
5      ,QTY * PRICE AS TOTALPRICE
6      ,ORD_DATE
7 FROM      B26_CUST_ORD_DET
8           ,B26_PROD
9           ,B26_CUST
10 WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
11    AND B26_CUST_ORD_DET.CID = B26_CUST.CID
12*    AND TO_CHAR(B26_CUST_ORD_DET.ORD_dATE,'DD-MON-YY')='07-JUN-18'
SQL> /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
SURESH	MILK	4	20	80	07-JUN-
SURESH	CURD	2	30	60	07-JUN-
KRISHNA	CURD	3	30	90	07-JUN-
KRISHNA	MILK	5	20	100	07-JUN-

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT CNAME
2      ,PNAME
3      ,QTY
4      ,PRICE
5      ,QTY * PRICE AS TOTALPRICE
6      ,ORD_DATE
7 FROM      B26_CUST_ORD_DET--7

```

```

8          ,B26_PROD --2
9          ,B26_CUST --3
10* WHERE B26_CUST_ORD_DET.PID = B26_PROD.PID
11 /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
-					
TEJASREE	MILK	4	20	80	07-JUN-
18					
KRISHNA	MILK	4	20	80	07-JUN-
18					
SURESH	MILK	4	20	80	07-JUN-
18					
TEJASREE	CURD	2	30	60	07-JUN-
18					
KRISHNA	CURD	2	30	60	07-JUN-
18					
SURESH	CURD	2	30	60	07-JUN-
18					
TEJASREE	CURD	3	30	90	07-JUN-
18					
KRISHNA	CURD	3	30	90	07-JUN-
18					
SURESH	CURD	3	30	90	07-JUN-
18					
TEJASREE	MILK	5	20	100	07-JUN-
18					
KRISHNA	MILK	5	20	100	07-JUN-
18					
SURESH	MILK	5	20	100	07-JUN-
18					
TEJASREE	MILK	20	20	400	07-JUN-
18					
KRISHNA	MILK	20	20	400	07-JUN-
18					
SURESH	MILK	20	20	400	07-JUN-
18					
TEJASREE	MILK	10	20	200	08-JUN-
18					
KRISHNA	MILK	10	20	200	08-JUN-
18					
SURESH	MILK	10	20	200	08-JUN-
18					

18 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT CNAME
2          ,PNAME
3          ,QTY
4          ,PRICE
5          ,QTY * PRICE AS TOTALPRICE
6          ,ORD_DATE
7  FROM      B26_CUST_ORD_DET--7
8          ,B26_PROD --2
9*          ,B26_CUST --3
10 /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
-					

SURESH 18	MILK	4	20	80	07-JUN-
SURESH 18	MILK	2	20	40	07-JUN-
SURESH 18	MILK	100	20	2000	07-JUN-
SURESH 18	MILK	3	20	60	07-JUN-
SURESH 18	MILK	5	20	100	07-JUN-
SURESH 18	MILK	20	20	400	07-JUN-
SURESH 18	MILK	10	20	200	08-JUN-
KRISHNA 18	MILK	4	20	80	07-JUN-
KRISHNA 18	MILK	2	20	40	07-JUN-
KRISHNA 18	MILK	100	20	2000	07-JUN-
KRISHNA 18	MILK	3	20	60	07-JUN-
KRISHNA 18	MILK	5	20	100	07-JUN-
KRISHNA 18	MILK	20	20	400	07-JUN-
KRISHNA 18	MILK	10	20	200	08-JUN-
TEJASREE 18	MILK	4	20	80	07-JUN-
TEJASREE 18	MILK	2	20	40	07-JUN-
TEJASREE 18	MILK	100	20	2000	07-JUN-
TEJASREE 18	MILK	3	20	60	07-JUN-
TEJASREE 18	MILK	5	20	100	07-JUN-
TEJASREE 18	MILK	20	20	400	07-JUN-
TEJASREE 18	MILK	10	20	200	08-JUN-
SURESH 18	CURD	4	30	120	07-JUN-
SURESH 18	CURD	2	30	60	07-JUN-
SURESH 18	CURD	100	30	3000	07-JUN-
SURESH 18	CURD	3	30	90	07-JUN-
SURESH 18	CURD	5	30	150	07-JUN-
SURESH 18	CURD	20	30	600	07-JUN-
SURESH 18	CURD	10	30	300	08-JUN-
KRISHNA 18	CURD	4	30	120	07-JUN-
KRISHNA 18	CURD	2	30	60	07-JUN-
KRISHNA 18	CURD	100	30	3000	07-JUN-
KRISHNA 18	CURD	3	30	90	07-JUN-

```
SQL> ED
Wrote file afiedt.buf
```

$$\begin{array}{r} 1 \\ \hline 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{array}$$

```
SQL> ED
Wrote file afiedt.buf
```

[illegible]

```

1
1

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT 1
2  FROM      B26_CUST_ORD_DET--7
3            ,B26_PROD
4* WHERE  PID = PID
SQL> /
WHERE  PID = PID
      *
ERROR at line 4:
ORA-00918: column ambiguously defined

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT 1
2  FROM      B26_CUST_ORD_DET COD
3            ,B26_PROD PR
4* WHERE  COD.PID = PR.PID
SQL> /

```

```

1
-----
1
1
1
1
1
1
1
1

```

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT 1
2  FROM      B26_CUST_ORD_DET COD
3            ,B26_PROD PR
4            ,B26_CUST CU
5* WHERE  COD.PID = PR.PID
SQL> /

```

```

1
-----
1
1
1
1
1
1
1
1
1
1
1
1
1
1
1
1

```


1
1
1
1
1

18 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT 1
2  FROM      B26_CUST_ORD_DET COD
3            ,B26_PROD  PR
4            ,B26_CUST  CU
5  WHERE COD.PID = PR.PID
6*  AND COD.CID  = CU.CID
SQL> /

```

```

-----
1
1
1
1
1
1
1

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME
2         ,PR.PNAME
3         ,COD.QTY
4         ,PR.PRICE
5         ,COD.QTY * PR.PRICE AS TOTAALPRICE
6         ,COD.ORD_DATE
7  FROM      B26_CUST_ORD_DET COD
8            ,B26_PROD  PR
9            ,B26_CUST  CU
10 WHERE COD.PID = PR.PID
11*  AND COD.CID  = CU.CID
SQL> /

```

CNAME ORD_DATE	PNAME	QTY	PRICE	TOTAALPRICE
SURESH 18	MILK	4	20	80 07-JUN-
SURESH 18	CURD	2	30	60 07-JUN-
KRISHNA 18	CURD	3	30	90 07-JUN-
KRISHNA 18	MILK	5	20	100 07-JUN-
KRISHNA 18	MILK	10	20	200 08-JUN-

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME
2         ,PR.PNAME
3         ,COD.QTY

```

```

4      ,PR.PRICE
5      ,COD.QTY * PR.PRICE AS TOTAALPRICE
6      ,COD.ORD_DATE
7 FROM      B26_CUST_ORD_DET COD
8           ,B26_PROD PR
9           ,B26_CUST CU
10 WHERE COD.PID = PR.PID
11 AND COD.CID = CU.CID
12* ORDER BY COD.ORD_DATE
SQL> /

```

CNAME ORD_DATE	PNAME	QTY	PRICE	TOTAALPRICE
--				
SURESH 18	MILK	4	20	80 07-JUN-
SURESH 18	CURD	2	30	60 07-JUN-
KRISHNA 18	CURD	3	30	90 07-JUN-
KRISHNA 18	MILK	5	20	100 07-JUN-
KRISHNA 18	MILK	10	20	200 08-JUN-

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT CU.CNAME
2      ,PR.PNAME
3      ,COD.QTY
4      ,PR.PRICE
5      ,COD.QTY * PR.PRICE AS TOTAALPRICE
6      ,COD.ORD_DATE
7 FROM      B26_CUST_ORD_DET COD
8           ,B26_PROD PR
9           ,B26_CUST CU
10 WHERE COD.PID = PR.PID
11 AND COD.CID = CU.CID
12* ORDER BY COD.ORD_DATE DESC
SQL> /

```

CNAME ORD_DATE	PNAME	QTY	PRICE	TOTAALPRICE
--				
KRISHNA 18	MILK	10	20	200 08-JUN-
KRISHNA 18	MILK	5	20	100 07-JUN-
KRISHNA 18	CURD	3	30	90 07-JUN-
SURESH 18	CURD	2	30	60 07-JUN-
SURESH 18	MILK	4	20	80 07-JUN-

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT CU.CNAME
2      ,PR.PNAME
3      ,COD.QTY

```

```

4      ,PR.PRICE
5      ,COD.QTY * PR.PRICE AS TOTAALPRICE
6      ,COD.ORD_DATE
7 FROM      B26_CUST_ORD_DET COD
8           ,B26_PROD PR
9           ,B26_CUST CU
10 WHERE COD.PID = PR.PID
11 AND COD.CID = CU.CID
12 AND TO_CHAR(COD.ORD_DATE,'DD-MON-YY') ='07-JUN-18'
13* ORDER BY COD.ORD_DATE DESC
SQL> /

```

CNAME ORD_DATE	PNAME	QTY	PRICE	TOTAALPRICE
-----	-----	-----	-----	-----
--				
KRISHNA 18	MILK	5	20	100 07-JUN-
KRISHNA 18	CURD	3	30	90 07-JUN-
SURESH 18	CURD	2	30	60 07-JUN-
SURESH 18	MILK	4	20	80 07-JUN-

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT CU.CNAME
2      ,PR.PNAME
3      ,COD.QTY
4      ,PR.PRICE
5      ,COD.QTY * PR.PRICE AS TOTAALPRICE
6      ,TO_CHAR(COD.ORD_DATE,'DD-MON-YY HH24 MI SS') ORD_DATE
7 FROM      B26_CUST_ORD_DET COD
8           ,B26_PROD PR
9           ,B26_CUST CU
10 WHERE COD.PID = PR.PID
11 AND COD.CID = CU.CID
12 AND TO_CHAR(COD.ORD_DATE,'DD-MON-YY') ='07-JUN-18'
13* ORDER BY COD.ORD_DATE DESC
SQL> /

```

CNAME ORD_DATE	PNAME	QTY	PRICE	TOTAALPRICE
-----	-----	-----	-----	-----

KRISHNA 18 08 45 18	MILK	5	20	100 07-JUN-
KRISHNA 18 08 45 13	CURD	3	30	90 07-JUN-
SURESH 18 08 43 49	CURD	2	30	60 07-JUN-
SURESH 18 08 43 27	MILK	4	20	80 07-JUN-

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT CNAME
2      ,PNAME
3      ,QTY
4      ,PRICE
5      ,QTY * PRICE AS TOTAALPRICE

```

```

6      ,TO_CHAR(ORD_DATE,'DD-MON-YY HH24 MI SS') ORD_DATE
7 FROM      B26_CUST_ORD_DET COD
8           ,B26_PROD PR
9           ,B26_CUST CU
10 WHERE COD.PID = PR.PID
11 AND COD.CID = CU.CID
12 AND TO_CHAR(COD.ORD_DATE,'DD-MON-YY') ='07-JUN-18'
13* ORDER BY COD.ORD_DATE DESC
SQL> /

```

CNAME	PNAME	QTY	PRICE	TOTAALPRICE
ORD_DATE				
-----	-----	-----	-----	-----

KRISHNA	MILK	5	20	100 07-JUN-
18 08 45 18				
KRISHNA	CURD	3	30	90 07-JUN-
18 08 45 13				
SURESH	CURD	2	30	60 07-JUN-
18 08 43 49				
SURESH	MILK	4	20	80 07-JUN-
18 08 43 27				

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT CNAME
2      ,PNAME
3      ,QTY
4      ,PRICE
5      ,QTY * PRICE AS TOTAALPRICE
6      ,TO_CHAR(ORD_DATE,'DD-MON-YY HH24 MI SS') ORD_DATE
7 FROM      B26_CUST_ORD_DET COD
8           ,B26_PROD PR
9           ,B26_CUST CU
10 WHERE COD.PID = PR.PID
11 AND CID = CID
12 AND TO_CHAR(COD.ORD_DATE,'DD-MON-YY') ='07-JUN-18'
13* ORDER BY COD.ORD_DATE DESC
SQL> /
AND CID = CID
*
```

```

ERROR at line 11:
ORA-00918: column ambiguously defined

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT CU.CNAME
2      ,PR.PNAME
3      ,COD.QTY
4      ,PR.PRICE
5      ,COD.QTY * PR.PRICE AS TOTAALPRICE
6      ,TO_CHAR(COD.ORD_DATE,'DD-MON-YY HH24 MI SS') ORD_DATE
7 FROM      B26_CUST_ORD_DET COD
8           ,B26_PROD PR
9           ,B26_CUST CU
10 WHERE COD.PID = PR.PID
11 AND COD.CID = CU.CID
12 AND TO_CHAR(COD.ORD_DATE,'DD-MON-YY') ='07-JUN-18'
13* ORDER BY COD.ORD_DATE DESC
SQL> /

```

CNAME ORD_DATE	PNAME	QTY	PRICE	TOTAALPRICE
-----	-----	-----	-----	-----
-----	-----	-----	-----	-----
KRISHNA 18 08 45 18	MILK	5	20	100 07-JUN-
KRISHNA 18 08 45 13	CURD	3	30	90 07-JUN-
SURESH 18 08 43 49	CURD	2	30	60 07-JUN-
SURESH 18 08 43 27	MILK	4	20	80 07-JUN-

```
SQL> L SCR
SP2-0224: invalid starting line number
SQL> CL SCR
```

```
SQL> SELECT
  2 FROM B26_CUST_ORD_DET COD;
FROM B26_CUST_ORD_DET COD
*
ERROR at line 2:
ORA-00936: missing expression
```

```
SQL> ED
Wrote file afiedt.buf
```

```
  1 SELECT *
  2* FROM B26_CUST_ORD_DET COD
SQL> /
```

CID	PID	QTY	ORD_DATE
-----	-----	-----	-----
1	1	4	07-JUN-18
1	2	2	07-JUN-18
1		100	07-JUN-18
2	2	3	07-JUN-18
2	1	5	07-JUN-18
	1	20	07-JUN-18
2	1	10	08-JUN-18

7 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
  1 SELECT COD.CID
  2      ,PR.PNAME
  3      ,COD.QTY
  4      ,PR.PRICE
  5      ,COD.ORD_DATE
  6 FROM B26_CUST_ORD_DET COD
  7      ,B26_PROD PR
  8* WHERE COD.PID = PR.PID
SQL> /
```

CID	PNAME	QTY	PRICE	ORD_DATE
-----	-----	-----	-----	-----
1	MILK	4	20	07-JUN-18
1	CURD	2	30	07-JUN-18
2	CURD	3	30	07-JUN-18
2	MILK	5	20	07-JUN-18
	MILK	20	20	07-JUN-18

2 MILK

10

20 08-JUN-18

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,COD.QTY
4      ,PR.PRICE
5      ,COD.ORD_DATE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8*  WHERE COD.PID = PR.PID(+)
SQL> /

```

CID	PNAME	QTY	PRICE	ORD_DATE
2	MILK	10	20	08-JUN-18
	MILK	20	20	07-JUN-18
2	MILK	5	20	07-JUN-18
1	MILK	4	20	07-JUN-18
2	CURD	3	30	07-JUN-18
1	CURD	2	30	07-JUN-18
1		100		07-JUN-18

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,COD.QTY
4      ,PR.PRICE
5      ,COD.ORD_DATE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8*  WHERE PR.PID = COD.PID
SQL> /

```

CID	PNAME	QTY	PRICE	ORD_DATE
1	MILK	4	20	07-JUN-18
1	CURD	2	30	07-JUN-18
2	CURD	3	30	07-JUN-18
2	MILK	5	20	07-JUN-18
	MILK	20	20	07-JUN-18
2	MILK	10	20	08-JUN-18

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,COD.QTY
4      ,PR.PRICE
5      ,COD.ORD_DATE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8*  WHERE PR.PID(+) = COD.PID

```

SQL> /

CID	PNAME	QTY	PRICE	ORD_DATE
2	MILK	10	20	08-JUN-18
	MILK	20	20	07-JUN-18
2	MILK	5	20	07-JUN-18
1	MILK	4	20	07-JUN-18
2	CURD	3	30	07-JUN-18
1	CURD	2	30	07-JUN-18
1		100		07-JUN-18

7 rows selected.

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,COD.QTY
4      ,PR.PRICE
5      ,COD.ORD_DATE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8*  WHERE COD.PID = PR.PID(+)
SQL> /

```

CID	PNAME	QTY	PRICE	ORD_DATE
2	MILK	10	20	08-JUN-18
	MILK	20	20	07-JUN-18
2	MILK	5	20	07-JUN-18
1	MILK	4	20	07-JUN-18
2	CURD	3	30	07-JUN-18
1	CURD	2	30	07-JUN-18
1		100		07-JUN-18

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,NVL( PR.PNAME,'NO PRODUCT')
3      ,COD.QTY
4      ,PR.PRICE
5      ,COD.ORD_DATE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8*  WHERE COD.PID = PR.PID(+)
SQL> /

```

CID	NVL(PR.PNAME, 'NOPROD	QTY	PRICE	ORD_DATE
2	MILK	10	20	08-JUN-18
	MILK	20	20	07-JUN-18
2	MILK	5	20	07-JUN-18
1	MILK	4	20	07-JUN-18
2	CURD	3	30	07-JUN-18
1	CURD	2	30	07-JUN-18
1	NO PRODUCT	100		07-JUN-18

7 rows selected.

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,NVL( PR.PNAME,'NO PRODUCT')
3      ,COD.QTY
4      ,PR.PRICE
5      ,COD.ORD_DATE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8*  WHERE PR.PID(+)  = COD.PID
SQL> /

```

CID	NVL(PR.PNAME, 'NOPROD	QTY	PRICE	ORD_DATE
2	MILK	10	20	08-JUN-18
	MILK	20	20	07-JUN-18
2	MILK	5	20	07-JUN-18
1	MILK	4	20	07-JUN-18
2	CURD	3	30	07-JUN-18
1	CURD	2	30	07-JUN-18
1	NO PRODUCT	100		07-JUN-18

7 rows selected.

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,NVL( PR.PNAME,'NO PRODUCT')
3      ,COD.QTY
4      ,PR.PRICE
5      ,COD.ORD_DATE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8*  WHERE PR.PID  = COD.PID
SQL> /

```

CID	NVL(PR.PNAME, 'NOPROD	QTY	PRICE	ORD_DATE
1	MILK	4	20	07-JUN-18
1	CURD	2	30	07-JUN-18
2	CURD	3	30	07-JUN-18
2	MILK	5	20	07-JUN-18
	MILK	20	20	07-JUN-18
2	MILK	10	20	08-JUN-18

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME
2      ,NVL( PR.PNAME,'NO PRODUCT')
3      ,COD.QTY
4      ,PR.PRICE
5      ,COD.ORD_DATE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8      ,B26_CUST CU

```



```

9   WHERE PR.PID = COD.PID
10* AND    CU.CID = COD.CID
SQL> /

```

CNAME	NVL(PR.PNAME, 'NOPROD	QTY	PRICE	ORD_DATE
SURESH	MILK	4	20	07-JUN-18
SURESH	CURD	2	30	07-JUN-18
KRISHNA	CURD	3	30	07-JUN-18
KRISHNA	MILK	5	20	07-JUN-18
KRISHNA	MILK	10	20	08-JUN-18

```

SQL> EDIT
Wrote file afiedt.buf

```

```

1  SELECT CU.CNAME
2         ,NVL( PR.PNAME, 'NO PRODUCT')
3         ,COD.QTY
4         ,PR.PRICE
5         ,COD.ORD_DATE
6  FROM B26_CUST_ORD_DET COD
7       ,B26_PROD PR
8       ,B26_CUST CU
9  WHERE PR.PID(+) = COD.PID
10* AND    CU.CID(+) = COD.CID
SQL> /

```

CNAME	NVL(PR.PNAME, 'NOPROD	QTY	PRICE	ORD_DATE
SURESH	NO PRODUCT	100		07-JUN-18
SURESH	CURD	2	30	07-JUN-18
SURESH	MILK	4	20	07-JUN-18
KRISHNA	CURD	3	30	07-JUN-18
KRISHNA	MILK	5	20	07-JUN-18
KRISHNA	MILK	10	20	08-JUN-18
	MILK	20	20	07-JUN-18

7 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT NVL(CU.CNAME, 'NO CUSTOMER) CNAME
2         ,NVL( PR.PNAME, 'NO PRODUCT') PNAME
3         ,COD.QTY
4         ,PR.PRICE
5         ,COD.ORD_DATE
6  FROM B26_CUST_ORD_DET COD
7       ,B26_PROD PR
8       ,B26_CUST CU
9  WHERE PR.PID(+) = COD.PID
10* AND    CU.CID(+) = COD.CID
SQL> /
ERROR:
ORA-01756: quoted string not properly terminated

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT NVL(CU.CNAME, 'NO CUSTOMER') CNAME
2         ,NVL( PR.PNAME, 'NO PRODUCT') PNAME
3         ,COD.QTY
4         ,PR.PRICE

```

```

5          ,COD.ORD_DATE
6 FROM B26_CUST_ORD_DET COD
7          ,B26_PROD PR
8          ,B26_CUST CU
9 WHERE PR.PID(+) = COD.PID
10* AND      CU.CID(+) = COD.CID
SQL> /

```

CNAME	PNAME	QTY	PRICE	ORD_DATE
SURESH	NO PRODUCT	100		07-JUN-18
SURESH	CURD	2	30	07-JUN-18
SURESH	MILK	4	20	07-JUN-18
KRISHNA	CURD	3	30	07-JUN-18
KRISHNA	MILK	5	20	07-JUN-18
KRISHNA	MILK	10	20	08-JUN-18
NO CUSTOMER	MILK	20	20	07-JUN-18

7 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT NVL(CU.CNAME,'NO CUSTOMER') CNAME
2          ,NVL( PR.PNAME,'NO PRODUCT') PNAME
3          ,COD.QTY
4          ,PR.PRICE
5          ,COD.QTY * PR.PRICE AS TOTALPRICE
6          ,COD.ORD_DATE
7 FROM B26_CUST_ORD_DET COD
8          ,B26_PROD PR
9          ,B26_CUST CU
10 WHERE PR.PID(+) = COD.PID
11* AND      CU.CID(+) = COD.CID
SQL> /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
-					
SURESH	NO PRODUCT	100			07-JUN-
18					
SURESH	CURD	2	30	60	07-JUN-
18					
SURESH	MILK	4	20	80	07-JUN-
18					
KRISHNA	CURD	3	30	90	07-JUN-
18					
KRISHNA	MILK	5	20	100	07-JUN-
18					
KRISHNA	MILK	10	20	200	08-JUN-
18					
NO CUSTOMER	MILK	20	20	400	07-JUN-
18					

7 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT NVL(CU.CNAME,'NO CUSTOMER') CNAME
2          ,NVL( PR.PNAME,'NO PRODUCT') PNAME
3          ,COD.QTY
4          ,PR.PRICE
5          ,COD.QTY * NVL(PR.PRICE,15) AS TOTALPRICE

```

```

6          ,COD.ORD_DATE
7 FROM B26_CUST_ORD_DET COD
8          ,B26_PROD PR
9          ,B26_CUST CU
10 WHERE PR.PID(+) = COD.PID
11* AND      CU.CID(+) = COD.CID
SQL> /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
SURESH	NO PRODUCT	100		1500	07-JUN-
18					
SURESH	CURD	2	30	60	07-JUN-
18					
SURESH	MILK	4	20	80	07-JUN-
18					
KRISHNA	CURD	3	30	90	07-JUN-
18					
KRISHNA	MILK	5	20	100	07-JUN-
18					
KRISHNA	MILK	10	20	200	08-JUN-
18					
NO CUSTOMER	MILK	20	20	400	07-JUN-
18					

7 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT NVL(CU.CNAME,'NO CUSTOMER') CNAME
2          ,NVL( PR.PNAME,'NO PRODUCT') PNAME
3          ,COD.QTY
4          ,NVL(PR.PRICE,15) AS PRICE
5          ,COD.QTY * NVL(PR.PRICE,15) AS TOTALPRICE
6          ,COD.ORD_DATE
7 FROM B26_CUST_ORD_DET COD
8          ,B26_PROD PR
9          ,B26_CUST CU
10 WHERE PR.PID(+) = COD.PID
11* AND      CU.CID(+) = COD.CID
SQL> /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
SURESH	NO PRODUCT	100	15	1500	07-JUN-
18					
SURESH	CURD	2	30	60	07-JUN-
18					
SURESH	MILK	4	20	80	07-JUN-
18					
KRISHNA	CURD	3	30	90	07-JUN-
18					
KRISHNA	MILK	5	20	100	07-JUN-
18					
KRISHNA	MILK	10	20	200	08-JUN-
18					
NO CUSTOMER	MILK	20	20	400	07-JUN-
18					

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT NVL(CU.CNAME,'NO CUSTOMER') CNAME
2          ,NVL( PR.PNAME,'NO PRODUCT') PNAME
3          ,COD.QTY
4          ,NVL(PR.PRICE,15) AS PRICE
5          ,COD.QTY * NVL(PR.PRICE,15) AS TOTALPRICE
6          ,COD.ORD_DATE
7  FROM B26_CUST_ORD_DET COD
8       ,B26_PROD PR
9       ,B26_CUST CU
10 WHERE PR.PID(+)  = COD.PID
11 AND    CU.CID(+) = COD.CID
12* AND TRUNC(ORD_DATE) = '07-JUN-18'
SQL> /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
SURESH	NO PRODUCT	100	15	1500	07-JUN-18
SURESH	CURD	2	30	60	07-JUN-18
SURESH	MILK	4	20	80	07-JUN-18
KRISHNA	CURD	3	30	90	07-JUN-18
KRISHNA	MILK	5	20	100	07-JUN-18
NO CUSTOMER	MILK	20	20	400	07-JUN-18

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT NVL(CU.CNAME,'NO CUSTOMER') CNAME
2          ,NVL( PR.PNAME,'NO PRODUCT') PNAME
3          ,COD.QTY
4          ,NVL(PR.PRICE,15) AS PRICE
5          ,COD.QTY * NVL(PR.PRICE,15) AS TOTALPRICE
6          ,COD.ORD_DATE
7  FROM B26_CUST_ORD_DET COD
8       ,B26_PROD PR
9       ,B26_CUST CU
10 WHERE PR.PID(+)  = COD.PID
11 AND    CU.CID(+) = COD.CID
12 AND TRUNC(ORD_DATE) = '07-JUN-18'
13* ORDER BY COD.ORD_DATE DESC
SQL> /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
NO CUSTOMER	MILK	20	20	400	07-JUN-18
KRISHNA	MILK	5	20	100	07-JUN-18
KRISHNA	CURD	3	30	90	07-JUN-18
SURESH	NO PRODUCT	100	15	1500	07-JUN-18

```

SURESH          CURD          2          30          60 07-JUN-
18
SURESH          MILK          4          20          80 07-JUN-
18

```

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT NVL(CU.CNAME,'NO CUSTOMER') CNAME
 2          ,NVL( PR.PNAME,'NO PRODUCT') PNAME
 3          ,COD.QTY
 4          ,NVL(PR.PRICE,15) AS PRICE
 5          ,COD.QTY * NVL(PR.PRICE,15) AS TOTALPRICE
 6          ,COD.ORD_DATE
 7  FROM B26_CUST_ORD_DET COD
 8       ,B26_PROD PR
 9       ,B26_CUST CU
10  WHERE PR.PID(+)  = COD.PID
11  AND    CU.CID(+)  = COD.CID
12  AND TRUNC(COD.ORD_DATE) = '07-JUN-18'
13* ORDER BY COD.ORD_DATE  DESC
SQL> /

```

CNAME	PNAME	QTY	PRICE	TOTALPRICE	ORD_DATE
NO CUSTOMER	MILK	20	20	400	07-JUN-18
KRISHNA	MILK	5	20	100	07-JUN-18
KRISHNA	CURD	3	30	90	07-JUN-18
SURESH	NO PRODUCT	100	15	1500	07-JUN-18
SURESH	CURD	2	30	60	07-JUN-18
SURESH	MILK	4	20	80	07-JUN-18

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT NVL(CU.CNAME,'NO CUSTOMER')
 2          || ' ON '
 3          || COD.ORD_DATE
 4          || ' ORDERED '
 5          || COD.QTY
 6          || NVL( PR.PNAME,'NO PRODUCT')
 7          || ' PACKETS '
 8          || ' AND IT COSTED HIM '
 9          || COD.QTY * NVL(PR.PRICE,15) AS ORD_INFORMATION
10  ,
11  FROM B26_CUST_ORD_DET COD
12       ,B26_PROD PR
13       ,B26_CUST CU
14  WHERE PR.PID(+)  = COD.PID
15  AND    CU.CID(+)  = COD.CID
16  AND TRUNC(COD.ORD_DATE) = '07-JUN-18'
17* ORDER BY COD.ORD_DATE  DESC
SQL> ED

```

Wrote file afiedt.buf

```

1  SELECT NVL(CU.CNAME,'NO CUSTOMER')
2      || ' ON '
3      || COD.ORD_DATE
4      || ' ORDERED '
5      || COD.QTY
6      || NVL( PR.PNAME,'NO PRODUCT')
7      || ' PACKETS '
8      || ' AND IT COSTED HIM '
9      || COD.QTY * NVL(PR.PRICE,15) AS ORD_INFORMATION
10 FROM B26_CUST_ORD_DET COD
11      ,B26_PROD PR
12      ,B26_CUST CU
13 WHERE PR.PID(+) = COD.PID
14 AND     CU.CID(+) = COD.CID
15 AND TRUNC(COD.ORD_DATE) = '07-JUN-18'
16* ORDER BY COD.ORD_DATE DESC
SQL> /

```

ORD_INFORMATION

```

-----
-----
-----
NO CUSTOMER ON  07-JUN-18 ORDERED 20MILK PACKETS  AND IT COSTED HIM 400
KRISHNA ON  07-JUN-18 ORDERED 5MILK PACKETS  AND IT COSTED HIM 100
KRISHNA ON  07-JUN-18 ORDERED 3CURD PACKETS  AND IT COSTED HIM 90
SURESH ON  07-JUN-18 ORDERED 100NO PRODUCT PACKETS  AND IT COSTED HIM 1500
SURESH ON  07-JUN-18 ORDERED 2CURD PACKETS  AND IT COSTED HIM 60
SURESH ON  07-JUN-18 ORDERED 4MILK PACKETS  AND IT COSTED HIM 80

```

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT NVL(CU.CNAME,'NO CUSTOMER')
2      || ' ON '
3      || COD.ORD_DATE
4      || ' ORDERED '
5      || COD.QTY
6      || ' '
7      || NVL( PR.PNAME,'NO PRODUCT')
8      || ' PACKETS '
9      || ' AND IT COSTED HIM '
10     || COD.QTY * NVL(PR.PRICE,15) AS ORD_INFORMATION
11 FROM B26_CUST_ORD_DET COD
12     ,B26_PROD PR
13     ,B26_CUST CU
14 WHERE PR.PID(+) = COD.PID
15 AND     CU.CID(+) = COD.CID
16 AND TRUNC(COD.ORD_DATE) = '07-JUN-18'
17* ORDER BY COD.ORD_DATE DESC
SQL> /

```

ORD_INFORMATION

```

-----
-----
-----
NO CUSTOMER ON  07-JUN-18 ORDERED 20 MILK PACKETS  AND IT COSTED HIM 400
KRISHNA ON  07-JUN-18 ORDERED 5 MILK PACKETS  AND IT COSTED HIM 100
KRISHNA ON  07-JUN-18 ORDERED 3 CURD PACKETS  AND IT COSTED HIM 90
SURESH ON  07-JUN-18 ORDERED 100 NO PRODUCT PACKETS  AND IT COSTED HIM 1500
SURESH ON  07-JUN-18 ORDERED 2 CURD PACKETS  AND IT COSTED HIM 60

```

SURESH ON 07-JUN-18 ORDERED 4 MILK PACKETS AND IT COSTED HIM 80

6 rows selected.

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT NVL(CU.CNAME,'NO CUSTOMER')
2      || ' ON '
3      || COD.ORD_DATE
4      || ' ORDERED '
5      || COD.QTY
6      || ' '
7      || NVL( PR.PNAME,'NO PRODUCT')
8      || ' PACKETS '
9      || ' AND IT COSTED HIM '
10     || COD.QTY * NVL(PR.PRICE,15) AS ORD_INFORMATION
11 FROM B26_CUST_ORD_DET COD
12     ,B26_PROD PR
13     ,B26_CUST CU
14 WHERE PR.PID(+) = COD.PID
15 AND     CU.CID(+) = COD.CID
16 AND TRUNC(COD.ORD_DATE) = '07-JUN-18'
17* ORDER BY COD.ORD_DATE DESC

```

SQL>

SQL>

SQL>

SQL> CL SCR

```

SQL> SELECT *
2  FROM B26_CUST;

```

	CID CNAME	CMOB	CADDR	CGEN
1	SURESH	12345	JAKKASANDRA	MALE
2	KRISHNA	12346	HSR	MALE
3	TEJASREE	12348	TEACHERSCOLONY	FEMALE

```

SQL> SELECT *
2  FROM B26_CUST_ORD_DET;

```

	CID	PID	QTY	ORD_DATE
1	1	1	4	07-JUN-18
1	2	2	2	07-JUN-18
1			100	07-JUN-18
2	2	2	3	07-JUN-18
2	1	1	5	07-JUN-18
		1	20	07-JUN-18
2	1	1	10	08-JUN-18

7 rows selected.

```

SQL> INSERT INTO B26_CUST_ORD_DET
2  VALUES
3  (
4  3
5  ,1
6  ,4
7  ,'07-JUN-18'
8  );

```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> EDIT
```

Wrote file afiedt.buf

```

1  SELECT NVL(CU.CNAME,'NO CUSTOMER')
2      || ' ON '
3      || COD.ORD_DATE
4      || ' ORDERED '
5      || COD.QTY
6      || ' '
7      || NVL( PR.PNAME,'NO PRODUCT')
8      || ' PACKETS '
9      || ' AND IT COSTED HIM '
10     || COD.QTY * NVL(PR.PRICE,15) AS ORD_INFORMATION
11 FROM B26_CUST_ORD_DET COD
12     ,B26_PROD PR
13     ,B26_CUST CU
14 WHERE PR.PID(+) = COD.PID
15 AND     CU.CID(+) = COD.CID
16 AND TRUNC(COD.ORD_DATE) = '07-JUN-18'
17* ORDER BY COD.ORD_DATE DESC
SQL> /
```

ORD_INFORMATION

```

-----
-----
-----
NO CUSTOMER ON 07-JUN-18 ORDERED 20 MILK PACKETS AND IT COSTED HIM 400
KRISHNA ON 07-JUN-18 ORDERED 5 MILK PACKETS AND IT COSTED HIM 100
KRISHNA ON 07-JUN-18 ORDERED 3 CURD PACKETS AND IT COSTED HIM 90
SURESH ON 07-JUN-18 ORDERED 100 NO PRODUCT PACKETS AND IT COSTED HIM 1500
SURESH ON 07-JUN-18 ORDERED 2 CURD PACKETS AND IT COSTED HIM 60
SURESH ON 07-JUN-18 ORDERED 4 MILK PACKETS AND IT COSTED HIM 80
TEJASREE ON 07-JUN-18 ORDERED 4 MILK PACKETS AND IT COSTED HIM 80
```

7 rows selected.

```
SQL> ED
```

Wrote file afiedt.buf

```

1  SELECT NVL(CU.CNAME,'NO CUSTOMER')
2      || ' ON '
3      || COD.ORD_DATE
4      || ' ORDERED '
5      || COD.QTY
6      || ' '
7      || NVL( PR.PNAME,'NO PRODUCT')
8      || ' PACKETS '
9      || ' AND IT COSTED '
10     || CASE WHEN CU.GEN ='MALE' THEN 'HIM '
11            ELSE 'HER '
12            END
13     || COD.QTY * NVL(PR.PRICE,15) AS ORD_INFORMATION
14 FROM B26_CUST_ORD_DET COD
15     ,B26_PROD PR
16     ,B26_CUST CU
17 WHERE PR.PID(+) = COD.PID
18 AND     CU.CID(+) = COD.CID
19 AND TRUNC(COD.ORD_DATE) = '07-JUN-18'
```



```

20* ORDER BY COD.ORD_DATE DESC
SQL> /
      || CASE WHEN CU.GEN ='MALE' THEN 'HIM '
      *

```

ERROR at line 10:
ORA-00904: "CU"."GEN": invalid identifier

```

SQL> ED
Wrote file afiedt.buf

```

```

 1  SELECT NVL(CU.CNAME,'NO CUSTOMER')
 2      || ' ON '
 3      || COD.ORD_DATE
 4      || ' ORDERED '
 5      || COD.QTY
 6      || ' '
 7      || NVL( PR.PNAME,'NO PRODUCT')
 8      || ' PACKETS '
 9      || ' AND IT COSTED '
10      || CASE WHEN CU.GENDER ='MALE' THEN 'HIM '
11          ELSE 'HER '
12      END
13      || COD.QTY * NVL(PR.PRICE,15) AS ORD_INFORMATION
14 FROM B26_CUST_ORD_DET COD
15      ,B26_PROD PR
16      ,B26_CUST CU
17 WHERE PR.PID(+) = COD.PID
18 AND     CU.CID(+) = COD.CID
19 AND TRUNC(COD.ORD_DATE) = '07-JUN-18'
20* ORDER BY COD.ORD_DATE DESC
SQL> /
      || CASE WHEN CU.GENDER ='MALE' THEN 'HIM '
      *

```

ERROR at line 10:
ORA-00904: "CU"."GENDER": invalid identifier

```

SQL> DESC B26_CUST

```

```

Name
Null?    Type

```

```

-----
-----
-----
CID
NOT NULL NUMBER
CNAME
VARCHAR2(20)
CMOB
NUMBER(10)
CADDR
VARCHAR2(20)
CGEN
VARCHAR2(10)

```

```

SQL> ED
Wrote file afiedt.buf

```

```

 1  SELECT NVL(CU.CNAME,'NO CUSTOMER')
 2      || ' ON '
 3      || COD.ORD_DATE
 4      || ' ORDERED '
 5      || COD.QTY
 6      || ' '

```

```

7          ||NVL( PR.PNAME,'NO PRODUCT')
8          ||' PACKETS '
9          ||' AND IT COSTED '
10         || CASE WHEN CU.CGENR ='MALE' THEN 'HIM '
11             ELSE 'HER '
12         END
13         ||COD.QTY * NVL(PR.PRICE,15) AS ORD_INFORMATION
14 FROM B26_CUST_ORD_DET COD
15       ,B26_PROD PR
16       ,B26_CUST CU
17 WHERE PR.PID(+)  = COD.PID
18 AND      CU.CID(+)  = COD.CID
19 AND TRUNC(COD.ORD_DATE) = '07-JUN-18'
20* ORDER BY COD.ORD_DATE DESC
SQL> /
      || CASE WHEN CU.CGENR ='MALE' THEN 'HIM '
      *

```

ERROR at line 10:
ORA-00904: "CU"."CGENR": invalid identifier

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT NVL(CU.CNAME,'NO CUSTOMER')
2         ||' ON '
3         ||COD.ORD_DATE
4         ||' ORDERED '
5         ||COD.QTY
6         ||' '
7         ||NVL( PR.PNAME,'NO PRODUCT')
8         ||' PACKETS '
9         ||' AND IT COSTED '
10        || CASE WHEN CU.CGEN ='MALE' THEN 'HIM '
11            ELSE 'HER '
12        END
13        ||COD.QTY * NVL(PR.PRICE,15) AS ORD_INFORMATION
14 FROM B26_CUST_ORD_DET COD
15       ,B26_PROD PR
16       ,B26_CUST CU
17 WHERE PR.PID(+)  = COD.PID
18 AND      CU.CID(+)  = COD.CID
19 AND TRUNC(COD.ORD_DATE) = '07-JUN-18'
20* ORDER BY COD.ORD_DATE DESC
SQL> /

```

ORD_INFORMATION

```

-----
-----
-----
NO CUSTOMER ON  07-JUN-18 ORDERED 20 MILK PACKETS  AND IT COSTED  HER 400
KRISHNA ON  07-JUN-18 ORDERED 5 MILK PACKETS  AND IT COSTED  HIM 100
KRISHNA ON  07-JUN-18 ORDERED 3 CURD PACKETS  AND IT COSTED  HIM 90
SURESH ON  07-JUN-18 ORDERED 100 NO PRODUCT PACKETS  AND IT COSTED  HIM 1500
SURESH ON  07-JUN-18 ORDERED 2 CURD PACKETS  AND IT COSTED  HIM 60
SURESH ON  07-JUN-18 ORDERED 4 MILK PACKETS  AND IT COSTED  HIM 80
TEJASREE ON  07-JUN-18 ORDERED 4 MILK PACKETS  AND IT COSTED  HER 80

```

7 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT NVL(CU.CNAME,'NO CUSTOMER')

```

```

2      || ' ON '
3      || COD.ORD_DATE
4      || ' ORDERED '
5      || COD.QTY
6      || ' '
7      || NVL( PR.PNAME, 'NO PRODUCT')
8      || ' PACKETS '
9      || ' AND IT COSTED '
10     || CASE WHEN CU.CGEN = 'MALE' THEN 'HIM '
11         ELSE 'HER '
12     END
13     || COD.QTY * NVL(PR.PRICE,15) AS ORD_INFORMATION
14 FROM B26_CUST_ORD_DET COD
15     ,B26_PROD PR
16     ,B26_CUST CU
17 WHERE PR.PID(+) = COD.PID
18 AND     CU.CID(+) = COD.CID
19 AND TRUNC(COD.ORD_DATE) = '07-JUN-18'
20* ORDER BY COD.ORD_DATE DESC

```

SQL>

SQL>

SQL>

SQL>

SQL> ED

Wrote file afiedt.buf

```

1  SELECT *
2  FROM B26_CUST_ORD_DET COD
3* ,B26_PROD
SQL> /

```

PRICE	CID	PID	QTY	ORD_DATE	PID	PNAME	PMFR
20	1	1	4	07-JUN-18	1	MILK	NANDINI
20	1	2	2	07-JUN-18	1	MILK	NANDINI
20	1		100	07-JUN-18	1	MILK	NANDINI
20	2	2	3	07-JUN-18	1	MILK	NANDINI
20	2	1	5	07-JUN-18	1	MILK	NANDINI
20		1	20	07-JUN-18	1	MILK	NANDINI
20	2	1	10	08-JUN-18	1	MILK	NANDINI
20	3	1	4	07-JUN-18	1	MILK	NANDINI
30	1	1	4	07-JUN-18	2	CURD	DODLA
30	1	2	2	07-JUN-18	2	CURD	DODLA
30	1		100	07-JUN-18	2	CURD	DODLA
30	2	2	3	07-JUN-18	2	CURD	DODLA
30	2	1	5	07-JUN-18	2	CURD	DODLA
30		1	20	07-JUN-18	2	CURD	DODLA

```

          2          1          10 08-JUN-18          2 CURD          DODLA
30
          3          1          4 07-JUN-18          2 CURD          DODLA
30

```

16 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.*
2      ,PR.PRICE
3  FROM B26_CUST_ORD_DET COD
4* ,B26_PROD PR
SQL> /

```

CID	PID	QTY	ORD_DATE	PRICE
1	1	4	07-JUN-18	20
1	2	2	07-JUN-18	20
1		100	07-JUN-18	20
2	2	3	07-JUN-18	20
2	1	5	07-JUN-18	20
	1	20	07-JUN-18	20
2	1	10	08-JUN-18	20
3	1	4	07-JUN-18	20
1	1	4	07-JUN-18	30
1	2	2	07-JUN-18	30
1		100	07-JUN-18	30
2	2	3	07-JUN-18	30
2	1	5	07-JUN-18	30
	1	20	07-JUN-18	30
2	1	10	08-JUN-18	30
3	1	4	07-JUN-18	30

16 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,COD.CNAME
3      ,COD.PID
4      ,COD.QTY
5      ,PR.PRICE
6  FROM B26_CUST_ORD_DET COD
7* ,B26_PROD PR
SQL> /
      ,COD.CNAME
      *

```

ERROR at line 2:

ORA-00904: "COD"."CNAME": invalid identifier

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,COD.PID
4      ,COD.QTY
5      ,PR.PRICE
6  FROM B26_CUST_ORD_DET COD
7* ,B26_PROD PR

```

SQL> /

CID	PNAME	PID	QTY	PRICE
1	MILK	1	4	20
1	MILK	2	2	20
1	MILK		100	20
2	MILK	2	3	20
2	MILK	1	5	20
	MILK	1	20	20
2	MILK	1	10	20
3	MILK	1	4	20
1	CURD	1	4	30
1	CURD	2	2	30
1	CURD		100	30
2	CURD	2	3	30
2	CURD	1	5	30
	CURD	1	20	30
2	CURD	1	10	30
3	CURD	1	4	30

16 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,COD.PID
4      ,COD.QTY
5      ,PR.PRICE
6  FROM B26_CUST_ORD_DET COD
7* ,B26_PROD PR

```

SQL>

SQL>

SQL>

SQL> INSERT INTO B26_PROD

2 VALUES

3 (

4 3

5

SQL> DESC B26_PROD

Name

Null? Type

PID

NOT NULL NUMBER

PNAME

VARCHAR2(20)

PMFR

VARCHAR2(20)

PRICE

NUMBER

SQL> INSERT INTO B26_PROD

2 VALUES

3 (

4 3

5 , 'ICE'

6 , 'ARUN'

7 ,40

8);

1 row created.

SQL> COMMIT;

Commit complete.

SQL> CL SCR

SQL> SELECT * FROM B26_PROD;

PID	PNAME	PMFR	PRICE
1	MILK	NANDINI	20
2	CURD	DODLA	30
3	ICE	ARUN	40

SQL> SELECT * FROM B26_CUST_ORD_DET;

CID	PID	QTY	ORD_DATE
1	1	4	07-JUN-18
1	2	2	07-JUN-18
1		100	07-JUN-18
2	2	3	07-JUN-18
2	1	5	07-JUN-18
	1	20	07-JUN-18
2	1	10	08-JUN-18
3	1	4	07-JUN-18

8 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,COD.PID
4      ,COD.QTY
5      ,PR.PRICE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8*  WHERE COD.PID = PR.PID
SQL> /

```

CID	PNAME	PID	QTY	PRICE
1	MILK	1	4	20
1	CURD	2	2	30
2	CURD	2	3	30
2	MILK	1	5	20
	MILK	1	20	20
2	MILK	1	10	20
3	MILK	1	4	20

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,COD.PID
4      ,COD.QTY

```

```

5          ,PR.PRICE
6 FROM B26_CUST_ORD_DET COD
7          ,B26_PROD PR
8*  WHERE COD.PID(+) = PR.PID
SQL> /

```

CID	PNAME	PID	QTY	PRICE
1	MILK	1	4	20
1	CURD	2	2	30
2	CURD	2	3	30
2	MILK	1	5	20
	MILK	1	20	20
2	MILK	1	10	20
3	MILK	1	4	20
	ICE			40

8 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COD.CID
2          ,PR.PNAME
3          ,NVL( COD.PID,PR.PID) AS PID
4          ,COD.QTY
5          ,PR.PRICE
6 FROM B26_CUST_ORD_DET COD
7          ,B26_PROD PR
8*  WHERE COD.PID(+) = PR.PID
SQL> /

```

CID	PNAME	PID	QTY	PRICE
1	MILK	1	4	20
1	CURD	2	2	30
2	CURD	2	3	30
2	MILK	1	5	20
	MILK	1	20	20
2	MILK	1	10	20
3	MILK	1	4	20
	ICE	3		40

8 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT COD.CID
2          ,PR.PNAME
3          ,NVL( COD.PID,PR.PID) AS PID
4          ,COD.QTY
5          ,PR.PRICE
6 FROM B26_CUST_ORD_DET COD
7          ,B26_PROD PR
8*  WHERE COD.PID = PR.PID
SQL> /

```

CID	PNAME	PID	QTY	PRICE
1	MILK	1	4	20
1	CURD	2	2	30
2	CURD	2	3	30
2	MILK	1	5	20

MILK	1	20	20
2 MILK	1	10	20
3 MILK	1	4	20

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,NVL( COD.PID,PR.PID) AS PID
4      ,COD.QTY
5      ,PR.PRICE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8*   WHERE COD.PID = PR.PID(+)

```

SQL> /

CID	PNAME	PID	QTY	PRICE
3	MILK	1	4	20
2	MILK	1	10	20
	MILK	1	20	20
2	MILK	1	5	20
1	MILK	1	4	20
2	CURD	2	3	30
1	CURD	2	2	30
1			100	

8 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,NVL( COD.PID,PR.PID) AS PID
4      ,COD.QTY
5      ,PR.PRICE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8*   WHERE COD.PID(+) = PR.PID(+)

```

SQL> /

```

WHERE COD.PID(+) = PR.PID(+)
      *

```

ERROR at line 8:

ORA-01468: a predicate may reference only one outer-joined table

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,NVL( COD.PID,PR.PID) AS PID
4      ,COD.QTY
5      ,PR.PRICE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8   WHERE COD.PID = PR.PID(+)
9  UNION ALL
10 SELECT COD.CID
11      ,PR.PNAME

```



```

12      ,NVL( COD.PID,PR.PID) AS PID
13      ,COD.QTY
14      ,PR.PRICE
15 FROM B26_CUST_ORD_DET COD
16      ,B26_PROD PR
17* WHERE COD.PID(+) = PR.PID
18 /

```

CID	PNAME	PID	QTY	PRICE
3	MILK	1	4	20
2	MILK	1	10	20
	MILK	1	20	20
2	MILK	1	5	20
1	MILK	1	4	20
2	CURD	2	3	30
1	CURD	2	2	30
1			100	
1	MILK	1	4	20
1	CURD	2	2	30
2	CURD	2	3	30
2	MILK	1	5	20
	MILK	1	20	20
2	MILK	1	10	20
3	MILK	1	4	20
	ICE	3		40

16 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,NVL( COD.PID,PR.PID) AS PID
4      ,COD.QTY
5      ,PR.PRICE
6 FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8 WHERE COD.PID = PR.PID(+)
9 UNION
10 SELECT COD.CID
11      ,PR.PNAME
12      ,NVL( COD.PID,PR.PID) AS PID
13      ,COD.QTY
14      ,PR.PRICE
15 FROM B26_CUST_ORD_DET COD
16      ,B26_PROD PR
17* WHERE COD.PID(+) = PR.PID
SQL> /

```

CID	PNAME	PID	QTY	PRICE
1	CURD	2	2	30
1	MILK	1	4	20
1			100	
2	CURD	2	3	30
2	MILK	1	5	20
2	MILK	1	10	20
3	MILK	1	4	20
	ICE	3		40
	MILK	1	20	20

9 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,NVL( COD.PID,PR.PID) AS PID
4      ,COD.QTY
5      ,PR.PRICE
6  FROM B26_CUST_ORD_DET COD
7      FULL OUTER JOIN B26_PROD PR
8*   ON COD.PID = PR.PID
SQL> /
```

CID	PNAME	PID	QTY	PRICE
1	MILK	1	4	20
1	CURD	2	2	30
1			100	
2	CURD	2	3	30
2	MILK	1	5	20
	MILK	1	20	20
2	MILK	1	10	20
3	MILK	1	4	20
	ICE	3		40

9 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,NVL( COD.PID,PR.PID) AS PID
4      ,COD.QTY
5      ,PR.PRICE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
8*   WHERE COD.PID(+) = PR.PID
SQL> /
```

CID	PNAME	PID	QTY	PRICE
1	MILK	1	4	20
1	CURD	2	2	30
2	CURD	2	3	30
2	MILK	1	5	20
	MILK	1	20	20
2	MILK	1	10	20
3	MILK	1	4	20
	ICE	3		40

8 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT COD.CID
2      ,PR.PNAME
3      ,NVL( COD.PID,PR.PID) AS PID
4      ,COD.QTY
5      ,PR.PRICE
6  FROM B26_CUST_ORD_DET COD
7      ,B26_PROD PR
```

```

      8      WHERE COD.PID(+) = PR.PID
      9*      AND COD.PID = PR.PID(+)
SQL> /
      WHERE COD.PID(+) = PR.PID
      *
```

ERROR at line 8:
ORA-01416: two tables cannot be outer-joined to each other

```

SQL> EDIT
Wrote file afiedt.buf
```

```

      1  SELECT COD.CID
      2          ,PR.PNAME
      3          ,NVL( COD.PID,PR.PID) AS PID
      4          ,COD.QTY
      5          ,PR.PRICE
      6  FROM B26_CUST_ORD_DET COD
      7          ,B26_PROD PR
      8      WHERE COD.PID(+) = PR.PID
      9*      AND COD.PID = PR.PID(+)
SQL>
SQL>
SQL>
SQL> CL SCR
```

```

SQL> CREATE TABLE B26_PROD_GRADE
      2  (
      3  GID  NUMBER
      4  ,LOPRICE NUMBER
      5  ,HIPRICE NUMBER
      6  );
```

Table created.

```

SQL> INSERT INTO B26_PROD_GRADE
      2  VALUES
      3  (
      4  1
      5  ,
      6  ,1
      7  ,10
      8  );
,1
*
```

ERROR at line 6:
ORA-00936: missing expression

```

SQL> ED
Wrote file afiedt.buf
```

```

      1  INSERT INTO B26_PROD_GRADE
      2  VALUES
      3  (
      4  1
      5  ,1
      6  ,10
      7* )
SQL> /
```

1 row created.

```

SQL> ED
```

Wrote file afiedt.buf

```
1  INSERT INTO B26_PROD_GRADE
2  VALUES
3  (
4  2
5  ,11
6  ,20
7* )
SQL> /
```

1 row created.

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_PROD_GRADE
2  VALUES
3  (
4  3
5  ,21
6  ,30
7* )
SQL> /
```

1 row created.

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_PROD_GRADE
2  VALUES
3  (
4  4
5  ,31
6  ,40
7* )
SQL> /
```

1 row created.

SQL> ED

Wrote file afiedt.buf

```
1  INSERT INTO B26_PROD_GRADE
2  VALUES
3  (
4  5
5  ,41
6  ,99
7* )
SQL> /
```

1 row created.

SQL> COMMIT;

Commit complete.

SQL> CL SCR

SQL> SELECT * FROM B26_PROD;

PID PNAME

PMFR

PRICE

```

-----
      1 MILK                NANDINI                20
      2 CURD                DODLA                  30
      3 ICE                 ARUN                   40

```

```
SQL> SELECT * FROM B26_PROD_GRADE;
```

```

      GID      LOPRICE      HIPRICE
-----
      1         1         10
      2        11         20
      3        21         30
      4        31         40
      5        41         99

```

```
SQL> PNAME GRADEW
```

```
SP2-0734: unknown command beginning "PNAME GR..." - rest of line ignored.
```

```
SQL>
```

```
SQL>
```

```
SQL>
```

```
SQL> SELECT PNAME
```

```
2      ,PRICE
```

```
3      ,GRADE
```

```
4 FROM B26_PROD PR
```

```
5      ,B26_PROD_GRADE PRG
```

```
6 WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE;
```

```
      ,GRADE
```

```
      *
```

```
ERROR at line 3:
```

```
ORA-00904: "GRADE": invalid identifier
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT PNAME
2      ,PRICE
3      ,GID
4  FROM B26_PROD PR
5      ,B26_PROD_GRADE PRG
6* WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
SQL> /

```

```

PNAME                PRICE                GID
-----
MILK                  20                  2
CURD                  30                  3
ICE                   40                  4

```

```
SQL> CL SCR
```

```
SQL> DESC B26_CUST
```

```
  Name
```

```
Null?    Type
```

```
-----
      CID
```

```
NOT NULL NUMBER
```

```
      CNAME
```

```
VARCHAR2(20)
```

```
      CMOB
```

```
NUMBER(10)
```

```
CADDR
VARCHAR2(20)
CGEN
VARCHAR2(10)
```

```
SQL> DESC B26_PROD
```

```
  Name
Null?   Type
```

```
-----
-----
-----
  PID
NOT NULL NUMBER
  PNAME
VARCHAR2(20)
  PMFR
VARCHAR2(20)
  PRICE
NUMBER
```

```
SQL> DESC B26_PROD_GRADE
```

```
  Name
Null?   Type
```

```
-----
-----
-----
  GID
NUMBER
  LOPRICE
NUMBER
  HIPRICE
NUMBER
```

```
SQL> DESC B26_CUST_ORD_DET
```

```
  Name
Null?   Type
```

```
-----
-----
-----
  CID
NUMBER
  PID
NUMBER
  QTY
NOT NULL NUMBER
  ORD_DATE
DATE
```

```
SQL> SELECT CU.CNAME AS CUSTOMERNAME
  2          ,PR.PNAME  AS PRODNAME
  3          ,PRG.GID   AS PROD_GRADE
  4          ,PR.PIRCE  AS PRICE
  5          ,COD.QTY   AS QTY
  6          ,COD.QTY * PR.PRICE AS TOTALPRICE
  7 FROM B26_CUST_ORD_DET COD
  8          ,B26_PROD PR
  9          ,B26_PROD_GRADE PRG
 10          ,B26_CUST CU;
          ,PR.PIRCE AS PRICE
          *
```

```
ERROR at line 4:
ORA-00904: "PR"."PIRCE": invalid identifier
```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10*   ,B26_CUST CU
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				
-----	-----	-----	-----	-----
--				
SURESH	MILK	1	20	4
80				
SURESH	MILK	1	20	2
40				
SURESH	MILK	1	20	100
2000				
SURESH	MILK	1	20	3
60				
SURESH	MILK	1	20	5
100				
SURESH	MILK	1	20	20
400				
SURESH	MILK	1	20	10
200				
SURESH	MILK	1	20	4
80				
SURESH	MILK	2	20	4
80				
SURESH	MILK	2	20	2
40				
SURESH	MILK	2	20	100
2000				
SURESH	MILK	2	20	3
60				
SURESH	MILK	2	20	5
100				
SURESH	MILK	2	20	20
400				
SURESH	MILK	2	20	10
200				
SURESH	MILK	2	20	4
80				
SURESH	MILK	3	20	4
80				
SURESH	MILK	3	20	2
40				
SURESH	MILK	3	20	100
2000				
SURESH	MILK	3	20	3
60				
SURESH	MILK	3	20	5
100				
SURESH	MILK	3	20	20
400				

SURESH 200	MILK	3	20	10
SURESH 80	MILK	3	20	4
SURESH 80	MILK	4	20	4
SURESH 40	MILK	4	20	2
SURESH 2000	MILK	4	20	100
SURESH 60	MILK	4	20	3
SURESH 100	MILK	4	20	5
SURESH 400	MILK	4	20	20
SURESH 200	MILK	4	20	10
SURESH 80	MILK	4	20	4
SURESH 80	MILK	5	20	4
SURESH 40	MILK	5	20	2
SURESH 2000	MILK	5	20	100
SURESH 60	MILK	5	20	3
SURESH 100	MILK	5	20	5
SURESH 400	MILK	5	20	20
SURESH 200	MILK	5	20	10
SURESH 80	MILK	5	20	4
KRISHNA 80	MILK	1	20	4
KRISHNA 40	MILK	1	20	2
KRISHNA 2000	MILK	1	20	100
KRISHNA 60	MILK	1	20	3
KRISHNA 100	MILK	1	20	5
KRISHNA 400	MILK	1	20	20
KRISHNA 200	MILK	1	20	10
KRISHNA 80	MILK	1	20	4
KRISHNA 80	MILK	2	20	4
KRISHNA 40	MILK	2	20	2
KRISHNA 2000	MILK	2	20	100
KRISHNA 60	MILK	2	20	3
KRISHNA 100	MILK	2	20	5
KRISHNA 400	MILK	2	20	20

KRISHNA 200	MILK	2	20	10
KRISHNA 80	MILK	2	20	4
KRISHNA 80	MILK	3	20	4
KRISHNA 40	MILK	3	20	2
KRISHNA 2000	MILK	3	20	100
KRISHNA 60	MILK	3	20	3
KRISHNA 100	MILK	3	20	5
KRISHNA 400	MILK	3	20	20
KRISHNA 200	MILK	3	20	10
KRISHNA 80	MILK	3	20	4
KRISHNA 80	MILK	4	20	4
KRISHNA 40	MILK	4	20	2
KRISHNA 2000	MILK	4	20	100
KRISHNA 60	MILK	4	20	3
KRISHNA 100	MILK	4	20	5
KRISHNA 400	MILK	4	20	20
KRISHNA 200	MILK	4	20	10
KRISHNA 80	MILK	4	20	4
KRISHNA 80	MILK	5	20	4
KRISHNA 40	MILK	5	20	2
KRISHNA 2000	MILK	5	20	100
KRISHNA 60	MILK	5	20	3
KRISHNA 100	MILK	5	20	5
KRISHNA 400	MILK	5	20	20
KRISHNA 200	MILK	5	20	10
KRISHNA 80	MILK	5	20	4
TEJASREE 80	MILK	1	20	4
TEJASREE 40	MILK	1	20	2
TEJASREE 2000	MILK	1	20	100
TEJASREE 60	MILK	1	20	3
TEJASREE 100	MILK	1	20	5
TEJASREE 400	MILK	1	20	20

TEJASREE 200	MILK	1	20	10
TEJASREE 80	MILK	1	20	4
TEJASREE 80	MILK	2	20	4
TEJASREE 40	MILK	2	20	2
TEJASREE 2000	MILK	2	20	100
TEJASREE 60	MILK	2	20	3
TEJASREE 100	MILK	2	20	5
TEJASREE 400	MILK	2	20	20
TEJASREE 200	MILK	2	20	10
TEJASREE 80	MILK	2	20	4
TEJASREE 80	MILK	3	20	4
TEJASREE 40	MILK	3	20	2
TEJASREE 2000	MILK	3	20	100
TEJASREE 60	MILK	3	20	3
TEJASREE 100	MILK	3	20	5
TEJASREE 400	MILK	3	20	20
TEJASREE 200	MILK	3	20	10
TEJASREE 80	MILK	3	20	4
TEJASREE 80	MILK	4	20	4
TEJASREE 40	MILK	4	20	2
TEJASREE 2000	MILK	4	20	100
TEJASREE 60	MILK	4	20	3
TEJASREE 100	MILK	4	20	5
TEJASREE 400	MILK	4	20	20
TEJASREE 200	MILK	4	20	10
TEJASREE 80	MILK	4	20	4
TEJASREE 80	MILK	5	20	4
TEJASREE 40	MILK	5	20	2
TEJASREE 2000	MILK	5	20	100
TEJASREE 60	MILK	5	20	3
TEJASREE 100	MILK	5	20	5
TEJASREE 400	MILK	5	20	20

TEJASREE 200	MILK	5	20	10
TEJASREE 80	MILK	5	20	4
SURESH 120	CURD	1	30	4
SURESH 60	CURD	1	30	2
SURESH 3000	CURD	1	30	100
SURESH 90	CURD	1	30	3
SURESH 150	CURD	1	30	5
SURESH 600	CURD	1	30	20
SURESH 300	CURD	1	30	10
SURESH 120	CURD	1	30	4
SURESH 120	CURD	2	30	4
SURESH 60	CURD	2	30	2
SURESH 3000	CURD	2	30	100
SURESH 90	CURD	2	30	3
SURESH 150	CURD	2	30	5
SURESH 600	CURD	2	30	20
SURESH 300	CURD	2	30	10
SURESH 120	CURD	2	30	4
SURESH 120	CURD	3	30	4
SURESH 60	CURD	3	30	2
SURESH 3000	CURD	3	30	100
SURESH 90	CURD	3	30	3
SURESH 150	CURD	3	30	5
SURESH 600	CURD	3	30	20
SURESH 300	CURD	3	30	10
SURESH 120	CURD	3	30	4
SURESH 120	CURD	4	30	4
SURESH 60	CURD	4	30	2
SURESH 3000	CURD	4	30	100
SURESH 90	CURD	4	30	3
SURESH 150	CURD	4	30	5
SURESH 600	CURD	4	30	20

SURESH 300	CURD	4	30	10
SURESH 120	CURD	4	30	4
SURESH 120	CURD	5	30	4
SURESH 60	CURD	5	30	2
SURESH 3000	CURD	5	30	100
SURESH 90	CURD	5	30	3
SURESH 150	CURD	5	30	5
SURESH 600	CURD	5	30	20
SURESH 300	CURD	5	30	10
SURESH 120	CURD	5	30	4
KRISHNA 120	CURD	1	30	4
KRISHNA 60	CURD	1	30	2
KRISHNA 3000	CURD	1	30	100
KRISHNA 90	CURD	1	30	3
KRISHNA 150	CURD	1	30	5
KRISHNA 600	CURD	1	30	20
KRISHNA 300	CURD	1	30	10
KRISHNA 120	CURD	1	30	4
KRISHNA 120	CURD	2	30	4
KRISHNA 60	CURD	2	30	2
KRISHNA 3000	CURD	2	30	100
KRISHNA 90	CURD	2	30	3
KRISHNA 150	CURD	2	30	5
KRISHNA 600	CURD	2	30	20
KRISHNA 300	CURD	2	30	10
KRISHNA 120	CURD	2	30	4
KRISHNA 120	CURD	3	30	4
KRISHNA 60	CURD	3	30	2
KRISHNA 3000	CURD	3	30	100
KRISHNA 90	CURD	3	30	3
KRISHNA 150	CURD	3	30	5
KRISHNA 600	CURD	3	30	20

KRISHNA 300	CURD	3	30	10
KRISHNA 120	CURD	3	30	4
KRISHNA 120	CURD	4	30	4
KRISHNA 60	CURD	4	30	2
KRISHNA 3000	CURD	4	30	100
KRISHNA 90	CURD	4	30	3
KRISHNA 150	CURD	4	30	5
KRISHNA 600	CURD	4	30	20
KRISHNA 300	CURD	4	30	10
KRISHNA 120	CURD	4	30	4
KRISHNA 120	CURD	5	30	4
KRISHNA 60	CURD	5	30	2
KRISHNA 3000	CURD	5	30	100
KRISHNA 90	CURD	5	30	3
KRISHNA 150	CURD	5	30	5

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				

--				
KRISHNA 600	CURD	5	30	20
KRISHNA 300	CURD	5	30	10
KRISHNA 120	CURD	5	30	4
TEJASREE 120	CURD	1	30	4
TEJASREE 60	CURD	1	30	2
TEJASREE 3000	CURD	1	30	100
TEJASREE 90	CURD	1	30	3
TEJASREE 150	CURD	1	30	5
TEJASREE 600	CURD	1	30	20
TEJASREE 300	CURD	1	30	10
TEJASREE 120	CURD	1	30	4
TEJASREE 120	CURD	2	30	4
TEJASREE 60	CURD	2	30	2
TEJASREE 3000	CURD	2	30	100

SURESH 120	ICE	1	40	3
SURESH 200	ICE	1	40	5
SURESH 800	ICE	1	40	20
SURESH 400	ICE	1	40	10
SURESH 160	ICE	1	40	4
SURESH 160	ICE	2	40	4
SURESH 80	ICE	2	40	2
SURESH 4000	ICE	2	40	100
SURESH 120	ICE	2	40	3
SURESH 200	ICE	2	40	5
SURESH 800	ICE	2	40	20
SURESH 400	ICE	2	40	10
SURESH 160	ICE	2	40	4
SURESH 160	ICE	3	40	4
SURESH 80	ICE	3	40	2
SURESH 4000	ICE	3	40	100
SURESH 120	ICE	3	40	3
SURESH 200	ICE	3	40	5
SURESH 800	ICE	3	40	20
SURESH 400	ICE	3	40	10
SURESH 160	ICE	3	40	4
SURESH 160	ICE	4	40	4
SURESH 80	ICE	4	40	2
SURESH 4000	ICE	4	40	100
SURESH 120	ICE	4	40	3
SURESH 200	ICE	4	40	5
SURESH 800	ICE	4	40	20
SURESH 400	ICE	4	40	10
SURESH 160	ICE	4	40	4
SURESH 160	ICE	5	40	4
SURESH 80	ICE	5	40	2
SURESH 4000	ICE	5	40	100

SURESH 120	ICE	5	40	3
SURESH 200	ICE	5	40	5
SURESH 800	ICE	5	40	20
SURESH 400	ICE	5	40	10
SURESH 160	ICE	5	40	4
KRISHNA 160	ICE	1	40	4
KRISHNA 80	ICE	1	40	2
KRISHNA 4000	ICE	1	40	100
KRISHNA 120	ICE	1	40	3
KRISHNA 200	ICE	1	40	5
KRISHNA 800	ICE	1	40	20
KRISHNA 400	ICE	1	40	10
KRISHNA 160	ICE	1	40	4
KRISHNA 160	ICE	2	40	4
KRISHNA 80	ICE	2	40	2
KRISHNA 4000	ICE	2	40	100
KRISHNA 120	ICE	2	40	3
KRISHNA 200	ICE	2	40	5
KRISHNA 800	ICE	2	40	20
KRISHNA 400	ICE	2	40	10
KRISHNA 160	ICE	2	40	4
KRISHNA 160	ICE	3	40	4
KRISHNA 80	ICE	3	40	2
KRISHNA 4000	ICE	3	40	100
KRISHNA 120	ICE	3	40	3
KRISHNA 200	ICE	3	40	5
KRISHNA 800	ICE	3	40	20
KRISHNA 400	ICE	3	40	10
KRISHNA 160	ICE	3	40	4
KRISHNA 160	ICE	4	40	4
KRISHNA 80	ICE	4	40	2
KRISHNA 4000	ICE	4	40	100

KRISHNA 120	ICE	4	40	3
KRISHNA 200	ICE	4	40	5
KRISHNA 800	ICE	4	40	20
KRISHNA 400	ICE	4	40	10
KRISHNA 160	ICE	4	40	4
KRISHNA 160	ICE	5	40	4
KRISHNA 80	ICE	5	40	2
KRISHNA 4000	ICE	5	40	100
KRISHNA 120	ICE	5	40	3
KRISHNA 200	ICE	5	40	5
KRISHNA 800	ICE	5	40	20
KRISHNA 400	ICE	5	40	10
KRISHNA 160	ICE	5	40	4
TEJASREE 160	ICE	1	40	4
TEJASREE 80	ICE	1	40	2
TEJASREE 4000	ICE	1	40	100
TEJASREE 120	ICE	1	40	3
TEJASREE 200	ICE	1	40	5
TEJASREE 800	ICE	1	40	20
TEJASREE 400	ICE	1	40	10
TEJASREE 160	ICE	1	40	4
TEJASREE 160	ICE	2	40	4
TEJASREE 80	ICE	2	40	2
TEJASREE 4000	ICE	2	40	100
TEJASREE 120	ICE	2	40	3
TEJASREE 200	ICE	2	40	5
TEJASREE 800	ICE	2	40	20
TEJASREE 400	ICE	2	40	10
TEJASREE 160	ICE	2	40	4
TEJASREE 160	ICE	3	40	4
TEJASREE 80	ICE	3	40	2
TEJASREE 4000	ICE	3	40	100

TEJASREE	ICE	3	40	3
120				
TEJASREE	ICE	3	40	5
200				
TEJASREE	ICE	3	40	20
800				
TEJASREE	ICE	3	40	10
400				
TEJASREE	ICE	3	40	4
160				
TEJASREE	ICE	4	40	4
160				
TEJASREE	ICE	4	40	2
80				
TEJASREE	ICE	4	40	100
4000				
TEJASREE	ICE	4	40	3
120				
TEJASREE	ICE	4	40	5
200				
TEJASREE	ICE	4	40	20
800				
TEJASREE	ICE	4	40	10
400				
TEJASREE	ICE	4	40	4
160				
TEJASREE	ICE	5	40	4
160				
TEJASREE	ICE	5	40	2
80				
TEJASREE	ICE	5	40	100
4000				
TEJASREE	ICE	5	40	3
120				
TEJASREE	ICE	5	40	5
200				
TEJASREE	ICE	5	40	20
800				
TEJASREE	ICE	5	40	10
400				
TEJASREE	ICE	5	40	4
160				

360 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11* WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				

--				
SURESH 80	MILK	2	20	4
SURESH 40	MILK	2	20	2
SURESH 2000	MILK	2	20	100
SURESH 60	MILK	2	20	3
SURESH 100	MILK	2	20	5
SURESH 400	MILK	2	20	20
SURESH 200	MILK	2	20	10
SURESH 80	MILK	2	20	4
KRISHNA 80	MILK	2	20	4
KRISHNA 40	MILK	2	20	2
KRISHNA 2000	MILK	2	20	100
KRISHNA 60	MILK	2	20	3
KRISHNA 100	MILK	2	20	5
KRISHNA 400	MILK	2	20	20
KRISHNA 200	MILK	2	20	10
KRISHNA 80	MILK	2	20	4
TEJASREE 80	MILK	2	20	4
TEJASREE 40	MILK	2	20	2
TEJASREE 2000	MILK	2	20	100
TEJASREE 60	MILK	2	20	3
TEJASREE 100	MILK	2	20	5
TEJASREE 400	MILK	2	20	20
TEJASREE 200	MILK	2	20	10
TEJASREE 80	MILK	2	20	4
SURESH 120	CURD	3	30	4
SURESH 60	CURD	3	30	2
SURESH 3000	CURD	3	30	100
SURESH 90	CURD	3	30	3
SURESH 150	CURD	3	30	5
SURESH 600	CURD	3	30	20
SURESH 300	CURD	3	30	10

SURESH 120	CURD	3	30	4
KRISHNA 120	CURD	3	30	4
KRISHNA 60	CURD	3	30	2
KRISHNA 3000	CURD	3	30	100
KRISHNA 90	CURD	3	30	3
KRISHNA 150	CURD	3	30	5
KRISHNA 600	CURD	3	30	20
KRISHNA 300	CURD	3	30	10
KRISHNA 120	CURD	3	30	4
TEJASREE 120	CURD	3	30	4
TEJASREE 60	CURD	3	30	2
TEJASREE 3000	CURD	3	30	100
TEJASREE 90	CURD	3	30	3
TEJASREE 150	CURD	3	30	5
TEJASREE 600	CURD	3	30	20
TEJASREE 300	CURD	3	30	10
TEJASREE 120	CURD	3	30	4
SURESH 160	ICE	4	40	4
SURESH 80	ICE	4	40	2
SURESH 4000	ICE	4	40	100
SURESH 120	ICE	4	40	3
SURESH 200	ICE	4	40	5
SURESH 800	ICE	4	40	20
SURESH 400	ICE	4	40	10
SURESH 160	ICE	4	40	4
KRISHNA 160	ICE	4	40	4
KRISHNA 80	ICE	4	40	2
KRISHNA 4000	ICE	4	40	100
KRISHNA 120	ICE	4	40	3
KRISHNA 200	ICE	4	40	5
KRISHNA 800	ICE	4	40	20
KRISHNA 400	ICE	4	40	10

KRISHNA	ICE	4	40	4
160				
TEJASREE	ICE	4	40	4
160				
TEJASREE	ICE	4	40	2
80				
TEJASREE	ICE	4	40	100
4000				
TEJASREE	ICE	4	40	3
120				
TEJASREE	ICE	4	40	5
200				
TEJASREE	ICE	4	40	20
800				
TEJASREE	ICE	4	40	10
400				
TEJASREE	ICE	4	40	4
160				

72 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1* SELECT * FROM B26_CUST_ORD_DET
SQL> /

```

CID	PID	QTY	ORD_DATE
1	1	4	07-JUN-18
1	2	2	07-JUN-18
1		100	07-JUN-18
2	2	3	07-JUN-18
2	1	5	07-JUN-18
	1	20	07-JUN-18
2	1	10	08-JUN-18
3	1	4	07-JUN-18

8 rows selected.

SQL> CL SCR

SQL> R

```

1* SELECT * FROM B26_CUST_ORD_DET

```

CID	PID	QTY	ORD_DATE
1	1	4	07-JUN-18
1	2	2	07-JUN-18
1		100	07-JUN-18
2	2	3	07-JUN-18
2	1	5	07-JUN-18
	1	20	07-JUN-18
2	1	10	08-JUN-18
3	1	4	07-JUN-18

8 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1 SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME AS PRODNAME
3      ,PRG.GID AS PROD_GRADE

```

```

4      ,PR.PRICE AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7 FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11 WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12*    AND COD.PID = PR.PID
SQL> /

```

CUSTOMERNAME TOTALPRICE	PRODNAME	PROD_GRADE	PRICE	QTY
-----	-----	-----	-----	-----
--				
TEJASREE	MILK	2	20	4
80				
KRISHNA	MILK	2	20	4
80				
SURESH	MILK	2	20	4
80				
TEJASREE	CURD	3	30	2
60				
KRISHNA	CURD	3	30	2
60				
SURESH	CURD	3	30	2
60				
TEJASREE	CURD	3	30	3
90				
KRISHNA	CURD	3	30	3
90				
SURESH	CURD	3	30	3
90				
TEJASREE	MILK	2	20	5
100				
KRISHNA	MILK	2	20	5
100				
SURESH	MILK	2	20	5
100				
TEJASREE	MILK	2	20	20
400				
KRISHNA	MILK	2	20	20
400				
SURESH	MILK	2	20	20
400				
TEJASREE	MILK	2	20	10
200				
KRISHNA	MILK	2	20	10
200				
SURESH	MILK	2	20	10
200				
TEJASREE	MILK	2	20	4
80				
KRISHNA	MILK	2	20	4
80				
SURESH	MILK	2	20	4
80				

21 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT CU.CNAME AS CUSTOMERNAME

```

```

2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7 FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11* WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12 /

```

CUSTOMERNAME TOTALPRICE	PRODNAME	PROD_GRADE	PRICE	QTY
-----	-----	-----	-----	-----
--				
SURESH	MILK	2	20	4
80				
SURESH	MILK	2	20	2
40				
SURESH	MILK	2	20	100
2000				
SURESH	MILK	2	20	3
60				
SURESH	MILK	2	20	5
100				
SURESH	MILK	2	20	20
400				
SURESH	MILK	2	20	10
200				
SURESH	MILK	2	20	4
80				
KRISHNA	MILK	2	20	4
80				
KRISHNA	MILK	2	20	2
40				
KRISHNA	MILK	2	20	100
2000				
KRISHNA	MILK	2	20	3
60				
KRISHNA	MILK	2	20	5
100				
KRISHNA	MILK	2	20	20
400				
KRISHNA	MILK	2	20	10
200				
KRISHNA	MILK	2	20	4
80				
TEJASREE	MILK	2	20	4
80				
TEJASREE	MILK	2	20	2
40				
TEJASREE	MILK	2	20	100
2000				
TEJASREE	MILK	2	20	3
60				
TEJASREE	MILK	2	20	5
100				
TEJASREE	MILK	2	20	20
400				
TEJASREE	MILK	2	20	10
200				
TEJASREE	MILK	2	20	4
80				

SURESH 120	CURD	3	30	4
SURESH 60	CURD	3	30	2
SURESH 3000	CURD	3	30	100
SURESH 90	CURD	3	30	3
SURESH 150	CURD	3	30	5
SURESH 600	CURD	3	30	20
SURESH 300	CURD	3	30	10
SURESH 120	CURD	3	30	4
KRISHNA 120	CURD	3	30	4
KRISHNA 60	CURD	3	30	2
KRISHNA 3000	CURD	3	30	100
KRISHNA 90	CURD	3	30	3
KRISHNA 150	CURD	3	30	5
KRISHNA 600	CURD	3	30	20
KRISHNA 300	CURD	3	30	10
KRISHNA 120	CURD	3	30	4
TEJASREE 120	CURD	3	30	4
TEJASREE 60	CURD	3	30	2
TEJASREE 3000	CURD	3	30	100
TEJASREE 90	CURD	3	30	3
TEJASREE 150	CURD	3	30	5
TEJASREE 600	CURD	3	30	20
TEJASREE 300	CURD	3	30	10
TEJASREE 120	CURD	3	30	4
SURESH 160	ICE	4	40	4
SURESH 80	ICE	4	40	2
SURESH 4000	ICE	4	40	100
SURESH 120	ICE	4	40	3
SURESH 200	ICE	4	40	5
SURESH 800	ICE	4	40	20
SURESH 400	ICE	4	40	10
SURESH 160	ICE	4	40	4

KRISHNA 160	ICE	4	40	4
KRISHNA 80	ICE	4	40	2
KRISHNA 4000	ICE	4	40	100
KRISHNA 120	ICE	4	40	3
KRISHNA 200	ICE	4	40	5
KRISHNA 800	ICE	4	40	20
KRISHNA 400	ICE	4	40	10
KRISHNA 160	ICE	4	40	4
TEJASREE 160	ICE	4	40	4
TEJASREE 80	ICE	4	40	2
TEJASREE 4000	ICE	4	40	100
TEJASREE 120	ICE	4	40	3
TEJASREE 200	ICE	4	40	5
TEJASREE 800	ICE	4	40	20
TEJASREE 400	ICE	4	40	10
TEJASREE 160	ICE	4	40	4

72 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10 WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
11*   AND COD.PID  = PR.PID

```

SQL> /

```

SELECT CU.CNAME AS CUSTOMERNAME
      *

```

ERROR at line 1:

ORA-00904: "CU"."CNAME": invalid identifier

SQL> ED

Wrote file afiedt.buf

```

1  SELECT --CU.CNAME AS CUSTOMERNAME
2      PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY

```

```

6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7 FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10 WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
11*    AND COD.PID = PR.PID
SQL> /

```

PRODNAME	PROD_GRADE	PRICE	QTY	TOTALPRICE
MILK	2	20	4	80
CURD	3	30	2	60
CURD	3	30	3	90
MILK	2	20	5	100
MILK	2	20	20	400
MILK	2	20	10	200
MILK	2	20	4	80

7 rows selected.

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      PR.PNAME  AS PRODNAME
3      ,PRG.GID  AS PROD_GRADE
4      ,PR.PRICE AS PRICE
5      ,COD.QTY  AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7 FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11 WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12*    AND COD.PID = PR.PID
13 /
      PR.PNAME  AS PRODNAME
      *

```

ERROR at line 2:

ORA-00923: FROM keyword not found where expected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID  AS PROD_GRADE
4      ,PR.PRICE AS PRICE
5      ,COD.QTY  AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7 FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11 WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12*    AND COD.PID = PR.PID
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				

```

-----
--
TEJASREE      MILK      2      20      4
80
KRISHNA       MILK      2      20      4
80
SURESH        MILK      2      20      4
80
TEJASREE      CURD      3      30      2
60
KRISHNA       CURD      3      30      2
60
SURESH        CURD      3      30      2
60
TEJASREE      CURD      3      30      3
90
KRISHNA       CURD      3      30      3
90
SURESH        CURD      3      30      3
90
TEJASREE      MILK      2      20      5
100
KRISHNA       MILK      2      20      5
100
SURESH        MILK      2      20      5
100
TEJASREE      MILK      2      20      20
400
KRISHNA       MILK      2      20      20
400
SURESH        MILK      2      20      20
400
TEJASREE      MILK      2      20      10
200
KRISHNA       MILK      2      20      10
200
SURESH        MILK      2      20      10
200
TEJASREE      MILK      2      20      4
80
KRISHNA       MILK      2      20      4
80
SURESH        MILK      2      20      4
80

```

21 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME   AS PRODNAME
3      ,PR.GID    AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12*   AND COD.PID = PR.PID
SQL>
SQL> /

```

CUSTOMERNAME TOTALPRICE	PRODNAME	PROD_GRADE	PRICE	QTY

--				
TEJASREE 80	MILK	2	20	4
KRISHNA 80	MILK	2	20	4
SURESH 80	MILK	2	20	4
TEJASREE 60	CURD	3	30	2
KRISHNA 60	CURD	3	30	2
SURESH 60	CURD	3	30	2
TEJASREE 90	CURD	3	30	3
KRISHNA 90	CURD	3	30	3
SURESH 90	CURD	3	30	3
TEJASREE 100	MILK	2	20	5
KRISHNA 100	MILK	2	20	5
SURESH 100	MILK	2	20	5
TEJASREE 400	MILK	2	20	20
KRISHNA 400	MILK	2	20	20
SURESH 400	MILK	2	20	20
TEJASREE 200	MILK	2	20	10
KRISHNA 200	MILK	2	20	10
SURESH 200	MILK	2	20	10
TEJASREE 80	MILK	2	20	4
KRISHNA 80	MILK	2	20	4
SURESH 80	MILK	2	20	4

21 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2         ,PR.PNAME  AS PRODNAME
3         ,PRG.GID   AS PROD_GRADE
4         ,PR.PRICE  AS PRICE
5         ,COD.QTY   AS QTY
6         ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8         ,B26_PROD PR
9         ,B26_PROD_GRADE PRG
10        ,B26_CUST CU
11 WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE

```

```

12      AND COD.PID = PR.PID
13*      AND COD.CID = CU.CID
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				
-----	-----	-----	-----	-----
--				
SURESH	MILK	2	20	4
80				
SURESH	CURD	3	30	2
60				
KRISHNA	CURD	3	30	3
90				
KRISHNA	MILK	2	20	5
100				
KRISHNA	MILK	2	20	10
200				
TEJASREE	MILK	2	20	4
80				

6 rows selected.

```
SQL> CL SCR
```

```
SQL> R
```

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2         ,PR.PNAME AS PRODNAME
3         ,PRG.GID  AS PROD_GRADE
4         ,PR.PRICE AS PRICE
5         ,COD.QTY  AS QTY
6         ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8         ,B26_PROD PR
9         ,B26_PROD_GRADE PRG
10        ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12         AND COD.PID = PR.PID
13*      AND COD.CID = CU.CID

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				
-----	-----	-----	-----	-----
--				
SURESH	MILK	2	20	4
80				
SURESH	CURD	3	30	2
60				
KRISHNA	CURD	3	30	3
90				
KRISHNA	MILK	2	20	5
100				
KRISHNA	MILK	2	20	10
200				
TEJASREE	MILK	2	20	4
80				

6 rows selected.

```
SQL> SET AUTOTRACE ON
```

```
ERROR:
```

```
ORA-28002: the password will expire within 369 days
```

```

SP2-0619: Error while connecting
SP2-0611: Error enabling STATISTICS report
SQL> RR
SP2-0042: unknown command "RR" - rest of line ignored.
SQL> R

```

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME   AS PRODNAME
3      ,PRG.GID    AS PROD_GRADE
4      ,PR.PRICE   AS PRICE
5      ,COD.QTY    AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID = PR.PID
13*    AND COD.CID = CU.CID

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				
-----	-----	-----	-----	-----
--				
SURESH	MILK	2	20	4
80				
SURESH	CURD	3	30	2
60				
KRISHNA	CURD	3	30	3
90				
KRISHNA	MILK	2	20	5
100				
KRISHNA	MILK	2	20	10
200				
TEJASREE	MILK	2	20	4
80				

6 rows selected.

Execution Plan

Plan hash value: 3461138243

Id	Operation	Name	Rows	Bytes	Cost
0	SELECT STATEMENT		1	141	9
(12)	00:00:01				
1	NESTED LOOPS				
2	NESTED LOOPS		1	141	9
(12)	00:00:01				
* 3	HASH JOIN		1	116	8
(13)	00:00:01				
4	NESTED LOOPS		1	77	5
(0)	00:00:01				
5	TABLE ACCESS FULL	B26_PROD	3	114	2
(0)	00:00:01				
* 6	TABLE ACCESS FULL	B26_PROD_GRADE	1	39	1
(0)	00:00:01				

```

| 7 | TABLE ACCESS FULL | B26_CUST_ORD_DET | 8 | 312 | 2
(0)| 00:00:01 |
|* 8 | INDEX UNIQUE SCAN | B26CUST_CID_PK | 1 | | 0
(0)| 00:00:01 |
| 9 | TABLE ACCESS BY INDEX ROWID | B26_CUST | 1 | 25 | 1
(0)| 00:00:01 |
-----
-----

```

Predicate Information (identified by operation id):

```

3 - access("COD"."PID"="PR"."PID")
6 - filter("PR"."PRICE">="PRG"."LOPRICE" AND "PR"."PRICE"<="PRG"."HIPPRICE")
8 - access("COD"."CID"="CU"."CID")

```

Note

- dynamic sampling used for this statement (level=2)

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2         ,PR.PNAME AS PRODNAME
3         ,PRG.GID AS PROD_GRADE
4         ,PR.PRICE AS PRICE
5         ,COD.QTY AS QTY
6         ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8         ,B26_PROD PR
9         ,B26_PROD_GRADE PRG
10        ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12         AND COD.PID = PR.PID
13*        AND COD.CID = CU.CID

```

SQL>

SQL>

SQL>

SQL> SET AUTOTRACE OFF

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2         ,PR.PNAME AS PRODNAME
3         ,PRG.GID AS PROD_GRADE
4         ,PR.PRICE AS PRICE
5         ,COD.QTY AS QTY
6         ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8         ,B26_PROD PR
9         ,B26_PROD_GRADE PRG
10        ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12         AND COD.PID = PR.PID
13*        AND COD.CID = CU.CID

```

SQL> /

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				

```

-----
--
SURESH          MILK          2          20          4
80
SURESH          CURD          3          30          2
60
KRISHNA         CURD          3          30          3
90
KRISHNA         MILK          2          20          5
100
KRISHNA         MILK          2          20          10
200
TEJASREE        MILK          2          20          4
80

```

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME   AS PRODNAME
3      ,PRG.GID    AS PROD_GRADE
4      ,PR.PRICE   AS PRICE
5      ,COD.QTY    AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID = PR.PID
13*    AND COD.CID = CU.CID
SQL>
SQL> /

```

```

CUSTOMERNAME      PRODNAME      PROD_GRADE      PRICE      QTY
TOTALPRICE
-----
--
SURESH          MILK          2          20          4
80
SURESH          CURD          3          30          2
60
KRISHNA         CURD          3          30          3
90
KRISHNA         MILK          2          20          5
100
KRISHNA         MILK          2          20          10
200
TEJASREE        MILK          2          20          4
80

```

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME   AS PRODNAME
3      ,PRG.GID    AS PROD_GRADE
4      ,PR.PRICE   AS PRICE
5      ,COD.QTY    AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE

```



```

7 FROM B26_CUST_ORD_DET COD
8     ,B26_PROD PR
9     ,B26_PROD_GRADE PRG
10    ,B26_CUST CU
11 WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID = PR.PID(+)
13*    AND COD.CID = CU.CID
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				
----	-----	-----	-----	-----
--				
SURESH	MILK	2	20	4
80				
SURESH	CURD	3	30	2
60				
KRISHNA	CURD	3	30	3
90				
KRISHNA	MILK	2	20	5
100				
KRISHNA	MILK	2	20	10
200				
TEJASREE	MILK	2	20	4
80				

6 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT CU.CNAME AS CUSTOMERNAME
2     ,PR.PNAME AS PRODNAME
3     ,PRG.GID AS PROD_GRADE
4     ,PR.PRICE AS PRICE
5     ,COD.QTY AS QTY
6     ,COD.QTY * PR.PRICE AS TOTALPRICE
7 FROM B26_CUST_ORD_DET COD
8     ,B26_PROD PR
9     ,B26_PROD_GRADE PRG
10    ,B26_CUST CU
11 WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID = PR.PID(+)
13*    AND COD.CID = CU.CID(+)
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				
----	-----	-----	-----	-----
--				
SURESH	MILK	2	20	4
80				
SURESH	CURD	3	30	2
60				
KRISHNA	CURD	3	30	3
90				
KRISHNA	MILK	2	20	5
100				
	MILK	2	20	20
400				
KRISHNA	MILK	2	20	10
200				
TEJASREE	MILK	2	20	4
80				

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID = PR.PID
13*    AND COD.CID = CU.CID(+)
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				
--				
SURESH	MILK	2	20	4
80				
SURESH	CURD	3	30	2
60				
KRISHNA	CURD	3	30	3
90				
KRISHNA	MILK	2	20	5
100				
	MILK	2	20	20
400				
KRISHNA	MILK	2	20	10
200				
TEJASREE	MILK	2	20	4
80				

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID = PR.PID
13*    AND COD.CID = CU.CID
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				
--				

SURESH 80	MILK	2	20	4
SURESH 60	CURD	3	30	2
KRISHNA 90	CURD	3	30	3
KRISHNA 100	MILK	2	20	5
KRISHNA 200	MILK	2	20	10
TEJASREE 80	MILK	2	20	4

6 rows selected.

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID = PR.PID
13*    AND COD.CID = CU.CID

```

SQL>

SQL>

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CL SCR

```

SQL> SELECT *
2  FROM B26_CUST_ORD_DET;

```

CID	PID	QTY	ORD_DATE
1	1	4	07-JUN-18
1	2	2	07-JUN-18
1		100	07-JUN-18
2	2	3	07-JUN-18
2	1	5	07-JUN-18
	1	20	07-JUN-18
2	1	10	08-JUN-18
3	1	4	07-JUN-18

8 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR

```

```

9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11 WHERE ( PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE OR PR.PRICE NULL)
12     AND COD.PID = PR.PID(+)
13*    AND COD.CID = CU.CID
SQL> /
WHERE ( PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE OR PR.PRICE NULL)
      *
ERROR at line 11:
ORA-00920: invalid relational operator

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME AS PRODNAME
3      ,PRG.GID AS PROD_GRADE
4      ,PR.PRICE AS PRICE
5      ,COD.QTY AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11 WHERE ( ( PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE ) OR PR.PRICE NULL)
12     AND COD.PID = PR.PID(+)
13*    AND COD.CID = CU.CID
SQL> /
WHERE ( ( PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE ) OR PR.PRICE NULL)
      *
ERROR at line 11:
ORA-00920: invalid relational operator

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME AS PRODNAME
3      ,PRG.GID AS PROD_GRADE
4      ,PR.PRICE AS PRICE
5      ,COD.QTY AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11 WHERE NVL(PR.PRICE ,15) BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID = PR.PID(+)
13*    AND COD.CID = CU.CID
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TEJASREE	MILK	2	20	4
KRISHNA	MILK	2	20	10
KRISHNA	MILK	2	20	5

SURESH 80	MILK	2	20	4
KRISHNA 90	CURD	3	30	3
SURESH 60	CURD	3	30	2
SURESH		2		100

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID  = PR.PID(+)
13*    AND COD.CID  = CU.CID
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY	TOTALPRICE
--					
SURESH 80	MILK	2	20	4	
SURESH 60	CURD	3	30	2	
KRISHNA 90	CURD	3	30	3	
KRISHNA 100	MILK	2	20	5	
KRISHNA 200	MILK	2	20	10	
TEJASREE 80	MILK	2	20	4	

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID  = PR.PID(+)
13*    AND COD.CID  = CU.CID
SQL>

```

SQL> /

CUSTOMERNAME TOTALPRICE	PRODNAME	PROD_GRADE	PRICE	QTY
-----	-----	-----	-----	-----
--				
SURESH 80	MILK	2	20	4
SURESH 60	CURD	3	30	2
KRISHNA 90	CURD	3	30	3
KRISHNA 100	MILK	2	20	5
KRISHNA 200	MILK	2	20	10
TEJASREE 80	MILK	2	20	4

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE (PR.PRICE IS NULL OR (PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE ) )
12     AND COD.PID = PR.PID(+)
13*    AND COD.CID = CU.CID
SQL> /

```

CUSTOMERNAME TOTALPRICE	PRODNAME	PROD_GRADE	PRICE	QTY
-----	-----	-----	-----	-----
--				
TEJASREE 80	MILK	2	20	4
KRISHNA 200	MILK	2	20	10
KRISHNA 100	MILK	2	20	5
SURESH 80	MILK	2	20	4
KRISHNA 90	CURD	3	30	3
SURESH 60	CURD	3	30	2
SURESH		1		100
SURESH		2		100
SURESH		3		100
SURESH		4		100
SURESH		5		100

11 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME   AS PRODNAME
3      ,PRG.GID    AS PROD_GRADE
4      ,PR.PRICE   AS PRICE
5      ,COD.QTY    AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE (PR.PRICE IS NULL OR (PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE ) )
12     AND COD.PID = PR.PID(+)
13*    AND COD.CID = CU.CID
SQL>
SQL>
SQL>
SQL> CL SCR

```

```
SQL> CREATE TABLE B26_T1
```

```

2  (
3  ID NUMBER
4  )L
5

```

```
SQL>
```

```
SQL> CREATE TABLE B26_T1
```

```

2  (
3  ID NUMBER
4  );

```

```
CREATE TABLE B26_T1
```

```
*
```

ERROR at line 1:

ORA-00955: name is already used by an existing object

```
SQL> ED
```

Wrote file afiedt.buf

```

1  CREATE TABLE B26_T2
2  (
3  ID NUMBER
4* )

```

```
SQL> /
```

```
CREATE TABLE B26_T2
```

```
*
```

ERROR at line 1:

ORA-00955: name is already used by an existing object

```
SQL> ED
```

Wrote file afiedt.buf

```

1  CREATE TABLE B26_T3
2  (
3  ID NUMBER
4* )

```

```
SQL> /
```

```
CREATE TABLE B26_T3
```

```
*
```

ERROR at line 1:

ORA-00955: name is already used by an existing object

```
SQL> ED
Wrote file afiedt.buf

  1* DROP TABLE B26_T2
SQL> /
```

Table dropped.

```
SQL> ED
Wrote file afiedt.buf

  1 CREATE TABLE B26_T2
  2 (
  3 ID NUMBER
  4* )
SQL> /
```

Table created.

```
SQL> CL SCR
```

```
SQL> DESC B26_T2
```

Name	
Null?	Type

ID	NUMBER
----	--------

```
SQL> DESC B26_T1
```

Name	
Null?	Type

ID	NUMBER
NAME	VARCHAR2(20)
MOB	NUMBER(10)

```
SQL> DESC B26_T3
```

Name	
Null?	Type

ID	NUMBER
NAME	VARCHAR2(20)

```
SQL> CL SCR
```

```
SQL> DESC B26_T1
```

Name	
Null?	Type

ID	NUMBER
----	--------


```
NAME
VARCHAR2(20)
MOB
NUMBER(10)
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 CREATE TABLE B26_T2
2 (
3 ID NUMBER
4* )
```

```
SQL>
```

```
SQL> /
```

```
CREATE TABLE B26_T2
*
```

```
ERROR at line 1:
ORA-00955: name is already used by an existing object
```

```
SQL> DESC B26_T2
```

```
Name
```

```
Null?      Type
```

```
-----
-----
-----
```

```
ID
NUMBER
```

```
SQL> DESC B26_T1
```

```
Name
```

```
Null?      Type
```

```
-----
-----
-----
```

```
ID
NUMBER
```

```
NAME
```

```
VARCHAR2(20)
```

```
MOB
```

```
NUMBER(10)
```

```
SQL> DROP TABLE B26_T1;
```

```
Table dropped.
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 CREATE TABLE B26_T1
2 (
3 ID NUMBER
4* )
```

```
SQL> /
```

```
Table created.
```

```
SQL> CL SCR
```

```
SQL> DESC B26_T1
```

```
Name
```

```
Null?      Type
```

```

-----
-----
-----
ID
NUMBER

```

```
SQL> DESC B26_T2
```

```

Name
Null?    Type
-----
-----
-----

```

```

ID
NUMBER

```

```

SQL> INSERT INTO B26_T1
2  VALUES
3  (
4  1
5  );

```

```
1 row created.
```

```

SQL> R
1  INSERT INTO B26_T1
2  VALUES
3  (
4  1
5* )

```

```
1 row created.
```

```

SQL> R
1  INSERT INTO B26_T1
2  VALUES
3  (
4  1
5* )

```

```
1 row created.
```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  INSERT INTO B26_T2
2  VALUES
3  (
4  1
5* )
SQL> /

```

```
1 row created.
```

```

SQL> R
1  INSERT INTO B26_T2
2  VALUES
3  (
4  1
5* )

```

```
1 row created.
```

```
SQL> COMMIT;
```

Commit complete.

SQL> CL SCR

SQL> SELECT * FROM B26_T1;

```

          ID
-----
          1
          1
          1

```

SQL> SELECT * FROM B26_T2;

```

          ID
-----
          1
          1

```

```

SQL> SELECT T1.ID
2         T2.ID
3        FROM B26_T1
4         .
5

```

SQL>

SQL> ED

Wrote file afiedt.buf

```

1  SELECT T1.ID
2         T2.ID
3        FROM B26_T1 T1
4*       ,B26_T2 T2
SQL> /
          T2.ID
          *

```

ERROR at line 2:

ORA-00923: FROM keyword not found where expected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT T1.ID
2         T2.ID
3        FROM B26_T1 T1
4*       ,B26_T2 T2
SQL> /
          T2.ID
          *

```

ERROR at line 2:

ORA-00923: FROM keyword not found where expected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT T1.ID
2         ,T2.ID
3        FROM B26_T1 T1
4*       ,B26_T2 T2
SQL> /

```

```

          ID          ID
-----

```

```
SQL> ED
Wrote file afiedt.buf
```

ID	ID
-----	-----

1	1
1	1
1	1
1	1
1	1
1	1
1	2
1	2
1	2
1	2
1	2
1	2

12 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT T1.ID
2      ,T2.ID
3  FROM B26_T1 T1
4      ,B26_T2 T2
5*  WHERE T1.ID = T2.ID
SQL> /

```

ID	ID
1	1
1	1
1	1
1	1
1	1
1	1

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT *
2*  FROM DEPT
SQL> /

```

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME AS PRODNAME
3      ,PRG.GID AS PROD_GRADE
4      ,PR.PRICE AS PRICE
5      ,COD.QTY AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE (PR.PRICE IS NULL OR (PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE ) )

```

```

12      AND COD.PID  = PR.PID(+)
13*      AND COD.CID = CU.CID
14  /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				
----	-----	-----	-----	-----
--				
TEJASREE	MILK	2	20	4
80				
KRISHNA	MILK	2	20	10
200				
KRISHNA	MILK	2	20	5
100				
SURESH	MILK	2	20	4
80				
KRISHNA	CURD	3	30	3
90				
SURESH	CURD	3	30	2
60				
SURESH		1		100
SURESH		2		100
SURESH		3		100
SURESH		4		100
SURESH		5		100

11 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2         ,PR.PNAME AS PRODNAME
3         ,PRG.GID  AS PROD_GRADE
4         ,PR.PRICE AS PRICE
5         ,COD.QTY  AS QTY
6         ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8         ,B26_PROD PR
9         ,B26_PROD_GRADE PRG
10        ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12      AND COD.PID  = PR.PID(+)
13*      AND COD.CID = CU.CID
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				
----	-----	-----	-----	-----
--				
SURESH	MILK	2	20	4
80				
SURESH	CURD	3	30	2
60				
KRISHNA	CURD	3	30	3
90				
KRISHNA	MILK	2	20	5
100				
KRISHNA	MILK	2	20	10
200				
TEJASREE	MILK	2	20	4
80				

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE NVL(PR.PRICE,15) BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID = PR.PID(+)
13*    AND COD.CID = CU.CID
14  /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				
TEJASREE	MILK	2	20	4
80				
KRISHNA	MILK	2	20	10
200				
KRISHNA	MILK	2	20	5
100				
SURESH	MILK	2	20	4
80				
KRISHNA	CURD	3	30	3
90				
SURESH	CURD	3	30	2
60				
SURESH		2		100

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE( ( PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE )
12         OR PR.PRICE IS NULL
13        )
14     AND COD.PID = PR.PID(+)
15*    AND COD.CID = CU.CID
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
TOTALPRICE				
TEJASREE	MILK	2	20	4
80				
KRISHNA	MILK	2	20	10
200				
KRISHNA	MILK	2	20	5
100				
SURESH	MILK	2	20	4
80				
KRISHNA	CURD	3	30	3
90				
SURESH	CURD	3	30	2
60				
SURESH		2		100

TEJASREE 80	MILK	2	20	4
KRISHNA 200	MILK	2	20	10
KRISHNA 100	MILK	2	20	5
SURESH 80	MILK	2	20	4
KRISHNA 90	CURD	3	30	3
SURESH 60	CURD	3	30	2
SURESH		1		100
SURESH		2		100
SURESH		3		100
SURESH		4		100
SURESH		5		100

11 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME  AS PRODNAME
3      ,PRG.GID   AS PROD_GRADE
4      ,PR.PRICE  AS PRICE
5      ,COD.QTY   AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE( ( PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE )
12         OR ( PR.PRICE IS NULL AND COD.PID IS NULL)
13         )
14  AND COD.PID = PR.PID(+)
15*  AND COD.CID = CU.CID
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY	TOTALPRICE
TEJASREE 80	MILK	2	20	4	
KRISHNA 200	MILK	2	20	10	
KRISHNA 100	MILK	2	20	5	
SURESH 80	MILK	2	20	4	
KRISHNA 90	CURD	3	30	3	
SURESH 60	CURD	3	30	2	
SURESH		1		100	
SURESH		2		100	
SURESH		3		100	
SURESH		4		100	
SURESH		5		100	

11 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME   AS PRODNAME
3      ,PRG.GID    AS PROD_GRADE
4      ,PR.PRICE   AS PRICE
5      ,COD.QTY    AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE( ( PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE )
12         )
13     AND COD.PID = PR.PID(+)
14*    AND COD.CID = CU.CID

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME   AS PRODNAME
3      ,PRG.GID    AS PROD_GRADE
4      ,PR.PRICE   AS PRICE
5      ,COD.QTY    AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11  WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID = PR.PID
13*    AND COD.CID = CU.CID

```

SQL> /

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY
SURESH	MILK	2	20	4
SURESH	CURD	3	30	2
KRISHNA	CURD	3	30	3
KRISHNA	MILK	2	20	5
KRISHNA	MILK	2	20	10
TEJASREE	MILK	2	20	4

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME   AS PRODNAME
3      ,PRG.GID    AS PROD_GRADE
4      ,PR.PRICE   AS PRICE
5      ,COD.QTY    AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD

```

```

8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11 WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12     AND COD.PID = PR.PID
13*    AND COD.CID = CU.CID
SQL>
SQL> /

```

CUSTOMERNAME	PRODNAME	PROD_GRADE	PRICE	QTY	TOTALPRICE
-----	-----	-----	-----	-----	-----
--					
SURESH	MILK	2	20	4	
80					
SURESH	CURD	3	30	2	
60					
KRISHNA	CURD	3	30	3	
90					
KRISHNA	MILK	2	20	5	
100					
KRISHNA	MILK	2	20	10	
200					
TEJASREE	MILK	2	20	4	
80					

6 rows selected.

SQL> CL SCR

SQL> ED

Wrote file afiedt.buf

```

1  SELECT CU.CNAME AS CUSTOMERNAME
2      ,PR.PNAME AS PRODNAME
3      ,PRG.GID AS PROD_GRADE
4      ,PR.PRICE AS PRICE
5      ,COD.QTY AS QTY
6      ,COD.QTY * PR.PRICE AS TOTALPRICE
7  FROM B26_CUST_ORD_DET COD
8      ,B26_PROD PR
9      ,B26_PROD_GRADE PRG
10     ,B26_CUST CU
11 WHERE PR.PRICE BETWEEN PRG.LOPRICE AND PRG.HIPRICE
12*    AND COD.PID = PR.PID
13 CL SCR
14
SQL>
SQL> \.
SP2-0042: unknown command "\." - rest of line ignored.
SQL>
SQL> spool off

```

11-Jun-2018 Joins.2 Notes

CROSS JOIN

EQUI JOIN OR INNER JOIN OR SIMPLE JOIN

OUTER JOINS

LOJ

ROJ

FOJ--

NON EQUI JOIN

SELF JOINS--

SELF RELATION

PK FK

PARENT COLUMN WILL BE THERE IN ONE TABLE

CHILD COLUMN TWILL BE THERE IN TABLE

BUT IN SELF RELATION BOHT PARENT AND CHILD COLUMN WILL BE THERE SAME TABLE

STUDENT

SID	SNAME	RID
1	KRISHNA	NULL
2	SURESH	1
3	TEJASREE	1
4	HIMANSHU	3
5	ABHISHEK	4

1) WRITE A SELECT STATEMENT TO GET BELOW OUTPUT?

STUDENTNAME	WHOREFERED/REFERED
KRISHNA	NO ONE
SURESH	KRISHNA
TEJASREE	KRISHNA
HIMANSHU	TEJASREE
ABHISHEK	HIMANSHU

```
SELECT SNAME
      ,RID AS REFEREDBY
FROM B26_STD_SR
WHERE RID = SID
```

- 1) WHENEVER YOU HAVE TO WRITE JOIN CONDITIONS ON SELF RELATION TABLE
YOU HAVE TO TAKE SAME TABLE TWICE IN FROM CLAUSE WITH DIFFERENT ALIAS NAME

```
FROM B26_STD_SR A
      ,B26_STD_SR B
```

- 2) TAKE A PRIMARY KEY COLUMN FROM ONE ALIAS AND FOREIGN KEY COLUMN FROM ANOTHER ALIAS
AND MAKE A JOIN CONDITION

```
FROM B26_STD_SR A

,B26_STD_SR B

WHERE A.RID = B.ID
```

3) IDENTIFY FY PARENT AND CHILD IN YOUR BUSINESS REQUIREMENT

```
STUDENTNAME--CHILD

REFEREDBY-- PARENT
```

```
EMPLOYEE--CHILD

MANAGERNAME-- PARENT
```

```
LAPTOP ---PARENT

MOUSE --CHILD

BAG --CHILD
```

4) FROM PRIMARY KEY COLUMN ALIAS TABLE WHATEVER YOU SELECT THAT WILL ACT AS PARENT DETAILS

FROM FOREIGN KEY COLUMN ALIAS TABLE WHATEVEVE YOU SELECT THAT WILL ACT AS CHILD DETAILS.

```
SELECT A.SNAME AS STUDENTNAME

,B.SNAME AS REFEREDBY

FROM B26_STD_SR A
```

```
,B26_STD_SR B
WHERE A.RID = B.ID
```

ANSI JOINS

B26_T1

ID

1
2
3
4
1
1
2
3
2

B26_T2

ID

4
3
2
1
1
1
2
5

EMPLOYEE	MANAGER
----------	---------

NAME	NAME
------	------

KING	NULL
------	------

BLAKE	KING
-------	------

CLARK	KING
-------	------

.	
---	--

.	
---	--

.	
---	--

.	
---	--

.	
---	--

.	
---	--

MILLER	CLARK
--------	-------

WRITE A SELECT STATEMENT TO GET LIST OF EMPLOYEES WHO ARE EARNING MORE SALARY THAN

THEIR MANAGERS

11-Jun-2018 Joins.2 Examples

```
SQL>
```

```
SQL>
```

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B26_Y1;
```

```
SELECT * FROM B26_Y1
```

```
      *
```

```
ERROR at line 1:
```

```
ORA-00942: table or view does not exist
```

```
SQL> SELECT * FROM B26_T1;
```

```
      ID
```

```
-----
```

```
1
```

```
2
```

```
3
```

```
4
```

```
1
```

```
1
```

```
2
```

3
2

9 rows selected.

```
SQL> SELECT * FROM B26_T2;
```

```
      ID
-----
      4
      3
      2
      1
      1
      1
      2
      5
```

8 rows selected.

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B26_T1;
```

```
      ID
-----
      1
      2
      3
      4
      1
      1
      2
      3
      2
```

9 rows selected.

```
SQL> SELECT * FROM B26_T2;
```

```
      ID
-----
      4
      3
      2
      1
      1
      1
      2
      5
```

8 rows selected.

```
SQL> SELECT T1.ID
      2      ,T2.ID
      3      FROM B26_T1
      4      .
      5
```

```
SQL>
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT T1.ID
2      ,T2.ID
```


3 FROM B26_T1 T1
4* ,B26_T2 T2
5 /

ID	ID
1	4
2	4
3	4
4	4
1	4
1	4
2	4
3	4
2	4
1	3
2	3

ID	ID
3	3
4	3
1	3
1	3
2	3
3	3
2	3
1	2
2	2
3	2
4	2

ID	ID
1	2
1	2
2	2
3	2
2	2
1	1
2	1
3	1
4	1
1	1
1	1

ID	ID
2	1
3	1
2	1
1	1
2	1
3	1
4	1
1	1
1	1
2	1
3	1

ID	ID
2	1
1	1

2	1
3	1
4	1
1	1
1	1
2	1
3	1
2	1
1	2

ID	ID
2	2
3	2
4	2
1	2
1	2
2	2
3	2
2	2
1	5
2	5
3	5

ID	ID
4	5
1	5
1	5
2	5
3	5
2	5

72 rows selected.

SQL> SET PAGESIZE 200

SQL> R

```

1  SELECT T1.ID
2      ,T2.ID
3  FROM B26_T1 T1
4*      ,B26_T2 T2

```

ID	ID
1	4
2	4
3	4
4	4
1	4
1	4
2	4
3	4
2	4
1	3
2	3
3	3
4	3
1	3
1	3
2	3
3	3
2	3
1	2
2	2

3	2
4	2
1	2
1	2
2	2
3	2
2	2
1	1
2	1
3	1
4	1
1	1
1	1
2	1
3	1
2	1
1	1
2	1
3	1
4	1
1	1
1	1
2	1
3	1
2	1
1	1
2	1
3	1
4	1
1	1
1	1
2	1
3	1
2	1
1	2
2	2
3	2
4	2
1	2
1	2
2	2
3	2
2	2
1	5
2	5
3	5
4	5
1	5
1	5
2	5
3	5
2	5

72 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT T1.ID
2      ,T2.ID
3  FROM B26_T1 T1
4      ,B26_T2 T2
5*  WHERE T1.ID = T2.ID
SQL>
```

```
SQL>
```

```
SQL> SELECT * FROM B26_T1;
```

ID
1
2
3
4
1
1
2
3
2

```
9 rows selected.
```

```
SQL> SELECT * FROM B26_T2;
```

ID
4
3
2
1
1
1
2
5

```
8 rows selected.
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1      SELECT T1.ID
2          ,T2.ID
3      FROM B26_T1 T1
4          ,B26_T2 T2
5*     WHERE T1.ID = T2.ID
6      /
```

ID	ID
1	1
1	1
1	1
2	2
2	2
3	3
4	4
1	1
1	1
1	1
1	1
1	1
1	1
1	1
2	2
2	2
3	3
2	2
2	2

```
18 rows selected.
```

```
SQL> ED
Wrote file afiedt.buf

  1      SELECT T1.ID
  2          ,T2.ID
  3      FROM B26_T1 T1
  4          ,B26_T2 T2
  5      WHERE T1.ID = T2.ID
  6*      AND T1.ID =1
SQL> /
```

ID	ID
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1

9 rows selected.

```
SQL> CL SCR
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```
  1      SELECT T1.ID
  2          ,T2.ID
  3      FROM B26_T1 T1
  4          ,B26_T2 T2
  5      WHERE T1.ID = T2.ID
  6*      AND T1.ID =1
```

```
SQL>
```

```
SQL>
```

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B26_T1;
```

ID
1
2
3
4
1
1
2
3
2

9 rows selected.

```
SQL> SELECT * FROM B26_T2;
```

ID
4
3
2

1
1
1
2
5

8 rows selected.

SQL> ED

Wrote file afiedt.buf

SP2-0223: No lines in SQL buffer.

SQL>

SQL> CL SCR

SQL> SELECT * FROM B26_T1 WHERE ID =1;

ID
1
1
1

SQL> SELECT * FROM B26_T2;

ID
4
3
2
1
1
1
1
2
5

8 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1      SELECT T1.ID
2          ,T2.ID
3      FROM B26_T1 T1
4          ,B26_T2 T2
5      WHERE T1.ID = T2.ID
6*      AND T1.ID =1
7      /

```

ID	ID
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1
1	1

9 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1      SELECT T1.ID
2          ,T2.ID
3      FROM B26_T1 T1
4          ,B26_T2 T2
5*     WHERE T1.ID = T2.ID
6  /
```

ID	ID
1	1
1	1
1	1
2	2
2	2
3	3
4	4
1	1
1	1
1	1
1	1
1	1
1	1
1	1
2	2
2	2
3	3
2	2
2	2

18 rows selected.

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CL SCR

SQL> SELECT * FROM B26_T1;

ID
1
2
3
4
1
1
2
3
2

9 rows selected.

SQL> SELECT * FROM B26_T2;

ID
4
3
2
1
1
1
2
5

8 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT T1.ID
2      ,T2.ID
3  FROM B26_T1 T1
4      ,B26_T2 T2
5* WHERE T1.ID <> T2.ID
SQL> /
```

ID	ID
1	4
2	4
3	4
1	4
1	4
2	4
3	4
2	4
1	3
2	3
4	3
1	3
1	3
2	3
2	3
1	2
3	2
4	2
1	2
1	2
3	2
2	1
3	1
4	1
2	1
3	1
2	1
2	1
3	1
4	1
2	1
3	1
2	1
2	1
3	1
4	1
2	1
3	1
2	1
1	2
3	2
4	2
1	2
1	2
3	2
1	5
2	5
3	5
4	5

1	5
1	5
2	5
3	5
2	5

54 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT T1.ID
2      ,T2.ID
3  FROM B26_T1 T1
4      ,B26_T2 T2
5  WHERE T1.ID <> T2.ID
6* AND T1.ID =1
SQL> /

```

ID	ID
1	4
1	3
1	2
1	2
1	5
1	4
1	3
1	2
1	2
1	5
1	4
1	3
1	2
1	2
1	5

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DISTINCT T1.ID
2      ,T2.ID
3  FROM B26_T1 T1
4      ,B26_T2 T2
5  WHERE T1.ID <> T2.ID
6* AND T1.ID =1
SQL> /

```

ID	ID
1	4
1	2
1	3
1	5

SQL> ED

Wrote file afiedt.buf

```

1  SELECT T1.ID
2      ,T2.ID
3  FROM B26_T1 T1
4      ,B26_T2 T2

```

```

5 WHERE T1.ID <> T2.ID
6* AND T1.ID =1
SQL> /

```

ID	ID
1	4
1	3
1	2
1	2
1	5
1	4
1	3
1	2
1	2
1	5
1	4
1	3
1	2
1	2
1	5

15 rows selected.

```
SQL> CL SCR
```

```
SQL> SELECT * FROM B26_T1;
```

ID
1
2
3
4
1
1
2
3
2

9 rows selected.

```
SQL> SELECT * FROM B26_T2;
```

ID
4
3
2
1
1
1
2
5

8 rows selected.

```

SQL> SELECT T1.ID
2      ,T2.ID
3      FROM B26_T1 T1
4      ,B26_T2 T2
5      WHERE T1.ID = T2.ID;

```

ID	ID
----	----

```

-----
      1      1
      1      1
      1      1
      2      2
      2      2
      3      3
      4      4
      1      1
      1      1
      1      1
      1      1
      1      1
      1      1
      2      2
      2      2
      3      3
      2      2
      2      2

```

18 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT T1.ID AS T1ID
 2      ,T2.ID AS T2ID
 3      FROM B26_T1 T1
 4      ,B26_T2 T2
 5*  WHERE T1.ID(+) = T2.ID
SQL> /

```

```

      T1ID      T2ID
-----
      1      1
      1      1
      1      1
      2      2
      2      2
      3      3
      4      4
      1      1
      1      1
      1      1
      1      1
      1      1
      1      1
      1      1
      2      2
      2      2
      3      3
      2      2
      2      2
           5

```

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT T1.ID AS T1ID
 2      ,T2.ID AS T2ID
 3      FROM B26_T1 T1
 4      ,B26_T2 T2
 5  WHERE T1.ID(+) = T2.ID

```

```

6*      AND T1.ID IS NULL
SQL> /

```

T1ID	T2ID
-----	-----
	5

```
SQL> CL SCR
```

```
SQL> SELECT *
      2  FROM B26_T1;
```

ID

1
2
3
4
1
1
2
3
2

```
9 rows selected.
```

```
SQL> SELECT *
      2  FROM B26_T2;
```

ID

4
3
2
1
1
1
2
5

```
8 rows selected.
```

```
SQL> EDI
Wrote file afiedt.buf
```

```

1  SELECT T1.ID
2      ,T2.ID
3  FROM B26_T1 T1
4      ,B26_T2 T2
5* WHERE T1.ID > T2.ID
SQL> /

```

ID	ID
-----	-----
2	1
2	1
2	1
3	1
3	1
4	1
2	1
2	1
2	1
3	1

3	1
4	1
2	1
2	1
2	1
3	1
3	1
4	1
3	2
3	2
4	2
3	2
3	2
4	2
4	3

25 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT T1.ID
2      ,T2.ID
3  FROM B26_T1 T1
4      ,B26_T2 T2
5* WHERE T1.ID > T2.ID
SQL> /

```

ID	ID
2	1
2	1
2	1
3	1
3	1
4	1
2	1
2	1
2	1
3	1
3	1
4	1
2	1
2	1
2	1
3	1
3	1
4	1
3	2
3	2
4	2
3	2
3	2
4	2
4	3

25 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT T1.ID
2      ,T2.ID
3  FROM B26_T1 T1

```

```

4      ,B26_T2 T2
5* WHERE T1.ID = T2.ID
SQL> /

```

ID	ID
1	1
1	1
1	1
2	2
2	2
3	3
4	4
1	1
1	1
1	1
1	1
1	1
2	2
2	2
3	3
2	2
2	2

18 rows selected.

```
SQL> CL SCR
```

```

SQL> CREATE TABLE B26_STD_SR
2  (
3  SID NUMBER
4  ,SNAME VARCHAR2(20)
5  ,RID  NUMBER
6  ,CONSTRAINT B26STDSR_SID_PK PRIMARY KEY(SID)
7  ,CONSTRAINT B26STDSR_RID_FK FOREIGN KEY(RID) REFERENCES B26_STD_SR(SID)
8
SQL> .
SQL>
SQL> ED
Wrote file afiedt.buf

```

```

1  CREATE TABLE B26_STD_SR
2  (
3  SID NUMBER CONSTRAINT B26STDSR_SID_PK PRIMARY KEY
4  ,SNAME VARCHAR2(20)
5  ,RID  NUMBER CONSTRAINT B26STDSR_RID_FK REFERENCES B26_STD_SR(SID)
6* )
7  /

```

Table created.

```

SQL> INSERT INTO B26_STD_SR
2  VALUES
3  (
4  1
5  , 'KRISHNA'
6  , NULL
7  );

```

1 row created.

```

SQL> ED
Wrote file afiedt.buf

```

```
1  INSERT INTO B26_STD_SR
2  VALUES
3  (
4  2
5  , 'SURESH'
6  , 1
7* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_STD_SR
2  VALUES
3  (
4  3
5  , 'TEJASREE'
6  , 1
7* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_STD_SR
2  VALUES
3  (
4  4
5  , 'HIMANSHU'
6  , 3
7* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO B26_STD_SR
2  VALUES
3  (
4  5
5  , 'ABHISHEK'
6  , 4
7* )
SQL> /
```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT *
2  FROM B26_STD_SR;
```

SID SNAME

RID

```

-----
      1 KRISHNA
      2 SURESH                      1
      3 TEJASREE                    1
      4 HIMANSHU                    3
      5 ABHISHEK                    4

```

```

SQL> SELECT SNAME
      2      ,RID AS REFEREDBY
      3 FROM B26_STD_SR
      4 WHERE RID = SID
      5

```

```
SQL>
```

```
SQL>
```

```

SQL> SELECT SNAME AS STUDENTNAME
      2      ,RID
      3

```

```
SQL>
```

```
SQL> CL SCR
```

```

SQL> SELECT *
      2 FROM B26_STD_SR;

```

```

      SID SNAME                      RID
-----
      1 KRISHNA
      2 SURESH                      1
      3 TEJASREE                    1
      4 HIMANSHU                    3
      5 ABHISHEK                    4

```

```

SQL> SELECT SNAME
      2      ,RID AS REFEREDBY
      3 FROM B26_STD_SR
      4 WHERE RID = SID;

```

no rows selected

```
SQL> EDIT
```

Wrote file afiedt.buf

```

      1 SELECT SNAME
      2      ,RID AS REFEREDBY
      3 FROM B26_STD_SR
      4*  WHERE RID = SID

```

```
SQL>
```

```
SQL>
```

```

SQL> INSERT INTO B26_STD_SR
      2 VALUES
      3 (
      4 6
      5 , 'ARJUN'
      6 , 6
      7 );

```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT *
```



```
2 FROM B26_STD_SR;
```

SID	SNAME	RID
1	KRISHNA	
2	SURESH	1
3	TEJASREE	1
4	HIMANSHU	3
5	ABHISHEK	4
6	ARJUN	6

6 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT SNAME
2      ,RID AS REFEREDBY
3 FROM B26_STD_SR
4*  WHERE RID = SID
SQL> /
```

SNAME	REFEREDBY
ARJUN	6

```
SQL> CL SCR
```

```
SQL> DESC B26_STD_SR
```

Name	Null?	Type
SID	NOT NULL	NUMBER
SNAME		VARCHAR2(20)
RID		NUMBER

```
SQL> SELECT A.SNAME AS STUDENTNAME
2      ,B.SNAME AS REFEREDBY
3 FROM B26_STD_SR A
4      ,B26_STD_SR B
5 WHERE A.RID = B.ID
6 ;
      WHERE A.RID = B.ID
              *
```

```
ERROR at line 5:
ORA-00904: "B"."ID": invalid identifier
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT A.SNAME AS STUDENTNAME
2      ,B.SNAME AS REFEREDBY
3 FROM B26_STD_SR A
4      ,B26_STD_SR B
5*  WHERE A.RID = B.SID
6 /
```

STUDENTNAME	REFEREDBY
TEJASREE	KRISHNA
SURESH	KRISHNA
HIMANSHU	TEJASREE
ABHISHEK	HIMANSHU
ARJUN	ARJUN

```
SQL> ED
Wrote file afiedt.buf
```

```

1      SELECT A.SNAME AS STUDENTNAME
2          ,B.SNAME  AS REFEREDBY
3      FROM B26_STD_SR  A
4          ,B26_STD_SR  B
5*      WHERE A.RID  = B.SID(+)
SQL> /
```

STUDENTNAME	REFEREDBY
TEJASREE	KRISHNA
SURESH	KRISHNA
HIMANSHU	TEJASREE
ABHISHEK	HIMANSHU
ARJUN	ARJUN
KRISHNA	

6 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1      SELECT A.SNAME AS STUDENTNAME
2          ,NVL( B.SNAME , 'NO ONE ' ) AS REFEREDBY
3      FROM B26_STD_SR  A
4          ,B26_STD_SR  B
5*      WHERE A.RID  = B.SID(+)
SQL> /
```

STUDENTNAME	REFEREDBY
TEJASREE	KRISHNA
SURESH	KRISHNA
HIMANSHU	TEJASREE
ABHISHEK	HIMANSHU
ARJUN	ARJUN
KRISHNA	NO ONE

6 rows selected.

```
SQL> CL SCR
```

```
SQL> SELECT *
2  FROM B26_STD_SR;
```

SID	SNAME	RID
1	KRISHNA	
2	SURESH	1
3	TEJASREE	1
4	HIMANSHU	3
5	ABHISHEK	4
6	ARJUN	6

6 rows selected.

```
SQL> CL SCR
```

```
SQL>
SQL>
SQL> ED
```

Wrote file afiedt.buf

```

1  SELECT A.SNAME AS STUDENAME
2      ,B.SNAME AS REFEREDBY
3  FROM B26_STD_SR  A
4      ,B26_std_SR B
5*   WHERE A.RID   = B.SID
SQL> /

```

STUDENAME	REFEREDBY
TEJASREE	KRISHNA
SURESH	KRISHNA
HIMANSHU	TEJASREE
ABHISHEK	HIMANSHU
ARJUN	ARJUN

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT A.SNAME AS STUDENAME
2      ,B.SNAME AS REFEREDBY
3  FROM B26_STD_SR  A
4      ,B26_std_SR B
5*   WHERE A.RID   = B.SID
SQL>
SQL>
SQL> DELETE FROM B26_STD_SR WHERE ID =6;
DELETE FROM B26_STD_SR WHERE ID =6
                                *

```

ERROR at line 1:
ORA-00904: "ID": invalid identifier

SQL> DELETE FROM B26_STD_SR WHERE SID =6;

1 row deleted.

SQL> COMMIT;

Commit complete.

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT A.SNAME AS STUDENAME
2      ,B.SNAME AS REFEREDBY
3  FROM B26_STD_SR  A
4      ,B26_std_SR B
5*   WHERE A.RID   = B.SID
SQL> /

```

STUDENAME	REFEREDBY
TEJASREE	KRISHNA
SURESH	KRISHNA
HIMANSHU	TEJASREE
ABHISHEK	HIMANSHU

SQL> ED

Wrote file afiedt.buf

```

1  SELECT B.SNAME AS REFEREDBY
2  FROM B26_STD_SR  A
3       ,B26_std_SR B
4*  WHERE A.RID  = B.SID
SQL> /

```

REFEREDBY

KRISHNA
KRISHNA
TEJASREE
HIMANSHU

SQL> ED

Wrote file afiedt.buf

```

1  SELECT B.SNAME AS REFEREDBY
2       ,COUNT(1) AS TOTCNT
3  FROM B26_STD_SR  A
4       ,B26_std_SR B
5  WHERE A.RID  = B.SID
6*  GROUP BY B.SNAME
SQL> /

```

REFEREDBY	TOTCNT
-----	-----
KRISHNA	2
TEJASREE	1
HIMANSHU	1

SQL> ED

Wrote file afiedt.buf

```

1  SELECT B.SNAME AS REFEREDBY
2       ,COUNT(1) AS TOTCNT
3  FROM B26_STD_SR  A
4       ,B26_std_SR B
5  WHERE A.RID  = B.SID(+)
6*  GROUP BY B.SNAME
SQL> /

```

REFEREDBY	TOTCNT
-----	-----
	1
KRISHNA	2
TEJASREE	1
HIMANSHU	1

SQL> ED

Wrote file afiedt.buf

```

1  SELECT B.SNAME AS REFEREDBY
2       ,COUNT(1) AS TOTCNT
3  FROM B26_STD_SR  A
4       ,B26_std_SR B
5  WHERE A.RID(+)  = B.SID
6*  GROUP BY B.SNAME
SQL> /

```

REFEREDBY	TOTCNT
-----	-----
KRISHNA	2

```

TEJASREE          1
HIMANSHU          1
ABHISHEK          1
SURESH            1

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT B.SNAME AS REFEREDBY
2      ,COUNT(1) AS TOTCNT
3  FROM B26_STD_SR A
4      ,B26_std_SR B
5  WHERE A.RID = B.SID
6* GROUP BY B.SNAME
SQL> /

```

REFEREDBY	TOTCNT
KRISHNA	2
TEJASREE	1
HIMANSHU	1

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT B.SNAME AS REFEREDBY
2      ,COUNT(1) AS TOTCNT
3  FROM B26_STD_SR A
4      ,B26_std_SR B
5  WHERE A.RID = B.SID
6* GROUP BY B.SNAME
SQL>
SQL> /

```

REFEREDBY	TOTCNT
KRISHNA	2
TEJASREE	1
HIMANSHU	1

SQL> ED

Wrote file afiedt.buf

```

1  SELECT A.SNAME AS STUDENTNAME
2      ,B.SNAME AS REFEREDBY
3  FROM B26_STD_SR A
4      ,B26_std_SR B
5*  WHERE A.RID = B.SID
6  /

```

STUDENTNAME	REFEREDBY
TEJASREE	KRISHNA
SURESH	KRISHNA
HIMANSHU	TEJASREE
ABHISHEK	HIMANSHU

SQL> ED

Wrote file afiedt.buf

```

1  SELECT A.SNAME AS STUDENTNAME
2      ,B.SNAME AS REFEREDBY

```

```

3 FROM B26_STD_SR A
4      ,B26_std_SR B
5* WHERE A.RID(+) = B.SID
SQL> /

```

STUDENTNAME	REFEREDBY
SURESH	KRISHNA
TEJASREE	KRISHNA
HIMANSHU	TEJASREE
ABHISHEK	HIMANSHU
	ABHISHEK
	SURESH

6 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT B.SNAME AS REFEREDBY
2      ,COUNT(1) AS TOTCNT
3 FROM B26_STD_SR A
4      ,B26_std_SR B
5 WHERE A.RID(+) = B.SID
6* GROUP BY B.SNAME
SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT B.SNAME AS REFEREDBY
2      ,COUNT(1) AS TOTCNT
3 FROM B26_STD_SR A
4      ,B26_std_SR B
5 WHERE A.RID(+) = B.SID
6* GROUP BY B.SNAME
SQL> /

```

REFEREDBY	TOTCNT
KRISHNA	2
TEJASREE	1
HIMANSHU	1
ABHISHEK	1
SURESH	1

```

SQL>
SQL> /

```

REFEREDBY	TOTCNT
KRISHNA	2
TEJASREE	1
HIMANSHU	1
ABHISHEK	1
SURESH	1

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT B.SNAME AS REFEREDBY
2      ,COUNT(A.SNAME) AS TOTCNT
3 FROM B26_STD_SR A
4      ,B26_std_SR B
5 WHERE A.RID(+) = B.SID
6* GROUP BY B.SNAME

```

SQL> /

REFEREDBY	TOTCNT
-----	-----
KRISHNA	2
TEJASREE	1
HIMANSHU	1
ABHISHEK	0
SURESH	0

SQL> ED

Wrote file afiedt.buf

```

1  SELECT B.SNAME AS REFEREDBY
2      ,COUNT(A.SNAME) AS TOTCNT
3  FROM B26_STD_SR  A
4      ,B26_std_SR B
5  WHERE A.RID(+)  = B.SID
6  GROUP BY B.SNAME
7*  HAVING COUNT(A.SNAME ) > 1
SQL> /

```

REFEREDBY	TOTCNT
-----	-----
KRISHNA	2

SQL> CL SCR

SQL> R

```

1  SELECT B.SNAME AS REFEREDBY
2      ,COUNT(A.SNAME) AS TOTCNT
3  FROM B26_STD_SR  A
4      ,B26_std_SR B
5  WHERE A.RID(+)  = B.SID
6  GROUP BY B.SNAME
7*  HAVING COUNT(A.SNAME ) > 1

```

REFEREDBY	TOTCNT
-----	-----
KRISHNA	2

SQL> CL SCR

SQL> SET PAGESIZE 200

SQL> SET LINESIZE 200

SQL>

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CL SCR

SQL> COLUMN ENAME FORMAT A10

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CL SCR

SQL> SELECT EMPNO

2 ,ENAME

3 ,MGR

4 FROM EMP;

EMPNO	ENAME	MGR
-----	-----	-----

7839	KING	
7698	BLAKE	7839
7782	CLARK	7839
7566	JONES	7839
7788	SCOTT	7566
7902	FORD	7566
7369	SMITH	7902
7499	ALLEN	7698
7521	WARD	7698
7654	MARTIN	7698
7844	TURNER	7698
7876	ADAMS	7788
7900	JAMES	7698
7934	MILLER	7782

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.ENAME AS EMPLOYEENAME
2      ,E2.ENAME AS MANAGERNAME
3  FROM EMP E1
4      ,EMP E2
5*  WHERE E1.MGR = E2.EMPNO
SQL> /

```

EMPLOYEENAME	MANAGERNAME
JONES	KING
CLARK	KING
BLAKE	KING
JAMES	BLAKE
TURNER	BLAKE
MARTIN	BLAKE
WARD	BLAKE
ALLEN	BLAKE
MILLER	CLARK
FORD	JONES
SCOTT	JONES
ADAMS	SCOTT
SMITH	FORD

13 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.ENAME AS EMPLOYEENAME
2      ,E2.ENAME AS MANAGERNAME
3  FROM EMP E1
4      ,EMP E2
5*  WHERE E1.MGR = E2.EMPNO (+)
SQL> /

```

EMPLOYEENAME	MANAGERNAME
JONES	KING
CLARK	KING
BLAKE	KING
JAMES	BLAKE
TURNER	BLAKE
MARTIN	BLAKE
WARD	BLAKE

ALLEN	BLAKE
MILLER	CLARK
FORD	JONES
SCOTT	JONES
ADAMS	SCOTT
SMITH	FORD
KING	

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.ENAME AS EMPLOYEENAME
2      , NVL(E2.ENAME, 'NO MANAGER') AS MANAGERNAME
3  FROM EMP E1
4      , EMP E2
5* WHERE E1.MGR = E2.EMPNO(+)
SQL> /

```

EMPLOYEENAME	MANAGERNAME
JONES	KING
CLARK	KING
BLAKE	KING
JAMES	BLAKE
TURNER	BLAKE
MARTIN	BLAKE
WARD	BLAKE
ALLEN	BLAKE
MILLER	CLARK
FORD	JONES
SCOTT	JONES
ADAMS	SCOTT
SMITH	FORD
KING	NO MANAGER

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT NVL(E2.ENAME, 'NO MANAGER') AS MANAGERNAME
2      , COUNT(E1.ENAME ) AS TOTCNT
3  FROM EMP E1
4      , EMP E2
5  WHERE E1.MGR = E2.EMPNO(+)
6* GROUP BY NVL(E2.ENAME, 'NO MANAGER')
SQL> /

```

FROM EMP E1

*

ERROR at line 3:

ORA-00936: missing expression

SQL> ED

Wrote file afiedt.buf

```

1  SELECT NVL(E2.ENAME, 'NO MANAGER') AS MANAGERNAME
2      , COUNT(E1.ENAME ) AS TOTCNT
3  FROM EMP E1
4      , EMP E2
5  WHERE E1.MGR = E2.EMPNO(+)
6* GROUP BY NVL(E2.ENAME, 'NO MANAGER')

```

SQL> /

MANAGERNAME	TOTCNT
JONES	2
FORD	1
CLARK	1
SCOTT	1
NO MANAGER	1
KING	3
BLAKE	5

7 rows selected.

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT NVL(E2.ENAME,'NO MANAGER') AS MANAGERNAME
2      ,COUNT(E1.ENAME ) AS TOTCNT
3  FROM EMP E1
4      ,EMP E2
5  WHERE E1.MGR = E2.EMPNO(+)
6  GROUP BY NVL(E2.ENAME,'NO MANAGER')
7* ORDER BY 2 DESC

```

SQL> /

MANAGERNAME	TOTCNT
BLAKE	5
KING	3
JONES	2
NO MANAGER	1
CLARK	1
FORD	1
SCOTT	1

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT NVL(E2.ENAME,'NO MANAGER') AS MANAGERNAME
2      ,COUNT(E1.ENAME ) AS TOTCNT
3  FROM EMP E1
4      ,EMP E2
5  WHERE E1.MGR = E2.EMPNO(+)
6  GROUP BY NVL(E2.ENAME,'NO MANAGER')
7  HAVING COUNT(E1.ENAME) > 1
8* ORDER BY 2 DESC

```

SQL> /

MANAGERNAME	TOTCNT
BLAKE	5
KING	3
JONES	2

SQL> CL SCR

SQL> R

```

1  SELECT NVL(E2.ENAME,'NO MANAGER') AS MANAGERNAME
2      ,COUNT(E1.ENAME ) AS TOTCNT
3  FROM EMP E1
4      ,EMP E2

```

```

5  WHERE E1.MGR = E2.EMPNO(+)
6  GROUP BY NVL(E2.ENAME, 'NO MANAGER')
7  HAVING COUNT(E1.ENAME) > 1
8* ORDER BY 2 DESC

```

MANAGERNAME	TOTCNT
BLAKE	5
KING	3
JONES	2

```

SQL> SELECT EMPNO, ENAME, SAL, MGR
2  FROM EMP;

```

EMPNO	ENAME	SAL	MGR
7839	KING	845	
7698	BLAKE	2850	7839
7782	CLARK	2450	7839
7566	JONES	2975	7839
7788	SCOTT	4791.6	7566
7902	FORD	3600	7566
7369	SMITH	800	7902
7499	ALLEN	1600	7698
7521	WARD	1250	7698
7654	MARTIN	1250	7698
7844	TURNER	1500	7698
7876	ADAMS	1100	7788
7900	JAMES	950	7698
7934	MILLER	1300	7782

14 rows selected.

```
SQL> CL SCR
```

```

SQL> R
1  SELECT EMPNO, ENAME, SAL, MGR
2* FROM EMP

```

EMPNO	ENAME	SAL	MGR
7839	KING	845	
7698	BLAKE	2850	7839
7782	CLARK	2450	7839
7566	JONES	2975	7839
7788	SCOTT	4791.6	7566
7902	FORD	3600	7566
7369	SMITH	800	7902
7499	ALLEN	1600	7698
7521	WARD	1250	7698
7654	MARTIN	1250	7698
7844	TURNER	1500	7698
7876	ADAMS	1100	7788
7900	JAMES	950	7698
7934	MILLER	1300	7782

14 rows selected.

```
SQL> CL SCR
```

```

SQL> R
1  SELECT EMPNO, ENAME, SAL, MGR
2* FROM EMP

```

EMPNO	ENAME	SAL	MGR
7839	KING	845	
7698	BLAKE	2850	7839
7782	CLARK	2450	7839
7566	JONES	2975	7839
7788	SCOTT	4791.6	7566
7902	FORD	3600	7566
7369	SMITH	800	7902
7499	ALLEN	1600	7698
7521	WARD	1250	7698
7654	MARTIN	1250	7698
7844	TURNER	1500	7698
7876	ADAMS	1100	7788
7900	JAMES	950	7698
7934	MILLER	1300	7782

14 rows selected.

SQL> EDI

Wrote file afiedt.buf

```

1  SELECT E1.ENAME EMPLOYEENAME
2      ,E1.SAL EMPLOYEESALARY
3      ,E2.ENAME MANAGERNAME
4      ,E2.SAL MANAGERSALARY
5  FROM EMP E1
6      ,EMP E2
7*  WHERE E1.MGR = E2.EMPNO
SQL> /

```

EMPLOYEENAME MANAGERSALARY	EMPLOYEESALARY	MANAGERNAME
-------------------------------	----------------	-------------

JONES	2975	KING
845		
CLARK	2450	KING
845		
BLAKE	2850	KING
845		
JAMES	950	BLAKE
2850		
TURNER	1500	BLAKE
2850		
MARTIN	1250	BLAKE
2850		
WARD	1250	BLAKE
2850		
ALLEN	1600	BLAKE
2850		
MILLER	1300	CLARK
2450		
FORD	3600	JONES
2975		
SCOTT	4791.6	JONES
2975		
ADAMS	1100	SCOTT
4791.6		
SMITH	800	FORD
3600		

13 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT E1.ENAME EMPLOYEENAME
2      ,E1.SAL EMPLOYEESALARY
3      ,E2.ENAME MANAGERNAME
4      ,E2.SAL MANAGERSALARY
5  FROM EMP E1
6      ,EMP E2
7  WHERE E1.MGR = E2.EMPNO
8*   AND  E1.SAL > E2.SAL
SQL> /
```

EMPLOYEENAME MANAGERSALARY	EMPLOYEESALARY	MANAGERNAME

JONES	2975	KING
845		
CLARK	2450	KING
845		
BLAKE	2850	KING
845		
FORD	3600	JONES
2975		
SCOTT	4791.6	JONES
2975		

```
SQL> CL SCR
```

```
SQL> R
1  SELECT E1.ENAME EMPLOYEENAME
2      ,E1.SAL EMPLOYEESALARY
3      ,E2.ENAME MANAGERNAME
4      ,E2.SAL MANAGERSALARY
5  FROM EMP E1
6      ,EMP E2
7  WHERE E1.MGR = E2.EMPNO
8*   AND  E1.SAL > E2.SAL
```

EMPLOYEENAME MANAGERSALARY	EMPLOYEESALARY	MANAGERNAME

JONES	2975	KING
845		
CLARK	2450	KING
845		
BLAKE	2850	KING
845		
FORD	3600	JONES
2975		
SCOTT	4791.6	JONES
2975		

```
SQL> CL SCR
```

```
SQL> SELECT ENAME, SAL, DEPTNO
2  FROM EMP;
```

ENAME	SAL	DEPTNO

KING	845	10
BLAKE	2850	30

CLARK	2450	10
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

SQL> SELECT * FROM DEPT;

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

SQL> SELECT ENAME, SAL, DNAME, LOC
2 FROM EMP
3 , DEPT;

ENAME	SAL	DNAME	LOC
KING	845	ACCOUNTING	NEW YORK
BLAKE	2850	ACCOUNTING	NEW YORK
CLARK	2450	ACCOUNTING	NEW YORK
JONES	2975	ACCOUNTING	NEW YORK
SCOTT	4791.6	ACCOUNTING	NEW YORK
FORD	3600	ACCOUNTING	NEW YORK
SMITH	800	ACCOUNTING	NEW YORK
ALLEN	1600	ACCOUNTING	NEW YORK
WARD	1250	ACCOUNTING	NEW YORK
MARTIN	1250	ACCOUNTING	NEW YORK
TURNER	1500	ACCOUNTING	NEW YORK
ADAMS	1100	ACCOUNTING	NEW YORK
JAMES	950	ACCOUNTING	NEW YORK
MILLER	1300	ACCOUNTING	NEW YORK
KING	845	RESEARCH	DALLAS
BLAKE	2850	RESEARCH	DALLAS
CLARK	2450	RESEARCH	DALLAS
JONES	2975	RESEARCH	DALLAS
SCOTT	4791.6	RESEARCH	DALLAS
FORD	3600	RESEARCH	DALLAS
SMITH	800	RESEARCH	DALLAS
ALLEN	1600	RESEARCH	DALLAS
WARD	1250	RESEARCH	DALLAS
MARTIN	1250	RESEARCH	DALLAS
TURNER	1500	RESEARCH	DALLAS
ADAMS	1100	RESEARCH	DALLAS
JAMES	950	RESEARCH	DALLAS
MILLER	1300	RESEARCH	DALLAS
KING	845	SALES	CHICAGO
BLAKE	2850	SALES	CHICAGO
CLARK	2450	SALES	CHICAGO
JONES	2975	SALES	CHICAGO
SCOTT	4791.6	SALES	CHICAGO
FORD	3600	SALES	CHICAGO

SMITH	800	SALES	CHICAGO
ALLEN	1600	SALES	CHICAGO
WARD	1250	SALES	CHICAGO
MARTIN	1250	SALES	CHICAGO
TURNER	1500	SALES	CHICAGO
ADAMS	1100	SALES	CHICAGO
JAMES	950	SALES	CHICAGO
MILLER	1300	SALES	CHICAGO
KING	845	OPERATIONS	BOSTON
BLAKE	2850	OPERATIONS	BOSTON
CLARK	2450	OPERATIONS	BOSTON
JONES	2975	OPERATIONS	BOSTON
SCOTT	4791.6	OPERATIONS	BOSTON
FORD	3600	OPERATIONS	BOSTON
SMITH	800	OPERATIONS	BOSTON
ALLEN	1600	OPERATIONS	BOSTON
WARD	1250	OPERATIONS	BOSTON
MARTIN	1250	OPERATIONS	BOSTON
TURNER	1500	OPERATIONS	BOSTON
ADAMS	1100	OPERATIONS	BOSTON
JAMES	950	OPERATIONS	BOSTON
MILLER	1300	OPERATIONS	BOSTON

56 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME, SAL, DNAME, LOC
2  FROM EMP
3*      CROSS JOIN DEPT
SQL> /

```

ENAME	SAL	DNAME	LOC
KING	845	ACCOUNTING	NEW YORK
BLAKE	2850	ACCOUNTING	NEW YORK
CLARK	2450	ACCOUNTING	NEW YORK
JONES	2975	ACCOUNTING	NEW YORK
SCOTT	4791.6	ACCOUNTING	NEW YORK
FORD	3600	ACCOUNTING	NEW YORK
SMITH	800	ACCOUNTING	NEW YORK
ALLEN	1600	ACCOUNTING	NEW YORK
WARD	1250	ACCOUNTING	NEW YORK
MARTIN	1250	ACCOUNTING	NEW YORK
TURNER	1500	ACCOUNTING	NEW YORK
ADAMS	1100	ACCOUNTING	NEW YORK
JAMES	950	ACCOUNTING	NEW YORK
MILLER	1300	ACCOUNTING	NEW YORK
KING	845	RESEARCH	DALLAS
BLAKE	2850	RESEARCH	DALLAS
CLARK	2450	RESEARCH	DALLAS
JONES	2975	RESEARCH	DALLAS
SCOTT	4791.6	RESEARCH	DALLAS
FORD	3600	RESEARCH	DALLAS
SMITH	800	RESEARCH	DALLAS
ALLEN	1600	RESEARCH	DALLAS
WARD	1250	RESEARCH	DALLAS
MARTIN	1250	RESEARCH	DALLAS
TURNER	1500	RESEARCH	DALLAS
ADAMS	1100	RESEARCH	DALLAS
JAMES	950	RESEARCH	DALLAS
MILLER	1300	RESEARCH	DALLAS
KING	845	SALES	CHICAGO

BLAKE	2850	SALES	CHICAGO
CLARK	2450	SALES	CHICAGO
JONES	2975	SALES	CHICAGO
SCOTT	4791.6	SALES	CHICAGO
FORD	3600	SALES	CHICAGO
SMITH	800	SALES	CHICAGO
ALLEN	1600	SALES	CHICAGO
WARD	1250	SALES	CHICAGO
MARTIN	1250	SALES	CHICAGO
TURNER	1500	SALES	CHICAGO
ADAMS	1100	SALES	CHICAGO
JAMES	950	SALES	CHICAGO
MILLER	1300	SALES	CHICAGO
KING	845	OPERATIONS	BOSTON
BLAKE	2850	OPERATIONS	BOSTON
CLARK	2450	OPERATIONS	BOSTON
JONES	2975	OPERATIONS	BOSTON
SCOTT	4791.6	OPERATIONS	BOSTON
FORD	3600	OPERATIONS	BOSTON
SMITH	800	OPERATIONS	BOSTON
ALLEN	1600	OPERATIONS	BOSTON
WARD	1250	OPERATIONS	BOSTON
MARTIN	1250	OPERATIONS	BOSTON
TURNER	1500	OPERATIONS	BOSTON
ADAMS	1100	OPERATIONS	BOSTON
JAMES	950	OPERATIONS	BOSTON
MILLER	1300	OPERATIONS	BOSTON

56 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME, SAL, DNAME, LOC
2  FROM EMP
3*      CROSS JOIN DEPT
SQL> /

```

ENAME	SAL	DNAME	LOC
KING	845	ACCOUNTING	NEW YORK
BLAKE	2850	ACCOUNTING	NEW YORK
CLARK	2450	ACCOUNTING	NEW YORK
JONES	2975	ACCOUNTING	NEW YORK
SCOTT	4791.6	ACCOUNTING	NEW YORK
FORD	3600	ACCOUNTING	NEW YORK
SMITH	800	ACCOUNTING	NEW YORK
ALLEN	1600	ACCOUNTING	NEW YORK
WARD	1250	ACCOUNTING	NEW YORK
MARTIN	1250	ACCOUNTING	NEW YORK
TURNER	1500	ACCOUNTING	NEW YORK
ADAMS	1100	ACCOUNTING	NEW YORK
JAMES	950	ACCOUNTING	NEW YORK
MILLER	1300	ACCOUNTING	NEW YORK
KING	845	RESEARCH	DALLAS
BLAKE	2850	RESEARCH	DALLAS
CLARK	2450	RESEARCH	DALLAS
JONES	2975	RESEARCH	DALLAS
SCOTT	4791.6	RESEARCH	DALLAS
FORD	3600	RESEARCH	DALLAS
SMITH	800	RESEARCH	DALLAS
ALLEN	1600	RESEARCH	DALLAS
WARD	1250	RESEARCH	DALLAS
MARTIN	1250	RESEARCH	DALLAS

TURNER	1500	RESEARCH	DALLAS
ADAMS	1100	RESEARCH	DALLAS
JAMES	950	RESEARCH	DALLAS
MILLER	1300	RESEARCH	DALLAS
KING	845	SALES	CHICAGO
BLAKE	2850	SALES	CHICAGO
CLARK	2450	SALES	CHICAGO
JONES	2975	SALES	CHICAGO
SCOTT	4791.6	SALES	CHICAGO
FORD	3600	SALES	CHICAGO
SMITH	800	SALES	CHICAGO
ALLEN	1600	SALES	CHICAGO
WARD	1250	SALES	CHICAGO
MARTIN	1250	SALES	CHICAGO
TURNER	1500	SALES	CHICAGO
ADAMS	1100	SALES	CHICAGO
JAMES	950	SALES	CHICAGO
MILLER	1300	SALES	CHICAGO
KING	845	OPERATIONS	BOSTON
BLAKE	2850	OPERATIONS	BOSTON
CLARK	2450	OPERATIONS	BOSTON
JONES	2975	OPERATIONS	BOSTON
SCOTT	4791.6	OPERATIONS	BOSTON
FORD	3600	OPERATIONS	BOSTON
SMITH	800	OPERATIONS	BOSTON
ALLEN	1600	OPERATIONS	BOSTON
WARD	1250	OPERATIONS	BOSTON
MARTIN	1250	OPERATIONS	BOSTON
TURNER	1500	OPERATIONS	BOSTON
ADAMS	1100	OPERATIONS	BOSTON
JAMES	950	OPERATIONS	BOSTON
MILLER	1300	OPERATIONS	BOSTON

56 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME, SAL, DNAME, LOC
2  FROM EMP E
3       , DEPT D
4*   WHERE E.DEPTNO = D.DEPTNO
SQL> /

```

ENAME	SAL	DNAME	LOC
KING	845	ACCOUNTING	NEW YORK
BLAKE	2850	SALES	CHICAGO
CLARK	2450	ACCOUNTING	NEW YORK
JONES	2975	RESEARCH	DALLAS
SCOTT	4791.6	RESEARCH	DALLAS
FORD	3600	RESEARCH	DALLAS
SMITH	800	RESEARCH	DALLAS
ALLEN	1600	SALES	CHICAGO
WARD	1250	SALES	CHICAGO
MARTIN	1250	SALES	CHICAGO
TURNER	1500	SALES	CHICAGO
ADAMS	1100	RESEARCH	DALLAS
JAMES	950	SALES	CHICAGO
MILLER	1300	ACCOUNTING	NEW YORK

14 rows selected.

SQL> CL SCR

```
SQL> R
1  SELECT ENAME, SAL, DNAME, LOC
2  FROM EMP E
3      , DEPT D
4*  WHERE E.DEPTNO = D.DEPTNO
```

ENAME	SAL	DNAME	LOC
KING	845	ACCOUNTING	NEW YORK
BLAKE	2850	SALES	CHICAGO
CLARK	2450	ACCOUNTING	NEW YORK
JONES	2975	RESEARCH	DALLAS
SCOTT	4791.6	RESEARCH	DALLAS
FORD	3600	RESEARCH	DALLAS
SMITH	800	RESEARCH	DALLAS
ALLEN	1600	SALES	CHICAGO
WARD	1250	SALES	CHICAGO
MARTIN	1250	SALES	CHICAGO
TURNER	1500	SALES	CHICAGO
ADAMS	1100	RESEARCH	DALLAS
JAMES	950	SALES	CHICAGO
MILLER	1300	ACCOUNTING	NEW YORK

14 rows selected.

```
SQL> EDIT
Wrote file afiedt.buf
```

```
1  SELECT ENAME
2      , SAL
3      , E.DEPTNO
4      , DNAME
5      , LOC
6  FROM EMP E
7      , DEPT D
8*  WHERE E.DEPTNO = D.DEPTNO
SQL> /
```

ENAME	SAL	DEPTNO	DNAME	LOC
KING	845	10	ACCOUNTING	NEW YORK
BLAKE	2850	30	SALES	CHICAGO
CLARK	2450	10	ACCOUNTING	NEW YORK
JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
MILLER	1300	10	ACCOUNTING	NEW YORK

14 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT ENAME
2      , SAL
3      , E.DEPTNO
```

```

4      ,DNAME
5      ,LOC
6 FROM EMP E
7      JOIN DEPT D
8*   ON E.DEPTNO = D.DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DNAME	LOC
KING	845	10	ACCOUNTING	NEW YORK
BLAKE	2850	30	SALES	CHICAGO
CLARK	2450	10	ACCOUNTING	NEW YORK
JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
MILLER	1300	10	ACCOUNTING	NEW YORK

14 rows selected.

```

SQL> CLS CR
SP2-0042: unknown command "CLS CR" - rest of line ignored.
SQL> CL SCR

```

```

SQL> EIDT
SP2-0042: unknown command "EIDT" - rest of line ignored.
SQL> EDIT
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6 FROM EMP E
7      JOIN DEPT D
8*   ON E.DEPTNO = D.DEPTNO
SQL>
SQL>
SQL> CL SCR

```

```

SQL> SELECT ENAME,SAL,DEPTNO
2  FROM EMP;

```

ENAME	SAL	DEPTNO
KING	845	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20

```
JAMES          950          30
MILLER         1300         10
```

14 rows selected.

SQL> SELECT * FROM DEPT;

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      JOIN DEPT D
8*   ON E.DEPTNO = D.DEPTNO
SQL> /
```

ENAME	SAL	DEPTNO	DNAME	LOC
KING	845	10	ACCOUNTING	NEW YORK
BLAKE	2850	30	SALES	CHICAGO
CLARK	2450	10	ACCOUNTING	NEW YORK
JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
MILLER	1300	10	ACCOUNTING	NEW YORK

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      , DEPT D
8*   WHERE E.DEPTNO(+) = D.DEPTNO
SQL> /
```

ENAME	SAL	DEPTNO	DNAME	LOC
KING	845	10	ACCOUNTING	NEW YORK
BLAKE	2850	30	SALES	CHICAGO
CLARK	2450	10	ACCOUNTING	NEW YORK

JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
MILLER	1300	10	ACCOUNTING OPERATIONS	NEW YORK BOSTON

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      RIGHT OUTER JOIN DEPT D
8*  WHERE E.DEPTNO = D.DEPTNO

```

SQL> \/

SP2-0042: unknown command "\/" - rest of line ignored.

SQL> /

```

WHERE E.DEPTNO = D.DEPTNO
*

```

ERROR at line 8:

ORA-00905: missing keyword

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      RIGHT OUTER JOIN DEPT D
8*  ON E.DEPTNO = D.DEPTNO

```

SQL> /

ENAME	SAL	DEPTNO	DNAME	LOC
KING	845	10	ACCOUNTING	NEW YORK
BLAKE	2850	30	SALES	CHICAGO
CLARK	2450	10	ACCOUNTING	NEW YORK
JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
MILLER	1300	10	ACCOUNTING	NEW YORK

OPERATIONS

BOSTON

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      RIGHT OUTER JOIN DEPT D
8*   ON D.DEPTNO = E.DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DNAME	LOC
KING	845	10	ACCOUNTING	NEW YORK
BLAKE	2850	30	SALES	CHICAGO
CLARK	2450	10	ACCOUNTING	NEW YORK
JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
MILLER	1300	10	ACCOUNTING	NEW YORK
			OPERATIONS	BOSTON

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E      RIGHT OUTER JOIN DEPT D
7*   ON D.DEPTNO = E.DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DNAME	LOC
KING	845	10	ACCOUNTING	NEW YORK
BLAKE	2850	30	SALES	CHICAGO
CLARK	2450	10	ACCOUNTING	NEW YORK
JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO

MILLER	1300	10 ACCOUNTING OPERATIONS	NEW YORK BOSTON
--------	------	-----------------------------	--------------------

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E      LEFT OUTER JOIN DEPT D
7*   ON D.DEPTNO = E.DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DNAME	LOC
MILLER	1300	10	ACCOUNTING	NEW YORK
CLARK	2450	10	ACCOUNTING	NEW YORK
KING	845	10	ACCOUNTING	NEW YORK
ADAMS	1100	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
JONES	2975	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
ALLEN	1600	30	SALES	CHICAGO
BLAKE	2850	30	SALES	CHICAGO

14 rows selected.

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E      LEFT OUTER JOIN DEPT D
7*   ON D.DEPTNO = E.DEPTNO
SQL>

```

SQL>

SQL>

```

SQL> UPDATE EMP
2  SET DEPTNO =11
3  WHERE DEPTNO =10
4  AND ROWNUM=2;

```

0 rows updated.

SQL> ED

Wrote file afiedt.buf

```

1  UPDATE EMP
2  SET DEPTNO =11
3  WHERE DEPTNO =10
4* AND ROWNUM<=2
SQL> /

```

2 rows updated.

SQL> COMMIT;

Commit complete.

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CL SCR

SQL> SELECT ENAME, SAL, DEPTNO
2 FROM EMP;

ENAME	SAL	DEPTNO
KING	845	11
BLAKE	2850	30
CLARK	2450	11
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

SQL> SELECT *
2 FROM DEPT;

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E      JOIN DEPT D
7*   ON D.DEPTNO = E.DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DNAME	LOC
BLAKE	2850	30	SALES	CHICAGO
JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO

TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
MILLER	1300	10	ACCOUNTING	NEW YORK

12 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      ,DEPT D
8*  WHERE D.DEPTNO = E.DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DNAME	LOC
BLAKE	2850	30	SALES	CHICAGO
JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
MILLER	1300	10	ACCOUNTING	NEW YORK

12 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      ,DEPT D
8*  WHERE D.DEPTNO(+) = E.DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DNAME	LOC
MILLER	1300	10	ACCOUNTING	NEW YORK
ADAMS	1100	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
JONES	2975	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
ALLEN	1600	30	SALES	CHICAGO
BLAKE	2850	30	SALES	CHICAGO

```
CLARK          2450          11
KING           845          11
```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      ,DEPT D
8*  WHERE D.DEPTNO(+) = E.DEPTNO(+)
```

SQL> /

```
WHERE D.DEPTNO(+) = E.DEPTNO(+)
      *
```

ERROR at line 8:

ORA-01468: a predicate may reference only one outer-joined table

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      JOIN DEPT D
8*  ON D.DEPTNO = E.DEPTNO
```

SQL> /

ENAME	SAL	DEPTNO	DNAME	LOC
BLAKE	2850	30	SALES	CHICAGO
JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
MILLER	1300	10	ACCOUNTING	NEW YORK

12 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      LEFT OUTER JOIN DEPT D
8*  ON D.DEPTNO = E.DEPTNO
```

SQL> /

ENAME	SAL	DEPTNO	DNAME	LOC
MILLER	1300	10	ACCOUNTING	NEW YORK
ADAMS	1100	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
JONES	2975	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
ALLEN	1600	30	SALES	CHICAGO
BLAKE	2850	30	SALES	CHICAGO
CLARK	2450	11		
KING	845	11		

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      RIGHT OUTER JOIN DEPT D
8*   ON D.DEPTNO = E.DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DNAME	LOC
BLAKE	2850	30	SALES	CHICAGO
JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
MILLER	1300	10	ACCOUNTING OPERATIONS	NEW YORK BOSTON

13 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      FULL OUTER JOIN DEPT D
8*   ON D.DEPTNO = E.DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DNAME	LOC
KING	845	11		
BLAKE	2850	30	SALES	CHICAGO
CLARK	2450	11		
JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
MILLER	1300	10	ACCOUNTING OPERATIONS	NEW YORK BOSTON

15 rows selected.

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT ENAME
2      ,SAL
3      ,E.DEPTNO
4      ,DNAME
5      ,LOC
6  FROM EMP E
7      FULL OUTER JOIN DEPT D
8*   ON D.DEPTNO = E.DEPTNO
SQL>
SQL> /

```

ENAME	SAL	DEPTNO	DNAME	LOC
KING	845	11		
BLAKE	2850	30	SALES	CHICAGO
CLARK	2450	11		
JONES	2975	20	RESEARCH	DALLAS
SCOTT	4791.6	20	RESEARCH	DALLAS
FORD	3600	20	RESEARCH	DALLAS
SMITH	800	20	RESEARCH	DALLAS
ALLEN	1600	30	SALES	CHICAGO
WARD	1250	30	SALES	CHICAGO
MARTIN	1250	30	SALES	CHICAGO
TURNER	1500	30	SALES	CHICAGO
ADAMS	1100	20	RESEARCH	DALLAS
JAMES	950	30	SALES	CHICAGO
MILLER	1300	10	ACCOUNTING OPERATIONS	NEW YORK BOSTON

15 rows selected.

SQL> SPOOL OFF

SUBQUERIES

WHAT IS SUBQUERY ?

ANY SELECT STATEMENT WITHIN A () IS CALLED SUBQUERY

WHY WE NEED A SUBQUERY ?

TO ANSWER MULTIPART QUESTIONS.

TO CLUB MORE THAN ONE SELECT STATEMENT RESULTS

AS A COLUMN WISE WE WILL GO FOR SUBQUERIES.

TYPES OF SUBQUERIES

3 TYPES OF SUBQUERY BASED ON WHERE WE WRITE

NESTED SUBQUERIES--1 MORE EXAMPLE

A SQ IN WHERE CLAUSE

INLINE VIEW

A SQ IN FROM CLAUSE which will act AS A TABLE

SUB SELECT

A SQ IN SELECT CLAUSE which will act AS A COLUMN

NOTE: SUB SELECT SHOULD RETURN ONLY ONE ROW AND ONE COLUMN AS OUTPUT

3 TYPES OF SUBQUERIES BASED HOW MANY ROWS/COLUMNS IT WRITENS

SINGLE ROW SUBQUERIES

A SQ WHICH RETURNS ONLY ONE ROW

Single row subquery operators are

= < > <= >= <> !=

MULTI ROW SUBQUERIES

A SQ WHICH RETURNS MORE THAN ONE ROW

Multi row subquery operators are

IN

ANY

>ANY--> >MIN

MORE THAN THE MINIMUM VALUE IN THE LIST

<ANY--- < MAX

LESS THAN THE MAXIMUM VALUE IN THE LIST

=ANY

IS NOTHITNG BUT "IN" OPERATOR

ALL

>ALL -- > MAX

MORE THAN THE MAXIMUM VALUE IN THE LIST

<ALL -- < MIN

LESS THAN THE MINIMUM VALUE IN THE LIST

=ALL

HOW MANY VALUES ARE THERE IN THE LIST EACH VALUE SHOULD BE SAME

THEN =ALL WILL RETURN THAT VALUE

I.E. ON EMPLOYEE SALARY SHOULD BE SAME AS ALL LIST OF SALARY DETAILS WHICH

ARE COMMING FROM SUBQUERY.

NOTE: YOU CAN AVOID USING THIS ANY AND ALL IF YOU CONVERT MULTI ROW SUBQURIES INTO SINGLE ROW SUBQUERIES BY USING AGGR FUNCTIONS

MULTI COLUMN SUBQUERIES

A SQ WHICH RETURNS MORE THAN ONE COLUMN

2 TYPES OF SUBQUERIES BASED ON HOW THEY EXECUTE

NORMAL SUBQUEIRES

FIRST SUBQUERY(INNERQUERY) WILL EXECUTE

SECOND OUTER QUERY WILL EXECUTE.

CORELATED SUBQUERIES

Corelated subqueries will execute in 4 STEPS

1) FIRST OUTER QUERY WILL EXECUTE

2) IT WILL PASS REQUIRED INFORMATION TO THE INNER QUERY

3) innerquery will execute and it will send TRUE/FALSE TO the outer query

4) OUTER QUERY WILL DECIDE WHETHER THAT ROW HAS TO BE CONSIDERED OR

NOT BASED ON RESULTS FROM INNER QUERY

Advantage of corealted subquries ?

Whnenever same kind of data comes it will not execute inner query again and again instead it will give you value from CACHE

NOTE: IF THERE IS ANY CHANCE USE EXISTS INSTEAD OF "IN"

USE NOT EXISTS INSTEAD ON "NOT IN"

DIFF BETWEEN NORMAL SUBQUERIES AND CORELATED SUBQUERIES ?

SET OPERATORS --

VIEWS AND MVIEWS

SEQUENCES

OLAP FUNCTIONS

IN -- EXISTS

NOT IN -- NOT EXISTS\

SELECT */COLUMNS

FROM TABLENAME O

WHERE EXISTS/NOT EXISTS (SELECT SOMETHING-- SOMETHING CAN BE ANYTHING IT IS UP TO YOU

FROM TABLENAME I

WHERE I.COLUMNNAME = O.COLUMNNAME

)

EXISTS -- 1 Y

NOT EXISTS -- 0 N

T1

1

2

3

4

4

5

T2

1

2

3

6

SEQUENCES--1

SET OPER-30

VIEWS--1

MVIEWS

OLAP--2

DDL

DML

DCL--2 2.30

INDEXES AND PARTIONGS--2

ENAME	SAL	DEPTNO	TOTCNT	TOTSAL	PERCENTA	HISAL	LS
-----	-----	-----					
VENU	4000	10	18	62712	8000	800	
BAGYA	4700	10	18	62712			
VENU	5000		18	62712			
RAJ	8000		18	62712			
KING	6050	10	18	62712			
BLAKE	4275	30	18	62712			
CLARK	3675	10	18	62712			
SCOTT	3300	20	18	62712			
FORD	3300	20	18	62712			
SMITH	1200	20	18	62712			
ALLEN	2400	30	18	62712			
WARD	1400	30	18	62712			
MARTIN	1875	30	18	62712			
ADAMS	1650	20	18	62712			
JAMES	1425	30	18	62712			
JONES	4462.5	20	18	62712			
BALU	3000	10	18	62712			
SURENDARKUMAR		3000					

ENAME	SAL	DEPTNO	DTOTCNT
-------	-----	--------	---------

VENU	4000	10	5
BAGYA	4700	10	5
VENU	5000		
RAJ	8000		
KING	6050	10	5
BLAKE	4275	30	
CLARK	3675	10	5
SCOTT	3300	20	5
FORD	3300	20	5
SMITH	1200	20	
ALLEN	2400	30	
WARD	1400	30	
MARTIN	1875	30	
ADAMS	1650	20	
JAMES	1425	30	
JONES	4462.5	20	
BALU	3000	10	5
SURENDARKUMAR	3000		

deptno totcnt

10

20

30

DEPTNO10 DEPTNO20 DEPTNO30 DEPTNONULL

5 5 5 3

EMPCNT DEPTCNT SALGRADECNT

18 5 7

TABlename ROWCNT

EMP 18

DEPT 5

SALGRADE 7

ENAME SAL TOTEMPCNT TOTSAL PERCNENTAGE HISAL

KHAJA 2000 15 46850 2000/46850*100 7000

MAJULA 2500 15 46850 7000

INDRA 3000 15

INDRA 2200 15

SAI 7000 15

CLARK 5675 15

SCOTT 3300 15

FORD 3300 15

SMITH 1200 15

WARD 1400 15

MARTIN 1200 15

ADAMS	1650	15		
JAMES	1425	15		
JONES	3000	15	46850	70000

ENAME	SAL	DEPTNO	DEPTCNT	DEPTSAL	TOTCNT
-----	-----	-----	-----	-----	-----
INDRA	3000	10	4	16675	16
SAI	7000	10	4	16675	16
BALUE	1000	10	4	16675	16
CLARK	5675	10	4	16675	
SCOTT	3300	20	9		
FORD	3300	20	9		
SMITH	1200	20	9		
SAI	7000	20	9		
INDRA	2200	20	9		
JONES	3000	20	9		
KHAJA	2000	20	9		
MAJULA	2500	20	9		
ADAMS	1650	20	9		
JAMES	1425	30	3	4025	
MARTIN	1200	30	3	4025	
WARD	1400	30	3	4025	16

2) IN YOUR NAME, WHICH ALPHABET IS BE

IN JOINS

IN EQUI JOIN WE WILL USE = OPERATOR IF YOU ARE USING < > <> BE CAREFUL ON HOW IT WILL EXECUTE RESULTS...

16-jun-2018 Subquireies Examples

```
SQL> SELECT ENAME  
      2 FROM EMP;
```

```
ENAME  
-----  
SURYA  
BLAKE  
CLARK  
JONES  
SCOTT  
FORD  
SMITH  
ALLEN  
WARD  
MARTIN  
TURNER  
ADAMS  
JAMES  
MILLER
```

14 rows selected.

```
SQL> ED  
Wrote file afiedt.buf
```

```
1  (  
2  SELECT ENAME  
3  FROM EMP  
4* )  
5  /
```

```
ENAME  
-----  
SURYA  
BLAKE  
CLARK  
JONES  
SCOTT  
FORD  
SMITH  
ALLEN  
WARD  
MARTIN  
TURNER  
ADAMS  
JAMES  
MILLER
```

14 rows selected.

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
```

```
2  ,SAL
3  FROM EMP;
```

ENAME	SAL
SURYA	2000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT ENAME
2  ,SAL
3  FROM EMP
4* WHERE ENAME ='ALLEN'
SQL> /
```

ENAME	SAL
ALLEN	1600

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT ENAME
2  ,SAL
3  FROM EMP
4* WHERE ENAME ='allen'
SQL> /
```

no rows selected

```
SQL> ed
Wrote file afiedt.buf
```

```
1  SELECT ENAME
2  ,SAL
3  FROM EMP
4* WHERE ENAME ='ALLEN'
SQL> /
```

ENAME	SAL
ALLEN	1600

```
SQL> CL SCR
```

```
SQL> R
1  SELECT ENAME
2  ,SAL
```



```

3  FROM EMP
4* WHERE ENAME ='ALLEN'

```

ENAME	SAL
ALLEN	1600

```

SQL> EDIT
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3* FROM EMP
4  /

```

ENAME	SAL
SURYA	2000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

```
SQL> CL SCR
```

```

SQL> R
1  SELECT ENAME
2  ,SAL
3* FROM EMP

```

ENAME	SAL
SURYA	2000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

```

SQL> SELECT *
2  FROM EMP
3  WHERE SAL > SELECT SAL FROM EMP WHERE ENAME ='ALLEN';
WHERE SAL > SELECT SAL FROM EMP WHERE ENAME ='ALLEN'

```

*

ERROR at line 3:
ORA-00936: missing expression

SQL> CL SCR

```
SQL> SELECT ENAME
2      ,SAL
3  FROM EMP;
```

ENAME	SAL
SURYA	2000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

```
SQL> SELECT SAL
2      FROM EMP
3  WHERE ENAME = 'ALLEN';
```

SAL
1600

```
SQL> SELECT ENAME
2      ,SAL
3  FROM EMP
4  WHERE SAL > 1600;
```

ENAME	SAL
SURYA	2000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600

6 rows selected.

SQL> ED
Wrote file afiedt.buf

```
1  SELECT ENAME
2      ,SAL
3  FROM EMP
4  WHERE SAL > SELECT SAL
5              FROM EMP
6*              WHERE ENAME = 'ALLEN'
SQL> /
```

```
WHERE SAL > SELECT SAL
```

```
*
```

```
ERROR at line 4:
```

```
ORA-00936: missing expression
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL >( SELECT SAL
5                FROM EMP
6                WHERE ENAME ='ALLEN'
7*              )
SQL> /
```

ENAME	SAL
SURYA	2000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600

```
6 rows selected.
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  ( SELECT SAL
2      FROM EMP
3      WHERE ENAME ='ALLEN'
4*  )
SQL> C
SP2-0025: Invalid change string.
SQL> /
```

SAL
1600

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL >( SELECT SAL
5                FROM EMP
6                WHERE ENAME ='ALLEN'
7*              )
SQL> /
```

ENAME	SAL
SURYA	2000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600

6 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL  >( SELECT SAL
5                FROM EMP
6                WHERE ENAME ='ALLEN'
7*              )

```

ENAME	SAL
SURYA	2000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL  <( SELECT SAL
5                FROM EMP
6                WHERE ENAME ='ALLEN'
7*              )

```

SQL> /

ENAME	SAL
SMITH	800
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL  NOT >( SELECT SAL
5                    FROM EMP
6                    WHERE ENAME ='ALLEN'
7*                  )

```

SQL> /

```

WHERE SAL  NOT >( SELECT SAL
                  *

```

ERROR at line 4:

ORA-00920: invalid relational operator

```
SQL> ED
Wrote file afiedt.buf
```

```
 1  SELECT ENAME
 2  ,SAL
 3  FROM EMP
 4  WHERE SAL < ( SELECT SAL
 5                  FROM EMP
 6                  WHERE ENAME = 'ALLEN'
 7*                )
SQL> /
```

ENAME	SAL
SMITH	800
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

7 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
 1  SELECT ENAME
 2  ,SAL
 3  FROM EMP
 4  WHERE SAL < ( SELECT SAL
 5                  FROM EMP
 6                  WHERE ENAME = 'ALLEN'
 7                )
 8* ORDER BY SAL DESC
SQL> /
```

ENAME	SAL
TURNER	1500
MILLER	1300
WARD	1250
MARTIN	1250
ADAMS	1100
JAMES	950
SMITH	800

7 rows selected.

```
SQL> CL SCR
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```
 1  SELECT ENAME
 2  ,SAL
 3  FROM EMP
 4  WHERE SAL < ( SELECT SAL
 5                  FROM EMP
 6                  WHERE ENAME = 'ALLEN'
 7                )
 8* ORDER BY SAL DESC
```

```
SQL>
SQL> /
```

ENAME	SAL
TURNER	1500
MILLER	1300
WARD	1250
MARTIN	1250
ADAMS	1100
JAMES	950
SMITH	800

7 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL  < ( SELECT SAL
5                  FROM EMP
6                  WHERE ENAME = 'ALLEN'
7                  )
8  ORDER BY SAL DESC
9* ,ENAME ASC
SQL> /
```

ENAME	SAL
TURNER	1500
MILLER	1300
MARTIN	1250
WARD	1250
ADAMS	1100
JAMES	950
SMITH	800

7 rows selected.

```
SQL> /ED
SP2-0042: unknown command "/ED" - rest of line ignored.
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL  < ( SELECT SAL
5                  FROM EMP
6                  WHERE ENAME = 'ALLEN'
7                  )
8  ORDER BY SAL DESC
9* ,ENAME ASC
SQL>
SQL>
SQL> CL SCR
```

```
SQL> SELECT ENAEM
2  ,
3
SQL>
SQL> CL SCR
```

```
SQL> SELECT ENAME
2      ,SAL
3  FROM EMP;
```

ENAME	SAL
SURYA	2000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

```
SQL> SELECT EANME
2
SQL>
SQL> CL SCR
```

```
SQL> SELECT ENAME
2      ,SAL
3  FROM EMP;
```

ENAME	SAL
SURYA	2000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

```
SQL> SELECT ENAME
2      ,SAL
3  FROM EMP
4  WHERE SAL = MAX(SAL);
WHERE SAL = MAX(SAL)
*
```

ERROR at line 4:
ORA-00934: group function is not allowed here

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT MAX(SAL)
2*   FROM EMP
SQL> /
```

```
MAX(SAL)
-----
4791.6
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT ENAME
2   ,SAL
3   FROM EMP
4*  WHERE SAL = 4791.6
SQL> /
```

```
ENAME          SAL
-----
SCOTT          4791.6
```

```
SQL> ED
Wrote file afiedt.buf
```

SP2-0223: No lines in SQL buffer.

```
SQL>
```

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
2   ,SAL
3   FROM EMP
4   WHERE SAL = 4791.6
5   /
```

```
ENAME          SAL
-----
SCOTT          4791.6
```

```
SQL> INSERT INTO EMP(EMPNO,ENAME,SAL)
2  VALUES
3  (
4  501,'ARJUN',50000);
```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```
1  SELECT ENAME
2   ,SAL
3   FROM EMP
4*  WHERE SAL = 4791.6
SQL> /
```

```
ENAME          SAL
-----
SCOTT          4791.6
```



```
SQL> SELECT SAL
```

```
2
```

```
SQL>
```

```
SQL> SELECT MAX(SAL)
```

```
2 FROM EMP;
```

```
MAX(SAL)
```

```
-----
```

```
50000
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT ENAME
```

```
2 ,SAL
```

```
3 FROM EMP
```

```
4* WHERE SAL = 5000
```

```
SQL> /
```

```
no rows selected
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT ENAME
```

```
2 ,SAL
```

```
3 FROM EMP
```

```
4* WHERE SAL = 50000
```

```
SQL> /
```

```
ENAME SAL
```

```
-----
```

```
ARJUN 50000
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT ENAME
```

```
2 ,SAL
```

```
3 FROM EMP
```

```
4 WHERE SAL = (SELECT MAX(SAL)
```

```
5 FROM EMP
```

```
6* )
```

```
SQL> /
```

```
ENAME SAL
```

```
-----
```

```
ARJUN 50000
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT ENAME
```

```
2 ,SAL
```

```
3 FROM EMP
```

```
4 WHERE SAL = (SELECT MAX(SAL)
```

```
5 FROM EMP
```

```
6* )
```

```
SQL>
```

```
SQL> /
```

```
ENAME SAL
```

```
-----
```

```
ARJUN          50000
```

```
SQL> CL SCR
```

```
SQL> R
```

```
1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL = (SELECT MAX(SAL)
5                FROM EMP
6*                )
```

```
ENAME          SAL
-----
ARJUN          50000
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  (SELECT AVG(SAL)
2                FROM EMP
3*                )
```

```
SQL> /
```

```
AVG(SAL)
-----
5227.77333
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL > (SELECT AVG(SAL)
5                FROM EMP
6*                )
```

```
SQL> /
```

```
ENAME          SAL
-----
ARJUN          50000
```

```
SQL> UPDATE EMP
```

```
2  SET SAL = 5000
3  WHERE ENAME ='ARJUN'
4  ;
```

```
1 row updated.
```

```
SQL> COMMIT;
```

```
Commit complete.
```

```
SQL> CL SCR
```

```
SQL> EDIT
```

```
Wrote file afiedt.buf
```

```
1  SELECT AVG(sAL)
2* FROM EMP
```

```
SQL> /
```

```
AVG(SAL)
```

2227.77333

SQL> ED
Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL > (SELECT AVG(sAL)
5                FROM EMP
6*             )
SQL> /
```

ENAME	SAL
-----	-----
ARJUN	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600

6 rows selected.

SQL> ED
Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL < (SELECT AVG(sAL)
5                FROM EMP
6*             )
SQL> /
```

ENAME	SAL
-----	-----
SURYA	2000
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

9 rows selected.

SQL> ED
Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL < (SELECT AVG(sAL)
5                FROM EMP
6*             )
SQL>
SQL> /
```

ENAME	SAL
-----	-----

SURYA	2000
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

9 rows selected.

SQL> CL SCR

```
SQL> SELECT ENAME
       2  ,SAL
       3  FROM EMP;
```

ENAME	SAL
-----	-----
SURYA	2000
ARJUN	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

15 rows selected.

```
SQL> SELECT MAX(SAL)
       2  FROM EMP;
```

MAX(SAL)

5000

SQL> ED

Wrote file afiedt.buf

```
       1  ( SELECT MAX(SAL)
       2  FROM EMP
       3*      )
SQL> /
```

MAX(SAL)

5000

SQL> ED

Wrote file afiedt.buf

```
       1  SELECT SAL
       2  FROM EMP
       3  WHERE SAL < ( SELECT MAX(SAL)
       4  FROM EMP
```

```

5*
SQL> /

```

```

      SAL
-----
      2000
      2850
      2450
      2975
    4791.6
      3600
       800
      1600
      1250
      1250
      1500
      1100
       950
      1300

```

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT MAX(SAL)
2  FROM EMP
3  WHERE SAL < ( SELECT MAX(SAL)
4                FROM EMP
5*                )
SQL> /

```

```

      MAX(SAL)
-----
    4791.6

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME,MAX(SAL)
2  FROM EMP
3  WHERE SAL < ( SELECT MAX(SAL)
4                FROM EMP
5*                )
6* GROUP BY ENAME
SQL> /

```

ENAME	MAX(SAL)
SURYA	2000
JONES	2975
ALLEN	1600
FORD	3600
CLARK	2450
MILLER	1300
SMITH	800
WARD	1250
SCOTT	4791.6
MARTIN	1250
TURNER	1500
ADAMS	1100
BLAKE	2850
JAMES	950

14 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME,MAX(SAL)
2    FROM EMP
3  WHERE SAL < ( SELECT MAX(SAL)
4                FROM EMP
5                )
6* GROUP BY ENAME
```

ENAME	MAX(SAL)
SURYA	2000
JONES	2975
ALLEN	1600
FORD	3600
CLARK	2450
MILLER	1300
SMITH	800
WARD	1250
SCOTT	4791.6
MARTIN	1250
TURNER	1500
ADAMS	1100
BLAKE	2850
JAMES	950

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,MAX(SAL)
2    FROM EMP
3  WHERE SAL <( SELECT MAX(SAL)
4                FROM EMP
5                )
6* GROUP BY ENAME
SQL> /
```

ENAME	MAX(SAL)
SURYA	2000
JONES	2975
ALLEN	1600
FORD	3600
CLARK	2450
MILLER	1300
SMITH	800
WARD	1250
SCOTT	4791.6
MARTIN	1250
TURNER	1500
ADAMS	1100
BLAKE	2850
JAMES	950

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT MAX(SAL
2    FROM EMP
3  WHERE SAL < ( SELECT MAX(SAL)
4                FROM EMP
5*                )
6  /
FROM EMP
*
```

ERROR at line 2:

ORA-00907: missing right parenthesis

SQL> ED

Wrote file afiedt.buf

```

1  SELECT MAX(SAL)
2    FROM EMP
3  WHERE SAL < ( SELECT MAX(SAL)
4                FROM EMP
5*                )
SQL> /
```

```

      MAX(SAL)
-----
      4791.6
```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT MAX(SAL)
2    FROM EMP
3  WHERE SAL < ( SELECT MAX(SAL)
4                FROM EMP
5*                )
```

SQL>

SQL>

SQL>

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CL SCR

SQL> R

```

1  SELECT MAX(SAL)
2    FROM EMP
3  WHERE SAL < ( SELECT MAX(SAL)
4                FROM EMP
5*                )
```

```

      MAX(SAL)
-----
      4791.6
```

SQL> SELECT ENAME

```

2    ,SAL
3  FROM EMP
4  WHERE SAL = 4791.6;
```

```

ENAME          SAL
-----
SCOTT          4791.6
```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL =(SELECT MAX(SAL)
5                FROM EMP
6                WHERE SAL <( SELECT MAX(SAL)
7                              FROM EMP
8                              )
9*          )
10 /

```

ENAME	SAL
SCOTT	4791.6

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL =(SELECT MAX(SAL)
5                FROM EMP
6                WHERE SAL <( SELECT MAX(SAL)
7                              FROM EMP
8                              WHERE SAL < ( SELECT MAX(SAL)
9                                          FROM EMP
10                                         )
11                                         )
12*          )
SQL> /

```

ENAME	SAL
FORD	3600

```

SQL> CL SCR

```

```

SQL> EDIT
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL =(SELECT MAX(SAL)
5                FROM EMP
6                WHERE SAL <( SELECT MAX(SAL)
7                              FROM EMP
8                              WHERE SAL < ( SELECT MAX(SAL)
9                                          FROM EMP
10                                         )
11                                         )
12*          )
SQL>
SQL> /

```

ENAME	SAL
FORD	3600

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME

```



```

26                                     FROM EMP
27                                     )
28                                     )
29*                                )

```

ENAME	SAL
ARJUN	5000
FORD	3600
SCOTT	4791.6

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2  ,SAL
 3  FROM EMP
 4  WHERE SAL =(SELECT MAX(SAL)
 5                FROM EMP
 6                )
 7  UNION
 8  SELECT ENAME
 9  ,SAL
10  FROM EMP
11  WHERE SAL =(SELECT MAX(SAL)
12                FROM EMP
13                WHERE SAL <( SELECT MAX(SAL)
14                              FROM EMP
15                              )
16                )
17  UNION
18  SELECT ENAME
19  ,SAL
20  FROM EMP
21  WHERE SAL =(SELECT MAX(SAL)
22                FROM EMP
23                WHERE SAL <( SELECT MAX(SAL)
24                              FROM EMP
25                              WHERE SAL < ( SELECT MAX(SAL)
26                                            FROM EMP
27                                            )
28                              )
29                )
30  UNION
31  SELECT ENAME
32  ,SAL
33  FROM EMP
34  WHERE SAL =(SELECT MAX(SAL)
35                FROM EMP
36                WHERE SAL <( SELECT MAX(SAL)
37                              FROM EMP
38                              WHERE SAL < ( SELECT MAX(SAL)
39                                            FROM EMP
40                                            WHERE SAL < ( SELECT MAX(SAL)
41                                                            FROM EMP)
42                              )
43                )
SQL> /

```

ENAME	SAL
ARJUN	5000
FORD	3600

```
JONES          2975
SCOTT          4791.6
```

```
SQL> ED
Wrote file afiedt.buf
```

```

 1  SELECT ENAME
 2  ,SAL
 3  FROM EMP
 4  WHERE SAL =(SELECT MAX(SAL)
 5                FROM EMP
 6                )
 7  UNION
 8  SELECT ENAME
 9  ,SAL
10  FROM EMP
11  WHERE SAL =(SELECT MAX(SAL)
12                FROM EMP
13                WHERE SAL <( SELECT MAX(SAL)
14                              FROM EMP
15                              )
16                )
17  UNION
18  SELECT ENAME
19  ,SAL
20  FROM EMP
21  WHERE SAL =(SELECT MAX(SAL)
22                FROM EMP
23                WHERE SAL <( SELECT MAX(SAL)
24                              FROM EMP
25                              WHERE SAL < ( SELECT MAX(SAL)
26                                                FROM EMP
27                                                )
28                              )
29                )
30  UNION
31  SELECT ENAME
32  ,SAL
33  FROM EMP
34  WHERE SAL =(SELECT MAX(SAL)
35                FROM EMP
36                WHERE SAL <( SELECT MAX(SAL)
37                              FROM EMP
38                              WHERE SAL < ( SELECT MAX(SAL)
39                                                FROM EMP
40                                                WHERE SAL < ( SELECT MAX(SAL)
41                                                                FROM EMP)
42                              )
43                )
44* ORDER BY 2 DESC
SQL> /
```

ENAME	SAL
ARJUN	5000
SCOTT	4791.6
FORD	3600
JONES	2975

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
2  ,SAL
```

```

3  ,DEPTNO
4  FROM EMP;

```

ENAME	SAL	DEPTNO
SURYA	2000	
ARJUN	5000	
BLAKE	2850	30
CLARK	2450	11
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

15 rows selected.

```

SQL> SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL =(SELECT MAX(SAL)
5                FROM EMP
6                )
7  UNION
8  SELECT ENAME
9  ,SAL
10 FROM EMP
11 WHERE SAL =(SELECT MAX(SAL)
12                FROM EMP
13                WHERE SAL <( SELECT MAX(SAL)
14                               FROM EMP
15                               )
16                )
17 UNION
18 SELECT ENAME
19 ,SAL
20 FROM EMP
21 WHERE SAL =(SELECT MAX(SAL)
22                FROM EMP
23                WHERE SAL <( SELECT MAX(SAL)
24                               FROM EMP
25                               WHERE SAL < ( SELECT MAX(SAL)
26                                              FROM EMP
27                                              )
28                               )
29                )
30 UNION
31 SELECT ENAME
32 ,SAL
33 FROM EMP
34 WHERE SAL =(SELECT MAX(SAL)
35                FROM EMP
36                WHERE SAL <( SELECT MAX(SAL)
37                               FROM EMP
38                               WHERE SAL < ( SELECT MAX(SAL)
39                                              FROM EMP
40                                              WHERE SAL < ( SELECT MAX(SAL)
FROM EMP)

```

```

41                                     )
42                                     )
43                                     )
44 /

```

ENAME	SAL
ARJUN	5000
FORD	3600
JONES	2975
SCOTT	4791.6

```

SQL> SELECT ENAME
  2  ,SAL
  3  FROM EMP
  4  WHERE SAL =(SELECT MAX(SAL)
  5                FROM EMP
  6                )
  7  UNION
  8  SELECT ENAME
  9  ,SAL
 10  FROM EMP
 11  WHERE SAL =(SELECT MAX(SAL)
 12                FROM EMP
 13                WHERE SAL <( SELECT MAX(SAL)
 14                          FROM EMP
 15                          )
 16                )
 17  UNION
 18  SELECT ENAME
 19  ,SAL
 20  FROM EMP
 21  WHERE SAL =(SELECT MAX(SAL)
 22                FROM EMP
 23                WHERE SAL <( SELECT MAX(SAL)
 24                          FROM EMP
 25                          WHERE SAL < ( SELECT MAX(SAL)
 26                                    FROM EMP
 27                                    )
 28                          )
 29                )
 30  UNION
 31  SELECT ENAME
 32  ,SAL
 33  FROM EMP
 34  WHERE SAL =(SELECT MAX(SAL)
 35                FROM EMP
 36                WHERE SAL <( SELECT MAX(SAL)
 37                          FROM EMP
 38                          WHERE SAL < ( SELECT MAX(SAL)
 39                                    FROM EMP
 40                                    WHERE SAL < ( SELECT MAX(SAL)
 41                                            FROM EMP)
 42                                            )
 43                                            )
 44 /

```

ENAME	SAL
ARJUN	5000
FORD	3600
JONES	2975
SCOTT	4791.6

```
SQL> CL SCR
```

```
SQL> SELECT DEPTNO
2      ,COUNT(1)
3  FROM EMP
4  GROUP BY DEPTNO;
```

DEPTNO	COUNT(1)
2	2
30	6
11	1
20	5
10	1

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT COUNT(1)
2  FROM EMP
3*  GROUP BY DEPTNO
SQL> /
```

COUNT(1)
2
6
1
5
1

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT MAX(COUNT(1))
2  FROM EMP
3*  GROUP BY DEPTNO
SQL> /
```

MAX(COUNT(1))
6

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT DEPTNO
2      ,COUNT(1)
3  FROM EMP
4*  GROUP BY DEPTNO
SQL> /
```

DEPTNO	COUNT(1)
2	2
30	6
11	1
20	5
10	1

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT DEPTNO
2          ,1
3* FROM EMP
SQL> /

```

DEPTNO	1
1	1
1	1
30	1
11	1
20	1
20	1
20	1
20	1
30	1
30	1
30	1
30	1
20	1
30	1
10	1

15 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT DEPTNO
2          ,COUNT(1)
3  FROM EMP
4* GROUP BY DEPTNO
SQL> /

```

DEPTNO	COUNT(1)
2	
30	6
11	1
20	5
10	1

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT DEPTNO
2          ,MAX(COUNT(1))
3  FROM EMP
4* GROUP BY DEPTNO
SQL> /

```

```

SELECT DEPTNO
      *

```

ERROR at line 1:
ORA-00937: not a single-group group function

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT DEPTNO
2          ,MAX(COUNT(1))
3  FROM EMP
4* GROUP BY DEPTNO,MAX(COUNT(1))
SQL> /

```

```
GROUP BY DEPTNO,MAX(COUNT(1))
```

```
*
```

```
ERROR at line 4:
```

```
ORA-00934: group function is not allowed here
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT DEPTNO
2      ,COUNT(1)
3  FROM EMP
4* GROUP BY DEPTNO
SQL> /
```

DEPTNO	COUNT(1)
	2
30	6
11	1
20	5
10	1

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT COUNT(1)
2  FROM EMP
3* GROUP BY DEPTNO
SQL> /
```

COUNT(1)
2
6
1
5
1

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT MAX(COUNT(1))
2  FROM EMP
3* GROUP BY DEPTNO
SQL> /
```

MAX(COUNT(1))
6

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1  SELECT DEPTNO
2      ,COUNT(1)
3  FROM EMP
4* GROUP BY DEPTNO
SQL> /
```

DEPTNO	COUNT(1)
	2
30	6

11	1
20	5
10	1

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DEPTNO
2  ,COUNT(1)
3  FROM EMP
4  GROUP BY DEPTNO
5*  HAVING COUNT(1) = 6
SQL> /

```

DEPTNO	COUNT(1)
30	6

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DEPTNO
2  ,COUNT(1)
3  FROM EMP
4  GROUP BY DEPTNO
5  HAVING COUNT(1) =(SELECT MAX(COUNT(1))
6                      FROM EMP
7                      GROUP BY DEPTNO
8                      )
9* /
10 /
/
*

```

ERROR at line 9:

ORA-00936: missing expression

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DEPTNO
2  ,COUNT(1)
3  FROM EMP
4  GROUP BY DEPTNO
5  HAVING COUNT(1) =(SELECT MAX(COUNT(1))
6                      FROM EMP
7                      GROUP BY DEPTNO
8*                      )
SQL> /

```

DEPTNO	COUNT(1)
30	6

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DEPTNO
2  FROM EMP
3  GROUP BY DEPTNO
4  HAVING COUNT(1) =(SELECT MAX(COUNT(1))
5                      FROM EMP
6                      GROUP BY DEPTNO
7*                      )

```

SQL> /

```

      DEPTNO
-----
      30

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DEPTNO
2  ,COUNT(1)
3  FROM EMP
4  GROUP BY DEPTNO
5  HAVING COUNT(1) =(SELECT MAX(COUNT(1))
6                      FROM EMP
7                      GROUP BY DEPTNO
8*                          )

```

SQL> /

```

      DEPTNO      COUNT(1)
-----
      30          6

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DEPTNO
2  FROM EMP
3  GROUP BY DEPTNO
4  HAVING COUNT(1) =(SELECT MAX(COUNT(1))
5                      FROM EMP
6                      GROUP BY DEPTNO
7*                          )
8  /

```

```

      DEPTNO
-----
      30

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE DEPTNO = (SELECT DEPTNO
6                      FROM EMP
7                      GROUP BY DEPTNO
8                      HAVING COUNT(1) =(SELECT MAX(COUNT(1))
9                      FROM EMP
10                     GROUP BY DEPTNO
11                     )
12*                          )

```

SQL> /

ENAME	SAL	DEPTNO
BLAKE	2850	30
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
JAMES	950	30

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE DEPTNO = (SELECT DEPTNO
6                  FROM EMP
7                  GROUP BY DEPTNO
8                  HAVING COUNT(1) =(SELECT MIN(COUNT(1))
9                                     FROM EMP
10                                    GROUP BY DEPTNO
11                                   )
12* )

```

SQL> /

```

WHERE DEPTNO = (SELECT DEPTNO
                *

```

ERROR at line 5:

ORA-01427: single-row subquery returns more than one row

SQL> ED

Wrote file afiedt.buf

```

1  SELECT MIN(COUNT(1))
2  FROM EMP
3* GROU BY DEPTNO

```

SQL> /

```

GROU BY DEPTNO
*

```

ERROR at line 3:

ORA-00933: SQL command not properly ended

SQL> ED

Wrote file afiedt.buf

```

1  SELECT MIN(COUNT(1))
2  FROM EMP
3* GROUP BY DEPTNO

```

SQL> /

```

MIN(COUNT(1))
-----

```

1

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DEPTNO
2  ,COUNT(1)
3  FROM EMP
4  GROUP BY DEPTNO
5  HAVING COUNT(1) = (
6  SELECT MIN(COUNT(1))
7  FROM EMP
8  GROUP BY DEPTNO
9* )

```

SQL> /

DEPTNO	COUNT(1)
11	1
10	1

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE DEPTNO = (
6  SELECT DEPTNO
7  ,COUNT(1)
8  FROM EMP
9  GROUP BY DEPTNO
10 HAVING COUNT(1) = (
11 SELECT MIN(COUNT(1))
12   FROM EMP
13 GROUP BY DEPTNO
14 )
15* )
SQL> /
SELECT DEPTNO
*
ERROR at line 6:
ORA-00913: too many values

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE DEPTNO = (
6  SELECT DEPTNO
7  FROM EMP
8  GROUP BY DEPTNO
9  HAVING COUNT(1) = (
10 SELECT MIN(COUNT(1))
11   FROM EMP
12 GROUP BY DEPTNO
13 )
14* )
SQL> /
SELECT DEPTNO
*
ERROR at line 6:
ORA-01427: single-row subquery returns more than one row

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE DEPTNO IN (
6  SELECT DEPTNO
7  FROM EMP

```

```

8  GROUP BY DEPTNO
9  HAVING COUNT(1) = (
10 SELECT MIN(COUNT(1))
11   FROM EMP
12 GROUP BY DEPTNO
13 )
14* )
SQL> /

```

ENAME	SAL	DEPTNO
MILLER	1300	10
CLARK	2450	11

```
SQL> CL SCR
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE DEPTNO = (SELECT DEPTNO
6                   FROM EMP
7                   GROUP BY DEPTNO
8                   HAVING COUNT(1) =(SELECT MIN(COUNT(1))
9                                     FROM EMP
10                                    GROUP BY DEPTNO
11                                    )
12*          )

```

```
SQL> ED
Wrote file afiedt.buf
```

```

1  (SELECT ENAME,SAL,DEPTNO
2         FROM EMP
3         GROUP BY DEPTNO
4         HAVING COUNT(1) =(SELECT MIN(COUNT(1))
5                           FROM EMP
6                           GROUP BY DEPTNO
7                           )
8*          )

```

```
SQL> /
(SELECT ENAME,SAL,DEPTNO
*
```

```
ERROR at line 1:
ORA-00979: not a GROUP BY expression
```

```
SQL> ED
Wrote file afiedt.buf
```

```

1  (SELECT ENAME,SAL,DEPTNO
2         FROM EMP
3         GROUP BY DEPTNO,ENAME,SAL
4         HAVING COUNT(1) =(SELECT MIN(COUNT(1))
5                           FROM EMP
6                           GROUP BY DEPTNO
7                           )
8*          )

```

```
SQL> /
```

ENAME	SAL	DEPTNO
-------	-----	--------

JAMES	950	30
MILLER	1300	10
ARJUN	5000	
SMITH	800	20
ALLEN	1600	30
SCOTT	4791.6	20
FORD	3600	20
SURYA	2000	
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
BLAKE	2850	30
CLARK	2450	11
JONES	2975	20

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  (SELECT ENAME,SAL,DEPTNO
2      FROM EMP
3      GROUP BY DEPTNO,ENAME,SAL
4      HAVING COUNT(1) =(SELECT MIN(COUNT(1))
5                          FROM EMP
6                          GROUP BY DEPTNO
7                          )
8*      )

```

SQL>

SQL> /

ENAME	SAL	DEPTNO
JAMES	950	30
MILLER	1300	10
ARJUN	5000	
SMITH	800	20
ALLEN	1600	30
SCOTT	4791.6	20
FORD	3600	20
SURYA	2000	
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
BLAKE	2850	30
CLARK	2450	11
JONES	2975	20

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  (SELECT ENAME,SAL,DEPTNO
2      FROM EMP
3      GROUP BY DEPTNO,ENAME,SAL
4      HAVING COUNT(1) =(SELECT MIN(COUNT(1))
5                          FROM EMP
6                          GROUP BY DEPTNO
7                          )
8*      )

```

SQL>

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
2
```

```
SQL>
```

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
```

```
2 ,SAL
```

```
3 ,DEPTNO
```

```
4 FROM EMP;
```

ENAME	SAL	DEPTNO
SURYA	2000	
ARJUN	5000	
BLAKE	2850	30
CLARK	2450	11
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

```
15 rows selected.
```

```
SQL> SELECT DEPTNOCOUNT(1)
```

```
2
```

```
SQL>
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT DEPTNO
```

```
2 ,COUNT(1)
```

```
3 FROM EMP
```

```
4* GROUP BY DEPTNO
```

```
SQL> /
```

DEPTNO	COUNT(1)
	2
30	6
11	1
20	5
10	1

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT DEPTNO
```

```
2 ,COUNT(1)
```

```
3 FROM EMP
```

```
4 WHERE COUNT(1) > 3
```

```
5* GROUP BY DEPTNO
```

```
SQL> /
```

```
WHERE COUNT(1) > 3
```

```
*
```

```
ERROR at line 4:
```

ORA-00934: group function is not allowed here

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DEPTNO
2      ,COUNT(1)
3  FROM EMP
4  GROUP BY DEPTNO
5*  HAVING COUNT(1) > 3
SQL> /

```

DEPTNO	COUNT(1)
30	6
20	5

SQL> CL SCR

SQL> R

```

1  SELECT DEPTNO
2      ,COUNT(1)
3  FROM EMP
4  GROUP BY DEPTNO
5*  HAVING COUNT(1) > 3

```

DEPTNO	COUNT(1)
30	6
20	5

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DEPTNO
2      ,COUNT(1)
3  FROM EMP
4*  WHERE COUNT(1) > 3
SQL> /
WHERE COUNT(1) > 3
*
```

ERROR at line 4:

ORA-00934: group function is not allowed here

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DEPTNO
2      ,COUNT(1)
3  FROM EMP
4*  HAVING COUNT(1) > 3
SQL> /
SELECT DEPTNO
*
```

ERROR at line 1:

ORA-00937: not a single-group group function

SQL> ED

Wrote file afiedt.buf

```

1  SELECT DEPTNO

```



```

2          ,COUNT(1)
3 FROM EMP
4 GROUP BY DEPTNO
5* HAVING COUNT(1) > 3
SQL> /

```

DEPTNO	COUNT(1)
30	6
20	5

```
SQL> CL SCR
```

```

SQL> R
1 SELECT DEPTNO
2          ,COUNT(1)
3 FROM EMP
4 GROUP BY DEPTNO
5* HAVING COUNT(1) > 3

```

DEPTNO	COUNT(1)
30	6
20	5

```

SQL> SELECT ENAME
2      ,SAL
3 FROM EMP;

```

ENAME	SAL
SURYA	2000
ARJUN	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

15 rows selected.

```
SQL> CL SCR
```

```

SQL> R
1 SELECT ENAME
2      ,SAL
3* FROM EMP

```

ENAME	SAL
SURYA	2000
ARJUN	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6

FORD	3600
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

15 rows selected.

SQL> ED
Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3* FROM EMP

```

SQL>

SQL>

SQL> CL SCR

```

SQL> SELECT COUNT(1)
2  FROM EMP;

```

```

COUNT(1)
-----
15

```

```

SQL> SELECT ENAME
2  ,SAL
3  FROM EMP;

```

ENAME	SAL
SURYA	2000
ARJUN	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

15 rows selected.

SQL> ED
Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,15 AS TOTCNT
4* FROM EMP

```

SQL> /

ENAME	SAL	TOTCNT
-------	-----	--------

SURYA	2000	15
ARJUN	5000	15
BLAKE	2850	15
CLARK	2450	15
JONES	2975	15
SCOTT	4791.6	15
FORD	3600	15
SMITH	800	15
ALLEN	1600	15
WARD	1250	15
MARTIN	1250	15
TURNER	1500	15
ADAMS	1100	15
JAMES	950	15
MILLER	1300	15

15 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,15 AS TOTCNT
4* FROM EMP

```

ENAME	SAL	TOTCNT
SURYA	2000	15
ARJUN	5000	15
BLAKE	2850	15
CLARK	2450	15
JONES	2975	15
SCOTT	4791.6	15
FORD	3600	15
SMITH	800	15
ALLEN	1600	15
WARD	1250	15
MARTIN	1250	15
TURNER	1500	15
ADAMS	1100	15
JAMES	950	15
MILLER	1300	15

15 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COUNT(1) AS TOTCNT
4* FROM EMP

```

SQL>

SQL> //

```

SELECT ENAME
      *

```

ERROR at line 1:

ORA-00937: not a single-group group function

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COUNT(1) AS TOTCNT
4  FROM EMP
5  GROUP BY ENAME
6* ,SAL
SQL> /

```

ENAME	SAL	TOTCNT
-----	-----	-----
JONES	2975	1
ARJUN	5000	1
ALLEN	1600	1
SCOTT	4791.6	1
MARTIN	1250	1
SURYA	2000	1
MILLER	1300	1
TURNER	1500	1
ADAMS	1100	1
JAMES	950	1
FORD	3600	1
SMITH	800	1
WARD	1250	1
BLAKE	2850	1
CLARK	2450	1

15 rows selected.

```

SQL> EDIT
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,COUNT(1) AS TOTCNT
4  FROM EMP
5  GROUP BY ENAME
6* ,SAL
SQL>
SQL>
SQL> CL SCR

```

```

SQL> R
1  SELECT ENAME
2  ,SAL
3  ,COUNT(1) AS TOTCNT
4  FROM EMP
5  GROUP BY ENAME
6* ,SAL

```

ENAME	SAL	TOTCNT
-----	-----	-----
JONES	2975	1
ARJUN	5000	1
ALLEN	1600	1
SCOTT	4791.6	1
MARTIN	1250	1
SURYA	2000	1
MILLER	1300	1
TURNER	1500	1
ADAMS	1100	1
JAMES	950	1
FORD	3600	1
SMITH	800	1
WARD	1250	1

```
BLAKE          2850          1
CLARK          2450          1
```

15 rows selected.

```
SQL> INSERT INTO EMP (EMPNO, ENAME, SAL)
      2 VALUES
      3 (
      4 501
      5 , 'FORD'
      6 , 3600
      7 );
```

```
INSERT INTO EMP (EMPNO, ENAME, SAL)
```

*

ERROR at line 1:

ORA-00001: unique constraint (SYSTEM.PK_EMP) violated

```
SQL> ED
```

Wrote file afiedt.buf

```
      1 INSERT INTO EMP (EMPNO, ENAME, SAL)
      2 VALUES
      3 (
      4 601
      5 , 'FORD'
      6 , 3600
      7* )
SQL> /
```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT EMPNO, ENAME, SAL
      2 FROM EMP
      3
```

```
SQL> SELECT EMPNO, ENAME, SAL
      2 FROM EMP
      3 ORDER BY 2, 3;
```

EMPNO	ENAME	SAL
7876	ADAMS	1100
7499	ALLEN	1600
501	ARJUN	5000
7698	BLAKE	2850
7782	CLARK	2450
601	FORD	3600
7902	FORD	3600
7900	JAMES	950
7566	JONES	2975
7654	MARTIN	1250
7934	MILLER	1300
7788	SCOTT	4791.6
7369	SMITH	800
7839	SURYA	2000
7844	TURNER	1500
7521	WARD	1250

16 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,COUNT(1) AS TOTCNT
4  FROM EMP
5  GROUP BY ENAME
6* ,SAL
SQL> /

```

ENAME	SAL	TOTCNT
JONES	2975	1
ARJUN	5000	1
ALLEN	1600	1
SCOTT	4791.6	1
MARTIN	1250	1
SURYA	2000	1
MILLER	1300	1
TURNER	1500	1
ADAMS	1100	1
JAMES	950	1
FORD	3600	2
SMITH	800	1
WARD	1250	1
BLAKE	2850	1
CLARK	2450	1

15 rows selected.

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,SELECT COUNT(1) FROM EMP
4* FROM EMP
SQL> /

```

```

,SELECT COUNT(1) FROM EMP
*

```

ERROR at line 3:
ORA-00936: missing expression

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,(SELECT COUNT(1) FROM EMP) AS TOTCNT
4* FROM EMP
SQL> /

```

ENAME	SAL	TOTCNT
FORD	3600	16
SURYA	2000	16
ARJUN	5000	16
BLAKE	2850	16

CLARK	2450	16
JONES	2975	16
SCOTT	4791.6	16
FORD	3600	16
SMITH	800	16
ALLEN	1600	16
WARD	1250	16
MARTIN	1250	16
TURNER	1500	16
ADAMS	1100	16
JAMES	950	16
MILLER	1300	16

16 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  , (SELECT COUNT(1) FROM EMP) AS TOTCNT
4* FROM EMP

```

ENAME	SAL	TOTCNT
-----	-----	-----
FORD	3600	16
SURYA	2000	16
ARJUN	5000	16
BLAKE	2850	16
CLARK	2450	16
JONES	2975	16
SCOTT	4791.6	16
FORD	3600	16
SMITH	800	16
ALLEN	1600	16
WARD	1250	16
MARTIN	1250	16
TURNER	1500	16
ADAMS	1100	16
JAMES	950	16
MILLER	1300	16

16 rows selected.

SQL> DELETE FROM EMP WHERE MGR IS NULL AND EMPNO <> 7839;

2 rows deleted.

SQL> COMMIT;

Commit complete.

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  , (SELECT COUNT(1) FROM EMP) AS TOTCNT
4* FROM EMP

```

SQL> /

ENAME	SAL	TOTCNT
-------	-----	--------

```

-----
SURYA          2000          14
BLAKE          2850          14
CLARK          2450          14
JONES          2975          14
SCOTT          4791.6        14
FORD           3600          14
SMITH           800          14
ALLEN          1600          14
WARD           1250          14
MARTIN         1250          14
TURNER         1500          14
ADAMS          1100          14
JAMES           950          14
MILLER         1300          14

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  , (SELECT COUNT(1) FROM EMP) AS TOTCNT
4* FROM EMP

```

SQL>

SQL>

SQL>

SQL> CL SCR

SQL> SELECT SUM(SAL)

2 FROM EMP;

SUM(SAL)

```

-----
28416.6

```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  , (SELECT COUNT(1) FROM EMP) AS TOTCNT
4  ,28416.6 AS TOTSAL
5* FROM EMP

```

SQL> /

```

ENAME          SAL          TOTCNT          TOTSAL
-----
SURYA          2000          14          28416.6
BLAKE          2850          14          28416.6
CLARK          2450          14          28416.6
JONES          2975          14          28416.6
SCOTT          4791.6        14          28416.6
FORD           3600          14          28416.6
SMITH           800          14          28416.6
ALLEN          1600          14          28416.6
WARD           1250          14          28416.6
MARTIN         1250          14          28416.6
TURNER         1500          14          28416.6
ADAMS          1100          14          28416.6
JAMES           950          14          28416.6
MILLER         1300          14          28416.6

```


14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2  ,SAL
 3  , (SELECT COUNT(1) FROM EMP) AS TOTCNT
 4  , (SELECT SUM(sAL) FROM EMP) AS TOTSAL
 5* FROM EMP
SQL> /

```

ENAME	SAL	TOTCNT	TOTSAL
SURYA	2000	14	28416.6
BLAKE	2850	14	28416.6
CLARK	2450	14	28416.6
JONES	2975	14	28416.6
SCOTT	4791.6	14	28416.6
FORD	3600	14	28416.6
SMITH	800	14	28416.6
ALLEN	1600	14	28416.6
WARD	1250	14	28416.6
MARTIN	1250	14	28416.6
TURNER	1500	14	28416.6
ADAMS	1100	14	28416.6
JAMES	950	14	28416.6
MILLER	1300	14	28416.6

14 rows selected.

SQL> CL SCR

SQL> R

```

 1  SELECT ENAME
 2  ,SAL
 3  , (SELECT COUNT(1) FROM EMP) AS TOTCNT
 4  , (SELECT SUM(sAL) FROM EMP) AS TOTSAL
 5* FROM EMP

```

ENAME	SAL	TOTCNT	TOTSAL
SURYA	2000	14	28416.6
BLAKE	2850	14	28416.6
CLARK	2450	14	28416.6
JONES	2975	14	28416.6
SCOTT	4791.6	14	28416.6
FORD	3600	14	28416.6
SMITH	800	14	28416.6
ALLEN	1600	14	28416.6
WARD	1250	14	28416.6
MARTIN	1250	14	28416.6
TURNER	1500	14	28416.6
ADAMS	1100	14	28416.6
JAMES	950	14	28416.6
MILLER	1300	14	28416.6

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT ENAME
 2  ,SAL

```

```

3  , (SELECT COUNT(1) FROM EMP) AS TOTCNT
4  , (SELECT SUM(sAL) FROM EMP) AS TOTSAL
5  , (SELECT MIN(SAL) FROM EMP) AS LOSAL
6  , (SELECT MAX(SAL) FROM EMP) AS HISAL
7* FROM EMP
SQL> /

```

ENAME	SAL	TOTCNT	TOTSAL	LOSAL	HISAL
SURYA	2000	14	28416.6	800	4791.6
BLAKE	2850	14	28416.6	800	4791.6
CLARK	2450	14	28416.6	800	4791.6
JONES	2975	14	28416.6	800	4791.6
SCOTT	4791.6	14	28416.6	800	4791.6
FORD	3600	14	28416.6	800	4791.6
SMITH	800	14	28416.6	800	4791.6
ALLEN	1600	14	28416.6	800	4791.6
WARD	1250	14	28416.6	800	4791.6
MARTIN	1250	14	28416.6	800	4791.6
TURNER	1500	14	28416.6	800	4791.6
ADAMS	1100	14	28416.6	800	4791.6
JAMES	950	14	28416.6	800	4791.6
MILLER	1300	14	28416.6	800	4791.6

14 rows selected.

SQL> CL SCR

```

SQL> R
1  SELECT ENAME
2  ,SAL
3  , (SELECT COUNT(1) FROM EMP) AS TOTCNT
4  , (SELECT SUM(sAL) FROM EMP) AS TOTSAL
5  , (SELECT MIN(SAL) FROM EMP) AS LOSAL
6  , (SELECT MAX(SAL) FROM EMP) AS HISAL
7* FROM EMP

```

ENAME	SAL	TOTCNT	TOTSAL	LOSAL	HISAL
SURYA	2000	14	28416.6	800	4791.6
BLAKE	2850	14	28416.6	800	4791.6
CLARK	2450	14	28416.6	800	4791.6
JONES	2975	14	28416.6	800	4791.6
SCOTT	4791.6	14	28416.6	800	4791.6
FORD	3600	14	28416.6	800	4791.6
SMITH	800	14	28416.6	800	4791.6
ALLEN	1600	14	28416.6	800	4791.6
WARD	1250	14	28416.6	800	4791.6
MARTIN	1250	14	28416.6	800	4791.6
TURNER	1500	14	28416.6	800	4791.6
ADAMS	1100	14	28416.6	800	4791.6
JAMES	950	14	28416.6	800	4791.6
MILLER	1300	14	28416.6	800	4791.6

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  , (SELECT COUNT(1) FROM EMP) AS TOTCNT
4  , (SELECT SUM(sAL) FROM EMP) AS TOTSAL
5  , (SELECT MIN(SAL) FROM EMP) AS LOSAL

```

```

6  , (SELECT MAX(SAL) FROM EMP) AS HISAL
7  ,TOTSAL / SAL * 100 AS YOUR_PERCENTAGE
8* FROM EMP
SQL> /
,TOTSAL / SAL * 100 AS YOUR_PERCENTAGE
*
```

ERROR at line 7:

ORA-00904: "TOTSAL": invalid identifier

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  , (SELECT COUNT(1) FROM EMP) AS TOTCNT
4  , (SELECT SUM(sAL) FROM EMP) AS TOTSAL
5  , (SELECT MIN(SAL) FROM EMP) AS LOSAL
6  , (SELECT MAX(SAL) FROM EMP) AS HISAL
7  , (SELECT SUM(sAL) FROM EMP) / SAL * 100 AS YOUR_PERCENTAGE
8* FROM EMP
SQL> /
```

ENAME	SAL	TOTCNT	TOTSAL	LOSAL	HISAL	YOUR_PERCENTAGE
SURYA	2000	14	28416.6	800	4791.6	1420.83
BLAKE	2850	14	28416.6	800	4791.6	997.073684
CLARK	2450	14	28416.6	800	4791.6	1159.86122
JONES	2975	14	28416.6	800	4791.6	955.179832
SCOTT	4791.6	14	28416.6	800	4791.6	593.050338
FORD	3600	14	28416.6	800	4791.6	789.35
SMITH	800	14	28416.6	800	4791.6	3552.075
ALLEN	1600	14	28416.6	800	4791.6	1776.0375
WARD	1250	14	28416.6	800	4791.6	2273.328
MARTIN	1250	14	28416.6	800	4791.6	2273.328
TURNER	1500	14	28416.6	800	4791.6	1894.44
ADAMS	1100	14	28416.6	800	4791.6	2583.32727
JAMES	950	14	28416.6	800	4791.6	2991.22105
MILLER	1300	14	28416.6	800	4791.6	2185.89231

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  , (SELECT COUNT(1) FROM EMP) AS TOTCNT
4  , (SELECT SUM(sAL) FROM EMP) AS TOTSAL
5  , (SELECT MIN(SAL) FROM EMP) AS LOSAL
6  , (SELECT MAX(SAL) FROM EMP) AS HISAL
7  , ( (SELECT SUM(sAL) FROM EMP) / SAL ) * 100 AS YOUR_PERCENTAGE
8* FROM EMP
SQL> /
```

ENAME	SAL	TOTCNT	TOTSAL	LOSAL	HISAL	YOUR_PERCENTAGE
SURYA	2000	14	28416.6	800	4791.6	1420.83
BLAKE	2850	14	28416.6	800	4791.6	997.073684
CLARK	2450	14	28416.6	800	4791.6	1159.86122
JONES	2975	14	28416.6	800	4791.6	955.179832
SCOTT	4791.6	14	28416.6	800	4791.6	593.050338
FORD	3600	14	28416.6	800	4791.6	789.35
SMITH	800	14	28416.6	800	4791.6	3552.075

ALLEN	1600	14	28416.6	800	4791.6	1776.0375
WARD	1250	14	28416.6	800	4791.6	2273.328
MARTIN	1250	14	28416.6	800	4791.6	2273.328
TURNER	1500	14	28416.6	800	4791.6	1894.44
ADAMS	1100	14	28416.6	800	4791.6	2583.32727
JAMES	950	14	28416.6	800	4791.6	2991.22105
MILLER	1300	14	28416.6	800	4791.6	2185.89231

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,(SELECT COUNT(1) FROM EMP) AS TOTCNT
4  ,(SELECT SUM(sAL) FROM EMP) AS TOTSAL
5  ,(SELECT MIN(SAL) FROM EMP) AS LOSAL
6  ,(SELECT MAX(SAL) FROM EMP) AS HISAL
7  ,( SAL/(SELECT SUM(sAL) FROM EMP) ) * 100 AS YOUR_PERCENTAGE
8* FROM EMP
SQL> /

```

ENAME	SAL	TOTCNT	TOTSAL	LOSAL	HISAL	YOUR_PERCENTAGE
SURYA	2000	14	28416.6	800	4791.6	7.03813968
BLAKE	2850	14	28416.6	800	4791.6	10.029349
CLARK	2450	14	28416.6	800	4791.6	8.62172111
JONES	2975	14	28416.6	800	4791.6	10.4692328
SCOTT	4791.6	14	28416.6	800	4791.6	16.861975
FORD	3600	14	28416.6	800	4791.6	12.6686514
SMITH	800	14	28416.6	800	4791.6	2.81525587
ALLEN	1600	14	28416.6	800	4791.6	5.63051174
WARD	1250	14	28416.6	800	4791.6	4.3988373
MARTIN	1250	14	28416.6	800	4791.6	4.3988373
TURNER	1500	14	28416.6	800	4791.6	5.27860476
ADAMS	1100	14	28416.6	800	4791.6	3.87097682
JAMES	950	14	28416.6	800	4791.6	3.34311635
MILLER	1300	14	28416.6	800	4791.6	4.57479079

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,(SELECT COUNT(1) FROM EMP) AS TOTCNT
4  ,(SELECT SUM(sAL) FROM EMP) AS TOTSAL
5  ,(SELECT MIN(SAL) FROM EMP) AS LOSAL
6  ,(SELECT MAX(SAL) FROM EMP) AS HISAL
7  ,ROUND ( ( SAL/(SELECT SUM(sAL) FROM EMP) ) * 100 ) AS YOUR_PERCENTAGE
8* FROM EMP
SQL> /

```

ENAME	SAL	TOTCNT	TOTSAL	LOSAL	HISAL	YOUR_PERCENTAGE
SURYA	2000	14	28416.6	800	4791.6	7
BLAKE	2850	14	28416.6	800	4791.6	10
CLARK	2450	14	28416.6	800	4791.6	9
JONES	2975	14	28416.6	800	4791.6	10
SCOTT	4791.6	14	28416.6	800	4791.6	17
FORD	3600	14	28416.6	800	4791.6	13
SMITH	800	14	28416.6	800	4791.6	3

ALLEN	1600	14	28416.6	800	4791.6	6
WARD	1250	14	28416.6	800	4791.6	4
MARTIN	1250	14	28416.6	800	4791.6	4
TURNER	1500	14	28416.6	800	4791.6	5
ADAMS	1100	14	28416.6	800	4791.6	4
JAMES	950	14	28416.6	800	4791.6	3
MILLER	1300	14	28416.6	800	4791.6	5

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,(SELECT COUNT(1) FROM EMP) AS TOTCNT
4  ,(SELECT SUM(sAL) FROM EMP) AS TOTSAL
5  ,(SELECT MIN(SAL) FROM EMP) AS LOSAL
6  ,(SELECT MAX(SAL) FROM EMP) AS HISAL
7  ,ROUND ( ( SAL/(SELECT SUM(sAL) FROM EMP) ) * 100 ) AS YOUR_PERCENTAGE
8  FROM EMP
9* ORDER BY YOUR_PERCENTAGE DESC
SQL> /

```

ENAME	SAL	TOTCNT	TOTSAL	LOSAL	HISAL	YOUR_PERCENTAGE
SCOTT	4791.6	14	28416.6	800	4791.6	17
FORD	3600	14	28416.6	800	4791.6	13
JONES	2975	14	28416.6	800	4791.6	10
BLAKE	2850	14	28416.6	800	4791.6	10
CLARK	2450	14	28416.6	800	4791.6	9
SURYA	2000	14	28416.6	800	4791.6	7
ALLEN	1600	14	28416.6	800	4791.6	6
MILLER	1300	14	28416.6	800	4791.6	5
TURNER	1500	14	28416.6	800	4791.6	5
MARTIN	1250	14	28416.6	800	4791.6	4
WARD	1250	14	28416.6	800	4791.6	4
ADAMS	1100	14	28416.6	800	4791.6	4
JAMES	950	14	28416.6	800	4791.6	3
SMITH	800	14	28416.6	800	4791.6	3

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,(SELECT COUNT(1),SUM(SAL),MIN(SAL),MAX(SAL) FROM EMP) AS TOTCNT
4* FROM EMP
5  /
,(SELECT COUNT(1),SUM(SAL),MIN(SAL),MAX(SAL) FROM EMP) AS TOTCNT
*

```

ERROR at line 3:

ORA-00913: too many values

SQL> CL SCR

```

SQL> SELECT MIN(SAL) AS LOSAL
2      ,MAX(sAL) AS HISAL
3      ,COUNT(1) AS TOTCNT
4      ,SUM(SAL) AS TOTSAL
5  FROM EMP;

```

-----	-----	-----	-----
LOSAL	HISAL	TOTCNT	TOTSAL
800	4791.6	14	28416.6

SQL> ED

Wrote file afiedt.buf

```

1  SELECT MIN(SAL) AS LOSAL
2      ,MAX(sAL) AS HISAL
3      ,COUNT(1) AS TOTCNT
4      ,SUM(SAL) AS TOTSAL
5* FROM EMP

```

SQL>

SQL>

SQL> CL SCR

SQL> SELECT ENAME

```

2  ,SAL
3  FRO MEMP;
FRO MEMP
*
```

ERROR at line 3:

ORA-00923: FROM keyword not found where expected

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CLS CR

SP2-0042: unknown command "CLS CR" - rest of line ignored.

SQL> CL SCR

SQL> SELECT ENAEM

```

2  ,SAL
3

```

SQL> CL SCR

SQL> SELECT ENAME

```

2  ,SAL
3  FROM EMP;

```

-----	-----
ENAME	SAL
SURYA	2000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	4791.6
FORD	3600
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT MIN(SAL) AS LOSAL

```

```

2      ,MAX(sAL) AS HISAL
3      ,COUNT(1) AS TOTCNT
4      ,SUM(SAL) AS TOTSAL
5* FROM EMP
SQL> /

```

-----	-----	-----	-----
LOSAL	HISAL	TOTCNT	TOTSAL
800	4791.6	14	28416.6

```

SQL> ED
Wrote file afiedt.buf

```

```

1  (
2  SELECT MIN(SAL) AS LOSAL
3      ,MAX(sAL) AS HISAL
4      ,COUNT(1) AS TOTCNT
5      ,SUM(SAL) AS TOTSAL
6  FROM EMP
7* )
SQL> /

```

-----	-----	-----	-----
LOSAL	HISAL	TOTCNT	TOTSAL
800	4791.6	14	28416.6

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT *
2* FROM E1
SQL> /
FROM E1
*
```

```

ERROR at line 2:
ORA-00942: table or view does not exist

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT *
2  FROM (
3      SELECT MIN(SAL) AS LOSAL
4      ,MAX(sAL) AS HISAL
5      ,COUNT(1) AS TOTCNT
6      ,SUM(SAL) AS TOTSAL
7      FROM EMP
8*  ) E1
9  /

```

-----	-----	-----	-----
LOSAL	HISAL	TOTCNT	TOTSAL
800	4791.6	14	28416.6

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E2.ENAME
2      ,E2.SAL
3      ,E2.DEPTNO
4      ,E1.*
5  FROM (
6      SELECT MIN(SAL) AS LOSAL

```

```

7          ,MAX(sAL) AS HISAL
8          ,COUNT(1) AS TOTCNT
9          ,SUM(sAL) AS TOTSAL
10         FROM EMP
11       ) E1
12*  ,EMP E2
13  /

```

ENAME	SAL	DEPTNO	LOSAL	HISAL	TOTCNT	TOTSAL
SURYA	2000		800	4791.6	14	28416.6
BLAKE	2850	30	800	4791.6	14	28416.6
CLARK	2450	11	800	4791.6	14	28416.6
JONES	2975	20	800	4791.6	14	28416.6
SCOTT	4791.6	20	800	4791.6	14	28416.6
FORD	3600	20	800	4791.6	14	28416.6
SMITH	800	20	800	4791.6	14	28416.6
ALLEN	1600	30	800	4791.6	14	28416.6
WARD	1250	30	800	4791.6	14	28416.6
MARTIN	1250	30	800	4791.6	14	28416.6
TURNER	1500	30	800	4791.6	14	28416.6
ADAMS	1100	20	800	4791.6	14	28416.6
JAMES	950	30	800	4791.6	14	28416.6
MILLER	1300	10	800	4791.6	14	28416.6

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E2.ENAME
2      ,E2.SAL
3      ,E2.DEPTNO
4      ,E1.*
5      ,E2.SAL /E1.TOTSAL AS PERCENTAGE
6  FROM (
7      SELECT MIN(sAL) AS LOSAL
8      ,MAX(sAL) AS HISAL
9      ,COUNT(1) AS TOTCNT
10     ,SUM(sAL) AS TOTSAL
11     FROM EMP
12   ) E1
13*  ,EMP E2
SQL> /

```

ENAME	SAL	DEPTNO	LOSAL	HISAL	TOTCNT	TOTSAL
SURYA	2000		800	4791.6	14	28416.6
BLAKE	2850	30	800	4791.6	14	28416.6
CLARK	2450	11	800	4791.6	14	28416.6
JONES	2975	20	800	4791.6	14	28416.6
SCOTT	4791.6	20	800	4791.6	14	28416.6
FORD	3600	20	800	4791.6	14	28416.6
SMITH	800	20	800	4791.6	14	28416.6

ALLEN	1600	30	800	4791.6	14	28416.6
.056305117						
WARD	1250	30	800	4791.6	14	28416.6
.043988373						
MARTIN	1250	30	800	4791.6	14	28416.6
.043988373						
TURNER	1500	30	800	4791.6	14	28416.6
.052786048						
ADAMS	1100	20	800	4791.6	14	28416.6
.038709768						
JAMES	950	30	800	4791.6	14	28416.6
.033431163						
MILLER	1300	10	800	4791.6	14	28416.6
.045747908						

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E2.ENAME
2      ,E2.SAL
3      ,E2.DEPTNO
4      ,E1.*
5      ,E2.SAL /E1.TOTSAL * 100 AS PERCENTAGE
6  FROM (
7      SELECT MIN(SAL) AS LOSAL
8      ,MAX(SAL) AS HISAL
9      ,COUNT(1) AS TOTCNT
10     ,SUM(SAL) AS TOTSAL
11     FROM EMP
12     ) E1
13* ,EMP E2
SQL> /

```

ENAME	SAL	DEPTNO	LOSAL	HISAL	TOTCNT	TOTSAL
PERCENTAGE						
-----	-----	-----	-----	-----	-----	-----

SURYA	2000		800	4791.6	14	28416.6
7.03813968						
BLAKE	2850	30	800	4791.6	14	28416.6
10.029349						
CLARK	2450	11	800	4791.6	14	28416.6
8.62172111						
JONES	2975	20	800	4791.6	14	28416.6
10.4692328						
SCOTT	4791.6	20	800	4791.6	14	28416.6
16.861975						
FORD	3600	20	800	4791.6	14	28416.6
12.6686514						
SMITH	800	20	800	4791.6	14	28416.6
2.81525587						
ALLEN	1600	30	800	4791.6	14	28416.6
5.63051174						
WARD	1250	30	800	4791.6	14	28416.6
4.3988373						
MARTIN	1250	30	800	4791.6	14	28416.6
4.3988373						
TURNER	1500	30	800	4791.6	14	28416.6
5.27860476						
ADAMS	1100	20	800	4791.6	14	28416.6
3.87097682						

```

JAMES          950          30          800          4791.6          14          28416.6
3.34311635
MILLER         1300         10          800          4791.6          14          28416.6
4.57479079

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT E2.ENAME
 2          ,E2.SAL
 3          ,E2.DEPTNO
 4          ,E1.*
 5          ,ROUND(E2.SAL /E1.TOTSAL * 100 ) AS PERCENTAGE
 6  FROM (
 7      SELECT MIN(SAL) AS LOSAL
 8          ,MAX(sAL) AS HISAL
 9          ,COUNT(1) AS TOTCNT
10          ,SUM(SAL) AS TOTSAL
11      FROM EMP
12  ) E1
13* ,EMP E2
SQL> /

```

ENAME PERCENTAGE	SAL	DEPTNO	LOSAL	HISAL	TOTCNT	TOTSAL
-----	-----	-----	-----	-----	-----	-----

SURYA 7	2000		800	4791.6	14	28416.6
BLAKE 10	2850	30	800	4791.6	14	28416.6
CLARK 9	2450	11	800	4791.6	14	28416.6
JONES 10	2975	20	800	4791.6	14	28416.6
SCOTT 17	4791.6	20	800	4791.6	14	28416.6
FORD 13	3600	20	800	4791.6	14	28416.6
SMITH 3	800	20	800	4791.6	14	28416.6
ALLEN 6	1600	30	800	4791.6	14	28416.6
WARD 4	1250	30	800	4791.6	14	28416.6
MARTIN 4	1250	30	800	4791.6	14	28416.6
TURNER 5	1500	30	800	4791.6	14	28416.6
ADAMS 4	1100	20	800	4791.6	14	28416.6
JAMES 3	950	30	800	4791.6	14	28416.6
MILLER 5	1300	10	800	4791.6	14	28416.6

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT E2.ENAME

```

```

2      ,E2.SAL
3      ,E2.DEPTNO
4      ,E1.*
5      ,ROUND(E2.SAL /E1.TOTSAL * 100 ) AS PERCENTAGE
6 FROM (
7      SELECT MIN(SAL) AS LOSAL
8              ,MAX(sAL) AS HISAL
9              ,COUNT(1) AS TOTCNT
10             ,SUM(SAL) AS TOTSAL
11      FROM EMP
12      ) E1
13      ,EMP E2
14* ORDER BY PERCENTAGE
SQL> /

```

ENAME PERCENTAGE	SAL	DEPTNO	LOSAL	HISAL	TOTCNT	TOTSAL
SMITH	800	20	800	4791.6	14	28416.6
JAMES	950	30	800	4791.6	14	28416.6
ADAMS	1100	20	800	4791.6	14	28416.6
MARTIN	1250	30	800	4791.6	14	28416.6
WARD	1250	30	800	4791.6	14	28416.6
TURNER	1500	30	800	4791.6	14	28416.6
MILLER	1300	10	800	4791.6	14	28416.6
ALLEN	1600	30	800	4791.6	14	28416.6
SURYA	2000		800	4791.6	14	28416.6
CLARK	2450	11	800	4791.6	14	28416.6
JONES	2975	20	800	4791.6	14	28416.6
BLAKE	2850	30	800	4791.6	14	28416.6
FORD	3600	20	800	4791.6	14	28416.6
SCOTT	4791.6	20	800	4791.6	14	28416.6

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E2.ENAME
2      ,E2.SAL
3      ,E2.DEPTNO
4      ,E1.*
5      ,ROUND(E2.SAL /E1.TOTSAL * 100 ) AS PERCENTAGE
6 FROM (
7      SELECT MIN(SAL) AS LOSAL
8              ,MAX(sAL) AS HISAL
9              ,COUNT(1) AS TOTCNT
10             ,SUM(SAL) AS TOTSAL
11      FROM EMP

```

```

12      ) E1
13      ,EMP E2
14* ORDER BY PERCENTAGE DESC
SQL> /

```

ENAME PERCENTAGE	SAL	DEPTNO	LOSAL	HISAL	TOTCNT	TOTSAL
-----	-----	-----	-----	-----	-----	-----

SCOTT	4791.6	20	800	4791.6	14	28416.6
17						
FORD	3600	20	800	4791.6	14	28416.6
13						
JONES	2975	20	800	4791.6	14	28416.6
10						
BLAKE	2850	30	800	4791.6	14	28416.6
10						
CLARK	2450	11	800	4791.6	14	28416.6
9						
SURYA	2000		800	4791.6	14	28416.6
7						
ALLEN	1600	30	800	4791.6	14	28416.6
6						
MILLER	1300	10	800	4791.6	14	28416.6
5						
TURNER	1500	30	800	4791.6	14	28416.6
5						
MARTIN	1250	30	800	4791.6	14	28416.6
4						
WARD	1250	30	800	4791.6	14	28416.6
4						
ADAMS	1100	20	800	4791.6	14	28416.6
4						
JAMES	950	30	800	4791.6	14	28416.6
3						
SMITH	800	20	800	4791.6	14	28416.6
3						

14 rows selected.

```

SQL> R
1  SELECT E2.ENAME
2      ,E2.SAL
3      ,E2.DEPTNO
4      ,E1.*
5      ,ROUND(E2.SAL /E1.TOTSAL * 100 ) AS PERCENTAGE
6  FROM (
7      SELECT MIN(SAL) AS LOSAL
8      ,MAX(sAL) AS HISAL
9      ,COUNT(1) AS TOTCNT
10     ,SUM(SAL) AS TOTSAL
11     FROM EMP
12     ) E1
13     ,EMP E2
14* ORDER BY PERCENTAGE DESC

```

ENAME PERCENTAGE	SAL	DEPTNO	LOSAL	HISAL	TOTCNT	TOTSAL
-----	-----	-----	-----	-----	-----	-----

SCOTT	4791.6	20	800	4791.6	14	28416.6
17						
FORD	3600	20	800	4791.6	14	28416.6
13						

JONES	2975	20	800	4791.6	14	28416.6
10						
BLAKE	2850	30	800	4791.6	14	28416.6
10						
CLARK	2450	11	800	4791.6	14	28416.6
9						
SURYA	2000		800	4791.6	14	28416.6
7						
ALLEN	1600	30	800	4791.6	14	28416.6
6						
MILLER	1300	10	800	4791.6	14	28416.6
5						
TURNER	1500	30	800	4791.6	14	28416.6
5						
MARTIN	1250	30	800	4791.6	14	28416.6
4						
WARD	1250	30	800	4791.6	14	28416.6
4						
ADAMS	1100	20	800	4791.6	14	28416.6
4						
JAMES	950	30	800	4791.6	14	28416.6
3						
SMITH	800	20	800	4791.6	14	28416.6
3						

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMP.ENAME
2         ,EMP.SAL
3         ,EMP.DEPTNO
4         ,E1.*
5         ,ROUND(EMP.SAL /E1.TOTSAL * 100 ) AS PERCENTAGE
6  FROM (
7      SELECT MIN(SAL) AS LOSAL
8             ,MAX(SAL) AS HISAL
9             ,COUNT(1) AS TOTCNT
10            ,SUM(SAL) AS TOTSAL
11      FROM EMP
12     ) E1
13  ,EMP
14* ORDER BY PERCENTAGE DESC
SQL> /

```

ENAME	SAL	DEPTNO	LOSAL	HISAL	TOTCNT	TOTSAL
PERCENTAGE						
-----	-----	-----	-----	-----	-----	-----

SCOTT	4791.6	20	800	4791.6	14	28416.6
17						
FORD	3600	20	800	4791.6	14	28416.6
13						
JONES	2975	20	800	4791.6	14	28416.6
10						
BLAKE	2850	30	800	4791.6	14	28416.6
10						
CLARK	2450	11	800	4791.6	14	28416.6
9						
SURYA	2000		800	4791.6	14	28416.6
7						
ALLEN	1600	30	800	4791.6	14	28416.6
6						

MILLER 5	1300	10	800	4791.6	14	28416.6
TURNER 5	1500	30	800	4791.6	14	28416.6
MARTIN 4	1250	30	800	4791.6	14	28416.6
WARD 4	1250	30	800	4791.6	14	28416.6
ADAMS 4	1100	20	800	4791.6	14	28416.6
JAMES 3	950	30	800	4791.6	14	28416.6
SMITH 3	800	20	800	4791.6	14	28416.6

14 rows selected.

SQL> CL SCR

SQL> SELECT *
2 FROM EMP;

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM
DEPTNO	LAST_APPR					
30	7839 SURYA				2000	
11	7698 BLAKE	MANAGER	7839	01-MAY-81	2850	
20	7782 CLARK	MANAGER	7839	09-JUN-81	2450	
20	7566 JONES	MANAGER	7839	02-APR-81	2975	
20	7788 SCOTT	ANALYST	7566	19-APR-87	4791.6	
20	7902 FORD	ANALYST	7566	03-DEC-81	3600	
20	07-JUN-18 7369 SMITH	CLERK	7902	17-DEC-80	800	
30	7499 ALLEN	SALESMAN	7698	20-FEB-81	1600	300
30	7521 WARD	SALESMAN	7698	22-FEB-81	1250	500
30	7654 MARTIN	SALESMAN	7698	28-SEP-81	1250	1400
30	7844 TURNER	SALESMAN	7698	08-SEP-81	1500	0
20	7876 ADAMS	CLERK	7788	23-MAY-87	1100	
30	7900 JAMES	CLERK	7698	03-DEC-81	950	
10	7934 MILLER	CLERK	7782	23-JAN-82	1300	

14 rows selected.

SQL> UPDATE EMP
2 SET DEPTNO =10
3 WHERE DEPTNO IS NULL;

1 row updated.

SQL> COMMIT;

Commit complete.

SQL> CL SCR

```
SQL> SELECT ENAME
      2  ,SAL
      3  ,DEPTNO
      4  FROM EMP
      5  ORDER BY DEPTNO;
```

ENAME	SAL	DEPTNO
MILLER	1300	10
SURYA	2000	10
CLARK	2450	11
FORD	3600	20
SCOTT	4791.6	20
JONES	2975	20
SMITH	800	20
ADAMS	1100	20
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
JAMES	950	30
ALLEN	1600	30
BLAKE	2850	30

14 rows selected.

SQL> CL SCR

```
SQL> R
      1  SELECT ENAME
      2  ,SAL
      3  ,DEPTNO
      4  FROM EMP
      5* ORDER BY DEPTNO
```

ENAME	SAL	DEPTNO
MILLER	1300	10
SURYA	2000	10
CLARK	2450	11
FORD	3600	20
SCOTT	4791.6	20
JONES	2975	20
SMITH	800	20
ADAMS	1100	20
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
JAMES	950	30
ALLEN	1600	30
BLAKE	2850	30

14 rows selected.

SQL> EDIT

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```
      1  SELECT ENAME
      2  ,SAL
      3  ,DEPTNO
      4  FROM EMP
```

```

5* ORDER BY DEPTNO
SQL>
SQL>
SQL> CL SCR

```

```

SQL> R
1  SELECT ENAME
2    ,SAL
3    ,DEPTNO
4  FROM EMP
5* ORDER BY DEPTNO

```

ENAME	SAL	DEPTNO
MILLER	1300	10
SURYA	2000	10
CLARK	2450	11
FORD	3600	20
SCOTT	4791.6	20
JONES	2975	20
SMITH	800	20
ADAMS	1100	20
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
JAMES	950	30
ALLEN	1600	30
BLAKE	2850	30

14 rows selected.

```

SQL> SELECT DEPTNO
2    ,COUNT(1)
3  FROM EMP
4  GROUP BY DEPTNO;

```

DEPTNO	COUNT(1)
30	6
11	1
20	5
10	2

```

SQL> ED
Wrote file afiedt.buf

```

```

1  (
2  SELECT DEPTNO
3    ,COUNT(1)
4  FROM EMP
5  GROUP BY DEPTNO
6* )
SQL> /

```

DEPTNO	COUNT(1)
30	6
11	1
20	5
10	2

```

SQL> ED
Wrote file afiedt.buf

```



```

1  SELECT E1.ENAME
2          ,E1.SAL
3          ,E1.DEPTNO
4          ,E2.DTOTCNT
5  FROM EMP E1
6          , (
7          SELECT DEPTNO
8              ,COUNT(1) AS DTOTCNT
9          FROM EMP
10         GROUP BY  DEPTNO
11        ) E2
12* WHERE E1.DEPTNO = E2.DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DTOTCNT
BLAKE	2850	30	6
SCOTT	4791.6	20	5
WARD	1250	30	6
SMITH	800	20	5
MARTIN	1250	30	6
SURYA	2000	10	2
ALLEN	1600	30	6
MILLER	1300	10	2
CLARK	2450	11	1
JONES	2975	20	5
ADAMS	1100	20	5
FORD	3600	20	5
TURNER	1500	30	6
JAMES	950	30	6

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E1.ENAME
2          ,E1.SAL
3          ,E1.DEPTNO
4          ,E2.DTOTCNT
5  FROM EMP E1
6          , (
7          SELECT DEPTNO
8              ,COUNT(1) AS DTOTCNT
9          FROM EMP
10         GROUP BY  DEPTNO
11        ) E2
12  WHERE E1.DEPTNO = E2.DEPTNO
13* ORDER BY DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DTOTCNT
SURYA	2000	10	2
MILLER	1300	10	2
CLARK	2450	11	1
JONES	2975	20	5
SCOTT	4791.6	20	5
FORD	3600	20	5
SMITH	800	20	5
ADAMS	1100	20	5
BLAKE	2850	30	6
ALLEN	1600	30	6
WARD	1250	30	6

MARTIN	1250	30	6
TURNER	1500	30	6
JAMES	950	30	6

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.ENAME
2      ,E1.SAL
3      ,E1.DEPTNO
4      ,E2.DTOTCNT
5  FROM EMP E1
6      , (
7      SELECT DEPTNO
8      ,COUNT(1) AS DTOTCNT
9      FROM EMP
10     GROUP BY DEPTNO
11   ) E2
12 WHERE DEPTNO = DEPTNO
13* ORDER BY DEPTNO

```

SQL> /

WHERE DEPTNO = DEPTNO

*

ERROR at line 12:

ORA-00918: column ambiguously defined

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.ENAME
2      ,E1.SAL
3      ,E1.DEPTNO
4      ,E2.DTOTCNT
5  FROM EMP E1
6      , (
7      SELECT DEPTNO
8      ,COUNT(1) AS DTOTCNT
9      FROM EMP
10     GROUP BY DEPTNO
11   ) E2
12 WHERE E1.DEPTNO = E2.DEPTNO
13* ORDER BY DEPTNO

```

SQL> /

ENAME	SAL	DEPTNO	DTOTCNT
SURYA	2000	10	2
MILLER	1300	10	2
CLARK	2450	11	1
JONES	2975	20	5
SCOTT	4791.6	20	5
FORD	3600	20	5
SMITH	800	20	5
ADAMS	1100	20	5
BLAKE	2850	30	6
ALLEN	1600	30	6
WARD	1250	30	6
MARTIN	1250	30	6
TURNER	1500	30	6
JAMES	950	30	6

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT E1.ENAME
 2          ,E1.SAL
 3          ,DEPTNO
 4          ,E2.DTOTCNT
 5  FROM EMP E1
 6          , (
 7          SELECT DEPTNO
 8              ,COUNT(1) AS DTOTCNT
 9          FROM EMP
10          GROUP BY DEPTNO
11          ) E2
12  WHERE E1.DEPTNO = E2.DEPTNO
13* ORDER BY DEPTNO
SQL> /
      ,DEPTNO
      *
```

ERROR at line 3:

ORA-00918: column ambiguously defined

SQL> ED

Wrote file afiedt.buf

```

 1  SELECT E1.ENAME
 2          ,E1.SAL
 3          ,E1.DEPTNO
 4          ,E2.DTOTCNT
 5  FROM EMP E1
 6          , (
 7          SELECT DEPTNO
 8              ,COUNT(1) AS DTOTCNT
 9          FROM EMP
10          GROUP BY DEPTNO
11          ) E2
12  WHERE E1.DEPTNO = E2.DEPTNO
13* ORDER BY DEPTNO
SQL> /
```

ENAME	SAL	DEPTNO	DTOTCNT
SURYA	2000	10	2
MILLER	1300	10	2
CLARK	2450	11	1
JONES	2975	20	5
SCOTT	4791.6	20	5
FORD	3600	20	5
SMITH	800	20	5
ADAMS	1100	20	5
BLAKE	2850	30	6
ALLEN	1600	30	6
WARD	1250	30	6
MARTIN	1250	30	6
TURNER	1500	30	6
JAMES	950	30	6

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.ENAME
2      ,E1.SAL
3      ,E1.DEPTNO
4      ,E2.DTOTCNT
5      ,E3.*
6  FROM EMP E1
7      , (
8      SELECT DEPTNO
9      ,COUNT(1) AS DTOTCNT
10     FROM EMP
11     GROUP BY DEPTNO
12   ) E2
13  , ( SELECT COUNT(1) AS TOTCNT
14     FROM EMP
15   ) E3
16 WHERE E1.DEPTNO = E2.DEPTNO
17* ORDER BY DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DTOTCNT	TOTCNT
MILLER	1300	10	2	14
SURYA	2000	10	2	14
CLARK	2450	11	1	14
FORD	3600	20	5	14
SCOTT	4791.6	20	5	14
JONES	2975	20	5	14
SMITH	800	20	5	14
ADAMS	1100	20	5	14
WARD	1250	30	6	14
MARTIN	1250	30	6	14
TURNER	1500	30	6	14
JAMES	950	30	6	14
ALLEN	1600	30	6	14
BLAKE	2850	30	6	14

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E1.ENAME
2      ,E1.SAL
3      ,E1.DEPTNO
4      ,E2.DTOTCNT
5      ,E2.DTOTSAL
6      ,E3.*
7  FROM EMP E1
8      , (
9      SELECT DEPTNO
10     ,COUNT(1) AS DTOTCNT
11     ,SUM(SAL) AS DTOTSAL
12   FROM EMP
13   GROUP BY DEPTNO
14 ) E2
15 , ( SELECT COUNT(1) AS TOTCNT
16   FROM EMP
17 ) E3
18 WHERE E1.DEPTNO = E2.DEPTNO
19* ORDER BY DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DTOTCNT	DTOTSAL	TOTCNT
-------	-----	--------	---------	---------	--------

ENAME	SAL	DEPTNO	DTOTCNT	DTOTSAL	TOTCNT	TOTSAL
MILLER	1300	10	2	3300	14	28416.6
SURYA	2000	10	2	3300	14	28416.6
CLARK	2450	11	1	2450	14	28416.6
FORD	3600	20	5	13266.6	14	28416.6
SCOTT	4791.6	20	5	13266.6	14	28416.6
JONES	2975	20	5	13266.6	14	28416.6
SMITH	800	20	5	13266.6	14	28416.6
ADAMS	1100	20	5	13266.6	14	28416.6
WARD	1250	30	6	9400	14	28416.6
MARTIN	1250	30	6	9400	14	28416.6
TURNER	1500	30	6	9400	14	28416.6
JAMES	950	30	6	9400	14	28416.6
ALLEN	1600	30	6	9400	14	28416.6
BLAKE	2850	30	6	9400	14	28416.6

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.ENAME
2      ,E1.SAL
3      ,E1.DEPTNO
4      ,E2.DTOTCNT
5      ,E2.DTOTSAL
6      ,E3.*
7  FROM EMP E1
8      , (
9      SELECT DEPTNO
10         ,COUNT(1) AS DTOTCNT
11         ,SUM(SAL) AS DTOTSAL
12      FROM EMP
13      GROUP BY DEPTNO
14      ) E2
15  , ( SELECT COUNT(1) AS TOTCNT
16      ,SUM(SAL) AS TOTSAL
17      FROM EMP
18      ) E3
19  WHERE E1.DEPTNO = E2.DEPTNO
20* ORDER BY DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO	DTOTCNT	DTOTSAL	TOTCNT	TOTSAL
MILLER	1300	10	2	3300	14	28416.6
SURYA	2000	10	2	3300	14	28416.6
CLARK	2450	11	1	2450	14	28416.6
FORD	3600	20	5	13266.6	14	28416.6
SCOTT	4791.6	20	5	13266.6	14	28416.6
JONES	2975	20	5	13266.6	14	28416.6
SMITH	800	20	5	13266.6	14	28416.6
ADAMS	1100	20	5	13266.6	14	28416.6
WARD	1250	30	6	9400	14	28416.6
MARTIN	1250	30	6	9400	14	28416.6
TURNER	1500	30	6	9400	14	28416.6
JAMES	950	30	6	9400	14	28416.6
ALLEN	1600	30	6	9400	14	28416.6
BLAKE	2850	30	6	9400	14	28416.6

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.ENAME
2      ,E1.SAL
3      ,E1.DEPTNO
4      ,E2.DTOTCNT
5      ,E2.DTOTSAL
6      ,E3.*
7      ,E2.DTOTSAL / E3.TOTSAL* 100 AS DPERCENTAGE
8  FROM EMP E1
9      , (
10     SELECT DEPTNO
11           ,COUNT(1) AS DTOTCNT
12           ,SUM(SAL) AS DTOTSAL
13     FROM EMP
14     GROUP BY DEPTNO
15   ) E2
16  , ( SELECT COUNT(1) AS TOTCNT
17      ,SUM(SAL) AS TOTSAL
18    FROM EMP
19  ) E3
20 WHERE E1.DEPTNO = E2.DEPTNO
21* ORDER BY DEPTNO
SQL> /

```

ENAME DPERCENTAGE	SAL	DEPTNO	DTOTCNT	DTOTSAL	TOTCNT	TOTSAL
-----	-----	-----	-----	-----	-----	-----

MILLER	1300	10	2	3300	14	28416.6
11.6129305						
SURYA	2000	10	2	3300	14	28416.6
11.6129305						
CLARK	2450	11	1	2450	14	28416.6
8.62172111						
FORD	3600	20	5	13266.6	14	28416.6
46.6860919						
SCOTT	4791.6	20	5	13266.6	14	28416.6
46.6860919						
JONES	2975	20	5	13266.6	14	28416.6
46.6860919						
SMITH	800	20	5	13266.6	14	28416.6
46.6860919						
ADAMS	1100	20	5	13266.6	14	28416.6
46.6860919						
WARD	1250	30	6	9400	14	28416.6
33.0792565						
MARTIN	1250	30	6	9400	14	28416.6
33.0792565						
TURNER	1500	30	6	9400	14	28416.6
33.0792565						
JAMES	950	30	6	9400	14	28416.6
33.0792565						
ALLEN	1600	30	6	9400	14	28416.6
33.0792565						
BLAKE	2850	30	6	9400	14	28416.6
33.0792565						

14 rows selected.

SQL> SPOOL OFF

17-jun-2018 Subquery conti Notes

DEPTNO TOTCNT

10	3
20	5
30	6

DEPTNO10	DEPTNO30	DEPTNO30
3	5	6

- 1) SUB SELECT
- 2) INLINE VIEWS
- 3) PIVOT
- 4) WITH CLAUSE

TOTCNT
14

TOT_EMP_EARNING_COMM	TOT_EMP_NOT_EARNING_COMM
4	10

IS NULL
IS NOT NULL

- 1) WRITE A SELECT STATEMENT TO GET THE HIGHEST EARNING EMPLOYEE DETAILS ?
- 2) WRITE A SELECT STATEMENT TO GET THE HIGHEST EARNING EMPLOYEE DETAILS IN EACH DEPARTMENT NUMBER ?

```
SELECT ENAME, SAL, DEPTNO
FROM EMP
WHERE (DEPTNO, SAL) IN (SELECT DEPTNO
                        , MAX(SAL)
                        FROM EMP
                        GROUP BY DEPTNO
                        )
/
```

```
SELECT ENAME
, SAL
, DEPTNO
FROM EMP
WHERE SAL = ( SELECT MAX(SAL)
              FROM EMP
              )
/
```

```
1) ( SELECT MAX(SAL)
      FROM EMP
    )
```

4791.6 AS RESULTS

```
SELECT ENAME
, SAL
, DEPTNO
FROM EMP
WHERE SAL = 4791.6
```

2) FROM EMP TABLE IT WILL TAKE EACH ROW SALARY VALUE AND IT WILL CHECK IS IT MATCHING WITH = 4791.6

IF IT IS MATCHING THEN THAT ROW WILL BE DISPLAYED
ELSE
THAT WILL NOT COME IN FINAL RESULTS

```
SELECT ENAME, SAL, DEPTNO
FROM EMP
WHERE (DEPTNO, SAL) IN (SELECT DEPTNO
                        , MAX(SAL)
                        FROM EMP
                        GROUP BY DEPTNO
                      )
```

1) AFTER EXECUTING BELOW QUERY IT WILL PREPARE RESULTS LIKE

```
( (10,2450), (20,4791.6), (30,2850) )
(SELECT DEPTNO
      , MAX(SAL)
      FROM EMP
      GROUP BY DEPTNO
    )
```

```
SELECT ENAME, SAL, DEPTNO
FROM EMP
WHERE (DEPTNO, SAL) IN ( (10,2450), (20,4791.6), (30,2850) )
```

2) IT WILL TAKE FROM EACH ROW DEPTNO AND SALARY INFORMATION IT WILL CHECK

WITH WHICH PAIR IT IS MATCHING IF IT IS NOT MATCHING THEN THAT ROW WILL NOT COME

IF IT IS MATCHING WITH ANY ONE OF THE PAIR THEN THAT ROW WILL COME

3) WRITE A SELECT STATMENT TO GET LIST OF EMPLOYEES WHO ARE EARNING MORE SALARY THAN ANY ONE OF THE CLERK?

4) WRITE A SELECT STATMENT TO GET LIST OF EMPLOYEES WHO ARE EARNING MORE SALARY THAN ALL CLERKS ?

5) WRITE A SELECT STATEMENT TO GET LIST OF EMPLOYEE DETAILS WHOS DEPARTMENT DETAILS ARE AVAILABLE IN DEPT TABLE

6) WRITE A SELECT STATEMENT TO GET LIST OF EMPLOYEE DETAILS WHOS DEPARTMENT DETAILS ARE NOT AVAILABLE IN DEPT TABLE

17-jun-2018 Subquery conti Examples

```
SQL> SELECT ENAME  
2 FROM EMP;
```

```
ENAME  
-----  
SURYA  
BLAKE  
CLARK  
JONES  
SCOTT  
FORD  
SMITH  
ALLEN  
WARD  
MARTIN  
TURNER  
ADAMS  
JAMES  
MILLER
```

14 rows selected.

```
SQL> ED  
Wrote file afiedt.buf
```

```
1 (  
2 SELECT ENAME  
3 FROM EMP  
4* )  
SQL> /
```

```
ENAME  
-----  
SURYA  
BLAKE  
CLARK  
JONES  
SCOTT  
FORD  
SMITH  
ALLEN
```

WARD
MARTIN
TURNER
ADAMS
JAMES
MILLER

14 rows selected.

SQL> CL SCR

SQL> SELECT ENAME,DEPTNO
2 FROM EMP;

ENAME	DEPTNO
-----	-----
SURYA	10
BLAKE	30
CLARK	11
JONES	20
SCOTT	20
FORD	20
SMITH	20
ALLEN	30
WARD	30
MARTIN	30
TURNER	30
ADAMS	20
JAMES	30
MILLER	10

14 rows selected.

SQL> UPDATE EMP
2 SET DEPTNO =10
3 WHERE DEPTNO =11;

1 row updated.

SQL> COMMIT;

Commit complete.

SQL> CL SCR

SQL> SELECT ENAME,DEPTNO
2 FROM EMP;

ENAME	DEPTNO
-----	-----
SURYA	10
BLAKE	30
CLARK	10
JONES	20
SCOTT	20
FORD	20
SMITH	20
ALLEN	30
WARD	30
MARTIN	30
TURNER	30
ADAMS	20
JAMES	30
MILLER	10

14 rows selected.

```
SQL> SELECT DEPTNO
      2 ,COUNT(1) AS TOTCNT
      3 FROM EMP
      4 GROUP BY DEPTNO;
```

DEPTNO	TOTCNT
30	6
20	5
10	3

```
SQL> ED
Wrote file afiedt.buf
```

```
      1 SELECT DEPTNO
      2 ,COUNT(1) AS TOTCNT
      3 FROM EMP
      4 GROUP BY DEPTNO
      5* ORDER BY DEPTNO
SQL> /
```

DEPTNO	TOTCNT
10	3
20	5
30	6

```
SQL> CL SCR
```

```
SQL> SELECT 3 AS DEPTNO10
      2 ,5 AS DEPTNO20
      3 ,6 AS DEPTNO30
      4 FROM DUAL;
```

DEPTNO10	DEPTNO20	DEPTNO30
3	5	6

```
SQL> ED
Wrote file afiedt.buf
```

```
      1* SELECT COUNT(1) FROM EMP WHERE DEPTNO = 10
SQL> /
```

COUNT(1)
3

```
SQL> ED
Wrote file afiedt.buf
```

```
      1 SELECT
      2 (SELECT COUNT(1) FROM EMP WHERE DEPTNO = 10) AS DEPTNO10
      3 , (SELECT COUNT(1) FROM EMP WHERE DEPTNO = 20) AS DEPTNO20
      4 , (SELECT COUNT(1) FROM EMP WHERE DEPTNO = 30) AS DEPTNO30
      5* FROM EMP
SQL> /
```

DEPTNO10	DEPTNO20	DEPTNO30
3	5	6

3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT
2      (SELECT COUNT(1) FROM EMP WHERE DEPTNO = 10) AS DEPTNO10
3      , (SELECT COUNT(1) FROM EMP WHERE DEPTNO = 20) AS DEPTNO20
4      , (SELECT COUNT(1) FROM EMP WHERE DEPTNO = 30) AS DEPTNO30
5* FROM EMP
SQL> /

```

DEPTNO10	DEPTNO20	DEPTNO30
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6
3	5	6

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT
2      (SELECT COUNT(1) FROM EMP WHERE DEPTNO = 10) AS DEPTNO10
3      , (SELECT COUNT(1) FROM EMP WHERE DEPTNO = 20) AS DEPTNO20
4      , (SELECT COUNT(1) FROM EMP WHERE DEPTNO = 30) AS DEPTNO30
5* FROM DUAL
SQL> /

```

DEPTNO10	DEPTNO20	DEPTNO30
3	5	6

SQL> CL SCR

SQL> R

```

1  SELECT
2      (SELECT COUNT(1) FROM EMP WHERE DEPTNO = 10) AS DEPTNO10

```

```

3          , (SELECT COUNT(1) FROM EMP WHERE DEPTNO = 20) AS DEPTNO20
4          , (SELECT COUNT(1) FROM EMP WHERE DEPTNO = 30) AS DEPTNO30
5* FROM DUAL

```

DEPTNO10	DEPTNO20	DEPTNO30
3	5	6

SQL>

SQL>

SQL> ED

Wrote file afiedt.buf

```

1 SELECT *
2 FROM   (SELECT COUNT(1) DEPTNO10 FROM EMP WHERE DEPTNO = 10) D10
3        , (SELECT COUNT(1) DEPTNO20 FROM EMP WHERE DEPTNO = 20) D20
4*      , (SELECT COUNT(1) DEPTNO30 FROM EMP WHERE DEPTNO = 30) D30

```

SQL> /

DEPTNO10	DEPTNO20	DEPTNO30
3	5	6

SQL> CL SCR

SQL> WITH E1

2 AS

3 (

4 SELECT ENAME,DEPTNO

5 FROM EMP

6)

7 ,E2

8 AS

9 (

10 SELECT COUNT(1) AS DEPTNO10

11 FROM EMP

12 .

SQL>

SQL> ED

Wrote file afiedt.buf

```

1 WITH E1
2 AS
3 (
4     SELECT ENAME,DEPTNO
5     FROM EMP
6 )
7 ,E2
8 AS
9 (
10    SELECT COUNT(1) AS DEPTNO10
11    FROM E1
12    WHERE DEPTNO =10
13 )
14 ,E3
15 AS
16 (
17    SELECT COUNT(1) AS DEPTNO10
18    FROM E1
19    WHERE DEPTNO =20
20 )
21 ,E4
22 AS
23 (

```

```

24      SELECT COUNT(1) AS DEPTNO10
25      FROM E1
26      WHERE DEPTNO =30
27  )
28  SELECT E2.*,E3.*,E4.*
29* FROM E2,E3,E4
30  /

```

DEPTNO10	DEPTNO10	DEPTNO10
3	5	6

SQL> ED

Wrote file afiedt.buf

```

1  WITH E1
2  AS
3  (
4      SELECT ENAME,DEPTNO
5      FROM EMP
6  )
7  ,E2
8  AS
9  (
10     SELECT COUNT(1) AS DEPTNO10
11     FROM E1
12     WHERE DEPTNO =10
13  )
14  ,E3
15  AS
16  (
17     SELECT COUNT(1) AS DEPTNO10
18     FROM E1
19     WHERE DEPTNO =20
20  )
21  ,E4
22  AS
23  (
24     SELECT COUNT(1) AS DEPTNO10
25     FROM E1
26     WHERE DEPTNO =30
27  )
28  SELECT E2.*,E3.*,E4.*
29* FROM E2,E3,E4
30  /

```

DEPTNO10	DEPTNO10	DEPTNO10
3	5	6

SQL> ED

Wrote file afiedt.buf

```

1  WITH E1
2  AS
3  (
4      SELECT ENAME,DEPTNO
5      FROM EMP
6  )
7  ,E2
8  AS
9  (
10     SELECT COUNT(1) AS DEPTNO10
11     FROM E1

```

```

12     WHERE DEPTNO =10
13   )
14 ,E3
15 AS
16   (
17     SELECT COUNT(1) AS DEPTNO10
18     FROM E1
19     WHERE DEPTNO =20
20   )
21 ,E4
22 AS
23   (
24     SELECT COUNT(1) AS DEPTNO10
25     FROM E1
26     WHERE DEPTNO =30
27   )
28 SELECT E2.*,E3.*,E4.*
29* FROM E2,E3,E4
SQL>
SQL> /

```

DEPTNO10	DEPTNO10	DEPTNO10
3	5	6

```
SQL> CL SCR
```

```

SQL> SELECT ENAME
2     ,SAL
3     ,DEPTNO
4     ,COMM
5 FROM EMP;

```

ENAME	SAL	DEPTNO	COMM
SURYA	2000	10	
BLAKE	2850	30	
CLARK	2450	10	
JONES	2975	20	
SCOTT	4791.6	20	
FORD	3600	20	
SMITH	800	20	
ALLEN	1600	30	300
WARD	1250	30	500
MARTIN	1250	30	1400
TURNER	1500	30	0
ADAMS	1100	20	
JAMES	950	30	
MILLER	1300	10	

```
14 rows selected.
```

```

SQL> SELECT COUNT(1) AS TOTCNT
2 FROM EMP;

```

TOTCNT
14

```

SQL> SELECT ( SELECT COUNT(1) FROM EMP WHERE COMM IS NOT NULL) AS
TOT_EMP_EARNING_COMM
2     ,(SELECT COUNT(1) FROM EMP WHERE COMM IS NULL) AS TOT_EMP_NOT_EARNING_COMM
3 FROM EMP;

```

[illegible]

4	10
4	10
4	10
4	10
4	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT (SELECT COUNT(1) FROM EMP WHERE COMM IS NOT NULL) AS
TOT_EMP_EARNING_COMM
2      , (SELECT COUNT(1) FROM EMP WHERE COMM IS NULL) AS TOT_EMP_NOT_EARNING_COMM
3* FROM EMP
SQL> /

```

TOT_EMP_EARNING_COMM	TOT_EMP_NOT_EARNING_COMM
-----	-----
4	10
4	10
4	10
4	10
4	10
4	10
4	10
4	10
4	10
4	10
4	10
4	10
4	10
4	10
4	10
4	10
4	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT (SELECT COUNT(1) FROM EMP WHERE COMM IS NOT NULL) AS
TOT_EMP_EARNING_COMM
2      , (SELECT COUNT(1) FROM EMP WHERE COMM IS NULL) AS TOT_EMP_NOT_EARNING_COMM
3* FROM DUAL
SQL> /

```

TOT_EMP_EARNING_COMM	TOT_EMP_NOT_EARNING_COMM
-----	-----
4	10

SQL> CL SCR

SQL> R

```

1  SELECT (SELECT COUNT(1) FROM EMP WHERE COMM IS NOT NULL) AS
TOT_EMP_EARNING_COMM
2      , (SELECT COUNT(1) FROM EMP WHERE COMM IS NULL) AS TOT_EMP_NOT_EARNING_COMM
3* FROM DUAL

```

TOT_EMP_EARNING_COMM	TOT_EMP_NOT_EARNING_COMM
-----	-----
4	10

SQL>

SQL>

SQL> SELECT COUNT(1)

```
2 FROM EMP;
```

```
COUNT(1)
```

```
-----  
14
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT COUNT(1)
```

```
2 , COUNT(COMM)
```

```
3* FROM EMP
```

```
SQL> /
```

```
COUNT(1) COUNT(COMM)
```

```
-----  
14 4
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT COUNT(1) - COUNT(COMM) AS TOT_EMP_NOT_EARNING_COMM
```

```
2 , COUNT(COMM) AS TOT_EMP_EARNING_COMM
```

```
3* FROM EMP
```

```
SQL> /
```

```
TOT_EMP_NOT_EARNING_COMM TOT_EMP_EARNING_COMM
```

```
-----  
10 4
```

```
SQL> CL SCR
```

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
```

```
2 , SAL
```

```
3
```

```
SQL> CL SCR
```

```
SQL> SELECT MAX(SAL)
```

```
2 FROM EMP;
```

```
MAX(SAL)
```

```
-----  
4791.6
```

```
SQL> ED
```

```
Wrote file afiedt.buf
```

```
1 SELECT *
```

```
2 FROM EMP
```

```
3 WHERE SAL = (SELECT MAX(SAL)
```

```
4 FROM EMP
```

```
5* )
```

```
SQL> /
```

```
EMPNO ENAME JOB MGR HIREDATE SAL COMM  
DEPTNO LAST_APPR
```

```
-----  
-- -----  
7788 SCOTT ANALYST 7566 19-APR-87 4791.6
```

```
20
```

```
SQL> ED
```

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE SAL = (SELECT MAX(SAL)
4                FROM EMP
5*              )
SQL> /

```

ENAME	SAL	DEPTNO
SCOTT	4791.6	20

SQL> CL SCR

```

SQL> R
1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE SAL = (SELECT MAX(SAL)
4                FROM EMP
5*              )

```

ENAME	SAL	DEPTNO
SCOTT	4791.6	20

```

SQL> SELECT DEPTNO
2  ,MAX(SAL)
3  FROM EMP
4  GROUP BY DEPTNO;

```

DEPTNO	MAX(SAL)
30	2850
20	4791.6
10	2450

SQL> ED

Wrote file afiedt.buf

```

1  (SELECT DEPTNO
2        ,MAX(SAL)
3        FROM EMP
4        GROUP BY DEPTNO
5*      )
SQL> /

```

DEPTNO	MAX(SAL)
30	2850
20	4791.6
10	2450

SQL> CL SCR

```

SQL> R
1  (SELECT DEPTNO
2        ,MAX(SAL)
3        FROM EMP
4        GROUP BY DEPTNO
5*      )

```

DEPTNO	MAX(SAL)
--------	----------

30	2850
20	4791.6
10	2450

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE DEPTNO = 10
4* AND SAL = 2450
SQL> /
```

ENAME	SAL	DEPTNO
CLARK	2450	10

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE DEPTNO = 20
4* AND SAL = 4791.6
SQL> /
```

ENAME	SAL	DEPTNO
SCOTT	4791.6	20

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3* WHERE (DEPTNO,SAL) = (10,2450)
SQL> /
WHERE (DEPTNO,SAL) = (10,2450)
                        *
```

ERROR at line 3:

ORA-00920: invalid relational operator

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3* WHERE (DEPTNO,SAL) IN (10,2450)
SQL> /
WHERE (DEPTNO,SAL) IN (10,2450)
                        *
```

ERROR at line 3:

ORA-00920: invalid relational operator

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3* WHERE (DEPTNO,SAL) IN ((10,2450))
```

SQL> /

ENAME	SAL	DEPTNO
CLARK	2450	10

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT DEPTNO
2  ,MAX(SAL)
3  FROM EMP
4* GROUP BY DEPTNO
SQL> /

```

DEPTNO	MAX(SAL)
30	2850
20	4791.6
10	2450

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3* WHERE (DEPTNO,SAL) IN ((10,2450))
SQL> /

```

ENAME	SAL	DEPTNO
CLARK	2450	10

SQL> D

SP2-0042: unknown command "D" - rest of line ignored.

SQL> D

SP2-0042: unknown command "D" - rest of line ignored.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3* WHERE (DEPTNO,SAL) IN ((10,2450), (20,4791.6) ,( 30,2850) )
SQL> /

```

ENAME	SAL	DEPTNO
BLAKE	2850	30
CLARK	2450	10
SCOTT	4791.6	20

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE (DEPTNO,SAL) IN (SELECT DEPTNO
4                           ,MAX(SAL)
5                           FROM EMP
6                           GROUP BY DEPTNO
7*                          )
SQL> /

```

ENAME	SAL	DEPTNO
BLAKE	2850	30
SCOTT	4791.6	20
CLARK	2450	10

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE (DEPTNO,SAL)  =(SELECT DEPTNO
4                        ,MAX(SAL)
5                        FROM EMP
6                        GROUP BY DEPTNO
7*                        )

```

SQL> /

```

WHERE (DEPTNO,SAL)  =(SELECT DEPTNO
                      *

```

ERROR at line 3:

ORA-01427: single-row subquery returns more than one row

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE (DEPTNO,SAL)  IN (SELECT DEPTNO
4                          ,MAX(SAL)
5                          FROM EMP
6                          GROUP BY DEPTNO
7*                          )

```

SQL> /

ENAME	SAL	DEPTNO
BLAKE	2850	30
SCOTT	4791.6	20
CLARK	2450	10

SQL> EDE

SP2-0042: unknown command "EDE" - rest of line ignored.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE DEPTNO  IN (SELECT DEPTNO
4                    ,MAX(SAL)
5                    FROM EMP
6                    GROUP BY DEPTNO
7*                    )

```

SQL> /

```

WHERE DEPTNO  IN (SELECT DEPTNO
                  *

```

ERROR at line 3:

ORA-00913: too many values

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE DEPTNO,SAL    IN (SELECT DEPTNO
4                          ,MAX(SAL)
5                          FROM EMP
6                          GROUP BY DEPTNO
7*                          )
SQL> /
WHERE DEPTNO,SAL    IN (SELECT DEPTNO
                        *
ERROR at line 3:
ORA-00920: invalid relational operator

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE (DEPTNO,SAL )  IN (SELECT DEPTNO
4                          ,MAX(SAL)
5                          FROM EMP
6                          GROUP BY DEPTNO
7*                          )
SQL> /

```

ENAME	SAL	DEPTNO
BLAKE	2850	30
SCOTT	4791.6	20
CLARK	2450	10

```
SQL> CL SCR
```

```

SQL> EDIT
Wrote file afiedt.buf

```

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE (DEPTNO )  = (SELECT DEPTNO
4                          ,MAX(SAL)
5                          FROM EMP
6                          GROUP BY DEPTNO
7*                          )
SQL> /
WHERE (DEPTNO )  = (SELECT DEPTNO
                    *
ERROR at line 3:
ORA-00913: too many values

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE (DEPTNO ,SAL)  = (SELECT DEPTNO
4                          ,MAX(SAL)
5                          FROM EMP
6                          GROUP BY DEPTNO
7*                          )
SQL> /
WHERE (DEPTNO ,SAL)  = (SELECT DEPTNO
                        *

```

ERROR at line 3:
ORA-01427: single-row subquery returns more than one row

SQL> ED
Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE (DEPTNO ,SAL)  IN (SELECT DEPTNO
4                          ,MAX(SAL)
5                          FROM EMP
6                          GROUP BY DEPTNO
7*                          )
SQL> /

```

ENAME	SAL	DEPTNO
BLAKE	2850	30
SCOTT	4791.6	20
CLARK	2450	10

SQL> CL SCR

```

SQL> SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP;

```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

SQL> ED
Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE SAL = ( SELECT MAX(SAL)
6                FROM EMP
7*            )
8  /

```

ENAME	SAL	DEPTNO
SCOTT	4791.6	20


```
SQL>
SQL> CL SCR

SQL> SET PAGESIZE 200
SQL> SET LINESIZE 200
SQL> COLUMN ENAME FORMAT A10
SQL> CL SCR
```

```
SQL> ED
SP2-0107: Nothing to save.
SQL> SELECT *
      2  FROM DUAL;
```

```
D
-
X
```

```
SQL> EDIT
Wrote file afiedt.buf
```

```
      1  SELECT ENAME, SAL, DEPTNO
      2* FROM EMP
SQL> /
```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
      1  (SELECT DEPTNO
      2
      3              , MAX(SAL)
      4              FROM EMP
      5*              GROUP BY DEPTNO
      6  )
```

DEPTNO	MAX(SAL)
30	2850
20	4791.6
10	2450

```
SQL> ED
Wrote file afiedt.buf
```

```
      1  SELECT ENAME, SAL, DEPTNO
      2  FROM EMP
```

```

3  WHERE (DEPTNO,SAL )  IN (SELECT DEPTNO
4                          ,MAX(SAL)
5                          FROM EMP
6                          GROUP BY DEPTNO
7*                          )
SQL> /

```

ENAME	SAL	DEPTNO
BLAKE	2850	30
SCOTT	4791.6	20
CLARK	2450	10

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3* WHERE (DEPTNO,SAL )  IN ( (10,2450) , (20,4791.6) , (30,2850) )
SQL> /

```

ENAME	SAL	DEPTNO
BLAKE	2850	30
CLARK	2450	10
SCOTT	4791.6	20

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE DEPTNO =10
4*    AND SAL = 2450
SQL> /

```

ENAME	SAL	DEPTNO
CLARK	2450	10

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3  WHERE(  DEPTNO =10
4          AND SAL = 2450
5          )
6  OR    ( DEPTNO =20
7          AND SAL = 4791.6
8          )
9  OR    ( DEPTNO = 30
10         AND SAL = 2850
11*        )
SQL> /

```

ENAME	SAL	DEPTNO
BLAKE	2850	30
CLARK	2450	10
SCOTT	4791.6	20

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3* WHERE (DEPTNO,SAL) IN ( SELECT DEPTNO,MAX(SAL) FROM EMP GROUP BY DEPTNO)
SQL> /

```

ENAME	SAL	DEPTNO
BLAKE	2850	30
SCOTT	4791.6	20
CLARK	2450	10

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3* WHERE (DEPTNO,SAL) = ( SELECT DEPTNO,MAX(SAL) FROM EMP WHERE DEPTNO =10 GROUP
BY DEPTNO)
SQL> /

```

ENAME	SAL	DEPTNO
CLARK	2450	10

```
SQL> CL SCR
```

```

SQL> SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP;

```

ENAME	SAL	JOB
SURYA	2000	
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

```

SQL> UPDATE EMP
2  SET JOB ='PRESIDENT'
3  WHERE JOB IS NULL;

```

1 row updated.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT ENAME
```

```

2  ,SAL
3  ,JOB
4  FROM EMP;

```

ENAME	SAL	JOB
SURYA	2000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

```

SQL> SELECT SAL
2  FROM EMP
3  WHERE JOB ='CLERK';

```

SAL
800
1100
950
1300

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL > (SELECT SAL
6                FROM EMP
7                WHERE JOB ='CLERK'
8*              )

```

```

SQL> /
WHERE SAL > (SELECT SAL
              *

```

ERROR at line 5:
ORA-01427: single-row subquery returns more than one row

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL >ANY (SELECT SAL
6                  FROM EMP
7                  WHERE JOB ='CLERK'
8*                )

```

```

SQL> /

```

ENAME	SAL	JOB
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
JONES	2975	MANAGER
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
SURYA	2000	PRESIDENT
ALLEN	1600	SALESMAN
TURNER	1500	SALESMAN
MILLER	1300	CLERK
MARTIN	1250	SALESMAN
WARD	1250	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK

13 rows selected.

SQL> ED
Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4* FROM EMP
SQL> /

```

ENAME	SAL	JOB
SURYA	2000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

SQL> CL SCR

```

SQL> R
1  SELECT ENAME
2  ,SAL
3  ,JOB
4* FROM EMP

```

ENAME	SAL	JOB
SURYA	2000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
SMITH	800	CLERK

ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  (SELECT SAL
2
3      FROM EMP
4*      WHERE JOB ='CLERK'
5  )
5  /

```

SAL
800
1100
950
1300

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5* WHERE SAL  >ANY (800,1100,950,1300)
SQL> /

```

ENAME	SAL	JOB
SURYA	2000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

13 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL  >(SELECT SAL
6
7      FROM EMP
8      WHERE JOB ='CLERK'
9  )
SQL> /

```

```

      8*
SQL> /
WHERE SAL  >(SELECT SAL
              *
ERROR at line 5:
ORA-01427: single-row subquery returns more than one row

```

```

SQL> ED
Wrote file afiedt.buf

```

```

  1  SELECT ENAME
  2  ,SAL
  3  ,JOB
  4  FROM EMP
  5  WHERE SAL  > (SELECT MIN(SAL)
  6                  FROM EMP
  7                  WHERE JOB ='CLERK'
  8*
SQL> /

```

ENAME	SAL	JOB
SURYA	2000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

13 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

  1  SELECT ENAME
  2  ,SAL
  3  ,JOB
  4* FROM EMP
SQL> /

```

ENAME	SAL	JOB
SURYA	2000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

SQL> CL SCR

SQL> R

```
1  SELECT ENAME
2  ,SAL
3  ,JOB
4* FROM EMP
```

ENAME	SAL	JOB
SURYA	2000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

SQL> EDIT

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5* WHERE SAL >ANY ( SELECT SAL FROM EMP WHERE JOB ='CLERK' ORDER BY SAL DESC)
SQL> /
WHERE SAL >ANY ( SELECT SAL FROM EMP WHERE JOB ='CLERK' ORDER BY SAL DESC)
*
```

ERROR at line 5:

ORA-00907: missing right parenthesis

SQL> ED

Wrote file afiedt.buf

```
1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5* WHERE SAL >ANY ( SELECT SAL FROM EMP WHERE JOB ='CLERK' )
SQL> /
```

ENAME	SAL	JOB
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
JONES	2975	MANAGER
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
SURYA	2000	PRESIDENT
ALLEN	1600	SALESMAN

TURNER	1500	SALESMAN
MILLER	1300	CLERK
MARTIN	1250	SALESMAN
WARD	1250	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK

13 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4* FROM EMP
5  /

```

ENAME	SAL	JOB
SURYA	2000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4* FROM EMP

```

ENAME	SAL	JOB
SURYA	2000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
SMITH	800	CLERK
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

14 rows selected.

```
SQL> ED
Wrote file afiedt.buf

 1  SELECT ENAME
 2  ,SAL
 3  ,JOB
 4  FROM EMP
 5* /WHERE SAL  >ANY(1300,1100,950,800)
SQL> /
/WHERE SAL  >ANY(1300,1100,950,800)
*
ERROR at line 5:
ORA-00933: SQL command not properly ended
```

```
SQL> ED
Wrote file afiedt.buf

 1  SELECT ENAME
 2  ,SAL
 3  ,JOB
 4  FROM EMP
 5* WHERE SAL  >ANY(1300,1100,950,800)
SQL> /
```

ENAME	SAL	JOB
SURYA	2000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

13 rows selected.

```
SQL> ED
Wrote file afiedt.buf

 1  SELECT ENAME
 2  ,SAL
 3  ,JOB
 4  FROM EMP
 5  WHERE SAL  >ANY ( SELECT SAL
 6                      FROM EMP
 7                      WHERE JOB ='CLERK'
 8*                      )
SQL> /
```

ENAME	SAL	JOB
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
JONES	2975	MANAGER
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
SURYA	2000	PRESIDENT

ALLEN	1600	SALESMAN
TURNER	1500	SALESMAN
MILLER	1300	CLERK
MARTIN	1250	SALESMAN
WARD	1250	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK

13 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL  > ( SELECT MIN(SAL)
6                  FROM EMP
7                  WHERE JOB ='CLERK'
8*                )
SQL> /

```

ENAME	SAL	JOB
SURYA	2000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

13 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL  <ANY ( SELECT SAL
6                  FROM EMP
7                  WHERE JOB ='CLERK'
8*                )
SQL> /

```

ENAME	SAL	JOB
SMITH	800	CLERK
JAMES	950	CLERK
ADAMS	1100	CLERK
MARTIN	1250	SALESMAN
WARD	1250	SALESMAN

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL < ( SELECT SAL
6                FROM EMP
7                WHERE JOB ='CLERK'
8*              )
SQL> /
WHERE SAL < ( SELECT SAL
              *
ERROR at line 5:
ORA-01427: single-row subquery returns more than one row

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL < ( SELECT SAL
6                FROM EMP
7                WHERE JOB ='CLERK'
8*              )
SQL> /
WHERE SAL < ( SELECT SAL
              *
ERROR at line 5:
ORA-01427: single-row subquery returns more than one row

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL >ANY ( SELECT (SAL)
6                  FROM EMP
7                  WHERE JOB ='CLERK'
8*                )
SQL> /

```

ENAME	SAL	JOB
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
JONES	2975	MANAGER
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
SURYA	2000	PRESIDENT
ALLEN	1600	SALESMAN
TURNER	1500	SALESMAN
MILLER	1300	CLERK
MARTIN	1250	SALESMAN
WARD	1250	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK

13 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL  > ( SELECT MIN(SAL)
6                  FROM EMP
7                  WHERE JOB ='CLERK'
8*                )
SQL> /
```

ENAME	SAL	JOB
SURYA	2000	PRESIDENT
BLAKE	2850	MANAGER
CLARK	2450	MANAGER
JONES	2975	MANAGER
SCOTT	4791.6	ANALYST
FORD	3600	ANALYST
ALLEN	1600	SALESMAN
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
TURNER	1500	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

13 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL  <ANY ( SELECT (SAL)
6                  FROM EMP
7                  WHERE JOB ='CLERK'
8*                )
SQL> /
```

ENAME	SAL	JOB
SMITH	800	CLERK
JAMES	950	CLERK
ADAMS	1100	CLERK
MARTIN	1250	SALESMAN
WARD	1250	SALESMAN

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL  < ( SELECT MAX(SAL)
6                  FROM EMP
7                  WHERE JOB ='CLERK'
8*                )
```

SQL> /

ENAME	SAL	JOB
SMITH	800	CLERK
WARD	1250	SALESMAN
MARTIN	1250	SALESMAN
ADAMS	1100	CLERK
JAMES	950	CLERK

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL =ANY ( SELECT SAL
6                    FROM EMP
7                    WHERE JOB ='CLERK'
8*                  )
SQL> /

```

ENAME	SAL	JOB
SMITH	800	CLERK
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL IN ( SELECT SAL
6                  FROM EMP
7                  WHERE JOB ='CLERK'
8*                )
SQL> /

```

ENAME	SAL	JOB
SMITH	800	CLERK
ADAMS	1100	CLERK
JAMES	950	CLERK
MILLER	1300	CLERK

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL >ALL ( SELECT SAL
6                   FROM EMP
7                   WHERE JOB ='CLERK'
8*                 )
SQL> /

```

ENAME	SAL	JOB
-------	-----	-----

```

-----
TURNER          1500 SALESMAN
ALLEN           1600 SALESMAN
SURYA           2000 PRESIDENT
CLARK           2450 MANAGER
BLAKE           2850 MANAGER
JONES           2975 MANAGER
FORD            3600 ANALYST
SCOTT           4791.6 ANALYST

```

8 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL >( SELECT MAX(SAL)
6                FROM EMP
7                WHERE JOB ='CLERK'
8*                )
SQL> /

```

```

ENAME          SAL  JOB
-----
SURYA           2000 PRESIDENT
BLAKE           2850 MANAGER
CLARK           2450 MANAGER
JONES           2975 MANAGER
SCOTT           4791.6 ANALYST
FORD            3600 ANALYST
ALLEN           1600 SALESMAN
TURNER          1500 SALESMAN

```

8 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL <ALL ( SELECT (SAL)
6                FROM EMP
7                WHERE JOB ='CLERK'
8*                )
SQL> /

```

no rows selected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL < ( SELECT MIN(SAL)
6                FROM EMP
7                WHERE JOB ='CLERK'
8*                )

```

SQL> /

no rows selected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL =ALL ( SELECT (SAL)
6                    FROM EMP
7                    WHERE JOB ='CLERK'
8*                  )
SQL> /

```

no rows selected

SQL> EED

SP2-0042: unknown command "EED" - rest of line ignored.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5* WHERE SAL = 4791.6
SQL> /

```

ENAME	SAL	JOB
SCOTT	4791.6	ANALYST

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5* WHERE SAL =ALL( 4791.6)
SQL> /

```

ENAME	SAL	JOB
SCOTT	4791.6	ANALYST

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5* WHERE SAL =ALL( 4791.6,4791.6)
SQL> /

```

ENAME	SAL	JOB
SCOTT	4791.6	ANALYST

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5* WHERE SAL =ALL( 4791.6,4791.6,4791.6)
SQL> /

```

ENAME	SAL	JOB
SCOTT	4791.6	ANALYST

SQL> ED
Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5* WHERE SAL =ALL( 4791.6,4791.6,4791.6,2850)
SQL> /

```

no rows selected

SQL> ED
Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,JOB
4  FROM EMP
5  WHERE SAL = 4791.6
6  AND SAL =4791.6
7  AND SAL =4791.6
8* AND SAL = 2850
SQL> /

```

no rows selected

SQL> CL SCR

```

SQL> SELECT ENAME
2  ,SAL
3  FROM EMP
4  WHERE SAL = ( SELECT MIN(SAL) FROM EMP);

```

ENAME	SAL
SMITH	800

SQL> CL SCR

```

SQL> SELECT ENAME,SAL
2
SQL>
SQL> SELECT ENAME,SAL,DEPTNO
2  FROM EMP;

```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
CLARK	2450	10

JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

```
SQL> UPDATE EMP
  2  SET DEPTNO =11
  3  WHERE DEPTNO =10
  4  AND SAL =2450;
```

1 row updated.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT ENAME, SAL, DEPTNO
  2  FROM EMP;
```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
CLARK	2450	11
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

```
SQL> SELECT * FROM DEPT;
```

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

```
SQL> SELECT ENAME
  2  , SAL
  3  , DEPTNO
  4  FROM EMP E
  5  , DEPT D
  6  WHERE E.DEPTNO = D.DEPTNO;
```

```
,DEPTNO
*
ERROR at line 3:
ORA-00918: column ambiguously defined
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT ENAME
2  ,SAL
3  ,E.DEPTNO
4  FROM EMP E
5  ,DEPT D
6* WHERE E.DEPTNO = D.DEPTNO
SQL> /
```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

13 rows selected.

```
SQL> SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE DEPTNO NOT IN ( SELECT DEPTNO FROM DEPT);
```

ENAME	SAL	DEPTNO
CLARK	2450	11

```
SQL> ED
Wrote file afiedt.buf
```

```
1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5* WHERE DEPTNO IN ( SELECT DEPTNO FROM DEPT)
SQL> /
```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30

WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

13 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5* WHERE DEPTNO NOT IN ( SELECT DEPTNO FROM DEPT)
SQL> /

```

ENAME	SAL	DEPTNO
CLARK	2450	11

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,E.DEPTNO
4  FROM EMP E
5  ,DEPT D
6* WHERE E.DEPTNO = D.DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

13 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,E.DEPTNO
4  FROM EMP E
5  ,DEPT D
6* WHERE E.DEPTNO <> D.DEPTNO
SQL> /

```

ENAME	SAL	DEPTNO
-------	-----	--------

```

-----
BLAKE          2850          30
CLARK          2450          11
JONES          2975          20
SCOTT         4791.6          20
FORD           3600          20
SMITH           800          20
ALLEN          1600          30
WARD           1250          30
MARTIN          1250          30
TURNER          1500          30
ADAMS           1100          20
JAMES           950          30
SURYA           2000          10
BLAKE          2850          30
CLARK          2450          11
ALLEN          1600          30
WARD           1250          30
MARTIN          1250          30
TURNER          1500          30
JAMES           950          30
MILLER          1300          10
SURYA           2000          10
CLARK          2450          11
JONES          2975          20
SCOTT         4791.6          20
FORD           3600          20
SMITH           800          20
ADAMS           1100          20
MILLER          1300          10
SURYA           2000          10
BLAKE          2850          30
CLARK          2450          11
JONES          2975          20
SCOTT         4791.6          20
FORD           3600          20
SMITH           800          20
ALLEN          1600          30
WARD           1250          30
MARTIN          1250          30
TURNER          1500          30
ADAMS           1100          20
JAMES           950          30
MILLER          1300          10

```

43 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,E.DEPTNO
4  FROM EMP E
5  ,DEPT D
6* WHERE E.DEPTNO = D.DEPTNO
SQL> /

```

```

ENAME          SAL          DEPTNO
-----
SURYA           2000          10
BLAKE          2850          30
JONES          2975          20
SCOTT         4791.6          20

```

FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

13 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,E.DEPTNO
4  FROM EMP E
5  ,DEPT D
6* WHERE E.DEPTNO = D.DEPTNO(+)
SQL> /

```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
CLARK	2450	11
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,E.DEPTNO
4  ,D.DEPTNO
5  FROM EMP E
6  ,DEPT D
7* WHERE E.DEPTNO = D.DEPTNO(+)
SQL> /

```

ENAME	SAL	DEPTNO	DEPTNO
SURYA	2000	10	10
BLAKE	2850	30	30
CLARK	2450	11	
JONES	2975	20	20
SCOTT	4791.6	20	20
FORD	3600	20	20
SMITH	800	20	20
ALLEN	1600	30	30

WARD	1250	30	30
MARTIN	1250	30	30
TURNER	1500	30	30
ADAMS	1100	20	20
JAMES	950	30	30
MILLER	1300	10	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,E.DEPTNO
4  ,D.DEPTNO
5  FROM EMP E
6  ,DEPT D
7  WHERE E.DEPTNO = D.DEPTNO(+)
8* AND D.DEPTNO IS NULL
SQL> /

```

ENAME	SAL	DEPTNO	DEPTNO
CLARK	2450	11	

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE DEPTNO IN ( SELECT DEPTNO
6                    FROM DEPT
7*                  )
SQL> /

```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

13 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE DEPTNO IN ( SELECT DEPTNO

```


ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

13 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP E
5  WHERE EXISTS ( SELECT *
6                  FROM DEPT D
7                  WHERE D.DEPTNO = E.DEPTNO
8*                )
9  /

```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

13 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP E
5  WHERE NOT EXISTS ( SELECT *
6                     FROM DEPT D
7                     WHERE D.DEPTNO = E.DEPTNO
8*                   )
SQL> /

```

ENAME	SAL	DEPTNO
-------	-----	--------

```
-----
CLARK          2450          11
```

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP E
5  WHERE NOT EXISTS ( SELECT  'ARJUN'
6                      FROM DEPT D
7                      WHERE D.DEPTNO = E.DEPTNO
8*                      )
SQL> /
```

```

ENAME          SAL          DEPTNO
-----
CLARK          2450          11
```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP E
5  WHERE NOT EXISTS ( SELECT  'HI RAM HOW ARE YOU'
6                      FROM DEPT D
7                      WHERE D.DEPTNO = E.DEPTNO
8*                      )
SQL> /
```

```

ENAME          SAL          DEPTNO
-----
CLARK          2450          11
```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP E
5  WHERE NOT EXISTS ( SELECT D.*,D.DNAME,D.LOC
6                      FROM DEPT D
7                      WHERE D.DEPTNO = E.DEPTNO
8*                      )
SQL> /
```

```

ENAME          SAL          DEPTNO
-----
CLARK          2450          11
```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP E
5  WHERE NOT EXISTS ( SELECT
6                      FROM DEPT D
7                      WHERE D.DEPTNO = E.DEPTNO
```

```

      8*                )
SQL> /
                FROM DEPT D
      *
ERROR at line 6:
ORA-00936: missing expression

```

```

SQL> ED
Wrote file afiedt.buf

```

```

  1  SELECT ENAME
  2  ,SAL
  3  ,DEPTNO
  4  FROM EMP E
  5  WHERE NOT EXISTS ( SELECT 0
  6                      FROM DEPT D
  7                      WHERE D.DEPTNO = E.DEPTNO
  8*                )
SQL> /

```

ENAME	SAL	DEPTNO
CLARK	2450	11

```

SQL> ED
Wrote file afiedt.buf

```

```

  1  SELECT ENAME
  2  ,SAL
  3  ,DEPTNO
  4  FROM EMP E
  5  WHERE EXISTS ( SELECT 1
  6                  FROM DEPT D
  7                  WHERE D.DEPTNO = E.DEPTNO
  8*                )
SQL> /

```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

13 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

  1  SELECT ENAME
  2  ,SAL
  3  ,DEPTNO
  4  FROM EMP E
  5  WHERE EXISTS ( SELECT 'Y'

```

```

6          FROM DEPT D
7          WHERE D.DEPTNO = E.DEPTNO
8*      )
SQL> /

```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

13 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP E
5  WHERE NOT EXISTS ( SELECT 'N'
6                      FROM DEPT D
7                      WHERE D.DEPTNO = E.DEPTNO
8*      )
SQL> /

```

ENAME	SAL	DEPTNO
CLARK	2450	11

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP E
5  WHERE EXISTS ( SELECT 1
6                  FROM DEPT D
7                  WHERE D.DEPTNO = E.DEPTNO
8*      )
SQL> /

```

ENAME	SAL	DEPTNO
SURYA	2000	10
BLAKE	2850	30
JONES	2975	20
SCOTT	4791.6	20
FORD	3600	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30

TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

13 rows selected.

SQL> CL SCR

SQL> CL SCR

SQL> SPOOL OFF

25-jun-2018 Hierarchial Query Notes

hierarchical queries

	7839	KING---	LEVEL 1
LEVEL 2	7698	BLAKE	7782 CLARK
			7566 JONES-----
LEVEL 3	(L3) 7499	ALLEN	7934 MILLER(L3)
			7788 SCOTT ---
- LEVEL 4	7521	WARD	7876 ADAMS --
LEVEL 3	7654	MARTIN	7902 FORD-----
-- LEVEL 4	7844	TURNER	7369 SMITH---
	7900	JAMES	

FROM WHEREE YOU START HIERARCHY THAT WILL BE LEVEL 1 HIRERARCY
FROM THERE IT WILL GIVE YOU NEXT LEVEL 2 ,3 .. AND SO ON

IF YOU WANT TO GET THE HIERARCHIAL QUERIES THEN PK AND FK DATA SHOULD BE IN SAME
TABLE
OR BY USING INLINE VIEWS YOU CAN GET DATA FROM MULTIPLE TABLES AND MAKE IT AS A
SINGLE TABLE IN RUNTIME

CONNECT BY PKCOLUMN = FKCOLUMN

PRIOR KEYWORD IF YOU KEEP WITH PK COLUMN DATA FROM WHERE YOU ARE STARTING THE
HIERARCHY
FROM THAT PLACE BELOW HIERARCHY IT WILL GIVE YOU

PRIOR KEY WORD IF YOU KEEP WITH FK COLUMN DATA FROM WHERE START THE HIERRACHY FROM
THAT PLACE TO
ABOVE HIERARCH KEY IT WILL KEY WILL GIVE YOU

```
START WITH---
CONNECT BY---
PRIOR---
CONNECT BY  ROOT--
(
P1
)
```

IT WILL GIVE YOU ROOT ROW COLUMN INFORMATION FOR EACH AND EVERY ROW

```
SYS_CONNECT_BY_PATH---
(
P1
,P2
)
```

IT WILL GIVE YOU FROM CURRENT PLACE TO TILL ROOT NODE COMPLETE HIERRARCHY IT WILL GIVE YOU

P1 FROM CURRENT PLACE TO TILL ROOT NODE WHICH INFORMATION YOU WANT TO SHOW

P2 IS FORMAT SPEICIFER HOW YOU WANT TO SEPARTE THE DATA

```
/
=>
<=
```

```
ISCYCLE
CONNECT_BY_ISLEAF
IT WILL GIVE YOU
0--IT IS NOT A LEAF NODE
1-- IT IS A LEAF NODE
LEVEL----
SIBLINGS
NOCYCLE--
```

```
X
ARJUN
```

```
101    ARJUN    103
102    SURESH   101
103    ABHI     102
```

```
LEVEL
----
```

IN SQL IT WILL WORK LIKE AUTO INCREMENTED LOOP

IT STARTS WITH 1 AND IT INCREMENT BY 1 AND CHECK CONDITION WHAT IS MENTIONED IN CONNECT BY

IF IT IS GETTING IT CONTINUED DO THE SAME THING...

1) 1 TO 10 NUMBERS

- 2) 1 TO 10 EVEN
- 3) 1 TO 10 ODD
- 4) 1 TO 10 ODD AND EVEN NUMBERS

5) WRITE A SELECT STATEMENT TO ACCEPT ONE NAME AS INPUT PARAMETER
&

INPUT : SURESH

OUTPUT :

S
U
R
E
S
H

INPUT : SURESH

OUTPUT :

SURESH
SURES
SURE
SUR
SU
S

INPUT : SURESH

OUTPUT :

S
SU
SUR
SURE
SURES
SURESH

INPUT:SURESH

OUTPUT:

*
**

INPUT:SURESH

OUTPUT:

**
*

3) WRITE A SELECT STATEMENT TO GET FROM TODAY TILL THIS MONTH END WHAT EVER DATES
ARE REMAINING

25-JUN-18
26-JUN-18
27-JUN-18
28-JUN-18
29-JUN-18
30-JUN-18

4) WRITE A SELECT STATEEMTN TO GET THIS MONTH ALL DATES

```
01-JUN-2018
02-JUN-208
.
.
.
30-JUN-2018
```

5) WRITE A SELECT STATEEMTN TO GET CURRENT MONTH ALL SATURDAY AND SUNDAY DATES ?

```
6) WRITE A SELECT STATEMENT TO GET FROM CURRENT MONTH BACK 12 MOHTS
201806
201805
201804
.
.
.
201705
```

7) WRITE A SELECT STATEMEN TO GIVE CURRENT MONTH WEEKS INFORMATION

OUTPUT:

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SAT=RUDA	SUNDAY
4	4	5	5	4	5	4

```
LEVEL
DATE FUNCTIONS
COUNT
CASE
```

22-Jun-2018 ROWNUM Functions

ROWNUM WILL BE CHANGINDG

ROWID WILL NOT CHANGE

1) IF YOU WANT TO GET ONE ROW ROWNUM =1

2) IF YOU WANT TO GET FIRST N NUMBER OF ROWS ROWNUM <= N

3) IF YOU WANT TO GET NTH ROW ?

NOTE DIRECTLY ON ROWNUM WE CAN NOT USE ROWNUM = NTH ROW

FIRST GET THE ROWNUM FOR ALL RECORDS AND GIVE A ONE ALIAS FOR THE ROWNUM

CONVERT THOSE SELECT STATEMENT INTO A INLINE VIEW QUERY

IN INLINE ALIAS WILL ACT AS A COLUMN AND ON THAT COLUMN

YOU CAN KEEP WHATEVER FILTERS WE WANT

4) WRITE A SELECT STATEMENT TO GET THE 1ST HIGHEST SALARY EMPLOYEE DETAILS ?

5) WRITE A SELECT STATEMENT TO GET 1..10 NUMBERS ?

1) SET OPERATORS--

2) BY USING ROWNUM WE CAN GET PROVIDED THAT THAT MANY ROWS ARE IN THE TABLE

3) BY USING LEVEL IS THE BEST METHOD

ROWNUM BY DEFAULT VALUE WILL BE 1

ON ROWNUM DIRECTLY WE CAN CHECK ONLY BELOW CONDITIONS

1) ROWNUM = 1

2) ROWNUM <= N

3) ROWNUM >= 1 THIS IS NOTHING BUT ALL ROWS FROM A TABLE

DONT USE ROWNUM AND ORDER BY IN SAME SELECT STATEMENT

IF YOU USE NO ERRORS BUT YOU WILL NOT EXPECTED RESULTS

BECAUSE

1) FIRST ROWNUM WILL EXECUTE

2) SECOND ORDER BY WILL EXECUTE

ROWID

IT IS A FIXED ADDRESS OF A ROWS

IT CONTAINS 18 CHARACTERS LENGTH

WHY ?

1) IT WILL IMPROVE THE PERFORMANCE OF QUERIES

2) TO DELETE DUPLICATE RECORDS FROM A TABLE

DELETE FROM TABLENAME

WHERE ROWID NOT IN (SELECT MIN(ROWID)/MAX(ROWID)

FROM TABLENAME

GROUP BY COLUMNS LIST(BASED ON WHICH COLUMNS YOU ARE TELLING IT IS DUPLICATED

);

1) ANDY DUPLICATES IN EMP TABLE

ON WHICH COLUMN YOU ARE TELLIN ?

3) HOW TO IDENTIFY THE DUPLICATE RECORDS WHICH ARE THERE IN TABLE

1) BY USING AGGR FUNCTIONS, GROUP BY HAVING

2) BY USING ROWID

KRISHNA 1000

ARJUN 1000

HIMANSH 1000

ABHI 2000

DELETE

2 FROM STD

3 WHERE ROWID NOT IN (

4 SELECT ROWID

5 FROM (

6 SELECT NAME

7 ,FEES

8 ,ROWID RID

9 ,ROW_NUMBER() OVER(PARTITION BY FEES ORDER BY FEES) DR

10 ,COUNT(1) OVER(PARTITION BY FEES) CNT

11 FROM STD

12)

13 WHERE((DR =2 AND CNT > 1)

14 OR (DR =1 AND CNT = 1)

15)

16*)

SQL> /

22-Jun-2018 ROWNUM Functions spool

SQL>

SQL> CL SCR

```
SQL> CONNECT SYSTEM/arjun
Connected.
```

```
SQL> SELECT ENAME, SAL
       2 FROM EMP;
```

ENAME	SAL
KING	5000
BLAKE	2850
CLARK	2450
JONES	2975
SCOTT	3000
FORD	3000
SMITH	800
ALLEN	1600
WARD	1250
MARTIN	1250
TURNER	1500
ADAMS	1100
JAMES	950
MILLER	1300

```
14 rows selected.
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT ROWNUM
2 ,ENAME
3 ,SAL
4* FROM EMP
SQL> /
```

ROWNUM	ENAME	SAL
1	KING	5000
2	BLAKE	2850
3	CLARK	2450
4	JONES	2975
5	SCOTT	3000
6	FORD	3000
7	SMITH	800
8	ALLEN	1600
9	WARD	1250
10	MARTIN	1250
11	TURNER	1500
12	ADAMS	1100
13	JAMES	950
14	MILLER	1300

```
14 rows selected.
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT ROWNUM
2 ,ENAME
3 ,SAL
4 ,DEPTNO
5* FROM EMP
SQL> /
```

ROWNUM	ENAME	SAL	DEPTNO
1	KING	5000	10
2	BLAKE	2850	30
3	CLARK	2450	10
4	JONES	2975	20
5	SCOTT	3000	20
6	FORD	3000	20
7	SMITH	800	20
8	ALLEN	1600	30
9	WARD	1250	30
10	MARTIN	1250	30
11	TURNER	1500	30
12	ADAMS	1100	20
13	JAMES	950	30
14	MILLER	1300	10

14 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5* FROM EMP

```

ROWNUM	ENAME	SAL	DEPTNO
1	KING	5000	10
2	BLAKE	2850	30
3	CLARK	2450	10
4	JONES	2975	20
5	SCOTT	3000	20
6	FORD	3000	20
7	SMITH	800	20
8	ALLEN	1600	30
9	WARD	1250	30
10	MARTIN	1250	30
11	TURNER	1500	30
12	ADAMS	1100	20
13	JAMES	950	30
14	MILLER	1300	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  ,ROWID
6* FROM EMP
SQL> /

```

ROWNUM	ENAME	SAL	DEPTNO	ROWID
1	KING	5000	10	AAAE5SAABAAAK+hAAA
2	BLAKE	2850	30	AAAE5SAABAAAK+hAAB
3	CLARK	2450	10	AAAE5SAABAAAK+hAAC
4	JONES	2975	20	AAAE5SAABAAAK+hAAD

5	SCOTT	3000	20	AAAE5SAABAAAK+hAAE
6	FORD	3000	20	AAAE5SAABAAAK+hAAF
7	SMITH	800	20	AAAE5SAABAAAK+hAAG
8	ALLEN	1600	30	AAAE5SAABAAAK+hAAH
9	WARD	1250	30	AAAE5SAABAAAK+hAAI
10	MARTIN	1250	30	AAAE5SAABAAAK+hAAJ
11	TURNER	1500	30	AAAE5SAABAAAK+hAAK
12	ADAMS	1100	20	AAAE5SAABAAAK+hAAL
13	JAMES	950	30	AAAE5SAABAAAK+hAAM
14	MILLER	1300	10	AAAE5SAABAAAK+hAAN

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  ,ROWID
6* FROM EMP
SQL>
SQL>
SQL> SELECT LENGTH('AAAE5SAABAAAK+hAAN')
2  FROM DUAL;
```

LENGTH('AAAE5SAABAAAK+HAAN')

18

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  ,ROWID
6* FROM EMP
SQL> /
```

ROWNUM	ENAME	SAL	DEPTNO	ROWID
1	KING	5000	10	AAAE5SAABAAAK+hAAA
2	BLAKE	2850	30	AAAE5SAABAAAK+hAAB
3	CLARK	2450	10	AAAE5SAABAAAK+hAAC
4	JONES	2975	20	AAAE5SAABAAAK+hAAD
5	SCOTT	3000	20	AAAE5SAABAAAK+hAAE
6	FORD	3000	20	AAAE5SAABAAAK+hAAF
7	SMITH	800	20	AAAE5SAABAAAK+hAAG
8	ALLEN	1600	30	AAAE5SAABAAAK+hAAH
9	WARD	1250	30	AAAE5SAABAAAK+hAAI
10	MARTIN	1250	30	AAAE5SAABAAAK+hAAJ
11	TURNER	1500	30	AAAE5SAABAAAK+hAAK
12	ADAMS	1100	20	AAAE5SAABAAAK+hAAL
13	JAMES	950	30	AAAE5SAABAAAK+hAAM
14	MILLER	1300	10	AAAE5SAABAAAK+hAAN

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  ,ROWID
6  FROM EMP
7* WHERE DEPTNO = 20
SQL> /

```

ROWNUM	ENAME	SAL	DEPTNO	ROWID
1	JONES	2975	20	AAAE5SAABAAAK+hAAD
2	SCOTT	3000	20	AAAE5SAABAAAK+hAAE
3	FORD	3000	20	AAAE5SAABAAAK+hAAF
4	SMITH	800	20	AAAE5SAABAAAK+hAAG
5	ADAMS	1100	20	AAAE5SAABAAAK+hAAL

SQL> CL SCR

SQL> CL SCR

```

SQL> SELECT ENAME,SAL,DEPTNO
2  FROM EMP;

```

ENAME	SAL	DEPTNO
KING	5000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	3000	20
FORD	3000	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO
2  FROM EMP
3* WHERE ROWNUM =1
SQL> /

```

ENAME	SAL	DEPTNO
KING	5000	10

SQL> CK SCR

SP2-0042: unknown command "CK SCR" - rest of line ignored.

SQL> CL SCR

SQL> EDIT

Wrote file afiedt.buf

```

1  SELECT ENAME, SAL, DEPTNO
2* FROM EMP
3  /

```

ENAME	SAL	DEPTNO
KING	5000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	3000	20
FORD	3000	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME, SAL, DEPTNO
2  FROM EMP
3* WHERE ROWNUM = 1
SQL> /

```

ENAME	SAL	DEPTNO
KING	5000	10

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME, SAL, DEPTNO
2  FROM EMP
3* WHERE ROWNUM = 2
SQL> /

```

no rows selected

```

SQL>
SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME, SAL, DEPTNO
2  FROM EMP
3* WHERE ROWNUM <= 2
SQL> /

```

ENAME	SAL	DEPTNO
KING	5000	10
BLAKE	2850	30

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME, SAL, DEPTNO
2  FROM EMP

```



```

3* WHERE ROWNUM <= 5
SQL> /

```

ENAME	SAL	DEPTNO
KING	5000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	3000	20

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME, SAL, DEPTNO
2  FROM EMP
3* WHERE ROWNUM <= 10
SQL> /

```

ENAME	SAL	DEPTNO
KING	5000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	3000	20
FORD	3000	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30

10 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME, SAL, DEPTNO
2  FROM EMP
3* WHERE ROWNUM <= 100
SQL> /

```

ENAME	SAL	DEPTNO
KING	5000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	3000	20
FORD	3000	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME, SAL, DEPTNO
2  FROM EMP
3* WHERE ROWNUM =2
SQL> /

```

no rows selected

```
SQL> CL SCR
```

```
SQL> EDIT
```

Wrote file afiedt.buf

```

1  SELECT ENAME, SAL, DEPTNO
2* FROM EMP
3  /

```

ENAME	SAL	DEPTNO
KING	5000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	3000	20
FORD	3000	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

```
SQL> ED
```

Wrote file afiedt.buf

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  FROM EMP
6* WHERE ROWNUM =2
SQL> /

```

no rows selected

```
SQL> ED
```

Wrote file afiedt.buf

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5* FROM EMP
6  /

```

ROWNUM	ENAME	SAL	DEPTNO
1	KING	5000	10
2	BLAKE	2850	30
3	CLARK	2450	10
4	JONES	2975	20

5	SCOTT	3000	20
6	FORD	3000	20
7	SMITH	800	20
8	ALLEN	1600	30
9	WARD	1250	30
10	MARTIN	1250	30
11	TURNER	1500	30
12	ADAMS	1100	20
13	JAMES	950	30
14	MILLER	1300	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  FROM EMP
6* WHERE ROWNUM = 2
SQL> /

```

no rows selected

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  FROM EMP
6* WHERE ROWNUM >=1
SQL> /

```

ROWNUM	ENAME	SAL	DEPTNO
1	KING	5000	10
2	BLAKE	2850	30
3	CLARK	2450	10
4	JONES	2975	20
5	SCOTT	3000	20
6	FORD	3000	20
7	SMITH	800	20
8	ALLEN	1600	30
9	WARD	1250	30
10	MARTIN	1250	30
11	TURNER	1500	30
12	ADAMS	1100	20
13	JAMES	950	30
14	MILLER	1300	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  FROM EMP

```

```

6* WHERE ROWNUM = 1
SQL> /

```

ROWNUM	ENAME	SAL	DEPTNO
1	KING	5000	10

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  FROM EMP
6* WHERE ROWNUM <= 3
SQL> /

```

ROWNUM	ENAME	SAL	DEPTNO
1	KING	5000	10
2	BLAKE	2850	30
3	CLARK	2450	10

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  FROM EMP
6* WHERE ROWNUM = 3
SQL> /

```

no rows selected

```

SQL> CL SCR

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5* FROM EMP
6  /

```

ROWNUM	ENAME	SAL	DEPTNO
1	KING	5000	10
2	BLAKE	2850	30
3	CLARK	2450	10
4	JONES	2975	20
5	SCOTT	3000	20
6	FORD	3000	20
7	SMITH	800	20
8	ALLEN	1600	30
9	WARD	1250	30
10	MARTIN	1250	30
11	TURNER	1500	30
12	ADAMS	1100	20
13	JAMES	950	30

```

14 MILLER          1300          10

```

```

14 rows selected.

```

```

SQL> ED

```

```

Wrote file afiedt.buf

```

```

1  SELECT ROWNUM RN
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5* FROM EMP
SQL> /

```

	RN	ENAME	SAL	DEPTNO
	1	KING	5000	10
	2	BLAKE	2850	30
	3	CLARK	2450	10
	4	JONES	2975	20
	5	SCOTT	3000	20
	6	FORD	3000	20
	7	SMITH	800	20
	8	ALLEN	1600	30
	9	WARD	1250	30
	10	MARTIN	1250	30
	11	TURNER	1500	30
	12	ADAMS	1100	20
	13	JAMES	950	30
	14	MILLER	1300	10

```

14 rows selected.

```

```

SQL> ED

```

```

Wrote file afiedt.buf

```

```

1  SELECT ROWNUM C1
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5* FROM EMP
SQL> /

```

	C1	ENAME	SAL	DEPTNO
	1	KING	5000	10
	2	BLAKE	2850	30
	3	CLARK	2450	10
	4	JONES	2975	20
	5	SCOTT	3000	20
	6	FORD	3000	20
	7	SMITH	800	20
	8	ALLEN	1600	30
	9	WARD	1250	30
	10	MARTIN	1250	30
	11	TURNER	1500	30
	12	ADAMS	1100	20
	13	JAMES	950	30
	14	MILLER	1300	10

```

14 rows selected.

```

```

SQL> ED

```

```

Wrote file afiedt.buf

```

```

1  SELECT ROWNUM SNO
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5* FROM EMP
SQL> /

```

SNO	ENAME	SAL	DEPTNO
1	KING	5000	10
2	BLAKE	2850	30
3	CLARK	2450	10
4	JONES	2975	20
5	SCOTT	3000	20
6	FORD	3000	20
7	SMITH	800	20
8	ALLEN	1600	30
9	WARD	1250	30
10	MARTIN	1250	30
11	TURNER	1500	30
12	ADAMS	1100	20
13	JAMES	950	30
14	MILLER	1300	10

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  (
2  SELECT ROWNUM SNO
3  ,ENAME
4  ,SAL
5  ,DEPTNO
6  FROM EMP
7* )
8  /

```

SNO	ENAME	SAL	DEPTNO
1	KING	5000	10
2	BLAKE	2850	30
3	CLARK	2450	10
4	JONES	2975	20
5	SCOTT	3000	20
6	FORD	3000	20
7	SMITH	800	20
8	ALLEN	1600	30
9	WARD	1250	30
10	MARTIN	1250	30
11	TURNER	1500	30
12	ADAMS	1100	20
13	JAMES	950	30
14	MILLER	1300	10

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E1.*
2  FROM
3  (

```

```

4      SELECT ROWNUM SNO
5              ,ENAME
6              ,SAL
7              ,DEPTNO
8      FROM EMP
9*    ) E1
10  /

```

SNO	ENAME	SAL	DEPTNO
1	KING	5000	10
2	BLAKE	2850	30
3	CLARK	2450	10
4	JONES	2975	20
5	SCOTT	3000	20
6	FORD	3000	20
7	SMITH	800	20
8	ALLEN	1600	30
9	WARD	1250	30
10	MARTIN	1250	30
11	TURNER	1500	30
12	ADAMS	1100	20
13	JAMES	950	30
14	MILLER	1300	10

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E1.*
2  FROM
3      (
4          SELECT ROWNUM SNO
5                  ,ENAME
6                  ,SAL
7                  ,DEPTNO
8          FROM EMP
9      ) E1
10* WHERE E1.SNO = 2
SQL> /

```

SNO	ENAME	SAL	DEPTNO
2	BLAKE	2850	30

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E1.*
2  FROM
3      (
4          SELECT ROWNUM SNO
5                  ,ENAME
6                  ,SAL
7                  ,DEPTNO
8          FROM EMP
9      ) E1
10* WHERE E1.SNO = 4
SQL> /

```

SNO	ENAME	SAL	DEPTNO
4	JONES	2975	20

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT E1.*
2  FROM
3      (
4          SELECT ROWNUM SNO
5              ,ENAME
6              ,SAL
7              ,DEPTNO
8          FROM EMP
9      ) E1
10* WHERE E1.SNO = 6
SQL> /
```

SNO	ENAME	SAL	DEPTNO
6	FORD	3000	20

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT E1.*
2  FROM
3      (
4          SELECT ROWNUM SNO
5              ,ENAME
6              ,SAL
7              ,DEPTNO
8          FROM EMP
9      ) E1
10* WHERE E1.SNO IN (2,4,6)
SQL> /
```

SNO	ENAME	SAL	DEPTNO
2	BLAKE	2850	30
4	JONES	2975	20
6	FORD	3000	20

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT E1.*
2  FROM
3      (
4          SELECT ROWNUM SNO
5              ,ENAME
6              ,SAL
7              ,DEPTNO
8          FROM EMP
9      ) E1
10* WHERE MOD(E1.SNO,2) =0
SQL> /
```

SNO	ENAME	SAL	DEPTNO
2	BLAKE	2850	30
4	JONES	2975	20
6	FORD	3000	20
8	ALLEN	1600	30
10	MARTIN	1250	30
12	ADAMS	1100	20

14 MILLER 1300 10

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.*
2  FROM
3      (
4          SELECT ROWNUM SNO
5              ,ENAME
6              ,SAL
7              ,DEPTNO
8          FROM EMP
9      ) E1
10* WHERE MOD(E1.SNO,2) =1
SQL> /

```

SNO	ENAME	SAL	DEPTNO
1	KING	5000	10
3	CLARK	2450	10
5	SCOTT	3000	20
7	SMITH	800	20
9	WARD	1250	30
11	TURNER	1500	30
13	JAMES	950	30

7 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.*
2  FROM
3      (
4          SELECT ROWNUM SNO
5              ,ENAME
6              ,SAL
7              ,DEPTNO
8          FROM EMP
9*      ) E1
10 /

```

SNO	ENAME	SAL	DEPTNO
1	KING	5000	10
2	BLAKE	2850	30
3	CLARK	2450	10
4	JONES	2975	20
5	SCOTT	3000	20
6	FORD	3000	20
7	SMITH	800	20
8	ALLEN	1600	30
9	WARD	1250	30
10	MARTIN	1250	30
11	TURNER	1500	30
12	ADAMS	1100	20
13	JAMES	950	30
14	MILLER	1300	10

14 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT E1.*
2  FROM
3      (
4          SELECT ROWNUM SNO
5              ,ENAME
6              ,SAL
7              ,DEPTNO
8          FROM EMP
9      ) E1
10* WHERE MOD(E1.SNO,2)  =1
SQL> /
```

SNO	ENAME	SAL	DEPTNO
1	KING	5000	10
3	CLARK	2450	10
5	SCOTT	3000	20
7	SMITH	800	20
9	WARD	1250	30
11	TURNER	1500	30
13	JAMES	950	30

7 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT E1.*
2  FROM
3      (
4          SELECT ROWNUM SNO
5              ,ENAME
6              ,SAL
7              ,DEPTNO
8          FROM EMP
9      ) E1
10 WHERE MOD(E1.SNO,2)  =1
11 UNION
12 SELECT E1.*
13 FROM
14     (
15         SELECT ROWNUM SNO
16             ,ENAME
17             ,SAL
18             ,DEPTNO
19         FROM EMP
20     ) E1
21* WHERE MOD(E1.SNO,2)  =0
SQL> /
```

SNO	ENAME	SAL	DEPTNO
1	KING	5000	10
2	BLAKE	2850	30
3	CLARK	2450	10
4	JONES	2975	20
5	SCOTT	3000	20
6	FORD	3000	20
7	SMITH	800	20
8	ALLEN	1600	30
9	WARD	1250	30

10	MARTIN	1250	30
11	TURNER	1500	30
12	ADAMS	1100	20
13	JAMES	950	30
14	MILLER	1300	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.*
2  FROM
3      (
4          SELECT ROWNUM SNO
5              ,ENAME
6              ,SAL
7              ,DEPTNO
8          FROM EMP
9      ) E1
10 WHERE MOD(E1.SNO,2)  =1
11 UNION  ALL
12 SELECT E1.*
13 FROM
14      (
15          SELECT ROWNUM SNO
16              ,ENAME
17              ,SAL
18              ,DEPTNO
19          FROM EMP
20      ) E1
21* WHERE MOD(E1.SNO,2)  =0
SQL> /

```

SNO	ENAME	SAL	DEPTNO
1	KING	5000	10
3	CLARK	2450	10
5	SCOTT	3000	20
7	SMITH	800	20
9	WARD	1250	30
11	TURNER	1500	30
13	JAMES	950	30
2	BLAKE	2850	30
4	JONES	2975	20
6	FORD	3000	20
8	ALLEN	1600	30
10	MARTIN	1250	30
12	ADAMS	1100	20
14	MILLER	1300	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.*
2  FROM
3      (
4          SELECT ROWNUM SNO
5              ,ENAME
6              ,SAL
7              ,DEPTNO
8          FROM EMP

```

```

9          ) E1
10 WHERE MOD(E1.SNO,2)  =1
11 UNION  ALL
12 SELECT E1.*
13 FROM
14      (
15          SELECT ROWNUM SNO
16                ,ENAME
17                ,SAL
18                ,DEPTNO
19          FROM EMP
20      ) E1
21* WHERE MOD(E1.SNO,2)  =0
SQL>
SQL> CL SCR

```

```

SQL> SELECT MAX(SAL(
2
SQL>
SQL> SELECT MAX(SAL)
2 FROM EMP;

```

```

      MAX(SAL)
-----
      5000

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SEECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE SAL = (SELECT MAX(SAL)
6                FROM EMP
7*              )
SQL> /
SEECT ENAME
*
ERROR at line 1:
ORA-00900: invalid SQL statement

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SEECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5  WHERE SAL = (SELECT MAX(SAL)
6                FROM EMP
7*              )
SQL> /
SEECT ENAME
*
ERROR at line 1:
ORA-00900: invalid SQL statement

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME

```


ENAME	SAL	DEPTNO
JONES	2975	20

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4* FROM EMP
SQL> /

```

ENAME	SAL	DEPTNO
KING	5000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	3000	20
FORD	3000	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5* ORDER BY SAL DESC
SQL> /

```

ENAME	SAL	DEPTNO
KING	5000	10
SCOTT	3000	20
FORD	3000	20
JONES	2975	20
BLAKE	2850	30
CLARK	2450	10
ALLEN	1600	30
TURNER	1500	30
MILLER	1300	10
MARTIN	1250	30
WARD	1250	30
ADAMS	1100	20
JAMES	950	30
SMITH	800	20

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  FROM EMP
6* ORDER BY SAL DESC
SQL> /

```

ROWNUM	ENAME	SAL	DEPTNO
1	KING	5000	10
5	SCOTT	3000	20
6	FORD	3000	20
4	JONES	2975	20
2	BLAKE	2850	30
3	CLARK	2450	10
8	ALLEN	1600	30
11	TURNER	1500	30
14	MILLER	1300	10
10	MARTIN	1250	30
9	WARD	1250	30
12	ADAMS	1100	20
13	JAMES	950	30
7	SMITH	800	20

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5* ORDER BY SAL
SQL> /

```

ENAME	SAL	DEPTNO
SMITH	800	20
JAMES	950	30
ADAMS	1100	20
MARTIN	1250	30
WARD	1250	30
MILLER	1300	10
TURNER	1500	30
ALLEN	1600	30
CLARK	2450	10
BLAKE	2850	30
JONES	2975	20
FORD	3000	20
SCOTT	3000	20
KING	5000	10

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP

```

```

5* ORDER BY SAL  ASC
SQL> /

```

ENAME	SAL	DEPTNO
SMITH	800	20
JAMES	950	30
ADAMS	1100	20
MARTIN	1250	30
WARD	1250	30
MILLER	1300	10
TURNER	1500	30
ALLEN	1600	30
CLARK	2450	10
BLAKE	2850	30
JONES	2975	20
FORD	3000	20
SCOTT	3000	20
KING	5000	10

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  FROM EMP
6* ORDER BY SAL  DESC
SQL> /

```

ROWNUM	ENAME	SAL	DEPTNO
1	KING	5000	10
5	SCOTT	3000	20
6	FORD	3000	20
4	JONES	2975	20
2	BLAKE	2850	30
3	CLARK	2450	10
8	ALLEN	1600	30
11	TURNER	1500	30
14	MILLER	1300	10
10	MARTIN	1250	30
9	WARD	1250	30
12	ADAMS	1100	20
13	JAMES	950	30
7	SMITH	800	20

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5* FROM EMP
6  /

```

ROWNUM	ENAME	SAL	DEPTNO
--------	-------	-----	--------

1	KING	5000	10
2	BLAKE	2850	30
3	CLARK	2450	10
4	JONES	2975	20
5	SCOTT	3000	20
6	FORD	3000	20
7	SMITH	800	20
8	ALLEN	1600	30
9	WARD	1250	30
10	MARTIN	1250	30
11	TURNER	1500	30
12	ADAMS	1100	20
13	JAMES	950	30
14	MILLER	1300	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ROWNUM
2  ,ENAME
3  ,SAL
4  ,DEPTNO
5  FROM EMP
6* ORDER BY SAL DESC
SQL> /

```

ROWNUM	ENAME	SAL	DEPTNO
1	KING	5000	10
5	SCOTT	3000	20
6	FORD	3000	20
4	JONES	2975	20
2	BLAKE	2850	30
3	CLARK	2450	10
8	ALLEN	1600	30
11	TURNER	1500	30
14	MILLER	1300	10
10	MARTIN	1250	30
9	WARD	1250	30
12	ADAMS	1100	20
13	JAMES	950	30
7	SMITH	800	20

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME
2  ,SAL
3  ,DEPTNO
4  FROM EMP
5* ORDER BY SAL DESC
SQL> /

```

ENAME	SAL	DEPTNO
KING	5000	10
SCOTT	3000	20
FORD	3000	20
JONES	2975	20
BLAKE	2850	30

CLARK	2450	10
ALLEN	1600	30
TURNER	1500	30
MILLER	1300	10
MARTIN	1250	30
WARD	1250	30
ADAMS	1100	20
JAMES	950	30
SMITH	800	20

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.*,ROWNUM
2  FROM (
3      SELECT ENAME
4             ,SAL
5             ,DEPTNO
6      FROM EMP
7      ORDER BY SAL DESC
8*   ) E1
SQL> /

```

ENAME	SAL	DEPTNO	ROWNUM
KING	5000	10	1
SCOTT	3000	20	2
FORD	3000	20	3
JONES	2975	20	4
BLAKE	2850	30	5
CLARK	2450	10	6
ALLEN	1600	30	7
TURNER	1500	30	8
MILLER	1300	10	9
MARTIN	1250	30	10
WARD	1250	30	11
ADAMS	1100	20	12
JAMES	950	30	13
SMITH	800	20	14

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.*,ROWNUM
2  FROM (
3      SELECT ENAME
4             ,SAL
5             ,DEPTNO
6      FROM EMP
7      ORDER BY SAL DESC
8*   ) E1
9* WHERE ROWNUM =1
SQL> /

```

ENAME	SAL	DEPTNO	ROWNUM
KING	5000	10	1

SQL> ED

Wrote file afiedt.buf

```

1  SELECT E1.*,ROWNUM
2  FROM (
3      SELECT ENAME
4             ,SAL
5             ,DEPTNO
6      FROM EMP
7      ORDER BY SAL DESC
8  ) E1
9* WHERE ROWNUM <=2
SQL> /

```

ENAME	SAL	DEPTNO	ROWNUM
KING	5000	10	1
SCOTT	3000	20	2

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E1.*,ROWNUM
2  FROM (
3      SELECT ENAME
4             ,SAL
5             ,DEPTNO
6      FROM EMP
7      ORDER BY SAL DESC
8  ) E1
9* WHERE ROWNUM <=3
SQL> /

```

ENAME	SAL	DEPTNO	ROWNUM
KING	5000	10	1
SCOTT	3000	20	2
FORD	3000	20	3

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E1.*,ROWNUM
2  FROM (
3      SELECT ENAME
4             ,SAL
5             ,DEPTNO
6      FROM EMP
7      ORDER BY SAL DESC
8  ) E1
9* WHERE ROWNUM <=4
SQL> /

```

ENAME	SAL	DEPTNO	ROWNUM
KING	5000	10	1
SCOTT	3000	20	2
FORD	3000	20	3
JONES	2975	20	4

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E1.*,ROWNUM
2  FROM (
3      SELECT ENAME

```

```

4          ,SAL
5          ,DEPTNO
6      FROM EMP
7      ORDER BY SAL DESC
8  ) E1
9* WHERE ROWNUM <=5
SQL> /

```

ENAME	SAL	DEPTNO	ROWNUM
KING	5000	10	1
SCOTT	3000	20	2
FORD	3000	20	3
JONES	2975	20	4
BLAKE	2850	30	5

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E1.*,ROWNUM
2  FROM (
3      SELECT ENAME
4          ,SAL
5          ,DEPTNO
6      FROM EMP
7      ORDER BY SAL DESC
8  ) E1
9* WHERE ROWNUM = 2
SQL> /

```

no rows selected

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT E1.*
2      ,ROWNUM RN
3  FROM (
4      SELECT ENAME
5          ,SAL
6          ,DEPTNO
7      FROM EMP
8      ORDER BY SAL DESC
9*      ) E1
10 /

```

ENAME	SAL	DEPTNO	RN
KING	5000	10	1
SCOTT	3000	20	2
FORD	3000	20	3
JONES	2975	20	4
BLAKE	2850	30	5
CLARK	2450	10	6
ALLEN	1600	30	7
TURNER	1500	30	8
MILLER	1300	10	9
MARTIN	1250	30	10
WARD	1250	30	11
ADAMS	1100	20	12
JAMES	950	30	13
SMITH	800	20	14

14 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT *
2  FROM (
3      SELECT E1.*
4          ,ROWNUM RN
5      FROM (
6          SELECT ENAME
7              ,SAL
8              ,DEPTNO
9          FROM EMP
10         ORDER BY SAL DESC
11     ) E1
12 ) E2
13* WHERE E2.RN = 2
SQL> /
```

ENAME	SAL	DEPTNO	RN
SCOTT	3000	20	2

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT *
2  FROM (
3      SELECT E1.*
4          ,ROWNUM RN
5      FROM (
6          SELECT ENAME
7              ,SAL
8              ,DEPTNO
9          FROM EMP
10         ORDER BY SAL DESC
11     ) E1
12 ) E2
13* WHERE E2.RN = 3
SQL> /
```

ENAME	SAL	DEPTNO	RN
FORD	3000	20	3

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT *
2  FROM (
3      SELECT E1.*
4          ,ROWNUM RN
5      FROM (
6          SELECT ENAME
7              ,SAL
8              ,DEPTNO
9          FROM EMP
10         ORDER BY SAL DESC
11     ) E1
12 ) E2
13* WHERE E2.RN = 4
SQL> /
```

ENAME	SAL	DEPTNO	RN
-------	-----	--------	----

```
-----
JONES          2975          20          4
```

SQL> ED

Wrote file afiedt.buf

```

1  SELECT *
2  FROM (
3      SELECT E1.*
4          ,ROWNUM RN
5      FROM (
6          SELECT ENAME
7              ,SAL
8              ,DEPTNO
9          FROM EMP
10         ORDER BY SAL DESC
11     ) E1
12 ) E2
13* WHERE E2.RN <=3
SQL> /
```

ENAME	SAL	DEPTNO	RN
KING	5000	10	1
SCOTT	3000	20	2
FORD	3000	20	3

SQL> ED

Wrote file afiedt.buf

```

1  SELECT *
2  FROM (
3      SELECT E1.*
4          ,ROWNUM RN
5      FROM (
6          SELECT ENAME
7              ,SAL
8              ,DEPTNO
9          FROM EMP
10         ORDER BY SAL ASC
11     ) E1
12 ) E2
13* WHERE E2.RN <=3
SQL> /
```

ENAME	SAL	DEPTNO	RN
SMITH	800	20	1
JAMES	950	30	2
ADAMS	1100	20	3

SQL> ED

Wrote file afiedt.buf

```

1  SELECT *
2  FROM (
3      SELECT E1.*
4          ,ROWNUM RN
5      FROM (
6          SELECT ENAME
7              ,SAL
8              ,DEPTNO
9          FROM EMP
10         ORDER BY SAL ASC
```

```

11          ) E1
12      ) E2
13 WHERE E2.RN <=3
14 UNION ALL
15 SELECT *
16 FROM (
17     SELECT E1.*
18           ,ROWNUM RN
19     FROM (
20         SELECT ENAME
21              ,SAL
22              ,DEPTNO
23     FROM EMP
24     ORDER BY SAL DESC
25     ) E1
26 ) E2
27* WHERE E2.RN <=3
28 /

```

ENAME	SAL	DEPTNO	RN
SMITH	800	20	1
JAMES	950	30	2
ADAMS	1100	20	3
KING	5000	10	1
SCOTT	3000	20	2
FORD	3000	20	3

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT ENAME,SAL,DEPTNO,'BOTTOM '||RN AS TOP_BOT
2  FROM (
3      SELECT E1.*
4           ,ROWNUM RN
5      FROM (
6          SELECT ENAME
7              ,SAL
8              ,DEPTNO
9      FROM EMP
10     ORDER BY SAL ASC
11     ) E1
12 ) E2
13 WHERE E2.RN <=3
14 UNION ALL
15 SELECT ENAME,SAL,DEPTNO,'TOP '||RN AS TOP_BOT
16 FROM (
17     SELECT E1.*
18           ,ROWNUM RN
19     FROM (
20         SELECT ENAME
21              ,SAL
22              ,DEPTNO
23     FROM EMP
24     ORDER BY SAL DESC
25     ) E1
26 ) E2
27* WHERE E2.RN <=3
SQL> /

```

ENAME	SAL	DEPTNO	TOP_BOT

SMITH	800	20 BOTTOM 1
JAMES	950	30 BOTTOM 2
ADAMS	1100	20 BOTTOM 3
KING	5000	10 TOP 1
SCOTT	3000	20 TOP 2
FORD	3000	20 TOP 3

6 rows selected.

SQL> CL SCR

```
SQL> SELECT LIMIT 3 *
      2 FROM EMP
      3 ORDER BY SAL DESC;
SELECT LIMIT 3 *
              *
```

ERROR at line 1:
ORA-00923: FROM keyword not found where expected

SQL> CL SCR

SQL> EDIT
Wrote file afiedt.buf

SP2-0223: No lines in SQL buffer.
SQL> /
SP2-0103: Nothing in SQL buffer to run.
SQL> CL SCR

```
SQL> SELECT ENAME\
      2
SQL>
SQL> SELECT ENAME
      2 ,SAL
      3 FROM EMP
      4 WHERE ROWNUM <=3;
```

ENAME	SAL
-----	-----
KING	5000
BLAKE	2850
CLARK	2450

```
SQL> SELECT 1
      2 FROM DUAL
      3 UNION ALL
      4 SELECT 122
      5
```

SQL>

SQL> ED
Wrote file afiedt.buf

```
      1 SELECT 1
      2 FROM DUAL
      3 UNION ALL
      4 SELECT 2
      5 FROM DUAL
      6 UNION ALL
      7 SELECT 3
      8 FROM DUAL
      9 UNION ALL
     10 SELECT 3
     11 FROM DUAL
```



```
12 UNION ALL
13 SELECT 5
14* FROM DUAL
SQL> /
```

```
1
-----
1
2
3
3
5
```

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT ROWNUM
2 ,ENAME
3* FROM EMP
SQL> /
```

```
ROWNUM ENAME
-----
1 KING
2 BLAKE
3 CLARK
4 JONES
5 SCOTT
6 FORD
7 SMITH
8 ALLEN
9 WARD
10 MARTIN
11 TURNER
12 ADAMS
13 JAMES
14 MILLER
```

14 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
1 SELECT ROWNUM
2* FROM EMP
SQL> /
```

```
ROWNUM
-----
1
2
3
4
5
6
7
8
9
10
11
12
13
14
```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT ROWNUM
  2  FROM EMP
  3* WHERE ROWNUM <=10
SQL> /
```

ROWNUM

1
2
3
4
5
6
7
8
9
10

10 rows selected.

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT ROWNUM
  2  FROM EMP
  3* WHERE ROWNUM <=20
SQL> /
```

ROWNUM

1
2
3
4
5
6
7
8
9
10
11
12
13
14

14 rows selected.

SQL> CL SCR

```
SQL> SELECT ENAME, SAL, DEPTNO
  2  FROM EMP;
```

ENAME	SAL	DEPTNO
-----	-----	-----
KING	5000	10
BLAKE	2850	30
CLARK	2450	10
JONES	2975	20
SCOTT	3000	20

FORD	3000	20
SMITH	800	20
ALLEN	1600	30
WARD	1250	30
MARTIN	1250	30
TURNER	1500	30
ADAMS	1100	20
JAMES	950	30
MILLER	1300	10

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO,ENAME,SAL,DEPTNO,ROWID
2* FROM EMP
SQL> /

```

EMPNO	ENAME	SAL	DEPTNO	ROWID
7839	KING	5000	10	AAAE5SAABAAAK+hAAA
7698	BLAKE	2850	30	AAAE5SAABAAAK+hAAB
7782	CLARK	2450	10	AAAE5SAABAAAK+hAAC
7566	JONES	2975	20	AAAE5SAABAAAK+hAAD
7788	SCOTT	3000	20	AAAE5SAABAAAK+hAAE
7902	FORD	3000	20	AAAE5SAABAAAK+hAAF
7369	SMITH	800	20	AAAE5SAABAAAK+hAAG
7499	ALLEN	1600	30	AAAE5SAABAAAK+hAAH
7521	WARD	1250	30	AAAE5SAABAAAK+hAAI
7654	MARTIN	1250	30	AAAE5SAABAAAK+hAAJ
7844	TURNER	1500	30	AAAE5SAABAAAK+hAAK
7876	ADAMS	1100	20	AAAE5SAABAAAK+hAAL
7900	JAMES	950	30	AAAE5SAABAAAK+hAAM
7934	MILLER	1300	10	AAAE5SAABAAAK+hAAN

14 rows selected.

SQL> CL SCR

SQL> R

```

1  SELECT EMPNO,ENAME,SAL,DEPTNO,ROWID
2* FROM EMP

```

EMPNO	ENAME	SAL	DEPTNO	ROWID
7839	KING	5000	10	AAAE5SAABAAAK+hAAA
7698	BLAKE	2850	30	AAAE5SAABAAAK+hAAB
7782	CLARK	2450	10	AAAE5SAABAAAK+hAAC
7566	JONES	2975	20	AAAE5SAABAAAK+hAAD
7788	SCOTT	3000	20	AAAE5SAABAAAK+hAAE
7902	FORD	3000	20	AAAE5SAABAAAK+hAAF
7369	SMITH	800	20	AAAE5SAABAAAK+hAAG
7499	ALLEN	1600	30	AAAE5SAABAAAK+hAAH
7521	WARD	1250	30	AAAE5SAABAAAK+hAAI
7654	MARTIN	1250	30	AAAE5SAABAAAK+hAAJ
7844	TURNER	1500	30	AAAE5SAABAAAK+hAAK
7876	ADAMS	1100	20	AAAE5SAABAAAK+hAAL
7900	JAMES	950	30	AAAE5SAABAAAK+hAAM
7934	MILLER	1300	10	AAAE5SAABAAAK+hAAN

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO,ENAME,SAL,DEPTNO,ROWID
2  FROM EMP
3* WHERE EMPNO = 7521
SQL> /

```

EMPNO	ENAME	SAL	DEPTNO	ROWID
7521	WARD	1250	30	AAAE5SAABAAAK+hAAI

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO,ENAME,SAL,DEPTNO,ROWID
2  FROM EMP
3* WHERE ROWID = 'AAAE5SAABAAAK+hAAI '
SQL> /

```

EMPNO	ENAME	SAL	DEPTNO	ROWID
7521	WARD	1250	30	AAAE5SAABAAAK+hAAI

SQL> CL SCR

SQL> SELECT * FROM EMPM;

SELECT * FROM EMPM

*

ERROR at line 1:

ORA-00942: table or view does not exist

SQL> SELECT * FROM EMP;

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM
7839	KING	PRESIDENT		17-NOV-81	5000	
7698	BLAKE	MANAGER	7839	01-MAY-81	2850	
7782	CLARK	MANAGER	7839	09-JUN-81	2450	
7566	JONES	MANAGER	7839	02-APR-81	2975	
7788	SCOTT	ANALYST	7566	19-APR-87	3000	
7902	FORD	ANALYST	7566	03-DEC-81	3000	
7369	SMITH	CLERK	7902	17-DEC-80	800	
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0
7876	ADAMS	CLERK	7788	23-MAY-87	1100	
7900	JAMES	CLERK	7698	03-DEC-81	950	

```

      7934 MILLER      CLERK              7782 23-JAN-82      1300
10

```

14 rows selected.

SQL> CL SCR

SQL> R

```

1* SELECT * FROM EMP

```

DEPTNO	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM
--	7839	KING	PRESIDENT		17-NOV-81	5000	
10	7698	BLAKE	MANAGER	7839	01-MAY-81	2850	
30	7782	CLARK	MANAGER	7839	09-JUN-81	2450	
10	7566	JONES	MANAGER	7839	02-APR-81	2975	
20	7788	SCOTT	ANALYST	7566	19-APR-81	3000	
20	7902	FORD	ANALYST	7566	03-DEC-81	3000	
20	7369	SMITH	CLERK	7902	17-DEC-80	800	
20	7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300
30	7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
30	7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400
30	7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0
30	7876	ADAMS	CLERK	7788	23-MAY-81	1100	
20	7900	JAMES	CLERK	7698	03-DEC-81	950	
30	7934	MILLER	CLERK	7782	23-JAN-82	1300	

14 rows selected.

SQL> INSERT INTO EMP

```

2 SELECT *
3 FROM EMP
4 WHERE EMPNO =7521;

```

INSERT INTO EMP

*

ERROR at line 1:

ORA-00001: unique constraint (SYSTEM.PK_EMP) violated

SQL> ALTER TABLE EMP

```

2 DISABLE CONSTRAINT PK_EMP;

```

Table altered.

SQL> INSERT INTO EMP

```

2 SELECT *
3 FROM EMP
4 WHERE EMPNO =7521;

```

1 row created.

SQL> R

```
1  INSERT INTO EMP
2  SELECT *
3  FROM EMP
4* WHERE EMPNO =7521
```

2 rows created.

SQL> COMMIT;

Commit complete.

SQL> CL SCR

```
SQL> SELECT *
2  FROM EMP
3  ORDER BY EMPNO;
```

DEPTNO	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM
--	7369	SMITH	CLERK	7902	17-DEC-80	800	
20	7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300
30	7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
30	7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
30	7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
30	7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
30	7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
20	7566	JONES	MANAGER	7839	02-APR-81	2975	
30	7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400
30	7698	BLAKE	MANAGER	7839	01-MAY-81	2850	
10	7782	CLARK	MANAGER	7839	09-JUN-81	2450	
20	7788	SCOTT	ANALYST	7566	19-APR-87	3000	
10	7839	KING	PRESIDENT		17-NOV-81	5000	
30	7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0
20	7876	ADAMS	CLERK	7788	23-MAY-87	1100	
30	7900	JAMES	CLERK	7698	03-DEC-81	950	
20	7902	FORD	ANALYST	7566	03-DEC-81	3000	
10	7934	MILLER	CLERK	7782	23-JAN-82	1300	

17 rows selected.

```
SQL> INSERT INTO EMP (EMPNO, ENAME, SAL)
2  VALUES
```

```

3  (
4
SQL> ED
Wrote file afiedt.buf

```

```

1  INSERT INTO EMP (EMPNO, ENAME, SAL)
2  VALUES
3  (7521
4  , 'ARJUN'
5  , 2000
6* )
SQL> /

```

1 row created.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  INSERT INTO EMP (EMPNO, ENAME, SAL)
2  VALUES
3  (101
4  , 'ARJUN'
5  , 3000
6* )
SQL> /

```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT * FROM EMP;
```

DEPTNO	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM
--	7839	KING	PRESIDENT		17-NOV-81	5000	
10	7698	BLAKE	MANAGER	7839	01-MAY-81	2850	
30	7782	CLARK	MANAGER	7839	09-JUN-81	2450	
10	7566	JONES	MANAGER	7839	02-APR-81	2975	
20	7788	SCOTT	ANALYST	7566	19-APR-87	3000	
20	7902	FORD	ANALYST	7566	03-DEC-81	3000	
20	7369	SMITH	CLERK	7902	17-DEC-80	800	
20	7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300
30	7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
30	7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400
30	7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0
30	7876	ADAMS	CLERK	7788	23-MAY-87	1100	

30	7900 JAMES	CLERK	7698 03-DEC-81	950	
10	7934 MILLER	CLERK	7782 23-JAN-82	1300	
30	7521 WARD	SALESMAN	7698 22-FEB-81	1250	500
30	7521 WARD	SALESMAN	7698 22-FEB-81	1250	500
30	7521 WARD	SALESMAN	7698 22-FEB-81	1250	500
	7521 ARJUN			2000	
	101 ARJUN			3000	

19 rows selected.

SQL> CL SCR

```
SQL> SELECT EMPNO,ENAME
2 FROM EMP
3 ORDER BY EMPNO;
```

EMPNO	ENAME
101	ARJUN
7369	SMITH
7499	ALLEN
7521	WARD
7521	WARD
7521	WARD
7521	ARJUN
7521	WARD
7566	JONES
7654	MARTIN
7698	BLAKE
7782	CLARK
7788	SCOTT
7839	KING
7844	TURNER
7876	ADAMS
7900	JAMES
7902	FORD
7934	MILLER

19 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1 SELECT ENAME
2* FROM EMP
3 /
```

ENAME
KING
BLAKE
CLARK
JONES
SCOTT
FORD
SMITH
ALLEN
WARD
MARTIN


```
TURNER
ADAMS
JAMES
MILLER
WARD
WARD
WARD
ARJUN
ARJUN
```

19 rows selected.

```
SQL> CL SCR
```

```
SQL> R
  1  SELECT ENAME
  2* FROM EMP
```

```
ENAME
-----
KING
BLAKE
CLARK
JONES
SCOTT
FORD
SMITH
ALLEN
WARD
MARTIN
TURNER
ADAMS
JAMES
MILLER
WARD
WARD
WARD
ARJUN
ARJUN
```

19 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT ENAME,COUNT(1)
  2  FROM EMP
  3* GROUP BY ENAME
SQL> /
```

ENAME	COUNT(1)
-----	-----
JONES	1
ALLEN	1
FORD	1
ARJUN	2
CLARK	1
MILLER	1
SMITH	1
WARD	4
SCOTT	1
MARTIN	1
TURNER	1
ADAMS	1

```
KING                1
BLAKE               1
JAMES              1
```

15 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```
  1  SELECT ENAME,COUNT(1)
  2  FROM EMP
  3  GROUP BY ENAME
  4*  HAVING COUNT(1) > 1
SQL> /
```

```
ENAME              COUNT(1)
-----
ARJUN              2
WARD              4
```

```
SQL> CL SCR
```

```
SQL> R
  1  SELECT ENAME,COUNT(1)
  2  FROM EMP
  3  GROUP BY ENAME
  4*  HAVING COUNT(1) > 1
```

```
ENAME              COUNT(1)
-----
ARJUN              2
WARD              4
```

```
SQL> DELETE FROM EMP WHERE ENAME ='ARJUN' AND ROWNUM =1;
```

1 row deleted.

```
SQL> DELETE FROM EMP WHERE ENAME ='WARD' AND ROWNUM <=3;
```

3 rows deleted.

```
SQL> SELECT * FROM EMP;
```

DEPTNO	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM
--	7839	KING	PRESIDENT		17-NOV-81	5000	
10	7698	BLAKE	MANAGER	7839	01-MAY-81	2850	
30	7782	CLARK	MANAGER	7839	09-JUN-81	2450	
10	7566	JONES	MANAGER	7839	02-APR-81	2975	
20	7788	SCOTT	ANALYST	7566	19-APR-87	3000	
20	7902	FORD	ANALYST	7566	03-DEC-81	3000	
20	7369	SMITH	CLERK	7902	17-DEC-80	800	
20	7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300
30							

30	7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400
30	7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0
20	7876	ADAMS	CLERK	7788	23-MAY-87	1100	
30	7900	JAMES	CLERK	7698	03-DEC-81	950	
10	7934	MILLER	CLERK	7782	23-JAN-82	1300	
30	7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
	101	ARJUN				3000	

15 rows selected.

SQL> ROLLBACK;

Rollback complete.

SQL> CL SCR

SQL> SELECT * FROM EMP;

DEPTNO	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM
--	7839	KING	PRESIDENT		17-NOV-81	5000	
10	7698	BLAKE	MANAGER	7839	01-MAY-81	2850	
30	7782	CLARK	MANAGER	7839	09-JUN-81	2450	
10	7566	JONES	MANAGER	7839	02-APR-81	2975	
20	7788	SCOTT	ANALYST	7566	19-APR-87	3000	
20	7902	FORD	ANALYST	7566	03-DEC-81	3000	
20	7369	SMITH	CLERK	7902	17-DEC-80	800	
30	7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300
30	7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
30	7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400
30	7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0
20	7876	ADAMS	CLERK	7788	23-MAY-87	1100	
30	7900	JAMES	CLERK	7698	03-DEC-81	950	
10	7934	MILLER	CLERK	7782	23-JAN-82	1300	
30	7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
30	7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
30	7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
	7521	ARJUN				2000	
	101	ARJUN				3000	

19 rows selected.

```
SQL> SELECT ENAME,ROWID
       2 FROM EMP
       3 ORDER BY ENAME;
```

ENAME	ROWID
ADAMS	AAAE5SAABAAAK+hAAL
ALLEN	AAAE5SAABAAAK+hAAH
ARJUN	AAAE5SAABAAAK+hAAR
ARJUN	AAAE5SAABAAAK+hAAS
BLAKE	AAAE5SAABAAAK+hAAB
CLARK	AAAE5SAABAAAK+hAAC
FORD	AAAE5SAABAAAK+hAAF
JAMES	AAAE5SAABAAAK+hAAM
JONES	AAAE5SAABAAAK+hAAD
KING	AAAE5SAABAAAK+hAAA
MARTIN	AAAE5SAABAAAK+hAAJ
MILLER	AAAE5SAABAAAK+hAAN
SCOTT	AAAE5SAABAAAK+hAAE
SMITH	AAAE5SAABAAAK+hAAG
TURNER	AAAE5SAABAAAK+hAAK
WARD	AAAE5SAABAAAK+hAAO
WARD	AAAE5SAABAAAK+hAAP
WARD	AAAE5SAABAAAK+hAAQ
WARD	AAAE5SAABAAAK+hAAI

19 rows selected.

How to delete the duplicate records in the table through Rownum

```
SQL> ED
Wrote file afiedt.buf
```

```
   1 SELECT ENAME,MIN(ROWID)
   2 FROM EMP
   3* GROUP BY ENAME
SQL> /
```

ENAME	MIN(ROWID)
JONES	AAAE5SAABAAAK+hAAD
ALLEN	AAAE5SAABAAAK+hAAH
FORD	AAAE5SAABAAAK+hAAF
ARJUN	AAAE5SAABAAAK+hAAR
CLARK	AAAE5SAABAAAK+hAAC
MILLER	AAAE5SAABAAAK+hAAN
SMITH	AAAE5SAABAAAK+hAAG
WARD	AAAE5SAABAAAK+hAAI
SCOTT	AAAE5SAABAAAK+hAAE
MARTIN	AAAE5SAABAAAK+hAAJ
TURNER	AAAE5SAABAAAK+hAAK
ADAMS	AAAE5SAABAAAK+hAAL
KING	AAAE5SAABAAAK+hAAA
BLAKE	AAAE5SAABAAAK+hAAB
JAMES	AAAE5SAABAAAK+hAAM

15 rows selected.

```
SQL> ED
Wrote file afiedt.buf
```

```

1  SELECT MIN(ROWID)
2  FROM EMP
3*  GROUP BY ENAME
SQL> /

```

MIN(ROWID)

```

-----
AAAE5SAABAAAK+hAAD
AAAE5SAABAAAK+hAAH
AAAE5SAABAAAK+hAAF
AAAE5SAABAAAK+hAAR
AAAE5SAABAAAK+hAAC
AAAE5SAABAAAK+hAAN
AAAE5SAABAAAK+hAAG
AAAE5SAABAAAK+hAAI
AAAE5SAABAAAK+hAAE
AAAE5SAABAAAK+hAAJ
AAAE5SAABAAAK+hAAK
AAAE5SAABAAAK+hAAL
AAAE5SAABAAAK+hAAA
AAAE5SAABAAAK+hAAB
AAAE5SAABAAAK+hAAM

```

15 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT *
2  FROM EMP
3  WHERE ROWID NOT IN (
4                      SELECT MIN(ROWID)
5                      FROM EMP
6                      GROUP BY ENAME
7*                      )
SQL> /

```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
101	ARJUN				3000	
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500

```

SQL> ED
Wrote file afiedt.buf

```

```

1  DELETE
2  FROM EMP
3  WHERE ROWID NOT IN (
4                      SELECT MIN(ROWID)
5                      FROM EMP
6                      GROUP BY ENAME
7*                      )
SQL> /

```

4 rows deleted.

```

SQL> COMMIT;

```

Commit complete.

```
SQL> CL SCR
```

```
SQL> CREATE TABLE STD
2  (
3  NAME VARCHAR2(20)
4  ,FEES NUMBER
5  );
```

Table created.

```
SQL> INSERT INTO STD
2  VALUES
3  (
4  'KRISHNA'
5  ,1000
6  );
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO STD
2  VALUES
3  (
4  'ARJUN'
5  ,1000
6* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO STD
2  VALUES
3  (
4  'HIMANSH'
5  ,1000
6* )
SQL> /
```

1 row created.

```
SQL> ED
Wrote file afiedt.buf
```

```
1  INSERT INTO STD
2  VALUES
3  (
4  'ABHI'
5  ,2000
6* )
SQL> /
```

1 row created.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> CL SCR
```

```
SQL> SELECT *
      2  FROM STD;
```

NAME	FEES
KRISHNA	1000
ARJUN	1000
HIMANSH	1000
ABHI	2000

```
SQL> SELECT *
      2  FROM STD
      3  WHERE ROWID NOT IN ( SELECT MIN(ROWID)
      4                        FROM STD
      5                        GROUP BY FEES
      6                        );
```

NAME	FEES
ARJUN	1000
HIMANSH	1000

```
SQL> ED
Wrote file afiedt.buf
```

```
      1  SELECT NAME
      2  ,FEES
      3  ,ROWID
      4* FROM STD
SQL> /
```

NAME	FEES	ROWID
KRISHNA	1000	AAAE5xAABAAAK+pAAA
ARJUN	1000	AAAE5xAABAAAK+pAAB
HIMANSH	1000	AAAE5xAABAAAK+pAAC
ABHI	2000	AAAE5xAABAAAK+pAAD

```
SQL> ED
Wrote file afiedt.buf
```

```
      1  SELECT NAME
      2  ,FEES
      3  ,ROWID
      4  FROM STD
      5* ORDER BY FEES
SQL> /
```

NAME	FEES	ROWID
KRISHNA	1000	AAAE5xAABAAAK+pAAA
ARJUN	1000	AAAE5xAABAAAK+pAAB
HIMANSH	1000	AAAE5xAABAAAK+pAAC
ABHI	2000	AAAE5xAABAAAK+pAAD

```
SQL> ED
Wrote file afiedt.buf
```

```
      1  SELECT NAME
      2  ,FEES
      3  ,ROWID
```

```

4 ,ROWNUM RN
5 FROM STD
6* ORDER BY FEES
SQL> /

```

NAME	FEES	ROWID	RN
KRISHNA	1000	AAAE5xAABAAAK+pAAA	1
ARJUN	1000	AAAE5xAABAAAK+pAAB	2
HIMANSH	1000	AAAE5xAABAAAK+pAAC	3
ABHI	2000	AAAE5xAABAAAK+pAAD	4

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT NAME
2 ,FEES
3 ,ROWID RID
4 ,ROWNUM RN
5 FROM STD
6* ORDER BY FEES
SQL> /

```

NAME	FEES	RID	RN
KRISHNA	1000	AAAE5xAABAAAK+pAAA	1
ARJUN	1000	AAAE5xAABAAAK+pAAB	2
HIMANSH	1000	AAAE5xAABAAAK+pAAC	3
ABHI	2000	AAAE5xAABAAAK+pAAD	4

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT NAME
2 ,FEES
3 ,ROWID RID
4 ,DENSE_RANK() OVER(PARTITION BY FEES) DR
5* FROM STD
SQL> /

```

```

,DENSE_RANK() OVER(PARTITION BY FEES) DR
*

```

```

ERROR at line 4:
ORA-30485: missing ORDER BY expression in the window specification

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1 SELECT NAME
2 ,FEES
3 ,ROWID RID
4 ,DENSE_RANK() OVER(PARTITION BY FEES ORDER BY FEES) DR
5* FROM STD
SQL> /

```

NAME	FEES	RID	DR
KRISHNA	1000	AAAE5xAABAAAK+pAAA	1
ARJUN	1000	AAAE5xAABAAAK+pAAB	1
HIMANSH	1000	AAAE5xAABAAAK+pAAC	1
ABHI	2000	AAAE5xAABAAAK+pAAD	1

```

SQL> ED
Wrote file afiedt.buf

```



```

1  SELECT NAME
2  ,FEES
3  ,ROWID RID
4  ,ROW_NUMBER() OVER(PARTITION BY FEES ORDER BY FEES) DR
5* FROM STD
SQL> /

```

NAME	FEES	RID	DR
KRISHNA	1000	AAAE5xAABAAAK+pAAA	1
ARJUN	1000	AAAE5xAABAAAK+pAAB	2
HIMANSH	1000	AAAE5xAABAAAK+pAAC	3
ABHI	2000	AAAE5xAABAAAK+pAAD	1

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT *
2  FROM (
3  SELECT NAME
4  ,FEES
5  ,ROWID RID
6  ,ROW_NUMBER() OVER(PARTITION BY FEES ORDER BY FEES) DR
7  ,COUNT(1) OVER(PARTITION BY FEES) CNT
8  FROM STD
9* )
SQL> /

```

NAME	FEES	RID	DR	CNT
KRISHNA	1000	AAAE5xAABAAAK+pAAA	1	3
ARJUN	1000	AAAE5xAABAAAK+pAAB	2	3
HIMANSH	1000	AAAE5xAABAAAK+pAAC	3	3
ABHI	2000	AAAE5xAABAAAK+pAAD	1	1

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT *
2  FROM (
3  SELECT NAME
4  ,FEES
5  ,ROWID RID
6  ,ROW_NUMBER() OVER(PARTITION BY FEES ORDER BY FEES) DR
7  ,COUNT(1) OVER(PARTITION BY FEES) CNT
8  FROM STD
9  )
10 WHERE( (DR =2 AND CNT > 1)
11         OR DR =1
12*       )
SQL> /

```

NAME	FEES	RID	DR	CNT
KRISHNA	1000	AAAE5xAABAAAK+pAAA	1	3
ARJUN	1000	AAAE5xAABAAAK+pAAB	2	3
ABHI	2000	AAAE5xAABAAAK+pAAD	1	1

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT *
2  FROM (

```

```

3  SELECT NAME
4  ,FEES
5  ,ROWID RID
6  ,ROW_NUMBER() OVER(PARTITION BY FEES ORDER BY FEES) DR
7  ,COUNT(1) OVER(PARTITION BY FEES) CNT
8  FROM STD
9  )
10 WHERE( (DR =2 AND CNT  > 1)
11         OR ( DR =1 AND CNT = 1)
12*      )
SQL> /

```

NAME	FEES	RID	DR	CNT
ARJUN	1000	AAAE5xAABAAAK+pAAB	2	3
ABHI	2000	AAAE5xAABAAAK+pAAD	1	1

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT  ROWID
2  FROM (
3  SELECT NAME
4  ,FEES
5  ,ROWID RID
6  ,ROW_NUMBER() OVER(PARTITION BY FEES ORDER BY FEES) DR
7  ,COUNT(1) OVER(PARTITION BY FEES) CNT
8  FROM STD
9  )
10 WHERE( (DR =2 AND CNT  > 1)
11         OR ( DR =1 AND CNT = 1)
12*      )
SQL> /

```

```

ROWID
-----
AAAE5xAABAAAK+pAAB
AAAE5xAABAAAK+pAAD

```

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT *
2  FROM STD
3  WHERE ROWID NOT IN (
4  SELECT  ROWID
5  FROM (
6  SELECT NAME
7  ,FEES
8  ,ROWID RID
9  ,ROW_NUMBER() OVER(PARTITION BY FEES ORDER BY FEES) DR
10 ,COUNT(1) OVER(PARTITION BY FEES) CNT
11 FROM STD
12 )
13 WHERE( (DR =2 AND CNT  > 1)
14         OR ( DR =1 AND CNT = 1)
15         )
16* )
SQL> /

```

NAME	FEES
KRISHNA	1000
HIMANSH	1000

```

SQL> ED
Wrote file afiedt.buf

  1  DELETE
  2  FROM STD
  3  WHERE ROWID NOT IN (
  4  SELECT  ROWID
  5  FROM (
  6  SELECT NAME
  7  ,FEES
  8  ,ROWID RID
  9  ,ROW_NUMBER() OVER(PARTITION BY FEES ORDER BY FEES) DR
 10  ,COUNT(1) OVER(PARTITION BY FEES) CNT
 11  FROM STD
 12  )
 13  WHERE( (DR =2 AND CNT  > 1)
 14          OR ( DR =1 AND CNT = 1)
 15          )
 16* )
SQL> /

```

2 rows deleted.

```
SQL> SELECT * FROM STD;
```

NAME	FEES
ARJUN	1000
ABHI	2000

```

SQL> ED
Wrote file afiedt.buf

```

```

  1* SELECT * FROM STD
SQL> SPOOL OFF

```

25-jun-2018 Hierarchial querie spool

```

SQL> SELECT EMPNO
  2  ,ENAME
  3  ,MGR
  4  FROM EMP;

```

EMPNO	ENAME	MGR
7839	KING	
7698	BLAKE	7839
7782	CLARK	7839
7566	JONES	7839
7788	SCOTT	7566
7902	FORD	7566
7369	SMITH	7902
7499	ALLEN	7698
7521	WARD	7698
7654	MARTIN	7698
7844	TURNER	7698
7876	ADAMS	7788
7900	JAMES	7698
7934	MILLER	7782

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,MGR
4  FROM EMP
5* CONNECT BY EMPNO = MGR
SQL> /

```

EMPNO	ENAME	MGR
7839	KING	
7698	BLAKE	7839
7782	CLARK	7839
7566	JONES	7839
7788	SCOTT	7566
7902	FORD	7566
7369	SMITH	7902
7499	ALLEN	7698
7521	WARD	7698
7654	MARTIN	7698
7844	TURNER	7698
7876	ADAMS	7788
7900	JAMES	7698
7934	MILLER	7782

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,MGR
4  FROM EMP
5  START WITH ENAME ='KING'
6* CONNECT BY EMPNO = MGR
SQL> /

```

EMPNO	ENAME	MGR
7839	KING	

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,MGR
4  FROM EMP
5  START WITH ENAME ='KING'
6* CONNECT BY PRIOR EMPNO = MGR
SQL> /

```

EMPNO	ENAME	MGR
7839	KING	
7566	JONES	7839
7788	SCOTT	7566
7876	ADAMS	7788

7902	FORD	7566
7369	SMITH	7902
7698	BLAKE	7839
7499	ALLEN	7698
7521	WARD	7698
7654	MARTIN	7698
7844	TURNER	7698
7900	JAMES	7698
7782	CLARK	7839
7934	MILLER	7782

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  FROM EMP
6  START WITH ENAME ='KING'
7  CONNECT BY PRIOR EMPNO = MGR
8* ORDER BY SAL
SQL> /

```

EMPNO	ENAME	SAL	MGR
7369	SMITH	800	7902
7900	JAMES	950	7698
7876	ADAMS	1100	7788
7654	MARTIN	1250	7698
7521	WARD	1250	7698
7934	MILLER	1300	7782
7844	TURNER	1500	7698
7499	ALLEN	1600	7698
7782	CLARK	2450	7839
7698	BLAKE	2850	7839
7566	JONES	2975	7839
7788	SCOTT	3000	7566
7902	FORD	3000	7566
7839	KING	5000	

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  FROM EMP
6  START WITH ENAME ='KING'
7  CONNECT BY PRIOR EMPNO = MGR
8* ORDER SIBLIGS BY SAL
SQL> /

```

ORDER SIBLIGS BY SAL

*

ERROR at line 8:

ORA-00924: missing BY keyword

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  FROM EMP
6  START WITH ENAME ='KING'
7  CONNECT BY PRIOR EMPNO = MGR
8* ORDER SIBLINGS BY SAL
SQL> /

```

EMPNO	ENAME	SAL	MGR
7839	KING	5000	
7782	CLARK	2450	7839
7934	MILLER	1300	7782
7698	BLAKE	2850	7839
7900	JAMES	950	7698
7521	WARD	1250	7698
7654	MARTIN	1250	7698
7844	TURNER	1500	7698
7499	ALLEN	1600	7698
7566	JONES	2975	7839
7788	SCOTT	3000	7566
7876	ADAMS	1100	7788
7902	FORD	3000	7566
7369	SMITH	800	7902

14 rows selected.

SQL> ED
Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  FROM EMP
6  START WITH ENAME ='KING'
7  CONNECT BY PRIOR EMPNO = MGR
8* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR
7839	KING	5000	
7566	JONES	2975	7839
7788	SCOTT	3000	7566
7876	ADAMS	1100	7788
7902	FORD	3000	7566
7369	SMITH	800	7902
7698	BLAKE	2850	7839
7499	ALLEN	1600	7698
7844	TURNER	1500	7698
7521	WARD	1250	7698
7654	MARTIN	1250	7698
7900	JAMES	950	7698
7782	CLARK	2450	7839
7934	MILLER	1300	7782

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  FROM EMP
6  START WITH ENAME ='BLAKE'
7  CONNECT BY PRIOR EMPNO = MGR
8* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR
7698	BLAKE	2850	7839
7499	ALLEN	1600	7698
7844	TURNER	1500	7698
7521	WARD	1250	7698
7654	MARTIN	1250	7698
7900	JAMES	950	7698

6 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  FROM EMP
6  START WITH ENAME ='BLAKE'
7  CONNECT BY EMPNO = MGR PRIOR
8* ORDER SIBLINGS BY SAL DESC
SQL> /

```

```

CONNECT BY EMPNO = MGR PRIOR
              *
```

ERROR at line 7:
ORA-00933: SQL command not properly ended

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  FROM EMP
6  START WITH ENAME ='BLAKE'
7  CONNECT BY EMPNO = PRIOR MGR
8* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR
7698	BLAKE	2850	7839
7839	KING	5000	

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
```

```

3  ,SAL
4  ,MGR
5  FROM EMP
6  START WITH ENAME ='SMITH'
7  CONNECT BY EMPNO = PRIOR  MGR
8* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR
7369	SMITH	800	7902
7902	FORD	3000	7566
7566	JONES	2975	7839
7839	KING	5000	

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,CONNECT_BY_ROOT(ENAME)
6  FROM EMP
7  START WITH ENAME ='KING'
8  CONNECT BY PRIOR  EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR	CONNECT_BY
7839	KING	5000	KING	
7566	JONES	2975	7839	KING
7788	SCOTT	3000	7566	KING
7876	ADAMS	1100	7788	KING
7902	FORD	3000	7566	KING
7369	SMITH	800	7902	KING
7698	BLAKE	2850	7839	KING
7499	ALLEN	1600	7698	KING
7844	TURNER	1500	7698	KING
7521	WARD	1250	7698	KING
7654	MARTIN	1250	7698	KING
7900	JAMES	950	7698	KING
7782	CLARK	2450	7839	KING
7934	MILLER	1300	7782	KING

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,CONNECT_BY_ROOT(SAL)
6  FROM EMP
7  START WITH ENAME ='KING'
8  CONNECT BY PRIOR  EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR	CONNECT_BY_ROOT(SAL)
-------	-------	-----	-----	----------------------

7839	KING	5000		5000
7566	JONES	2975	7839	5000
7788	SCOTT	3000	7566	5000
7876	ADAMS	1100	7788	5000
7902	FORD	3000	7566	5000
7369	SMITH	800	7902	5000
7698	BLAKE	2850	7839	5000
7499	ALLEN	1600	7698	5000
7844	TURNER	1500	7698	5000
7521	WARD	1250	7698	5000
7654	MARTIN	1250	7698	5000
7900	JAMES	950	7698	5000
7782	CLARK	2450	7839	5000
7934	MILLER	1300	7782	5000

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,CONNECT_BY_ROOT(SAL,ENAME)
6  FROM EMP
7  START WITH ENAME ='KING'
8  CONNECT BY PRIOR EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC

```

SQL> /

```

,CONNECT_BY_ROOT(SAL,ENAME)
      *

```

ERROR at line 5:

ORA-00907: missing right parenthesis

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,CONNECT_BY_ROOT(SAL)
6  ,CONNECT_BY_ROOT(ENAME)
7  FROM EMP
8  START WITH ENAME ='KING'
9  CONNECT BY PRIOR EMPNO = MGR
10* ORDER SIBLINGS BY SAL DESC

```

SQL> /

EMPNO	ENAME	SAL	MGR	CONNECT_BY_ROOT(SAL)	CONNECT_BY
7839	KING	5000		5000	KING
7566	JONES	2975	7839	5000	KING
7788	SCOTT	3000	7566	5000	KING
7876	ADAMS	1100	7788	5000	KING
7902	FORD	3000	7566	5000	KING
7369	SMITH	800	7902	5000	KING
7698	BLAKE	2850	7839	5000	KING
7499	ALLEN	1600	7698	5000	KING
7844	TURNER	1500	7698	5000	KING
7521	WARD	1250	7698	5000	KING
7654	MARTIN	1250	7698	5000	KING

7900 JAMES	950	7698	5000 KING
7782 CLARK	2450	7839	5000 KING
7934 MILLER	1300	7782	5000 KING

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,SYS_CONNECT_BY_PATH(ENAME, '<=') PATH
6  FROM EMP
7  START WITH ENAME ='KING'
8  CONNECT BY PRIOR EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC
SQL> COLUMN PATH FORMAT A30
SQL> R

```

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,SYS_CONNECT_BY_PATH(ENAME, '<=') PATH
6  FROM EMP
7  START WITH ENAME ='KING'
8  CONNECT BY PRIOR EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC

```

EMPNO	ENAME	SAL	MGR	PATH
7839	KING	5000		<=KING
7566	JONES	2975	7839	<=KING<=JONES
7788	SCOTT	3000	7566	<=KING<=JONES<=SCOTT
7876	ADAMS	1100	7788	<=KING<=JONES<=SCOTT<=ADAMS
7902	FORD	3000	7566	<=KING<=JONES<=FORD
7369	SMITH	800	7902	<=KING<=JONES<=FORD<=SMITH
7698	BLAKE	2850	7839	<=KING<=BLAKE
7499	ALLEN	1600	7698	<=KING<=BLAKE<=ALLEN
7844	TURNER	1500	7698	<=KING<=BLAKE<=TURNER
7521	WARD	1250	7698	<=KING<=BLAKE<=WARD
7654	MARTIN	1250	7698	<=KING<=BLAKE<=MARTIN
7900	JAMES	950	7698	<=KING<=BLAKE<=JAMES
7782	CLARK	2450	7839	<=KING<=CLARK
7934	MILLER	1300	7782	<=KING<=CLARK<=MILLER

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,SYS_CONNECT_BY_PATH(ENAME, ',') PATH
6  FROM EMP
7  START WITH ENAME ='KING'
8  CONNECT BY PRIOR EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR	PATH
-------	-------	-----	-----	------

```

-----
      7839 KING                5000                ,KING
      7566 JONES              2975              7839 ,KING,JONES
      7788 SCOTT              3000              7566 ,KING,JONES,SCOTT
      7876 ADAMS              1100              7788 ,KING,JONES,SCOTT,ADAMS
      7902 FORD               3000              7566 ,KING,JONES,FORD
      7369 SMITH              800               7902 ,KING,JONES,FORD,SMITH
      7698 BLAKE              2850              7839 ,KING,BLAKE
      7499 ALLEN              1600              7698 ,KING,BLAKE,ALLEN
      7844 TURNER             1500              7698 ,KING,BLAKE,TURNER
      7521 WARD               1250              7698 ,KING,BLAKE,WARD
      7654 MARTIN             1250              7698 ,KING,BLAKE,MARTIN
      7900 JAMES              950               7698 ,KING,BLAKE,JAMES
      7782 CLARK              2450              7839 ,KING,CLARK
      7934 MILLER             1300              7782 ,KING,CLARK,MILLER

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,SYS_CONNECT_BY_PATH(ENAME,'=>') PATH
6  FROM EMP
7  START WITH ENAME ='KING'
8  CONNECT BY PRIOR EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC
SQL> /

```

```

      EMPNO ENAME                SAL                MGR PATH
-----
      7839 KING                5000                =>KING
      7566 JONES              2975              7839 =>KING=>JONES
      7788 SCOTT              3000              7566 =>KING=>JONES=>SCOTT
      7876 ADAMS              1100              7788 =>KING=>JONES=>SCOTT=>ADAMS
      7902 FORD               3000              7566 =>KING=>JONES=>FORD
      7369 SMITH              800               7902 =>KING=>JONES=>FORD=>SMITH
      7698 BLAKE              2850              7839 =>KING=>BLAKE
      7499 ALLEN              1600              7698 =>KING=>BLAKE=>ALLEN
      7844 TURNER             1500              7698 =>KING=>BLAKE=>TURNER
      7521 WARD               1250              7698 =>KING=>BLAKE=>WARD
      7654 MARTIN             1250              7698 =>KING=>BLAKE=>MARTIN
7900 JAMES              950              7698 =>KING=>BLAKE=>JAMES
      7782 CLARK              2450              7839 =>KING=>CLARK
      7934 MILLER             1300              7782 =>KING=>CLARK=>MILLER

```

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,SYS_CONNECT_BY_PATH(SAL,'=>') PATH
6  FROM EMP
7  START WITH ENAME ='KING'
8  CONNECT BY PRIOR EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR	PATH
7839	KING	5000		=>5000
7566	JONES	2975	7839	=>5000=>2975
7788	SCOTT	3000	7566	=>5000=>2975=>3000
7876	ADAMS	1100	7788	=>5000=>2975=>3000=>1100
7902	FORD	3000	7566	=>5000=>2975=>3000
7369	SMITH	800	7902	=>5000=>2975=>3000=>800
7698	BLAKE	2850	7839	=>5000=>2850
7499	ALLEN	1600	7698	=>5000=>2850=>1600
7844	TURNER	1500	7698	=>5000=>2850=>1500
7521	WARD	1250	7698	=>5000=>2850=>1250
7654	MARTIN	1250	7698	=>5000=>2850=>1250
7900	JAMES	950	7698	=>5000=>2850=>950
7782	CLARK	2450	7839	=>5000=>2450
7934	MILLER	1300	7782	=>5000=>2450=>1300

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,SYS_CONNECT_BY_PATH(SAL,'=>') PATH
6  ,SYS_CONNECT_BY_PATH(ENAME,'=>') PATH
7  FROM EMP
8  START WITH ENAME ='KING'
9  CONNECT BY PRIOR EMPNO = MGR
10* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR	PATH
7839	KING	5000		=>5000
7566	JONES	2975	7839	=>5000=>2975
7788	SCOTT	3000	7566	=>5000=>2975=>3000
7876	ADAMS	1100	7788	=>5000=>2975=>3000=>1100
7902	FORD	3000	7566	=>5000=>2975=>3000
7369	SMITH	800	7902	=>5000=>2975=>3000=>800
7698	BLAKE	2850	7839	=>5000=>2850
7499	ALLEN	1600	7698	=>5000=>2850=>1600
7844	TURNER	1500	7698	=>5000=>2850=>1500

```
=>KING=>BLAKE=>TURNER
```

```
      7521 WARD          1250      7698 =>5000=>2850=>1250
=>KING=>BLAKE=>WARD
```

```
      7654 MARTIN       1250      7698 =>5000=>2850=>1250
=>KING=>BLAKE=>MARTIN
```

```
      7900 JAMES        950      7698 =>5000=>2850=>950
=>KING=>BLAKE=>JAMES
```

```
      7782 CLARK        2450      7839 =>5000=>2450
=>KING=>CLARK
```

```
      7934 MILLER       1300      7782 =>5000=>2450=>1300
=>KING=>CLARK=>MILLER
```

14 rows selected.

```
SQL> ED
```

Wrote file afiedt.buf

```
 1  SELECT EMPNO
 2  ,SYS_CONNECT_BY_PATH(SAL,'=>') PATH
 3  ,SYS_CONNECT_BY_PATH(ENAME,'=>') PATH
 4  FROM EMP
 5  START WITH ENAME ='KING'
 6  CONNECT BY PRIOR EMPNO = MGR
 7* ORDER SIBLINGS BY SAL DESC
SQL> /
```

EMPNO	PATH	PATH
7839	=>5000	=>KING
7566	=>5000=>2975	=>KING=>JONES
7788	=>5000=>2975=>3000	=>KING=>JONES=>SCOTT
7876	=>5000=>2975=>3000=>1100	=>KING=>JONES=>SCOTT=>ADAMS
7902	=>5000=>2975=>3000	=>KING=>JONES=>FORD
7369	=>5000=>2975=>3000=>800	=>KING=>JONES=>FORD=>SMITH
7698	=>5000=>2850	=>KING=>BLAKE
7499	=>5000=>2850=>1600	=>KING=>BLAKE=>ALLEN
7844	=>5000=>2850=>1500	=>KING=>BLAKE=>TURNER
7521	=>5000=>2850=>1250	=>KING=>BLAKE=>WARD
7654	=>5000=>2850=>1250	=>KING=>BLAKE=>MARTIN
7900	=>5000=>2850=>950	=>KING=>BLAKE=>JAMES
7782	=>5000=>2450	=>KING=>CLARK
7934	=>5000=>2450=>1300	=>KING=>CLARK=>MILLER

14 rows selected.

```
SQL> ED
```

Wrote file afiedt.buf

```
 1  SELECT EMPNO
 2  ,ENAME
 3  ,SAL
 4  ,MGR
 5  ,SYS_CONNECT_BY_PATH(ENAME||' '||SAL,'=>') PATH
 6  FROM EMP
 7  START WITH ENAME ='KING'
 8  CONNECT BY PRIOR EMPNO = MGR
 9* ORDER SIBLINGS BY SAL DESC
SQL> /
```

EMPNO	ENAME	SAL	MGR	PATH
7839	KING	5000		=>KING 5000
7566	JONES	2975	7839	=>KING 5000=>JONES 2975
7788	SCOTT	3000	7566	=>KING 5000=>JONES 2975=>SCOTT 3000
7876	ADAMS	1100	7788	=>KING 5000=>JONES 2975=>SCOTT 3000=>ADAMS 1100
7902	FORD	3000	7566	=>KING 5000=>JONES 2975=>FORD 3000
7369	SMITH	800	7902	=>KING 5000=>JONES 2975=>FORD 3000=>SMITH 800
7698	BLAKE	2850	7839	=>KING 5000=>BLAKE 2850
7499	ALLEN	1600	7698	=>KING 5000=>BLAKE 2850=>ALLEN 1600
7844	TURNER	1500	7698	=>KING 5000=>BLAKE 2850=>TURNER 1500
7521	WARD	1250	7698	=>KING 5000=>BLAKE 2850=>WARD 1250
7654	MARTIN	1250	7698	=>KING 5000=>BLAKE 2850=>MARTIN 1250
7900	JAMES	950	7698	=>KING 5000=>BLAKE 2850=>JAMES 950
7782	CLARK	2450	7839	=>KING 5000=>CLARK 2450
7934	MILLER	1300	7782	=>KING 5000=>CLARK 2450=>MILLER 1300

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SYS_CONNECT_BY_PATH(ENAME||' '||SAL,'=>') PATH
4  FROM EMP
5  START WITH ENAME ='KING'
6  CONNECT BY PRIOR EMPNO = MGR
7* ORDER SIBLINGS BY SAL DESC
8  /

```

EMPNO	ENAME	PATH
7839	KING	=>KING 5000
7566	JONES	=>KING 5000=>JONES 2975
7788	SCOTT	=>KING 5000=>JONES 2975=>SCOTT 3000
7876	ADAMS	=>KING 5000=>JONES 2975=>SCOTT 3000=>ADAMS 1100
7902	FORD	=>KING 5000=>JONES 2975=>FORD 3000

```

7369 SMITH      =>KING 5000=>JONES 2975=>FORD
                3000=>SMITH 800

7698 BLAKE      =>KING 5000=>BLAKE 2850
7499 ALLEN      =>KING 5000=>BLAKE 2850=>ALLEN
                1600

7844 TURNER     =>KING 5000=>BLAKE 2850=>TURNER
                R 1500

7521 WARD       =>KING 5000=>BLAKE 2850=>WARD
                1250

7654 MARTIN     =>KING 5000=>BLAKE 2850=>MARTIN
                N 1250

7900 JAMES      =>KING 5000=>BLAKE 2850=>JAMES
                950

7782 CLARK      =>KING 5000=>CLARK 2450
7934 MILLER     =>KING 5000=>CLARK 2450=>MILLER
                R 1300

```

14 rows selected.

SQL> COLUMN PATH FORMAT A50

SQL> SET LINESIZE 100

SQL> R

```

1  SELECT EMPNO
2  ,ENAME
3  ,SYS_CONNECT_BY_PATH(ENAME||' '||SAL,'=>') PATH
4  FROM EMP
5  START WITH ENAME ='KING'
6  CONNECT BY PRIOR EMPNO = MGR
7* ORDER SIBLINGS BY SAL DESC

```

EMPNO	ENAME	PATH
7839	KING	=>KING 5000
7566	JONES	=>KING 5000=>JONES 2975
7788	SCOTT	=>KING 5000=>JONES 2975=>SCOTT 3000
7876	ADAMS	=>KING 5000=>JONES 2975=>SCOTT 3000=>ADAMS 1100
7902	FORD	=>KING 5000=>JONES 2975=>FORD 3000
7369	SMITH	=>KING 5000=>JONES 2975=>FORD 3000=>SMITH 800
7698	BLAKE	=>KING 5000=>BLAKE 2850
7499	ALLEN	=>KING 5000=>BLAKE 2850=>ALLEN 1600
7844	TURNER	=>KING 5000=>BLAKE 2850=>TURNER 1500
7521	WARD	=>KING 5000=>BLAKE 2850=>WARD 1250
7654	MARTIN	=>KING 5000=>BLAKE 2850=>MARTIN 1250
7900	JAMES	=>KING 5000=>BLAKE 2850=>JAMES 950
7782	CLARK	=>KING 5000=>CLARK 2450
7934	MILLER	=>KING 5000=>CLARK 2450=>MILLER 1300

14 rows selected.

SQL> ED

Wrote file afiedt.buf

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR

```

```

5  ,CONNECT_BY_ISLEAF
6  FROM EMP
7  START WITH ENAME ='KING'
8  CONNECT BY PRIOR  EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR	CONNECT_BY_ISLEAF
7839	KING	5000		0
7566	JONES	2975	7839	0
7788	SCOTT	3000	7566	0
7876	ADAMS	1100	7788	1
7902	FORD	3000	7566	0
7369	SMITH	800	7902	1
7698	BLAKE	2850	7839	0
7499	ALLEN	1600	7698	1
7844	TURNER	1500	7698	1
7521	WARD	1250	7698	1
7654	MARTIN	1250	7698	1
7900	JAMES	950	7698	1
7782	CLARK	2450	7839	0
7934	MILLER	1300	7782	1

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT *
2  FROM (
3  SELECT EMPNO
4  ,ENAME
5  ,SAL
6  ,MGR
7  ,CONNECT_BY_ISLEAF LV
8  FROM EMP
9  START WITH ENAME ='KING'
10 CONNECT BY PRIOR  EMPNO = MGR
11 ORDER SIBLINGS BY SAL DESC
12 )
13* WHERE LV=1
SQL> /

```

EMPNO	ENAME	SAL	MGR	LV
7876	ADAMS	1100	7788	1
7369	SMITH	800	7902	1
7499	ALLEN	1600	7698	1
7844	TURNER	1500	7698	1
7521	WARD	1250	7698	1
7654	MARTIN	1250	7698	1
7900	JAMES	950	7698	1
7934	MILLER	1300	7782	1

8 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT *
2  FROM (
3  SELECT EMPNO
4  ,ENAME

```



```

5  ,SAL
6  ,MGR
7  ,CONNECT_BY_ISLEAF LV
8  FROM EMP
9  START WITH ENAME ='KING'
10 CONNECT BY PRIOR  EMPNO = MGR
11 ORDER SIBLINGS BY SAL DESC
12 )
13* WHERE LV=0
SQL> /

```

EMPNO	ENAME	SAL	MGR	LV
7839	KING	5000		0
7566	JONES	2975	7839	0
7788	SCOTT	3000	7566	0
7902	FORD	3000	7566	0
7698	BLAKE	2850	7839	0
7782	CLARK	2450	7839	0

6 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,LEVEL
6  FROM EMP
7  START WITH ENAME ='KING'
8  CONNECT BY PRIOR  EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR	LEVEL
7839	KING	5000		1
7566	JONES	2975	7839	2
7788	SCOTT	3000	7566	3
7876	ADAMS	1100	7788	4
7902	FORD	3000	7566	3
7369	SMITH	800	7902	4
7698	BLAKE	2850	7839	2
7499	ALLEN	1600	7698	3
7844	TURNER	1500	7698	3
7521	WARD	1250	7698	3
7654	MARTIN	1250	7698	3
7900	JAMES	950	7698	3
7782	CLARK	2450	7839	2
7934	MILLER	1300	7782	3

14 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,LEVEL
6  FROM EMP

```

```

7  START WITH ENAME ='BLAKE'
8  CONNECT BY PRIOR   EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR	LEVEL
7698	BLAKE	2850	7839	1
7499	ALLEN	1600	7698	2
7844	TURNER	1500	7698	2
7521	WARD	1250	7698	2
7654	MARTIN	1250	7698	2
7900	JAMES	950	7698	2

6 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,CONNECT_BY_ISCYCLE
6  FROM EMP
7  START WITH ENAME ='BLAKE'
8  CONNECT BY PRIOR   EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC
SQL> /

```

```

CONNECT BY PRIOR   EMPNO = MGR
                *

```

ERROR at line 8:

ORA-30930: NOCYCLE keyword is required with CONNECT_BY_ISCYCLE pseudocolumn

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT EMPNO
2  ,ENAME
3  ,SAL
4  ,MGR
5  ,CONNECT_BY_ISCYCLE
6  FROM EMP
7  START WITH ENAME ='BLAKE'
8  CONNECT BY NOCYCLE PRIOR   EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR	CONNECT_BY_ISCYCLE
7698	BLAKE	2850	7839	0
7499	ALLEN	1600	7698	0
7844	TURNER	1500	7698	0
7521	WARD	1250	7698	0
7654	MARTIN	1250	7698	0
7900	JAMES	950	7698	0

6 rows selected.

```

SQL> ED
Wrote file afiedt.buf

```

```

1  SELECT EMPNO

```

```

2 ,ENAME
3 ,SAL
4 ,MGR
5 ,CONNECT_BY_ISCYCLE
6 FROM EMP
7 START WITH ENAME ='KING'
8 CONNECT BY NOCYCLE PRIOR EMPNO = MGR
9* ORDER SIBLINGS BY SAL DESC
SQL> /

```

EMPNO	ENAME	SAL	MGR	CONNECT_BY_ISCYCLE
7839	KING	5000		0
7566	JONES	2975	7839	0
7788	SCOTT	3000	7566	0
7876	ADAMS	1100	7788	0
7902	FORD	3000	7566	0
7369	SMITH	800	7902	0
7698	BLAKE	2850	7839	0
7499	ALLEN	1600	7698	0
7844	TURNER	1500	7698	0
7521	WARD	1250	7698	0
7654	MARTIN	1250	7698	0
7900	JAMES	950	7698	0
7782	CLARK	2450	7839	0
7934	MILLER	1300	7782	0

14 rows selected.

```

SQL> UPDATE EMP
2 SET MGR=7369
3 WHERE EMPNO=7839;

```

1 row updated.

```

SQL> SELECT EMPNO
2 ,ENAME
3 ,SAL
4 ,MGR
5 ,CONNECT_BY_ISCYCLE
6 FROM EMP
7 START WITH ENAME ='KING'
8 CONNECT BY NOCYCLE PRIOR EMPNO = MGR
9 ORDER SIBLINGS BY SAL DESC;

```

EMPNO	ENAME	SAL	MGR	CONNECT_BY_ISCYCLE
7839	KING	5000	7369	0
7566	JONES	2975	7839	0
7788	SCOTT	3000	7566	0
7876	ADAMS	1100	7788	0
7902	FORD	3000	7566	0
7369	SMITH	800	7902	1
7698	BLAKE	2850	7839	0
7499	ALLEN	1600	7698	0
7844	TURNER	1500	7698	0
7521	WARD	1250	7698	0
7654	MARTIN	1250	7698	0
7900	JAMES	950	7698	0
7782	CLARK	2450	7839	0
7934	MILLER	1300	7782	0

14 rows selected.

```
SQL> CL SCR
```

```
SQL> SELECT ROWNUM
       2  FROM EMP;
```

```
      ROWNUM
-----
         1
         2
         3
         4
         5
         6
         7
         8
         9
        10
        11
        12
        13
        14
```

```
14 rows selected.
```

```
SQL> ED
Wrote file afiedt.buf
```

```
   1  SELECT ROWNUM
   2  FROM EMP
   3* WHERE ROWNUM <=10
SQL> /
```

```
      ROWNUM
-----
         1
         2
         3
         4
         5
         6
         7
         8
         9
        10
```

```
10 rows selected.
```

```
SQL> CL SCR
```

```
SQL> SELECT ROWNUM
       2  FROM DUAL;
```

```
      ROWNUM
-----
         1
```

```
SQL> ED
Wrote file afiedt.buf
```

```
   1  SELECT LEVEL
   2  FROM DUAL
   3* CONNECT BY LEVEL <=10
SQL> /
```

```
      LEVEL
-----
        1
        2
        3
        4
        5
        6
        7
        8
        9
       10
```

10 rows selected.

SQL> ED

Wrote file afiedt.buf

```
   1  SELECT LEVEL
   2  FROM DUAL
   3* CONNECT BY LEVEL <=100
SQL> /
```

```
      LEVEL
-----
        1
        2
        3
        4
        5
        6
        7
        8
        9
       10
       11
       12
       13
       14
       15
       16
       17
       18
       19
       20
       21
       22
       23
       24
       25
       26
       27
       28
       29
       30
       31
       32
       33
       34
       35
       36
       37
       38
       39
```

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41
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74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89

90

91
92
93
94

95

96
97

LEVEL

98

99
100

100 rows selected.

SQL> ED

Wrote file afiedt.buf

```
  1  SELECT LEVEL
  2  FROM DUAL
  3  WHERE MOD(LEVEL,2)=0
  4* CONNECT BY LEVEL <=100
SQL> /
```

```
      LEVEL
-----
         2
         4
         6
         8
        10
        12
       14
       16
       18
       20
       22
      24
      26
      28
      30
      32
     34
     36
     38
     40
     42
     44
     46
     48
     50
     52
     54
     56
     58
     60
     62
     64
     66
     68
     70
     72
     74
     76
     78
     80
     82
     84
     86
     88
     90
     92
     94
     96
     98
    100
```

50 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT LEVEL
2  FROM DUAL
3  WHERE MOD(LEVEL,2)=0
4* CONNECT BY LEVEL <=10
SQL> /
```

```
      LEVEL
-----
         2
         4
         6
         8
        10
```

SQL> ED

Wrote file afiedt.buf

```
1  SELECT LEVEL
2  FROM DUAL
3  WHERE MOD(LEVEL,2)=0
4  CONNECT BY LEVEL <=10
5  UNION
6  SELECT LEVEL
7  FROM DUAL
8  WHERE MOD(LEVEL,2)=1
9* CONNECT BY LEVEL <=10
SQL> /
```

```
      LEVEL
-----
         1
         2
         3
         4
         5
         6
         7
         8
         9
        10
```

10 rows selected.

SQL> ED

Wrote file afiedt.buf

```
1  SELECT LEVEL
2  FROM DUAL
3  WHERE MOD(LEVEL,2)=0
4  CONNECT BY LEVEL <=10
5  UNION ALL
6  SELECT LEVEL
7  FROM DUAL
8  WHERE MOD(LEVEL,2)=1
9* CONNECT BY LEVEL <=10
SQL> /
```

```
      LEVEL
```



```
-----  
      2  
      4  
      6  
      8  
     10  
      1  
      3  
      5  
      7  
      9
```

10 rows selected.

SQL> SPOOL OFF